

Curvature-Controlled Moiré Scaling and a Microscopic Magic–No-Magic Dichotomy in Hyperbolic Twisted Bilayers

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We formulate a microscopic hyperbolic twisted bilayer in a circuit-QED resonator network, where the programmed hopping matrix realizes the Bolza surface geometry and twist enters through the interlayer distance. The exact square five-state first-star model provides a Euclidean control and has no positive quadratic magic root. A centered surface-group hyperbolic bilayer instead supports a positive simple root that persists for the full exponential tunneling kernel. In the regular-octagon symmetry class the same coupling cancels the complete Hodge kinetic tensor, linking the microscopic interference pattern to multidirectional dispersion suppression. The non-Abelian representation sectors determine the complete finite-cover spectrum and separate local kinetic cancellation from global bandwidth, isolation, density-of-states, and coherence observables. Finite-cover no-loss, no-pollution, and cross-cover estimates connect these observables to the infinite hyperbolic operator. The resulting spectroscopy and photon-mediated interaction signatures provide direct experimental probes of curvature-controlled, non-Abelian magic in superconducting resonator arrays.

I. PHYSICAL PLATFORM AND MICROSCOPIC HAMILTONIAN

We realize the negatively curved twisted bilayer as a passive superconducting microwave network whose elementary degrees of freedom are bosonic normal modes of coplanar-waveguide resonators. This choice is essential for the physical interpretation of the construction. Networks of superconducting resonators have already been used to realize tight-binding graphs with effective negative curvature, and recent circuit architectures encode the target hyperbolic geometry directly in a designed capacitive coupling network [1–3]. Large coplanar-waveguide lattices have also been integrated with multiple transmon qubits, demonstrating that the same hardware family can support spectroscopic measurements and photon-mediated interactions [4]. The Euclidean chip layout realizes the negatively curved model through its one-particle hopping matrix. A resonator represents a lattice site, a capacitor or parametric coupler represents a hopping edge, and the programmed coupling matrix represents the metric and connectivity of the target hyperbolic lattice. The Euclidean embedding of the metal traces is therefore an engineering layout; the effective geometry is determined by the graph, the mode frequencies, and the calibrated coupling amplitudes. The construction below separates four logically distinct structures: the circuit Lagrangian, the hyperbolic single-layer graph, the synthetic relative twist, and the resulting bilayer hopping matrix. This separation allows us to identify every twist-dependent spectral change with a specified microscopic circuit element rather than with a phenomenological post-processing parameter. Throughout this section, hopping parameters such as t and w are angular frequencies. The physical Hamiltonian is written as $\hat{H} = \hbar\hat{\mathcal{H}}$,

where $\hat{\mathcal{H}}$ has units of angular frequency.

A. Superconducting resonator lattice

Consider one superconducting coplanar-waveguide resonator labelled by a composite index

$$\alpha = (\ell, i), \quad \ell \in \{1, 2\},$$

where ℓ is the synthetic layer and i is the lattice-site label. Let x denote the coordinate along the transmission line. The distributed flux field is expanded in normal modes,

$$\Phi_\alpha(x, t) = \sum_{n \geq 0} u_{\alpha n}(x) \phi_{\alpha n}(t). \quad (1)$$

The mode functions satisfy the transmission-line eigenvalue problem

$$-\frac{d}{dx} \left[\frac{1}{l_\alpha(x)} \frac{du_{\alpha n}}{dx} \right] = \omega_{\alpha n}^2 c_\alpha(x) u_{\alpha n}(x), \quad (2)$$

where $c_\alpha(x)$ and $l_\alpha(x)$ are the capacitance and inductance per unit length. We normalize the selected mode so that

$$C_\alpha = \int dx c_\alpha(x) u_\alpha^2(x), \quad (3)$$

and

$$\frac{1}{L_\alpha} = \int dx \frac{1}{l_\alpha(x)} \left[\frac{du_\alpha(x)}{dx} \right]^2. \quad (4)$$

The isolated-mode Lagrangian is therefore

$$\mathcal{L}_\alpha^{(0)} = \frac{C_\alpha}{2} \dot{\phi}_\alpha^2 - \frac{1}{2L_\alpha} \phi_\alpha^2, \quad (5)$$

with bare angular frequency

$$\omega_\alpha = \frac{1}{\sqrt{L_\alpha C_\alpha}}. \quad (6)$$

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The single-mode truncation is controlled by the separation Δ_{mode} from the nearest unused resonator mode. If J_{max} denotes the largest hopping scale and z_{eff} the largest number of appreciable couplings emerging from one resonator, we require

$$z_{\text{eff}} J_{\text{max}} \ll \Delta_{\text{mode}}. \quad (7)$$

Under this condition, virtual transitions to unused CPW modes produce only perturbative corrections of order

$$\|\delta h_{\text{multimode}}\| = O\left(\frac{z_{\text{eff}}^2 J_{\text{max}}^2}{\Delta_{\text{mode}}}\right). \quad (8)$$

Let the selected resonator modes be collected into the real flux vector $\phi = (\phi_1, \dots, \phi_{N_{\text{tot}}})^T$. A coupling capacitor $C_{\alpha\beta}^c \geq 0$ placed between voltage antinodes of modes α and β contributes

$$\mathcal{L}_{\alpha\beta}^c = \frac{C_{\alpha\beta}^c}{2} \left(\nu_{\alpha\beta}^{(\alpha)} \dot{\phi}_\alpha - \nu_{\alpha\beta}^{(\beta)} \dot{\phi}_\beta \right)^2. \quad (9)$$

The dimensionless factors $\nu_{\alpha\beta}^{(\alpha)}$ and $\nu_{\alpha\beta}^{(\beta)}$ are the normalized values of the two mode functions at the connection points. Their product determines the sign of the effective hopping. For coupling at equivalent voltage antinodes, both factors may be set to unity. Absorbing the mode-function factors into an effective coupling capacitance, the complete quadratic circuit Lagrangian is

$$\mathcal{L} = \frac{1}{2} \dot{\phi}^T \mathbf{C} \dot{\phi} - \frac{1}{2} \phi^T \mathbf{L}^{-1} \phi, \quad (10)$$

where $\mathbf{L}^{-1} = \text{diag}(L_1^{-1}, \dots, L_{N_{\text{tot}}}^{-1})$ and

$$\mathbf{C} = \text{diag}(C_1, \dots, C_{N_{\text{tot}}}) + \sum_{\alpha < \beta} C_{\alpha\beta}^c (\mathbf{e}_\alpha - \mathbf{e}_\beta)(\mathbf{e}_\alpha - \mathbf{e}_\beta)^T. \quad (11)$$

Because every term in the second contribution is positive semidefinite and every isolated resonator has $C_\alpha > 0$, the capacitance matrix is strictly positive definite: $\mathbf{C} > 0$. The canonical charge vector is

$$\mathbf{Q} = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \mathbf{C} \dot{\phi}. \quad (12)$$

The exact linear-circuit Hamiltonian is consequently

$$H_{\text{circ}} = \frac{1}{2} \mathbf{Q}^T \mathbf{C}^{-1} \mathbf{Q} + \frac{1}{2} \phi^T \mathbf{L}^{-1} \phi. \quad (13)$$

This expression is exact within the selected distributed-mode subspace. The tight-binding Hamiltonian follows from a controlled weak-coupling expansion rather than from an assumed graph model. Define the loaded modal capacitance

$$\bar{C}_\alpha = C_\alpha + \sum_{\beta \neq \alpha} C_{\alpha\beta}^c. \quad (14)$$

Write

$$\mathbf{C} = \mathbf{D} - \mathbf{C}_{\text{off}}, \quad \mathbf{D} = \text{diag}(\bar{C}_\alpha), \quad (15)$$

where

$$(\mathbf{C}_{\text{off}})_{\alpha\beta} = \begin{cases} C_{\alpha\beta}^c, & \alpha \neq \beta, \\ 0, & \alpha = \beta. \end{cases} \quad (16)$$

Introduce the dimensionless loading parameter

$$\epsilon_C = \left\| \mathbf{D}^{-1/2} \mathbf{C}_{\text{off}} \mathbf{D}^{-1/2} \right\|. \quad (17)$$

For $\epsilon_C < 1$, the Neumann expansion gives

$$\mathbf{C}^{-1} = \mathbf{D}^{-1} + \mathbf{D}^{-1} \mathbf{C}_{\text{off}} \mathbf{D}^{-1} + O(\epsilon_C^2). \quad (18)$$

Hence, for $\alpha \neq \beta$,

$$(\mathbf{C}^{-1})_{\alpha\beta} = \frac{C_{\alpha\beta}^c}{\bar{C}_\alpha \bar{C}_\beta} + O(\epsilon_C^2). \quad (19)$$

We compensate the diagonal capacitive loading by choosing

$$L_\alpha = \frac{1}{\omega_0^2 \bar{C}_\alpha}, \quad (20)$$

or, equivalently, by adjusting the physical resonator length. The uncoupled loaded frequency is then the common value ω_0 . Residual fabrication errors are retained later as explicit onsite detunings rather than being hidden inside the nominal hopping matrix. The loaded impedance is

$$Z_\alpha = \sqrt{\frac{L_\alpha}{\bar{C}_\alpha}} = \frac{1}{\omega_0 \bar{C}_\alpha}. \quad (21)$$

Canonical quantization is performed with

$$\hat{\phi}_\alpha = \sqrt{\frac{\hbar Z_\alpha}{2}} (\hat{a}_\alpha + \hat{a}_\alpha^\dagger), \quad (22)$$

and

$$\hat{Q}_\alpha = -i \sqrt{\frac{\hbar}{2 Z_\alpha}} (\hat{a}_\alpha - \hat{a}_\alpha^\dagger). \quad (23)$$

The photon operators obey

$$[\hat{a}_\alpha, \hat{a}_\beta^\dagger] = \delta_{\alpha\beta}, \quad [\hat{a}_\alpha, \hat{a}_\beta] = 0. \quad (24)$$

Substituting Eq. (23) into the off-diagonal part of Eq. (13), and retaining the number-conserving terms, gives

$$\frac{\hat{H}_{\text{RWA}}}{\hbar} = \sum_\alpha \omega_0 \hat{a}_\alpha^\dagger \hat{a}_\alpha + \sum_{\alpha < \beta} J_{\alpha\beta} (\hat{a}_\alpha^\dagger \hat{a}_\beta + \hat{a}_\beta^\dagger \hat{a}_\alpha) + O(\epsilon_C^2 \omega_0), \quad (25)$$

with

$$J_{\alpha\beta} = \frac{(\mathbf{C}^{-1})_{\alpha\beta}}{2\sqrt{Z_\alpha Z_\beta}} = \frac{\omega_0 C_{\alpha\beta}^c}{2\sqrt{\bar{C}_\alpha \bar{C}_\beta}} + O(\epsilon_C^2 \omega_0). \quad (26)$$

For nonidentical resonators, the leading expression becomes

$$J_{\alpha\beta} = \frac{1}{2} \sqrt{\omega_\alpha \omega_\beta} \frac{C_{\alpha\beta}^c}{\sqrt{C_\alpha C_\beta}} + O(\epsilon_C^2 \omega_0). \quad (27)$$

Equation (26) gives the direct hardware-to-Hamiltonian calibration. For identical loaded capacitances, the capacitor required to implement a target hopping $J_{\alpha\beta}^{\text{tar}}$ is

$$C_{\alpha\beta}^c = \frac{2\bar{C}}{\omega_0} |J_{\alpha\beta}^{\text{tar}}| + O\left[\bar{C} \left(\frac{J_{\alpha\beta}^{\text{tar}}}{\omega_0}\right)^2\right]. \quad (28)$$

The sign of the hopping may be selected through the signs of the resonator mode functions at the connection points, through a parametric coupler, or through a sublattice gauge transformation when the target graph is bipartite. The rotating-wave approximation is controlled by

$$\epsilon_{\text{RWA}} = \frac{z_{\text{eff}} J_{\text{max}}}{\omega_0} \ll 1. \quad (29)$$

Counter-rotating processes then generate corrections of order

$$\|\delta h_{\text{CR}}\| = O\left(\frac{z_{\text{eff}}^2 J_{\text{max}}^2}{\omega_0}\right). \quad (30)$$

The low-disorder and weak-coupling regime required here is compatible with existing superconducting resonator arrays [1, 3, 5]. For one mode per site, the baseline single-layer Hamiltonian is

$$\frac{\hat{H}_\ell}{\hbar} = \omega_0 \sum_i \hat{a}_{\ell i}^\dagger \hat{a}_{\ell i} - t \sum_{\langle i,j \rangle_\ell} \left(\hat{a}_{\ell i}^\dagger \hat{a}_{\ell j} + \hat{a}_{\ell j}^\dagger \hat{a}_{\ell i} \right), \quad \ell = 1, 2. \quad (31)$$

The sign convention in Eq. (31) fixes $t > 0$. On a bipartite graph, changing the sign of every nearest-neighbour hopping is implemented by the unitary sublattice transformation

$$\hat{a}_{\ell i} \mapsto (-1)^{p(i)} \hat{a}_{\ell i}, \quad p(i) \in \{0, 1\}. \quad (32)$$

The capacitive-circuit sign and the tight-binding sign convention are therefore physically equivalent for the bipartite finite covers selected below. The physical excitations are bosonic photons, not electrons. Nevertheless, the complete one-particle spectral problem is statistics independent. For any number-conserving quadratic Hamiltonian

$$\hat{H} = \hbar \sum_{\alpha, \beta} h_{\alpha\beta} \hat{a}_\alpha^\dagger \hat{a}_\beta, \quad (33)$$

its restriction to the one-photon sector is exactly the matrix h . Replacing the bosonic operators by fermionic operators leaves that matrix unchanged. Consequently, every one-particle operator identity, norm estimate, spectral projector, and finite-cover result established for the

quadratic single-particle matrix remains applicable to the resonator platform. Bosonic statistics become essential only when multiple excitations and nonlinear circuit elements are included; neither is required for the linear band-structure problem defined here.

B. Genus-two regular-octagon/Bolza-surface lattice

We use the Poincaré disk representation of the hyperbolic plane with curvature

$$K = -\frac{1}{R^2} < 0. \quad (34)$$

The metric is

$$ds^2 = \frac{4R^2 |dz|^2}{(1 - |z|^2)^2}, \quad |z| < 1. \quad (35)$$

The primitive compact quotient is obtained by identifying opposite sides of a regular hyperbolic octagon whose interior angles are all $\alpha_8 = \frac{\pi}{4}$. The resulting smooth quotient has genus two. The area follows directly from the hyperbolic polygon formula:

$$\begin{aligned} A_O &= R^2 [(8 - 2)\pi - 8\alpha_8] \\ &= R^2 (6\pi - 2\pi) = 4\pi R^2. \end{aligned}$$

This agrees with Gauss–Bonnet,

$$A_O = 4\pi R^2 (g - 1) = 4\pi R^2, \quad g = 2. \quad (36)$$

Divide the octagon into sixteen congruent right hyperbolic triangles by joining the centre to the vertices and side midpoints. Each triangle has angles $(\frac{\pi}{8}, \frac{\pi}{8}, \frac{\pi}{2})$. Let r_{in} denote the octagon inradius. The hyperbolic right-triangle identity gives

$$\cosh\left(\frac{r_{\text{in}}}{R}\right) = \frac{\cos(\pi/8)}{\sin(\pi/8)} = \cot\left(\frac{\pi}{8}\right) = 1 + \sqrt{2}. \quad (37)$$

Hence

$$r_{\text{in}} = R \operatorname{arcosh}\left(1 + \sqrt{2}\right). \quad (38)$$

The orbit lattice used below places one resonator at the centre of each octagon in the universal $\{8, 8\}$ tessellation. Two neighbouring resonators lie in octagons sharing one side. Their separation is twice the inradius:

$$a_B = 2r_{\text{in}} = 2R \operatorname{arcosh}\left(1 + \sqrt{2}\right). \quad (39)$$

Because $\{8, 8\}$ is self-dual, this centre-to-centre distance is also equal to the edge length of the regular octagon. The dimensionless microscopic curvature of the exact Bolza geometry is therefore fixed:

$$\kappa_B := \frac{a_B}{R} = 2 \operatorname{arcosh}\left(1 + \sqrt{2}\right) \simeq 3.057141839. \quad (40)$$

This identity will be important when defining the curvature scan: an exact regular Bolza octagon does not permit a_B/R to be varied continuously while preserving all of its regularity conditions. Let r_{circ} be the hyperbolic distance from the centre to an octagon vertex. The same right triangle yields

$$\cosh\left(\frac{r_{\text{circ}}}{R}\right) = \cot^2\left(\frac{\pi}{8}\right) = (1 + \sqrt{2})^2. \quad (41)$$

A point at hyperbolic radius r_{circ} has Poincaré-disk radius

$$\begin{aligned} \varrho_v &= \tanh\left(\frac{r_{\text{circ}}}{2R}\right) = \sqrt{\frac{\cosh(r_{\text{circ}}/R) - 1}{\cosh(r_{\text{circ}}/R) + 1}} \\ &= 2^{-1/4}. \end{aligned} \quad (42)$$

We may therefore choose the eight vertices as

$$z_\nu = 2^{-1/4} \exp\left[i\frac{(2\nu+1)\pi}{8}\right], \quad \nu = 0, \dots, 7. \quad (43)$$

Let the centre of the octagon be $o = 0$. A hyperbolic translation along the real diameter that moves the origin through the nearest-neighbour distance a_B is

$$\tau_\eta(z) = \frac{z + \eta}{\eta z + 1}, \quad (44)$$

where

$$\begin{aligned} \eta &= \tanh\left(\frac{a_B}{2R}\right) = \tanh\left[\text{arcosh}\left(1 + \sqrt{2}\right)\right] \\ &= \sqrt{2(\sqrt{2} - 1)}. \end{aligned} \quad (45)$$

Let $r_\varphi(z) = e^{i\varphi}z$ be the centred rotation. The eight maps to neighbouring octagon centres can be represented by

$$\mathbf{s}_\nu = r_{\nu\pi/4} \circ \tau_\eta \circ r_{-\nu\pi/4}, \quad \nu = 0, \dots, 7. \quad (46)$$

They satisfy $\mathbf{s}_{\nu+4} = \mathbf{s}_\nu^{-1}$. Four independent side-pairing generators may be chosen from this set. After the standard Nielsen relabelling, the resulting torsion-free surface group has the presentation

$$\Gamma_B = \langle \mathbf{a}_1, \mathbf{b}_1, \mathbf{a}_2, \mathbf{b}_2 \mid [\mathbf{a}_1, \mathbf{b}_1][\mathbf{a}_2, \mathbf{b}_2] = e \rangle. \quad (47)$$

Define the symmetric first-shell generating set

$$\mathcal{S}_8 = \{\mathbf{a}_1^{\pm 1}, \mathbf{b}_1^{\pm 1}, \mathbf{a}_2^{\pm 1}, \mathbf{b}_2^{\pm 1}\}. \quad (48)$$

The infinite single-layer resonator lattice is the centre orbit

$$\Lambda_B = \{x_\gamma = \gamma o : \gamma \in \Gamma_B\}. \quad (49)$$

Two orbit sites are nearest neighbours precisely when $\gamma^{-1}\gamma' \in \mathcal{S}_8$. Every site has coordination number $z = 8$. The adjacency operator on the universal orbit lattice is

$$A_\infty = \sum_{\mathbf{s} \in \mathcal{S}_8} \lambda_{\Gamma_B}(\mathbf{s}), \quad (50)$$

where λ_{Γ_B} is the left regular representation. Because the generating set is closed under inversion, $A_\infty^\dagger = A_\infty$. An experimental circuit must contain finitely many resonators. We choose a finite-index normal subgroup $\Gamma_N \triangleleft \Gamma_B$ and define the finite quotient

$$Q_N = \Gamma_B/\Gamma_N, \quad N = |Q_N|. \quad (51)$$

One resonator is assigned to every coset $q \in Q_N$. Let $R_N(\mathbf{s})$ denote right multiplication by a generator,

$$R_N(\mathbf{s})|q\rangle = |qs\rangle. \quad (52)$$

The finite adjacency matrix is

$$A_N = \sum_{\mathbf{s} \in \mathcal{S}_8} R_N(\mathbf{s}). \quad (53)$$

The intralayer one-particle matrix is

$$h_{0,N} = \omega_0 I_N - t A_N. \quad (54)$$

Every row of A_N contains eight unit entries, so $\|A_N\| \leq 8$. Consequently,

$$\text{Spec}(h_{0,N}) \subset [\omega_0 - 8|t|, \omega_0 + 8|t|]. \quad (55)$$

The normalized uniform vector

$$|u_+\rangle = \frac{1}{\sqrt{N}} \sum_{q \in Q_N} |q\rangle \quad (56)$$

satisfies

$$A_N|u_+\rangle = 8|u_+\rangle. \quad (57)$$

The infinite Cayley graph is bipartite. To see this explicitly, define the homomorphism $\pi : \Gamma_B \rightarrow \mathbb{Z}_2$ by assigning

$$\pi(\mathbf{a}_1) = \pi(\mathbf{b}_1) = \pi(\mathbf{a}_2) = \pi(\mathbf{b}_2) = 1. \quad (58)$$

The defining surface relation contains an even number of generator letters and therefore maps to zero in $\mathbb{Z}_2$. Hence the map is well defined. We select finite-cover subgroups satisfying $\Gamma_N \subset \ker \pi$. The corresponding quotient graph remains bipartite. The staggered vector

$$|u_-\rangle = \frac{1}{\sqrt{N}} \sum_{q \in Q_N} (-1)^{\pi(q)} |q\rangle \quad (59)$$

then obeys

$$A_N|u_-\rangle = -8|u_-\rangle. \quad (60)$$

Therefore the finite connected bipartite monolayer has the exact single-particle bandwidth $W_{\text{mono}} = 16|t|$. This value supplies a noninteracting kinetic reference scale independent of the later bilayer reconstruction. Each layer contains N resonators and

$$E_{\parallel}^{(1)} = \frac{zN}{2} = 4N \quad (61)$$

undirected intralayer edges. The two-layer device therefore requires $N_{\text{res}} = 2N$ resonators and $N_{\parallel}^c = 8N$ intralayer couplers. If every site couples appreciably to $z_{\perp}$ sites in the opposite layer, the number of interlayer couplers is $N_{\perp}^c = Nz_{\perp}$, and the total coupler count is

$$N_{\text{coup}} = N(8 + z_{\perp}). \quad (62)$$

These formulas convert group-theoretic cover degree directly into hardware complexity. The centred rotation $c_8 : z \mapsto e^{i\pi/4}z$ cyclically permutes the eight first-shell directions. It induces a group automorphism $\varphi_8 : \mathcal{S}_8 \rightarrow \mathcal{S}_8$. We choose quotient subgroups satisfying $\varphi_8(\Gamma_N) = \Gamma_N$. The automorphism then descends to a unitary permutation $\mathcal{R}_{8,N}$ on the finite circuit:

$$\mathcal{R}_{8,N}|q\rangle = |\varphi_8(q)\rangle. \quad (63)$$

Because φ_8 permutes the generator set,

$$\mathcal{R}_{8,N}A_N\mathcal{R}_{8,N}^\dagger = A_N. \quad (64)$$

Thus

$$[h_{0,N}, \mathcal{R}_{8,N}] = 0. \quad (65)$$

If the quotient is also invariant under a reflection of the regular octagon and all couplings are real, the unitary point group is enlarged from C_8 to D_8 . We retain C_8 as the symmetry guaranteed by the centred rotational construction and treat reflection symmetry as an additional device constraint.

C. Synthetic hyperbolic twist

We separate the intrinsic hyperbolic geometry from the synthetic layer direction by introducing

$$\mathcal{M}_R = \mathbb{H}_R^2 \times \mathbb{R}_z \quad (66)$$

with product metric

$$ds_{\mathcal{M}}^2 = ds_{\mathbb{H}_R^2}^2 + dz^2. \quad (67)$$

The two synthetic layers are located at

$$z_1 = -\frac{h}{2}, \quad z_2 = +\frac{h}{2}, \quad (68)$$

where $h > 0$ is an effective layer separation. In a multi-layer chip, h may be related to a geometric separation. In a purely synthetic implementation, h is a calibrated parameter that fixes the normalization and radial profile of the programmed interlayer couplings. Let $g_\theta(z) = e^{i\theta}z$ be the exact centred hyperbolic rotation in the Poincaré disk. The single-layer site coordinates are

$$x_\gamma = \gamma o, \quad \gamma \in \Gamma_B. \quad (69)$$

The two bilayer point sets are

$$\Lambda_1 = \Lambda_B, \quad \Lambda_2(\theta) = g_\theta \Lambda_B. \quad (70)$$

The corresponding product-space coordinates are

$$X_\gamma^{(1)} = \left(x_\gamma, -\frac{h}{2}\right), \quad X_\gamma^{(2)} = \left(g_\theta x_\gamma, +\frac{h}{2}\right). \quad (71)$$

Because g_θ is an isometry,

$$d_{\mathbb{H}}(g_\theta x_\gamma, g_\theta x_{\gamma'}) = d_{\mathbb{H}}(x_\gamma, x_{\gamma'}). \quad (72)$$

Every intralayer bond length is therefore independent of the rigid twist. The isolated second-layer Hamiltonian is unitarily equivalent to the first one. The resonators may be placed at arbitrary convenient Euclidean chip coordinates. We do not identify those coordinates with $x_\gamma \in \mathbb{H}_R^2$. Instead, for each target twist θ , we calculate the desired matrix

$$T_N(\theta) = \left[T_{qq'}^{(N)}(\theta)\right]_{q, q' \in Q_N} \quad (73)$$

from the synthetic hyperbolic distances and then fabricate or program the corresponding couplers through Eq. (28). The operation $\theta \mapsto T_N(\theta)$ is the physical implementation of the twist. No mechanical rotation of a curved object is required. To define the finite-device distance without depending on arbitrary choices of lifts, let $\tilde{q}$ and $\tilde{q}'$ be reference lifts of two quotient sites. We define the paired-cover in-plane separation by

$$d_{N,\theta}(q, q') = \min_{\gamma, \gamma' \in \Gamma_N} d_{\mathbb{H}}(\gamma x_{\tilde{q}}, g_\theta \gamma' x_{\tilde{q}'}). \quad (74)$$

Changing either reference lift only relabels the minimizing pair (γ, γ') , so Eq. (74) is well defined. The corresponding product-space distance is

$$D_{N,\theta}(q, q') = \sqrt{h^2 + d_{N,\theta}^2(q, q')}. \quad (75)$$

For a hopping cutoff D_c , only pairs satisfying

$$D_{N,\theta}(q, q') \leq D_c \quad (76)$$

are connected. If the cover injectivity radius satisfies $r_{\text{inj}}^{(N)} > D_c$, the relevant minimizing image is unique inside every coupling neighbourhood, and the finite circuit reproduces the universal-cover interlayer geometry without wraparound. Let

$$U_\theta : \ell^2(\Lambda_B) \rightarrow \ell^2(g_\theta \Lambda_B) \quad (77)$$

act as

$$U_\theta |x_\gamma\rangle = |g_\theta x_\gamma\rangle. \quad (78)$$

The map is unitary because g_θ is bijective. If h_1 and h_2 are the isolated-layer matrices, distance invariance gives $h_2 = U_\theta h_1 U_\theta^\dagger$. In the original layer coordinates, the one-particle bilayer matrix has the form

$$h_{\text{BL}}^{\text{lab}} = \begin{pmatrix} h_1 & T_\theta \\ T_\theta^\dagger & h_2 \end{pmatrix}. \quad (79)$$

Define the block unitary

$$\mathcal{U}_\theta = \begin{pmatrix} I & 0 \\ 0 & U_\theta^\dagger \end{pmatrix}. \quad (80)$$

The co-rotating Hamiltonian is

$$h_{\text{BL}}(\theta) = \mathcal{U}_\theta h_{\text{BL}}^{\text{lab}} \mathcal{U}_\theta^\dagger = \begin{pmatrix} h_0 & \tau_\theta \\ \tau_\theta^\dagger & h_0 \end{pmatrix}, \quad (81)$$

where

$$h_0 = h_1, \quad \tau_\theta = T_\theta U_\theta. \quad (82)$$

Both diagonal blocks are now identical. Every dependence on the rigid twist is confined to τ_θ :

$$\frac{\partial h_0}{\partial \theta} = 0, \quad \frac{\partial h_{\text{BL}}}{\partial \theta} = \begin{pmatrix} 0 & \partial_\theta \tau_\theta \\ \partial_\theta \tau_\theta^\dagger & 0 \end{pmatrix}. \quad (83)$$

For any two twist angles θ_1 and θ_2 , define

$$\delta\tau = \tau_{\theta_1} - \tau_{\theta_2}. \quad (84)$$

The Hamiltonian difference is

$$\delta h = h_{\text{BL}}(\theta_1) - h_{\text{BL}}(\theta_2) = \begin{pmatrix} 0 & \delta\tau \\ \delta\tau^\dagger & 0 \end{pmatrix}. \quad (85)$$

Squaring gives

$$(\delta h)^2 = \begin{pmatrix} \delta\tau \delta\tau^\dagger & 0 \\ 0 & \delta\tau^\dagger \delta\tau \end{pmatrix}. \quad (86)$$

Therefore

$$\|h_{\text{BL}}(\theta_1) - h_{\text{BL}}(\theta_2)\| = \|\tau_{\theta_1} - \tau_{\theta_2}\|. \quad (87)$$

The complete one-particle spectral reconstruction generated by a rigid twist is exactly equal, in operator norm, to the reconstruction of the interlayer coupling matrix. The regular-octagon lattice is invariant under a rotation by $\pi/4$. For symmetry-compatible finite covers,

$$\text{Spec } h_{\text{BL}}\left(\theta + \frac{\pi}{4}\right) = \text{Spec } h_{\text{BL}}(\theta). \quad (88)$$

Layer exchange and reflection imply

$$\text{Spec } h_{\text{BL}}(-\theta) = \text{Spec } h_{\text{BL}}(\theta). \quad (89)$$

Hence inequivalent centred relative twists may be restricted to $0 \leq \theta \leq \frac{\pi}{8}$. This is the hyperbolic C_8 analogue of the irreducible $0 \leq \theta \leq \pi/4$ interval for two square lattices.

D. Interlayer coupling

The exact physical separation between a first-layer site i and a second-layer site j is

$$D_{ij}(\theta) = \sqrt{h^2 + d_{\mathbb{H}}^2(x_i, g_\theta x_j)}. \quad (90)$$

The minimum possible separation is $D_{\min} = h$, attained by vertically aligned synthetic sites. We choose the baseline interlayer hopping law

$$T_{ij}(\theta) = w \exp\left[-\frac{D_{ij}(\theta) - h}{\lambda_\perp}\right], \quad (91)$$

where $T(h) = w$ and $\lambda_\perp > 0$ is the synthetic tunnelling-decay length. A finite circuit implements

$$T_{ij}^{(D_c)}(\theta) = T_{ij}(\theta) \Theta(D_c - D_{ij}(\theta)), \quad (92)$$

or a smooth compactly supported replacement of the Heaviside cutoff. Using Eq. (28), the corresponding interlayer coupling capacitor is

$$C_{ij}^\perp(\theta) = \frac{2\bar{C}w}{\omega_0} \exp\left[-\frac{D_{ij}(\theta) - h}{\lambda_\perp}\right] + O\left[\bar{C}\left(\frac{w}{\omega_0}\right)^2\right]. \quad (93)$$

Thus the synthetic hyperbolic metric is converted into an explicit fabrication or programming rule for each interlayer coupler. For identical resonators, the ratio of interlayer to intralayer coupling capacitors is, to leading order,

$$\frac{C_{ij}^\perp(\theta)}{C_{\parallel}^c} = \frac{w}{t} \exp\left[-\frac{D_{ij}(\theta) - h}{\lambda_\perp}\right]. \quad (94)$$

This ratio is independent of the common loaded resonator capacitance and therefore provides a direct dimensionless hardware calibration. Let a reference site lie at intrinsic radial distance $r = d_{\mathbb{H}}(o, x)$. For the same site label in the two layers, the exact twist-induced in-plane separation is

$$d_\theta(r) = 2R \operatorname{arsinh}\left[\sinh\left(\frac{r}{R}\right) \left|\sin\left(\frac{\theta}{2}\right)\right|\right]. \quad (95)$$

The product-space separation is

$$D_{\text{loc}}(r, \theta) = \sqrt{h^2 + 4R^2 \operatorname{arsinh}^2\left[\sinh\left(\frac{r}{R}\right) \left|\sin\left(\frac{\theta}{2}\right)\right|\right]}. \quad (96)$$

The same-label hopping profile is therefore

$$T_{\text{loc}}(r, \theta) = w \exp\left\{-\frac{1}{\lambda_\perp} [D_{\text{loc}}(r, \theta) - h]\right\}. \quad (97)$$

This is an explicit microscopic transfer of the hyperbolic twist geometry into a circuit hopping amplitude. Define

$$u_r = \sinh\left(\frac{r}{R}\right). \quad (98)$$

For small signed twist,

$$d_\theta(r) = Ru_r\theta - \frac{Ru_r(1 + u_r^2)}{24}\theta^3 + O(\theta^5). \quad (99)$$

Hence

$$d_\theta^2(r) = R^2 u_r^2 \theta^2 - \frac{R^2 u_r^2 (1 + u_r^2)}{12} \theta^4 + O(\theta^6). \quad (100)$$

Expanding the product-space distance gives

$$D_{\text{loc}}(r, \theta) = h + A_2(r)\theta^2 + A_4(r)\theta^4 + O(\theta^6), \quad (101)$$

where

$$A_2(r) = \frac{R^2 u_r^2}{2h}, \quad (102)$$

and

$$A_4(r) = -\frac{R^2 u_r^2(1 + u_r^2)}{24h} - \frac{R^4 u_r^4}{8h^3}. \quad (103)$$

Substitution into the exponential kernel yields

$$\begin{aligned} \frac{T_{\text{loc}}(r, \theta)}{w} &= 1 - \frac{A_2(r)}{\lambda_{\perp}} \theta^2 \\ &+ \left[-\frac{A_4(r)}{\lambda_{\perp}} + \frac{A_2^2(r)}{2\lambda_{\perp}^2} \right] \theta^4 + O(\theta^6). \end{aligned} \quad (104)$$

In particular,

$$\left. \frac{\partial T_{\text{loc}}}{\partial \theta} \right|_{\theta=0} = 0, \quad (105)$$

and

$$\left. \frac{\partial^2 T_{\text{loc}}}{\partial \theta^2} \right|_{\theta=0} = -\frac{wR^2}{h\lambda_{\perp}} \sinh^2\left(\frac{r}{R}\right). \quad (106)$$

The leading twist response of a scalar distance-dependent interlayer coupling is quadratic, spatially nonuniform, and exponentially amplified with hyperbolic radius. For a Euclidean bilayer, the corresponding result is

$$\left. \frac{\partial^2 T_{\text{loc}}^{(E)}}{\partial \theta^2} \right|_{\theta=0} = -\frac{wr^2}{h\lambda_{\perp}}. \quad (107)$$

The exact curvature-amplification factor is therefore

$$\mathcal{A}_{\text{tw}}(r; R) = \frac{\partial_{\theta}^2 T_{\text{loc}}^{(H)}|_0}{\partial_{\theta}^2 T_{\text{loc}}^{(E)}|_0} = \left[\frac{R \sinh(r/R)}{r} \right]^2. \quad (108)$$

Its weak-curvature expansion is

$$\mathcal{A}_{\text{tw}}(r; R) = 1 + \frac{r^2}{3R^2} + \frac{2r^4}{45R^4} + O\left(\frac{r^6}{R^6}\right), \quad (109)$$

whereas for $r/R \gg 1$,

$$\mathcal{A}_{\text{tw}}(r; R) \sim \frac{e^{2r/R}}{4(r/R)^2}. \quad (110)$$

The leading mixed curvature–twist correction follows by subtracting the Euclidean response:

$$T_{\text{loc}}^{(H)} - T_{\text{loc}}^{(E)} = -\frac{wr^4}{6h\lambda_{\perp}R^2} \theta^2 + O\left(\frac{wr^6\theta^2}{h\lambda_{\perp}R^4}\right) + O(\theta^4). \quad (111)$$

Equation (111) provides a direct perturbative benchmark for the circuit implementation: after the bare hopping parameters are held fixed, the first non-Euclidean change of the same-label interlayer coupling occurs at order $|K|r^4\theta^2$. The exponential kernel is a concrete baseline rather than an assumed universal law. Let

$$T(D) = wf(D), \quad f(h) = 1, \quad (112)$$

where f is smooth and monotonically decreasing near $D = h$. Define the dimensionless logarithmic slope

$$\zeta_{\perp} = -h \frac{f'(h)}{f(h)} = -hf'(h) > 0. \quad (113)$$

Taylor expansion gives

$$\frac{T_{\text{loc}}(r, \theta) - w}{w} = -\frac{\zeta_{\perp}}{2h^2} R^2 \sinh^2\left(\frac{r}{R}\right) \theta^2 + O(\theta^4). \quad (114)$$

Thus every smooth radial profile with the same vertical amplitude w and the same logarithmic slope $\zeta_{\perp}$ has the same leading geometry-controlled twist response. Dependence on $f''(h)$ first enters at quartic order. This provides a precise separation between the universal geometric susceptibility and nonuniversal short-distance profile curvature. For the exponential kernel,

$$f(D) = e^{-(D-h)/\lambda_{\perp}}, \quad \zeta_{\perp} = \frac{h}{\lambda_{\perp}}. \quad (115)$$

Equation (114) then reproduces Eq. (106). The number of orbit sites in a hyperbolic shell grows exponentially. For a cocompact lattice in $\mathbb{H}_R^2$, there exist finite constants $C_{\Gamma} > 0$ and $\rho > 0$ such that

$$\#\{j : n\rho \leq d_{\mathbb{H}}(x_i, x_j) < (n+1)\rho\} \leq C_{\Gamma} e^{n\rho/R}. \quad (116)$$

For an exponentially decaying kernel

$$|T(D)| \leq we^{-\mu(D-h)}, \quad \mu = \lambda_{\perp}^{-1}, \quad (117)$$

the interlayer row sum satisfies

$$\begin{aligned} \sum_j |T_{ij}| &\leq C_{\Gamma} w \sum_{n=0}^{\infty} e^{n\rho/R} e^{-\mu n\rho} \\ &= C_{\Gamma} w \sum_{n=0}^{\infty} e^{-(\mu - R^{-1})n\rho}. \end{aligned} \quad (118)$$

A sufficient condition for convergence is

$$\mu > \frac{1}{R} \iff \lambda_{\perp} < R. \quad (119)$$

Under this condition, the Schur test gives

$$\|\tau_{\theta}\| \leq \sup_i \sum_j |T_{ij}| < \infty. \quad (120)$$

Let $L > D_{\min}$ be a geodesic in-plane cutoff. The omitted tail obeys

$$\mathcal{E}_{\perp}(L) := \sup_i \sum_{j: d_{\mathbb{H}}(x_i, g_{\theta} x_j) > L} |T_{ij}|. \quad (121)$$

Using the shell estimate gives

$$\mathcal{E}_\perp(L) \leq \frac{C_\Gamma w e^{-(\mu-R^{-1})L}}{1 - e^{-(\mu-R^{-1})\rho}}. \quad (122)$$

Hence the truncated and full operators satisfy

$$\left\| \tau_\theta^{(\infty)} - \tau_\theta^{(L)} \right\| \leq \mathcal{E}_\perp(L). \quad (123)$$

The finite-range circuit therefore approaches the full exponential model at an explicitly controlled exponential rate whenever $\lambda_\perp < R$.

E. Complete Hamiltonian

For a finite quotient containing N sites per layer, collect the annihilation operators into

$$\hat{\mathbf{a}} = (\hat{a}_{1,1}, \dots, \hat{a}_{1,N}, \hat{a}_{2,1}, \dots, \hat{a}_{2,N})^\top. \quad (124)$$

The complete lossless baseline Hamiltonian is

$$\begin{aligned} \frac{\hat{H}(\theta)}{\hbar} = & \omega_0 \sum_{\ell=1}^2 \sum_{i=1}^N \hat{a}_{\ell i}^\dagger \hat{a}_{\ell i} \\ & - t \sum_{\ell=1}^2 \sum_{\langle i,j \rangle_\ell} \left(\hat{a}_{\ell i}^\dagger \hat{a}_{\ell j} + \hat{a}_{\ell j}^\dagger \hat{a}_{\ell i} \right) \\ & + \sum_{i,j=1}^N \left[T_{ij}(\theta) \hat{a}_{1i}^\dagger \hat{a}_{2j} + T_{ij}^*(\theta) \hat{a}_{2j}^\dagger \hat{a}_{1i} \right]. \end{aligned} \quad (125)$$

Equivalently,

$$\hat{H}(\theta) = \hbar \hat{\mathbf{a}}^\dagger h_N(\theta) \hat{\mathbf{a}}, \quad (126)$$

where

$$h_N(\theta) = \omega_0 I_{2N} + \begin{pmatrix} -tA_N & T_N(\theta) \\ T_N^\dagger(\theta) & -tA_N \end{pmatrix}. \quad (127)$$

Equation (127) is the microscopic Hamiltonian used in all subsequent physical calculations. The Hamiltonian is Hermitian:

$$h_N^\dagger(\theta) = h_N(\theta). \quad (128)$$

The total photon number

$$\hat{N}_{\text{ph}} = \sum_{\ell,i} \hat{a}_{\ell i}^\dagger \hat{a}_{\ell i} \quad (129)$$

is conserved:

$$[\hat{H}, \hat{N}_{\text{ph}}] = 0. \quad (130)$$

For real couplings, complex conjugation defines spinless time reversal,

$$\Theta = \mathcal{K}, \quad \Theta^2 = +1, \quad (131)$$

and

$$\Theta h_N(\theta) \Theta^{-1} = h_N(\theta). \quad (132)$$

The baseline model therefore carries no synthetic magnetic flux and no nonzero Chern number. Complex phase-engineered couplings may be added as an independent extension, but they are not required for the curvature-controlled hybridization mechanism. At perfect alignment, $\theta = 0$. For a real radial interlayer profile, $T_N(0) = T_N^\dagger(0)$. Define the layer bonding–antibonding unitary

$$W_L = \frac{1}{\sqrt{2}} \begin{pmatrix} I_N & I_N \\ I_N & -I_N \end{pmatrix}. \quad (133)$$

Direct multiplication gives

$$W_L^\dagger h_N(0) W_L = \begin{pmatrix} \omega_0 I_N - tA_N + T_N(0) & 0 \\ 0 & \omega_0 I_N - tA_N - T_N(0) \end{pmatrix}. \quad (134)$$

Thus the aligned device separates exactly into layer-even and layer-odd sectors. For the purely vertical benchmark $T_N(0) = wI_N$, if

$$A_N |u_n\rangle = \lambda_n |u_n\rangle, \quad (135)$$

then the bilayer frequencies are

$$\omega_{n,\pm} = \omega_0 - t\lambda_n \pm w. \quad (136)$$

Equation (136) is an exact implementation benchmark: the aligned numerical spectrum must consist of two rigidly shifted copies of the monolayer spectrum whenever the interlayer matrix is wI_N . Let

$$P_L = \begin{pmatrix} 0 & I_N \\ I_N & 0 \end{pmatrix} \quad (137)$$

exchange the two layers. Distance invariance implies

$$T_N(-\theta) = T_N^\top(\theta) \quad (138)$$

for a real distance-dependent kernel. Therefore

$$P_L h_N(\theta) P_L^{-1} = h_N(-\theta). \quad (139)$$

The spectra are even:

$$\text{Spec } h_N(\theta) = \text{Spec } h_N(-\theta). \quad (140)$$

Every nondegenerate eigenfrequency is consequently an even function of the signed twist in a neighbourhood of alignment. Let

$$h_N(\theta) |u_n(\theta)\rangle = \omega_n(\theta) |u_n(\theta)\rangle \quad (141)$$

define a normalized simple eigenmode. Decompose it into layer components,

$$|u_n\rangle = \begin{pmatrix} |u_{n1}\rangle \\ |u_{n2}\rangle \end{pmatrix}. \quad (142)$$

The Hellmann–Feynman identity applied to the off-diagonal twist derivative gives

$$\frac{d\omega_n}{d\theta} = 2 \operatorname{Re} \langle u_{n1} | \partial_\theta T_N | u_{n2} \rangle. \quad (143)$$

The first derivative of every eigenfrequency is therefore an explicitly interlayer hybridization observable. The second derivative is

$$\frac{d^2\omega_n}{d\theta^2} = \langle u_n | \partial_\theta^2 h_N | u_n \rangle + 2 \sum_{m \neq n} \frac{|\langle u_m | \partial_\theta h_N | u_n \rangle|^2}{\omega_n - \omega_m}.$$

The two terms have distinct physical meanings. The first is the direct curvature of the programmed coupling matrix. The second is the virtual hybridization with every remote bilayer mode. A local quadratic flattening condition therefore need not be controlled by a single two-level crossing; it generally contains the complete remote-mode spectrum. At alignment, a real radial coupling is even in θ , so

$$\partial_\theta T_N|_{\theta=0} = 0. \quad (144)$$

Consequently,

$$\left. \frac{d\omega_n}{d\theta} \right|_{\theta=0} = 0, \quad (145)$$

and

$$\left. \frac{d^2\omega_n}{d\theta^2} \right|_{\theta=0} = 2 \operatorname{Re} \langle u_{n1}(0) | \partial_\theta^2 T_N |_{\theta=0} | u_{n2}(0) \rangle. \quad (146)$$

The spatially resolved susceptibility Eq. (106) therefore enters the measurable frequency curvature directly. The eigenvalue equation associated with Eq. (127) is

$$(h_0 - \omega) |u_1\rangle + T_N |u_2\rangle = 0, \quad (147)$$

and

$$T_N^\dagger |u_1\rangle + (h_0 - \omega) |u_2\rangle = 0, \quad (148)$$

where

$$h_0 = \omega_0 I_N - t A_N. \quad (149)$$

Whenever $\omega \notin \operatorname{Spec}(h_0)$, Eq. (148) gives

$$|u_2\rangle = (\omega - h_0)^{-1} T_N^\dagger |u_1\rangle. \quad (150)$$

Substitution into Eq. (147) yields the exact energy-dependent effective problem

$$[h_0 + \Sigma_1(\omega, \theta)] |u_1\rangle = \omega |u_1\rangle, \quad (151)$$

with interlayer self-energy

$$\Sigma_1(\omega, \theta) = T_N(\theta) (\omega - h_0)^{-1} T_N^\dagger(\theta). \quad (152)$$

Similarly,

$$\Sigma_2(\omega, \theta) = T_N^\dagger(\theta) (\omega - h_0)^{-1} T_N(\theta). \quad (153)$$

The self-energy satisfies

$$\|\Sigma_1(\omega, \theta)\| \leq \frac{\|T_N(\theta)\|^2}{\operatorname{dist}[\omega, \operatorname{Spec}(h_0)]}. \quad (154)$$

This formula identifies the two ingredients required for strong spectral reconstruction: appreciable interlayer matrix elements and a small energy denominator associated with resonant monolayer modes. The twist changes both the matrix structure and the selection rules of $T_N(\theta)$, while the monolayer spectrum remains fixed under a rigid rotation. The exact causal chain of the microscopic construction is therefore

$$(K, \theta) \longrightarrow d_{\mathbb{H}} \longrightarrow D_{ij}(\theta) \longrightarrow C_{ij}^\perp(\theta) \longrightarrow T_N(\theta) \longrightarrow h_N(\theta). \quad (155)$$

F. Parameter space and controlled comparison protocols

Let the microscopic resonator spacing a and intralayer hopping t define the reference length and frequency scales. Introduce

$$\kappa_a = a\sqrt{|K|} = \frac{a}{R}, \quad \omega = \frac{w}{t}, \quad \eta_\perp = \frac{\lambda_\perp}{a}, \quad \eta_h = \frac{h}{a}. \quad (156)$$

For normalized in-plane distance

$$\bar{d}_{ij} = \frac{d_{\mathbb{H}}(x_i, g_\theta x_j)}{a}, \quad (157)$$

the dimensionless interlayer matrix is

$$[\mathbb{T}_N(\theta, \kappa_a, \eta_h, \eta_\perp)]_{ij} = \exp \left[-\frac{\sqrt{\eta_h^2 + \bar{d}_{ij}^2} - \eta_h}{\eta_\perp} \right]. \quad (158)$$

The normalized one-particle Hamiltonian is

$$\frac{h_N - \omega_0 I_{2N}}{t} = \begin{pmatrix} -A_N & \omega \mathbb{T}_N \\ \omega \mathbb{T}_N^\dagger & -A_N \end{pmatrix}. \quad (159)$$

The principal parameter set is therefore

$$\mathcal{P}_{\text{main}} = \left(\theta, K, \frac{w}{t}, \frac{\lambda_\perp}{a} \right), \quad (160)$$

with $\eta_h = h/a$ held fixed in the primary scan. Equivalently, the curvature may be represented by the dimensionless variable $\kappa_a = a\sqrt{|K|}$. An overall rescaling

$$(t, w, \delta, \kappa) \longmapsto c(t, w, \delta, \kappa) \quad (161)$$

rescales all detunings from ω_0 by c and changes the physical time unit $\tau_t = t^{-1}$, but it does not change the

eigenvectors or any dimensionless band geometry. Comparisons of flattening must therefore use scale-invariant quantities rather than an unnormalized bandwidth alone. For the exact regular-octagon/Bolza lattice, Eq. (40) fixes $\kappa_a = \kappa_B \simeq 3.057141839$. Consequently, varying K at fixed a while claiming to preserve an exact regular Bolza octagon would be inconsistent. We therefore separate two protocols. The exact-Bolza protocol is

$$\mathcal{P}_B = \left\{ \kappa_a = \kappa_B, \quad 0 \leq \theta \leq \frac{\pi}{8}, \quad \omega > 0, \quad \eta_\perp > 0, \quad \eta_h = \eta_h^{(0)} \right\}. \quad (162)$$

All statements that use exact Bolza symmetry are evaluated on this slice. To isolate a continuous curvature response, we additionally define a synthetic metric-deformation protocol. The site labels and finite circuit connectivity are held fixed, but the target distances entering Eq. (93) are recalculated using a one-parameter family of constant-curvature metrics: $0 \leq \kappa_a \leq \kappa_B$. The endpoint $\kappa_a = 0$ is the Euclidean control and $\kappa_a = \kappa_B$ is the exact Bolza metric. Intermediate members are synthetic metric interpolations implemented by coupling weights; they are not claimed to be distinct exact regular Bolza surfaces. This distinction prevents a continuous analog-simulation parameter from being conflated with an exact regular-tessellation identity. Under the bare-fixed curvature protocol, the microscopic circuit scales are held fixed:

$$\frac{dt}{dK} = \frac{dw}{dK} = \frac{d\lambda_\perp}{dK} = \frac{dh}{dK} = 0. \quad (163)$$

Every change of T_{ij} then arises from the curvature dependence of the programmed metric distance rather than from an explicit curvature-dependent change of the bare coupling law. Residual resonator-frequency disorder is represented by

$$\hat{H}_{\text{onsite}} = \hbar \sum_{\ell,i} \delta_{\ell i} \hat{a}_{\ell i}^\dagger \hat{a}_{\ell i}, \quad |\delta_{\ell i}| \leq \sigma_\omega. \quad (164)$$

Intralayer coupling disorder is written as

$$t_{ij} = t \left(1 + \varepsilon_{ij}^\parallel \right), \quad |\varepsilon_{ij}^\parallel| \leq \sigma_t, \quad (165)$$

and interlayer coupling disorder as

$$T_{ij} = T_{ij}^{(0)} \left(1 + \varepsilon_{ij}^\perp \right), \quad |\varepsilon_{ij}^\perp| \leq \sigma_\perp. \quad (166)$$

A uniform layer detuning is introduced by

$$\hat{H}_\Delta = \frac{\hbar \Delta_L}{2} \left(\hat{N}_1 - \hat{N}_2 \right), \quad (167)$$

where

$$\hat{N}_\ell = \sum_i \hat{a}_{\ell i}^\dagger \hat{a}_{\ell i}. \quad (168)$$

The corresponding one-particle matrix is

$$h_N(\theta, \Delta_L) = \omega_0 I_{2N} + \begin{pmatrix} -tA_N + \frac{\Delta_L}{2} I_N & T_N(\theta) \\ T_N^\dagger(\theta) & -tA_N - \frac{\Delta_L}{2} I_N \end{pmatrix}. \quad (169)$$

Photon loss is described by the Lindblad equation

$$\frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar} [\hat{H}, \hat{\rho}] + \sum_{\ell,i} \kappa_{\ell i} \mathcal{D}[\hat{a}_{\ell i}] \hat{\rho}, \quad (170)$$

where

$$\mathcal{D}[\hat{L}] \hat{\rho} = \hat{L} \hat{\rho} \hat{L}^\dagger - \frac{1}{2} \left\{ \hat{L}^\dagger \hat{L}, \hat{\rho} \right\}. \quad (171)$$

Linear-response poles are governed by the effective non-Hermitian matrix

$$h_{\text{eff}} = h_N - \frac{i}{2} \text{diag}(\kappa_{1,1}, \dots, \kappa_{2,N}). \quad (172)$$

Uniform loss shifts every pole by the same imaginary part and does not change the right eigenvectors. Nonuniform loss can modify mode profiles and is therefore retained as an explicit secondary perturbation. The finite-device parameters are

$$N = |Q_N|, \quad r_{\text{inj}}^{(N)}, \quad D_c. \quad (173)$$

The one-particle matrix dimension is $\dim h_N = 2N$. The local universal-cover hopping geometry is uncontaminated by periodic wraparound whenever $r_{\text{inj}}^{(N)} > D_c$. The complete scale hierarchy required by the single-mode, number-conserving tight-binding description is

$$\max \{8|t|, z_\perp |w|, |\Delta_L|, \sigma_\omega, \kappa_{\text{max}}\} \ll \min \{\omega_0, \Delta_{\text{mode}}\}. \quad (174)$$

For the infinite exponential model, we additionally impose

$$\eta_\perp = \frac{\lambda_\perp}{a} < \frac{R}{a} = \frac{1}{\kappa_a}. \quad (175)$$

At the exact Bolza point,

$$\eta_\perp < \kappa_B^{-1} \simeq 0.327102913 \quad (176)$$

is a sufficient thermodynamic summability condition. A finite-range device does not require this inequality for matrix boundedness, but the inequality is required if the device sequence is intended to converge to the untruncated exponential operator. We use four comparison protocols to prevent a trivial energy-scale change from being misidentified as a curvature effect. The primary bare-fixed protocol keeps t , w , h , $\lambda_\perp$ constant while changing only the target geometry. If lattices with different coordination numbers are compared, the fixed-local-second-moment control uses

$$t(z) = \frac{t_{\text{rms}}}{\sqrt{z}}, \quad (177)$$

so that

$$\langle i | (h_0 - \omega_0 I)^2 | i \rangle = t_{\text{rms}}^2 \quad (178)$$

is independent of z . The fixed-total-hybridization control replaces

$$T_N(\theta, K) \mapsto \frac{\mathcal{T}_0}{\|T_N(\theta, K)\|_F / \sqrt{N}} T_N(\theta, K), \quad (179)$$

where $\mathcal{T}_0$ is a fixed reference root-mean-square interlayer strength. Spectral changes that survive this normalization arise from the redistribution and matrix structure of the interlayer coupling, not from a change in its total Frobenius weight. Finally, the zero-hybridization control is $w = 0$. It preserves the two monolayer spectra but removes every twist-dependent term:

$$h_N(\theta)|_{w=0} = \begin{pmatrix} h_0 & 0 \\ 0 & h_0 \end{pmatrix}. \quad (180)$$

Any spectral feature that remains unchanged under Eq. (180) cannot be attributed to twist-induced interlayer reconstruction. The complete parameter hierarchy is therefore

$$\begin{aligned} \text{primary parameters:} & \quad (\theta, K, w/t, \lambda_\perp/a), \\ \text{fixed baseline:} & \quad h/a, \omega_0/t, \mathcal{S}_8, \\ \text{imperfections:} & \quad \sigma_\omega/t, \sigma_t, \sigma_\perp, \Delta_L/t, \kappa/t, \\ \text{finite-device data:} & \quad N, r_{\text{inj}}^{(N)}/a, D_c/a. \end{aligned} \quad (181)$$

Equations (28), (49), (91), and (127) close the construction from an experimentally specified superconducting circuit to a fully explicit twisted hyperbolic bilayer Hamiltonian. No band flattening, magic-angle condition, or non-Abelian spectral conclusion is assumed at this stage; those are quantitative consequences obtained directly from the Hamiltonian constructed here.

II. EXACT GEOMETRY OF TWISTED SQUARE BILAYERS

The distinction between the last two items is essential. A local moiré beat length is defined for every relative rotation, whereas an exact crystallographic supercell exists only at commensurate rotations. Even at an exactly commensurate angle, the local beat cell and the primitive coincidence cell need not coincide. Let the unrotated microscopic square lattice be

$$\Lambda = a\mathbb{Z}^2 = \left\{ a \begin{pmatrix} j \\ k \end{pmatrix} : j, k \in \mathbb{Z} \right\}, \quad (182)$$

where $a > 0$ is the microscopic lattice constant. We choose the primitive vectors

$$\mathbf{a}_1 = a \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{a}_2 = a \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (183)$$

so that the microscopic primitive area is

$$A_0 = |\det(\mathbf{a}_1 \ \mathbf{a}_2)| = a^2. \quad (184)$$

The first layer is $\Lambda_1 = \Lambda$, and the second layer is $\Lambda_2(\theta) = R_\theta \Lambda$, where

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (185)$$

The exact coincidence-site lattice is the intersection

$$\Lambda_M(\theta) = \Lambda \cap R_\theta \Lambda. \quad (186)$$

An exact finite moiré supercell exists precisely when Λ_M has finite index in Λ . We denote that index by

$$\Sigma(\theta) = [\Lambda : \Lambda_M(\theta)]. \quad (187)$$

Because each microscopic square cell contains one site per layer, a coincidence cell contains

$$N_{\text{sc}}^{(1)} = N_{\text{sc}}^{(2)} = \Sigma \quad (188)$$

sites in the two individual layers. The layer labels remain independent even when two sites occupy the same in-plane coordinate. The complete bilayer count is therefore $N_{\text{sc}} = 2\Sigma$. The exact supercell area is $A_{\text{sc}} = \Sigma a^2$.

Exact commensurability criterion

We first prove the equivalence

$$\begin{aligned} \Sigma(\theta) < \infty & \iff R_\theta \in SO(2, \mathbb{Q}) \iff \cos \theta, \sin \theta \in \mathbb{Q} \\ & \iff \tan \left(\frac{\theta}{2} \right) \in \mathbb{Q}, \end{aligned} \quad (189)$$

for a nontrivial rotation $0 < \theta < \pi$. Suppose first that $\Sigma(\theta) < \infty$. A finite-index subgroup of a rank-two lattice is again a rank-two lattice. We may therefore choose two linearly independent coincidence vectors $\mathbf{u}_1, \mathbf{u}_2 \in \Lambda_M$. Since they lie in the first lattice, there exists an integer matrix

$$U \in M_2(\mathbb{Z}), \quad \det U \neq 0, \quad (190)$$

such that

$$(\mathbf{u}_1 \ \mathbf{u}_2) = aU. \quad (191)$$

The same vectors belong to the rotated layer, so there exists

$$V \in M_2(\mathbb{Z}), \quad \det V \neq 0, \quad (192)$$

satisfying

$$(\mathbf{u}_1 \ \mathbf{u}_2) = aR_\theta V. \quad (193)$$

Equating the two expressions gives $U = R_\theta V$. Hence $R_\theta = UV^{-1}$. Since

$$V^{-1} = \frac{\text{adj } V}{\det V}, \quad (194)$$

every entry of V^{-1} is rational. Therefore $R_\theta \in M_2(\mathbb{Q})$. Because R_θ is orthogonal with determinant one, $R_\theta \in SO(2, \mathbb{Q})$. Inspection of Eq. (185) then yields $\cos \theta, \sin \theta \in \mathbb{Q}$. For $\theta \neq \pi$,

$$\tan\left(\frac{\theta}{2}\right) = \frac{\sin \theta}{1 + \cos \theta}, \quad (195)$$

and the half-angle tangent is rational. Conversely, suppose

$$\tan\left(\frac{\theta}{2}\right) \in \mathbb{Q}. \quad (196)$$

The half-angle identities

$$\cos \theta = \frac{1 - \tan^2(\theta/2)}{1 + \tan^2(\theta/2)}, \quad (197)$$

and

$$\sin \theta = \frac{2 \tan(\theta/2)}{1 + \tan^2(\theta/2)} \quad (198)$$

show that the entries of R_θ are rational. Choose a positive integer q that clears the denominators of R_θ . Since $R_\theta^{-1} = R_\theta^T$, the same integer clears the denominators of the inverse. For every $\mathbf{z} \in \mathbb{Z}^2$, $\mathbf{x} = aq\mathbf{z} \in \Lambda$. Moreover,

$$R_\theta^{-1}\mathbf{x} = a(qR_\theta^{-1})\mathbf{z} \in a\mathbb{Z}^2, \quad (199)$$

so $\mathbf{x} \in R_\theta\Lambda$. It follows that

$$aq\mathbb{Z}^2 \subseteq \Lambda \cap R_\theta\Lambda. \quad (200)$$

The lattice on the left has finite index q^2 in Λ , and therefore the coincidence lattice also has finite index. This completes the proof of Eq. (189). We now write the rational half-angle in lowest terms:

$$\tan\left(\frac{\theta}{2}\right) = \frac{n}{m}, \quad m > n > 0, \quad \gcd(m, n) = 1. \quad (201)$$

The exact trigonometric data are

$$\cos \theta = \frac{m^2 - n^2}{m^2 + n^2}, \quad (202)$$

and

$$\sin \theta = \frac{2mn}{m^2 + n^2}. \quad (203)$$

The rotation matrix is therefore

$$R_\theta = \frac{1}{m^2 + n^2} \begin{pmatrix} m^2 - n^2 & -2mn \\ 2mn & m^2 - n^2 \end{pmatrix}. \quad (204)$$

The corresponding angle is

$$\theta_{m,n} = 2 \arctan\left(\frac{n}{m}\right). \quad (205)$$

Two immediately visible coincidence vectors are

$$\mathbf{T}_1^{\text{naive}} = a \begin{pmatrix} m \\ n \end{pmatrix}, \quad \mathbf{T}_2^{\text{naive}} = a \begin{pmatrix} -n \\ m \end{pmatrix}. \quad (206)$$

Indeed,

$$R_\theta \begin{pmatrix} m \\ -n \end{pmatrix} = \begin{pmatrix} m \\ n \end{pmatrix}, \quad (207)$$

and

$$R_\theta \begin{pmatrix} n \\ m \end{pmatrix} = \begin{pmatrix} -n \\ m \end{pmatrix}. \quad (208)$$

The vectors in Eq. (206) are orthogonal:

$$\mathbf{T}_1^{\text{naive}} \cdot \mathbf{T}_2^{\text{naive}} = 0, \quad (209)$$

and have equal squared length

$$|\mathbf{T}_1^{\text{naive}}|^2 = |\mathbf{T}_2^{\text{naive}}|^2 = a^2(m^2 + n^2). \quad (210)$$

Their cell area is

$$A_{\text{naive}} = a^2(m^2 + n^2). \quad (211)$$

This cell is primitive only when m and n have opposite parity. The reduction is controlled by

$$d = \gcd(m^2 - n^2, 2mn, m^2 + n^2). \quad (212)$$

No odd prime can divide all three integers. To prove this, suppose an odd prime p divides $2mn$. Then $p \mid m$ or $p \mid n$. If $p \mid m$ and $p \mid m^2 - n^2$, then $p \mid n$, contradicting $\gcd(m, n) = 1$. The same contradiction follows when the roles of m and n are interchanged. Hence d is a power of two. If m and n have opposite parity, then

$$m^2 + n^2 \equiv 1 \pmod{2}, \quad (213)$$

and therefore $d = 1$. If m and n are both odd, then

$$m^2 \equiv n^2 \equiv 1 \pmod{8}, \quad (214)$$

and

$$m^2 + n^2 \equiv 2 \pmod{4}. \quad (215)$$

Exactly one factor of two is common, so $d = 2$. Thus

$$d = \begin{cases} 1, & m \not\equiv n \pmod{2}, \\ 2, & m \equiv n \equiv 1 \pmod{2}. \end{cases} \quad (216)$$

The primitive reduction can be proved exactly using Gaussian integers. Identify

$$\mathbb{Z}^2 \longleftrightarrow \mathbb{Z}[i], \quad \begin{pmatrix} x \\ y \end{pmatrix} \longleftrightarrow x + iy. \quad (217)$$

Introduce

$$\alpha = m + in, \quad \bar{\alpha} = m - in. \quad (218)$$

Equation (204) is equivalent to

$$e^{i\theta} = \frac{\alpha}{\bar{\alpha}}. \quad (219)$$

Let

$$g = \gcd(\alpha, \bar{\alpha})_{\mathbb{Z}[i]}, \quad (220)$$

and define

$$\alpha_0 = \frac{\alpha}{g}, \quad \beta_0 = \frac{\bar{\alpha}}{g}. \quad (221)$$

The reduced Gaussian integers are coprime:

$$\gcd(\alpha_0, \beta_0)_{\mathbb{Z}[i]} = 1 \quad (222)$$

up to Gaussian units. A Gaussian integer z belongs to the coincidence lattice precisely when

$$z \in \mathbb{Z}[i] \cap \frac{\alpha_0}{\beta_0} \mathbb{Z}[i]. \quad (223)$$

If

$$z = \frac{\alpha_0}{\beta_0} w, \quad w \in \mathbb{Z}[i], \quad (224)$$

then $\beta_0 z = \alpha_0 w$. Since α_0 and β_0 are coprime and $\mathbb{Z}[i]$ is a unique-factorization domain, Euclid's lemma gives $\alpha_0 \mid z$. Conversely, every $z = \alpha_0 u$, $u \in \mathbb{Z}[i]$, satisfies

$$z = \frac{\alpha_0}{\beta_0} (\beta_0 u), \quad (225)$$

and therefore belongs to both lattices. Hence

$$\Lambda_M \simeq a\alpha_0 \mathbb{Z}[i]. \quad (226)$$

For opposite parity, g is a Gaussian unit, so $N_{\mathbb{Z}[i]}(g) = 1$. For both-odd data, $1+i$ divides both α and $\bar{\alpha}$, but $(1+i)^2 = 2i$ cannot divide both because m and n are coprime. Thus, up to a unit,

$$g = 1+i, \quad N_{\mathbb{Z}[i]}(g) = 2. \quad (227)$$

The coincidence index is the Gaussian norm of the primitive generator:

$$\Sigma(m, n) = N_{\mathbb{Z}[i]}(\alpha_0) = \frac{m^2 + n^2}{d}. \quad (228)$$

For opposite parity, a primitive coincidence basis is

$$\mathbf{T}_1 = a \begin{pmatrix} m \\ n \end{pmatrix}, \quad \mathbf{T}_2 = a \begin{pmatrix} -n \\ m \end{pmatrix}. \quad (229)$$

For both-odd data, division by $1+i$ gives

$$\mathbf{T}_1 = \frac{a}{2} \begin{pmatrix} m+n \\ n-m \end{pmatrix}, \quad \mathbf{T}_2 = \frac{a}{2} \begin{pmatrix} m-n \\ m+n \end{pmatrix}. \quad (230)$$

In both cases, $\mathbf{T}_1 \cdot \mathbf{T}_2 = 0$, and

$$|\mathbf{T}_1|^2 = |\mathbf{T}_2|^2 = a^2 \Sigma. \quad (231)$$

The exact coincidence lattice is therefore itself square, with lattice constant

$$L_{\text{sc}} = a\sqrt{\Sigma}. \quad (232)$$

Its primitive area and bilayer site count are

$$A_{\text{sc}} = a^2 \Sigma, \quad N_{\text{sc}} = 2\Sigma = \frac{2(m^2 + n^2)}{d}. \quad (233)$$

It is useful to combine the two parity sectors into one integer pair. Define

$$(p, q) = \begin{cases} (m, n), & d = 1, \\ \left(\frac{m+n}{2}, \frac{n-m}{2}\right), & d = 2. \end{cases} \quad (234)$$

Then

$$p, q \in \mathbb{Z}, \quad \gcd(p, q) = 1, \quad p^2 + q^2 = \Sigma. \quad (235)$$

The primitive basis matrix becomes

$$C_M = (\mathbf{T}_1 \quad \mathbf{T}_2) = a \begin{pmatrix} p & -q \\ q & p \end{pmatrix}. \quad (236)$$

It obeys the exact analysis identities

$$C_M^T C_M = a^2 \Sigma I_2, \quad (237)$$

and

$$\det C_M = a^2 \Sigma. \quad (238)$$

Because $\gcd(p, q) = 1$, the Smith normal form of C_M/a is

$$\text{SNF} \left(\frac{C_M}{a} \right) = \text{diag}(1, \Sigma). \quad (239)$$

Therefore

$$\mathbb{Z}^2 / \left(\frac{\Lambda_M}{a} \right) \simeq \mathbb{Z}_\Sigma. \quad (240)$$

One convenient set of algebraic site representatives for either single-layer quotient is

$$\mathcal{R}_\Sigma = \{j\mathbf{e}_x : j = 0, \dots, \Sigma - 1\}. \quad (241)$$

To verify that these representatives are distinct, suppose $k\mathbf{e}_x \in (C_M/a)\mathbb{Z}^2$. Then

$$\left(\frac{C_M}{a} \right)^{-1} k\mathbf{e}_x = \frac{k}{\Sigma} \begin{pmatrix} p \\ -q \end{pmatrix} \in \mathbb{Z}^2. \quad (242)$$

Since

$$\gcd(p, \Sigma) = \gcd(q, \Sigma) = 1, \quad (243)$$

we must have $\Sigma \mid k$. Hence $\mathbf{e}_x$ has order exactly Σ in the quotient. A square lattice is invariant under rotations by integer multiples of $\pi/2$. Interchanging the two layers or reflecting the entire bilayer changes θ to $-\theta$. The physical equivalence relation is therefore generated by

$$\theta \sim \theta + k\frac{\pi}{2}, \quad \theta \sim -\theta, \quad k \in \mathbb{Z}. \quad (244)$$

Every relative rotation has a unique representative $0 \leq \hat{\theta} \leq \frac{\pi}{4}$. For $0 < \theta < \pi/2$,

$$\hat{\theta} = \min \left\{ \theta, \frac{\pi}{2} - \theta \right\}. \quad (245)$$

If

$$\tan \left(\frac{\theta}{2} \right) = \frac{n}{m}, \quad (246)$$

then the complementary angle

$$\theta^\# = \frac{\pi}{2} - \theta \quad (247)$$

satisfies

$$\begin{aligned} \tan \left(\frac{\theta^\#}{2} \right) &= \tan \left(\frac{\pi}{4} - \frac{\theta}{2} \right) \\ &= \frac{1 - \tan(\theta/2)}{1 + \tan(\theta/2)} = \frac{m - n}{m + n}. \end{aligned}$$

The reduced complementary integer pair is therefore

$$m^\# = \frac{m + n}{d}, \quad n^\# = \frac{m - n}{d}, \quad (248)$$

where d is the parity divisor of the original pair. The parity divisor of the partner is $d^\# = \frac{2}{d}$. A direct calculation gives

$$\begin{aligned} \frac{(m^\#)^2 + (n^\#)^2}{d^\#} &= \frac{2(m^2 + n^2)/d^2}{2/d} \\ &= \frac{m^2 + n^2}{d} = \Sigma. \end{aligned}$$

Thus complementary square-symmetry angles have the same primitive coincidence index and the same number of layer-resolved microscopic degrees of freedom.

A. The ten-site commensurate supercell

The geometry shown in Topic 1 contains two equal-length microscopic vectors symmetrically located about a reference axis:

$$\mathbf{v}_- = a \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \mathbf{v}_+ = a \begin{pmatrix} 2 \\ 1 \end{pmatrix}. \quad (249)$$

Their norms are

$$|\mathbf{v}_-| = |\mathbf{v}_+| = a\sqrt{5}. \quad (250)$$

Their scalar product gives

$$\cos \theta = \frac{\mathbf{v}_- \cdot \mathbf{v}_+}{|\mathbf{v}_-||\mathbf{v}_+|} = \frac{4 - 1}{5} = \frac{3}{5}. \quad (251)$$

The signed area gives

$$\sin \theta = \frac{|\det(\mathbf{v}_-, \mathbf{v}_+)|}{|\mathbf{v}_-||\mathbf{v}_+|} = \frac{4}{5}. \quad (252)$$

Hence

$$\tan \left(\frac{\theta}{2} \right) = \frac{\sin \theta}{1 + \cos \theta} = \frac{1}{2}. \quad (253)$$

In the notation $2r = \theta$,

$$r = \arctan \left(\frac{1}{2} \right) = 26.565051177 \dots^\circ, \quad (254)$$

and the angle drawn between the two layers is

$$2r = \theta = 2 \arctan \left(\frac{1}{2} \right) = 53.130102354 \dots^\circ. \quad (255)$$

The exact rotation matrix is

$$R_\theta^{(10)} = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}. \quad (256)$$

The integer data are

$$(m, n) = (2, 1), \quad d = 1. \quad (257)$$

Therefore

$$\Sigma_{10} = 2^2 + 1^2 = 5. \quad (258)$$

A primitive coincidence basis is

$$\mathbf{T}_1^{(10)} = a \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{T}_2^{(10)} = a \begin{pmatrix} -1 \\ 2 \end{pmatrix}. \quad (259)$$

The basis matrix is

$$C_{10} = a \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}. \quad (260)$$

Its exact geometric checks are

$$C_{10}^T C_{10} = 5a^2 I_2, \quad \det C_{10} = 5a^2. \quad (261)$$

Thus

$$L_{\text{sc}}^{(10)} = \sqrt{5} a, \quad A_{\text{sc}}^{(10)} = 5a^2. \quad (262)$$

The quotient is $\Lambda/\Lambda_M \simeq \mathbb{Z}_5$. Each layer therefore contributes exactly five microscopic sites. The complete bilayer count is $N_{\text{sc}}^{(10)} = 5 + 5 = 10$. At a coincidence point, the layer-one and layer-two orbitals occupy the same in-plane coordinate but remain orthogonal layer-resolved basis states: $\langle 1, \mathbf{r} | 2, \mathbf{r} \rangle = 0$. Their contribution to the Hilbert-space dimension is therefore two, not one.

This resolves the apparent “nine versus ten” ambiguity in the geometric drawing. The displayed angle exceeds the irreducible square-lattice bound $\pi/4$. Its symmetry-reduced partner is

$$\hat{\theta}_{10} = \frac{\pi}{2} - \theta = 36.869897646\dots^\circ. \quad (263)$$

Equation (248) maps $(2, 1) \mapsto (3, 1)$. For $(m, n) = (3, 1)$,

$$d = 2, \quad \Sigma = \frac{3^2 + 1^2}{2} = 5. \quad (264)$$

The reduced-angle rotation matrix is

$$R_{\hat{\theta}_{10}} = \frac{1}{5} \begin{pmatrix} 4 & -3 \\ 3 & 4 \end{pmatrix}. \quad (265)$$

A corresponding primitive basis is

$$\hat{\mathbf{T}}_1^{(10)} = a \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \hat{\mathbf{T}}_2^{(10)} = a \begin{pmatrix} 1 \\ 2 \end{pmatrix}. \quad (266)$$

The two descriptions are related by a square-lattice point-group transformation and represent the same coincidence class.

B. The thirty-four-site commensurate supercell

We now impose $N_{\text{sc}} = 34$. Using $N_{\text{sc}} = 2\Sigma$,

$$\Sigma_{34} = \frac{34}{2} = 17. \quad (267)$$

In the opposite-parity sector, the integer equation is $m^2 + n^2 = 17$. The positive coprime solution with $m > n$ is uniquely $(m, n) = (4, 1)$. Since the two integers have opposite parity, $d = 1$. The half-angle and relative twist are

$$r_{34} = \arctan\left(\frac{1}{4}\right) = 14.036243468\dots^\circ, \quad (268)$$

and

$$\theta_{34} = 2 \arctan\left(\frac{1}{4}\right) = 28.072486936\dots^\circ. \quad (269)$$

This angle already lies in $0 < \hat{\theta} < \pi/4$, so it is the irreducible representative. The exact trigonometric data are

$$\cos \theta_{34} = \frac{4^2 - 1^2}{4^2 + 1^2} = \frac{15}{17}, \quad (270)$$

and

$$\sin \theta_{34} = \frac{2(4)(1)}{4^2 + 1^2} = \frac{8}{17}. \quad (271)$$

The exact rotation matrix is

$$R_{\theta_{34}} = \frac{1}{17} \begin{pmatrix} 15 & -8 \\ 8 & 15 \end{pmatrix}. \quad (272)$$

A primitive coincidence basis is

$$\mathbf{T}_1^{(34)} = a \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \quad \mathbf{T}_2^{(34)} = a \begin{pmatrix} -1 \\ 4 \end{pmatrix}. \quad (273)$$

The corresponding matrix is

$$C_{34} = a \begin{pmatrix} 4 & -1 \\ 1 & 4 \end{pmatrix}. \quad (274)$$

It satisfies

$$C_{34}^\top C_{34} = 17a^2 I_2, \quad \det C_{34} = 17a^2. \quad (275)$$

Hence

$$L_{\text{sc}}^{(34)} = \sqrt{17} a, \quad A_{\text{sc}}^{(34)} = 17a^2, \quad (276)$$

and $N_{\text{sc}}^{(34)} = 2 \times 17 = 34$. The quotient $\Lambda/\Lambda_{\text{M}} \simeq \mathbb{Z}_{17}$ shows that each layer contributes seventeen inequivalent microscopic sites. The complementary square-symmetry representation follows from $(4, 1) \mapsto (5, 3)$. Both integers in the partner pair are odd, so

$$d^\# = 2, \quad \frac{5^2 + 3^2}{2} = 17. \quad (277)$$

The partner angle is

$$\theta_{34}^\# = 2 \arctan\left(\frac{3}{5}\right) = 61.927513064\dots^\circ, \quad (278)$$

and

$$\theta_{34} + \theta_{34}^\# = 90^\circ. \quad (279)$$

The two angles therefore represent the same crystallographic class under the fourfold point symmetry of the square lattice.

C. Complete classification below one hundred sites

The Gaussian-integer construction yields a complete arithmetic classification of all square-lattice coincidence indices. Let $\gamma = p + iq$ be a primitive Gaussian generator of the coincidence lattice, with $\gcd(p, q) = 1$ and opposite parity between p and q . The index is

$$\Sigma = N_{\mathbb{Z}[i]}(\gamma) = p^2 + q^2. \quad (280)$$

Because p and q have opposite parity, $2 \nmid \Sigma$. Now let ℓ be a rational prime satisfying $\ell \equiv 3 \pmod{4}$. Suppose $\ell \mid p^2 + q^2$. If $\ell \nmid q$, then q is invertible modulo ℓ , and

$$(pq^{-1})^2 \equiv -1 \pmod{\ell}. \quad (281)$$

However, -1 is not a quadratic residue modulo a prime congruent to three modulo four. Hence $\ell \mid q$. The divisibility $\ell \mid p^2 + q^2$ then implies $\ell \mid p$, contradicting $\gcd(p, q) = 1$. Therefore

$$\ell \equiv 3 \pmod{4} \implies \ell \nmid \Sigma. \quad (282)$$

Every nontrivial rational prime factor of a primitive coincidence index must consequently satisfy

$$p_j \equiv 1 \pmod{4}. \quad (283)$$

Thus

$$\Sigma = \prod_j p_j^{e_j}, \quad p_j \equiv 1 \pmod{4}, \quad e_j \in \mathbb{N}_0. \quad (284)$$

The converse also holds. A prime $p_j \equiv 1 \pmod{4}$ splits in the Gaussian integers: $p_j = \pi_j \bar{\pi}_j$. For an integer of the form Eq. (284), choose one Gaussian prime from each conjugate pair and define

$$\gamma = \prod_j \pi_j^{e_j}. \quad (285)$$

Then

$$N_{\mathbb{Z}[i]}(\gamma) = \prod_j N_{\mathbb{Z}[i]}(\pi_j)^{e_j} = \prod_j p_j^{e_j} = \Sigma. \quad (286)$$

Because no conjugate pair has been included simultaneously, γ and $\bar{\gamma}$ are coprime up to units. The rotation generated by their ratio therefore has primitive coincidence index Σ . This proves necessity and sufficiency. For fewer than one hundred bilayer sites,

$$N_{\text{sc}} = 2\Sigma < 100, \quad (287)$$

so $\Sigma < 50$. The trivial index $\Sigma = 1$ corresponds to a square-lattice symmetry and does not generate an enlarged moiré cell. Applying Eq. (284), the complete non-trivial list is

$$\Sigma \in \{5, 13, 17, 25, 29, 37, 41\}. \quad (288)$$

The corresponding bilayer counts are

$$N_{\text{sc}} \in \{10, 26, 34, 50, 58, 74, 82\}. \quad (289)$$

No other nontrivial square-bilayer supercell contains fewer than one hundred layer-resolved microscopic sites.

If angles are listed in the entire interval $0 < \theta < \pi/2$, rather than modulo square symmetry, the complete set is

$$\begin{aligned} \theta \in \{ & 12.680383492^\circ, 16.260204708^\circ, 18.924644416^\circ, 22.619864948^\circ, \\ & 28.072486936^\circ, 36.869897646^\circ, 43.602818973^\circ, 46.397181027^\circ, \\ & 53.130102354^\circ, 61.927513064^\circ, 67.380135052^\circ, 71.075355584^\circ, \\ & 73.739795292^\circ, 77.319616508^\circ \}. \end{aligned} \quad (290)$$

The angles occur in complementary pairs:

$$\theta + \theta^\# = \frac{\pi}{2}. \quad (291)$$

Modulo the square point group, only the seven angles in Table I are inequivalent. If crystallographically trivial rotations are included, one may also add

$$\theta = 0, \quad \theta = \frac{\pi}{2}, \quad (292)$$

for which $\Sigma = 1$, $N_{\text{sc}} = 2$. These rotations map the square lattice onto itself and do not generate a nontrivial moiré cell. An instructive counterexample is $\theta = \frac{\pi}{4}$. Although this angle is visually compatible with the diagonal directions of a square, it is not exactly commensurate because

$$\tan\left(\frac{\pi}{8}\right) = \sqrt{2} - 1 \notin \mathbb{Q}. \quad (293)$$

Thus point-group visual symmetry and translational commensurability are different statements. The sequence of counts in increasing-angle order is 82, 50, 74, 26, 34, 10, 58. It is not monotone. The failure of monotonicity is not a numerical artifact; it is a direct consequence of the arithmetic factor derived below.

D. Exact supercell scaling

We now separate the geometric twist dependence from the discrete arithmetic data. From

$$\tan\left(\frac{\hat{\theta}}{2}\right) = \frac{n}{m}, \quad (294)$$

we have

$$m = n \cot\left(\frac{\hat{\theta}}{2}\right). \quad (295)$$

Substitution into Eq. (228) gives

$$\Sigma = \frac{n^2}{d} \left[\cot^2\left(\frac{\hat{\theta}}{2}\right) + 1 \right] = \frac{n^2}{d} \csc^2\left(\frac{\hat{\theta}}{2}\right).$$

Therefore

$$N_{\text{sc}} = \frac{2n^2}{d} \csc^2\left(\frac{\hat{\theta}}{2}\right). \quad (296)$$

Define the arithmetic multiplier

$$\mathcal{A}_{m,n} = \frac{n^2}{d}. \quad (297)$$

$$\Sigma = \mathcal{A}_{m,n} \csc^2\left(\frac{\hat{\theta}}{2}\right), \quad (298)$$

and

$$N_{\text{sc}} = 2\mathcal{A}_{m,n} \csc^2\left(\frac{\hat{\theta}}{2}\right). \quad (299)$$

The first factor is continuous and geometric. The second is discrete and arithmetic. For small angle,

$$\csc^2 x = \frac{1}{x^2} + \frac{1}{3} + \frac{x^2}{15} + \frac{2x^4}{189} + O(x^6). \quad (300)$$

TABLE I. Complete set of nontrivial crystallographically inequivalent commensurate square-bilayer twists with $N_{\text{sc}} < 100$. The angle $\hat{\theta}$ lies in $0 < \hat{\theta} < \pi/4$. The final column gives the complementary representative in $(\pi/4, \pi/2)$.

Σ	N_{sc}	(m, n)	d	$(\cos \hat{\theta}, \sin \hat{\theta})$	$\hat{\theta}$	$(m^\sharp, n^\sharp); \theta^\sharp$
41	82	(9, 1)	2	$\left(\frac{40}{41}, \frac{9}{41}\right)$	12.680383492°	(5, 4); 77.319616508°
25	50	(7, 1)	2	$\left(\frac{24}{25}, \frac{7}{25}\right)$	16.260204708°	(4, 3); 73.739795292°
37	74	(6, 1)	1	$\left(\frac{35}{37}, \frac{23}{37}\right)$	18.924644416°	(7, 5); 71.075355584°
13	26	(5, 1)	2	$\left(\frac{12}{13}, \frac{5}{13}\right)$	22.619864948°	(3, 2); 67.380135052°
17	34	(4, 1)	1	$\left(\frac{15}{17}, \frac{8}{17}\right)$	28.072486936°	(5, 3); 61.927513064°
5	10	(3, 1)	2	$\left(\frac{4}{5}, \frac{3}{5}\right)$	36.869897646°	(2, 1); 53.130102354°
29	58	(5, 2)	1	$\left(\frac{21}{29}, \frac{20}{29}\right)$	43.602818973°	(7, 3); 46.397181027°

Setting $x = \hat{\theta}/2$ yields

$$\csc^2\left(\frac{\hat{\theta}}{2}\right) = \frac{4}{\hat{\theta}^2} + \frac{1}{3} + \frac{\hat{\theta}^2}{60} + \frac{\hat{\theta}^4}{1512} + O(\hat{\theta}^6). \quad (301)$$

Hence

$$\Sigma = \frac{4n^2}{d\hat{\theta}^2} + \frac{n^2}{3d} + \frac{n^2\hat{\theta}^2}{60d} + \frac{n^2\hat{\theta}^4}{1512d} + O(n^2\hat{\theta}^6), \quad (302)$$

and

$$N_{\text{sc}} = \frac{8n^2}{d\hat{\theta}^2} + \frac{2n^2}{3d} + \frac{n^2\hat{\theta}^2}{30d} + \frac{n^2\hat{\theta}^4}{756d} + O(n^2\hat{\theta}^6). \quad (303)$$

The exact linear size is

$$L_{\text{sc}} = a\sqrt{\Sigma} = \frac{an}{\sqrt{d}} \csc\left(\frac{\hat{\theta}}{2}\right). \quad (304)$$

Using

$$\csc x = \frac{1}{x} + \frac{x}{6} + \frac{7x^3}{360} + \frac{31x^5}{15120} + O(x^7), \quad (305)$$

we obtain

$$L_{\text{sc}} = \frac{an}{\sqrt{d}} \left[\frac{2}{\hat{\theta}} + \frac{\hat{\theta}}{12} + \frac{7\hat{\theta}^3}{2880} + \frac{31\hat{\theta}^5}{483840} + O(\hat{\theta}^7) \right]. \quad (306)$$

Along every commensurate sequence for which n remains bounded,

$$L_{\text{sc}} = \Theta(\hat{\theta}^{-1}), \quad A_{\text{sc}} = \Theta(\hat{\theta}^{-2}), \quad (307)$$

and

$$N_{\text{sc}} = \Theta(\hat{\theta}^{-2}). \quad (308)$$

This is the usual inverse-square moiré scaling, but it is only the bounded-arithmetic branch of the exact result. Since $n \geq 1$, $1 \leq d \leq 2$, we have

$$\frac{2n^2}{d} \geq 1. \quad (309)$$

Equation (296) therefore gives the universal lower bound

$$N_{\text{sc}} \geq \csc^2\left(\frac{\hat{\theta}}{2}\right). \quad (310)$$

At small angle,

$$N_{\text{sc}} \geq \frac{4}{\hat{\theta}^2} + \frac{1}{3} + O(\hat{\theta}^2). \quad (311)$$

The lower envelope is attained by $n = 1$, $d = 2$, which corresponds to odd m . On this branch,

$$N_{\text{sc}} = \csc^2\left(\frac{\hat{\theta}}{2}\right) = m^2 + 1. \quad (312)$$

The ten-site reduced representative $(m, n) = (3, 1)$ lies exactly on this lower envelope: $N_{\text{sc}} = 3^2 + 1 = 10$. The adjacent $n = 1$, opposite-parity branch has even m and $d = 1$. There,

$$N_{\text{sc}} = 2 \csc^2\left(\frac{\hat{\theta}}{2}\right) = 2(m^2 + 1). \quad (313)$$

The thirty-four-site structure $(m, n) = (4, 1)$ lies on this branch:

$$N_{\text{sc}} = 2(4^2 + 1) = 34. \quad (314)$$

There is no universal angle-only upper bound on the exact cell size. To make this statement explicit, consider the coprime sequence

$$n_j = j^q, \quad m_j = j^{q+1} + 1, \quad q \in \mathbb{N}_0. \quad (315)$$

Indeed,

$$\gcd(j^q, j^{q+1} + 1) = 1. \quad (316)$$

For large j ,

$$\tan\left(\frac{\hat{\theta}_j}{2}\right) = \frac{j^q}{j^{q+1} + 1} = \frac{1}{j} + O(j^{-q-2}), \quad (317)$$

so

$$\hat{\theta}_j = \frac{2}{j} + O(j^{-3}). \quad (318)$$

However,

$$N_{\text{sc},j} = \frac{2[(j^{q+1} + 1)^2 + j^{2q}]}{d_j} = \Theta(j^{2q+2}), \quad (319)$$

and therefore

$$N_{\text{sc},j} = \Theta(\widehat{\theta}_j^{-(2q+2)}). \quad (320)$$

Since q is arbitrary, exact commensurate sequences can have arithmetic growth exponents larger than two by an arbitrarily large amount. More generally, let $\widehat{\theta}_j \rightarrow 0$ be a primitive commensurate sequence and suppose

$$\sigma = \lim_{j \rightarrow \infty} \frac{\log n_j}{\log \widehat{\theta}_j^{-1}} \quad (321)$$

exists. Since $d_j \in \{1, 2\}$, Eq. (303) gives

$$\log N_{\text{sc},j} = 2 \log \widehat{\theta}_j^{-1} + 2 \log n_j + \log \left(\frac{8}{d_j} \right) + o(1). \quad (322)$$

Dividing by $\log \widehat{\theta}_j^{-1}$ yields

$$\lim_{j \rightarrow \infty} \frac{\log N_{\text{sc},j}}{\log \widehat{\theta}_j^{-1}} = 2 + 2\sigma. \quad (323)$$

Similarly,

$$\lim_{j \rightarrow \infty} \frac{\log L_{\text{sc},j}}{\log \widehat{\theta}_j^{-1}} = 1 + \sigma. \quad (324)$$

Thus the familiar exponents 2 and 1 are recovered precisely when the reduced numerator has subpower growth:

$$\log n_j = o(\log \widehat{\theta}_j^{-1}). \quad (325)$$

For a fixed arithmetic multiplier, the finite-angle geometric running exponent is

$$\beta_{\text{geom}}^{(N)}(\widehat{\theta}) = -\frac{d \log [\csc^2(\widehat{\theta}/2)]}{d \log \widehat{\theta}} = \widehat{\theta} \cot \left(\frac{\widehat{\theta}}{2} \right).$$

Its small-angle expansion is

$$\beta_{\text{geom}}^{(N)} = 2 - \frac{\widehat{\theta}^2}{6} - \frac{\widehat{\theta}^4}{360} - \frac{\widehat{\theta}^6}{15120} + O(\widehat{\theta}^8). \quad (326)$$

The corresponding linear-size exponent is

$$\beta_{\text{geom}}^{(L)} = \frac{\widehat{\theta}}{2} \cot \left(\frac{\widehat{\theta}}{2} \right) = 1 - \frac{\widehat{\theta}^2}{12} - \frac{\widehat{\theta}^4}{720} + O(\widehat{\theta}^6). \quad (327)$$

These derivatives describe the continuous geometric envelope; the exact crystallographic data remain discrete because (m, n) must be integral.

E. Euclidean moiré reciprocal lattice and moiré Brillouin zone

The primitive direct basis is

$$C_{\text{M}} = a \begin{pmatrix} p & -q \\ q & p \end{pmatrix}, \quad p^2 + q^2 = \Sigma. \quad (328)$$

The primitive reciprocal basis matrix is

$$B_{\text{M}} = 2\pi (C_{\text{M}}^{-1})^{\text{T}}. \quad (329)$$

Since

$$C_{\text{M}}^{-1} = \frac{1}{a\Sigma} \begin{pmatrix} p & q \\ -q & p \end{pmatrix}, \quad (330)$$

we obtain

$$B_{\text{M}} = \frac{2\pi}{a\Sigma} \begin{pmatrix} p & -q \\ q & p \end{pmatrix}. \quad (331)$$

Thus

$$\mathbf{G}_1 = \frac{2\pi}{a\Sigma} \begin{pmatrix} p \\ q \end{pmatrix}, \quad \mathbf{G}_2 = \frac{2\pi}{a\Sigma} \begin{pmatrix} -q \\ p \end{pmatrix}. \quad (332)$$

The direct and reciprocal bases satisfy

$$\mathbf{T}_i \cdot \mathbf{G}_j = 2\pi \delta_{ij}. \quad (333)$$

They also obey

$$B_{\text{M}}^{\text{T}} B_{\text{M}} = \frac{(2\pi)^2}{a^2 \Sigma} I_2. \quad (334)$$

Consequently,

$$|\mathbf{G}_1| = |\mathbf{G}_2| = \frac{2\pi}{a\sqrt{\Sigma}}, \quad (335)$$

and $\mathbf{G}_1 \cdot \mathbf{G}_2 = 0$. The exact moiré Brillouin zone is a square of area

$$A_{\text{mBZ}} = |\det B_{\text{M}}| = \frac{(2\pi)^2}{a^2 \Sigma}. \quad (336)$$

The area ratio to the microscopic Brillouin zone is

$$\frac{A_{\text{mBZ}}}{A_{\text{BZ}}^{(0)}} = \frac{1}{\Sigma}, \quad A_{\text{BZ}}^{(0)} = \left(\frac{2\pi}{a} \right)^2. \quad (337)$$

A convenient exact parametrization is

$$\mathbf{k} = \kappa_1 \mathbf{G}_1 + \kappa_2 \mathbf{G}_2, \quad -\frac{1}{2} \leq \kappa_1, \kappa_2 < \frac{1}{2}. \quad (338)$$

The high-symmetry points are

$$\begin{aligned} \Gamma &= \mathbf{0}, & X &= \frac{\mathbf{G}_1}{2} = \frac{\pi}{a\Sigma} \begin{pmatrix} p \\ q \end{pmatrix}, \\ Y &= \frac{\mathbf{G}_2}{2} = \frac{\pi}{a\Sigma} \begin{pmatrix} -q \\ p \end{pmatrix}, \end{aligned} \quad (339)$$

and

$$M = \frac{\mathbf{G}_1 + \mathbf{G}_2}{2} = \frac{\pi}{a\Sigma} \begin{pmatrix} p-q \\ p+q \end{pmatrix}. \quad (340)$$

Their distances from the zone centre are

$$|\Gamma X| = \frac{\pi}{a\sqrt{\Sigma}}, \quad |\Gamma M| = \frac{\sqrt{2}\pi}{a\sqrt{\Sigma}}. \quad (341)$$

The standard symmetry path is

$$\Gamma \longrightarrow X \longrightarrow M \longrightarrow \Gamma. \quad (342)$$

Because $\Lambda_M \subset \Lambda$, the reciprocal inclusion is reversed: $\Lambda^* \subset \Lambda_M^*$. The reciprocal quotient has order

$$[\Lambda_M^* : \Lambda^*] = \Sigma. \quad (343)$$

Thus the microscopic reciprocal lattice has exactly Σ distinct cosets inside the moiré reciprocal lattice. This reciprocal count is the geometric origin of the Σ folded momentum representatives per layer used in the subsequent Bloch Hamiltonian. For the displayed ten-site geometry,

$$C_{10} = a \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}, \quad (344)$$

and

$$\mathbf{G}_1^{(10)} = \frac{2\pi}{5a} \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{G}_2^{(10)} = \frac{2\pi}{5a} \begin{pmatrix} -1 \\ 2 \end{pmatrix}. \quad (345)$$

The reciprocal scale and mBZ area are

$$G_M^{(10)} = \frac{2\pi}{a\sqrt{5}}, \quad A_{\text{mBZ}}^{(10)} = \frac{(2\pi)^2}{5a^2}. \quad (346)$$

The high-symmetry corner is

$$M_{10} = \frac{\pi}{5a} \begin{pmatrix} 1 \\ 3 \end{pmatrix}. \quad (347)$$

For the thirty-four-site cell,

$$C_{34} = a \begin{pmatrix} 4 & -1 \\ 1 & 4 \end{pmatrix}, \quad (348)$$

and

$$\mathbf{G}_1^{(34)} = \frac{2\pi}{17a} \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \quad \mathbf{G}_2^{(34)} = \frac{2\pi}{17a} \begin{pmatrix} -1 \\ 4 \end{pmatrix}. \quad (349)$$

The reciprocal scale and mBZ area are

$$G_M^{(34)} = \frac{2\pi}{a\sqrt{17}}, \quad A_{\text{mBZ}}^{(34)} = \frac{(2\pi)^2}{17a^2}. \quad (350)$$

The zone corner is

$$M_{34} = \frac{\pi}{17a} \begin{pmatrix} 3 \\ 5 \end{pmatrix}. \quad (351)$$

The exact coincidence lattice must not be identified with the local beat lattice obtained by subtracting a pair of microscopic reciprocal vectors. After point-group reduction, let the relative angle be $\hat{\theta}$. The primitive reciprocal basis of the unrotated microscopic lattice is

$$B_0 = \frac{2\pi}{a} I_2. \quad (352)$$

The rotated basis is

$$B_{\hat{\theta}} = R_{\hat{\theta}} B_0. \quad (353)$$

The elementary reciprocal mismatch matrix is

$$\Delta B = B_{\hat{\theta}} - B_0 = \frac{2\pi}{a} (R_{\hat{\theta}} - I_2). \quad (354)$$

Using

$$R_{\hat{\theta}} - I_2 = 2 \sin\left(\frac{\hat{\theta}}{2}\right) R_{\hat{\theta}/2+\pi/2}, \quad (355)$$

we obtain

$$|\Delta \mathbf{b}_1| = |\Delta \mathbf{b}_2| = \frac{4\pi}{a} \sin\left(\frac{\hat{\theta}}{2}\right). \quad (356)$$

The reciprocal beat-cell area is

$$A_{\text{beat}}^* = \left(\frac{2\pi}{a}\right)^2 4 \sin^2\left(\frac{\hat{\theta}}{2}\right). \quad (357)$$

The corresponding real-space beat-cell area is

$$A_{\text{beat}} = \frac{(2\pi)^2}{A_{\text{beat}}^*} = \frac{a^2}{4 \sin^2(\hat{\theta}/2)}. \quad (358)$$

Because the mismatch matrix is a scaled rotation, the local beat cell is square, with linear scale

$$L_{\text{beat}} = \frac{a}{2 \sin(\hat{\theta}/2)}. \quad (359)$$

The exact coincidence-cell length is

$$L_{\text{sc}} = \frac{an}{\sqrt{d}} \csc\left(\frac{\hat{\theta}}{2}\right). \quad (360)$$

Therefore

$$\frac{L_{\text{sc}}}{L_{\text{beat}}} = \frac{2n}{\sqrt{d}}. \quad (361)$$

At the area level,

$$\frac{A_{\text{sc}}}{A_{\text{beat}}} = \frac{4n^2}{d}. \quad (362)$$

Equivalently, if

$$\Delta q = \frac{4\pi}{a} \sin\left(\frac{\hat{\theta}}{2}\right) \quad (363)$$

is the primary reciprocal peak splitting, then

$$\frac{\Delta q}{G_M} = \frac{2n}{\sqrt{d}}. \quad (364)$$

The reciprocal-area relation is

$$\frac{A_{\text{beat}}^*}{A_{\text{mBZ}}} = \frac{4n^2}{d}. \quad (365)$$

The factor

$$\frac{4n^2}{d} = \begin{cases} 4n^2, & d = 1, \\ 2n^2, & d = 2 \end{cases} \quad (366)$$

is an integer. It measures exactly how many primitive exact mBZ areas fit into the reciprocal beat-cell area. Hence reciprocal peak splitting determines the local geometric beat scale but does not determine the exact crystallographic cell without the arithmetic data n and d . The relative orientation also carries a parity signature. The direct beat lattice is oriented at

$$\phi_{\text{beat}} = \frac{\hat{\theta}}{2} \pmod{\frac{\pi}{2}}. \quad (367)$$

For $d = 1$, the exact coincidence basis Eq. (229) has orientation

$$\phi_{\text{sc}} = \arctan\left(\frac{n}{m}\right) = \frac{\hat{\theta}}{2}, \quad (368)$$

so the beat and coincidence squares are parallel. For $d = 2$, Eq. (230) gives

$$\tan \phi_{\text{sc}} = \frac{n - m}{m + n} = \tan\left(\frac{\hat{\theta}}{2} - \frac{\pi}{4}\right),$$

and therefore

$$\phi_{\text{sc}} = \frac{\hat{\theta}}{2} - \frac{\pi}{4} \pmod{\frac{\pi}{2}}. \quad (369)$$

Thus

$$\phi_{\text{sc}} - \phi_{\text{beat}} = \begin{cases} 0 \pmod{\pi/2}, & d = 1, \\ -\pi/4 \pmod{\pi/2}, & d = 2. \end{cases} \quad (370)$$

The parity divisor controls not only the primitive area reduction but also the relative orientation between the exact coincidence cell and the local beat pattern. For the symmetry-reduced ten-site structure, $n = 1$, $d = 2$. Hence

$$\frac{L_{\text{sc}}^{(10)}}{L_{\text{beat}}^{(10)}} = \sqrt{2}, \quad \frac{A_{\text{sc}}^{(10)}}{A_{\text{beat}}^{(10)}} = 2. \quad (371)$$

Explicitly,

$$L_{\text{beat}}^{(10)} = a\sqrt{\frac{5}{2}}, \quad L_{\text{sc}}^{(10)} = a\sqrt{5}. \quad (372)$$

For the thirty-four-site structure, $n = 1$, $d = 1$. Therefore

$$\frac{L_{\text{sc}}^{(34)}}{L_{\text{beat}}^{(34)}} = 2, \quad \frac{A_{\text{sc}}^{(34)}}{A_{\text{beat}}^{(34)}} = 4. \quad (373)$$

The two length scales are

$$L_{\text{beat}}^{(34)} = \frac{a\sqrt{17}}{2}, \quad L_{\text{sc}}^{(34)} = a\sqrt{17}. \quad (374)$$

For every coprime input pair (m, n) , $m > n > 0$, the complete exact geometry is obtained without numerical optimization:

1. compute

$$\theta = 2 \arctan\left(\frac{n}{m}\right); \quad (375)$$

2. determine

$$d = \begin{cases} 1, & m \not\equiv n \pmod{2}, \\ 2, & m \equiv n \equiv 1 \pmod{2}; \end{cases} \quad (376)$$

3. calculate

$$\Sigma = \frac{m^2 + n^2}{d}, \quad N_{\text{sc}} = 2\Sigma; \quad (377)$$

4. construct (p, q) from Eq. (234);

5. form

$$C_M = a \begin{pmatrix} p & -q \\ q & p \end{pmatrix}; \quad (378)$$

6. form

$$B_M = \frac{2\pi}{a\Sigma} \begin{pmatrix} p & -q \\ q & p \end{pmatrix}; \quad (379)$$

7. generate the Σ site classes from Eq. (241);

8. reduce the angle to $0 \leq \hat{\theta} \leq \pi/4$ using Eq. (248).

Every implementation must satisfy the exact identities

$$\begin{aligned} (m^2 + n^2)R_\theta &= \begin{pmatrix} m^2 - n^2 & -2mn \\ 2mn & m^2 - n^2 \end{pmatrix} \in M_2(\mathbb{Z}), \\ R_\theta^\top R_\theta &= I_2, \quad \det R_\theta = 1, \quad C_M^\top B_M = 2\pi I_2, \\ C_M^\top C_M &= a^2 \Sigma I_2, \quad \det C_M = a^2 \Sigma, \end{aligned} \quad (380)$$

and

$$\det B_M = \frac{(2\pi)^2}{a^2 \Sigma}. \quad (381)$$

These integer and matrix identities provide exact crystallographic certificates for the geometry generator before any tight-binding matrix is assembled.

III. EUCLIDEAN BANDS, DIFFRACTION, AND A PHYSICAL MAGIC-ANGLE BENCHMARK

The exact coincidence geometry derived in Sec. II determines the real-space supercell and the corresponding moiré Brillouin zone. Geometry alone, however, does not determine whether a twisted bilayer develops a physically meaningful flat band. A larger supercell necessarily produces more folded branches, but band multiplication is a crystallographic relabelling. Physical flattening requires a dynamical reconstruction of the eigenvalues and eigenvectors by interlayer hybridization. Throughout this section, the Hamiltonian eigenvalues are written as frequencies measured relative to a common resonator frequency ω_0 . Multiplication by $\hbar$ converts every frequency into an energy. The one-particle calculation is independent of whether the quadratic operators are represented by bosonic resonator operators or fermionic electronic operators.

A. Tight-binding Hamiltonian

Let

$$\Lambda_M = \{n_1 \mathbf{T}_1 + n_2 \mathbf{T}_2 : n_1, n_2 \in \mathbb{Z}\} \quad (382)$$

be the exact coincidence lattice of index

$$\Sigma = [\Lambda : \Lambda_M]. \quad (383)$$

Each layer contains Σ microscopic sites in one exact moiré cell. We choose representatives

$$\mathcal{R}_\ell = \{\mathbf{r}_{\ell s} : s = 1, \dots, \Sigma\}, \quad \ell = 1, 2, \quad (384)$$

so that every site of layer ℓ has a unique decomposition

$$\mathbf{r}_{\ell, \mathbf{R}, s} = \mathbf{R} + \mathbf{r}_{\ell s}, \quad \mathbf{R} \in \Lambda_M. \quad (385)$$

The second layer is obtained by rotating the microscopic square lattice, $\Lambda_2 = R_\theta \Lambda_1$. The two representative sets need not coincide pointwise, although both contain exactly Σ layer-resolved degrees of freedom. Let $\hat{a}_{\ell, \mathbf{R}, s}^\dagger$ create one excitation on site s of layer ℓ in moiré cell $\mathbf{R}$. The most general translation-covariant, number-conserving quadratic Hamiltonian is

$$\begin{aligned} \frac{\hat{H}_E}{\hbar} = & \omega_0 \sum_{\ell=1}^2 \sum_{\mathbf{R} \in \Lambda_M} \sum_{s=1}^{\Sigma} \hat{a}_{\ell, \mathbf{R}, s}^\dagger \hat{a}_{\ell, \mathbf{R}, s} \\ & + \sum_{\ell=1}^2 \sum_{\mathbf{R}, \mathbf{R}'} \sum_{s, s'} h_{ss'}^{(\ell)} (\mathbf{R}' - \mathbf{R}) \hat{a}_{\ell, \mathbf{R}, s}^\dagger \hat{a}_{\ell, \mathbf{R}', s'} \\ & + \sum_{\mathbf{R}, \mathbf{R}'} \sum_{s, s'} \left[T_{ss'} (\mathbf{R}' - \mathbf{R}) \hat{a}_{1, \mathbf{R}, s}^\dagger \hat{a}_{2, \mathbf{R}', s'} + \text{h.c.} \right]. \end{aligned} \quad (386)$$

Hermiticity requires

$$h_{ss'}^{(\ell)}(\mathbf{R}) = \left[h_{s's}^{(\ell)}(-\mathbf{R}) \right]^*. \quad (387)$$

For a radial intralayer hopping kernel $t_{\parallel}(r)$, we take

$$h_{ss'}^{(\ell)}(\mathbf{R}) = -t_{\parallel}(|\mathbf{r}_{\ell s} - \mathbf{r}_{\ell s'} - \mathbf{R}|). \quad (388)$$

For a physical layer separation h and a radial interlayer kernel $t_{\perp}(D)$,

$$T_{ss'}(\mathbf{R}) = t_{\perp} \left[\sqrt{h^2 + |\mathbf{r}_{1s} - \mathbf{r}_{2s'} - \mathbf{R}|^2} \right]. \quad (389)$$

A convenient smooth baseline is

$$t_{\perp}(D) = w \exp \left[-\frac{D - h}{\lambda_{\perp}} \right], \quad (390)$$

where $w = t_{\perp}(h)$ is the vertical interlayer hopping and $\lambda_{\perp}$ is the decay length. The nearest-neighbour monolayer limit is

$$t_{\parallel}(r) = \begin{cases} t, & r = a, \\ 0, & r \neq a. \end{cases} \quad (391)$$

The isolated square-lattice Hamiltonian is then

$$\frac{\hat{H}_0}{\hbar} = -t \sum_{\langle i, j \rangle} \left(\hat{a}_i^\dagger \hat{a}_j + \hat{a}_j^\dagger \hat{a}_i \right). \quad (392)$$

For a finite periodic sample containing N_c exact moiré cells, define

$$\hat{a}_{\ell s}(\mathbf{k}) = \frac{1}{\sqrt{N_c}} \sum_{\mathbf{R} \in \Lambda_M} e^{-i\mathbf{k} \cdot \mathbf{R}} \hat{a}_{\ell, \mathbf{R}, s}, \quad (393)$$

where $\mathbf{k} \in \text{mBZ}$. The inverse transformation is

$$\hat{a}_{\ell, \mathbf{R}, s} = \frac{1}{\sqrt{N_c}} \sum_{\mathbf{k} \in \text{mBZ}} e^{i\mathbf{k} \cdot \mathbf{R}} \hat{a}_{\ell s}(\mathbf{k}). \quad (394)$$

Substitution into Eq. (386) gives

$$\hat{H}_E = \hbar \sum_{\mathbf{k} \in \text{mBZ}} \hat{\mathbf{A}}^\dagger(\mathbf{k}) H_M(\mathbf{k}) \hat{\mathbf{A}}(\mathbf{k}), \quad (395)$$

where

$$\hat{\mathbf{A}}(\mathbf{k}) = \begin{pmatrix} \hat{a}_{11}(\mathbf{k}) \\ \vdots \\ \hat{a}_{1\Sigma}(\mathbf{k}) \\ \hat{a}_{21}(\mathbf{k}) \\ \vdots \\ \hat{a}_{2\Sigma}(\mathbf{k}) \end{pmatrix}. \quad (396)$$

The exact Euclidean bilayer Bloch Hamiltonian is

$$H_M(\mathbf{k}) = \omega_0 I_{2\Sigma} + \begin{pmatrix} h_1(\mathbf{k}) & T(\mathbf{k}) \\ T^\dagger(\mathbf{k}) & h_2(\mathbf{k}) \end{pmatrix}. \quad (397)$$

Its blocks are

$$[h_\ell(\mathbf{k})]_{ss'} = \sum_{\mathbf{R} \in \Lambda_M} h_{ss'}^{(\ell)}(\mathbf{R}) e^{i\mathbf{k} \cdot \mathbf{R}}, \quad (398)$$

and

$$[T(\mathbf{k})]_{ss'} = \sum_{\mathbf{R} \in \Lambda_M} T_{ss'}(\mathbf{R}) e^{i\mathbf{k} \cdot \mathbf{R}}. \quad (399)$$

The matrix dimension is exactly

$$\dim H_M(\mathbf{k}) = 2\Sigma = N_{sc}. \quad (400)$$

Thus real-space site counting and reciprocal-space band counting coincide exactly. The Bloch blocks satisfy

$$h_\ell^\dagger(\mathbf{k}) = h_\ell(\mathbf{k}), \quad (401)$$

and hence

$$H_M^\dagger(\mathbf{k}) = H_M(\mathbf{k}). \quad (402)$$

For real hoppings,

$$H_M^*(\mathbf{k}) = H_M(-\mathbf{k}), \quad (403)$$

so

$$\text{Spec } H_M(\mathbf{k}) = \text{Spec } H_M(-\mathbf{k}). \quad (404)$$

For nearest-neighbour hopping, the primitive monolayer dispersion is

$$\varepsilon_0(\mathbf{q}) = -2t [\cos(q_x a) + \cos(q_y a)]. \quad (405)$$

Its extrema are

$$\varepsilon_{\min} = -4t, \quad \varepsilon_{\max} = 4t, \quad (406)$$

so the primitive bandwidth is $W_0 = 8|t|$. The physical group velocity is

$$\mathbf{v}_0(\mathbf{q}) = \frac{1}{\hbar} \nabla_{\mathbf{q}} \varepsilon_0(\mathbf{q}) = \frac{2ta}{\hbar} \begin{pmatrix} \sin(q_x a) \\ \sin(q_y a) \end{pmatrix}.$$

The Hessian is

$$\nabla_{\mathbf{q}}^2 \varepsilon_0 = 2ta^2 \begin{pmatrix} \cos(q_x a) & 0 \\ 0 & \cos(q_y a) \end{pmatrix}. \quad (407)$$

At the zone centre,

$$\nabla_{\mathbf{q}}^2 \varepsilon_0(\mathbf{0}) = 2ta^2 I_2. \quad (408)$$

The bare effective mass is therefore

$$m_0^* = \frac{\hbar^2}{2ta^2}. \quad (409)$$

The expansion near Γ is

$$\begin{aligned} \varepsilon_0(\mathbf{q}) &= -4t + ta^2 (q_x^2 + q_y^2) - \frac{ta^4}{12} (q_x^4 + q_y^4) \\ &+ \frac{ta^6}{360} (q_x^6 + q_y^6) + O(a^8 q^8). \end{aligned} \quad (410)$$

We define

$$A = ta^2 = \frac{\hbar^2}{2m_0^*}. \quad (411)$$

Because $\Lambda_\ell^* \subset \Lambda_M^*$, the quotient $\Lambda_M^*/\Lambda_\ell^*$ contains exactly Σ elements. Choose representatives

$$\mathcal{Q}_\ell = \left\{ \mathbf{Q}_\alpha^{(\ell)} : \alpha = 1, \dots, \Sigma \right\}. \quad (412)$$

The translated moiré Brillouin zones tile the primitive Brillouin zone:

$$\text{BZ}_\ell = \bigsqcup_{\alpha=1}^{\Sigma} \left(\text{mBZ} + \mathbf{Q}_\alpha^{(\ell)} \right) \pmod{\Lambda_\ell^*}. \quad (413)$$

Define

$$[U_\ell(\mathbf{k})]_{s\alpha} = \frac{1}{\sqrt{\Sigma}} \exp \left[i \left(\mathbf{k} + \mathbf{Q}_\alpha^{(\ell)} \right) \cdot \mathbf{r}_{\ell s} \right]. \quad (414)$$

The quotient-character orthogonality relation is

$$\frac{1}{\Sigma} \sum_{s=1}^{\Sigma} e^{i(\mathbf{Q}_\beta^{(\ell)} - \mathbf{Q}_\alpha^{(\ell)}) \cdot \mathbf{r}_{\ell s}} = \delta_{\alpha\beta}. \quad (415)$$

Consequently,

$$U_\ell^\dagger(\mathbf{k}) U_\ell(\mathbf{k}) = I_\Sigma. \quad (416)$$

For the unrotated layer,

$$U_1^\dagger h_1 U_1 = \text{diag}_\alpha \left[\varepsilon_0 \left(\mathbf{k} + \mathbf{Q}_\alpha^{(1)} \right) \right]. \quad (417)$$

For the rotated layer,

$$U_2^\dagger h_2 U_2 = \text{diag}_\beta \left[\varepsilon_0 \left(R_{-\theta} \left[\mathbf{k} + \mathbf{Q}_\beta^{(2)} \right] \right) \right]. \quad (418)$$

We denote these diagonal matrices by $E_1^{\text{fold}}(\mathbf{k})$, $E_2^{\text{fold}}(\mathbf{k})$. The folded interlayer matrix is

$$\mathcal{T}(\mathbf{k}) = U_1^\dagger(\mathbf{k}) T(\mathbf{k}) U_2(\mathbf{k}). \quad (419)$$

With

$$\mathcal{U}(\mathbf{k}) = \begin{pmatrix} U_1(\mathbf{k}) & 0 \\ 0 & U_2(\mathbf{k}) \end{pmatrix}, \quad (420)$$

the folded Hamiltonian is

$$H_{\text{fold}}(\mathbf{k}) = \mathcal{U}^\dagger H_M(\mathbf{k}) \mathcal{U} = \begin{pmatrix} E_1^{\text{fold}}(\mathbf{k}) & \mathcal{T}(\mathbf{k}) \\ \mathcal{T}^\dagger(\mathbf{k}) & E_2^{\text{fold}}(\mathbf{k}) \end{pmatrix}. \quad (421)$$

Therefore,

$$\text{Spec } H_{\text{fold}}(\mathbf{k}) = \text{Spec } H_M(\mathbf{k}). \quad (422)$$

The site representation and the folded-momentum representation are two exactly equivalent descriptions of the same 2Σ -dimensional spectral problem.

B. Ten-site, thirty-four-site, and sub-hundred-site band structures

The exact cells obtained in Sec. II define a hierarchy of finite-dimensional Bloch benchmarks. The two smallest cells singled out in Topic 1 have

$$\Sigma_{10} = 5, \quad \Sigma_{34} = 17. \quad (423)$$

Their Bloch dimensions are

$$\dim H_{10}(\mathbf{k}) = 10, \quad \dim H_{34}(\mathbf{k}) = 34. \quad (424)$$

For a canonical square coincidence basis

$$C_M = a \begin{pmatrix} p & -q \\ q & p \end{pmatrix}, \quad p^2 + q^2 = \Sigma, \quad (425)$$

the reciprocal basis is

$$\mathbf{G}_1 = \frac{2\pi}{a\Sigma} \begin{pmatrix} p \\ q \end{pmatrix}, \quad \mathbf{G}_2 = \frac{2\pi}{a\Sigma} \begin{pmatrix} -q \\ p \end{pmatrix}. \quad (426)$$

Its magnitude is

$$G_M = \frac{2\pi}{a\sqrt{\Sigma}}. \quad (427)$$

A convenient cyclic set of folding representatives for the first layer is

$$\mathbf{Q}_j^{(1)} = j\mathbf{G}_1, \quad j = 0, \dots, \Sigma - 1, \quad (428)$$

understood modulo Λ_1^* . Since

$$\Sigma\mathbf{G}_1 = \frac{2\pi}{a} \begin{pmatrix} p \\ q \end{pmatrix} \in \Lambda_1^*, \quad (429)$$

and

$$\gcd(p, \Sigma) = \gcd(q, \Sigma) = 1, \quad (430)$$

no positive integer smaller than Σ returns the vector to the microscopic reciprocal lattice. Thus $\mathbf{G}_1$ has order exactly Σ in the reciprocal quotient. For each Σ , the unhybridized folded matrices can be written explicitly as

$$\left[E_1^{\text{fold}, \Sigma}(\mathbf{k}) \right]_{jj} = \varepsilon_0 (\mathbf{k} + j\mathbf{G}_1), \quad (431)$$

and

$$\left[E_2^{\text{fold}, \Sigma}(\mathbf{k}) \right]_{jj} = \varepsilon_0 \left[R_{-\theta} \left(\mathbf{k} + \mathbf{Q}_j^{(2)} \right) \right]. \quad (432)$$

The full matrix is

$$H_\Sigma(\mathbf{k}) = \begin{pmatrix} E_1^{\text{fold}, \Sigma}(\mathbf{k}) & \mathcal{T}_\Sigma(\mathbf{k}) \\ \mathcal{T}_\Sigma^\dagger(\mathbf{k}) & E_2^{\text{fold}, \Sigma}(\mathbf{k}) \end{pmatrix}. \quad (433)$$

For the ten-site cell,

$$\mathbf{G}_1^{(10)} = \frac{2\pi}{5a} \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{G}_2^{(10)} = \frac{2\pi}{5a} \begin{pmatrix} -1 \\ 2 \end{pmatrix}. \quad (434)$$

The five folded branches of the first layer are

$$E_{1j}^{(0,10)}(\mathbf{k}) = -2t \left\{ \cos \left[a \left(k_x + jG_{1x}^{(10)} \right) \right] + \cos \left[a \left(k_y + jG_{1y}^{(10)} \right) \right] \right\}, \quad (435)$$

with $j = 0, 1, 2, 3, 4$. For the thirty-four-site cell,

$$\mathbf{G}_1^{(34)} = \frac{2\pi}{17a} \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \quad \mathbf{G}_2^{(34)} = \frac{2\pi}{17a} \begin{pmatrix} -1 \\ 4 \end{pmatrix}, \quad (436)$$

and the corresponding first-layer folded branches are

$$E_{1j}^{(0,34)}(\mathbf{k}) = \varepsilon_0 \left(\mathbf{k} + j\mathbf{G}_1^{(34)} \right), \quad j = 0, \dots, 16. \quad (437)$$

The exact nontrivial sub-hundred-site benchmarks are summarized below.

For every cell, the conventional display path is

$$\Gamma \longrightarrow X \longrightarrow M \longrightarrow \Gamma, \quad (438)$$

with

$$\Gamma = \mathbf{0}, \quad X = \frac{\mathbf{G}_1}{2}, \quad M = \frac{\mathbf{G}_1 + \mathbf{G}_2}{2}. \quad (439)$$

A continuous parametrization is

$$\begin{aligned} \mathbf{k}_{\Gamma X}(s) &= \frac{s}{2} \mathbf{G}_1, \quad 0 \leq s \leq 1, \\ \mathbf{k}_{XM}(s) &= \frac{\mathbf{G}_1}{2} + \frac{s}{2} \mathbf{G}_2, \quad 0 \leq s \leq 1, \end{aligned} \quad (440)$$

and

$$\mathbf{k}_{M\Gamma}(s) = \frac{1-s}{2} (\mathbf{G}_1 + \mathbf{G}_2), \quad 0 \leq s \leq 1. \quad (441)$$

The path is used only for visual presentation. Every bandwidth, gap, velocity maximum, effective-mass analysis, and density of states entering the physical magic benchmark is evaluated over the complete two-dimensional domain

$$\mathbf{k} = u\mathbf{G}_1 + v\mathbf{G}_2, \quad -\frac{1}{2} \leq u, v < \frac{1}{2}. \quad (442)$$

A narrow appearance along $\Gamma-X-M-\Gamma$ is not accepted as evidence for a globally flat band.

C. $T = 0$: exact band folding without physical flattening

Set $T(\mathbf{k}) = 0$. The Hamiltonian becomes

$$H_{\text{fold}}^{(0)}(\mathbf{k}) = E_1^{\text{fold}}(\mathbf{k}) \oplus E_2^{\text{fold}}(\mathbf{k}). \quad (443)$$

Its eigenvalues are exactly

$$E_{1\alpha}^{(0)}(\mathbf{k}) = \varepsilon_0 \left(\mathbf{k} + \mathbf{Q}_\alpha^{(1)} \right), \quad (444)$$

and

$$E_{2\beta}^{(0)}(\mathbf{k}) = \varepsilon_0 \left[R_{-\theta} \left(\mathbf{k} + \mathbf{Q}_\beta^{(2)} \right) \right]. \quad (445)$$

TABLE II. Exact Euclidean band-structure benchmarks with $N_{\text{sc}} < 100$. Every layer contributes Σ folded branches, while the coupled bilayer Hamiltonian has dimension 2Σ .

Σ	N_{sc}	(m, n)	d	$\hat{\theta}$	$G_{\text{Ma}}/(2\pi)$	$\dim H_\Sigma$
5	10	(3, 1)	2	36.869897646°	$5^{-1/2}$	10
13	26	(5, 1)	2	22.619864948°	$13^{-1/2}$	26
17	34	(4, 1)	1	28.072486936°	$17^{-1/2}$	34
25	50	(7, 1)	2	16.260204708°	$25^{-1/2}$	50
29	58	(5, 2)	1	43.602818973°	$29^{-1/2}$	58
37	74	(6, 1)	1	18.924644416°	$37^{-1/2}$	74
41	82	(9, 1)	2	12.680383492°	$41^{-1/2}$	82

Differentiation gives

$$\nabla_{\mathbf{k}} E_{1\alpha}^{(0)} = \nabla_{\mathbf{q}} \varepsilon_0(\mathbf{q})|_{\mathbf{q}=\mathbf{k}+\mathbf{Q}_\alpha^{(1)}}, \quad (446)$$

and

$$\nabla_{\mathbf{k}} E_{2\beta}^{(0)} = R_\theta \nabla_{\mathbf{q}} \varepsilon_0(\mathbf{q})|_{\mathbf{q}=R_{-\theta}[\mathbf{k}+\mathbf{Q}_\beta^{(2)}]}. \quad (447)$$

Rotation preserves the norm, so

$$|\nabla_{\mathbf{k}} E_{2\beta}^{(0)}| = |\nabla_{\mathbf{q}} \varepsilon_0|. \quad (448)$$

Folding therefore does not suppress the primitive velocity field. It only partitions the same field among Σ branches in the reduced zone. This statement is also exact at the density-of-states level. Define the DOS per microscopic bilayer degree of freedom by

$$\rho_{\text{fold}}^{(0)}(E) = \frac{1}{2\Sigma A_{\text{mBZ}}} \sum_{\ell=1}^2 \sum_{\alpha=1}^{\Sigma} \int_{\text{mBZ}} d^2k \delta[E - E_{\ell\alpha}^{(0)}(\mathbf{k})]. \quad (449)$$

Since the translated mBZs tile the primitive BZ,

$$\sum_{\alpha=1}^{\Sigma} \int_{\text{mBZ}} d^2k F(\mathbf{k} + \mathbf{Q}_\alpha^{(\ell)}) = \int_{\text{BZ}_\ell} d^2q F(\mathbf{q}). \quad (450)$$

Moreover,

$$A_{\text{BZ}}^{(0)} = \Sigma A_{\text{mBZ}}. \quad (451)$$

Therefore

$$\begin{aligned} \rho_{\text{fold}}^{(0)}(E) &= \frac{1}{A_{\text{BZ}}^{(0)}} \int_{\text{BZ}^{(0)}} d^2q \delta[E - \varepsilon_0(\mathbf{q})] \\ &= \rho_0(E). \end{aligned} \quad (452)$$

Thus

$$\rho_{\text{fold}}^{(0)}(E) = \rho_0(E). \quad (453)$$

More generally, for every measurable function F ,

$$\frac{1}{2\Sigma A_{\text{mBZ}}} \sum_{\ell,\alpha} \int_{\text{mBZ}} d^2k F[E_{\ell\alpha}^{(0)}(\mathbf{k}), \nabla_{\mathbf{k}} E_{\ell\alpha}^{(0)}(\mathbf{k})]$$

$$= \frac{1}{A_{\text{BZ}}^{(0)}} \int_{\text{BZ}^{(0)}} d^2q F[\varepsilon_0(\mathbf{q}), \nabla_{\mathbf{q}} \varepsilon_0(\mathbf{q})]. \quad (454)$$

The complete joint distribution of energy and velocity is therefore unchanged by exact supercell relabelling. A single folded branch can appear narrow because it samples only one small piece of the primitive Brillouin zone. Define its displayed width by

$$W_\alpha^{\text{plot}} = \max_{\mathbf{k} \in \text{mBZ}} E_\alpha^{(0)}(\mathbf{k}) - \min_{\mathbf{k} \in \text{mBZ}} E_\alpha^{(0)}(\mathbf{k}). \quad (455)$$

This quantity may decrease as the supercell grows, even though the union of the Σ branches remains exactly the primitive band. The physical unfolded family has width

$$W_{\text{unfolded}} = \max_{\mathbf{q} \in \text{BZ}^{(0)}} \varepsilon_0(\mathbf{q}) - \min_{\mathbf{q} \in \text{BZ}^{(0)}} \varepsilon_0(\mathbf{q}) = 8|t|. \quad (456)$$

Hence

$$\text{small } W_\alpha^{\text{plot}} \not\Rightarrow \text{physical band flattening}. \quad (457)$$

At $T = 0$, every eigenstate belongs to one layer and one folded momentum: $P_L = \pm 1$, and the folded-momentum participation number defined below satisfies $N_{\text{part},k} = 1$. There is therefore no interlayer hybridization mechanism capable of producing a physical magic-angle state. The exact null chain is

$$T = 0 \Rightarrow \begin{cases} \text{primitive spectrum is only folded,} \\ \text{unfolded velocity field is unchanged,} \\ \rho_{\text{fold}} = \rho_0, \\ P_L = \pm 1, \\ N_{\text{part},k} = 1. \end{cases} \quad (458)$$

D. $T \neq 0$: genuine interlayer hybridization

At a fixed $\mathbf{k}$, write

$$H(\mathbf{k}) = \begin{pmatrix} A(\mathbf{k}) & T(\mathbf{k}) \\ T^\dagger(\mathbf{k}) & B(\mathbf{k}) \end{pmatrix}. \quad (459)$$

The unhybridized matrix is

$$H_0(\mathbf{k}) = A(\mathbf{k}) \oplus B(\mathbf{k}). \quad (460)$$

Since the interlayer block is off diagonal,

$$\text{Tr } H = \text{Tr } A + \text{Tr } B = \text{Tr } H_0. \quad (461)$$

The spectral centroid is unchanged by interlayer hopping. Squaring gives

$$H^2 = \begin{pmatrix} A^2 + TT^\dagger & AT + TB \\ T^\dagger A + BT^\dagger & B^2 + T^\dagger T \end{pmatrix}. \quad (462)$$

Therefore

$$\text{Tr } H^2 = \text{Tr } A^2 + \text{Tr } B^2 + 2 \text{Tr } TT^\dagger. \quad (463)$$

Define the normalized raw second moment

$$\mu_2(\mathbf{k}) = \frac{1}{2\Sigma} \text{Tr } H^2(\mathbf{k}). \quad (464)$$

Its exact hybridization-induced increment is

$$\Delta\mu_2(\mathbf{k}) = \mu_2(\mathbf{k}) - \mu_2^{(0)}(\mathbf{k}) = \frac{1}{\Sigma} \text{Tr } [T(\mathbf{k})T^\dagger(\mathbf{k})]. \quad (465)$$

Equivalently,

$$\Delta\mu_2(\mathbf{k}) = \frac{1}{\Sigma} \|T(\mathbf{k})\|_F^2 \geq 0. \quad (466)$$

This identity provides a basis-independent analysis of the implemented interlayer hybridization. If

$$T(\mathbf{k}) = w\mathcal{T}(\mathbf{k}), \quad (467)$$

then

$$\Delta\mu_2(\mathbf{k}) = \frac{w^2}{\Sigma} \|\mathcal{T}(\mathbf{k})\|_F^2. \quad (468)$$

The first nonzero raw-moment response is therefore exactly quadratic in the interlayer hopping amplitude. Consider two unhybridized folded states with energies $\varepsilon_1(\mathbf{k})$, $\varepsilon_2(\mathbf{k})$, coupled by

$$V(\mathbf{k}) = \langle 1, \mathbf{k} | T(\mathbf{k}) | 2, \mathbf{k} \rangle. \quad (469)$$

The exact two-level Hamiltonian is

$$H_2(\mathbf{k}) = \begin{pmatrix} \varepsilon_1(\mathbf{k}) & V(\mathbf{k}) \\ V^*(\mathbf{k}) & \varepsilon_2(\mathbf{k}) \end{pmatrix}. \quad (470)$$

Define

$$\bar{\varepsilon} = \frac{\varepsilon_1 + \varepsilon_2}{2}, \quad \Delta = \varepsilon_1 - \varepsilon_2. \quad (471)$$

The exact eigenvalues are

$$E_{\pm} = \bar{\varepsilon} \pm \Omega, \quad \Omega = \sqrt{\frac{\Delta^2}{4} + |V|^2}. \quad (472)$$

At resonance, $\Delta(\mathbf{k}_0) = 0$, the hybridization gap is

$$\Delta_{\text{hyb}} = E_+ - E_- = 2|V(\mathbf{k}_0)|. \quad (473)$$

Differentiating Eq. (472) gives

$$\nabla_{\mathbf{k}} E_{\pm} = \frac{\nabla_{\mathbf{k}} \varepsilon_1 + \nabla_{\mathbf{k}} \varepsilon_2}{2} \pm \frac{\frac{\Delta}{4} \nabla_{\mathbf{k}} \Delta + \text{Re}(V^* \nabla_{\mathbf{k}} V)}{\sqrt{\Delta^2/4 + |V|^2}}. \quad (474)$$

The corresponding velocities are

$$\mathbf{v}_{\pm} = \frac{1}{\hbar} \nabla_{\mathbf{k}} E_{\pm}. \quad (475)$$

At an exact resonance with locally momentum-independent coupling,

$$\nabla_{\mathbf{k}} V(\mathbf{k}_0) = 0, \quad (476)$$

we obtain

$$\mathbf{v}_{\pm}(\mathbf{k}_0) = \frac{\mathbf{v}_1(\mathbf{k}_0) + \mathbf{v}_2(\mathbf{k}_0)}{2}. \quad (477)$$

Hybridization therefore replaces the two crossing velocities by their average at the centre of the avoided crossing. For a simple full-band eigenvalue $E_n(\mathbf{k})$, the exact Hellmann–Feynman derivative is

$$\partial_{\mu} E_n = \langle u_n | \partial_{\mu} H | u_n \rangle. \quad (478)$$

The Hessian is

$$\begin{aligned} \partial_{\mu} \partial_{\nu} E_n &= \langle u_n | \partial_{\mu} \partial_{\nu} H | u_n \rangle \\ &+ 2 \text{Re} \sum_{m \neq n} \frac{\langle u_n | \partial_{\mu} H | u_m \rangle \langle u_m | \partial_{\nu} H | u_n \rangle}{E_n - E_m}. \end{aligned} \quad (479)$$

The second term explicitly shows how nearby remote bands renormalize the effective mass. Small energy denominators near an avoided-crossing network can generate a large Hessian correction even when the bare monolayer curvature is fixed. The block eigenvalue problem is

$$(A - E)|\psi_1\rangle + T|\psi_2\rangle = 0, \quad (480)$$

and

$$T^\dagger|\psi_1\rangle + (B - E)|\psi_2\rangle = 0. \quad (481)$$

Whenever $E \notin \text{Spec } B$, the second equation gives

$$|\psi_2\rangle = (E - B)^{-1} T^\dagger |\psi_1\rangle. \quad (482)$$

Substitution yields

$$[A + \Sigma_1(E, \mathbf{k})] |\psi_1\rangle = E |\psi_1\rangle, \quad (483)$$

with

$$\Sigma_1(E, \mathbf{k}) = T(\mathbf{k}) [E - B(\mathbf{k})]^{-1} T^\dagger(\mathbf{k}). \quad (484)$$

The operator norm obeys

$$\|\Sigma_1(E, \mathbf{k})\| \leq \frac{\|T(\mathbf{k})\|^2}{\text{dist}[E, \text{Spec } B(\mathbf{k})]}. \quad (485)$$

Strong reconstruction therefore requires both a non-negligible interlayer matrix and near-resonant remote states. Differentiating the nonlinear eigenproblem gives

$$\frac{dE}{dk_\mu} = Z_{\text{eff}} \langle \psi_1 | \partial_\mu A + (\partial_\mu \Sigma_1)_E | \psi_1 \rangle, \quad (486)$$

where

$$Z_{\text{eff}} = [\langle \psi_1 | I - \partial_E \Sigma_1 | \psi_1 \rangle]^{-1}. \quad (487)$$

Define

$$\mathbf{v}_{\text{bare}} = \frac{Z_{\text{eff}}}{\hbar} \langle \psi_1 | \nabla_{\mathbf{k}} A | \psi_1 \rangle, \quad (488)$$

and

$$\mathbf{v}_{\text{hyb}} = \frac{Z_{\text{eff}}}{\hbar} \langle \psi_1 | \nabla_{\mathbf{k}} \Sigma_1 | \psi_1 \rangle. \quad (489)$$

Then

$$\mathbf{v} = \mathbf{v}_{\text{bare}} + \mathbf{v}_{\text{hyb}}. \quad (490)$$

The exact local flattening mechanism is a cancellation between the bare kinetic covector and the interlayer self-energy covector. A normalized local cancellation residual is

$$\mathcal{R}_v(\mathbf{k}) = \frac{|\mathbf{v}_{\text{bare}} + \mathbf{v}_{\text{hyb}}|}{|\mathbf{v}_{\text{bare}}| + |\mathbf{v}_{\text{hyb}}|}. \quad (491)$$

It satisfies $0 \leq \mathcal{R}_v \leq 1$. Perfect local cancellation gives $\mathcal{R}_v = 0$. A global flat band requires

$$\mathcal{R}_{v,C}^{\text{max}} = \sup_{\mathbf{k} \in \text{mBZ}} \mathcal{R}_{v,C}(\mathbf{k}) \ll 1, \quad (492)$$

rather than cancellation at one isolated momentum. For a two-dimensional band $E_n(\mathbf{k})$, define the DOS per unit area by

$$D_n(E) = \frac{1}{(2\pi)^2} \int_{\text{mBZ}} d^2k \delta[E - E_n(\mathbf{k})]. \quad (493)$$

When E is a regular value, the coarea formula gives

$$D_n(E) = \frac{1}{(2\pi)^2} \oint_{E_n(\mathbf{k})=E} \frac{d\ell}{|\nabla_{\mathbf{k}} E_n(\mathbf{k})|}. \quad (494)$$

Thus suppression of the group velocity enhances the DOS whenever the constant-energy contour retains nonzero measure. For an isotropic quadratic extremum

$$E(\mathbf{k}) = E_\Gamma + A_{\text{eff}} k^2, \quad (495)$$

direct integration gives

$$D(E) = \frac{1}{4\pi |A_{\text{eff}}|} \quad (496)$$

on the allowed side of the extremum. Since

$$A_{\text{eff}} = \frac{\hbar^2}{2m_{\text{eff}}^*}, \quad (497)$$

we obtain

$$D(E) = \frac{|m_{\text{eff}}^*|}{2\pi\hbar^2}. \quad (498)$$

Effective-mass enhancement and DOS enhancement are therefore the same local phenomenon in the quadratic regime. Write a normalized folded eigenstate as

$$|u_{n\mathbf{k}}\rangle = \sum_{\alpha=1}^{\Sigma} u_{1\alpha}^{(n)}(\mathbf{k}) |1, \alpha; \mathbf{k}\rangle + \sum_{\beta=1}^{\Sigma} u_{2\beta}^{(n)}(\mathbf{k}) |2, \beta; \mathbf{k}\rangle. \quad (499)$$

The folded weights are

$$w_{1\alpha}^{(n)} = |u_{1\alpha}^{(n)}|^2, \quad w_{2\beta}^{(n)} = |u_{2\beta}^{(n)}|^2, \quad (500)$$

with

$$\sum_{\alpha} w_{1\alpha}^{(n)} + \sum_{\beta} w_{2\beta}^{(n)} = 1. \quad (501)$$

The folded-momentum inverse participation ratio is

$$\text{IPR}_k^{(n)} = \sum_{\alpha} \left(w_{1\alpha}^{(n)}\right)^2 + \sum_{\beta} \left(w_{2\beta}^{(n)}\right)^2. \quad (502)$$

The corresponding participation number is

$$N_{\text{part},k}^{(n)} = \frac{1}{\text{IPR}_k^{(n)}}. \quad (503)$$

A trivially folded state satisfies $N_{\text{part},k} \simeq 1$, whereas a state reconstructed from several resonant folded momenta satisfies $N_{\text{part},k} > 1$. Let P_1 and P_2 be the layer projectors. The layer polarization is

$$P_L^{(n)}(\mathbf{k}) = \langle u_{n\mathbf{k}} | P_1 - P_2 | u_{n\mathbf{k}} \rangle. \quad (504)$$

It satisfies $-1 \leq P_L \leq 1$. The layer participation number is

$$\mathcal{P}_L = \frac{2}{1 + P_L^2}. \quad (505)$$

It obeys $1 \leq \mathcal{P}_L \leq 2$. A layer-polarized state has $\mathcal{P}_L = 1$, whereas an equal-weight bilayer state has $\mathcal{P}_L = 2$. If

$$H = H_0 + \omega t V_\perp, \quad \omega = \frac{w}{t}, \quad (506)$$

Hellmann–Feynman gives

$$\frac{\partial E_n}{\partial \omega} = t \langle u_n | V_\perp | u_n \rangle. \quad (507)$$

A state with vanishing response to w/t cannot be identified as a hybridization-generated flat-band state without additional evidence.

E. Fourier transform and diffraction

Let the microscopic basis positions of one layer be τ_α , $\alpha = 1, \dots, n_b$. For the one-site square model, $n_b = 1$, $\tau_1 = \mathbf{0}$. The first-layer positions are

$$\mathbf{r}_{1,\mathbf{R},\alpha} = \mathbf{R} + \tau_\alpha, \quad (508)$$

and the second-layer positions are

$$\mathbf{r}_{2,\mathbf{R},\alpha} = R_\theta (\mathbf{R} + \tau_\alpha) + \mathbf{d}, \quad (509)$$

where $\mathbf{d}$ is a rigid in-plane displacement. The three-dimensional coordinates are

$$\mathbf{X}_{\ell,\mathbf{R},\alpha} = (\mathbf{r}_{\ell,\mathbf{R},\alpha}, z_\ell), \quad z_1 = -\frac{h}{2}, \quad z_2 = +\frac{h}{2}. \quad (510)$$

For scattering vector $\mathbf{Q} = (\mathbf{q}, q_z)$, the kinematic amplitude is

$$\mathcal{A}(\mathbf{Q}) = \sum_{\ell,\mathbf{R},\alpha} f_{\ell\alpha}(\mathbf{Q}) e^{-i\mathbf{Q} \cdot \mathbf{r}_{\ell,\mathbf{R},\alpha}} e^{-iq_z z_\ell}. \quad (511)$$

The unnormalized intensity is

$$I(\mathbf{Q}) = |\mathcal{A}(\mathbf{Q})|^2. \quad (512)$$

For identical unit form factors and purely in-plane scattering,

$$S(\mathbf{q}) = \frac{1}{N_{\text{tot}}} \left| \sum_{j=1}^{N_{\text{tot}}} e^{-i\mathbf{q} \cdot \mathbf{r}_j} \right|^2. \quad (513)$$

The unnormalized convention requested in Topic 1 is

$$S_{\text{unnorm}}(\mathbf{q}) = \left| \sum_j e^{i\mathbf{q} \cdot \mathbf{r}_j} \right|^2, \quad (514)$$

which differs only by normalization and Fourier-sign convention. Write

$$\mathcal{A} = \mathcal{A}_1 + \mathcal{A}_2. \quad (515)$$

Then

$$I = |\mathcal{A}_1|^2 + |\mathcal{A}_2|^2 + 2 \text{Re}(\mathcal{A}_1 \mathcal{A}_2^*). \quad (516)$$

The first two terms are the separate monolayer diffraction patterns. The third is the coherent bilayer interference term. For layer one, Poisson summation gives

$$\sum_{\mathbf{R} \in \Lambda} e^{-i\mathbf{Q} \cdot \mathbf{R}} = \frac{(2\pi)^2}{a^2} \sum_{\mathbf{G} \in \Lambda^*} \delta^{(2)}(\mathbf{Q} - \mathbf{G}). \quad (517)$$

Therefore the first-layer Bragg peaks lie on $\Lambda_1^* = \Lambda^*$. For the second layer, $\Lambda_2^* = R_\theta \Lambda^*$. The ideal rigid-bilayer linear-scattering amplitude therefore has support

$$\text{supp } \mathcal{A}_0 = \Lambda_1^* \cup \Lambda_2^*. \quad (518)$$

At an exact commensurate twist, $\Lambda_M^* = \Lambda_1^* + \Lambda_2^*$ contains both microscopic reciprocal lattices. Nevertheless, $\Lambda_M^* \setminus (\Lambda_1^* \cup \Lambda_2^*)$ is systematically extinct for a perfectly rigid bilayer in the linear density-scattering approximation. Thus translation-allowed moiré vector $\Rightarrow$ nonzero rigid-bilayer intensity. Let $\mathbf{G}$ be a reciprocal vector of magnitude G . The corresponding peak in the second layer is $R_\theta \mathbf{G}$. Define

$$\Delta \mathbf{G}_\theta = R_\theta \mathbf{G} - \mathbf{G}. \quad (519)$$

Its squared norm is

$$\begin{aligned} |\Delta \mathbf{G}_\theta|^2 &= |R_\theta \mathbf{G}|^2 + |\mathbf{G}|^2 - 2\mathbf{G} \cdot R_\theta \mathbf{G} \\ &= 2G^2(1 - \cos \theta) = 4G^2 \sin^2\left(\frac{\theta}{2}\right). \end{aligned}$$

Hence

$$\Delta q_G = 2G \left| \sin\left(\frac{\theta}{2}\right) \right|. \quad (520)$$

For the first square reciprocal shell,

$$G_0 = \frac{2\pi}{a}, \quad (521)$$

so

$$\Delta q_1 = \frac{4\pi}{a} \left| \sin\left(\frac{\theta}{2}\right) \right|. \quad (522)$$

The corresponding local beat length is

$$L_{\text{beat}} = \frac{2\pi}{\Delta q_1} = \frac{a}{2|\sin(\theta/2)|}. \quad (523)$$

For an exact commensurate twist,

$$G_M = \frac{2\pi}{a\sqrt{\Sigma}}. \quad (524)$$

Using

$$\Sigma = \frac{n^2}{d} \csc^2\left(\frac{\hat{\theta}}{2}\right), \quad (525)$$

we obtain

$$\frac{\Delta q_1}{G_M} = \frac{2n}{\sqrt{d}}. \quad (526)$$

Thus the primary atomic peak splitting is generally an arithmetic higher harmonic of the primitive exact moiré reciprocal lattice. The corresponding real-space ratio is

$$\frac{L_{\text{sc}}}{L_{\text{beat}}} = \frac{2n}{\sqrt{d}}, \quad (527)$$

and the area ratio is

$$\frac{A_{\text{sc}}}{A_{\text{beat}}} = \frac{4n^2}{d}. \quad (528)$$

Peak splitting determines the local geometric moiré length, but the primitive exact reciprocal spacing contains additional arithmetic information. Let a weak moiré-periodic displacement field satisfy

$$\mathbf{u}_\ell(\mathbf{r} + \mathbf{R}_M) = \mathbf{u}_\ell(\mathbf{r}). \quad (529)$$

Its Fourier expansion is

$$\mathbf{u}_\ell(\mathbf{r}) = \sum_{\mathbf{Q} \in \Lambda_M^*} \mathbf{u}_{\ell, \mathbf{Q}} e^{i\mathbf{Q} \cdot \mathbf{r}}. \quad (530)$$

For $|\mathbf{q} \cdot \mathbf{u}| \ll 1$, the scattering phase expands as

$$e^{-i\mathbf{q} \cdot (\mathbf{r} + \mathbf{u})} = e^{-i\mathbf{q} \cdot \mathbf{r}} \left[1 - i\mathbf{q} \cdot \mathbf{u} - \frac{1}{2}(\mathbf{q} \cdot \mathbf{u})^2 + \dots \right]. \quad (531)$$

The first-order satellite condition is

$$\mathbf{q}_{\text{sat}} = \mathbf{G}_\ell + \mathbf{Q}, \quad \mathbf{Q} \in \Lambda_M^*. \quad (532)$$

The leading amplitude obeys

$$\delta \mathcal{A}_\ell^{(1)} \propto -i(\mathbf{G}_\ell + \mathbf{Q}) \cdot \mathbf{u}_{\ell, \mathbf{Q}}, \quad (533)$$

and hence

$$\frac{I_{\mathbf{G}+\mathbf{Q}}}{I_{\mathbf{G}}} \propto |(\mathbf{G} + \mathbf{Q}) \cdot \mathbf{u}_{\mathbf{Q}}|^2. \quad (534)$$

The amplitude is linear in the reconstruction field, while the isolated satellite intensity is quadratic. A periodically modulated scattering weight,

$$f(\mathbf{r}) = f_0 [1 + \eta(\mathbf{r})], \quad (535)$$

with

$$\eta(\mathbf{r}) = \sum_{\mathbf{Q}} \eta_{\mathbf{Q}} e^{i\mathbf{Q} \cdot \mathbf{r}}, \quad (536)$$

produces the same satellite positions, with

$$\frac{I_{\mathbf{G}+\mathbf{Q}}}{I_{\mathbf{G}}} = |\eta_{\mathbf{Q}}|^2 + O(|\eta|^3) \quad (537)$$

after the common form factor is divided out. A direct bridge between diffraction and band reconstruction is available when the scattering contrast and the interlayer hopping respond to the same weak registry field $\eta(\mathbf{r})$. We assume

$$f(\mathbf{r}) = f_0 + g_f \eta(\mathbf{r}) + O(\eta^2), \quad (538)$$

and

$$T(\mathbf{r}) = T_0 + g_T \eta(\mathbf{r}) + O(\eta^2). \quad (539)$$

For every nonzero moiré harmonic,

$$T_{\mathbf{Q}} = g_T \eta_{\mathbf{Q}} + O(\eta^2), \quad (540)$$

whereas

$$\frac{I_{\mathbf{G}+\mathbf{Q}}}{I_{\mathbf{G}}} = \left| \frac{g_f}{f_0} \right|^2 |\eta_{\mathbf{Q}}|^2 + O(|\eta|^3). \quad (541)$$

Therefore

$$|T_{\mathbf{Q}}|^2 = \left| \frac{g_T f_0}{g_f} \right|^2 \frac{I_{\mathbf{G}+\mathbf{Q}}}{I_{\mathbf{G}}} + O(|\eta|^3). \quad (542)$$

For a scalar translationally covariant interlayer field, Parseval orthogonality gives

$$\frac{1}{\Sigma} \text{Tr} [T(\mathbf{k}) T^\dagger(\mathbf{k})] = \sum_{\mathbf{Q}} |T_{\mathbf{Q}}|^2. \quad (543)$$

Combining this identity with Eq. (465) yields

$$\Delta\mu_2 = \sum_{\mathbf{Q}} |T_{\mathbf{Q}}|^2. \quad (544)$$

Under the common-registry response assumption,

$$\Delta\mu_2 = \left| \frac{g_T f_0}{g_f} \right|^2 \sum_{\mathbf{Q}} \frac{I_{\mathbf{G}+\mathbf{Q}}}{I_{\mathbf{G}}} + O(|\eta|^3). \quad (545)$$

Thus satellite intensities can calibrate the total interlayer hybridization power entering the exact second spectral moment, provided the two probe channels are governed by the same registry field. This relation is model specific through g_T/g_f , but the Parseval and spectral-moment steps are exact. Each microscopic square reciprocal lattice has C_4 symmetry. Because centred rotations commute,

$$R_{\pi/2} R_\theta = R_\theta R_{\pi/2}. \quad (546)$$

Therefore

$$S(R_{\pi/2} \mathbf{q}) = S(\mathbf{q}) \quad (547)$$

for identical isotropic layer weights. At $\hat{\theta} = \frac{\pi}{4}$, a rotation by $\pi/4$ exchanges the two reciprocal lattices:

$$R_{\pi/4} (\Lambda_1^* \cup \Lambda_2^*) = \Lambda_1^* \cup \Lambda_2^*. \quad (548)$$

Hence the equal-weight diffraction pattern has

$$S(R_{\pi/4} \mathbf{q}) = S(\mathbf{q}). \quad (549)$$

The angle $\pi/4$ is not an exact square coincidence rotation because

$$\tan\left(\frac{\pi}{8}\right) = \sqrt{2} - 1 \notin \mathbb{Q}. \quad (550)$$

The eightfold diffraction pattern is therefore quasicrystalline rather than periodically crystallographic. For a finite observation window of linear size L , each reciprocal delta peak is broadened by

$$\delta q_{\text{win}} = O(L^{-1}). \quad (551)$$

The split peaks are resolvable only when

$$2G \left| \sin \left(\frac{\theta}{2} \right) \right| \gtrsim \delta q_{\text{win}}. \quad (552)$$

The minimum resolvable twist is

$$\theta_{\text{res}} = 2 \arcsin \left(\frac{\delta q_{\text{win}}}{2G} \right). \quad (553)$$

At small twist,

$$\theta_{\text{res}} \simeq \frac{\delta q_{\text{win}}}{G}. \quad (554)$$

For the first square reciprocal shell,

$$\theta_{\text{res}} \simeq \frac{a}{2\pi} \delta q_{\text{win}} = O \left(\frac{a}{L} \right). \quad (555)$$

A measured splitting uncertainty $\sigma_{\Delta q}$ propagates to

$$\sigma_{\theta} \simeq \frac{\sigma_{\Delta q}}{G \cos(\theta/2)}. \quad (556)$$

Peak position, integrated intensity, and linewidth therefore measure three distinct quantities:

$$\begin{aligned} \text{peak position} &\longrightarrow \text{modulation wavevector,} \\ \text{integrated intensity} &\longrightarrow \text{reconstruction amplitude,} \\ \text{linewidth} &\longrightarrow \text{coherence length.} \end{aligned} \quad (557)$$

F. Exact no-go theorem for the minimal first-star model

The weak-coupling expansion of a square-lattice band edge suggests that interlayer coupling can reduce the inverse effective mass. We now test whether the minimal equal-amplitude first-star Hamiltonian actually possesses a finite quadratic-flattening root. Measure momentum in units of G_M , energy in units of $E_M = AG_M^2$, and define

$$\mathbf{p} = \frac{\mathbf{k}}{G_M}, \quad \alpha = \frac{w_1}{AG_M^2}. \quad (558)$$

In the ordered basis

$$(|c\rangle, |x+\rangle, |x-\rangle, |y+\rangle, |y-\rangle), \quad (559)$$

define

$$d_0 = p_x^2 + p_y^2, \quad d_{x\pm} = p_x^2 + p_y^2 + 1 \pm 2p_x, \quad (560)$$

and

$$d_{y\pm} = p_x^2 + p_y^2 + 1 \pm 2p_y. \quad (561)$$

The exact five-state Hamiltonian is

$$H_{\square}(\mathbf{p}, \alpha) = \begin{pmatrix} d_0 & \alpha & \alpha & \alpha & \alpha \\ \alpha & d_{x+} & 0 & 0 & 0 \\ \alpha & 0 & d_{x-} & 0 & 0 \\ \alpha & 0 & 0 & d_{y+} & 0 \\ \alpha & 0 & 0 & 0 & d_{y-} \end{pmatrix}. \quad (562)$$

No expansion in α has been made. At $\mathbf{p} = \mathbf{0}$, define the symmetric satellite state

$$|s\rangle = \frac{1}{2} (|x+\rangle + |x-\rangle + |y+\rangle + |y-\rangle). \quad (563)$$

The three-dimensional orthogonal complement in the satellite subspace is dark and has eigenvalue one. On $\text{span}\{|c\rangle, |s\rangle\}$, the Hamiltonian reduces to

$$H_{cs} = \begin{pmatrix} 0 & 2\alpha \\ 2\alpha & 1 \end{pmatrix}. \quad (564)$$

Define

$$r(\alpha) = \sqrt{1 + 16\alpha^2}. \quad (565)$$

The complete point spectrum is

$$\lambda_- = \frac{1-r}{2}, \quad 1, 1, 1, \quad \lambda_+ = \frac{1+r}{2}. \quad (566)$$

The target branch is λ_- , which continuously approaches the central state at $\alpha = 0$. A normalized target eigenvector is

$$|\psi_-\rangle = \sqrt{\frac{r+1}{2r}} |c\rangle - \sqrt{\frac{r-1}{2r}} |s\rangle. \quad (567)$$

The distance to a dark level is

$$1 - \lambda_- = \frac{1+r}{2}. \quad (568)$$

The distance to λ_+ is r , and

$$r \geq \frac{1+r}{2} \quad (r \geq 1). \quad (569)$$

Thus the exact target isolation gap at $\mathbf{p} = \mathbf{0}$ is

$$\delta_0(\alpha) = \frac{1+r}{2} \geq 1. \quad (570)$$

To determine the dispersion exactly in α , write

$$E_-(\mathbf{p}, \alpha) = |\mathbf{p}|^2 + z(\mathbf{p}, \alpha). \quad (571)$$

Eliminating the four satellite amplitudes gives the exact Schur equation

$$\begin{aligned} 0 &= F(z, \mathbf{p}; \alpha) \\ &:= z - 2\alpha^2(z-1) \left[\frac{1}{(z-1)^2 - 4p_x^2} + \frac{1}{(z-1)^2 - 4p_y^2} \right]. \end{aligned} \quad (572)$$

At $\mathbf{p} = \mathbf{0}$,

$$F(z, \mathbf{0}; \alpha) = z - \frac{4\alpha^2}{z-1}. \quad (573)$$

The lower root is $z = \lambda_-$. Moreover,

$$\partial_z F(\lambda_-, \mathbf{0}; \alpha) = 1 + \frac{4\alpha^2}{(\lambda_- - 1)^2} > 0. \quad (574)$$

The implicit-function theorem therefore guarantees an analytic target branch near $\mathbf{p} = 0$. Expanding in momentum but not in α ,

$$F(z, \mathbf{p}; \alpha) = z - \frac{4\alpha^2}{z-1} - \frac{8\alpha^2}{(z-1)^3} |\mathbf{p}|^2 + O(|\mathbf{p}|^4). \quad (575)$$

Set

$$z = \lambda_- + \beta |\mathbf{p}|^2 + O(|\mathbf{p}|^4). \quad (576)$$

Using

$$4\alpha^2 = \lambda_- (\lambda_- - 1), \quad (577)$$

we obtain

$$\beta = -\frac{2(r-1)}{r(r+1)}. \quad (578)$$

Since $E_- = |\mathbf{p}|^2 + z$,

$$E_-(\mathbf{p}, \alpha) = \lambda_-(\alpha) + c_{\square}(\alpha) |\mathbf{p}|^2 + O(|\mathbf{p}|^4), \quad (579)$$

where

$$c_{\square}(\alpha) = 1 + \beta = \frac{r^2 - r + 2}{r(r+1)}. \quad (580)$$

We now minimize this coefficient exactly. Treating $r \geq 1$ as the independent variable,

$$c_{\square}(r) = \frac{r^2 - r + 2}{r^2 + r}. \quad (581)$$

Its derivative is

$$\frac{dc_{\square}}{dr} = \frac{2(r^2 - 2r - 1)}{(r^2 + r)^2}. \quad (582)$$

The only stationary point for $r \geq 1$ is

$$r_{\min} = 1 + \sqrt{2}. \quad (583)$$

The derivative is negative for $1 < r < 1 + \sqrt{2}$ and positive for $r > 1 + \sqrt{2}$, so this point is the global minimum. Substitution gives

$$\min_{\alpha \geq 0} c_{\square}(\alpha) = 4\sqrt{2} - 5 > 0. \quad (584)$$

The minimizing coupling satisfies

$$\alpha_{\min}^2 = \frac{r_{\min}^2 - 1}{16} = \frac{1 + \sqrt{2}}{8}. \quad (585)$$

Consequently,

$$c_{\square}(\alpha) \geq 4\sqrt{2} - 5 \quad \text{for every } \alpha \geq 0. \quad (586)$$

The exact five-state model has no finite quadratic-flattening root. The weak-coupling expansion is

$$c_{\square}(\alpha) = 1 - 8\alpha^2 + 128\alpha^4 - 1920\alpha^6 + O(\alpha^8). \quad (587)$$

Truncation after the quadratic term gives the formal zero

$$\alpha_{\text{pc}} = \frac{1}{\sqrt{8}}. \quad (588)$$

At that same coupling, the exact coefficient is

$$c_{\square} \left(\frac{1}{\sqrt{8}} \right) = 3 - \frac{4\sqrt{3}}{3} = 0.690598923 \dots > 0. \quad (589)$$

The perturbative zero is therefore not quantitatively close to an exact root; it is an extrapolation outside the regime in which the truncated series can determine the sign. We conclude perturbative inverse-mass suppression $\not\Rightarrow$ physical magic angle. A richer model must modify the exact momentum-dependent self-energy through additional tunnelling stars, satellite-satellite coupling, orbital matrix structure, sublattice structure, or anisotropic channels strong enough to produce a genuine sign change while the target remains spectrally isolated.

G. Euclidean physical magic benchmark

Let $C = \{a_1, \dots, a_{r_C}\}$ be a group of r_C bands tracked continuously over the entire mBZ and over the parameter range of interest. The corresponding spectral projector is

$$P_C(\mathbf{k}) = \sum_{a \in C} |u_{a\mathbf{k}}\rangle \langle u_{a\mathbf{k}}|. \quad (590)$$

At nearby parameter values λ and $\lambda + \delta\lambda$, the cluster is matched by minimizing the projector distance

$$D_P = \|P_C(\lambda + \delta\lambda) - P_C(\lambda)\|. \quad (591)$$

This prevents artificial branch exchange at avoided crossings and is invariant under unitary rotations inside a degenerate multiplet. Define

$$E_C^- = \inf_{\mathbf{k} \in \text{mBZ}, a \in C} E_a(\mathbf{k}), \quad (592)$$

and

$$E_C^+ = \sup_{\mathbf{k} \in \text{mBZ}, a \in C} E_a(\mathbf{k}). \quad (593)$$

The global Euclidean bandwidth is

$$W_{E,C} = E_C^+ - E_C^-. \quad (594)$$

It is evaluated over the complete mBZ, not only on a selected symmetry path. Let $E_{\text{below}}(\mathbf{k})$ be the highest band not belonging to C below the target cluster, and let $E_{\text{above}}(\mathbf{k})$ be the lowest band above it. Define

$$\Delta_{E,C}^- = \inf_{\mathbf{k}} \left[\min_{a \in C} E_a(\mathbf{k}) - E_{\text{below}}(\mathbf{k}) \right], \quad (595)$$

and

$$\Delta_{E,C}^+ = \inf_{\mathbf{k}} \left[E_{\text{above}}(\mathbf{k}) - \max_{a \in C} E_a(\mathbf{k}) \right]. \quad (596)$$

The signed global isolation gap is

$$\Delta_{E,C} = \min \left\{ \Delta_{E,C}^-, \Delta_{E,C}^+ \right\}. \quad (597)$$

The target cluster is globally isolated only if $\Delta_{E,C} > 0$. A narrow band touching another band is not accepted as an isolated magic-band state. For each target band,

$$\mathbf{v}_a(\mathbf{k}) = \frac{1}{\hbar} \nabla_{\mathbf{k}} E_a(\mathbf{k}). \quad (598)$$

The maximum physical velocity is

$$v_{\text{max},E,C} = \sup_{\mathbf{k} \in \text{mBZ}, a \in C} |\mathbf{v}_a(\mathbf{k})|. \quad (599)$$

The diameter of the square mBZ is

$$D_k = \sqrt{2} G_M. \quad (600)$$

For a differentiable connected band,

$$\begin{aligned} |E_a(\mathbf{k}_2) - E_a(\mathbf{k}_1)| &\leq \int_{\gamma} |\nabla_{\mathbf{k}} E_a| d\ell \\ &\leq \hbar v_{\text{max},E,a} D_k. \end{aligned} \quad (601)$$

Hence

$$W_{E,a} \leq \hbar D_k v_{\text{max},E,a}. \quad (602)$$

Small maximum velocity is therefore a sufficient global bandwidth control, whereas stationarity at one point is not. At a nondegenerate stationary point $\mathbf{k}_*$,

$$\nabla_{\mathbf{k}} E_a(\mathbf{k}_*) = 0, \quad (603)$$

define

$$\left[m_{E,a}^{*-1} \right]_{\mu\nu} = \frac{1}{\hbar^2} \frac{\partial^2 E_a}{\partial k_\mu \partial k_\nu} \bigg|_{\mathbf{k}=\mathbf{k}_*}. \quad (604)$$

For a C_4 -symmetric nondegenerate band at Γ ,

$$R_{\pi/2}^\top (\nabla_{\mathbf{k}}^2 E) R_{\pi/2} = \nabla_{\mathbf{k}}^2 E. \quad (605)$$

A real symmetric 2×2 matrix satisfying this relation is proportional to the identity:

$$\nabla_{\mathbf{k}}^2 E(\Gamma) = \beta I_2. \quad (606)$$

Thus

$$\frac{\partial^2 E}{\partial k_x^2} = \frac{\partial^2 E}{\partial k_y^2} = \beta, \quad \frac{\partial^2 E}{\partial k_x \partial k_y} = 0. \quad (607)$$

The local mass-cancellation angle is defined, when it exists, by

$$\theta_{\text{mass}} : \quad \beta(\theta_{\text{mass}}, \omega) = 0. \quad (608)$$

This condition removes the quadratic term near one high-symmetry point. It does not imply $W_{E,C} = 0$. For a global second-order diagnostic, define

$$\Lambda_{\text{max},E,C} = \sup_{\mathbf{k},a,j} |\lambda_j [\nabla_{\mathbf{k}}^2 E_a(\mathbf{k})]|. \quad (609)$$

The associated quadratic energy scale is

$$E_{\text{disp},E,C}^{(2)} = \frac{1}{2} D_k^2 \Lambda_{\text{max},E,C}. \quad (610)$$

The normalized target-cluster DOS is

$$\rho_{E,C}(E) = \frac{1}{r_C A_{\text{mBZ}}} \sum_{a \in C} \int_{\text{mBZ}} d^2 k \delta[E - E_a(\mathbf{k})]. \quad (611)$$

It satisfies

$$\int_{-\infty}^{\infty} \rho_{E,C}(E) dE = 1. \quad (612)$$

For finite numerical data, use the Lorentzian kernel

$$K_\eta(E) = \frac{1}{\pi} \frac{\eta}{E^2 + \eta^2}. \quad (613)$$

The broadened DOS is

$$\rho_{E,C,\eta}(E) = \frac{1}{r_C A_{\text{mBZ}}} \sum_{a \in C} \int_{\text{mBZ}} d^2 k K_\eta[E - E_a(\mathbf{k})]. \quad (614)$$

Its maximum is

$$\rho_{E,C,\eta}^{\text{max}} = \max_E \rho_{E,C,\eta}(E). \quad (615)$$

For a finite-width resolved band, the broadening hierarchy is $\epsilon_E \ll \eta \ll W_{E,C}$, where ϵ_E is the numerical eigenvalue uncertainty. Define the hard isolation condition

$$G_\Delta = \begin{cases} 1, & \Delta_{E,C} > 0, \\ 0, & \Delta_{E,C} \leq 0. \end{cases} \quad (616)$$

The energetic-flatness residual is

$$r_W^{(E)} = \frac{W_{E,C}}{\Delta_{E,C}}. \quad (617)$$

The first-order dispersion residual is

$$r_v^{(E)} = \frac{\hbar D_k v_{\text{max},E,C}}{\Delta_{E,C}}. \quad (618)$$

The strong second-order residual is

$$r_m^{(E)} = \frac{D_k^2 \Lambda_{\text{max},E,C}}{2 \Delta_{E,C}}. \quad (619)$$

Using the moiré kinetic scale E_M , define the DOS residual

$$r_\rho^{(E)} = \frac{1}{E_M \rho_{E,C,\eta}^{\text{max}}}. \quad (620)$$

The primary physical certificate is the vector

$$\mathcal{C}_{E,C} = (W_{E,C}, \Delta_{E,C}, v_{\max,E,C}, m_{E,C}^*, \rho_{E,C}, N_{\text{part},k}, P_L). \quad (621)$$

No scalar score replaces this vector. For ranking parameter points, define the minimal first-order score

$$\mathcal{M}_{E,C}^{(1)} = G_\Delta \left[\frac{1}{1+r_W^{(E)}} \frac{1}{1+r_v^{(E)}} \frac{1}{1+r_\rho^{(E)}} \right]^{1/3}, \quad (622)$$

and the strong second-order score

$$\mathcal{M}_{E,C}^{(2)} = G_\Delta \left[\frac{1}{1+r_W^{(E)}} \frac{1}{1+r_v^{(E)}} \frac{1}{1+r_m^{(E)}} \frac{1}{1+r_\rho^{(E)}} \right]^{1/4}. \quad (623)$$

Every factor lies between zero and one. The strict flat-band limit requires

$$r_W^{(E)} \rightarrow 0, \quad r_v^{(E)} \rightarrow 0, \quad r_m^{(E)} \rightarrow 0, \quad r_\rho^{(E)} \rightarrow 0, \quad (624)$$

while

$$\liminf \frac{\Delta_{E,C}}{E_{\text{kin}}} > 0. \quad (625)$$

The hybridization-causality conditions are imposed separately:

$$\begin{aligned} W_{E,C}(\omega) &< W_{E,C}(0), \\ v_{\max,E,C}(\omega) &< v_{\max,E,C}(0), \end{aligned} \quad (626)$$

and $\bar{N}_{\text{part},k} > 1$, where

$$\bar{N}_{\text{part},k} = \frac{1}{A_{\text{mBZ}}} \int_{\text{mBZ}} d^2k N_{\text{part},k}(\mathbf{k}). \quad (627)$$

A candidate failing these conditions is not classified as twist-hybridization-induced. For fixed $\omega = \frac{w}{t}$, the Euclidean physical magic-angle set is

$$\Theta_{\text{magic}}^{(E)}(\omega) = \arg \max_{\theta,C} \mathcal{M}_{E,C}^{(2)}(\theta, \omega), \quad (628)$$

subject to positive isolation, continuous projector tracking, and the hybridization-causality conditions above. A magic angle is therefore defined only after the microscopic ratio w/t and the target spectral family have been specified. Near the square-lattice Γ point,

$$\varepsilon_0(\mathbf{k}) + 4t = Ak^2 + O(a^4k^4), \quad A = ta^2. \quad (629)$$

The exact moiré reciprocal scale is

$$G_M = \frac{2\pi}{a\sqrt{\Sigma}}. \quad (630)$$

The characteristic moiré kinetic scale is

$$E_M = AG_M^2 = \frac{4\pi^2 t}{\Sigma}. \quad (631)$$

Using

$$\Sigma = \frac{n_c^2}{d} \csc^2 \left(\frac{\theta}{2} \right), \quad (632)$$

we obtain

$$E_M = \frac{4\pi^2 dt}{n_c^2} \sin^2 \left(\frac{\theta}{2} \right). \quad (633)$$

For small twist,

$$E_M = \frac{\pi^2 dt}{n_c^2} \theta^2 + O(t\theta^4). \quad (634)$$

Let w_1 be the first-star interlayer Fourier amplitude and write

$$w_1(\theta) = \gamma_1(\theta)w, \quad (635)$$

with

$$\gamma_1(\theta) = \gamma_0 + O(\theta^2). \quad (636)$$

The dimensionless Euclidean coupling is

$$\alpha_\square = \frac{w_1}{E_M} = \frac{\gamma_1(\theta)\omega\Sigma}{4\pi^2}. \quad (637)$$

Equivalently,

$$\alpha_\square = \frac{\gamma_1(\theta)n_c^2}{4\pi^2 d} \omega \csc^2 \left(\frac{\theta}{2} \right). \quad (638)$$

At small twist,

$$\alpha_\square = \frac{\gamma_0 n_c^2 \omega}{\pi^2 d \theta^2} [1 + O(\theta^2)]. \quad (639)$$

Introduce the rescaled momentum

$$\mathbf{p} = \frac{\mathbf{k}}{G_M}. \quad (640)$$

The low-energy Hamiltonian has the scaling form

$$\frac{H_{\text{low}} + 4t}{E_M} = \mathcal{H}_0(\mathbf{p}) + \alpha_\square \mathcal{V}_1 + O(\Sigma^{-1}), \quad (641)$$

provided the normalized tunnelling profile has a regular small-angle limit. Consequently, every dimensionless low-energy observable obeys

$$\mathcal{O}(\theta, \omega) = \mathcal{O}_0(\alpha_\square) + O(\Sigma^{-1}). \quad (642)$$

In particular,

$$\begin{aligned} \frac{W_{E,C}}{E_M} &= \mathcal{W}_C(\alpha_\square) + O(\Sigma^{-1}), \\ \frac{\Delta_{E,C}}{E_M} &= \mathcal{D}_C(\alpha_\square) + O(\Sigma^{-1}), \end{aligned} \quad (643)$$

and

$$E_M \rho_{E,C}(E) = \mathcal{R}_C \left(\frac{E - E_{C,0}}{E_M}, \alpha_\square \right) + O(\Sigma^{-1}). \quad (644)$$

Suppose a richer microscopic continuum Hamiltonian possesses an isolated physical optimum at $\alpha_\square = \alpha_\nu^*$. The physical angle then satisfies

$$\sin^2 \left(\frac{\theta_\nu^*}{2} \right) = \frac{\gamma_1(\theta_\nu^*) n_c^2}{4\pi^2 d \alpha_\nu^*} \frac{w}{t}. \quad (645)$$

Using $\gamma_1 = \gamma_0 + O(\theta^2)$,

$$\theta_\nu^* = 2 \arcsin \left[\frac{n_c}{2\pi \sqrt{d \alpha_\nu^*}} \sqrt{\gamma_0 \frac{w}{t}} \right] + O \left[\left(\frac{w}{t} \right)^{3/2} \right]. \quad (646)$$

Therefore

$$\theta_\nu^* = \frac{n_c}{\pi \sqrt{d \alpha_\nu^*}} \sqrt{\gamma_0 \frac{w}{t}} + O \left[\left(\frac{w}{t} \right)^{3/2} \right]. \quad (647)$$

The corresponding supercell index is

$$\Sigma_\nu^* = \frac{4\pi^2 \alpha_\nu^*}{\gamma_1(\theta_\nu^*)} \frac{t}{w}, \quad (648)$$

and

$$N_{sc,\nu}^* = \frac{8\pi^2 \alpha_\nu^*}{\gamma_1(\theta_\nu^*)} \frac{t}{w}. \quad (649)$$

More generally, if the relevant monolayer dispersion begins at order $p > 0$,

$$\varepsilon(\mathbf{k}_0 + \mathbf{q}) - \varepsilon(\mathbf{k}_0) = c_p |\mathbf{q}|^p + o(|\mathbf{q}|^p), \quad (650)$$

then

$$E_M^{(p)} = \Theta(t\theta^p), \quad \alpha_p = \Theta \left(\frac{w/t}{\theta^p} \right). \quad (651)$$

A finite dimensionless optimum implies

$$\theta_\nu^* = \Theta \left[\left(\frac{w}{t} \right)^{1/p} \right]. \quad (652)$$

The square-lattice band edge has $p = 2$, giving the square-root law. The equal-amplitude five-state model of Sec. III F does not possess the required finite quadratic root. Equations (647)–(652) are therefore conditional scaling relations for a richer microscopic Hamiltonian whose physical optimum has first been established from the complete spectral problem. At weak coupling, a general C_4 -symmetric tunnelling spectrum gives

$$\frac{A_{\text{eff}}}{A} = 1 - C_2 \alpha_\square^2 + O(\alpha_\square^4), \quad (653)$$

where

$$C_2 = 2 \sum_{\mathbf{g}} \frac{|\tau_{\mathbf{g}}|^2}{|\mathbf{g}|^4}. \quad (654)$$

For four equal first-star vectors, $C_2 = 8$. Within the quadratic regime,

$$\begin{aligned} \frac{m_0^*}{m_{\text{eff}}^*} &= 1 - C_2 \alpha_\square^2, & \frac{v}{v_0} &= |1 - C_2 \alpha_\square^2|, \\ \frac{W^{(2)}}{W_0^{(2)}} &= |1 - C_2 \alpha_\square^2|, & \frac{D(E)}{D_0(E)} &= \frac{1}{|1 - C_2 \alpha_\square^2|}. \end{aligned} \quad (655)$$

The initial changes of effective mass, velocity, quadratic bandwidth, and DOS are therefore four manifestations of the same hybridization-induced curvature renormalization. If a richer model genuinely reaches $A_{\text{eff}} = 0$ while its quartic coefficients remain finite, the leading C_4 dispersion becomes

$$E(\mathbf{k}) = E_* + B_1 (k_x^4 + k_y^4) + B_2 k_x^2 k_y^2 + O(k^6). \quad (656)$$

Since

$$|k_x|, |k_y| \leq \frac{G_M}{2}, \quad (657)$$

the quartic bandwidth obeys

$$W_{\text{quartic}} \leq G_M^4 \left(\frac{|B_1|}{8} + \frac{|B_2|}{16} \right) + O(G_M^6). \quad (658)$$

Because $G_M = \Theta(\theta)$, the bandwidth crosses from $W_{\text{quadratic}} = O(t\theta^2)$ to $W_{\text{quartic}} = O(t\theta^4)$. Along a branch with $\theta^* = O[(w/t)^{1/2}]$,

$$W_{\text{quartic}}^* = O \left[t \left(\frac{w}{t} \right)^2 \right]. \quad (659)$$

The local mass angle and the physical magic angle are therefore distinct:

$$\theta_{\text{mass}} : \beta(\theta_{\text{mass}}, \omega) = 0, \quad \theta_{\text{magic}} : \mathcal{M}_{E,C}^{(2)} \text{ is maximal.} \quad (660)$$

Agreement, $\theta_{\text{mass}} \simeq \theta_{\text{magic}}$, would show that the global flat-band optimum is controlled by a zone-centre effective-mass cancellation. A systematic difference, $\theta_{\text{mass}} \neq \theta_{\text{magic}}$, would identify a full-zone, higher-order, or multiresonant mechanism. The ten-site and thirty-four-site systems provide exact implementation checks, while the complete sub-hundred-site family tests crystallographic counting, reciprocal folding, symmetry, avoided crossings, DOS normalization, and the initial effective-mass trend. Small-angle physical magic scaling, when present in a richer microscopic model, must be established from a sequence of larger exact cells rather than inferred from one visually narrow finite-supercell band plot. For a symmetric energy window

$$\mathcal{I} = [E_0 - \delta E, E_0 + \delta E], \quad (661)$$

and a single-layer density of states that is four times continuously differentiable near E_0 , Taylor expansion gives

$$\int_{\mathcal{I}} \nu_0(E) dE = \int_{-\delta E}^{+\delta E} \left[\nu_0(E_0) + x \nu_0'(E_0) + \frac{x^2}{2} \nu_0''(E_0) \right]$$

$$\begin{aligned}
& + \frac{x^3}{6} \nu_0^{(3)}(E_0) + \frac{x^4}{24} \nu_0^{(4)}(E_0) + O(x^5) \Big] dx \\
& = 2\delta E \nu_0(E_0) + \frac{\delta E^3}{3} \nu_0''(E_0) \\
& + \frac{\delta E^5}{60} \nu_0^{(4)}(E_0) + O(\delta E^7). \quad (662)
\end{aligned}$$

Consequently,

$$\begin{aligned}
N_{\mathcal{I},\tau}^{\text{cand}} & = 2\delta E \nu_0(E_0) N_{\text{pair},\tau}^{\text{eff}} \\
& + \frac{\delta E^3}{3} \nu_0''(E_0) N_{\text{pair},\tau}^{\text{eff}} + O(\delta E^5 N_{\text{pair},\tau}^{\text{eff}}). \quad (663)
\end{aligned}$$

The leading candidate count is therefore

$$N_{\mathcal{I},\tau}^{\text{cand}} = 2\delta E \nu_0(E_0) N_{\text{pair},\tau}^{\text{eff}} + O(\delta E^3 N_{\text{pair},\tau}^{\text{eff}}). \quad (664)$$

At fixed nonzero negative curvature and fixed energy window, The hyperbolic active-pair scaling $N_{\text{pair},\tau}^{\text{eff}} = \Theta(|\theta|^{-1})$ therefore gives

$$N_{\mathcal{I},\tau}^{\text{cand,H}} = \Theta(|\theta|^{-1}), \quad (665)$$

whereas the Euclidean reference satisfies

$$N_{\mathcal{I},\tau}^{\text{cand,E}} = \Theta(|\theta|^{-2}). \quad (666)$$

This count measures the number of geometrically available monolayer modes inside the selected energy window. It does not assert that every candidate forms a resolved bilayer doublet. Spectral resonance must be tested independently. Let the normalized densities of states of the two uncoupled layers be $\nu_1(E)$, $\nu_2(E)$, with

$$\int_{-\infty}^{+\infty} \nu_\ell(E) dE = 1. \quad (667)$$

For an allowed energy mismatch $\Gamma_{\text{res}} > 0$, define the spectral-overlap fraction

$$\mathcal{O}_{\Gamma_{\text{res}}}(\mathcal{I}) = \int_{\mathcal{I}} dE \nu_1(E) \int_{\mathcal{I} \cap [E - \Gamma_{\text{res}}, E + \Gamma_{\text{res}}]} \nu_2(E') dE'. \quad (668)$$

It satisfies $0 \leq \mathcal{O}_{\Gamma_{\text{res}}}(\mathcal{I}) \leq 1$. A pre-hybridization estimate of the number of nearly resonant layer pairs is

$$N_{\text{near-res},\tau}^{\text{cand}} = N_{\text{pair},\tau}^{\text{eff}} \mathcal{O}_{\Gamma_{\text{res}}}(\mathcal{I}). \quad (669)$$

For identical layers,

$$\nu_1(E) = \nu_2(E) = \nu_0(E), \quad (670)$$

and for an energy window whose boundaries remain farther than Γ_{res} from the region of interest,

$$\begin{aligned}
\mathcal{O}_{\Gamma_{\text{res}}}(\mathcal{I}) & = 2\Gamma_{\text{res}} \int_{\mathcal{I}} \nu_0^2(E) dE \\
& + \frac{\Gamma_{\text{res}}^3}{3} \int_{\mathcal{I}} \nu_0(E) \nu_0''(E) dE + O(\Gamma_{\text{res}}^5). \quad (671)
\end{aligned}$$

Thus

$$\begin{aligned}
N_{\text{near-res},\tau}^{\text{cand}} & = 2\Gamma_{\text{res}} N_{\text{pair},\tau}^{\text{eff}} \int_{\mathcal{I}} \nu_0^2(E) dE \\
& + O(\Gamma_{\text{res}}^3 N_{\text{pair},\tau}^{\text{eff}}). \quad (672)
\end{aligned}$$

The square of the density of states enters because resonance requires simultaneous spectral availability in both layers. The exact resolved-channel number remains the singular-value count defined in Eq. (1235). Let $r_{\mathcal{I}} = \text{rank } P_{\mathcal{I}}$, and set $a_{\text{res}} = \frac{\kappa_{\text{res}}}{2}$. If $a_{\text{res}} < \|V_{\mathcal{I}}\|$, the number of singular values exceeding the threshold also obeys the lower estimate

$$N_{\text{res}} \geq \max \left\{ 0, \left\lceil \frac{\|V_{\mathcal{I}}\|_{\text{F}}^2 - r_{\mathcal{I}} a_{\text{res}}^2}{\|V_{\mathcal{I}}\|^2 - a_{\text{res}}^2} \right\rceil \right\}. \quad (673)$$

To prove this bound, let $n = N_{\text{res}}$ and denote the singular values by $\sigma_1 \geq \dots \geq \sigma_{r_{\mathcal{I}}}$. The n singular values above threshold satisfy

$$\sigma_j^2 \leq \|V_{\mathcal{I}}\|^2, \quad j \leq n, \quad (674)$$

while the remaining singular values obey

$$\sigma_j^2 < a_{\text{res}}^2, \quad j > n. \quad (675)$$

Hence

$$\|V_{\mathcal{I}}\|_{\text{F}}^2 = \sum_{j=1}^{r_{\mathcal{I}}} \sigma_j^2 \leq n \|V_{\mathcal{I}}\|^2 + (r_{\mathcal{I}} - n) a_{\text{res}}^2.$$

Solving for n gives Eq. (673). Combining the Frobenius-norm upper bound with Eq. (673) gives the established interval

$$\begin{aligned}
& \max \left\{ 0, \left\lceil \frac{\|V_{\mathcal{I}}\|_{\text{F}}^2 - r_{\mathcal{I}} a_{\text{res}}^2}{\|V_{\mathcal{I}}\|^2 - a_{\text{res}}^2} \right\rceil \right\} \\
& \leq N_{\text{res}} \leq \min \left\{ r_{\mathcal{I}}, \left\lfloor \frac{\|V_{\mathcal{I}}\|_{\text{F}}^2}{a_{\text{res}}^2} \right\rfloor \right\}. \quad (676)
\end{aligned}$$

This interval separates a large Frobenius norm carried by many active channels from the same norm concentrated in one dominant singular mode. The boundary-layer variables yield more than an asymptotic profile. They determine the complete probability distribution of the microscopic registry mismatch sampled by the hyperbolic area measure. Recall

$$y = L(1 - u) = \frac{\xi_{\text{M}} - r}{R}. \quad (677)$$

In the curvature-dominated limit,

$$d\mu_{\infty}(y) = e^{-y} dy, \quad 0 \leq y < \infty. \quad (678)$$

Therefore the fraction of the effective moiré area lying within a physical radial distance RY of the domain boundary is

$$\mathbb{P}(0 \leq \xi_{\text{M}} - r \leq RY) = \int_0^Y e^{-y} dy = 1 - e^{-Y}.$$

Consequently, 50% of the effective area lies within $R \ln 2$ of the outer registry boundary. Likewise, 90% of the effective area lies within $R \ln 10$ of the outer registry boundary. These physical shell thicknesses remain finite as $\xi_M \rightarrow \infty$; only their normalized thickness relative to ξ_M vanishes. Introduce $z = e^{-y}$. Since

$$dy = -\frac{dz}{z}, \quad (679)$$

we have $e^{-y} dy = -dz$. After reversing the integration limits, z is uniformly distributed on $0 \leq z \leq 1$. The limiting mismatch is

$$d = 2R \operatorname{arsinh}(bz), \quad b = \sinh\left(\frac{a}{2R}\right). \quad (680)$$

Inversion gives

$$z = \frac{\sinh(d/2R)}{\sinh(a/2R)}. \quad (681)$$

Therefore the exact cumulative mismatch distribution is

$$F_H(d) = \mathbb{P}(d_M \leq d) = \frac{\sinh(d/2R)}{\sinh(a/2R)}, \quad 0 \leq d \leq a. \quad (682)$$

Differentiation gives the probability density

$$p_H(d) = \frac{\cosh(d/2R)}{2R \sinh(a/2R)}, \quad 0 \leq d \leq a. \quad (683)$$

Its normalization follows directly:

$$\begin{aligned} \int_0^a p_H(d) dd &= \frac{1}{2R \sinh(a/2R)} \int_0^a \cosh\left(\frac{d}{2R}\right) dd \\ &= 1. \end{aligned} \quad (684)$$

For comparison, the Euclidean rescaled disk has

$$d_E = au, \quad p_E(u) = 2u. \quad (685)$$

Hence

$$p_E(d) = \frac{2d}{a^2}, \quad 0 \leq d \leq a. \quad (686)$$

The Euclidean domain weights large mismatch quadratically through the area element. The curvature-dominated hyperbolic domain instead approaches a nearly uniform mismatch distribution when $a/R \ll 1$. Let

$$\zeta = \frac{a}{2R}. \quad (687)$$

The exact mean hyperbolic mismatch is

$$\begin{aligned} \langle d \rangle_H &= \int_0^a d p_H(d) dd = \frac{2R}{\sinh \zeta} \int_0^\zeta x \cosh x dx \\ &= \frac{2R}{\sinh \zeta} [\zeta \sinh \zeta - \cosh \zeta + 1] \end{aligned}$$

$$= a - 2R \tanh\left(\frac{\zeta}{2}\right). \quad (688)$$

Thus

$$\langle d \rangle_H = a - 2R \tanh\left(\frac{a}{4R}\right). \quad (689)$$

The exact second moment is

$$\begin{aligned} \langle d^2 \rangle_H &= \frac{4R^2}{\sinh \zeta} \int_0^\zeta x^2 \cosh x dx \\ &= 4R^2 [\zeta^2 - 2\zeta \coth \zeta + 2]. \end{aligned} \quad (690)$$

Consequently,

$$\operatorname{Var}_H(d) = 4R^2 [\zeta^2 - 2\zeta \coth \zeta + 2] - \left[a - 2R \tanh\left(\frac{\zeta}{2}\right) \right]^2. \quad (691)$$

For $a/R \ll 1$,

$$\langle d \rangle_H = \frac{a}{2} + \frac{a^3}{96R^2} + O\left(\frac{a^5}{R^4}\right), \quad (692)$$

and

$$\langle d^2 \rangle_H = \frac{a^2}{3} + \frac{a^4}{90R^2} + O\left(\frac{a^6}{R^4}\right). \quad (693)$$

By contrast,

$$\langle d \rangle_E = \frac{2a}{3}, \quad \langle d^2 \rangle_E = \frac{a^2}{2}. \quad (694)$$

Thus negative curvature does not merely change the radius of the registry domain. It changes the complete statistical distribution of coupling distances sampled by that domain. For any radial interlayer kernel $T(D)$, the n -th coupling moment in the curvature-dominated limit is therefore

$$\langle T^n \rangle_H = \frac{1}{2R \sinh(a/2R)} \int_0^a \cosh\left(\frac{d}{2R}\right) \left[T\left(\sqrt{h^2 + d^2}\right) \right]^n dd. \quad (695)$$

The mean and variance follow from

$$\bar{T}_H = \langle T \rangle_H, \quad W_{\text{sc},\infty}^2 = \langle T^2 \rangle_H - \langle T \rangle_H^2. \quad (696)$$

Equation (695) reduces the limiting coupling statistics to a one-dimensional quadrature with no finite-cover or angular fitting parameter. Let $0 < \ell \leq a$ be any mismatch threshold. The fraction of the curvature-dominated effective area satisfying $d_M \leq \ell$ is

$$f_H(\ell) = \frac{\sinh(\ell/2R)}{\sinh(a/2R)}. \quad (697)$$

The corresponding Euclidean fraction is

$$f_E(\ell) = \left(\frac{\ell}{a}\right)^2. \quad (698)$$

For a coupling threshold τ , define the clipped mismatch

$$\ell_\tau^{\text{clip}} = \min\{\ell_\tau, a\}. \quad (699)$$

The fraction of effective layer pairs satisfying $T \geq \tau w$ is

$$f_\tau^{(\text{H})} = \frac{\sinh(\ell_\tau^{\text{clip}}/2R)}{\sinh(a/2R)}. \quad (700)$$

The effective strong-pair count can therefore be written in either of the equivalent forms

$$N_{\text{pair},\tau}^{\text{eff}} = \rho_0 A_\tau, \quad (701)$$

or

$$N_{\text{pair},\tau}^{\text{eff}} = f_\tau^{(\text{H})} N_{\text{pair},a}^{\text{eff}} \quad (702)$$

in the curvature-dominated regime. For two coupling thresholds $0 < \tau_1 < \tau_2 < 1$, the corresponding asymptotic active-area ratio is

$$\lim_{|\theta| \rightarrow 0} \frac{A_{\tau_1}}{A_{\tau_2}} = \frac{\sinh(\ell_{\tau_1}/2R)}{\sinh(\ell_{\tau_2}/2R)}. \quad (703)$$

In Euclidean geometry, the same ratio is

$$\frac{A_{\tau_1}^{(\text{E})}}{A_{\tau_2}^{(\text{E})}} = \left(\frac{\ell_{\tau_1}}{\ell_{\tau_2}}\right)^2. \quad (704)$$

The hierarchy of strong, intermediate, and weak interlayer links is therefore itself curvature dependent. Let T_{triv} denote the component of the interlayer operator that transforms in the trivial translation representation. For a translation-homogeneous finite quotient this is the group average. Define

$$\delta T = T - T_{\text{triv}}. \quad (705)$$

The nontrivial finite-device coupling power is

$$\mathcal{P}_{\text{sc}} = \text{Tr}(\delta T \delta T^\dagger) = \|\delta T\|_{\text{F}}^2. \quad (706)$$

For a finite normal quotient

$$Q_N = \Gamma/\Gamma_N, \quad |Q_N| = N, \quad (707)$$

let

$$\widehat{\delta T}(\rho) = \sum_{q \in Q_N} \delta T(q) \rho(q) \quad (708)$$

for every irreducible unitary representation

$$\rho : Q_N \longrightarrow U(d_\rho). \quad (709)$$

Define the normalized transform

$$\widetilde{\delta T}(\rho) = \frac{\sqrt{d_\rho}}{N} \widehat{\delta T}(\rho). \quad (710)$$

Finite-group Plancherel theory gives

$$\frac{1}{N} \sum_{q \in Q_N} |\delta T(q)|^2 = \sum_{\rho \in \widehat{Q_N}} \left\| \widetilde{\delta T}(\rho) \right\|_{\text{F}}^2. \quad (711)$$

Because the group average has been removed, $\widetilde{\delta T}(\mathbf{1}) = 0$. Hence

$$\frac{1}{N} \sum_{q \in Q_N} |\delta T(q)|^2 = \sum_{\rho \neq \mathbf{1}} \left\| \widetilde{\delta T}(\rho) \right\|_{\text{F}}^2. \quad (712)$$

The real-space variance of the interlayer network is exactly equal to the total power carried by nontrivial representation sectors. The same coupling power appears in the second spectral moment. For

$$H = \begin{pmatrix} h_1 & T \\ T^\dagger & h_2 \end{pmatrix} \quad (713)$$

and

$$H_0 = h_1 \oplus h_2, \quad (714)$$

direct multiplication gives

$$\text{Tr} H^2 - \text{Tr} H_0^2 = 2 \text{Tr} (T T^\dagger). \quad (715)$$

After subtracting the uniform bonding–antibonding contribution,

$$\Delta M_2^{\text{nontriv}} = 2 \text{Tr} (\delta T \delta T^\dagger) = 2 \mathcal{P}_{\text{sc}}. \quad (716)$$

Thus the curvature-controlled coupling variance is directly measurable through a basis-independent spectral moment. For a fixed-range coupling network with bounded local coordination and a converged per-pair variance,

$$\frac{\mathcal{P}_{\text{sc}}}{N_{\text{pair}}^{\text{eff}}} \longrightarrow W_{\text{sc},\infty}^2 > 0. \quad (717)$$

Consequently,

$$\mathcal{P}_{\text{sc}}^{(\text{H})} = \Theta(|\theta|^{-1}), \quad (718)$$

whereas the Euclidean reference gives

$$\mathcal{P}_{\text{sc}}^{(\text{E})} = \Theta(|\theta|^{-2}). \quad (719)$$

These are total-power laws. The normalized local hybridization scale remains finite and must be distinguished from the increasing number of participating channels. Not every nontrivial representation component couples efficiently to a chosen target band. Let P_ν project onto a target spectral manifold of rank

$$r_\nu = \text{rank } P_\nu, \quad (720)$$

and let Q_ν^{act} project onto the symmetry-compatible remote states retained as active hybridization channels. Define

$$V_\nu = P_\nu \delta T Q_\nu^{\text{act}}. \quad (721)$$

Let z_ν be the declared number of equivalent active channels per target state. The active RMS hybridization is

$$w_\nu^{\text{eff}} = \left[\frac{\text{Tr}(V_\nu V_\nu^\dagger)}{r_\nu z_\nu} \right]^{1/2}. \quad (722)$$

Its dimensionless form factor is

$$\gamma_\nu^{\text{eff}}(K, \theta) = \frac{w_\nu^{\text{eff}}(K, \theta)}{w}. \quad (723)$$

Define the active-power fraction

$$\eta_\nu^{\text{act}} = \frac{\text{Tr}(V_\nu V_\nu^\dagger)}{\text{Tr}(\delta T \delta T^\dagger)}. \quad (724)$$

Because orthogonal projection cannot increase the Hilbert–Schmidt norm, $0 \leq \eta_\nu^{\text{act}} \leq 1$. Therefore

$$(w_\nu^{\text{eff}})^2 = \frac{\eta_\nu^{\text{act}}}{r_\nu z_\nu} \mathcal{P}_{\text{sc}}. \quad (725)$$

Equation (725) separates the total geometric scattering power from the fraction that enters the resonant target sector. The local inverse registry length is

$$q_{\text{M}}^{\text{loc}} = \frac{2\pi}{\xi_{\text{M}}} = \frac{2\pi}{RL}, \quad L = \frac{\xi_{\text{M}}}{R}. \quad (726)$$

This quantity is not asserted to be a globally additive hyperbolic reciprocal vector. It is the inverse spatial scale of the registry reconstruction. To connect continuously to the exact square-lattice calibration, let the Euclidean reference branch be labelled by the coprime numerator n_c and parity divisor d . The primitive exact square-moiré reciprocal scale and the local beat scale obey

$$G_{\text{M}}^{(\text{E})} = \frac{\sqrt{d}}{2n_c} q_{\text{M}}^{\text{loc}, \text{E}}. \quad (727)$$

We therefore define the arithmetic-normalized local scale

$$Q_{\text{M}}^{\text{eff}} = \frac{\sqrt{d}}{2n_c} q_{\text{M}}^{\text{loc}} = \frac{\pi\sqrt{d}}{n_c RL}. \quad (728)$$

For a different Euclidean calibration, the fixed prefactor $\sqrt{d}/(2n_c)$ is replaced by the corresponding geometry-to-band matching factor; none of the curvature exponents below changes. Near a quadratic square-lattice band edge,

$$\varepsilon_0(\mathbf{k}) + 4t = ta^2 k^2 + O(ta^4 k^4). \quad (729)$$

The geometry-controlled kinetic scale is therefore

$$E_{\text{M}}^{\text{eff}} = ta^2 (Q_{\text{M}}^{\text{eff}})^2. \quad (730)$$

Substitution gives

$$E_{\text{M}}^{\text{eff}} = \frac{\pi^2 d t a^2}{n_c^2 R^2 L^2}. \quad (731)$$

With $\zeta = \frac{a}{2R}$, this becomes

$$E_{\text{M}}^{\text{eff}} = \frac{4\pi^2 d t}{n_c^2} \frac{\zeta^2}{L^2}. \quad (732)$$

In the Euclidean limit,

$$L = \frac{\zeta}{s} + O(\zeta^3), \quad (733)$$

so

$$E_{\text{M}}^{\text{eff}} \longrightarrow \frac{4\pi^2 d t}{n_c^2} s^2. \quad (734)$$

At small twist,

$$E_{\text{M}}^{(\text{E})} = \frac{\pi^2 d t}{n_c^2} \theta^2 + O(t\theta^4). \quad (735)$$

At fixed negative curvature,

$$L = \ln \left[\frac{4 \sinh \zeta}{|\theta|} \right] + o(1), \quad (736)$$

and therefore

$$E_{\text{M}}^{(\text{H})} \sim \frac{4\pi^2 d t \zeta^2}{n_c^2 \ln^2 [4 \sinh \zeta / |\theta|]}. \quad (737)$$

The Euclidean kinetic scale vanishes algebraically as θ^2 , whereas the hyperbolic scale vanishes only as an inverse logarithm squared. Define the dimensionless interlayer-to-intralayer ratio $\omega = \frac{w}{t}$. The active hybridization-to-kinetic ratio is

$$\alpha_\nu^{\text{eff}}(K, \theta, \omega) = \frac{w_\nu^{\text{eff}}(K, \theta)}{E_{\text{M}}^{\text{eff}}(K, \theta)}. \quad (738)$$

Using

$$w_\nu^{\text{eff}} = \gamma_\nu^{\text{eff}} w, \quad (739)$$

and Eq. (732), we obtain

$$\alpha_\nu^{\text{eff}} = \frac{\gamma_\nu^{\text{eff}}(K, \theta) n_c^2}{4\pi^2 d} \omega \left(\frac{L}{\zeta} \right)^2. \quad (740)$$

Every factor in Eq. (740) is directly calculable from the metric geometry and the projected interlayer matrix. In the Euclidean limit,

$$\frac{L}{\zeta} = \frac{1}{s} + O(\zeta^2), \quad (741)$$

so

$$\alpha_\nu^{\text{eff}, \text{E}} = \frac{\gamma_\nu^{\text{eff}, \text{E}} n_c^2}{4\pi^2 d} \omega \csc^2 \left(\frac{\theta}{2} \right) + O(|K|a^2). \quad (742)$$

At small twist,

$$\alpha_\nu^{\text{eff}, \text{E}} \sim \frac{\gamma_\nu^{\text{eff}, \text{E}} n_c^2}{\pi^2 d} \frac{\omega}{\theta^2}. \quad (743)$$

At fixed negative curvature, suppose

$$\gamma_\nu^{\text{eff}}(K, \theta) \longrightarrow \gamma_{\nu, \infty}^{\text{eff}}(K), \quad 0 < \gamma_{\nu, \infty}^{\text{eff}} < \infty. \quad (744)$$

Then

$$\alpha_{\nu}^{\text{eff,H}} \sim \frac{\gamma_{\nu,\infty}^{\text{eff}} n_c^2}{4\pi^2 d \zeta^2} \omega \ln^2 \left[\frac{4 \sinh \zeta}{|\theta|} \right]. \quad (745)$$

Thus

$$\alpha_{\nu}^{\text{eff}} : \quad \omega |\theta|^{-2} \longrightarrow \omega \ln^2 |\theta|^{-1}. \quad (746)$$

Negative curvature changes the functional scaling of the effective hybridization strength rather than merely modifying its prefactor. At the same θ , w/t , and active form factor,

$$\frac{\alpha_{\nu}^{\text{eff,H}}}{\alpha_{\nu}^{\text{eff,E}}} = \left[\frac{\xi_{\text{M}}^{(\text{H})}}{\xi_{\text{M}}^{(\text{E})}} \right]^2. \quad (747)$$

Allowing different active overlaps gives

$$\frac{\alpha_{\nu}^{\text{eff,H}}}{\alpha_{\nu}^{\text{eff,E}}} = \frac{\gamma_{\nu}^{\text{eff,H}}}{\gamma_{\nu}^{\text{eff,E}}} \left[\frac{\xi_{\text{M}}^{(\text{H})}}{\xi_{\text{M}}^{(\text{E})}} \right]^2. \quad (748)$$

Because $\xi_{\text{M}}^{(\text{H})} < \xi_{\text{M}}^{(\text{E})}$, pure geometry increases the characteristic momentum and kinetic scale and therefore reduces $\alpha_{\nu}^{\text{eff}}$ at the same twist. The active overlap $\gamma_{\nu}^{\text{eff}}$ supplies an independent curvature-dependent reconstruction channel. In the double-scaling limit,

$$\frac{\xi_{\text{M}}^{(\text{H})}}{\xi_{\text{M}}^{(\text{E})}} \longrightarrow \mathcal{G}(u) = u \operatorname{arsinh}(u^{-1}), \quad (749)$$

so

$$\frac{\alpha_{\nu}^{\text{eff,H}}}{\alpha_{\nu}^{\text{eff,E}}} \longrightarrow \frac{\gamma_{\nu}^{\text{eff,H}}}{\gamma_{\nu}^{\text{eff,E}}} \mathcal{G}^2(u). \quad (750)$$

Define the active-form-factor exponent

$$\eta_{\gamma,\nu} = -\frac{d \ln \gamma_{\nu}^{\text{eff}}}{d \ln s}. \quad (751)$$

Using Eq. (740), the exact running effective-coupling exponent is

$$\beta_{\alpha,\nu} = -\frac{d \ln \alpha_{\nu}^{\text{eff}}}{d \ln s} = 2\nu_{\xi} + \eta_{\gamma,\nu}. \quad (752)$$

For a regular active profile, $\eta_{\gamma,\nu} \longrightarrow 0$. The two geometric endpoints are then $\beta_{\alpha,\nu}^{(\text{E})} \longrightarrow 2$, and $\beta_{\alpha,\nu}^{(\text{H})} \longrightarrow 0$. The vanishing hyperbolic exponent is the fixed-point origin of the logarithmic effective-coupling law. The geometry determines $\alpha_{\nu}^{\text{eff}}(K, \theta, \omega)$, but it does not by itself prove that the dimensionless spectral problem possesses a flat-band optimum. We therefore state the following result conditionally. Assume that the fully specified dimensionless Hamiltonian has an isolated physical flattening point

$$\alpha_{\nu}^{\text{eff}} = \alpha_{\nu}^*, \quad 0 < \alpha_{\nu}^* < \infty, \quad (753)$$

and that the active form factor approaches the positive constant in Eq. (744). Define

$$c_{\nu} = \frac{2\pi\zeta}{n_c} \sqrt{\frac{d\alpha_{\nu}^*}{\gamma_{\nu,\infty}^{\text{eff}}}}. \quad (754)$$

The condition $\alpha_{\nu}^{\text{eff}} = \alpha_{\nu}^*$ then gives

$$L_{\nu}^* = \frac{c_{\nu}}{\sqrt{\omega}}. \quad (755)$$

Since

$$\sinh L = \frac{\sinh \zeta}{s}, \quad (756)$$

we obtain

$$s_{\nu}^* = \frac{\sinh \zeta}{\sinh(c_{\nu}/\sqrt{\omega})}. \quad (757)$$

Therefore the conditional hyperbolic magic-angle curve is

$$\theta_{\nu}^*(K, \omega) = 2 \arcsin \left[\frac{\sinh(a/2R)}{\sinh(c_{\nu}/\sqrt{\omega})} \right]. \quad (758)$$

This expression is valid whenever its argument does not exceed unity. For weak interlayer coupling, $\omega \longrightarrow 0^+$, we have

$$\frac{c_{\nu}}{\sqrt{\omega}} \longrightarrow \infty. \quad (759)$$

Using

$$\frac{1}{\sinh x} = 2e^{-x} [1 + e^{-2x} + O(e^{-4x})], \quad (760)$$

we find

$$\theta_{\nu}^*(K, \omega) = 4 \sinh \left(\frac{a}{2R} \right) \exp \left(-\frac{c_{\nu}}{\sqrt{\omega}} \right) \left[1 + O \left(e^{-2c_{\nu}/\sqrt{\omega}} \right) \right]. \quad (761)$$

The Euclidean condition obtained from Eq. (743) is

$$\theta_{\nu}^*(0, \omega) = \frac{n_c}{\pi} \sqrt{\frac{\gamma_{\nu}^{\text{eff,E}}}{d\alpha_{\nu}^*}} \sqrt{\omega} + O(\omega^{3/2}). \quad (762)$$

Thus the same finite dimensionless spectral root would generate

$$\theta_{\nu}^* : \quad \sqrt{\omega} \longrightarrow \exp \left(-\frac{c_{\nu}}{\sqrt{\omega}} \right) \quad (763)$$

when the geometry crosses from the Euclidean to the fixed-curvature hyperbolic regime. The effective moiré area along the conditional hyperbolic curve is obtained by substituting Eq. (755) into the disk-area formula:

$$A_{\text{M},\nu}^{\text{eff},*} = 2\pi R^2 \left[\cosh \left(\frac{c_{\nu}}{\sqrt{\omega}} \right) - 1 \right]. \quad (764)$$

At weak coupling,

$$A_{M,\nu}^{\text{eff},*} \sim \pi R^2 \exp\left(\frac{c_\nu}{\sqrt{\omega}}\right). \quad (765)$$

The corresponding area-normalized pair count is

$$N_{\text{pair},\nu}^{\text{eff},*} \sim \frac{\pi n_b R^2}{A_0(K)} \exp\left(\frac{c_\nu}{\sqrt{\omega}}\right). \quad (766)$$

For a coupling threshold τ ,

$$N_{\text{pair},\tau,\nu}^{\text{eff},*} \sim \frac{\pi n_b R^2}{A_0(K)} \frac{\sinh(\ell_\tau/2R)}{\sinh(a/2R)} \exp\left(\frac{c_\nu}{\sqrt{\omega}}\right). \quad (767)$$

The Euclidean conditional magic count instead satisfies

$$N_M^{\text{eff},E,*} = \Theta(\omega^{-1}). \quad (768)$$

Thus the geometry predicts the resource crossover

$$N_M^{\text{eff},*} : \quad \omega^{-1} \longrightarrow \exp\left(\frac{c_\nu}{\sqrt{\omega}}\right). \quad (769)$$

This is an effective geometric mode-count law. It is not an exact crystallographic index law; exact finite-device complexity additionally requires the commensurability analysis developed later. Let $[\alpha_{\nu,-}, \alpha_{\nu,+}]$ be a numerically established dimensionless interval containing the physical flattening regime. Suppose the accessible circuit parameters satisfy

$$\theta_{\min} \leq |\theta| \leq \theta_{\max}, \quad 0 < \omega \leq \omega_{\max}. \quad (770)$$

Define

$$L_{\max} = \text{arsinh}\left[\frac{\sinh \zeta}{\sin(\theta_{\min}/2)}\right], \quad (771)$$

and let

$$\gamma_\nu^{\max} = \sup_{\theta, K} \gamma_\nu^{\text{eff}}(K, \theta) \quad (772)$$

over the accessible domain. The maximum attainable effective coupling is bounded by

$$\alpha_{\nu,\max}^{\text{eff}} \leq \frac{\gamma_\nu^{\max} n_c^2}{4\pi^2 d} \omega_{\max} \left(\frac{L_{\max}}{\zeta}\right)^2. \quad (773)$$

Therefore

$$\alpha_{\nu,\max}^{\text{eff}} < \alpha_{\nu,-} \implies \text{the established flattening regime is inaccessible.} \quad (774)$$

This implication is rigorous and independent of any visual band inspection. Conversely, reaching

$$\alpha_\nu^{\text{eff}} \in [\alpha_{\nu,-}, \alpha_{\nu,+}] \quad (775)$$

is necessary but not sufficient. Positive spectral isolation, global velocity suppression, and complete representation-sector coverage remain independent requirements. We

now close the microscopic chain from the curvature-dependent coupling matrix to the band dispersion. Let P_ν project onto the target low-energy subspace and define $Q_\nu = I - P_\nu$. For a Hamiltonian $H(\lambda)$ depending on a generalized spectral coordinate λ , define

$$H_{PP} = P_\nu H P_\nu, \quad H_{QQ} = Q_\nu H Q_\nu, \quad (776)$$

and

$$V = P_\nu H Q_\nu. \quad (777)$$

Whenever

$$E \notin \text{Spec } H_{QQ}, \quad (778)$$

the exact Feshbach effective Hamiltonian is

$$H_{\text{eff}}(E, \lambda) = H_{PP}(\lambda) + \Sigma_\nu(E, \lambda), \quad (779)$$

with

$$\Sigma_\nu(E, \lambda) = V(\lambda) [E - H_{QQ}(\lambda)]^{-1} V^\dagger(\lambda). \quad (780)$$

Define the active spectral denominator

$$\Delta_\nu^{\text{act}}(E, \lambda) = \text{dist}[E, \text{Spec } H_{QQ}(\lambda)]. \quad (781)$$

Then

$$\|\Sigma_\nu\| \leq \frac{\|V\|^2}{\Delta_\nu^{\text{act}}}. \quad (782)$$

Let

$$R_Q(E, \lambda) = [E - H_{QQ}(\lambda)]^{-1}. \quad (783)$$

At fixed E , differentiation gives

$$\begin{aligned} \partial_\mu \Sigma_\nu &= (\partial_\mu V) R_Q V^\dagger + V R_Q (\partial_\mu V^\dagger) \\ &\quad + V R_Q (\partial_\mu H_{QQ}) R_Q V^\dagger. \end{aligned} \quad (784)$$

Therefore

$$\|\partial_\mu \Sigma_\nu\| \leq \frac{2\|V\|\|\partial_\mu V\|}{\Delta_\nu^{\text{act}}} + \frac{\|V\|^2\|\partial_\mu H_{QQ}\|}{(\Delta_\nu^{\text{act}})^2}. \quad (785)$$

This bound identifies three independent flattening controls: $\|V\|$, $\|\partial_\mu V\|$, Δ_ν^{act} . A large total coupling is not sufficient if the active derivative vanishes or the target is far off resonance. Conversely, a moderate coupling can produce a large dispersive correction near a small active denominator. Let

$$H_{\text{eff}}(E_\nu(\lambda), \lambda) |u_\nu(\lambda)\rangle = E_\nu(\lambda) |u_\nu(\lambda)\rangle. \quad (786)$$

Differentiation gives

$$\partial_\mu E_\nu = Z_\nu \langle u_\nu | \partial_\mu H_{PP} + (\partial_\mu \Sigma_\nu)_E | u_\nu \rangle, \quad (787)$$

where

$$Z_\nu = [1 - \langle u_\nu | \partial_E \Sigma_\nu | u_\nu \rangle]^{-1}. \quad (788)$$

Because

$$\partial_E \Sigma_\nu = -V R_Q^2 V^\dagger \preceq 0 \quad (789)$$

for real E outside the remote spectrum, $0 < Z_\nu \leq 1$. Define the bare kinetic covector

$$dE_\nu^{\text{bare}} = Z_\nu \langle u_\nu | dH_{PP} | u_\nu \rangle, \quad (790)$$

and the hybridization covector

$$dE_\nu^{\text{hyb}} = Z_\nu \langle u_\nu | (d\Sigma_\nu)_E | u_\nu \rangle. \quad (791)$$

Then

$$dE_\nu = dE_\nu^{\text{bare}} + dE_\nu^{\text{hyb}}. \quad (792)$$

The exact local velocity-cancellation residual is

$$\mathcal{R}_\nu^v(\lambda) = \frac{\|dE_\nu^{\text{bare}} + dE_\nu^{\text{hyb}}\|}{\|dE_\nu^{\text{bare}}\| + \|dE_\nu^{\text{hyb}}\|}. \quad (793)$$

It satisfies $0 \leq \mathcal{R}_\nu^v \leq 1$. Let the generalized spectral domain $\mathcal{D}$ have metric diameter $D_\mathcal{D}$. For a differentiable connected band,

$$\begin{aligned} W_\nu &= \sup_{\lambda_1, \lambda_2 \in \mathcal{D}} |E_\nu(\lambda_2) - E_\nu(\lambda_1)| \\ &\leq D_\mathcal{D} \sup_{\lambda \in \mathcal{D}} \|dE_\nu(\lambda)\|. \end{aligned} \quad (794)$$

If $\mathcal{R}_\nu^v(\lambda) \leq \varepsilon_v$ throughout $\mathcal{D}$, and

$$\|dE_\nu^{\text{bare}}\| + \|dE_\nu^{\text{hyb}}\| \leq V_\nu^{\text{sum}}, \quad (795)$$

then

$$W_\nu \leq D_\mathcal{D} V_\nu^{\text{sum}} \varepsilon_v. \quad (796)$$

Thus uniform self-energy cancellation implies global bandwidth suppression without an additional qualitative assumption. The complete microscopic mechanism is

$$\begin{aligned} (K, \theta) &\longrightarrow d_\theta(r) \longrightarrow \xi_M \longrightarrow p_L(u) \\ &\longrightarrow T_L(u) \longrightarrow \delta T \longrightarrow V_\nu \\ &\longrightarrow w_\nu^{\text{eff}} \longrightarrow \alpha_\nu^{\text{eff}} \longrightarrow \Sigma_\nu \\ &\longrightarrow dE_\nu^{\text{bare}} + dE_\nu^{\text{hyb}} \longrightarrow v_\nu \longrightarrow W_\nu. \end{aligned} \quad (797)$$

Suppose a controlled low-energy reduction exists in which the target spectrum can be written as

$$E_\nu(\lambda; K, \theta, \omega) = E_{\nu,0} + E_M^{\text{eff}} \Phi_\nu[\tilde{\lambda}, \alpha_\nu^{\text{eff}}] + \varepsilon_\nu, \quad (798)$$

where $\tilde{\lambda}$ is a dimensionless generalized momentum coordinate and $\|\varepsilon_\nu\| \leq \epsilon_\nu^{\text{red}}$ is an independently controlled reduction error. Define the dimensionless master bandwidth

$$\mathcal{W}_\nu(\alpha) = \sup_{\tilde{\lambda}} \Phi_\nu(\tilde{\lambda}, \alpha) - \inf_{\tilde{\lambda}} \Phi_\nu(\tilde{\lambda}, \alpha). \quad (799)$$

Then

$$|W_\nu - E_M^{\text{eff}} \mathcal{W}_\nu(\alpha_\nu^{\text{eff}})| \leq 2\epsilon_\nu^{\text{red}}. \quad (800)$$

Similarly, if $\mathcal{D}_\nu(\alpha)$ denotes the dimensionless isolation gap,

$$|\Delta_\nu - E_M^{\text{eff}} \mathcal{D}_\nu(\alpha_\nu^{\text{eff}})| \leq 2\epsilon_\nu^{\text{red}}. \quad (801)$$

A physical flat-band optimum requires a value α_ν^* for which

$$\mathcal{W}_\nu(\alpha_\nu^*) \ll \mathcal{D}_\nu(\alpha_\nu^*), \quad (802)$$

together with $\mathcal{D}_\nu(\alpha_\nu^*) > 0$, and

$$\epsilon_\nu^{\text{red}} \ll E_M^{\text{eff}} \mathcal{W}_\nu(\alpha_\nu^*) \quad (803)$$

for a finite nonzero bandwidth claim. The geometric calculation fixes the map

$$(K, \theta, \omega) \longmapsto \alpha_\nu^{\text{eff}}. \quad (804)$$

It does not determine $\mathcal{W}_\nu(\alpha)$, $\mathcal{D}_\nu(\alpha)$, α_ν^* without solving the actual microscopic Hamiltonian. The preceding formulas separate three quantitatively independent curvature effects. First, curvature changes the kinetic scale:

$$E_M^{(\text{E})} \propto t\theta^2, \quad E_M^{(\text{H})} \propto \frac{t}{\ln^2 |\theta|^{-1}}. \quad (805)$$

Second, curvature changes the complete distribution of interlayer couplings:

$$p_E(d) = \frac{2d}{a^2}, \quad p_H(d) = \frac{\cosh(d/2R)}{2R \sinh(a/2R)}. \quad (806)$$

This modifies γ_ν^{eff} , η_ν^{act} , $r_{\text{st}}(V_\mathcal{I})$. Third, curvature changes the number of geometrically and spectrally available channels:

$$N_{\text{res}}^{(\text{E})} = O(|\theta|^{-2}), \quad N_{\text{res}}^{(\text{H})} = O(|\theta|^{-1}) \quad (807)$$

under fixed energy, coupling, and resolution windows. These mechanisms enter the self-energy differently. The kinetic-scale change modifies the dimensionless denominator α_ν^{eff} . The coupling-distribution change modifies the projected matrix elements. The channel-count change modifies the rank and multiplicity of the resonant self-energy. None can be replaced by another through an overall energy rescaling. The exact results established in this chapter are $d_\theta(r)$, ξ_M , A_M^{eff} , N_M^{eff} , together with $p_L(u)$, $p_H(d)$, $W_{\text{sc},\infty}$, α_ν^{eff} . These results establish the geometry and coupling reconstruction completely. They do not imply, without additional spectral verification,

$$W_\nu = 0, \quad \Delta_\nu > 0, \quad \alpha_\nu^* \text{ exists}, \quad (808)$$

or

$$\theta_\nu^* \text{ is a physical magic angle.} \quad (809)$$

A physical magic-angle claim requires the same continuously tracked spectral manifold to satisfy

$$\frac{W_\nu}{\Delta_\nu} \ll 1, \quad \frac{D_{\mathcal{D}} \sup_\lambda \|dE_\nu\|}{\Delta_\nu} \ll 1, \quad \rho_\nu^{\max} \gg E_{\text{kin}}^{-1}, \quad (810)$$

and

$$N_{\text{res}} > 0, \quad \gamma_\nu^{\text{eff}} > 0, \quad (811)$$

with convergence under increasing finite-cover size, hopping cutoff, and representation-sector completion.

IV. MATHEMATICAL INTERFACE TO THE COMPANION APPENDIX

A. Hyperbolic moiré translation group

Let

$$G = \text{Isom}^+(\mathbb{H}_R^2) \simeq \text{PSL}(2, \mathbb{R}) \quad (812)$$

be the orientation-preserving isometry group of the hyperbolic plane. Let $\Gamma < G$ be the torsion-free cocompact Fuchsian group generating the first microscopic crystal. For one reference point $o \in \mathbb{H}_R^2$, the orbit lattice is $\Lambda_1 = \Gamma o$. If the primitive quotient contains n_b physical orbitals or resonator modes, the complete first-layer lattice consists of n_b copies of this orbit. Let $g_\theta \in G$ be the relative isometry defining the second layer. For the centred twist used here, $g_\theta o = o$, and g_θ is the rotation through angle θ about o . The second layer is $\Lambda_2 = g_\theta \Lambda_1$. Its translation group is

$$\Gamma_2 = g_\theta \Gamma g_\theta^{-1}. \quad (813)$$

An isometry $h \in G$ preserves the first layer precisely when $h \in \Gamma$. It preserves the second layer precisely when

$$h \in g_\theta \Gamma g_\theta^{-1}. \quad (814)$$

It therefore preserves both layers if and only if it belongs to the intersection

$$\Gamma_M(\theta) = \Gamma \cap g_\theta \Gamma g_\theta^{-1}. \quad (815)$$

The maximality of Eq. (815) is immediate but physically important. If $h \in \Gamma_M$, then h preserves both layers. Conversely, every common translation must lie in both layer groups and hence in their intersection. Therefore every common translation subgroup H satisfies $H \subseteq \Gamma_M$. Thus Γ_M is the unique maximal exact translation group of the bilayer. Let the bilayer Hilbert space be written relative to an exact Γ_M -cell as

$$\mathcal{H}_\infty = \ell^2(\Gamma_M) \otimes \mathbb{C}^{\nu_M}, \quad (816)$$

where

$$\nu_M = 2n_b q_M \quad (817)$$

will be derived in Sec. IV C. A basis vector is

$$|\gamma, \alpha\rangle, \quad \gamma \in \Gamma_M, \quad \alpha = 1, \dots, \nu_M. \quad (818)$$

The common translations act through the left regular representation,

$$U_\eta |\gamma, \alpha\rangle = |\eta\gamma, \alpha\rangle, \quad \eta \in \Gamma_M. \quad (819)$$

Exact bilayer translation symmetry is

$$[H, U_\eta] = 0 \quad \text{for every } \eta \in \Gamma_M. \quad (820)$$

Consequently,

$$\langle \gamma, \alpha | H | \gamma', \beta \rangle = \langle e, \alpha | H | \gamma^{-1} \gamma', \beta \rangle. \quad (821)$$

Define the matrix-valued hopping kernel

$$h_{\alpha\beta}(\delta) = \langle e, \alpha | H | \delta, \beta \rangle, \quad \delta \in \Gamma_M. \quad (822)$$

The Hamiltonian acts as the right-convolution operator

$$(H\psi)_\alpha(\gamma) = \sum_{\delta \in \Gamma_M} \sum_{\beta=1}^{\nu_M} h_{\alpha\beta}(\delta) \psi_\beta(\gamma\delta). \quad (823)$$

Hermiticity is equivalent to

$$h(\delta^{-1}) = h^\dagger(\delta). \quad (824)$$

In the co-rotating layer basis, the kernel has the explicit block form

$$h(\delta; \theta) = \begin{pmatrix} h_0(\delta) & \tau_\theta(\delta) \\ \tau_\theta^\dagger(\delta^{-1}) & h_0(\delta) \end{pmatrix}. \quad (825)$$

Rigid twisting leaves the isolated layers unitarily equivalent, and therefore

$$\partial_\theta h_0(\delta) = 0. \quad (826)$$

All twist dependence is carried by $\tau_\theta(\delta)$, whose entries are determined by the physical distance-dependent interlayer couplings derived previously. When the intersection Γ_M has infinite index in Γ , it still remains the exact common symmetry group, but its quotient does not define a finite moiré cell. Local geometry and the microscopic Hamiltonian remain well defined; only the finite exact crystallographic reduction fails. The maximal-common-subgroup statement and its complete proof are supplied by Mathematical Appendix, Lemma 1. The convolution representation and its Hermiticity properties are supplied by the generalized hyperbolic Bloch construction in Mathematical Appendix, Theorems 26–31.

B. Exact commensurability

Two subgroups $H_1, H_2 < G$ are commensurable when

$$[H_1 : H_1 \cap H_2] < \infty \quad (827)$$

and

$$[H_2 : H_1 \cap H_2] < \infty. \quad (828)$$

The commensurator of Γ is

$$\text{Comm}_G(\Gamma) = \{g \in G : \Gamma \sim g\Gamma g^{-1}\}. \quad (829)$$

Equivalently,

$$\text{Comm}_G(\Gamma) = \left\{g \in G : [\Gamma : \Gamma \cap g\Gamma g^{-1}] < \infty, \right. \\ \left. [g\Gamma g^{-1} : \Gamma \cap g\Gamma g^{-1}] < \infty \right\}. \quad (830)$$

The exact crystallographic criterion is a finite hyperbolic moiré supercell exists $\iff g_\theta \in \text{Comm}_G(\Gamma)$. To see how the two directions enter the physical construction, suppose first that

$$g_\theta \in \text{Comm}_G(\Gamma). \quad (831)$$

Then

$$\Gamma_M = \Gamma \cap g_\theta \Gamma g_\theta^{-1} \quad (832)$$

has finite index in both layer groups. A fundamental domain of Γ_M is therefore a finite union of primitive domains of either layer, so it is a finite common periodic cell. Conversely, suppose a finite common cell exists and let H be its translation group. Common periodicity implies

$$H \subseteq \Gamma \cap g_\theta \Gamma g_\theta^{-1} = \Gamma_M. \quad (833)$$

Finiteness of the cell implies $[\Gamma : H] < \infty$ and

$$[g_\theta \Gamma g_\theta^{-1} : H] < \infty. \quad (834)$$

The subgroup-index identity

$$[\Gamma : H] = [\Gamma : \Gamma_M][\Gamma_M : H] \quad (835)$$

then forces $[\Gamma : \Gamma_M] < \infty$, and the same argument applies to the second layer. Hence g_θ belongs to the commensurator. For two identical conjugate layers, the two indices are equal. Indeed, the two parent groups have equal volume:

$$\text{covol}(\Gamma) = \text{covol}(g_\theta \Gamma g_\theta^{-1}) = A_0. \quad (836)$$

If

$$q_1 = [\Gamma : \Gamma_M], \quad q_2 = [g_\theta \Gamma g_\theta^{-1} : \Gamma_M], \quad (837)$$

then

$$\text{covol}(\Gamma_M) = q_1 A_0 = q_2 A_0, \quad (838)$$

and therefore $q_1 = q_2$. For centred rotations, define the stabilizer of the twist centre,

$$K_o = \text{Stab}_G(o) \simeq SO(2), \quad (839)$$

and the fixed-centre commensurating subgroup

$$C_o(\Gamma) = \text{Comm}_G(\Gamma) \cap K_o. \quad (840)$$

The exact centred-twist condition is $g_\theta \in C_o(\Gamma)$. Because $C_o(\Gamma)$ is a subgroup of the circle group, it satisfies the finite-or-dense dichotomy: $C_o(\Gamma)$ is either finite or dense in K_o . Consequently,

$$\exists \theta_j \neq 0, \quad \theta_j \rightarrow 0, \quad g_{\theta_j} \in \text{Comm}_G(\Gamma) \quad (841)$$

if and only if $|C_o(\Gamma)| = \infty$. For a non-arithmetic cocompact Fuchsian lattice, the full commensurator is discrete. Its intersection with the compact group K_o is therefore finite:

$$\Gamma \text{ non-arithmetic} \implies |C_o(\Gamma)| < \infty. \quad (842)$$

When a nontrivial centred exact twist exists, the minimum exact angle is strictly positive:

$$\theta_{\min}^{\text{exact}}(o) = \min \{|\theta| : g_\theta \in C_o(\Gamma), \quad g_\theta \neq e\} > 0. \quad (843)$$

Arithmeticity removes the discreteness obstruction of the full commensurator, but a fixed-centre sequence still requires the additional condition $|C_o(\Gamma)| = \infty$. Density of the full commensurator alone does not automatically prove that an arbitrarily prescribed centre is active. Every distinct exact commensurating sequence approaching the identity has diverging coincidence index:

$$g_j \rightarrow e, \quad g_j \neq e, \quad g_j \in \text{Comm}_G(\Gamma) \implies q_M(g_j) \rightarrow \infty. \quad (844)$$

For a centred sequence,

$$\theta_j \rightarrow 0 \implies q_M(\theta_j) \rightarrow \infty. \quad (845)$$

This theorem proves divergence but does not fix its rate. The rate is an arithmetic property of the selected commensurating sequence. For a generic incommensurate twist,

$$g_\theta \notin \text{Comm}_G(\Gamma), \quad [\Gamma : \Gamma_M] = \infty. \quad (846)$$

In this case, the local distance field, the coupling matrix and the infinite bilayer Hamiltonian remain meaningful. We must use commensurate approximants or paired finite covers instead of an exact finite moiré quotient. The necessary-and-sufficient commensurability criterion is supplied by Mathematical Appendix, Theorem 5. The finite-or-dense centred-twist dichotomy, the non-arithmetic obstruction and the index-divergence theorem are supplied by Mathematical Appendix, Theorems 6–8.

C. Exact finite-device complexity

For every exact commensurating relative isometry, define

$$q_M(g) = [\Gamma : \Gamma \cap g\Gamma g^{-1}]. \quad (847)$$

For a centred twist,

$$q_M(\theta) = [\Gamma : \Gamma \cap g_\theta \Gamma g_\theta^{-1}]. \quad (848)$$

The exact common quotient is

$$\Sigma_M = \Gamma_M \backslash \mathbb{H}_R^2. \quad (849)$$

If $\Sigma_0 = \Gamma \backslash \mathbb{H}_R^2$ has area A_0 and genus g_0 , then

$$A_M^{\text{sc}} = q_M A_0. \quad (850)$$

The natural covering map $\Sigma_M \rightarrow \Sigma_0$ has degree $\deg p = q_M$. For closed orientable torsion-free quotients, Euler characteristic is multiplicative:

$$2 - 2g_M = q_M (2 - 2g_0). \quad (851)$$

Therefore

$$g_M = 1 + q_M (g_0 - 1). \quad (852)$$

If the primitive quotient contains n_b microscopic orbitals per layer, one exact common cell contains $N_{M,1}^{\text{sc}} = n_b q_M$ orbitals in either layer. Since coincident layer orbitals remain distinct Hilbert-space states, the bilayer contains

$$N_M^{\text{sc}} = 2n_b q_M. \quad (853)$$

Thus

$$q_M = \frac{A_M^{\text{sc}}}{A_0} = \frac{g_M - 1}{g_0 - 1} = \frac{N_M^{\text{sc}}}{2n_b}. \quad (854)$$

For a genus-two Bolza primitive quotient,

$$g_0 = 2, \quad A_0 = 4\pi R^2. \quad (855)$$

Hence

$$g_M = 1 + q_M, \quad A_M^{\text{sc}} = 4\pi R^2 q_M. \quad (856)$$

For one resonator mode per parent orbit, $n_b = 1$, so one exact bilayer cell contains

$$N_{\text{res}}^{\text{cell}} = 2q_M \quad (857)$$

resonators. To obtain a larger boundaryless finite device, choose $\Gamma_N \triangleleft \Gamma_M$ with finite quotient

$$Q_N = \Gamma_M / \Gamma_N, \quad n_N = |Q_N| = [\Gamma_M : \Gamma_N]. \quad (858)$$

The total one-particle Hilbert-space dimension is then

$$D_N = 2n_b q_M n_N. \quad (859)$$

Let every intralayer site have coordination number $z_{\parallel}$, and let every first-layer site couple appreciably to an average of $z_{\perp}$ sites in the second layer. The number of undirected intralayer couplers is

$$N_{\parallel}^c = z_{\parallel} n_b q_M n_N, \quad (860)$$

because the factor 1/2 from edge double counting cancels the factor of two from the two layers. The number of interlayer couplers is

$$N_{\perp}^c = z_{\perp} n_b q_M n_N. \quad (861)$$

Therefore

$$N_{\text{coup}} = n_b q_M n_N (z_{\parallel} + z_{\perp}). \quad (862)$$

For the regular-octagon orbit graph, $z_{\parallel} = 8$. The same formulas determine computational complexity. A sparse matrix-vector product requires

$$O[D_N (z_{\parallel} + z_{\perp})] \quad (863)$$

operations, whereas dense diagonalization scales as

$$O(D_N^3) = O[(2n_b q_M n_N)^3]. \quad (864)$$

The exact coincidence index is therefore simultaneously a hardware, memory and diagonalization-complexity multiplier. The index obeys three exact structural identities. Layer exchange gives

$$q_M(g^{-1}) = q_M(g). \quad (865)$$

Microscopic translations of either layer give

$$q_M(\gamma_L g \gamma_R) = q_M(g), \quad \gamma_L, \gamma_R \in \Gamma. \quad (866)$$

Thus q_M is naturally a function on $\Gamma \backslash \text{Comm}_G(\Gamma) / \Gamma$. Composition is submultiplicative:

$$q_M(gh) \leq q_M(g) q_M(h). \quad (867)$$

The normalizer

$$N_G(\Gamma) = \{g \in G : g\Gamma g^{-1} = \Gamma\} \quad (868)$$

corresponds precisely to index one:

$$q_M(g) = 1 \iff g \in N_G(\Gamma). \quad (869)$$

A genuinely enlarged exact moiré cell therefore requires

$$g \in \text{Comm}_G(\Gamma) \setminus N_G(\Gamma). \quad (870)$$

The exact cell must not be identified with the effective geometric registry domain. Define the locking factor

$$\mathcal{L}_{\text{arith}}(\theta) = \frac{A_M^{\text{sc}}(\theta)}{A_M^{\text{eff}}(\theta)} = \frac{q_M(\theta) A_0}{A_M^{\text{eff}}(\theta)}. \quad (871)$$

Then

$$N_M^{\text{sc}} = \mathcal{L}_{\text{arith}} N_M^{\text{eff}}. \quad (872)$$

Let

$$s = \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (873)$$

Define the geometric running exponent

$$\beta_M = -\frac{d \ln A_M^{\text{eff}}}{d \ln s} \quad (874)$$

and the arithmetic correction

$$\delta_{\text{arith}} = -\frac{d \ln \mathcal{L}_{\text{arith}}}{d \ln s}. \quad (875)$$

The exact crystallographic exponent is

$$\beta_{\text{sc}} = \beta_M + \delta_{\text{arith}}. \quad (876)$$

Therefore the geometric law $A_M^{\text{eff}} \propto |\theta|^{-1}$ does not imply $q_M \propto |\theta|^{-1}$ unless $\mathcal{L}_{\text{arith}}$ remains bounded above and below. If

$$0 < c_- \leq \mathcal{L}_{\text{arith}}(\theta_j) \leq c_+ < \infty \quad (877)$$

along an exact sequence, then

$$q_M(\theta_j) = \Theta(|\theta_j|^{-1}). \quad (878)$$

If instead

$$\mathcal{L}_{\text{arith}}(\theta_j) \sim C_{\text{arith}} |\theta_j|^{-\delta}, \quad (879)$$

then

$$q_M(\theta_j) \sim |\theta_j|^{-(1+\delta)}. \quad (880)$$

The effective geometric count and the exact hardware count must therefore be reported separately. The area, covering, genus and microscopic-count identities are supplied by Mathematical Appendix, Theorem 5. Index symmetry, double-coset invariance, submultiplicativity and normalizer equivalence are proved in the same exact-commensurability section. The locking-factor and running-exponent decomposition are proved in the geometric-arithmetic locking section of the Mathematical Appendix.

D. Abelian and non-Abelian representation sectors

The Euclidean Bloch variable $\mathbf{k}$ labels one-dimensional irreducible representations of the Abelian translation group $\mathbb{Z}^2$. The hyperbolic group Γ_M is non-Abelian, so the replacement is

$$\mathbf{k} \longrightarrow [\rho], \quad \rho : \Gamma_M \longrightarrow U(d_\rho). \quad (881)$$

Representations related by a common unitary conjugation describe the same generalized momentum:

$$\rho^W(\gamma) = W \rho(\gamma) W^{-1}. \quad (882)$$

For the matrix-valued convolution kernel $h(\delta)$, define

$$H[\rho] = \sum_{\delta \in \Gamma_M} h(\delta) \otimes \rho(\delta). \quad (883)$$

Its dimension is

$$\dim H[\rho] = \nu_M d_\rho = 2n_b q_M d_\rho. \quad (884)$$

Using

$$h(\delta^{-1}) = h^\dagger(\delta), \quad \rho(\delta^{-1}) = \rho^\dagger(\delta), \quad (885)$$

we obtain

$$\begin{aligned} H[\rho]^\dagger &= \sum_{\delta} h^\dagger(\delta) \otimes \rho^\dagger(\delta) \\ &= \sum_{\delta} h(\delta^{-1}) \otimes \rho(\delta^{-1}) = H[\rho]. \end{aligned}$$

Thus every unitary sector gives a Hermitian finite-dimensional spectral problem. If

$$\sum_{\delta \in \Gamma_M} \|h(\delta)\| < \infty, \quad (886)$$

then

$$\sup_{\rho} \|H[\rho]\| \leq \sum_{\delta} \|h(\delta)\|. \quad (887)$$

The same hopping-summability condition therefore controls all finite-dimensional unitary sectors simultaneously. Gauge-equivalent representations give unitarily equivalent Hamiltonians:

$$H[\rho^W] = (I_{\nu_M} \otimes W) H[\rho] (I_{\nu_M} \otimes W^{-1}). \quad (888)$$

Consequently,

$$\text{Spec } H[\rho^W] = \text{Spec } H[\rho]. \quad (889)$$

A one-dimensional representation

$$\chi : \Gamma_M \longrightarrow U(1) \quad (890)$$

annihilates every commutator and therefore factors through

$$\Gamma_M^{\text{ab}} = \Gamma_M / [\Gamma_M, \Gamma_M]. \quad (891)$$

For a genus- g_M surface group, $\Gamma_M^{\text{ab}} \simeq \mathbb{Z}^{2g_M}$. Choose the marked generators $a_1, b_1, \dots, a_{g_M}, b_{g_M}$. An Abelian character is specified by

$$\chi_{\mathbf{k}}(a_j) = e^{ik_{a_j}}, \quad \chi_{\mathbf{k}}(b_j) = e^{ik_{b_j}}. \quad (892)$$

The Abelian generalized Brillouin space is

$$\mathcal{B}_1 = \text{Hom}(\Gamma_M, U(1)) \simeq \mathbb{T}^{2g_M}. \quad (893)$$

Its dimension is

$$\dim_{\mathbb{R}} \mathcal{B}_1 = 2g_M = 2 + 2q_M(g_0 - 1). \quad (894)$$

Let

$$n : \Gamma_M \longrightarrow \mathbb{Z}^{2g_M} \quad (895)$$

be the abelianization map. The Abelian Bloch Hamiltonian is

$$H(\mathbf{k}) = \sum_{\delta \in \Gamma_M} h(\delta) e^{i\mathbf{k}^\top n(\delta)}. \quad (896)$$

If

$$\delta_1 \delta_2^{-1} \in [\Gamma_M, \Gamma_M], \quad (897)$$

then $\chi(\delta_1) = \chi(\delta_2)$ for every Abelian character. The entire Abelian theory is therefore blind to distinctions carried only by noncommuting translation words. For $d > 1$, define the generator holonomies

$$A_j = \rho(a_j), \quad B_j = \rho(b_j). \quad (898)$$

The surface relation imposes

$$\prod_{j=1}^{g_M} A_j B_j A_j^{-1} B_j^{-1} = I_d. \quad (899)$$

The irreducible rank- d generalized momentum space is

$$\mathcal{B}_d = \text{Hom}_{\text{irr}}(\Gamma_M, U(d)) / U(d). \quad (900)$$

At a smooth irreducible point,

$$\dim_{\mathbb{R}} \mathcal{B}_d = (2g_M - 2)d^2 + 2. \quad (901)$$

Using Eq. (852),

$$\dim_{\mathbb{R}} \mathcal{B}_d = 2q_M(g_0 - 1)d^2 + 2. \quad (902)$$

The determinant defines an Abelian shadow,

$$\det \rho : \Gamma_M \longrightarrow U(1). \quad (903)$$

Subtracting the Abelian base dimension gives the genuinely non-Abelian local dimension

$$D_{\text{NA}} = (2g_M - 2)(d^2 - 1) = 2q_M(g_0 - 1)(d^2 - 1). \quad (904)$$

For an irreducible representation with $d > 1$, the image must contain noncommuting matrices. Otherwise all unitary holonomies could be simultaneously diagonalized and the representation would decompose into one-dimensional characters. A translation-invariant operator cannot couple inequivalent irreducible translation sectors. If

$$T_{\rho\sigma} : V_\sigma \longrightarrow V_\rho \quad (905)$$

satisfies

$$\rho(\gamma) T_{\rho\sigma} = T_{\rho\sigma} \sigma(\gamma) \quad \text{for every } \gamma, \quad (906)$$

then Schur's lemma gives

$$\rho \not\sim \sigma \implies T_{\rho\sigma} = 0. \quad (907)$$

This is the exact non-Abelian replacement of crystal-momentum conservation. Choose a finite-index normal subgroup $\Gamma_N \triangleleft \Gamma_M$ and define

$$Q_N = \Gamma_M / \Gamma_N. \quad (908)$$

Let $\hat{Q}_N$ denote the complete set of inequivalent irreducible unitary representations of Q_N . Their dimensions satisfy

$$\sum_{\rho \in \hat{Q}_N} d_\rho^2 = |Q_N|. \quad (909)$$

The finite group algebra decomposes as

$$\mathbb{C}[Q_N] \simeq \bigoplus_{\rho \in \hat{Q}_N} V_\rho \otimes V_\rho^*. \quad (910)$$

The full finite-cover Hamiltonian decomposes exactly as

$$H_N \simeq \bigoplus_{\rho \in \hat{Q}_N} [I_{d_\rho} \otimes H[\rho]]. \quad (911)$$

The block $H[\rho]$ has dimension $\nu_M d_\rho$, and appears with multiplicity d_ρ . The number of states carried by that isotypic sector is $\nu_M d_\rho^2$. Summing gives

$$\sum_{\rho} \nu_M d_\rho^2 = \nu_M |Q_N| = D_N. \quad (912)$$

The exact isotypic projector is

$$P_\rho = \frac{d_\rho}{|Q_N|} \sum_{q \in Q_N} \chi_\rho(q^{-1}) U_q, \quad (913)$$

where

$$\chi_\rho(q) = \text{Tr } \rho(q). \quad (914)$$

These projectors satisfy

$$P_\rho^2 = P_\rho, \quad P_\rho^\dagger = P_\rho, \quad P_\rho P_\sigma = 0 \quad (\rho \neq \sigma), \quad (915)$$

and

$$\sum_{\rho \in \hat{Q}_N} P_\rho = I. \quad (916)$$

If $E_{n,\rho}$ are the eigenvalues of $H[\rho]$, the complete normalized finite-cover DOS is

$$\nu_N(E) = \frac{1}{\nu_M |Q_N|} \sum_{\rho \in \hat{Q}_N} d_\rho \sum_{n=1}^{\nu_M d_\rho} \delta(E - E_{n,\rho}). \quad (917)$$

The sector weight is proportional to d_ρ^2 , not to one sector per irrep. For a retained set $\mathcal{R}_N \subseteq \hat{Q}_N$, define the finite-level representation-coverage fraction

$$C_{\text{rep},N} = \frac{1}{|Q_N|} \sum_{\rho \in \mathcal{R}_N} d_\rho^2. \quad (918)$$

Exact finite-quotient completeness is $C_{\text{rep},N} = 1$. The total regular-representation weight carried by Abelian sectors is

$$C_{\text{Ab},N} = \frac{|\widehat{Q}_N^{(1)}|}{|Q_N|} = \frac{|Q_N^{\text{ab}}|}{|Q_N|}, \quad (919)$$

where $\widehat{Q}_N^{(1)}$ is the set of one-dimensional irreps. The genuinely non-Abelian weight is

$$C_{\text{NA},N} = 1 - \frac{|Q_N^{\text{ab}}|}{|Q_N|}. \quad (920)$$

Define

$$S_N^{\text{full}} = \text{Spec } H_N \quad (921)$$

and

$$S_N^{\text{Ab}} = \bigcup_{\rho \in \widehat{Q}_N, d_\rho=1} \text{Spec } H[\rho]. \quad (922)$$

The one-sided non-Abelian completion distance is

$$\delta_{\text{NA},N} = \sup_{E \in S_N^{\text{full}}} \text{dist}(E, S_N^{\text{Ab}}). \quad (923)$$

If $\delta_{\text{NA},N} > 0$ the Abelian sectors fail to reproduce the complete finite-cover spectrum. A supercell character $\chi : \Gamma_{\text{sc}} \rightarrow U(1)$ does not necessarily remain one dimensional when viewed from the primitive group. Its primitive-group content is

$$\rho_{\text{ind}} = \text{Ind}_{\Gamma_{\text{sc}}}^{\Gamma_M} \chi. \quad (924)$$

If

$$\rho_{\text{ind}} \simeq \bigoplus_{\rho} m_{\rho}(\chi) \rho, \quad (925)$$

then the supercell calculation accesses the corresponding non-Abelian primitive-cell sectors. Thus hyperbolic supercells are a controlled zone-folding mechanism for higher-dimensional representations. Finite-level completeness must not be confused with completeness of the infinite unitary dual:

$$C_{\text{rep},N} = 1 \not\Rightarrow \text{complete infinite-bulk harmonic analysis.} \quad (926)$$

The former means that every irrep of one finite quotient has been included with its exact multiplicity. Infinite-bulk convergence remains a separate question handled in Sec. IV F. The generalized Bloch reduction, Abelian commutator blindness, non-Abelian holonomy moduli, irrep conservation, induced-representation zone folding and finite-quotient Wedderburn decomposition are supplied by Mathematical Appendix, Theorems 26–31. The finite-level versus infinite-dual scope distinction is supplied by the representation completeness analysis in Mathematical Appendix.

E. Canonical Hodge tangent metric

A generalized band is not naturally a function of a two-dimensional wavevector. It is a function on a smooth stratum of $\mathcal{B}_d$. We therefore require a coordinate-independent tangent space and a physically normalized metric. Let

$$\rho_s : \Gamma_M \rightarrow U(d) \quad (927)$$

be a smooth path with $\rho_0 = \rho$. Define the right logarithmic derivative

$$X(\gamma) = \left. \frac{d\rho_s(\gamma)}{ds} \right|_{s=0} \rho(\gamma)^{-1}. \quad (928)$$

Differentiation of

$$\rho_s(\gamma\delta) = \rho_s(\gamma)\rho_s(\delta) \quad (929)$$

gives

$$X(\gamma\delta) = X(\gamma) + \text{Ad}_{\rho(\gamma)} X(\delta). \quad (930)$$

Thus X is a one-cocycle for the adjoint representation. A pure gauge path

$$\rho_s(\gamma) = e^{sY} \rho(\gamma) e^{-sY} \quad (931)$$

has cocycle

$$X_Y(\gamma) = Y - \text{Ad}_{\rho(\gamma)} Y. \quad (932)$$

The physical tangent space is therefore

$$T_{[\rho]} \mathcal{B}_d \simeq H^1(\Gamma_M, \text{Ad } \rho). \quad (933)$$

Let $E_\rho \rightarrow \Sigma_M$ be the rank- d flat unitary bundle defined by ρ , and let $\text{ad } E_\rho$ be its flat $\mathfrak{u}(d)$ -bundle. The equivalent twisted de Rham tangent space is

$$V_\rho = H^1(\Sigma_M; \text{ad } E_\rho). \quad (934)$$

Twisted Hodge theory assigns each class $X \in V_\rho$ a unique harmonic representative α_X . For $A, B \in \mathfrak{u}(d)$, we fix the normalized fiber pairing

$$\langle A, B \rangle_d = \frac{1}{d} \text{Re Tr}(A^\dagger B). \quad (935)$$

Let

$$A_M = \text{Area}(\Sigma_M). \quad (936)$$

The canonical intensive Hodge metric is

$$G_\rho^{\text{can}}(X, Y) = \frac{1}{A_M} \int_{\Sigma_M} \langle \alpha_X, \alpha_Y \rangle_d d \text{vol}_g. \quad (937)$$

Equivalently,

$$G_\rho^{\text{can}}(X, Y) = \frac{1}{dA_M} \text{Re} \int_{\Sigma_M} \text{Tr}(\alpha_X^\dagger \wedge \star \alpha_Y). \quad (938)$$

The factor d^{-1} prevents the norm from growing trivially with representation rank. The factor A_M^{-1} makes the metric intensive under finite covers. For dimensionless holonomy coordinates, $[G_\rho^{\text{can}}] = L^{-2}$. The representation-space Hamiltonian derivative along X is

$$\dot{H}_X[\rho] = \sum_{\delta \in \Gamma_M} h(\delta) \otimes X(\delta) \rho(\delta). \quad (939)$$

For a simple normalized eigenstate

$$H[\rho]|u_n\rangle = E_n[\rho]|u_n\rangle, \quad (940)$$

Hellmann–Feynman gives

$$dE_n[X] = \langle u_n | \dot{H}_X | u_n \rangle. \quad (941)$$

For a coboundary X_Y ,

$$\dot{H}_{X_Y} = [I_{\nu_M} \otimes Y, H[\rho]], \quad (942)$$

and therefore

$$dE_n[X_Y] = 0. \quad (943)$$

Thus dE_n is a well-defined covector on the physical cohomology space. For any covector $e \in V_\rho^*$, define the dual Hodge norm variationally:

$$\|e\|_{(G_\rho^{\text{can}})^{-1}} = \sup_{X \neq 0} \frac{|e(X)|}{\sqrt{G_\rho^{\text{can}}(X, X)}}. \quad (944)$$

The canonical generalized velocity is

$$\Omega_n^{\text{can}}(\rho) = \frac{1}{\hbar} \|dE_n(\rho)\|_{(G_\rho^{\text{can}})^{-1}}. \quad (945)$$

Its units are

$$[\Omega_n^{\text{can}}] = L/T. \quad (946)$$

It is therefore a velocity-like dispersion scale, not a frequency. The exact stationary condition is

$$dE_n = 0 \quad \text{on } H^1(\Gamma_M, \text{Ad } \rho). \quad (947)$$

This condition is metric independent. Vanishing of one path derivative,

$$\frac{dE_n[\rho(s)]}{ds} = 0, \quad (948)$$

tests only one tangent vector and does not imply Eq. (947). Let $\mathcal{D}$ be a connected declared automorphic domain in one smooth fixed-rank stratum. Its Hodge distance is

$$d_G(\rho_0, \rho_1) = \inf_{\gamma} \int_0^1 \sqrt{G_{\gamma(s)}^{\text{can}}(\dot{\gamma}, \dot{\gamma})} ds, \quad (949)$$

and its diameter is

$$D_G = \text{diam}_G(\mathcal{D}). \quad (950)$$

Define

$$\Omega_C^{\text{max}} = \sup_{\rho \in \mathcal{D}, a \in C} \frac{1}{\hbar} \|dE_a(\rho)\|_{(G_\rho^{\text{can}})^{-1}}. \quad (951)$$

Every continuously tracked branch obeys

$$\text{osc}_{\mathcal{D}} E_a \leq \hbar D_G \Omega_C^{\text{max}}. \quad (952)$$

For a cluster with internal splitting

$$w_*(C) = \max_{a, b \in C} |E_a(\rho_*) - E_b(\rho_*)|, \quad (953)$$

the total energy diameter satisfies

$$\text{diam} \{E_a(\rho) : \rho \in \mathcal{D}, a \in C\} \leq w_*(C) + 2\hbar D_G \Omega_C^{\text{max}}. \quad (954)$$

Derivative suppression does not remove internal multiplet splitting. At a stationary representation, let

$$M_n = \nabla^2 E_n \quad (955)$$

be the covariant spectral Hessian. Its coordinate-independent principal values are defined by the generalized eigenvalue problem

$$M_n v_j = \Lambda_{n,j} G_\rho^{\text{can}} v_j. \quad (956)$$

The quantities $\Lambda_{n,j}$ have units

$$[\Lambda_{n,j}] = EL^2. \quad (957)$$

When an effective-mass interpretation is appropriate, we define

$$(m_{n,j}^*)^{-1} = \frac{\Lambda_{n,j}}{\hbar^2}. \quad (958)$$

Complete quadratic cancellation requires

$$\Lambda_{n,j} = 0 \quad \text{for every physical tangent direction } j. \quad (959)$$

Vanishing of the trace alone is insufficient because positive and negative principal Hessian values can cancel in the trace. Under a constant metric rescaling

$$G' = cG, \quad c > 0, \quad (960)$$

we have

$$\Omega' = c^{-1/2} \Omega, \quad D_{G'} = c^{1/2} D_G, \quad (961)$$

and therefore

$$D_{G'} \Omega' = D_G \Omega. \quad (962)$$

The dimensionless global dispersion diagnostic

$$\mathcal{V}_C^{\text{can}} = \frac{\hbar D_G \Omega_C^{\text{max}}}{E_{\text{ref}}} \quad (963)$$

is invariant under constant normalization changes. Comparisons across different curvatures require a declared transport

$$\mathcal{T}_K : V_{\rho_0} \longrightarrow V_{\rho_K}. \quad (964)$$

The pulled-back covector and metric are

$$\bar{e}_K = \mathcal{T}_K^* dE_K, \quad \bar{G}_K = \mathcal{T}_K^* G_{\rho_K}^{\text{can}}. \quad (965)$$

A running-metric diagnostic is

$$\Omega_{\text{run}}(K) = \frac{1}{\hbar} \|\bar{e}_K\|_{\bar{G}_K}^{-1}, \quad (966)$$

whereas the fixed-metric diagnostic is

$$\Omega_{\text{fr}}(K) = \frac{1}{\hbar} \|\bar{e}_K\|_{\bar{G}_0}^{-1}. \quad (967)$$

The first mixes spectral and metric reconstruction; the second isolates spectral reconstruction relative to the fixed reference geometry. Uniform spectral convergence alone does not imply convergence of dE . A velocity claim requires

$$\sup_{\lambda \in \mathcal{D}} [\|H_N(\lambda) - H(\lambda)\| + \|DH_N(\lambda) - DH(\lambda)\|] \rightarrow 0 \quad (968)$$

on transported fibers, together with a uniform isolation gap and uniform convergence of the canonical metrics. Hessian claims require the corresponding C^2 comparison. The Hodge condition is used later as a mechanism test:

$$\theta_H : \quad dE(\theta_H) = 0 \quad \text{and, when claimed, } \Lambda_j(\theta_H) = 0 \quad \forall j. \quad (969)$$

It does not define the physical magic angle. The latter is determined independently from the full bandwidth, positive isolation gap, DOS, hybridization and finite-cover certificates. The cocycle tangent construction and representation-space differential are supplied by Mathematical Appendix, Theorems 30–31. The normalized fiber pairing, canonical intensive Hodge metric, coordinate covariance, velocity–oscillation inequality, Hessian convention and curvature transport are supplied by Mathematical Appendix, Definitions 8–9, Theorems 74–79 and Propositions 48–50.

F. Finite-cover convergence requirements

Let $\Gamma_N \triangleleft \Gamma_M$ be a finite-index normal subgroup. In the word metric of a fixed finite symmetric generating set, define

$$r_{\text{inj},N} = \frac{1}{2} \min \{|\gamma| : \gamma \in \Gamma_N \setminus \{e\}\}. \quad (970)$$

For a normal residual tower,

$$\Gamma_M \triangleright \Gamma_1 \triangleright \Gamma_2 \triangleright \cdots, \quad \bigcap_N \Gamma_N = \{e\}, \quad (971)$$

we have $r_{\text{inj},N} \rightarrow \infty$. If the truncated Hamiltonian has propagation radius L_0 , then every closed path contributing to the k -th spectral moment has length at most kL_0 . The exact local-lifting condition is

$$2r_{\text{inj},N} > kL_0. \quad (972)$$

Under this condition,

$$\tau_N(H_N^k) = \tau(H^k), \quad (973)$$

where τ and τ_N are the normalized infinite and finite-cover traces. Stabilization of every fixed moment gives weak normalized DOS convergence, but not global spectral convergence. Let $\Sigma = \text{Spec } H$ and $S_N = \text{Spec } H_N$. Define the one-sided loss error

$$\ell_N = \sup_{\lambda \in \Sigma} \text{dist}(\lambda, S_N). \quad (974)$$

A no-loss certificate is $\ell_N \rightarrow 0$. For a finite-propagation operator, every $\lambda \in \Sigma$ has a finitely supported approximate Weyl vector ψ satisfying

$$\|(H - \lambda)\psi\| < \varepsilon. \quad (975)$$

Once the support of ψ and its propagation neighborhood embed isometrically in the cover, its lift ψ_N satisfies

$$\|(H_N - \lambda)\psi_N\| = \|(H - \lambda)\psi\| < \varepsilon. \quad (976)$$

Therefore

$$\text{dist}(\lambda, S_N) < \varepsilon. \quad (977)$$

This proves no loss but gives no information about spurious finite-cover eigenvalues outside Σ . Define the pollution error

$$p_N = \sup_{s \in S_N} \text{dist}(s, \Sigma). \quad (978)$$

A no-pollution certificate is $p_N \rightarrow 0$. The Hausdorff error is

$$\eta_N = d_H(S_N, \Sigma) = \max\{\ell_N, p_N\}. \quad (979)$$

Any one of the following supplies an explicit no-pollution certificate. First, a direct upper inclusion:

$$p_N \leq \varepsilon_N, \quad \varepsilon_N \rightarrow 0. \quad (980)$$

Second, a uniform residual lower bound:

$$\inf_{\|v\|=1} \|(H_N - \lambda)v\| \geq \text{dist}(\lambda, \Sigma) - \varepsilon_N \quad (981)$$

for every real λ in a common spectral interval. Third, on a declared invariant retained sector $\mathcal{K}_N$, polynomial norm domination:

$$\limsup_{N \rightarrow \infty} \|p(H_N|_{\mathcal{K}_N})\| \leq \|p(H)\| \quad (982)$$

for every real polynomial p . Removing only the constant or trivial-representation mode does not prove Eq. (982); other non-tempered quotient sectors may remain. Weak DOS convergence cannot replace no pollution. A spectral outlier can carry normalized multiplicity tending to zero and therefore disappear from the weak DOS while remaining a finite distance from the infinite spectrum. Let

$I_C \subset \Sigma$ be an intrinsic spectral island with bandwidth $W_C = \text{diam } I_C$ and gap

$$\Delta_C = \text{dist}(I_C, \Sigma \setminus I_C) > 0. \quad (983)$$

Assume

$$\eta_N < \frac{\Delta_C}{3}. \quad (984)$$

Select the finite-cover island by the fixed intrinsic neighborhood

$$I_{C,N} = \left\{ s \in S_N : \text{dist}(s, I_C) \leq \frac{\Delta_C}{3} \right\}. \quad (985)$$

Then

$$|W_{C,N} - W_C| \leq 2\eta_N, \quad (986)$$

and

$$|\Delta_{C,N} - \Delta_C| \leq 2\eta_N. \quad (987)$$

In particular,

$$\Delta_{C,N} \geq \Delta_C - 2\eta_N > 0. \quad (988)$$

Both no loss and no pollution are required because the finite cluster and its nearest complementary spectrum must be transported simultaneously. For each finite quotient, representation completeness requires

$$C_{\text{rep},N} = \frac{1}{|Q_N|} \sum_{\rho \in \mathcal{R}_N} d_\rho^2 = 1. \quad (989)$$

The regular spectrum must be reconstructed with

$$H_N \simeq \bigoplus_{\rho \in \widehat{Q}_N} I_{d_\rho} \otimes H[\rho]. \quad (990)$$

A calculation restricted to one character, one rank, or one selected family does not satisfy the finite-level completeness condition. This requirement remains logically separate from $\eta_N \rightarrow 0$. Complete representation coverage reconstructs one finite quotient exactly; it does not prove that the quotient spectrum converges to the reduced infinite bulk. The physical interlayer kernel has infinite range. Let $H^{(L)}$ and $H_N^{(L)}$ be the infinite and finite-cover operators truncated at physical hopping range L . Define

$$\varepsilon_\infty^{(0)}(L) = \|H - H^{(L)}\|, \quad (991)$$

and

$$\varepsilon_N^{(0)}(L) = \|H_N - H_N^{(L)}\|. \quad (992)$$

Let

$$\eta_N^{\text{fr}}(L) = d_H \left(\text{Spec } H_N^{(L)}, \text{Spec } H^{(L)} \right) \quad (993)$$

contain both finite-range no-loss and no-pollution errors. The complete spectral error obeys

$$\eta_N^{\text{exp}}(L) \leq \varepsilon_N^{(0)}(L) + \eta_N^{\text{fr}}(L) + \varepsilon_\infty^{(0)}(L). \quad (994)$$

A valid diagonal sequence satisfies

$$L_N \rightarrow \infty, \quad \frac{L_N}{r_{\text{inj},N}} \rightarrow 0. \quad (995)$$

The first limit removes hopping truncation. The second leaves a parametrically large injectivity buffer around every retained hopping path. Suppose the group-shell count obeys

$$N_{\text{shell}}(r) \leq C_\Gamma e^{r/R} \quad (996)$$

and the hopping kernel satisfies

$$\|h(r)\| \leq C_h e^{-\mu r}. \quad (997)$$

If

$$\gamma = \mu - \frac{1}{R} > 0, \quad (998)$$

the C^0 tail satisfies

$$\varepsilon_\infty^{(0)}(L) \leq C_0 e^{-\gamma L}. \quad (999)$$

Representation derivatives introduce powers of the word or physical length. For derivative order j ,

$$\varepsilon_\infty^{(j)}(L) \leq C_j (1 + L)^j e^{-\gamma L}, \quad j = 1, 2. \quad (1000)$$

Thus C^1 and C^2 convergence require their own weighted tails. An explicit admissible range schedule is

$$L_N = \frac{1}{\gamma} \ln(1 + \gamma r_{\text{inj},N}). \quad (1001)$$

It satisfies

$$L_N \rightarrow \infty, \quad \frac{L_N}{r_{\text{inj},N}} \rightarrow 0, \quad (1002)$$

and

$$e^{-\gamma L_N} = \frac{1}{1 + \gamma r_{\text{inj},N}}. \quad (1003)$$

For a positive intrinsic energy scale $E_{\text{ref},N}$, the normalized C^0 convergence condition is

$$\frac{\varepsilon_N^{(0)}(L_N) + \eta_N^{\text{fr}}(L_N) + \varepsilon_\infty^{(0)}(L_N)}{E_{\text{ref},N}} \rightarrow 0. \quad (1004)$$

Velocity convergence additionally requires

$$\frac{\varepsilon_N^{(1)}(L_N) + \eta_N^{(1)}(L_N) + \varepsilon_\infty^{(1)}(L_N)}{E_{\text{ref},N} L_{\text{ref},N}} \rightarrow 0, \quad (1005)$$

and Hessian or effective-mass convergence requires

$$\frac{\varepsilon_N^{(2)}(L_N) + \eta_N^{(2)}(L_N) + \varepsilon_\infty^{(2)}(L_N)}{E_{\text{ref},N} L_{\text{ref},N}^2} \rightarrow 0. \quad (1006)$$

For each finite calculation, we record

$$\mathfrak{C}_N = \left(\frac{L_N}{r_{\text{inj},N}}, \ell_N, p_N, 1 - C_{\text{rep},N}, \varepsilon_N^{(0)}, \varepsilon_N^{(1)}, \varepsilon_N^{(2)} \right). \quad (1007)$$

A full spectral-bandwidth claim requires

$$\frac{L_N}{r_{\text{inj},N}} \rightarrow 0, \quad \ell_N \rightarrow 0, \quad p_N \rightarrow 0, \quad C_{\text{rep},N} = 1, \quad \varepsilon_N^{(0)} \rightarrow 0. \quad (1008)$$

A group-velocity claim additionally requires $\varepsilon_N^{(1)} \rightarrow 0$, and an effective-mass or Hodge-Hessian claim additionally requires $\varepsilon_N^{(2)} \rightarrow 0$. For one target island, define the total C^0 uncertainty

$$\varepsilon_{\text{tot},N}^{(0)} = \varepsilon_N^{(0)} + \eta_N^{\text{fr}} + \varepsilon_\infty^{(0)} + \varepsilon_{\text{solver},N} + \varepsilon_{\text{parameter},N}. \quad (1009)$$

The finite result may be identified with the intrinsic island only when

$$\varepsilon_{\text{tot},N}^{(0)} < \frac{\Delta_C}{3}. \quad (1010)$$

The established bandwidth and gap errors are then

$$|W_{C,N} - W_C| \leq 2\varepsilon_{\text{tot},N}^{(0)}, \quad (1011)$$

and

$$|\Delta_{C,N} - \Delta_C| \leq 2\varepsilon_{\text{tot},N}^{(0)}. \quad (1012)$$

For two independently constructed admissible towers A and B ,

$$|W_{C,N}^A - W_{C,M}^B| \leq 2\eta_N^A + 2\eta_M^B, \quad (1013)$$

and

$$|\Delta_{C,N}^A - \Delta_{C,M}^B| \leq 2\eta_N^A + 2\eta_M^B. \quad (1014)$$

Cross-tower agreement is therefore a consequence of common convergence to the intrinsic spectrum, not an independent replacement for no pollution. At fixed positive broadening $\eta > 0$, weak normalized DOS convergence implies convergence of

$$\rho_\eta(E) = \int K_\eta(E - s) d\nu(s) \quad (1015)$$

for every bounded continuous kernel K_η . It does not imply pointwise convergence of the unsmoothed DOS, convergence of its maximum, or a justified interchange of

$$N \rightarrow \infty \quad \text{and} \quad \eta \rightarrow 0. \quad (1016)$$

The permitted convergence logic is therefore deliberately sequential and must not be compressed into a stronger

statement than the available certificates justify. The condition $r_{\text{inj},N} \rightarrow \infty$ first implies stabilization of local paths and of all fixed-order moments. This local stabilization is sufficient to establish weak convergence of the normalized density of states together with the absence of spectral loss. Absence of loss alone, however, does not exclude spectral pollution; the latter must be established by an independent no-pollution certificate. Once such a certificate is available, and provided in addition that the representation-completeness condition $C_{\text{rep},N} = 1$ holds and that the mesoscopic scale satisfies $L_N \rightarrow \infty$ with $L_N/r_{\text{inj},N} \rightarrow 0$, the finite covers constitute a established approximation of the declared bulk spectral sector. Several stronger shortcuts are explicitly excluded from this argument. In particular, large system size N by itself does not imply global spectral convergence, and weak convergence of the density of states does not imply convergence of spectral edges or gaps. Likewise, absence of spectral loss does not imply absence of spectral pollution, while removing the trivial irreducible representation is not sufficient to establish tempered completeness. Finally, C^0 convergence of the relevant spectral objects controls only their values and does not, without additional regularity estimates, imply convergence of first- or second-order derivatives such as velocities or Hessians. Fixed-moment stabilization, weak DOS convergence, lifted-Weyl no loss, no-pollution certificates, island transport, admissible towers, cross-tower independence and diagonal hopping-shell convergence are supplied by Mathematical Appendix, Theorems 60–69 and Definitions 5–6. Representation-completeness scope is supplied by the finite-quotient and regular-bulk analysis in the Mathematical Appendix.

V. PHYSICAL OBSERVABLES AND NUMERICAL EVALUATION

We now define the complete set of observables used to distinguish a physical twist-induced flat band from band folding, symmetry multiplicity, a van Hove singularity, localization-induced level crowding, a finite-cover outlier, or an incomplete representation-space calculation. We write the primary control point as

$$\mathbf{p} = (\theta, K, \omega), \quad \omega = \frac{w}{t}, \quad K \leq 0. \quad (1017)$$

The decay length, layer separation, microscopic orbital content, hopping profile, disorder amplitudes, cover tower, and numerical tolerances are held fixed in the primary comparison unless explicitly promoted to additional control variables. For each declared spectral problem, we distinguish four objects:

$$\begin{aligned} &\text{Hamiltonian family : } H(\lambda; \mathbf{p}), \\ &\text{tracked spectral cluster : } P_C(\lambda; \mathbf{p}), \\ &\text{physical observables : } (W_C, \Delta_C, v_{\text{max},C}, \Lambda_{\text{max},C}, \varrho_C, \mathcal{H}_C), \\ &\text{verification data : } (\varepsilon_{\text{spec}}, \varepsilon_{\text{der}}, \varepsilon_{\text{cover}}, \varepsilon_{\text{shell}}, C_{\text{rep}}). \end{aligned} \quad (1018)$$

Here λ denotes the complete coordinate on the declared spectral domain. It may be an ordinary Euclidean momentum, an Abelian higher-genus holonomy, a point of a fixed-rank non-Abelian automorphic domain, or a discrete finite-quotient representation label. We never combine these distinct domains into a fictitious universal Brillouin zone. The complete computational chain is

$$\begin{aligned}
H(\lambda; \mathbf{p}) &\longrightarrow P_C(\lambda; \mathbf{p}) \\
&\longrightarrow (E_C^-, E_C^+, W_C, \Delta_C) \\
&\longrightarrow (dE_C, \nabla^2 E_C, v_{\max, C}, \Lambda_{\max, C}) \\
&\longrightarrow (\varrho_C, C_C^{\text{coh}}, \mathcal{H}_C) \\
&\longrightarrow \mathbf{r}_{\text{magic}, C} \\
&\longrightarrow \mathcal{M}_C^{\text{cert}} \\
&\longrightarrow \Theta_{\text{magic}}(K, \omega).
\end{aligned} \tag{1019}$$

A. Generalized band spectrum

We use four spectral realizations, each with a separate mathematical meaning. For the Euclidean square-bilayer benchmark,

$$\lambda = \mathbf{k}, \quad \mathbf{k} \in \text{mBZ}, \tag{1020}$$

and the band problem is

$$H_E(\mathbf{k}; \mathbf{p})|u_{n\mathbf{k}}\rangle = E_n^E(\mathbf{k}; \mathbf{p})|u_{n\mathbf{k}}\rangle. \tag{1021}$$

For an exactly commensurate hyperbolic bilayer and a finite-dimensional unitary representation

$$\rho : \Gamma_M \longrightarrow U(d_\rho), \tag{1022}$$

the generalized Bloch Hamiltonian is

$$H[\rho; \mathbf{p}] = \sum_{\delta \in \Gamma_M} h(\delta; \mathbf{p}) \otimes \rho(\delta). \tag{1023}$$

Its eigenvalue problem is

$$H[\rho; \mathbf{p}]|u_n[\rho; \mathbf{p}]\rangle = E_n[\rho; \mathbf{p}]|u_n[\rho; \mathbf{p}]\rangle. \tag{1024}$$

The matrix dimension is

$$D_\rho = 2n_b q_M d_\rho. \tag{1025}$$

For an Abelian character,

$$\rho = \chi_{\mathbf{k}}, \quad \mathbf{k} \in \mathbb{T}^{2g_M}, \tag{1026}$$

we obtain

$$H_{\text{Ab}}(\mathbf{k}; \mathbf{p}) = \sum_{\delta \in \Gamma_M} h(\delta; \mathbf{p}) e^{i\mathbf{k}^T n(\delta)}. \tag{1027}$$

The corresponding energy sheets are $E_n^{\text{Ab}}(\mathbf{k}; \mathbf{p})$. For a normal finite cover,

$$Q_N = \Gamma_M / \Gamma_N, \tag{1028}$$

the exact finite Hamiltonian decomposes as

$$H_N(\mathbf{p}) \simeq \bigoplus_{\rho \in \widehat{Q}_N} [I_{d_\rho} \otimes H_N[\rho; \mathbf{p}]]. \tag{1029}$$

Every eigenvalue of $H_N[\rho]$ appears in the full finite spectrum with multiplicity d_ρ . The state-count identity is

$$\sum_{\rho \in \widehat{Q}_N} d_\rho \dim H_N[\rho] = 2n_b q_M |Q_N|. \tag{1030}$$

The complete finite-quotient spectrum is

$$S_N^{\text{full}}(\mathbf{p}) = \bigcup_{\rho \in \widehat{Q}_N} \text{Spec } H_N[\rho; \mathbf{p}], \tag{1031}$$

with every eigenvalue retained with its exact regular-representation multiplicity. The Abelian subset is

$$S_N^{\text{Ab}}(\mathbf{p}) = \bigcup_{\rho \in \widehat{Q}_N, d_\rho=1} \text{Spec } H_N[\rho; \mathbf{p}]. \tag{1032}$$

We define the genuinely non-Abelian spectral remainder by

$$S_N^{\text{NA}} = S_N^{\text{full}} \setminus S_N^{\text{Ab}}, \tag{1033}$$

understood as a multiset difference when multiplicities are required. The directed Abelian-completion error is

$$\delta_{\text{NA}, N} = \sup_{E \in S_N^{\text{full}}} \text{dist}(E, S_N^{\text{Ab}}). \tag{1034}$$

A positive value proves that one-dimensional sectors do not reproduce the complete spectrum of that finite quotient. The reduced infinite-volume spectrum is a separate object:

$$\Sigma_{\text{red}}(\mathbf{p}) = \text{Spec } H_\lambda(\mathbf{p}), \tag{1035}$$

where H_λ is the faithful reduced regular operator. We do not identify Σ_{red} with the unrestricted union of all finite-dimensional automorphic spectra. A finite-cover spectrum is called a bulk approximation only after the no-loss and no-pollution conditions of Sec. V J have been established. Continuous parameter tracking requires all compared Hamiltonians to act on one identified Hilbert space. For a finite synthetic circuit, we therefore fix the resonator and coupler labels and vary only the matrix elements:

$$\mathbf{p} \longmapsto H_N(\mathbf{p}) \in \text{Herm}(D_N). \tag{1036}$$

This is the primary family used for continuous scans in θ , K , and ω . Exact commensurate cells at different values of θ can have different values of $q_M(\theta)$, $g_M(\theta)$, $D_\rho(\theta)$. We do not directly differentiate matrices of different dimensions. Instead, we use one of the following declared constructions:

1. a common finite synthetic device with continuously varying coupling weights;
2. a common larger cover into which both exact cells embed;
3. a unitary transport

$$\mathcal{U}_{\mathbf{p} \leftarrow \mathbf{p}_0} : \mathcal{H}_{\mathbf{p}_0} \longrightarrow \mathcal{H}_{\mathbf{p}} \quad (1037)$$

between equal-dimensional transported fibers;

4. separate exact-cell calculations compared only through spectral-set observables and established finite-cover limits.

No raw derivative is formed across an undeclared change of topology, rank, marking, or Hilbert-space dimension. For every finite representation block, we verify

$$\varepsilon_{\text{Herm},\rho} = \frac{\|H[\rho] - H[\rho]^\dagger\|_{\text{F}}}{\max\{1, \|H[\rho]\|_{\text{F}}\}}. \quad (1038)$$

A valid calculation requires $\varepsilon_{\text{Herm},\rho} \leq \varepsilon_{\text{Herm}}^{\text{tol}}$. For a computed normalized eigenpair $(\tilde{E}_{n,\rho}, |\tilde{u}_{n,\rho}\rangle)$, define the residual

$$r_{n,\rho} = H[\rho]|\tilde{u}_{n,\rho}\rangle - \tilde{E}_{n,\rho}|\tilde{u}_{n,\rho}\rangle. \quad (1039)$$

For a Hermitian matrix,

$$\text{dist}[\tilde{E}_{n,\rho}, \text{Spec } H[\rho]] \leq \|r_{n,\rho}\|. \quad (1040)$$

We define the established eigenvalue uncertainty

$$\varepsilon_{E,n,\rho} = \|r_{n,\rho}\| + \varepsilon_{\text{round},n,\rho} + \varepsilon_{\text{model},n,\rho}. \quad (1041)$$

If

$$\tilde{U}_C = (|\tilde{u}_1\rangle \cdots |\tilde{u}_{r_C}\rangle) \quad (1042)$$

is an orthonormal computed basis for a cluster and

$$\tilde{D}_C = \text{diag}(\tilde{E}_1, \dots, \tilde{E}_{r_C}), \quad (1043)$$

the block residual is

$$R_C = H\tilde{U}_C - \tilde{U}_C\tilde{D}_C. \quad (1044)$$

We record both $\|R_C\|$ and $\|R_C\|_{\text{F}}$. For a complete finite quotient, the representation decomposition is checked by

$$\sum_{\rho \in \hat{Q}_N} d_\rho^2 = |Q_N|. \quad (1045)$$

We also verify the trace identities

$$\text{Tr } H_N^m = \sum_{\rho \in \hat{Q}_N} d_\rho \text{Tr } H_N[\rho]^m, \quad m = 0, 1, \dots, m_{\text{max}}. \quad (1046)$$

The $m = 0$ analysis reproduces the total Hilbert-space dimension. The $m = 1$ and $m = 2$ checks test the spectral centroid and hopping power. Higher moments test non-Abelian closed-walk interference. We define the normalized moment residual

$$\varepsilon_m^{\text{mom}} = \frac{|\text{Tr } H_N^m - \sum_\rho d_\rho \text{Tr } H_N[\rho]^m|}{\max\{1, |\text{Tr } H_N^m|\}}. \quad (1047)$$

All subsequent full-spectrum observables require

$$\max_{0 \leq m \leq m_{\text{max}}} \varepsilon_m^{\text{mom}} \leq \varepsilon_{\text{mom}}^{\text{tol}}. \quad (1048)$$

B. Target-band tracking

A band index obtained by sorting eigenvalues independently at each parameter point is not a physical band label. Eigenvalue ordering changes at crossings and avoided crossings. We therefore track a spectral subspace rather than a sorted integer label. Choose an anchor point $(\lambda_0, \mathbf{p}_0)$ and a target cluster

$$C_0 = \{E_1^0, \dots, E_{r_C}^0\}. \quad (1049)$$

Let $\mathcal{C}_0 \subset \mathbb{C}$ be a positively oriented contour surrounding the target eigenvalues and no complementary spectrum. Define

$$\delta_{\mathcal{C},0} = \inf_{z \in \mathcal{C}_0} \text{dist}[z, \text{Spec } H(\lambda_0; \mathbf{p}_0)] > 0. \quad (1050)$$

The target projector is

$$P_C(\lambda_0; \mathbf{p}_0) = \frac{1}{2\pi i} \oint_{\mathcal{C}_0} [z - H(\lambda_0; \mathbf{p}_0)]^{-1} dz. \quad (1051)$$

Its rank is

$$r_C = \text{Tr } P_C. \quad (1052)$$

The Riesz definition remains valid through arbitrary internal crossings, exact degeneracies, and unitary basis rotations inside the target subspace. It fails only when the contour touches complementary spectrum or the cluster rank changes. Let

$$H_0 = H(\lambda_0; \mathbf{p}_0) \quad (1053)$$

and

$$H_1 = H(\lambda_1; \mathbf{p}_1). \quad (1054)$$

Set

$$\varepsilon_H = \|H_1 - H_0\|. \quad (1055)$$

If $\varepsilon_H < \delta_{\mathcal{C},0}$, then the same contour remains in the resolvent set of H_1 . Indeed, for $z \in \mathcal{C}_0$,

$$z - H_1 = [I - (H_1 - H_0)(z - H_0)^{-1}](z - H_0). \quad (1056)$$

Since

$$\|(H_1 - H_0)(z - H_0)^{-1}\| \leq \frac{\varepsilon_H}{\delta_{\mathcal{C},0}} < 1, \quad (1057)$$

the first factor is invertible by the Neumann lemma. The transported projector is

$$P_{C,1} = \frac{1}{2\pi i} \oint_{\mathcal{C}_0} (z - H_1)^{-1} dz. \quad (1058)$$

Its rank equals r_C . The resolvent identity gives

$$P_{C,1} - P_{C,0} = \frac{1}{2\pi i} \oint_{\mathcal{C}_0} (z - H_1)^{-1} (H_1 - H_0) (z - H_0)^{-1} dz. \quad (1059)$$

Consequently,

$$\|P_{C,1} - P_{C,0}\| \leq \frac{|\mathcal{C}_0|}{2\pi} \frac{\varepsilon_H}{\delta_{\mathcal{C},0} (\delta_{\mathcal{C},0} - \varepsilon_H)}. \quad (1060)$$

Here $|\mathcal{C}_0|$ denotes the contour length. We impose the adaptive step condition

$$\varepsilon_H \leq \zeta_{\text{step}} \delta_{\mathcal{C},0}, \quad 0 < \zeta_{\text{step}} < 1. \quad (1061)$$

If the condition fails, we reduce the parameter or representation-space step before continuing the cluster. For two rank- r_C orthogonal projectors P and Q , define the normalized projector overlap

$$\mathcal{O}(P, Q) = \frac{1}{r_C} \text{Tr}(PQ). \quad (1062)$$

It satisfies $0 \leq \mathcal{O}(P, Q) \leq 1$. If ϑ_j are the principal angles between the two subspaces, then

$$\mathcal{O}(P, Q) = \frac{1}{r_C} \sum_{j=1}^{r_C} \cos^2 \vartheta_j. \quad (1063)$$

The largest principal-angle distance is

$$d_{\text{sub}}(P, Q) = \|P - Q\| = \sin \vartheta_{\text{max}}. \quad (1064)$$

For a computed basis matrix U_0 of the first cluster and U_1 of a candidate neighboring cluster, the singular values of $U_0^\dagger U_1$ are $\cos \vartheta_1, \dots, \cos \vartheta_{r_C}$. This supplies a basis-independent numerical implementation. At a neighboring point with several candidate clusters $C_1, \dots, C_M$, we choose

$$C_{\text{next}} \in \arg \max_j \mathcal{O}(P_{C,\text{current}}, P_{C_j}), \quad (1065)$$

subject to fixed-rank, contour-gap, symmetry, and representation-family constraints. We do not use energy proximity alone. A robust matching score may be written as

$$\mathcal{S}_j = \alpha_P \mathcal{O}(P_{\text{current}}, P_{C_j})$$

$$\begin{aligned} & - \alpha_E \frac{|\overline{E}_{C_j} - \overline{E}_{\text{pred}}|}{E_{\text{kin}}} \\ & - \alpha_L |\overline{P}_{L,C_j} - \overline{P}_{L,\text{pred}}| \\ & - \alpha_{\text{sym}} d_{\text{sym}}(C_j, C_{\text{current}}), \end{aligned} \quad (1066)$$

where all coefficients are fixed before inspecting the target result. The projector-overlap term remains the primary label. On a fixed-rank isolated chart,

$$P_C(\mathbf{p}) = \frac{1}{2\pi i} \oint_{\mathcal{C}} R(z; \mathbf{p}) dz, \quad R(z; \mathbf{p}) = [z - H(\mathbf{p})]^{-1}. \quad (1067)$$

For a control coordinate p_μ ,

$$\partial_\mu R = R(\partial_\mu H)R. \quad (1068)$$

Hence

$$\partial_\mu P_C = \frac{1}{2\pi i} \oint_{\mathcal{C}} R(\partial_\mu H)R dz. \quad (1069)$$

The norm is bounded by

$$\|\partial_\mu P_C\| \leq \frac{|\mathcal{C}|}{2\pi} \frac{\|\partial_\mu H\|}{\delta_{\mathcal{C}}^2}. \quad (1070)$$

A large projector derivative signals rapid wave-function reconstruction even when the eigenvalue dispersion is small. The second derivative is

$$\begin{aligned} \partial_\mu \partial_\nu P_C &= \frac{1}{2\pi i} \oint_{\mathcal{C}} \left[R(\partial_\mu \partial_\nu H)R \right. \\ &\quad + R(\partial_\mu H)R(\partial_\nu H)R \\ &\quad \left. + R(\partial_\nu H)R(\partial_\mu H)R \right] dz. \end{aligned} \quad (1071)$$

Suppose a closed parameter-space loop is discretized as

$$\mathbf{p}_0 \rightarrow \mathbf{p}_1 \rightarrow \dots \rightarrow \mathbf{p}_M = \mathbf{p}_0. \quad (1072)$$

After transporting the cluster around the loop, define

$$\varepsilon_{\text{loop}} = \|P_C^{\text{return}} - P_C^{\text{initial}}\|. \quad (1073)$$

A reliable label assignment requires $\varepsilon_{\text{loop}} \leq \varepsilon_{\text{loop}}^{\text{tol}}$. A large defect identifies an unresolved branch exchange, contour failure, rank change, or topological monodromy that must be reported rather than silently relabeled. We terminate smooth cluster continuation when any of the following occurs:

$$\Delta_C = 0, \quad \text{rank } P_C \text{ changes}, \quad \delta_{\mathcal{C}} \leq \delta_{\mathcal{C}}^{\text{tol}},$$

$$\mathcal{O}(P_{\text{old}}, P_{\text{new}}) < \mathcal{O}_{\text{min}}, \quad \varepsilon_{\text{loop}} > \varepsilon_{\text{loop}}^{\text{tol}},$$

the declared symmetry or representation label changes. (1074)

A branch termination is a physical or mathematical event. It is not repaired by choosing whichever neighboring eigenvalue appears visually smoothest.

C. Global bandwidth

Let $\mathcal{D}$ be one declared spectral domain and let $P_C(\lambda; \mathbf{p})$ be a continuously tracked isolated cluster of fixed rank r_C . Define the local lower and upper cluster energies by

$$e_C^-(\lambda; \mathbf{p}) = \lambda_{\min} \left[P_C H P_C |_{\text{Ran } P_C} \right], \quad (1075)$$

and

$$e_C^+(\lambda; \mathbf{p}) = \lambda_{\max} \left[P_C H P_C |_{\text{Ran } P_C} \right]. \quad (1076)$$

The global edges are

$$E_C^-(\mathbf{p}) = \inf_{\lambda \in \mathcal{D}} e_C^-(\lambda; \mathbf{p}), \quad (1077)$$

and

$$E_C^+(\mathbf{p}) = \sup_{\lambda \in \mathcal{D}} e_C^+(\lambda; \mathbf{p}). \quad (1078)$$

The global cluster bandwidth is

$$W_C(\mathbf{p}) = E_C^+(\mathbf{p}) - E_C^-(\mathbf{p}). \quad (1079)$$

For a single simple band,

$$P_C = |u_n\rangle\langle u_n|, \quad (1080)$$

and

$$W_n = \sup_{\lambda \in \mathcal{D}} E_n(\lambda) - \inf_{\lambda \in \mathcal{D}} E_n(\lambda). \quad (1081)$$

For a multi-band cluster, define the maximum local internal splitting

$$w_C^{\text{int}} = \sup_{\lambda \in \mathcal{D}} [e_C^+(\lambda) - e_C^-(\lambda)]. \quad (1082)$$

The total width satisfies $W_C \geq w_C^{\text{int}}$. A vanishing derivative of a common cluster center does not eliminate w_C^{int} . Let

$$\mathcal{G}_h = \{\lambda_1, \dots, \lambda_{N_h}\} \subset \mathcal{D} \quad (1083)$$

be a grid with metric covering radius

$$h = \sup_{\lambda \in \mathcal{D}} \min_{\lambda_j \in \mathcal{G}_h} d_G(\lambda, \lambda_j). \quad (1084)$$

Suppose the two edge functions satisfy the common Lipschitz bound

$$|e_C^\pm(\lambda) - e_C^\pm(\lambda')| \leq L_C d_G(\lambda, \lambda'). \quad (1085)$$

Then

$$\max_j e_C^+(\lambda_j) \leq E_C^+ \leq \max_j e_C^+(\lambda_j) + L_C h, \quad (1086)$$

and

$$\min_j e_C^-(\lambda_j) - L_C h \leq E_C^- \leq \min_j e_C^-(\lambda_j). \quad (1087)$$

Define the sampled width

$$\widetilde{W}_{C,h} = \max_j e_C^+(\lambda_j) - \min_j e_C^-(\lambda_j). \quad (1088)$$

Then

$$\widetilde{W}_{C,h} \leq W_C \leq \widetilde{W}_{C,h} + 2L_C h. \quad (1089)$$

If each sampled eigenvalue has uncertainty at most ε_E , the conservative upper bandwidth becomes

$$W_C^U = \widetilde{W}_{C,h} + 2L_C h + 2\varepsilon_E. \quad (1090)$$

For a finite quotient, the representation set is discrete and complete. There is no continuous-domain grid error once every irrep block has been included. The remaining uncertainty is

$$\varepsilon_{W,N} = 2\varepsilon_{E,N} + \varepsilon_{\text{shell},N} + \varepsilon_{\text{cover},N}. \quad (1091)$$

We fix the kinetic reference energy from the untwisted Euclidean monolayer: $E_{\text{kin}} = W_0$. For the nearest-neighbor square lattice, $E_{\text{kin}} = 8|t|$. The normalized bandwidth is

$$\overline{W}_C = \frac{W_C}{E_{\text{kin}}}. \quad (1092)$$

The zero-hybridization reference is

$$W_C^{(0)}(\theta, K) = W_C(\theta, K, \omega = 0). \quad (1093)$$

The hybridization compression ratio is

$$R_{W,C}^{\text{hyb}} = \frac{W_C(\theta, K, \omega)}{W_C^{(0)}(\theta, K)}. \quad (1094)$$

A necessary condition for interlayer-induced narrowing is $R_{W,C}^{\text{hyb}} < 1$. This condition alone is not sufficient because a narrow band can be unisolated, localized, or incomplete in representation space.

D. Global isolation gap

A local direct separation at each generalized momentum does not imply a global energy gap. We therefore distinguish local direct isolation from global indirect isolation. Let

$$Q_C(\lambda) = I - P_C(\lambda). \quad (1095)$$

The pointwise direct cluster gap is

$$\delta_C^{\text{dir}}(\lambda) = \text{dist} \left[\text{Spec} \left(P_C H P_C |_{\text{Ran } P_C} \right), \text{Spec} \left(Q_C H Q_C |_{\text{Ran } Q_C} \right) \right]. \quad (1096)$$

Its global minimum is

$$\delta_{C,\min}^{\text{dir}} = \inf_{\lambda \in \mathcal{D}} \delta_C^{\text{dir}}(\lambda). \quad (1097)$$

Define the complementary lower and upper edges by

$$E_{\text{below},C} = \sup \{E : E \in \text{Spec}(Q_C H Q_C), E < E_C^-\}, \quad (1098)$$

and

$$E_{\text{above},C} = \inf \{E : E \in \text{Spec}(Q_C H Q_C), E > E_C^+\}. \quad (1099)$$

The lower and upper indirect gaps are

$$\Delta_C^- = E_C^- - E_{\text{below},C}, \quad (1100)$$

and

$$\Delta_C^+ = E_{\text{above},C} - E_C^+. \quad (1101)$$

The signed global isolation gap is

$$\Delta_C = \min \{\Delta_C^-, \Delta_C^+\}. \quad (1102)$$

The cluster is globally isolated if and only if $\Delta_C > 0$. It is possible to have $\delta_{C,\min}^{\text{dir}} > 0$ but $\Delta_C \leq 0$. This occurs when the target and complementary bands are separated at every common λ but overlap after their global energy ranges are compared. Let the target and complementary edge functions have Lipschitz constants $L_C^\pm$, $L_{\text{comp}}^\pm$. For a grid of covering radius h , define the sampled lower and upper gaps

$$\tilde{\Delta}_{C,h}^- = \min_j [e_C^-(\lambda_j) - e_{\text{below},C}(\lambda_j)], \quad (1103)$$

and

$$\tilde{\Delta}_{C,h}^+ = \min_j [e_{\text{above},C}(\lambda_j) - e_C^+(\lambda_j)]. \quad (1104)$$

Then

$$\Delta_C^- \geq \tilde{\Delta}_{C,h}^- - (L_C^- + L_{\text{comp}}^-) h, \quad (1105)$$

and

$$\Delta_C^+ \geq \tilde{\Delta}_{C,h}^+ - (L_C^+ + L_{\text{comp}}^+) h. \quad (1106)$$

If all eigenvalues have uncertainty at most ε_E , the conservative global lower gap is

$$\Delta_C^L = \min \left\{ \tilde{\Delta}_{C,h}^- - (L_C^- + L_{\text{comp}}^-) h, \right. \\ \left. \tilde{\Delta}_{C,h}^+ - (L_C^+ + L_{\text{comp}}^+) h \right\} - 2\varepsilon_E. \quad (1107)$$

Spectral isolation is numerically established only when $\Delta_C^L > 0$. Let

$$\tilde{H} = H + \delta H, \quad \|\delta H\| \leq \varepsilon_H. \quad (1108)$$

Weyl stability gives

$$d_H(\text{Spec } \tilde{H}, \text{Spec } H) \leq \varepsilon_H. \quad (1109)$$

For the same transported island,

$$|\tilde{W}_C - W_C| \leq 2\varepsilon_H, \quad (1110)$$

and

$$|\tilde{\Delta}_C - \Delta_C| \leq 2\varepsilon_H. \quad (1111)$$

Hence a sufficient robust-isolation condition is

$$\Delta_C > 2\varepsilon_H. \quad (1112)$$

For a finite cover approximating an intrinsic bulk island, the complete error $\varepsilon_{\text{tot},N}^{(0)}$ must include hopping truncation, angular approximation, finite-cover transport, no-pollution, parameter interpolation, and eigensolver uncertainty. The island is accepted only when

$$\varepsilon_{\text{tot},N}^{(0)} < \frac{\Delta_C}{3}. \quad (1113)$$

E. Generalized group velocity

For an ordinary Euclidean band,

$$E_n^E = E_n^E(\mathbf{k}; \mathbf{p}), \quad (1114)$$

the physical group velocity is

$$\mathbf{v}_n^E(\mathbf{k}; \mathbf{p}) = \frac{1}{\hbar} \nabla_{\mathbf{k}} E_n^E(\mathbf{k}; \mathbf{p}). \quad (1115)$$

For a simple band,

$$\partial_{k_\mu} E_n^E = \langle u_n | \partial_{k_\mu} H_E | u_n \rangle. \quad (1116)$$

For the Abelian Hamiltonian

$$H_{\text{Ab}}(\mathbf{k}) = \sum_{\delta} h(\delta) e^{i\mathbf{k}^\top n(\delta)}, \quad (1117)$$

the exact derivative is

$$\partial_{k_\mu} H_{\text{Ab}}(\mathbf{k}) = i \sum_{\delta} n_\mu(\delta) h(\delta) e^{i\mathbf{k}^\top n(\delta)}. \quad (1118)$$

The second derivative is

$$\partial_{k_\mu} \partial_{k_\nu} H_{\text{Ab}}(\mathbf{k}) = - \sum_{\delta} n_\mu(\delta) n_\nu(\delta) h(\delta) e^{i\mathbf{k}^\top n(\delta)}. \quad (1119)$$

Let

$$X \in H^1(\Gamma_M, \text{Ad } \rho) \quad (1120)$$

be represented by a cocycle $X(\delta)$. The representation-space Hamiltonian derivative is

$$D_X H[\rho] = \sum_{\delta} h(\delta) \otimes X(\delta) \rho(\delta). \quad (1121)$$

For a simple band,

$$dE_n[X] = \langle u_n | D_X H[\rho] | u_n \rangle. \quad (1122)$$

Let G_ρ^{can} be the canonical intensive Hodge metric defined in Sec. IV E. The local generalized velocity of a simple branch is

$$v_n^{\text{gen}}(\rho; \mathbf{p}) = \frac{1}{\hbar} \|dE_n(\rho; \mathbf{p})\|_{(G_\rho^{\text{can}})^{-1}}. \quad (1123)$$

At a degenerate or internally crossing cluster, individual derivatives depend on the basis used inside the cluster. Let P_C be the isolated cluster projector. For a tangent X , define the compressed first-order slope operator

$$\mathcal{V}_C[X] = P_C D_X H P_C|_{\text{Ran } P_C}. \quad (1124)$$

The first-order eigenvalue slopes generated by the perturbation X are the eigenvalues of $\mathcal{V}_C[X]$. We therefore define

$$v_C^{\text{gen}}(\rho) = \frac{1}{\hbar} \sup_{X \neq 0} \frac{\|\mathcal{V}_C[X]\|}{\sqrt{G_\rho^{\text{can}}(X, X)}}. \quad (1125)$$

For a rank-one projector, this reduces exactly to Eq. (1123). Choose a G_ρ^{can} -orthonormal tangent basis

$$X_1, \dots, X_{r_\rho}, \quad G_\rho^{\text{can}}(X_a, X_b) = \delta_{ab}. \quad (1126)$$

Set

$$A_a = P_C D_{X_a} H P_C. \quad (1127)$$

Then

$$v_C^{\text{gen}} = \frac{1}{\hbar} \sup_{\|\mathbf{x}\|_2=1} \left\| \sum_{a=1}^{r_\rho} x_a A_a \right\|. \quad (1128)$$

A computable upper bound is

$$v_C^{\text{gen}} \leq \frac{1}{\hbar} \left[\sum_{a=1}^{r_\rho} \|A_a\|^2 \right]^{1/2}. \quad (1129)$$

A lower bound is obtained from any finite set of unit directions $\{\mathbf{x}_j\}$:

$$v_C^{\text{gen}} \geq \frac{1}{\hbar} \max_j \left\| \sum_a x_{j,a} A_a \right\|. \quad (1130)$$

Adaptive angular refinement continues until the upper-lower bracket is below the declared tolerance. The global maximum is

$$v_{\text{max},C}(\mathbf{p}) = \sup_{\rho \in \mathcal{D}} v_C^{\text{gen}}(\rho; \mathbf{p}). \quad (1131)$$

For a finite quotient, the supremum is the maximum over all included irreps. For the Abelian sector, it is the supremum over the declared torus. For a non-Abelian continuous test domain, it is the supremum over that explicitly declared fixed-rank domain. Let

$$D_G = \text{diam}_G(\mathcal{D}) \quad (1132)$$

be the metric diameter of a connected spectral domain. For a simple branch,

$$\begin{aligned} |E_n(\rho_1) - E_n(\rho_0)| &\leq \int_\gamma |dE_n(\dot{\gamma})| ds \\ &\leq \hbar v_{\text{max},n} \text{Length}_G(\gamma). \end{aligned} \quad (1133)$$

Taking the infimum over connecting paths and the supremum over endpoints gives

$$W_n \leq \hbar D_G v_{\text{max},n}. \quad (1134)$$

For a cluster,

$$W_C \leq w_C^{\text{int}}(\rho_*) + 2\hbar D_G v_{\text{max},C}, \quad (1135)$$

where $w_C^{\text{int}}(\rho_*)$ is the internal splitting at a reference point. We define the first-order dispersion energy

$$E_{\text{disp},C}^{(1)} = \hbar D_G v_{\text{max},C}. \quad (1136)$$

Let $\gamma(s)$ be a unit-speed Hodge geodesic with

$$\gamma(0) = \rho, \quad \dot{\gamma}(0) = X, \quad G(X, X) = 1. \quad (1137)$$

The centered finite difference is

$$D_h E = \frac{E[\gamma(h)] - E[\gamma(-h)]}{2h}. \quad (1138)$$

If

$$\left| \frac{d^3}{ds^3} E[\gamma(s)] \right| \leq M_3 \quad (1139)$$

for $|s| \leq h$, then

$$|D_h E - dE[X]| \leq \frac{M_3 h^2}{6}. \quad (1140)$$

If each energy has absolute uncertainty ε_E , the total derivative error is bounded by

$$\varepsilon_{dE}(h) \leq \frac{2\varepsilon_E}{h} + \frac{M_3 h^2}{6}. \quad (1141)$$

The error-bound minimizing step is

$$h_{\text{opt}}^{(1)} = \left(\frac{6\varepsilon_E}{M_3} \right)^{1/3}. \quad (1142)$$

The analytic Hellmann–Feynman derivative and the geodesic finite difference are computed independently whenever both are available.

F. Effective mass and generalized Hessian

At a stationary representation, $dE_n = 0$, the leading dispersion is determined by the covariant Hessian

$$M_n = \nabla^2 E_n. \quad (1143)$$

For two tangent directions X and Y ,

$$M_n(X, Y) = \langle u_n | D_X D_Y H | u_n \rangle + 2 \operatorname{Re} \sum_{m \neq n} \frac{\langle u_n | D_X H | u_m \rangle \langle u_m | D_Y H | u_n \rangle}{E_n - E_m}. \quad (1144)$$

The first term is the direct second derivative of the Hamiltonian. The second term contains virtual mixing with every complementary state. Using the reduced resolvent

$$S_n = Q_n(E_n - H)^{-1}Q_n, \quad Q_n = I - P_n, \quad (1145)$$

we may write

$$M_n(X, Y) = \langle u_n | D_X D_Y H | u_n \rangle + \langle u_n | D_X H S_n D_Y H + D_Y H S_n D_X H | u_n \rangle. \quad (1146)$$

Suppose an isolated rank- r_C cluster is exactly degenerate at a reference representation:

$$P_C H P_C = E_{C,*} P_C. \quad (1147)$$

Set

$$Q_C = I - P_C \quad (1148)$$

and

$$S_C = Q_C(E_{C,*} - H)^{-1}Q_C. \quad (1149)$$

The basis-independent second-order effective operator is

$$\begin{aligned} \mathcal{M}_C(X, Y) &= P_C D_X D_Y H P_C \\ &+ P_C D_X H S_C D_Y H P_C \\ &+ P_C D_Y H S_C D_X H P_C. \end{aligned} \quad (1150)$$

The second-order band splittings are obtained from the eigenvalues of $\mathcal{M}_C(X, X)$. No arbitrary eigenvector basis inside the degenerate subspace is used. Choose a tangent basis and let

$$M_{ab} = M(X_a, X_b), \quad G_{ab} = G_\rho^{\text{can}}(X_a, X_b). \quad (1151)$$

The coordinate-invariant principal Hessian values solve

$$M \mathbf{v}_j = \Lambda_j G \mathbf{v}_j. \quad (1152)$$

Equivalently, they are the eigenvalues of $G^{-1/2} M G^{-1/2}$. For a simple branch, define the principal effective masses by

$$m_j^* = \frac{\hbar^2}{\Lambda_j}, \quad (1153)$$

whenever $\Lambda_j \neq 0$. A positive Λ_j corresponds to a locally electron-like minimum, whereas a negative Λ_j corresponds to a locally hole-like maximum. For a cluster, define

$$\Lambda_{\max, C} = \sup_{\rho \in \mathcal{D}, a \in C, j} |\Lambda_{a,j}(\rho)|. \quad (1154)$$

The maximal second-order energy variation is

$$E_{\text{disp}, C}^{(2)} = \frac{1}{2} D_G^2 \Lambda_{\max, C}. \quad (1155)$$

Let

$$r_\rho = \dim T_{[\rho]} \mathcal{D}. \quad (1156)$$

Define the metric trace

$$\bar{\Lambda} = \frac{1}{r_\rho} \operatorname{Tr} (G^{-1} M). \quad (1157)$$

The traceless Hessian is

$$M^{\text{tl}} = M - \bar{\Lambda} G. \quad (1158)$$

Its normalized anisotropy is

$$\mathcal{A}_M = \left\| G^{-1/2} M^{\text{tl}} G^{-1/2} \right\|_{\text{F}}. \quad (1159)$$

A vanishing trace, $\bar{\Lambda} = 0$, does not imply complete cancellation unless $\mathcal{A}_M = 0$. The complete quadratic cancellation condition is

$$G^{-1/2} M G^{-1/2} = 0, \quad (1160)$$

equivalently

$$\Lambda_j = 0 \quad \text{for every } j. \quad (1161)$$

For a C_4 - or C_8 -isotropic symmetry sector in which the tangent representation is irreducible over the relevant real subspace, symmetry can force $M = \bar{\Lambda} G$. Only under this independently verified symmetry condition does $\bar{\Lambda} = 0$ imply full quadratic cancellation. Along a unit-speed Hodge geodesic $\gamma(s)$, define

$$D_h^{(2)} E = \frac{E[\gamma(h)] - 2E[\gamma(0)] + E[\gamma(-h)]}{h^2}. \quad (1162)$$

If

$$\left| \frac{d^4}{ds^4} E[\gamma(s)] \right| \leq M_4 \quad (1163)$$

for $|s| \leq h$, then

$$\left| D_h^{(2)} E - \nabla^2 E(X, X) \right| \leq \frac{M_4 h^2}{12}. \quad (1164)$$

With eigenvalue uncertainty ε_E , the total error is

$$\varepsilon_M(h) \leq \frac{4\varepsilon_E}{h^2} + \frac{M_4 h^2}{12}. \quad (1165)$$

The error-bound minimizing step is

$$h_{\text{opt}}^{(2)} = \left(\frac{48\varepsilon_E}{M_4} \right)^{1/4}. \quad (1166)$$

Mixed Hessian components are evaluated by

$$D_h^{XY} E = \frac{1}{4h^2} \left[E(hX + hY) - E(hX - hY) - E(-hX + hY) + E(-hX - hY) \right]. \quad (1167)$$

The analytic reduced-resolvent Hessian and the finite-difference Hessian are compared before a second-order claim is accepted. To compare Hessians at different curvatures, we pull them to one reference tangent space using the declared transport $\mathcal{T}_K$. The pulled-back Hessian and metric are

$$\overline{M}_K = \mathcal{T}_K^* M_K, \quad \overline{G}_K = \mathcal{T}_K^* G_K. \quad (1168)$$

We report both the running-metric spectrum

$$\text{Spec} \left(\overline{G}_K^{-1/2} \overline{M}_K \overline{G}_K^{-1/2} \right) \quad (1169)$$

and the fixed-metric spectrum

$$\text{Spec} \left(\overline{G}_0^{-1/2} \overline{M}_K \overline{G}_0^{-1/2} \right). \quad (1170)$$

The fixed quantity isolates spectral reconstruction relative to the reference metric.

G. Density of states

The phrase density of states refers to different measures in different spectral realizations. We therefore specify the measure before computing any DOS. For a Euclidean target cluster,

$$\varrho_C^{\text{E}}(E) = \frac{1}{r_C A_{\text{mBZ}}} \sum_{a \in C} \int_{\text{mBZ}} d^2 k \delta[E - E_a(\mathbf{k})]. \quad (1171)$$

It is normalized:

$$\int_{\mathbb{R}} \varrho_C^{\text{E}}(E) dE = 1. \quad (1172)$$

For the Abelian hyperbolic torus

$$\mathcal{B}_1 \simeq \mathbb{T}^{2g_M}, \quad (1173)$$

let

$$d\mu_H(\mathbf{k}) = \frac{d^{2g_M} k}{(2\pi)^{2g_M}} \quad (1174)$$

be normalized Haar measure. The Abelian target DOS is

$$\varrho_C^{\text{Ab}}(E) = \frac{1}{r_C} \sum_{a \in C} \int_{\mathcal{B}_1} d\mu_H(\mathbf{k}) \delta[E - E_a^{\text{Ab}}(\mathbf{k})]. \quad (1175)$$

For a higher-rank automorphic test domain $\mathcal{D}_d \subset \mathcal{B}_d$, a density obtained using a chosen Hodge-volume measure is

denoted $\varrho_C^{\text{aut},d}$. It is a density over that declared test family. We do not identify it with the reduced-bulk DOS unless an independent representation-theoretic measure and completeness theorem supplies that identification. Let C_ρ be the target eigenvalue set inside the block $H_N[\rho]$, containing $r_C(\rho)$ block eigenvalues. The total number of physical finite-cover states in the cluster is

$$N_{C,N} = \sum_{\rho \in \hat{Q}_N} d_\rho r_C(\rho). \quad (1176)$$

The complete target DOS is

$$\varrho_{C,N}^{\text{full}}(E) = \frac{1}{N_{C,N}} \sum_{\rho \in \hat{Q}_N} d_\rho \sum_{a \in C_\rho} \delta[E - E_{a,\rho}]. \quad (1177)$$

The factor d_ρ is the regular-representation multiplicity. A DOS constructed by assigning equal weight to each inequivalent irrep is incorrect unless all irreps are one dimensional. The Abelian finite-quotient target DOS is

$$\varrho_{C,N}^{\text{Ab}}(E) = \frac{1}{N_{C,N}^{\text{Ab}}} \sum_{\rho \in \hat{Q}_N, d_\rho=1} \sum_{a \in C_\rho} \delta[E - E_{a,\rho}]. \quad (1178)$$

Let τ be the canonical normalized regular trace and let

$$P_C = \mathbf{1}_{I_C}(H) \quad (1179)$$

be the intrinsic functional-calculus projector. The normalized reduced-bulk cluster measure ν_C is defined by

$$\int_{\mathbb{R}} f(E) d\nu_C(E) = \frac{\tau[P_C f(H)]}{\tau(P_C)} \quad (1180)$$

for every bounded continuous f . If this measure has a density with respect to Lebesgue measure, we denote it by $\varrho_C^{\text{red}}(E)$. Existence of a density is not assumed merely from existence of the spectral measure. For numerical data, define the Lorentzian kernel

$$K_\eta(E) = \frac{1}{\pi} \frac{\eta}{E^2 + \eta^2}, \quad \eta > 0. \quad (1181)$$

The broadened DOS is

$$\varrho_{C,\eta}(E) = \int_{\mathbb{R}} K_\eta(E - E') d\nu_C(E'). \quad (1182)$$

For a finite cluster,

$$\varrho_{C,N,\eta}(E) = \frac{1}{N_{C,N}} \sum_{j \in C} K_\eta(E - E_j). \quad (1183)$$

Normalization is preserved:

$$\int_{\mathbb{R}} \varrho_{C,\eta}(E) dE = 1. \quad (1184)$$

For a resolved finite-width cluster, we require

$$\varepsilon_E \ll \eta \ll W_C. \quad (1185)$$

For an exactly flat cluster with $W_C = 0$, this hierarchy is impossible. We first establish exact or thermodynamic flatness and then take $N \rightarrow \infty$ before $\eta \rightarrow 0^+$. For a normalized exact flat band at E_* ,

$$\varrho_{C,\eta}(E) = K_\eta(E - E_*), \quad (1186)$$

and

$$\varrho_{C,\eta}^{\max} = \frac{1}{\pi\eta}. \quad (1187)$$

For every normalized density supported on an interval of width W_C ,

$$1 = \int_{I_C} \varrho_C(E) dE \leq W_C \|\varrho_C\|_\infty. \quad (1188)$$

Therefore

$$W_C \varrho_C^{\max} \geq 1. \quad (1189)$$

Bandwidth suppression forces DOS enhancement, but the converse fails: a saddle-point singularity can produce a large DOS while the global bandwidth remains finite. Let P_ℓ , $\ell = 1, 2$, be the layer projectors. The layer-resolved cluster measure is defined by

$$\int f(E) d\nu_{C,\ell}(E) = \frac{\text{Tr}[P_\ell P_C f(H)]}{\text{Tr} P_C}. \quad (1190)$$

Its broadened density is

$$\varrho_{C,\ell,\eta}(E) = \frac{1}{N_C} \sum_{n \in C} \langle u_n | P_\ell | u_n \rangle K_\eta(E - E_n). \quad (1191)$$

The two layers satisfy

$$\varrho_{C,1,\eta}(E) + \varrho_{C,2,\eta}(E) = \varrho_{C,\eta}(E). \quad (1192)$$

For a real-space orbital $|i\rangle$, the local DOS is

$$\varrho_{i,\eta}(E) = \sum_n |\langle i | u_n \rangle|^2 K_\eta(E - E_n). \quad (1193)$$

The layer- and site-resolved microwave response can therefore be compared directly with the calculated LDOS. For a large sparse Hamiltonian with spectral bounds $E_{\min} < E_{\max}$, define

$$a_H = \frac{E_{\max} - E_{\min}}{2 - \varepsilon_{\text{edge}}}, \quad b_H = \frac{E_{\max} + E_{\min}}{2}, \quad (1194)$$

where $\varepsilon_{\text{edge}} > 0$ leaves a numerical margin inside $[-1, 1]$. The rescaled Hamiltonian is

$$\tilde{H} = \frac{H - b_H I}{a_H}. \quad (1195)$$

Its spectrum lies strictly inside $[-1, 1]$. The Chebyshev moments of the target measure are

$$\mu_m^C = \frac{\text{Tr}[P_C T_m(\tilde{H})]}{\text{Tr} P_C}, \quad (1196)$$

where

$$T_0(x) = 1, \quad T_1(x) = x, \quad (1197)$$

and

$$T_{m+1}(x) = 2xT_m(x) - T_{m-1}(x). \quad (1198)$$

For expansion order M , the Jackson coefficients are

$$g_m^{(M)} = \frac{1}{M+2} \left[(M-m+1) \cos\left(\frac{\pi m}{M+2}\right) + \sin\left(\frac{\pi m}{M+2}\right) \cot\left(\frac{\pi}{M+2}\right) \right]. \quad (1199)$$

With

$$x = \frac{E - b_H}{a_H}, \quad (1200)$$

the Jackson-KPM approximation is

$$\varrho_{C,M}^{\text{KPM}}(E) = \frac{1}{\pi a_H \sqrt{1-x^2}} \left[g_0^{(M)} \mu_0^C + 2 \sum_{m=1}^M g_m^{(M)} \mu_m^C T_m(x) \right].$$

For stochastic trace estimation, choose random vectors

$$|\xi_r\rangle, \quad r = 1, \dots, N_{\text{rv}}, \quad (1201)$$

satisfying

$$\mathbb{E}[|\xi_r\rangle\langle\xi_r|] = I. \quad (1202)$$

Then

$$\hat{\mu}_m^C = \frac{\sum_r \langle \xi_r | P_C T_m(\tilde{H}) | \xi_r \rangle}{\sum_r \langle \xi_r | P_C | \xi_r \rangle} \quad (1203)$$

is a consistent estimator. We estimate stochastic uncertainty from independent random-vector batches rather than from a single realization. For an independent DOS analysis, start from a normalized random vector $|v_1\rangle$ and construct an m_L -step Lanczos tridiagonal matrix T_{m_L} . For a sufficiently regular function f ,

$$\langle v_1 | f(H) | v_1 \rangle \simeq \mathbf{e}_1^\top f(T_{m_L}) \mathbf{e}_1. \quad (1204)$$

If

$$T_{m_L} = Z \text{diag}(\vartheta_1, \dots, \vartheta_{m_L}) Z^\top, \quad (1205)$$

the corresponding quadrature measure is

$$\nu_{\text{SLQ}} = \sum_{j=1}^{m_L} |Z_{1j}|^2 \delta_{\vartheta_j}. \quad (1206)$$

Averaging over random vectors gives a stochastic approximation to the normalized trace measure. We require KPM and SLQ to agree within the combined uncertainty for:

$$\sup_E \left| F_C^{\text{KPM}}(E) - F_C^{\text{SLQ}}(E) \right|, \quad (1207)$$

where

$$F_C(E) = \int_{-\infty}^E \varrho_C(E') dE', \quad (1208)$$

and for the fixed-broadening density

$$\sup_E \left| \varrho_{C,\eta}^{\text{KPM}}(E) - \varrho_{C,\eta}^{\text{SLQ}}(E) \right|. \quad (1209)$$

The complete DOS uncertainty is recorded as

$$\varepsilon_\varrho = \varepsilon_{\text{KPM}} + \varepsilon_{\text{SLQ}} + \varepsilon_{\text{stoch}} + \varepsilon_{\text{grid}} + \varepsilon_{\text{cover}} + \varepsilon_{\text{shell}} + \varepsilon_\eta. \quad (1210)$$

Weak convergence or fixed- η convergence is not upgraded to an unsmoothed local-DOS theorem without a separate uniform regularity estimate.

H. Layer hybridization and coherence

Let

$$P_1 + P_2 = I, \quad P_1 P_2 = 0 \quad (1211)$$

be the two layer projectors. For a normalized eigenstate $|u_n\rangle$, define

$$w_1^{(n)} = \langle u_n | P_1 | u_n \rangle, \quad w_2^{(n)} = \langle u_n | P_2 | u_n \rangle. \quad (1212)$$

They satisfy

$$w_1^{(n)} + w_2^{(n)} = 1. \quad (1213)$$

The layer polarization is

$$P_L^{(n)} = w_1^{(n)} - w_2^{(n)}. \quad (1214)$$

The layer-mixing weight is

$$\mathcal{H}_L^{(n)} = 4w_1^{(n)}w_2^{(n)} = 1 - \left(P_L^{(n)}\right)^2. \quad (1215)$$

It satisfies $0 \leq \mathcal{H}_L^{(n)} \leq 1$. A layer-polarized state has $\mathcal{H}_L^{(n)} = 0$, whereas a layer-balanced state has $\mathcal{H}_L^{(n)} = 1$. The layer IPR is

$$\text{IPR}_L^{(n)} = \left(w_1^{(n)}\right)^2 + \left(w_2^{(n)}\right)^2 = \frac{1 + \left(P_L^{(n)}\right)^2}{2}. \quad (1216)$$

The effective layer participation number is

$$\mathcal{P}_L^{(n)} = \frac{2}{1 + \left(P_L^{(n)}\right)^2}. \quad (1217)$$

Layer balance alone does not prove dynamical hybridization. At $\omega = 0$, two exactly degenerate layer copies can

be arbitrarily rotated into balanced or polarized eigenvectors. We therefore supplement the state-level weights with projector- and response-based invariants. For an isolated rank- r_C cluster, define

$$W_{\ell,C} = \frac{\text{Tr}(P_\ell P_C)}{r_C}, \quad \ell = 1, 2. \quad (1218)$$

Then $W_{1,C} + W_{2,C} = 1$. The cluster polarization is

$$P_{L,C} = W_{1,C} - W_{2,C}, \quad (1219)$$

and the cluster layer-balance factor is

$$\mathcal{H}_{L,C} = 4W_{1,C}W_{2,C} = 1 - P_{L,C}^2. \quad (1220)$$

These quantities are invariant under every unitary basis rotation inside $\text{Ran } P_C$. Write the cluster projector in layer-block form:

$$P_C = \begin{pmatrix} A_C & B_C \\ B_C^\dagger & D_C \end{pmatrix}. \quad (1221)$$

The off-diagonal block is

$$B_C = P_1 P_C P_2. \quad (1222)$$

We define the basis-independent interlayer projector coherence

$$\mathcal{C}_{\perp,C} = \frac{4}{r_C} \|B_C\|_{\mathbb{F}}^2. \quad (1223)$$

Because $P_C^2 = P_C$, its upper-left block satisfies

$$A_C^2 + B_C B_C^\dagger = A_C. \quad (1224)$$

Therefore

$$\|B_C\|_{\mathbb{F}}^2 = \text{Tr}(A_C - A_C^2). \quad (1225)$$

The nonzero eigenvalues of A_C lie in $[0, 1]$, and there are at most r_C of them. Since

$$x(1-x) \leq \frac{1}{4}, \quad 0 \leq x \leq 1, \quad (1226)$$

we obtain

$$\|B_C\|_{\mathbb{F}}^2 \leq \frac{r_C}{4}. \quad (1227)$$

Hence $0 \leq \mathcal{C}_{\perp,C} \leq 1$. A block-diagonal layer projector has $\mathcal{C}_{\perp,C} = 0$. A maximally coherent balanced rank-one state has $\mathcal{C}_{\perp,C} = 1$. Write the Hamiltonian as

$$H(\omega) = H_0 + \omega t V_\perp, \quad (1228)$$

where

$$V_\perp = \frac{1}{t} \frac{\partial H}{\partial \omega} \quad (1229)$$

is dimensionless. For a simple eigenvalue, Hellmann–Feynman gives

$$\frac{\partial E_n}{\partial \omega} = t \langle u_n | V_\perp | u_n \rangle. \quad (1230)$$

Define the normalized single-state sensitivity

$$\chi_{\perp,n} = \frac{|\partial_\omega E_n|}{t \|V_\perp\|}. \quad (1231)$$

Then $0 \leq \chi_{\perp,n} \leq 1$. For a rank- r_C cluster, cancellation between bonding and antibonding slopes can make $\text{Tr}(P_C V_\perp)$ vanish even when each state is strongly sensitive. We therefore define

$$\chi_{\perp,C} = \frac{\text{Tr} |P_C V_\perp P_C|}{r_C \|V_\perp\|}. \quad (1232)$$

Here

$$|A| = \sqrt{A^\dagger A}. \quad (1233)$$

The quantity obeys $0 \leq \chi_{\perp,C} \leq 1$. Let $P_{\mathcal{I}}^{(1)}$, $P_{\mathcal{I}}^{(2)}$ project onto the selected energy windows in the two uncoupled layers. The restricted interlayer map is

$$V_{\mathcal{I}} = P_{\mathcal{I}}^{(1)} T P_{\mathcal{I}}^{(2)}. \quad (1234)$$

Let $\sigma_1 \geq \dots \geq \sigma_{r_{\mathcal{I}}} \geq 0$ be its singular values. For a spectroscopic linewidth or resolution threshold κ_{res} , define

$$N_{\text{res}} = \# \{j : 2\sigma_j > \kappa_{\text{res}}\}. \quad (1235)$$

The active RMS coupling is

$$w_{\text{rms},\mathcal{I}} = \frac{\|V_{\mathcal{I}}\|_{\text{F}}}{\sqrt{r_{\mathcal{I}}}}. \quad (1236)$$

The stable rank is

$$r_{\text{st}}(V_{\mathcal{I}}) = \frac{\|V_{\mathcal{I}}\|_{\text{F}}^2}{\|V_{\mathcal{I}}\|^2}. \quad (1237)$$

A large stable rank indicates that hybridization is distributed over many channels rather than concentrated in one singular mode. Consider a finite periodic cover with N_c translation cells. For a nondegenerate normalized eigenstate, define the probability in cell x :

$$P_{n;x} = \sum_{\alpha=1}^{\nu} |\psi_{n;x\alpha}|^2. \quad (1238)$$

The cell IPR is

$$\text{IPR}_n^{\text{cell}} = \sum_{x=1}^{N_c} P_{n;x}^2, \quad (1239)$$

and the normalized cell participation is

$$p_n^{\text{cell}} = \frac{1}{N_c \text{IPR}_n^{\text{cell}}}. \quad (1240)$$

A uniformly extended state has $p_n^{\text{cell}} = 1$. A state supported on $O(1)$ cells has

$$p_n^{\text{cell}} = O(N_c^{-1}) \rightarrow 0. \quad (1241)$$

For a degenerate cluster, define the cell projector density

$$Q_x^{(C)} = \frac{1}{r_C} \sum_{\alpha=1}^{\nu} \langle x, \alpha | P_C | x, \alpha \rangle. \quad (1242)$$

It satisfies

$$\sum_{x=1}^{N_c} Q_x^{(C)} = 1. \quad (1243)$$

The projector cell IPR is

$$\text{IPR}^{\text{cell}}[P_C] = \sum_{x=1}^{N_c} \left(Q_x^{(C)} \right)^2, \quad (1244)$$

and the normalized projector participation is

$$p^{\text{cell}}[P_C] = \frac{1}{N_c \text{IPR}^{\text{cell}}[P_C]}. \quad (1245)$$

This quantity is invariant under arbitrary basis rotations inside the cluster. Let $C_{\text{rep},C}$ be the representation coverage of the tracked family, and define the translation-covariance defect

$$\varepsilon_{\text{cov}}[P_C] = \max_{g \in \mathcal{G}_{\text{test}}} \frac{\| [P_C, U_g] \|_{\text{F}}}{2 \| P_C \|_{\text{F}}}. \quad (1246)$$

For a nondegenerate band, define

$$C_C^{\text{coh}} = p_C^{\text{cell}} C_{\text{rep},C}. \quad (1247)$$

For a degenerate cluster, define

$$C_C^{\text{coh}} = \frac{p^{\text{cell}}[P_C] C_{\text{rep},C}}{1 + \varepsilon_{\text{cov}}[P_C]}. \quad (1248)$$

In both cases, $0 \leq C_C^{\text{coh}} \leq 1$. The coherence-weighted DOS is

$$\varrho_{C,\eta}^{\text{coh}}(E) = C_C^{\text{coh}} \varrho_{C,\eta}(E). \quad (1249)$$

Its maximum is

$$\varrho_{C,\eta}^{\text{coh,max}} = \max_E \varrho_{C,\eta}^{\text{coh}}(E). \quad (1250)$$

A localized level crowding mechanism has $C_C^{\text{coh}} \rightarrow 0$ and is therefore suppressed even if the raw DOS is large. The generalized-velocity reference at zero interlayer coupling is

$$v_{\text{max},C}^{(0)} = v_{\text{max},C}(\theta, K, \omega = 0). \quad (1251)$$

Define

$$R_{v,C}^{\text{hyb}} = \frac{v_{\text{max},C}(\theta, K, \omega)}{v_{\text{max},C}^{(0)}(\theta, K)}. \quad (1252)$$

A causal twist-induced flattening candidate must satisfy

$$R_{W,C}^{\text{hyb}} < 1, \quad R_{v,C}^{\text{hyb}} < 1, \quad (1253)$$

together with at least one basis-independent nonzero hybridization diagnostic:

$$\mathcal{C}_{\perp,C} > 0, \quad \chi_{\perp,C} > 0, \quad N_{\text{res}} > 0. \quad (1254)$$

For finite numerical data, we use conservative margins

$$\begin{aligned} R_{W,C}^{\text{hyb},U} &< 1 - \delta_W, \\ R_{v,C}^{\text{hyb},U} &< 1 - \delta_v, \end{aligned} \quad (1255)$$

and

$$\max \left\{ \mathcal{C}_{\perp,C}^L, \chi_{\perp,C}^L, \frac{N_{\text{res}}^L}{\mathcal{I}} \right\} > \delta_{\perp}, \quad (1256)$$

where all margins are specified before evaluating the candidate optimum.

I. Physical flatness functional

The scalar flatness functional is a ranking device constructed from a more fundamental residual vector. We never report the scalar without its constituent observables. Define

$$G_{\Delta,C} = \begin{cases} 1, & \Delta_C > 0, \\ 0, & \Delta_C \leq 0. \end{cases} \quad (1257)$$

An unisolated cluster receives zero physical magic score regardless of its bandwidth or DOS. The energetic flatness residual is

$$r_{W,C} = \frac{W_C}{\Delta_C}. \quad (1258)$$

The first-order dispersion residual is

$$r_{v,C} = \frac{\hbar D_G v_{\text{max},C}}{\Delta_C}. \quad (1259)$$

The second-order dispersion residual is

$$r_{2,C} = \frac{D_G^2 \Lambda_{\text{max},C}}{2\Delta_C}. \quad (1260)$$

The coherent-DOS residual is

$$r_{\varrho,C} = \frac{1}{E_{\text{kin}} \varrho_{C,\eta}^{\text{coh,max}}}. \quad (1261)$$

The coherence residual is

$$r_{\text{coh},C} = \frac{1 - C_C^{\text{coh}}}{C_C^{\text{coh}}}. \quad (1262)$$

If $C_C^{\text{coh}} = 0$, we set $r_{\text{coh},C} = +\infty$. The minimal residual vector is

$$\mathbf{r}_{\text{magic},C}^{(1)} = (r_{W,C}, r_{v,C}, r_{2,C}, r_{\varrho,C}, r_{\text{coh},C}). \quad (1263)$$

The strong second-order vector is

$$\mathbf{r}_{\text{magic},C}^{(2)} = (r_{W,C}, r_{v,C}, r_{2,C}, r_{\varrho,C}, r_{\text{coh},C}). \quad (1264)$$

For $\Delta_C > 0$, $C_C^{\text{coh}} > 0$, define

$$\mathcal{M}_C^{(1)} = \left[\frac{1}{(1 + r_{W,C})(1 + r_{v,C})(1 + r_{2,C})(1 + r_{\varrho,C})(1 + r_{\text{coh},C})} \right]^{1/4}. \quad (1265)$$

Otherwise, $\mathcal{M}_C^{(1)} = 0$. Substitution gives

$$\begin{aligned} \mathcal{M}_C^{(1)} = G_{\Delta,C} &\left[\frac{\Delta_C}{\Delta_C + W_C} \frac{\Delta_C}{\Delta_C + \hbar D_G v_{\text{max},C}} \right. \\ &\times \left. \frac{E_{\text{kin}} \varrho_{C,\eta}^{\text{coh,max}}}{1 + E_{\text{kin}} \varrho_{C,\eta}^{\text{coh,max}}} C_C^{\text{coh}} \right]^{1/4}. \end{aligned} \quad (1266)$$

Every factor lies in $[0, 1]$, so $0 \leq \mathcal{M}_C^{(1)} \leq 1$. When the complete Hessian has been calculated and established, define

$$\mathcal{M}_C^{(2)} = \left[\frac{1}{(1 + r_{W,C})(1 + r_{v,C})(1 + r_{2,C})(1 + r_{\varrho,C})(1 + r_{\text{coh},C})} \right]^{1/5}. \quad (1267)$$

Equivalently,

$$\begin{aligned} \mathcal{M}_C^{(2)} = G_{\Delta,C} &\left[\frac{\Delta_C}{\Delta_C + W_C} \frac{\Delta_C}{\Delta_C + \hbar D_G v_{\text{max},C}} \right. \\ &\times \left. \frac{\Delta_C}{\Delta_C + \frac{1}{2} D_G^2 \Lambda_{\text{max},C}} \times \frac{E_{\text{kin}} \varrho_{C,\eta}^{\text{coh,max}}}{1 + E_{\text{kin}} \varrho_{C,\eta}^{\text{coh,max}}} C_C^{\text{coh}} \right]^{1/5}. \end{aligned}$$

The minimal functional detects a narrow, isolated, weakly dispersive, coherent spectral manifold. The strong functional additionally rejects a band that is first-order stationary but retains a large quadratic dispersion. Under a uniform positive rescaling

$$H \mapsto cH, \quad c > 0, \quad (1268)$$

we have

$$\begin{aligned} W_C &\mapsto cW_C, \quad \Delta_C \mapsto c\Delta_C, \\ v_{\text{max},C} &\mapsto cv_{\text{max},C}, \quad \Lambda_{\text{max},C} \mapsto c\Lambda_{\text{max},C}, \end{aligned} \quad (1269)$$

and

$$\varrho_{cH}(E) = \frac{1}{c} \varrho_H \left(\frac{E}{c} \right). \quad (1270)$$

The reference energy scales as $E_{\text{kin}} \mapsto cE_{\text{kin}}$. Therefore $r_{W,C}$, $r_{v,C}$, $r_{2,C}$, $r_{\varrho,C}$, $r_{\text{coh},C}$ are invariant, and hence

$$\mathcal{M}_C^{(1)} \mapsto \mathcal{M}_C^{(1)}, \quad \mathcal{M}_C^{(2)} \mapsto \mathcal{M}_C^{(2)}. \quad (1271)$$

A trivial reduction of the complete energy scale cannot increase the score. To test dependence on the equal-weight convention, choose

$$\alpha_W, \alpha_v, \alpha_\varrho, \alpha_{\text{coh}} > 0 \quad (1272)$$

with

$$\alpha_W + \alpha_v + \alpha_\varrho + \alpha_{\text{coh}} = 1. \quad (1273)$$

Define

$$\begin{aligned} \mathcal{M}_C^{(\alpha)} &= G_{\Delta, C} (1 + r_{W, C})^{-\alpha_W} (1 + r_{v, C})^{-\alpha_v} \\ &\times (1 + r_{\varrho, C})^{-\alpha_\varrho} (1 + r_{\text{coh}, C})^{-\alpha_{\text{coh}}}. \end{aligned} \quad (1274)$$

The exact limit $\mathbf{r}_{\text{magic}, C}^{(1)} \rightarrow \mathbf{0}$ is equivalent to $\mathcal{M}_C^{(\alpha)} \rightarrow 1$ for every strictly positive normalized weight vector. Define the objective vector

$$\mathbf{O}_C = \left(W_C, \hbar D_G v_{\text{max}, C}, -\varrho_{C, \eta}^{\text{coh}, \text{max}}, -\Delta_C, -C_C^{\text{coh}} \right). \quad (1275)$$

A candidate $\mathbf{p}_1$ dominates $\mathbf{p}_2$ when every objective component at $\mathbf{p}_1$ is no greater and at least one is strictly smaller. Every accepted optimum must lie on the resulting Pareto front. The scalar functional ranks the physically nondominated set; it does not replace the objective vector. Let $\mathfrak{C}_{\text{adm}}(\mathbf{p})$ be the set of clusters that satisfy the hard tracking, gap, convergence, and representation requirements. Define

$$\mathcal{M}(\mathbf{p}) = \max_{C \in \mathfrak{C}_{\text{adm}}(\mathbf{p})} \mathcal{M}_C^{(1)}(\mathbf{p}). \quad (1276)$$

The maximizing cluster is

$$C_*(\mathbf{p}) \in \arg \max_{C \in \mathfrak{C}_{\text{adm}}(\mathbf{p})} \mathcal{M}_C^{(1)}(\mathbf{p}). \quad (1277)$$

For fixed K and ω , the operational magic-angle set is

$$\Theta_{\text{magic}}(K, \omega) = \arg \max_{\theta} \mathcal{M}(\theta, K, \omega). \quad (1278)$$

The set can contain one point, several isolated points, or a finite interval within numerical uncertainty. We do not force uniqueness. The complete magic set in control space is

$$\mathcal{P}_{\text{magic}} = \arg \max_{(\theta, K, \omega)} \mathcal{M}(\theta, K, \omega). \quad (1279)$$

Inside a smooth region with one fixed maximizing cluster, positive gap, and fixed active extrema, define

$$E_{\text{disp}} = \hbar D_G v_{\text{max}, C}, \quad X_\varrho = E_{\text{kin}} \varrho_{C, \eta}^{\text{coh}, \text{max}}. \quad (1280)$$

For a control coordinate

$$p_a \in \{\theta, K, \omega\}, \quad (1281)$$

an interior stationary point satisfies

$$0 = -\frac{\Delta_C \partial_{p_a} W_C - W_C \partial_{p_a} \Delta_C}{\Delta_C (\Delta_C + W_C)}$$

$$\begin{aligned} & -\frac{\Delta_C \partial_{p_a} E_{\text{disp}} - E_{\text{disp}} \partial_{p_a} \Delta_C}{\Delta_C (\Delta_C + E_{\text{disp}})} \\ & + \frac{1}{1 + X_\varrho} \partial_{p_a} \ln \varrho_{C, \eta}^{\text{coh}, \text{max}} + \partial_{p_a} \ln C_C^{\text{coh}}. \end{aligned}$$

Thus a magic optimum is not generally the minimum of W_C alone. Bandwidth, gap, velocity, DOS, and coherence enter the balance simultaneously. Let the numerical estimates and uncertainties be $\widetilde{W}_C$, $\widetilde{\Delta}_C$, $\widetilde{v}_{\text{max}, C}$, $\widetilde{\Lambda}_{\text{max}, C}$, $\widetilde{\varrho}_{C, \eta}^{\text{coh}, \text{max}}$, $\widetilde{C}_C^{\text{coh}}$, and ε_W , ε_Δ , ε_v , ε_Λ , ε_ϱ , ε_{coh} . Define the conservative quantities

$$\begin{aligned} W_C^U &= \widetilde{W}_C + \varepsilon_W, & \Delta_C^L &= \widetilde{\Delta}_C - \varepsilon_\Delta, \\ v_C^U &= \widetilde{v}_{\text{max}, C} + \varepsilon_v, & \Lambda_C^U &= \widetilde{\Lambda}_{\text{max}, C} + \varepsilon_\Lambda, \\ \varrho_C^L &= \max \left\{ 0, \widetilde{\varrho}_{C, \eta}^{\text{coh}, \text{max}} - \varepsilon_\varrho \right\}, \end{aligned} \quad (1282)$$

and

$$C_C^L = \max \left\{ 0, \widetilde{C}_C^{\text{coh}} - \varepsilon_{\text{coh}} \right\}. \quad (1283)$$

If $\Delta_C^L > 0$, $C_C^L > 0$, the established minimal score is

$$\mathcal{M}_C^{\text{cert}, (1)} = \left[\frac{\Delta_C^L}{\Delta_C^L + W_C^U} \frac{\Delta_C^L}{\Delta_C^L + \hbar D_G v_C^U} \times \frac{E_{\text{kin}} \varrho_C^L}{1 + E_{\text{kin}} \varrho_C^L} C_C^L \right]^{1/4}.$$

Otherwise, $\mathcal{M}_C^{\text{cert}, (1)} = 0$. Monotonicity of every factor gives

$$\mathcal{M}_C^{(1)} \geq \mathcal{M}_C^{\text{cert}, (1)}. \quad (1284)$$

The established strong score is

$$\begin{aligned} \mathcal{M}_C^{\text{cert}, (2)} &= \left[\frac{\Delta_C^L}{\Delta_C^L + W_C^U} \frac{\Delta_C^L}{\Delta_C^L + \hbar D_G v_C^U} \right. \\ &\times \left. \frac{\Delta_C^L}{\Delta_C^L + \frac{1}{2} D_G^2 \Lambda_C^U} \times \frac{E_{\text{kin}} \varrho_C^L}{1 + E_{\text{kin}} \varrho_C^L} C_C^L \right]^{1/5}. \end{aligned}$$

J. Magic-state conditions

We separate the existence of a favorable numerical point from the verification of a physical magic-angle-like regime. Before evaluating the spectrum, we freeze Γ , Q_N , n_b , t , w , h , $\lambda_\perp$, $T(D)$, L_N , $\mathcal{D}$, $\mathcal{G}_h$, $\eta, \varepsilon_{\text{tol}}$. A model, cluster window, smoothing scale, or representation family is not changed after inspecting an unfavorable result without recording a new analysis branch. The same cluster must satisfy

$$\text{rank } P_C = r_C \quad (1285)$$

throughout the declared domain, together with

$$\delta_{\mathcal{C}} > \delta_{\mathcal{C}}^{\text{tol}}, \quad \mathcal{O}(P_{C, j}, P_{C, j+1}) \geq \mathcal{O}_{\text{min}}, \quad (1286)$$

and $\varepsilon_{\text{loop}} \leq \varepsilon_{\text{loop}}^{\text{tol}}$. A finite candidate requires $\Delta_C^L > 0$. A thermodynamic candidate requires

$$\liminf_{N \rightarrow \infty} \frac{\Delta_{C,N}}{E_{\text{kin}}} > 0. \quad (1287)$$

A sequence for which both $W_{C,N}$ and $\Delta_{C,N}$ vanish is not classified as an isolated magic band. The finite criterion is

$$\frac{W_C^U}{\Delta_C^L} \leq \varepsilon_W^{\text{acc}}, \quad (1288)$$

with a specified tolerance. The strict thermodynamic condition is

$$\frac{W_{C,N}}{\Delta_{C,N}} \rightarrow 0. \quad (1289)$$

The finite criterion is

$$\frac{\hbar D_G v_C^U}{\Delta_C^L} \leq \varepsilon_v^{\text{acc}}. \quad (1290)$$

The strict thermodynamic condition is

$$\frac{\hbar D_{G,N} v_{\text{max},C,N}}{\Delta_{C,N}} \rightarrow 0. \quad (1291)$$

The supremum must cover the complete declared physical tangent domain. A zero derivative along one selected path does not satisfy this condition. The strong finite criterion is

$$\frac{D_G^2 \Lambda_C^U}{2 \Delta_C^L} \leq \varepsilon_2^{\text{acc}}. \quad (1292)$$

The strict strong condition is

$$\frac{D_{G,N}^2 \Lambda_{\text{max},C,N}}{2 \Delta_{C,N}} \rightarrow 0. \quad (1293)$$

We report the full principal-value spectrum $\{\Lambda_{C,j}\}$ and the anisotropy $\mathcal{A}_{M,C}$ rather than only the Hessian trace. The finite coherent-DOS criterion is

$$E_{\text{kin}} \varrho_{C,\eta}^{\text{coh},L} \geq X_{\varrho}^{\text{acc}}. \quad (1294)$$

The strict limit is

$$E_{\text{kin}} \varrho_{C,\eta}^{\text{coh,max}} \rightarrow \infty \quad (1295)$$

under the specified order of thermodynamic and broadening limits. Coherent participation must satisfy, for a nondegenerate band,

$$\liminf_{N \rightarrow \infty} p_{C,N}^{\text{cell}} > 0. \quad (1296)$$

For a degenerate cluster, it is replaced by

$$\liminf_{N \rightarrow \infty} p^{\text{cell}}[P_{C,N}] > 0, \quad \varepsilon_{\text{cov}}[P_{C,N}] \rightarrow 0, \quad (1297)$$

and $C_{\text{rep},C,N} \rightarrow 1$. A twist-induced physical flat band must satisfy

$$\limsup_{N \rightarrow \infty} R_{W,C,N}^{\text{hyb}} < 1, \quad (1298)$$

and

$$\limsup_{N \rightarrow \infty} R_{v,C,N}^{\text{hyb}} < 1. \quad (1299)$$

At least one basis-independent hybridization measure must remain nonzero:

$$\liminf_{N \rightarrow \infty} \max \left\{ \mathcal{C}_{\perp,C,N}, \chi_{\perp,C,N}, \frac{N_{\text{res},N}}{r_{\mathcal{I},N}} \right\} > 0. \quad (1300)$$

This condition excludes pure folding, a pre-existing monolayer flat band, and an arbitrary layer-basis rotation at zero coupling. For a full finite-quotient conclusion, we require

$$\begin{aligned} C_{\text{rep},N} &= 1, & r_{\text{inj},N} &\rightarrow \infty, & L_N &\rightarrow \infty, \\ \frac{L_N}{r_{\text{inj},N}} &\rightarrow 0, & \ell_N &\rightarrow 0, & p_N &\rightarrow 0, \end{aligned} \quad (1301)$$

and the normalized C^0 , C^1 , and, when required, C^2 error budgets:

$$\varepsilon_N^{(0)} \rightarrow 0, \quad \varepsilon_N^{(1)} \rightarrow 0, \quad \varepsilon_N^{(2)} \rightarrow 0. \quad (1302)$$

For an incommensurate angular-approximant sequence $\theta_N \rightarrow \theta$, we additionally require

$$|\theta_N - \theta| \exp\left(\frac{L_N}{R}\right) \rightarrow 0. \quad (1303)$$

This prevents exponentially amplified hyperbolic registry errors from surviving inside the growing physical hopping shell. Two inequivalent admissible cover towers A and B must yield the same limiting score and optimizer set:

$$\liminf_{N \rightarrow \infty} \mathcal{M}_{C,N}^A = \liminf_{N \rightarrow \infty} \mathcal{M}_{C,N}^B, \quad (1304)$$

and

$$d_{\text{H}}[\Theta_{\text{magic},N}^A, \Theta_{\text{magic},N}^B] \rightarrow 0. \quad (1305)$$

Conditions A–E and G–I define the operational magic set; adding Condition F gives the strong set. Restriction to one-dimensional characters defines an Abelian-sector result, while the full non-Abelian conclusion uses all required representation sectors. Candidates produced by ordinary band folding, symmetry multiplicity, van Hove concentration, localization, boundary accumulation, gap closure, spectral pollution, shell truncation, or a one-dimensional path are excluded by the corresponding conditions.

We define the Abelian operational landscape by restricting the spectral domain to one-dimensional representations: $\mathcal{M}^{\text{Ab}}(\theta, K, \omega)$. Its angle set is

$$\Theta_{\text{magic}}^{\text{Ab}}(K, \omega) = \arg \max_{\theta} \mathcal{M}^{\text{Ab}}(\theta, K, \omega). \quad (1306)$$

For a complete finite quotient, we include every irrep with its exact multiplicity and define $\mathcal{M}_N^{\text{full}}(\theta, K, \omega)$. The full finite-quotient angle set is

$$\Theta_{\text{magic}, N}^{\text{full}}(K, \omega) = \arg \max_{\theta} \mathcal{M}_N^{\text{full}}(\theta, K, \omega). \quad (1307)$$

A thermodynamic full-spectrum set is defined only after a established bulk limit:

$$\Theta_{\text{magic}}^{\text{full}}(K, \omega) = \lim_{N \rightarrow \infty} \Theta_{\text{magic}, N}^{\text{full}}(K, \omega), \quad (1308)$$

where the limit is understood in Hausdorff distance for compact optimizer sets. The equality

$$\Theta_{\text{magic}}^{\text{Ab}} = \Theta_{\text{magic}}^{\text{full}} \quad (1309)$$

is not assumed. It is the central comparison performed after the observables and certificates defined here have been computed. The complete minimal magic-flat limit is

$$\begin{aligned} \frac{W_{C,N}}{\Delta_{C,N}} &\rightarrow 0, \quad \frac{\hbar D_{G,N} v_{\text{max},C,N}}{\Delta_{C,N}} \rightarrow 0, \\ E_{\text{kin}} \varrho_{C,N,\eta}^{\text{coh,max}} &\rightarrow \infty, \quad C_{C,N}^{\text{coh}} \rightarrow 1, \end{aligned} \quad (1310)$$

and

$$\liminf_{N \rightarrow \infty} \frac{\Delta_{C,N}}{E_{\text{kin}}} > 0. \quad (1311)$$

The strong limit additionally requires

$$\frac{D_{G,N}^2 \Lambda_{\text{max},C,N}}{2\Delta_{C,N}} \rightarrow 0. \quad (1312)$$

A twist-induced classification additionally requires Eqs. (1298)–(1300). The final certificate recorded at every accepted parameter point is

$$\begin{aligned} \mathfrak{C}_{\text{magic}, C}^{(2)} &= \left(W_C, \Delta_C, v_{\text{max},C}, \Lambda_{\text{max},C}, \varrho_{C,\eta}^{\text{coh,max}}, \right. \\ &\quad C_C^{\text{coh}}, \mathcal{C}_{\perp,C}, \chi_{\perp,C}, N_{\text{res}}, R_{W,C}^{\text{hyb}}, R_{v,C}^{\text{hyb}}, C_{\text{rep},C}, \\ &\quad \left. = \ell_N, p_N, \varepsilon_N^{(0)}, \varepsilon_N^{(1)}, \varepsilon_N^{(2)}, \mathcal{M}_C^{\text{cert},(1)}, \mathcal{M}_C^{\text{cert},(2)} \right). \end{aligned}$$

We therefore use the term physical magic-angle-like regime only when the same continuously tracked spectral manifold is globally narrow, positively isolated, weakly dispersive to the required derivative order, coherently extended, spectrally concentrated, genuinely interlayer hybridized, representation complete at the claimed level, and stable under the full finite-cover and hopping-shell convergence protocol.

VI. TWIST-CONTROLLED BAND RECONSTRUCTION

We now determine how a continuously varied relative twist reconstructs the spectrum of the hyperbolic

resonator bilayer. We separate three logically different statements:

$$\begin{aligned} \text{spectral reconstruction : } & H(\theta_2) \not\approx H(\theta_1), \\ \text{band narrowing : } & W_C(\theta_2) < W_C(\theta_1), \\ \text{physical flat-band formation : } & (W_C, \Delta_C, v_{\text{max},C}, \Lambda_{\text{max},C}, \varrho_C, C_C^{\text{coh}}) \end{aligned} \quad (1313)$$

The first condition can hold without either of the latter two. An avoided crossing, a gap opening, or a redistribution of the density of states is already a spectral reconstruction, even when the corresponding band remains strongly dispersive. We therefore compute the complete twist-dependent spectrum before applying the physical flat-band criterion. We use the curvature magnitude

$$\kappa = -K = \frac{1}{R^2} \geq 0 \quad (1314)$$

and the dimensionless interlayer coupling $\omega = \frac{w}{t}$. After subtracting the common resonator frequency and dividing by t , the co-rotating one-particle Hamiltonian is written as

$$\mathcal{H}(\kappa, \theta, \omega) = \begin{pmatrix} A(\kappa) & \omega \mathcal{T}(\kappa, \theta) \\ \omega \mathcal{T}^\dagger(\kappa, \theta) & A(\kappa) \end{pmatrix}. \quad (1315)$$

The physical eigenfrequencies are

$$E_n = \omega_0 + t e_n, \quad (1316)$$

where e_n is an eigenvalue of $\mathcal{H}$. Rigid rotation leaves the isolated layers unitarily equivalent. Consequently,

$$\frac{\partial A}{\partial \theta} = 0, \quad (1317)$$

and every twist derivative is carried by the interlayer block:

$$\mathcal{H}_\theta = \frac{\partial \mathcal{H}}{\partial \theta} = \omega \begin{pmatrix} 0 & \mathcal{T}_\theta \\ \mathcal{T}_\theta^\dagger & 0 \end{pmatrix}. \quad (1318)$$

Similarly,

$$\mathcal{H}_{\theta\theta} = \omega \begin{pmatrix} 0 & \mathcal{T}_{\theta\theta} \\ \mathcal{T}_{\theta\theta}^\dagger & 0 \end{pmatrix}. \quad (1319)$$

The block structure gives the exact norm identities

$$\|\mathcal{H}_\theta\| = |\omega| \|\mathcal{T}_\theta\|, \quad \|\mathcal{H}_{\theta\theta}\| = |\omega| \|\mathcal{T}_{\theta\theta}\|. \quad (1320)$$

For identical centered layers, layer exchange gives

$$P_L \mathcal{H}(\kappa, \theta, \omega) P_L = \mathcal{H}(\kappa, -\theta, \omega), \quad (1321)$$

where

$$P_L = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}. \quad (1322)$$

Therefore

$$\text{Spec } \mathcal{H}(\kappa, \theta, \omega) = \text{Spec } \mathcal{H}(\kappa, -\theta, \omega). \quad (1323)$$

Every analytic scalar spectral observable is even in θ :

$$\mathcal{O}(\kappa, \theta, \omega) = \mathcal{O}_0(\kappa, \omega) + \theta^2 \mathcal{O}_2(\kappa, \omega) + O(\theta^4). \quad (1324)$$

A linear term near $\theta = 0$ therefore signals explicit layer asymmetry, branch misidentification, a nonsymmetric truncation, or a numerical error.

A. Complete twist scan

For the centered regular-octagon lattice, the point symmetry identifies

$$\theta \sim \theta + \frac{\pi}{4} \quad (1325)$$

and layer exchange identifies $\theta \sim -\theta$. We therefore use the irreducible interval

$$\Theta_{\text{tw}} = \left[0, \frac{\pi}{8}\right]. \quad (1326)$$

For the Euclidean square benchmark, the corresponding interval is

$$\Theta_{\text{tw}}^{(\text{E})} = \left[0, \frac{\pi}{4}\right]. \quad (1327)$$

The primary scan keeps

$\kappa, \omega, h/a, \lambda_{\perp}/a, t,$ and the finite-cover graph (1328) fixed while varying only θ . This produces a norm-continuous family on one Hilbert space:

$$\theta \mapsto \mathcal{H}(\kappa, \theta, \omega) \in \text{Herm}(D_N). \quad (1329)$$

Exact commensurate cells at different angles can have different coincidence indices and different Hilbert-space dimensions. We do not differentiate such matrices directly. Exact-cell calculations are used for crystallographic benchmarks, whereas continuous spectral flow is evaluated on a common synthetic finite cover or on a declared transported fiber. Let two hyperbolic sites have polar data $(r_i, \phi_i), (r_j, \phi_j)$, and define

$$\Phi_{ij}(\theta) = \phi_i - \phi_j - \theta. \quad (1330)$$

Set

$$u_i = \sqrt{\kappa} r_i, \quad u_j = \sqrt{\kappa} r_j, \quad u_{ij} = \sqrt{\kappa} d_{ij}. \quad (1331)$$

The hyperbolic law of cosines gives

$$\begin{aligned} \cosh u_{ij} &= \cosh u_i \cosh u_j \\ &\quad - \sinh u_i \sinh u_j \cos \Phi_{ij}. \end{aligned} \quad (1332)$$

Define

$$B_{ij} = \sinh u_i \sinh u_j. \quad (1333)$$

Differentiation with respect to θ gives

$$\sqrt{\kappa} \sinh u_{ij} \frac{\partial d_{ij}}{\partial \theta} = -B_{ij} \sin \Phi_{ij}. \quad (1334)$$

Therefore

$$\frac{\partial d_{ij}}{\partial \theta} = -\frac{B_{ij} \sin \Phi_{ij}}{\sqrt{\kappa} \sinh(\sqrt{\kappa} d_{ij})}. \quad (1335)$$

A second differentiation yields

$$\begin{aligned} &\kappa \cosh(\sqrt{\kappa} d_{ij}) \left(\frac{\partial d_{ij}}{\partial \theta} \right)^2 \\ &+ \sqrt{\kappa} \sinh(\sqrt{\kappa} d_{ij}) \frac{\partial^2 d_{ij}}{\partial \theta^2} = B_{ij} \cos \Phi_{ij}. \end{aligned} \quad (1336)$$

Hence

$$\begin{aligned} \frac{\partial^2 d_{ij}}{\partial \theta^2} &= \frac{B_{ij} \cos \Phi_{ij}}{\sqrt{\kappa} \sinh(\sqrt{\kappa} d_{ij})} \\ &\quad - \sqrt{\kappa} \coth(\sqrt{\kappa} d_{ij}) \left(\frac{\partial d_{ij}}{\partial \theta} \right)^2. \end{aligned} \quad (1337)$$

The complete three-dimensional synthetic separation is

$$D_{ij} = \sqrt{h^2 + d_{ij}^2}. \quad (1338)$$

Its first derivative is

$$D_{ij,\theta} = \frac{d_{ij}}{D_{ij}} d_{ij,\theta}. \quad (1339)$$

The second derivative is

$$\begin{aligned} D_{ij,\theta\theta} &= \frac{d_{ij,\theta}^2 + d_{ij} d_{ij,\theta\theta}}{D_{ij}} - \frac{d_{ij}^2 d_{ij,\theta}^2}{D_{ij}^3} \\ &= \frac{h^2 d_{ij,\theta}^2 + d_{ij} D_{ij}^2 d_{ij,\theta\theta}}{D_{ij}^3}. \end{aligned} \quad (1340)$$

For a general radial hopping law

$$T_{ij} = T(D_{ij}), \quad (1341)$$

we obtain

$$T_{ij,\theta} = T'(D_{ij}) D_{ij,\theta}, \quad (1342)$$

and

$$T_{ij,\theta\theta} = T''(D_{ij}) D_{ij,\theta}^2 + T'(D_{ij}) D_{ij,\theta\theta}. \quad (1343)$$

For the exponential kernel

$$T(D) = w \exp \left[-\frac{D - h}{\lambda_{\perp}} \right], \quad (1344)$$

the derivatives become

$$T_{ij,\theta} = -\frac{T_{ij}}{\lambda_{\perp}} D_{ij,\theta}, \quad (1345)$$

and

$$T_{ij,\theta\theta} = T_{ij} \left[\frac{D_{ij,\theta}^2}{\lambda_\perp^2} - \frac{D_{ij,\theta\theta}}{\lambda_\perp} \right]. \quad (1346)$$

For the same site label at radial distance r ,

$$d_{\text{same}}(\theta) = 2R \operatorname{arsinh} \left[\sinh \left(\frac{r}{R} \right) \left| \sin \frac{\theta}{2} \right| \right]. \quad (1347)$$

Although the unsigned distance contains $|\sin(\theta/2)|$, the physical quantity D_{same}^2 and the radial hopping are even and analytic at $\theta = 0$. The exact local susceptibility is

$$\left. \frac{\partial^2 T_{\text{same}}}{\partial \theta^2} \right|_{\theta=0} = -\frac{wR^2}{h\lambda_\perp} \sinh^2 \left(\frac{r}{R} \right). \quad (1348)$$

The twist response is therefore spatially nonuniform and grows exponentially with hyperbolic radius. For a simple normalized eigenstate,

$$\mathcal{H}(\theta)|u_n(\theta)\rangle = e_n(\theta)|u_n(\theta)\rangle, \quad (1349)$$

Hellmann–Feynman gives

$$\frac{de_n}{d\theta} = \langle u_n | \mathcal{H}_\theta | u_n \rangle. \quad (1350)$$

The exact second derivative is

$$\frac{d^2 e_n}{d\theta^2} = \langle u_n | \mathcal{H}_{\theta\theta} | u_n \rangle + 2 \sum_{m \neq n} \frac{|\langle u_m | \mathcal{H}_\theta | u_n \rangle|^2}{e_n - e_m}.$$

The first term is the direct curvature of the programmed coupling network. The second term is the wave-function reconstruction generated by all remote modes and is enhanced by small spectral denominators. At alignment, twist parity gives

$$\left. \frac{de_n}{d\theta} \right|_{\theta=0} = 0 \quad (1351)$$

for every analytic nondegenerate branch. Therefore

$$e_n(\theta) = e_n(0) + \frac{\theta^2}{2} \left. \frac{d^2 e_n}{d\theta^2} \right|_0 + O(\theta^4). \quad (1352)$$

For an isolated cluster,

$$P_C(\theta) = \frac{1}{2\pi i} \oint_{\mathcal{C}} [z - \mathcal{H}(\theta)]^{-1} dz. \quad (1353)$$

Differentiation gives

$$P_{C,\theta} = \frac{1}{2\pi i} \oint_{\mathcal{C}} R(z) \mathcal{H}_\theta R(z) dz, \quad (1354)$$

where

$$R(z) = [z - \mathcal{H}]^{-1}. \quad (1355)$$

If

$$\delta_{\mathcal{C}} = \inf_{z \in \mathcal{C}} \operatorname{dist}[z, \operatorname{Spec} \mathcal{H}], \quad (1356)$$

then

$$\|P_{C,\theta}\| \leq \frac{|\mathcal{C}|}{2\pi} \frac{\|\mathcal{H}_\theta\|}{\delta_{\mathcal{C}}^2}. \quad (1357)$$

A large projector response can therefore occur near a narrow avoided crossing even if the corresponding energy shift is small. Let

$$0 = \theta_0 < \theta_1 < \dots < \theta_{N_\theta} = \theta_{\max} \quad (1358)$$

be the twist grid. On an interval $[\theta_j, \theta_{j+1}]$, define

$$L_{\theta,j} = \sup_{\theta \in [\theta_j, \theta_{j+1}]} \|\mathcal{H}_\theta(\theta)\|. \quad (1359)$$

The fundamental theorem of calculus gives

$$\|\mathcal{H}(\theta_{j+1}) - \mathcal{H}(\theta_j)\| \leq L_{\theta,j} \delta\theta_j, \quad (1360)$$

where

$$\delta\theta_j = \theta_{j+1} - \theta_j. \quad (1361)$$

Let $\delta_{C,j}$ be the minimum separation between the tracked cluster and its complement at the two interval endpoints. We impose

$$L_{\theta,j} \delta\theta_j \leq \zeta_{\text{track}} \delta_{C,j}, \quad 0 < \zeta_{\text{track}} < \frac{1}{2}. \quad (1362)$$

If the inequality fails, the interval is bisected before projector continuation. For any scalar observable $f(\theta)$ with

$$|f''(\theta)| \leq M_{f,2}, \quad (1363)$$

linear interpolation on one grid interval has error

$$|f(\theta) - f_{\text{lin}}(\theta)| \leq \frac{M_{f,2}}{8} \delta\theta_j^2. \quad (1364)$$

This error is included in the bandwidth, gap, velocity, and score budgets. We use a two-stage scan:

1. a uniform coarse scan over the full irreducible twist interval;
2. adaptive refinement around small gaps, rapid projector rotation, local maxima of the flatness score, and changes in representation or symmetry labels.

The refinement indicators are

$$\mathcal{R}_{\text{refine}} = \max \left\{ \frac{L_\theta \delta\theta}{\delta_C}, \|P_C(\theta + \delta\theta) - P_C(\theta)\|, \frac{|\Delta \mathcal{M}_C|}{\mathcal{M}_C + \varepsilon_{\mathcal{M}}}, \frac{|\Delta_C^{\text{dir}}|^{-1}}{\Delta_{\text{ref}}^{-1}} \right\} \quad (1365)$$

For a finite Hilbert space of dimension D , define the normalized empirical spectral measure

$$\nu_\theta = \frac{1}{D} \sum_{n=1}^D \delta_{e_n(\theta)}. \quad (1366)$$

The complete twist-reconstruction distance relative to alignment is

$$D_{\text{spec}}^{(\theta)}(\theta; \kappa, \omega) = W_1(\nu_\theta, \nu_0). \quad (1367)$$

For ordered eigenvalues,

$$D_{\text{spec}}^{(\theta)} = \frac{1}{D} \sum_{n=1}^D |e_n(\theta) - e_n(0)|. \quad (1368)$$

Weyl's inequality gives

$$D_{\text{spec}}^{(\theta)} \leq \|\mathcal{H}(\theta) - \mathcal{H}(0)\|. \quad (1369)$$

Since the diagonal blocks are twist independent,

$$D_{\text{spec}}^{(\theta)} \leq |\omega| \|\mathcal{T}(\kappa, \theta) - \mathcal{T}(\kappa, 0)\|. \quad (1370)$$

Thus the complete finite-dimensional spectral displacement is bounded entirely by the reconstruction of the interlayer coupling matrix. Define the normalized tunnelling matrix

$$\tilde{\mathcal{T}}(\theta) = \frac{\mathcal{T}(\theta)}{\|\mathcal{T}(\theta)\|_{\text{F}}}. \quad (1371)$$

The pure tunnelling-shape distance is

$$D_{\mathcal{T}}^{\text{shape}}(\theta) = \|\tilde{\mathcal{T}}(\theta) - \tilde{\mathcal{T}}(0)\|_{\text{F}}. \quad (1372)$$

It vanishes when twist changes only the total magnitude of the interlayer matrix. The normalized spectral density is

$$\varrho(E, \theta) = \frac{1}{D} \sum_{n=1}^D \delta[E - e_n(\theta)]. \quad (1373)$$

Differentiating gives

$$\frac{\partial \varrho}{\partial \theta} = -\frac{1}{D} \sum_n \frac{de_n}{d\theta} \delta'[E - e_n(\theta)]. \quad (1374)$$

Define the twist spectral current

$$J_\theta(E, \theta) = \frac{1}{D} \sum_n \frac{de_n}{d\theta} \delta[E - e_n(\theta)]. \quad (1375)$$

Then

$$\frac{\partial \varrho}{\partial \theta} + \frac{\partial J_\theta}{\partial E} = 0. \quad (1376)$$

Using Hellmann–Feynman,

$$J_\theta(E, \theta) = \frac{1}{D} \text{Tr} [\mathcal{H}_\theta \delta(E - \mathcal{H})]. \quad (1377)$$

Twist therefore transports states through energy space but does not create or remove states. For Lorentzian broadening,

$$\varrho_\eta(E, \theta) = -\frac{1}{\pi D} \text{Im} \text{Tr} [E + i\eta - \mathcal{H}(\theta)]^{-1}. \quad (1378)$$

The resolvent derivative gives

$$\frac{\partial \varrho_\eta}{\partial \theta} = -\frac{1}{\pi D} \text{Im} \text{Tr} [R_\eta \mathcal{H}_\theta R_\eta]. \quad (1379)$$

Since

$$\|R_\eta\| \leq \eta^{-1}, \quad (1380)$$

we obtain

$$\left| \frac{\partial \varrho_\eta}{\partial \theta} \right| \leq \frac{\|\mathcal{H}_\theta\|}{\pi \eta^2}. \quad (1381)$$

Define

$$\mu_m(\theta) = \frac{1}{D} \text{Tr} \mathcal{H}^m(\theta). \quad (1382)$$

Cyclicity of the trace gives

$$\frac{d\mu_m}{d\theta} = \frac{m}{D} \text{Tr} [\mathcal{H}^{m-1} \mathcal{H}_\theta]. \quad (1383)$$

For identical layer blocks of dimension M ,

$$\mu_1 = \frac{1}{M} \text{Tr} A, \quad (1384)$$

and

$$\mu_2 = \frac{1}{M} \text{Tr} A^2 + \frac{\omega^2}{M} \text{Tr} (\mathcal{T} \mathcal{T}^\dagger). \quad (1385)$$

Thus

$$\frac{d\mu_2}{d\theta} = \frac{2\omega^2}{M} \text{Re} \text{Tr} (\mathcal{T}_\theta \mathcal{T}^\dagger). \quad (1386)$$

The third moment contains the matrix orientation relative to the intralayer kinetic operator:

$$\mu_3 = \frac{1}{M} \text{Tr} A^3 + \frac{3\omega^2}{2M} \text{Tr} [A (\mathcal{T} \mathcal{T}^\dagger + \mathcal{T}^\dagger \mathcal{T})].$$

Consequently, a twist-dependent μ_3 at fixed $\text{Tr} \mathcal{T} \mathcal{T}^\dagger$ proves a genuine reorientation of the hybridization texture rather than a scalar change of its norm. At every angle, we record

$$\mathfrak{S}(\theta) = \left(\text{Spec } \mathcal{H}, P_C, D_{\text{spec}}^{(\theta)}, D_{\mathcal{T}}^{\text{shape}}, \mu_1, \mu_2, \mu_3, W_C, \Delta_C, v_{\text{max}, C}, \Lambda_{\text{max}, C}, \varrho_{C, \eta}, C_C^{\text{coh}}, \mathcal{C}_{\perp, C}, \chi_{\perp, C} \right).$$

No physical flat-band conclusion is drawn from the plotted eigenvalues alone.

B. Interlayer avoided-crossing network

Band flattening in a twisted bilayer generally arises from a network of hybridizations rather than from one

isolated two-level event. We first derive the exact local structure of one crossing and then assemble the crossings into a symmetry-resolved network. Let

$$|a, \rho, \lambda_{\text{sym}}\rangle, \quad |b, \sigma, \lambda'_{\text{sym}}\rangle \quad (1387)$$

be two locally tracked diabatic states, where ρ, σ are translation representations and $\lambda_{\text{sym}}, \lambda'_{\text{sym}}$ denote residual point-group labels. Translation invariance requires an interlayer coupling V_{ab} to be an intertwiner:

$$\rho(g)V_{ab} = V_{ab}\sigma(g) \quad \text{for every } g. \quad (1388)$$

Schur's lemma gives

$$\rho \not\simeq \sigma \implies V_{ab} = 0. \quad (1389)$$

An additional point-group mismatch can also force $V_{ab} = 0$. We therefore classify a degeneracy as follows:

$$\begin{aligned} \rho \not\simeq \sigma &\implies \text{symmetry-allowed exact crossing,} \\ \rho \simeq \sigma, \quad V_{ab} \neq 0 &\implies \text{generic avoided crossing,} \\ \rho \simeq \sigma, \quad V_{ab} = 0 &\implies \text{crossing protected by an additional symmetry.} \end{aligned} \quad (1390)$$

An exact crossing between distinct irreducible sectors is a representation-theoretic multiplicity event. It is not itself evidence for kinetic flattening. Let

$$P_{ab} = |a\rangle\langle a| + |b\rangle\langle b| \quad (1391)$$

and

$$Q_{ab} = I - P_{ab}. \quad (1392)$$

The exact two-state Feshbach operator is

$$H_{ab}^{\text{eff}}(E, \theta) = P_{ab}HP_{ab} + P_{ab}HQ_{ab}[E - Q_{ab}HQ_{ab}]^{-1}Q_{ab}HP_{ab}. \quad (1393)$$

Let

$$g_{\text{tail}} = \text{dist}[E, \text{Spec}(Q_{ab}HQ_{ab})], \quad (1394)$$

and

$$b_{\text{tail}} = \|P_{ab}HQ_{ab}\|. \quad (1395)$$

The remote self-energy obeys

$$\|\Sigma_{ab}^{\text{tail}}(E)\| \leq \frac{b_{\text{tail}}^2}{g_{\text{tail}}}. \quad (1396)$$

We accept a two-level interpretation only when this quantity is smaller than the extracted avoided-crossing gap and the declared error budget. After including the diagonal and off-diagonal self-energy corrections, we write

$$H_{ab}^{\text{eff}} = \bar{\varepsilon}I_2 + \begin{pmatrix} \Delta/2 & V \\ V^* & -\Delta/2 \end{pmatrix}, \quad (1397)$$

where all quantities may depend on $(E, \theta, \kappa, \omega)$. In an exact invariant two-state problem, there is no E -dependence. The exact static two-level eigenvalues are

$$E_{\pm} = \bar{\varepsilon} \pm \Omega, \quad \Omega = \sqrt{\frac{\Delta^2}{4} + |V|^2}. \quad (1398)$$

The level separation is

$$g_{ab}(\theta) = E_+ - E_- = \sqrt{\Delta^2(\theta) + 4|V(\theta)|^2}. \quad (1399)$$

A bare resonance satisfies $\Delta(\theta_{\text{res}}) = 0$. At that angle,

$$g_{ab}(\theta_{\text{res}}) = 2|V(\theta_{\text{res}})|. \quad (1400)$$

However, the minimum of the actual gap need not coincide exactly with the zero of Δ when V varies with twist. Differentiating g_{ab}^2 gives

$$\frac{d}{d\theta}g_{ab}^2 = 2\Delta\Delta_{\theta} + 8\text{Re}(V^*V_{\theta}). \quad (1401)$$

The exact minimum-gap condition is therefore

$$\Delta\Delta_{\theta} + 4\text{Re}(V^*V_{\theta}) = 0. \quad (1402)$$

Only when $V_{\theta} \simeq 0$ does the minimum occur at $\Delta = 0$. A true crossing requires

$$\Delta(\theta_*) = 0, \quad V(\theta_*) = 0. \quad (1403)$$

Write $V = |V|e^{i\varphi}$. Define the mixing angle by

$$\tan(2\vartheta) = \frac{2|V|}{\Delta}. \quad (1404)$$

The eigenvectors may be chosen as

$$|+\rangle = \cos \vartheta |a\rangle + e^{-i\varphi} \sin \vartheta |b\rangle, \quad (1405)$$

$$|-\rangle = -e^{i\varphi} \sin \vartheta |a\rangle + \cos \vartheta |b\rangle. \quad (1406)$$

When the two diabatic states are layer polarized, the basis-independent layer-mixing weight is

$$\mathcal{H}_{ab}(\theta) = \sin^2(2\vartheta) = \frac{4|V|^2}{\Delta^2 + 4|V|^2}. \quad (1407)$$

It obeys $0 \leq \mathcal{H}_{ab} \leq 1$. At resonance, $\mathcal{H}_{ab} = 1$. Assume locally

$$\Delta(\theta) = g_{\theta}(\theta - \theta_{\text{res}}) + O[(\theta - \theta_{\text{res}})^2], \quad (1408)$$

with

$$g_{\theta} = \left. \frac{d\Delta}{d\theta} \right|_{\theta_{\text{res}}}, \quad (1409)$$

and take V as constant to leading order. The half-maximum condition

$$\mathcal{H}_{ab} = \frac{1}{2} \quad (1410)$$

gives $|\Delta| = 2|V|$. The full angular width at half maximum is

$$\delta\theta_{ab}^{\text{FWHM}} = \frac{4|V|}{|g\theta|}. \quad (1411)$$

A small gap can therefore produce an extremely sharp exchange of eigenvector character when the diabatic detuning slope is large. The integrated mixing weight is

$$\begin{aligned} \mathcal{A}_{ab}^{\text{mix}} &= \int_{-\infty}^{+\infty} \mathcal{H}_{ab}(\theta) d\theta \\ &= \int_{-\infty}^{+\infty} \frac{4|V|^2}{g_\theta^2(\theta - \theta_{\text{res}})^2 + 4|V|^2} d\theta \\ &= \frac{2\pi|V|}{|g\theta|}. \end{aligned} \quad (1412)$$

This quantity measures the angular range over which the two diabatic states exchange spectral character. Let λ be any continuous physical spectral coordinate. It can be an ordinary momentum component or a non-Abelian tangent coordinate. Differentiating Eq. (1398) gives

$$d_\lambda E_\pm = d_\lambda \bar{\varepsilon} \pm \frac{\frac{\Delta}{4} d_\lambda \Delta + \text{Re}(V^* d_\lambda V)}{\sqrt{\Delta^2/4 + |V|^2}}.$$

At exact resonance,

$$d_\lambda E_\pm = \frac{d_\lambda \varepsilon_a + d_\lambda \varepsilon_b}{2} \pm \frac{\text{Re}(V^* d_\lambda V)}{|V|}. \quad (1413)$$

If $d_\lambda V = 0$,

$$d_\lambda E_\pm = \frac{d_\lambda \varepsilon_a + d_\lambda \varepsilon_b}{2}. \quad (1414)$$

Counterpropagating diabatic states satisfying

$$d_\lambda \varepsilon_a \simeq -d_\lambda \varepsilon_b \quad (1415)$$

therefore generate a strongly suppressed hybridized velocity. When $d_\lambda V = 0$, the difference between the hybridized slopes is

$$d_\lambda E_+ - d_\lambda E_- = \frac{\Delta}{\sqrt{\Delta^2 + 4|V|^2}} (d_\lambda \varepsilon_a - d_\lambda \varepsilon_b). \quad (1416)$$

Using Eq. (1407),

$$\frac{|\Delta|}{\sqrt{\Delta^2 + 4|V|^2}} = \sqrt{1 - \mathcal{H}_{ab}}. \quad (1417)$$

Thus the same mixing factor that measures layer hybridization also measures suppression of the diabatic slope contrast. Avoided crossing alone does not guarantee flatness. At resonance both hybridized branches inherit the average bare slope. If

$$d_\lambda \varepsilon_a + d_\lambda \varepsilon_b \neq 0, \quad (1418)$$

the two branches can remain strongly dispersive despite complete hybridization. For the eigenvector Eq. (1405), the Fubini-Study metric in the twist direction is

$$g_{\theta\theta}^{(+)} = \left(\frac{d\vartheta}{d\theta} \right)^2 + \frac{1}{4} \sin^2(2\vartheta) \left(\frac{d\varphi}{d\theta} \right)^2. \quad (1419)$$

The mixing-angle derivative is

$$\frac{d\vartheta}{d\theta} = \frac{\Delta \partial_\theta |V| - |V| \partial_\theta \Delta}{\Delta^2 + 4|V|^2}. \quad (1420)$$

At resonance with locally constant real coupling,

$$g_{\theta\theta}^{(+)} = \frac{(\partial_\theta \Delta)^2}{16|V|^2}. \quad (1421)$$

A narrow avoided-crossing gap therefore produces a large wave-function susceptibility even when the eigenvalues remain continuous. We define a symmetry-resolved avoided-crossing graph

$$\mathfrak{G}_{\text{AC}} = (\mathfrak{V}_{\text{AC}}, \mathfrak{E}_{\text{AC}}). \quad (1422)$$

Each vertex

$$v_a = (a, \rho_a, \lambda_{\text{sym},a}, P_a^{\text{dia}}) \quad (1423)$$

represents a locally tracked diabatic branch. An edge $e = (a, b)$ is added only when:

$$\begin{aligned} \rho_a &\simeq \rho_b, \quad \lambda_{\text{sym},a} \text{ and } \lambda_{\text{sym},b} \text{ permit coupling,} \\ \Delta_{ab}(\theta) &\text{ changes sign or reaches a established minimum,} \\ |V_{ab}| &> \varepsilon_{V,ab}, \quad g_{\text{tail},ab} > g_{\text{tail}}^{\min}. \end{aligned} \quad (1424)$$

Every edge carries the data

$$\begin{aligned} \mathfrak{e}_{ab} &= \left(\theta_{ab}^{\min}, g_{ab}^{\min}, \delta\theta_{ab}^{\text{FWHM}}, \mathcal{A}_{ab}^{\text{mix}}, \max g_{\theta\theta}^{(ab)}, \Delta_{\theta,ab}, \right. \\ &\quad \left. V_{ab}, V_{\theta,ab}, \mathbf{v}_a, \mathbf{v}_b, \mathbf{v}_+, \mathbf{v}_- \right). \end{aligned}$$

Crossings between inequivalent irreducible sectors are stored in a separate exact-crossing graph $\mathfrak{G}_{\text{X}}$. They are never merged into the dynamical hybridization graph. The cumulative mixing action is

$$\mathcal{A}_{\text{AC}} = \sum_{e \in \mathfrak{E}_{\text{AC}}} \mathcal{A}_e^{\text{mix}}. \quad (1425)$$

The minimum established network gap is

$$g_{\text{AC}}^{\min} = \min_{e \in \mathfrak{E}_{\text{AC}}} g_e^{\min}. \quad (1426)$$

The number of active crossings incident on the target family is

$$N_{\text{AC},C} = \# \{e \in \mathfrak{E}_{\text{AC}} : e \text{ intersects } C\}. \quad (1427)$$

A large value of $\mathcal{A}_{\text{AC}}$ proves extensive wave-function exchange, but does not by itself prove global flattening. The resulting branch must still satisfy the bandwidth, gap, velocity, Hessian, DOS, and coherence tests.

C. Joint analysis of bandwidth, gap, velocity, effective mass, and density of states

We now evaluate all physical observables on the same continuously tracked spectral cluster. We suppress the cluster subscript when no ambiguity arises. Suppose the upper and lower cluster edges are attained at unique non-degenerate points $\lambda_+(\theta)$, $\lambda_-(\theta)$. Then

$$W(\theta) = E_+[\lambda_+(\theta), \theta] - E_-[\lambda_-(\theta), \theta]. \quad (1428)$$

At each extremum, $d_\lambda E_\pm = 0$. The envelope theorem gives

$$\frac{dW}{d\theta} = \langle u_+ | H_\theta | u_+ \rangle - \langle u_- | H_\theta | u_- \rangle. \quad (1429)$$

Twist narrows the band locally when

$$\langle u_+ | H_\theta | u_+ \rangle < \langle u_- | H_\theta | u_- \rangle. \quad (1430)$$

Define the logarithmic twist-compression susceptibility

$$\chi_W^{(\theta)} = -\frac{\partial \ln W}{\partial \theta}. \quad (1431)$$

A positive value means that increasing twist locally suppresses the bandwidth. The extremum itself moves with twist. Let x^μ be local spectral coordinates and let

$$M_{\mu\nu} = \partial_\mu \partial_\nu E \quad (1432)$$

be the invertible Hessian at the extremum. Differentiation of

$$\partial_\mu E[x_*(\theta), \theta] = 0 \quad (1433)$$

gives

$$\frac{dx_*^\mu}{d\theta} = -(M^{-1})^{\mu\nu} \partial_\nu \partial_\theta E. \quad (1434)$$

The complete second twist derivative of the edge energy is

$$\frac{d^2 E_*}{d\theta^2} = \partial_\theta^2 E - (\partial_\theta \partial_\mu E)(M^{-1})^{\mu\nu} (\partial_\nu \partial_\theta E). \quad (1435)$$

The second term is the correction generated by motion of the extremum through the spectral domain. Suppose the active upper indirect gap is

$$\Delta^+ = E_{\text{above}}^{\min} - E_C^+. \quad (1436)$$

Its derivative is

$$\frac{d\Delta^+}{d\theta} = \langle u_{\text{above}, \min} | H_\theta | u_{\text{above}, \min} \rangle - \langle u_{C,+} | H_\theta | u_{C,+} \rangle. \quad (1437)$$

For the active lower gap,

$$\Delta^- = E_C^- - E_{\text{below}}^{\max}, \quad (1438)$$

we obtain

$$\frac{d\Delta^-}{d\theta} = \langle u_{C,-} | H_\theta | u_{C,-} \rangle - \langle u_{\text{below}, \max} | H_\theta | u_{\text{below}, \max} \rangle. \quad (1439)$$

The global gap is

$$\Delta = \min\{\Delta^-, \Delta^+\}. \quad (1440)$$

It remains continuous when the active edge switches, but it can develop a cusp at $\Delta^- = \Delta^+$. Define the twist flatness susceptibility

$$\chi_F^{(\theta)} = \frac{\partial}{\partial \theta} \ln \left(\frac{\Delta}{W} \right) = \frac{\partial_\theta \Delta}{\Delta} - \frac{\partial_\theta W}{W}. \quad (1441)$$

A positive value can arise from bandwidth suppression, gap opening, or both. The maximum generalized velocity is

$$v_{\max, C} = \sup_{\lambda \in \mathcal{D}} \frac{1}{\hbar} \|dE_C(\lambda)\|_{G^{-1}}. \quad (1442)$$

The velocity-bandwidth inequality gives

$$W_C \leq \hbar D_G v_{\max, C} + w_C^{\text{int}}, \quad (1443)$$

where w_C^{int} is the retained internal cluster splitting. At every avoided crossing, we evaluate the exact decomposition

$$dE_\pm = d\varepsilon \pm \frac{\frac{\Delta}{4} d\Delta + \text{Re}(V^* dV)}{\Omega}. \quad (1444)$$

The full numerical differential must agree with this expression within the two-level tail error

$$\varepsilon_{2\text{lev}} \leq \frac{b_{\text{tail}}^2}{g_{\text{tail}}} + \varepsilon_{\text{solver}} + \varepsilon_{\text{der}}. \quad (1445)$$

At a stationary point, the geometry-normalized principal Hessian values are

$$M\mathbf{v}_j = \Lambda_j G\mathbf{v}_j. \quad (1446)$$

We report

$$\Lambda_{\max} = \max_j |\Lambda_j|, \quad \Lambda_{\text{rms}} = \left[\frac{1}{D} \sum_{j=1}^D \Lambda_j^2 \right]^{1/2}, \quad (1447)$$

and

$$\mathcal{A}_M = \left\| G^{-1/2} \left[M - \frac{\text{Tr}(G^{-1}M)}{D} G \right] G^{-1/2} \right\|_{\text{F}}. \quad (1448)$$

A trace cancellation without small $\Lambda_{\max}$ and $\mathcal{A}_M$ is not accepted as multidirectional flattening. For a normalized cluster DOS with finite square integral, define

$$W_{g,C}^{\text{eff}} = \left[\int_{\mathbb{R}} g_C^2(E) dE \right]^{-1}. \quad (1449)$$

Cauchy–Schwarz gives $W_{\varrho,C}^{\text{eff}} \leq W_C$. Define the DOS concentration ratio

$$\eta_{\varrho,C} = \frac{W_{\varrho,C}^{\text{eff}}}{W_C}, \quad 0 < \eta_{\varrho,C} \leq 1. \quad (1450)$$

An order-one value indicates spectral weight distributed through the complete narrow band. A very small value indicates a sharp singularity embedded in a considerably wider band. Near a nondegenerate stationary point in a D -dimensional generalized spectral domain,

$$E - E_* = \frac{1}{2} x^\top M x + O(|x|^3). \quad (1451)$$

For a positive-definite Hessian,

$$\varrho(E) \propto \frac{(E - E_*)^{D/2-1}}{\sqrt{\det(G^{-1}M)}}. \quad (1452)$$

A soft Hessian increases the stationary-point DOS prefactor, but the global bandwidth and gap must still be checked independently. We collect the complete physical information into

$$\mathbf{O}_C(\theta) = \left(W_C, \Delta_C, v_{\max,C}, \Lambda_{\max,C}, \Lambda_{\text{rms},C}, \varrho_{C,\eta}^{\text{coh,max}}, \eta_{\varrho,C}, C_C^{\text{coh}}, \mathcal{C}_{\perp,C}, \chi_{\perp,C}, N_{\text{res},C}, D_{\text{spec}}^{(\theta)}, D_{\mathcal{T}}^{\text{shape}} \right). \quad (1453)$$

No component is inferred from another except through the rigorous one-way implication

$$v_{\max,C} \rightarrow 0 \implies W_C \rightarrow 0 \implies \varrho_C^{\max} \rightarrow \infty. \quad (1454)$$

The reverse implications do not hold, and $\Delta_C > 0$ remains independent. For a fixed (κ, ω) , define

$$\begin{aligned} \theta_W &\in \arg \min_{\theta} W_C(\theta), \\ \theta_v &\in \arg \min_{\theta} v_{\max,C}(\theta), \\ \theta_2 &\in \arg \min_{\theta} \Lambda_{\max,C}(\theta), \\ \theta_{\varrho} &\in \arg \max_{\theta} \varrho_{C,\eta}^{\text{coh,max}}(\theta), \end{aligned} \quad (1455)$$

and

$$\theta_F \in \arg \max_{\theta} \frac{\Delta_C(\theta)}{W_C(\theta)}. \quad (1456)$$

When each optimum is unique, define the synchronization spread

$$\Delta\theta_{\text{sync}} = \max \{ \theta_W, \theta_v, \theta_2, \theta_{\varrho}, \theta_F \} - \min \{ \theta_W, \theta_v, \theta_2, \theta_{\varrho}, \theta_F \}. \quad (1457)$$

A small value supports a common flattening mechanism. A large value shows that different observables optimize at different twists. It does not invalidate the operational optimum, because the physical score is a constrained multi-observable balance rather than a requirement that all individual extrema coincide.

D. Physical flat-band angle

We now define the physical flat-band angle from the complete multi-observable landscape rather than from the minimum bandwidth alone. Let $\mathcal{M}_C(\theta; \kappa, \omega)$ be the operational physical score defined in Sec. VI. Let $\mathfrak{C}_{\text{adm}}(\theta; \kappa, \omega)$ be the set of clusters satisfying the tracking, isolation, representation, cover, shell, coherence, and hybridization conditions. The parameter-level physical flatness landscape is

$$\mathcal{M}_{\text{flat}}(\theta; \kappa, \omega) = \max_{C \in \mathfrak{C}_{\text{adm}}} \mathcal{M}_C(\theta; \kappa, \omega). \quad (1458)$$

The physical flat-band angle set is

$$\Theta_{\text{flat}}(\kappa, \omega) = \arg \max_{\theta \in \Theta_{\text{tw}}} \mathcal{M}_{\text{flat}}(\theta; \kappa, \omega). \quad (1459)$$

If the maximizer is unique,

$$\Theta_{\text{flat}}(\kappa, \omega) = \{ \theta_{\text{flat}}(\kappa, \omega) \}. \quad (1460)$$

If several angles are indistinguishable within the complete uncertainty, we report a discrete set or interval rather than forcing a unique value. The angle θ_{flat} is generally different from

$$\theta_W = \arg \min_{\theta} W, \quad (1461)$$

from a local Hessian-cancellation angle, and from the angle maximizing the raw DOS. For one smooth branch C_ν , define

$$L_\nu = \ln \mathcal{M}_\nu. \quad (1462)$$

An interior candidate branch satisfies

$$F_\nu(\theta, \kappa, \omega) := \frac{\partial L_\nu}{\partial \theta} = 0. \quad (1463)$$

It is a local maximum when

$$F_{\nu,\theta} = \frac{\partial^2 L_\nu}{\partial \theta^2} < 0. \quad (1464)$$

Define

$$V_C = \hbar D_G v_{\max,C}, \quad X_C = E_{\text{kin}} \varrho_{C,\eta}^{\text{coh,max}}, \quad C_C = C_C^{\text{coh}}. \quad (1465)$$

The exact stationary condition is

$$\begin{aligned} 0 = & -\frac{\Delta \partial_\theta W - W \partial_\theta \Delta}{\Delta(\Delta + W)} - \frac{\Delta \partial_\theta V - V \partial_\theta \Delta}{\Delta(\Delta + V)} \\ & + \frac{1}{1+X} \partial_\theta \ln \varrho_{C,\eta}^{\text{coh,max}} + \partial_\theta \ln C_C. \end{aligned}$$

A bandwidth minimum is neither necessary nor sufficient for this equation. At a boundary optimum, $\theta = 0$ requires the appropriate one-sided condition $F_\nu(0) \leq 0$, whereas $\theta = \theta_{\max}$ requires $F_\nu(\theta_{\max}) \geq 0$. The previous chapter defined a conservative lower score

$$\mathcal{M}_C^L(\theta) = \mathcal{M}_C^{\text{cert}}(\theta). \quad (1466)$$

TABLE III. Quantitative classification of twist-induced spectral reconstruction. Every classification refers to one continuously tracked spectral cluster.

Regime	Required behavior	Interpretation
Pure spectral displacement	$D_{\text{spec}}^{(\theta)} > 0$, no joint compression	Spectrum changes but does not flatten
Avoided-crossing reconstruction	$g_{\text{min}} > 0$, $\mathcal{H}_{ab} > 0$	Symmetry-allowed interlayer level repulsion
DOS-only concentration	$\varrho_{\text{max}} \uparrow$, $W \downarrow$	Local van Hove or critical-point reconstruction
Bandwidth-dominated narrowing	$W \downarrow$, weak Δ response	Global energy compression without strong gap gain
Gap-dominated isolation	$\Delta \uparrow$, weak W response	Improved isolation without substantial flattening
Joint isolation and compression	$W \downarrow$, $\Delta \uparrow$, $v_{\text{max}} \downarrow$	Strong physical flat-band tendency
First-order stationary only	v_{max} small locally, Λ_{max} finite	Residual quadratic dispersion remains
Strong physical flattening	W/Δ , $E_{\text{disp}}^{(1)}/\Delta$, $E_{\text{disp}}^{(2)}/\Delta$ all small	Narrow, isolated, multidirectionally soft band

We also define an upper score using the favorable ends of every uncertainty interval:

$$\begin{aligned} W_C^L &= \max\{0, \tilde{W}_C - \varepsilon_W\}, & \Delta_C^U &= \tilde{\Delta}_C + \varepsilon_\Delta, \\ v_C^L &= \max\{0, \tilde{v}_C - \varepsilon_v\}, & \varrho_C^U &= \tilde{\varrho}_C + \varepsilon_\varrho, \end{aligned} \quad (1467)$$

and

$$C_C^U = \min\{1, \tilde{C}_C + \varepsilon_{\text{coh}}\}. \quad (1468)$$

Monotonicity of the physical functional gives

$$\mathcal{M}_C^L(\theta) \leq \mathcal{M}_C(\theta) \leq \mathcal{M}_C^U(\theta). \quad (1469)$$

Define

$$M_{\text{max}}^L = \sup_{\theta} \mathcal{M}_{\text{flat}}^L(\theta). \quad (1470)$$

Every true maximizing angle must belong to

$$\hat{\Theta}_{\text{flat}}^{\text{cert}} = \{\theta \in \Theta_{\text{tw}} : \mathcal{M}_{\text{flat}}^U(\theta) \geq M_{\text{max}}^L\}. \quad (1471)$$

Indeed, if

$$\mathcal{M}_{\text{flat}}^U(\theta) < M_{\text{max}}^L, \quad (1472)$$

then some other angle has a established physical score larger than every possible true score at θ , and θ cannot be a global optimizer. The established angular uncertainty is

$$\delta\theta_{\text{flat}}^{\text{cert}} = \frac{1}{2} \text{diam } \hat{\Theta}_{\text{flat}}^{\text{cert}}. \quad (1473)$$

Suppose the exact logarithmic score has a unique maximum θ_{flat} and satisfies

$$L(\theta_{\text{flat}}) - L(\theta) \geq \frac{\mu_\theta}{2} |\theta - \theta_{\text{flat}}|^2 \quad (1474)$$

in a neighborhood of the optimum, with $\mu_\theta > 0$. If a numerical score $\tilde{L}$ satisfies

$$\sup_{\theta} |\tilde{L}(\theta) - L(\theta)| \leq \varepsilon_L, \quad (1475)$$

and $\hat{\theta}_{\text{flat}}$ maximizes $\tilde{L}$, then

$$L(\theta_{\text{flat}}) - L(\hat{\theta}_{\text{flat}}) \leq 2\varepsilon_L. \quad (1476)$$

Therefore

$$|\hat{\theta}_{\text{flat}} - \theta_{\text{flat}}| \leq 2\sqrt{\frac{\varepsilon_L}{\mu_\theta}}. \quad (1477)$$

For cover N , define $\Theta_{\text{flat},N}(\kappa, \omega)$. A bulk physical angle requires Hausdorff convergence

$$d_H[\Theta_{\text{flat},N}, \Theta_{\text{flat},\infty}] \longrightarrow 0 \quad (1478)$$

with increasing injectivity radius, vanishing shell error, complete finite-level representation coverage, and independent no-pollution control. For two inequivalent cover towers A and B , we additionally require

$$d_H[\Theta_{\text{flat},N}^{(A)}, \Theta_{\text{flat},M}^{(B)}] \longrightarrow 0. \quad (1479)$$

E. $w = 0$ negative control

The zero-interlayer-coupling calculation is an exact null test of every twist-induced claim. At $\omega = 0$, the Hamiltonian is

$$\mathcal{H}(\kappa, \theta, 0) = \begin{pmatrix} A(\kappa) & 0 \\ 0 & A(\kappa) \end{pmatrix}. \quad (1480)$$

It is independent of twist:

$$\partial_\theta \mathcal{H}(\kappa, \theta, 0) = 0. \quad (1481)$$

Consequently,

$$\text{Spec } \mathcal{H}(\kappa, \theta, 0) = \text{Spec } A(\kappa) \cup \text{Spec } A(\kappa), \quad (1482)$$

and every scalar spectral observable is twist independent:

$$\partial_\theta (W, \Delta, v_{\text{max}}, \Lambda_{\text{max}}, \varrho, \mu_m) = 0. \quad (1483)$$

The spectral reconstruction distance vanishes:

$$D_{\text{spec}}^{(\theta)}(\theta; \kappa, 0) = 0. \quad (1484)$$

The twist spectral current vanishes:

$$J_\theta(E, \theta; \omega = 0) = 0. \quad (1485)$$

The restricted interlayer map is zero: $V_I = 0$, so

$$N_{\text{res}} = 0, \quad w_{\text{rms},I} = 0, \quad r_{\text{st}}(V_I) = 0. \quad (1486)$$

For the complete two-copy spectral projector,

$$P_C = \begin{pmatrix} P_{0,C} & 0 \\ 0 & P_{0,C} \end{pmatrix}, \quad (1487)$$

the interlayer projector coherence is $\mathcal{C}_{\perp,C} = 0$. The interlayer energy sensitivity is also zero: $\chi_{\perp,C} = 0$ after the exactly degenerate layer pair is treated as one complete cluster. An arbitrary basis rotation inside the twofold layer degeneracy can produce a layer-balanced individual eigenvector at $\omega = 0$. Such a vector is not a basis-independent hybridization signal. The complete projector remains block diagonal and $\mathcal{C}_{\perp,C} = 0$. The hybridization compression ratios satisfy

$$R_{W,C}^{\text{hyb}} = 1, \quad R_{v,C}^{\text{hyb}} = 1. \quad (1488)$$

Therefore the hybridization-causality condition fails exactly. A monolayer may possess an intrinsic flat band. Such a band can retain a high spectral flatness score at $\omega = 0$, but it is classified as an intrinsic monolayer flat band rather than a twist-induced bilayer magic state. Write

$$\mathcal{H}(\omega) = \mathcal{H}_0 + \omega \mathcal{V}_{\perp}. \quad (1489)$$

For a nondegenerate layer-polarized eigenstate,

$$\langle n | \mathcal{V}_{\perp} | n \rangle = 0, \quad (1490)$$

and the leading shift is

$$e_n(\omega) = e_n(0) + \omega^2 \sum_{m \neq n} \frac{|\langle m | \mathcal{V}_{\perp} | n \rangle|^2}{e_n(0) - e_m(0)} + O(\omega^4). \quad (1491)$$

Thus off-resonant reconstruction begins at

$$\delta E_{\text{off}} = O\left(\frac{w^2}{t}\right). \quad (1492)$$

Inside an exactly degenerate layer-pair subspace, we diagonalize $P \mathcal{V}_{\perp} P$. The splitting is linear:

$$\Delta E_{\text{res}} = O(w). \quad (1493)$$

A coupling sweep must therefore recover $\Delta_{\text{AC}} \propto \omega$ for resonant avoided crossings and $\delta W_{\text{off}} \propto \omega^2$ for off-resonant self-energy reconstruction. We define the zero-coupling numerical null residual

$$\varepsilon_0^{\text{tw}} = \max_{\theta \in \Theta_{\text{tw}}} \left\{ \frac{D_{\text{spec}}^{(\theta)}(\theta; \kappa, 0)}{E_{\text{kin}}}, \frac{|W(\theta) - W(0)|}{E_{\text{kin}}}, \frac{|\Delta(\theta) - \Delta(0)|}{E_{\text{kin}}}, \frac{|v_{\text{max}}(\theta) - v_{\text{max}}(0)|}{v_{\text{ref}}} \right\}. \quad (1494)$$

A valid implementation requires

$$\varepsilon_0^{\text{tw}} \leq \varepsilon_0^{\text{tol}}. \quad (1495)$$

Failure identifies a geometry, basis, truncation, labeling, or numerical error before any flat-band conclusion is considered.

F. Curvature dependence

The fixed-fiber Hamiltonian is

$$\mathcal{H} = \mathcal{H}(\kappa, \theta, \omega). \quad (1496)$$

Its curvature derivative is

$$\mathcal{H}_{\kappa} = \begin{pmatrix} A_{\kappa} & \omega \mathcal{T}_{\kappa} \\ \omega \mathcal{T}_{\kappa}^{\dagger} & A_{\kappa} \end{pmatrix}. \quad (1497)$$

Curvature changes the local Hamiltonian through A_{κ} and $\mathcal{T}_{\kappa}$. In parallel, exact commensurability changes the global representation structure:

$$(\kappa, \theta) \mapsto \Gamma_{\text{M}} \mapsto q_{\text{M}} \mapsto g_{\text{M}} \mapsto \mathcal{B}_d. \quad (1498)$$

A jump of q_{M} changes the exact cell and representation space, but it does not imply a discontinuity of the local fixed-fiber Hamiltonian. Define

$$\begin{aligned} \delta_{\kappa\theta} \mathcal{H} &= \mathcal{H}(\kappa, \theta, \omega) - \mathcal{H}(\kappa, 0, \omega) \\ &\quad - \mathcal{H}(0, \theta, \omega) + \mathcal{H}(0, 0, \omega). \end{aligned} \quad (1499)$$

In the co-rotating representation, the intralayer blocks cancel:

$$\delta_{\kappa\theta} \mathcal{H} = \omega \begin{pmatrix} 0 & \Delta_{\kappa\theta} \mathcal{T} \\ \Delta_{\kappa\theta} \mathcal{T}^{\dagger} & 0 \end{pmatrix}, \quad (1500)$$

where

$$\begin{aligned} \Delta_{\kappa\theta} \mathcal{T} &= \mathcal{T}(\kappa, \theta) - \mathcal{T}(\kappa, 0) \\ &\quad - \mathcal{T}(0, \theta) + \mathcal{T}(0, 0). \end{aligned} \quad (1501)$$

Therefore

$$\|\delta_{\kappa\theta} \mathcal{H}\| = |\omega| \|\Delta_{\kappa\theta} \mathcal{T}\|. \quad (1502)$$

Let ν_{full} be the spectrum of $\mathcal{H}(\kappa, \theta, \omega)$, and let ν_{add} be the spectrum of

$$\mathcal{H}_{\text{add}} = \mathcal{H}(\kappa, 0, \omega) + \mathcal{H}(0, \theta, \omega) - \mathcal{H}(0, 0, \omega).$$

The nonadditive spectral distance is

$$D_{\kappa\theta}^{\text{spec}} = W_1(\nu_{\text{full}}, \nu_{\text{add}}). \quad (1503)$$

It obeys

$$D_{\kappa\theta}^{\text{spec}} \leq |\omega| \|\Delta_{\kappa\theta} \mathcal{T}\|. \quad (1504)$$

For identical centered layers, scalar observables are even in twist. Therefore $\delta_{\kappa\theta} \mathcal{O} = O(\kappa\theta^2)$ near the Euclidean aligned point whenever the family is analytic. We also define the mixed observable responses

$$\begin{aligned} \delta_{\kappa\theta} W &= W(\kappa, \theta) - W(\kappa, 0) - W(0, \theta) + W(0, 0), \\ \delta_{\kappa\theta} \Delta &= \Delta(\kappa, \theta) - \Delta(\kappa, 0) - \Delta(0, \theta) + \Delta(0, 0), \end{aligned} \quad (1505)$$

and

$$\delta_{\kappa\theta} \ln \mathcal{M} = \ln \mathcal{M}(\kappa, \theta) - \ln \mathcal{M}(\kappa, 0) - \ln \mathcal{M}(0, \theta) + \ln \mathcal{M}(0, 0). \quad (1506)$$

A nonzero value isolates curvature-assisted twistrionic reconstruction from the sum of pure-curvature and pure-twist effects. For the same tracked cluster, define

$$R_W^K = \frac{W(\kappa, \theta, \omega)}{W(0, \theta, \omega)}, \quad R_\Delta^K = \frac{\Delta(\kappa, \theta, \omega)}{\Delta(0, \theta, \omega)}, \quad (1507)$$

when the denominator is nonzero,

$$R_v^K = \frac{v_{\max}(\kappa, \theta, \omega)}{v_{\max}(0, \theta, \omega)}, \quad (1508)$$

and

$$R_\rho^K = \frac{\rho_\eta^{\text{coh}, \max}(\kappa, \theta, \omega)}{\rho_\eta^{\text{coh}, \max}(0, \theta, \omega)}. \quad (1509)$$

The flatness-ratio enhancement is

$$R_F^K = \frac{[\Delta/W](\kappa, \theta, \omega)}{[\Delta/W](0, \theta, \omega)}. \quad (1510)$$

A uniform positive energy rescaling gives $R_F^K = 1$. Therefore $R_F^K \neq 1$ proves nonuniform spectral-shape reconstruction. Let

$$L_\nu(\theta, \kappa, \omega) = \ln \mathcal{M}_\nu(\theta, \kappa, \omega) \quad (1511)$$

and

$$F_\nu = \partial_\theta L_\nu. \quad (1512)$$

Suppose

$$F_\nu(\theta_\nu^*, \kappa, \omega) = 0 \quad (1513)$$

and

$$F_{\nu, \theta} = \partial_\theta^2 L_\nu \neq 0. \quad (1514)$$

The implicit-function theorem gives a unique smooth branch locally. Its curvature displacement is

$$\frac{\partial \theta_\nu^*}{\partial \kappa} = -\frac{\partial_\kappa \partial_\theta L_\nu}{\partial_\theta^2 L_\nu}. \quad (1515)$$

Similarly,

$$\frac{\partial \theta_\nu^*}{\partial \omega} = -\frac{\partial_\omega \partial_\theta L_\nu}{\partial_\theta^2 L_\nu}. \quad (1516)$$

A nonzero $\partial_\kappa \theta_\nu^*$ measures movement of the optimal angle. It does not imply an improvement of the optimized flat-band quality. Define the optimized branch score

$$\mathcal{M}_\nu^*(\kappa, \omega) = \mathcal{M}_\nu[\theta_\nu^*(\kappa, \omega), \kappa, \omega]. \quad (1517)$$

The envelope theorem gives

$$\frac{\partial \ln \mathcal{M}_\nu^*}{\partial \kappa} = \left. \frac{\partial \ln \mathcal{M}_\nu}{\partial \kappa} \right|_{\theta=\theta_\nu^*}. \quad (1518)$$

We classify a first-order pure shift when

$$\frac{\partial \theta_\nu^*}{\partial \kappa} \neq 0 \quad (1519)$$

but

$$\frac{\partial \ln \mathcal{M}_\nu^*}{\partial \kappa} = 0 \quad (1520)$$

within the established uncertainty. The optimized curvature-enhancement factor is

$$R_{\mathcal{M}, \nu}^K = \frac{\mathcal{M}_\nu^*(\kappa, \omega)}{\mathcal{M}_\nu^*(0, \omega)}. \quad (1521)$$

A curvature-enhanced flat band satisfies $R_{\mathcal{M}, \nu}^K > 1$ with a lower confidence bound exceeding one. The total score gain has the exact decomposition

$$\begin{aligned} & \ln \mathcal{M}_\nu[\theta_\nu^*(\kappa), \kappa] - \ln \mathcal{M}_\nu[\theta_\nu^*(0), 0] \\ &= \left\{ \ln \mathcal{M}_\nu[\theta_\nu^*(\kappa), \kappa] - \ln \mathcal{M}_\nu[\theta_\nu^*(0), \kappa] \right\} \\ &+ \left\{ \ln \mathcal{M}_\nu[\theta_\nu^*(0), \kappa] - \ln \mathcal{M}_\nu[\theta_\nu^*(0), 0] \right\}. \end{aligned} \quad (1522)$$

The first bracket is the reoptimization gain obtained by moving the angle after curvature is applied. The second is the intrinsic fixed-angle curvature response. We require the enhancement to survive: fixed bare t, w , fixed local second moment or coordination normalization, fixed total interlayer Frobenius norm, increasing finite-cover size, independent cover towers. Survival under fixed $\text{Tr } \mathcal{T} \mathcal{T}^\dagger$ proves that the enhancement is caused by the spatial and representation-space structure of the coupling rather than by a scalar increase in w . To determine whether curvature merely remaps a pre-existing Euclidean root, we define the fixed-root geometry-only reference. Let

$$\zeta = \frac{a\sqrt{\kappa}}{2}, \quad s = \sin\left(\frac{\theta}{2}\right). \quad (1523)$$

Assume that the relevant dimensionless spectral root α_ν^* and the normalized dominant tunnelling harmonic γ_0 are curvature independent. Define

$$\Lambda_\nu = \frac{2\pi}{n_c} \sqrt{\frac{d\alpha_\nu^*}{\gamma_0 \omega}}. \quad (1524)$$

The geometry-only branch satisfies

$$\text{arsinh}\left[\frac{\sinh \zeta}{s_\nu^*}\right] = \Lambda_\nu \zeta. \quad (1525)$$

Therefore

$$s_\nu^*(\kappa, \omega) = \frac{\sinh \zeta}{\sinh(\Lambda_\nu \zeta)}, \quad (1526)$$

and

$$\theta_\nu^{\text{geo}}(\kappa, \omega) = 2 \arcsin \left[\frac{\sinh \zeta}{\sinh(\Lambda_\nu \zeta)} \right]. \quad (1527)$$

The Euclidean limit is

$$\sin \left[\frac{\theta_\nu^{\text{geo}}(0, \omega)}{2} \right] = \Lambda_\nu^{-1}. \quad (1528)$$

To prove monotonicity, define

$$f(x) = x \coth x. \quad (1529)$$

Its derivative is

$$\begin{aligned} f'(x) &= \coth x - x \operatorname{csch}^2 x \\ &= \frac{\sinh x \cosh x - x}{\sinh^2 x} > 0 \end{aligned} \quad (1530)$$

for $x > 0$, because

$$\sinh x \cosh x = \frac{1}{2} \sinh(2x) > x. \quad (1531)$$

For $\Lambda_\nu > 1$,

$$\Lambda_\nu \coth(\Lambda_\nu \zeta) > \coth \zeta. \quad (1532)$$

Hence

$$\frac{d}{d\zeta} \ln s_\nu^* = \coth \zeta - \Lambda_\nu \coth(\Lambda_\nu \zeta) < 0. \quad (1533)$$

The geometry-only branch is single valued and strictly monotone. Therefore pure registry remapping cannot create an interior fold. For weak curvature,

$$\theta_\nu^{\text{geo}}(\kappa, \omega) = \theta_\nu^{\text{E}}(\omega) - \frac{a^2 \kappa}{12} \cot \left[\frac{\theta_\nu^{\text{E}}(\omega)}{2} \right] + O(\kappa^2 a^4). \quad (1534)$$

Equivalently, using $K = -\kappa$,

$$\theta_\nu^{\text{geo}}(K, \omega) = \theta_\nu^{\text{E}}(\omega) + \frac{a^2 K}{12} \cot \left[\frac{\theta_\nu^{\text{E}}(\omega)}{2} \right] + O(K^2 a^4). \quad (1535)$$

Within the fixed-root reference, negative curvature lowers the twist required to realize the same registry length. The physical branch need not obey this sign. Curvature can also modify the active tunnelling harmonic and the dimensionless spectral root. Let $\gamma_1(\kappa, \theta)$ be the active normalized tunnelling harmonic and let $\alpha_\nu^*(\kappa, \theta)$ be the dimensionless spectral root extracted from the full Hamiltonian. The reduced branch equation is

$$G_\nu(\theta, \kappa, \omega) = \operatorname{arsinh} \left[\frac{b}{s} \right] - \frac{2\pi\zeta}{n_c} \sqrt{\frac{d\alpha_\nu^*(\kappa, \theta)}{\gamma_1(\kappa, \theta)\omega}} = 0,$$

where

$$b = \sinh \zeta, \quad s = \sin \left(\frac{\theta}{2} \right), \quad c = \cos \left(\frac{\theta}{2} \right). \quad (1536)$$

An interior fold requires

$$G_\nu = 0, \quad G_{\nu, \theta} = 0. \quad (1537)$$

Direct differentiation gives the necessary condition

$$\partial_\theta \ln \left[\frac{\gamma_1}{\alpha_\nu^*} \right] = \frac{bc}{s\sqrt{s^2 + b^2} \operatorname{arsinh}(b/s)}. \quad (1538)$$

The right-hand side is strictly positive for $0 < \theta < \pi$. A fold therefore requires sufficiently strong angular reconstruction of the effective tunnelling-to-root ratio. Pure geometry with constant γ_1 and α_ν^* cannot satisfy the condition. Let

$$F_\nu = \partial_\theta L_\nu. \quad (1539)$$

A generic fold at $(\theta_c, \kappa_c, \omega_c)$ satisfies

$$F_\nu = 0, \quad F_{\nu, \theta} = 0, \quad F_{\nu, \kappa} \neq 0, \quad F_{\nu, \theta\theta} \neq 0. \quad (1540)$$

Let

$$x = \theta - \theta_c, \quad \delta\kappa = \kappa - \kappa_c. \quad (1541)$$

The local expansion is

$$F_\nu = F_{\nu, \kappa}^c \delta\kappa + \frac{1}{2} F_{\nu, \theta\theta}^c x^2 + O(|x|^3, |x\delta\kappa|, \delta\kappa^2). \quad (1542)$$

The stationary branches satisfy

$$x_\pm = \pm \sqrt{-\frac{2F_{\nu, \kappa}^c}{F_{\nu, \theta\theta}^c} \delta\kappa + O(\delta\kappa)}. \quad (1543)$$

They exist on the side for which

$$-\frac{F_{\nu, \kappa}^c \delta\kappa}{F_{\nu, \theta\theta}^c} > 0. \quad (1544)$$

The branch with

$$F_{\nu, \theta} \simeq F_{\nu, \theta\theta}^c x < 0 \quad (1545)$$

is a local maximum, while the other is an unstable separating minimum. The branch separation is

$$\Delta\theta_{\text{fold}} = 2 \sqrt{-\frac{2F_{\nu, \kappa}^c}{F_{\nu, \theta\theta}^c} \delta\kappa + O(\delta\kappa)}. \quad (1546)$$

Thus

$$\Delta\theta_{\text{fold}} \propto |\kappa - \kappa_c|^{1/2}. \quad (1547)$$

We classify a stable branch as curvature born only when:

no corresponding stable branch exists at $\kappa = 0$,

$F_\nu = F_{\nu, \theta} = 0$ at the converged critical point,

$F_{\nu, \kappa} \neq 0, \quad F_{\nu, \theta\theta} \neq 0, \quad \kappa_c > 0,$

$\Delta_\nu^L > 0, \quad C_{\nu}^{\text{coh}, L} > 0, \quad |\theta_\nu^* - \theta_c| \propto |\kappa - \kappa_c|^{1/2},$

$\kappa_{c, N}, F_{\nu, \kappa}^{(N)}, F_{\nu, \theta\theta}^{(N)}$ converge across independent cover towers. (1548)

The increase in the number of stable branches is

$$N_{\text{flat}}(\kappa > \kappa_c) = N_{\text{flat}}(0) + 1 \quad (1549)$$

locally for a generic fold. A candidate can enter the admissible set without a stationary-point fold. A gap-mediated onset satisfies $\Delta \rightarrow 0^+$ while $F_\theta \neq 0$. It is an isolation boundary rather than creation of a new stationary extremum. A coherence-mediated termination satisfies $C^{\text{coh}} \rightarrow 0$ while the stationary equation can remain regular. A crystallographic termination occurs when

$$g_\theta \notin \text{Comm}_G(\Gamma), \quad (1550)$$

so the exact finite-index cell ceases to exist. The local fixed-fiber spectral branch may nevertheless remain smooth. A hierarchy switch occurs when two stable branches satisfy $\mathcal{M}_\mu^* = \mathcal{M}_\nu^*$ at distinct angles, $\theta_\mu^* \neq \theta_\nu^*$, and exchange global score ordering. The optimized score remains continuous, but the globally selected flat-band angle jumps.

For each stable branch, we record

$$\begin{aligned} \mathfrak{B}_\nu = & \left(\theta_\nu^*, \mathcal{M}_\nu^*, W_\nu, \Delta_\nu, v_{\text{max},\nu}, \Lambda_{\text{max},\nu}, \right. \\ & \varrho_\nu^{\text{coh,max}}, C_\nu^{\text{coh}}, \mathcal{C}_{\perp,\nu}, N_{\text{res},\nu}, \\ & R_{W,\nu}^K, R_{\Delta,\nu}^K, R_{v,\nu}^K, R_{\varrho,\nu}^K, R_{\mathcal{M},\nu}^K, \\ & \left. q_{\text{M},\nu}, d_{\rho,\nu}, r_{C,\nu}, F_{\nu,\theta}, \varepsilon_{\text{tot},\nu} \right). \end{aligned} \quad (1551)$$

For a branch-creation event, we additionally record

$$\mathfrak{F}_\nu = (\kappa_c, \theta_c, \omega_c, F_{\nu,\kappa}^c, F_{\nu,\theta\theta}^c, A_{\text{fold}}, \Delta_\nu^c, C_\nu^{\text{coh},c}), \quad (1552)$$

where

$$A_{\text{fold}} = \left| \frac{8F_{\nu,\kappa}^c}{F_{\nu,\theta\theta}^c} \right|. \quad (1553)$$

The measured two-branch separation must satisfy

$$(\Delta\theta_{\text{fold}})^2 = A_{\text{fold}}|\kappa - \kappa_c| + o(|\kappa - \kappa_c|). \quad (1554)$$

The coefficient is calculated from local derivatives rather than fitted freely. The complete causal chain is

$$\begin{aligned} (\kappa, \theta, \omega) & \longrightarrow \mathcal{T}(\kappa, \theta) \longrightarrow \mathfrak{G}_{\text{AC}} \longrightarrow P_C(\theta) \longrightarrow (W, \Delta, v_{\text{max}}, \Lambda_{\text{max}}, \varrho, C^{\text{coh}}) \\ & \longrightarrow \mathcal{M}_{\text{flat}}(\theta; \kappa, \omega) \longrightarrow \Theta_{\text{flat}}(\kappa, \omega) \\ & \longrightarrow \{\text{shift, enhancement, curvature-born branch}\}. \end{aligned} \quad (1555)$$

We accept a physical flat-band angle only when the same projector-tracked spectral family is globally narrow, positively isolated, multidirectionally weakly dispersive, coherently extended, spectrally concentrated, genuinely interlayer hybridized, complete in the declared representation sense, and stable under angular refinement, hopping-shell extension, increasing injectivity radius, and independent finite-cover families.

VII. ABELIAN AND NON-ABELIAN MAGIC-ANGLE COMPARISON

We now perform the decisive comparison between the one-dimensional $U(1)$ representation theory and the complete representation-resolved spectrum of the hyperbolic moiré translation group. We ask whether the physical flat-band angle obtained from Abelian generalized Bloch theory, $\theta_{\text{flat}}^{U(1)}$, coincides with the angle obtained after all physically required non-Abelian Bloch sectors have been included, $\theta_{\text{flat}}^{\text{full}}$. The comparison is made at fixed

$$K, \quad \omega = \frac{w}{t}, \quad \frac{h}{a}, \quad \frac{\lambda_\perp}{a}, \quad (1556)$$

with the same microscopic hopping kernel, hopping cut-off, finite-cover family, solver tolerance, spectral broadening, target-band ancestry, and physical flatness functional. We do not compare an accurately converged Abelian calculation with an incompletely sampled non-Abelian calculation, or vice versa. Let the exact common hyperbolic translation group have the presentation

$$\Gamma_{\text{M}} = \left\langle a_1, b_1, \dots, a_{g_{\text{M}}}, b_{g_{\text{M}}} \left| \prod_{j=1}^{g_{\text{M}}} [a_j, b_j] = e \right. \right\rangle, \quad (1557)$$

where

$$[a, b] = aba^{-1}b^{-1}. \quad (1558)$$

For an exact normal finite cover, we write

$$Q_N = \Gamma_{\text{M}}/\Gamma_N, \quad \Gamma_N \triangleleft \Gamma_{\text{M}}, \quad (1559)$$

and denote the complete set of inequivalent irreducible unitary representations by $\widehat{Q}_N$. We separate it into

$$\widehat{Q}_N^{(1)} = \left\{ \rho \in \widehat{Q}_N : d_\rho = 1 \right\}, \quad (1560)$$

and

$$\widehat{Q}_N^{(>1)} = \left\{ \rho \in \widehat{Q}_N : d_\rho > 1 \right\}. \quad (1561)$$

We refer to the latter as the non-Abelian Bloch sectors, abbreviated as NABS. The regular-representation state fractions carried by the two families are

$$C_{\text{Ab},N} = \frac{1}{|Q_N|} \sum_{\rho \in \widehat{Q}_N^{(1)}} d_\rho^2 = \frac{|Q_N^{\text{ab}}|}{|Q_N|}, \quad (1562)$$

and

$$C_{\text{NA},N} = \frac{1}{|Q_N|} \sum_{\rho \in \widehat{Q}_N^{(>1)}} d_\rho^2 = 1 - C_{\text{Ab},N}. \quad (1563)$$

A small number of inequivalent higher-dimensional irreps can carry a large fraction of the physical Hilbert space because each irrep carries regular weight d_ρ^2 .

TABLE IV. Operational distinction between curvature-induced shift, enhancement, and creation of a new physical flat-band branch.

Outcome	established conditions	Physical conclusion
Regular shift only	$\partial_\kappa \theta^* \neq 0, \partial_\kappa \mathcal{M}^* = 0$	Curvature relocates the optimum without improving it
Shift and enhancement	$\partial_\kappa \theta^* \neq 0, R_{\mathcal{M}}^K > 1$	Curvature both moves and improves the flat band
Enhancement without leading shift	$\partial_\kappa \theta^* = 0, R_{\mathcal{M}}^K > 1$	Flat-band quality improves at the same angle
Geometry-only continuation	Single monotone $\theta_{\nu}^{\text{geo}}(\kappa)$	Registry remapping of an existing Euclidean root
Curvature-born branch	Converged finite- κ fold and $\Delta N_{\text{flat}} = 1$	A new stable flat-band optimum is created
Gap-mediated admissibility onset	Δ crosses zero, no fold	An existing stationary branch becomes isolated
Coherence termination	$C^{\text{coh}} \rightarrow 0$	A narrow branch loses bulk coherence
Hierarchy switch	Two stable branches exchange score order	Globally preferred angle jumps without branch creation

A. Abelian $U(1)$ calculation

Its dimension is

1. Complete Abelian generalized Brillouin torus

$$\dim H_{U(1)} = \nu_{\text{M}} = 2n_b q_{\text{M}}. \quad (1575)$$

Every one-dimensional unitary representation

Hermiticity follows from

$$\chi : \Gamma_{\text{M}} \longrightarrow U(1) \quad (1564)$$

$$h(\delta^{-1}) = h^\dagger(\delta). \quad (1576)$$

annihilates the commutator subgroup:

Indeed,

$$\chi([\Gamma_{\text{M}}, \Gamma_{\text{M}}]) = 1. \quad (1565)$$

$$\begin{aligned} H_{U(1)}^\dagger &= \sum_{\delta} h^\dagger(\delta) e^{-i\mathbf{k}^\top n(\delta)} \\ &= \sum_{\delta} h(\delta^{-1}) e^{i\mathbf{k}^\top n(\delta^{-1})} = H_{U(1)}. \end{aligned}$$

It therefore factors through the Abelianization

$$\Gamma_{\text{M}}^{\text{ab}} = \Gamma_{\text{M}} / [\Gamma_{\text{M}}, \Gamma_{\text{M}}] \simeq \mathbb{Z}^{2g_{\text{M}}}. \quad (1566)$$

We parameterize a character by

The Abelian eigenvalue problem is

$$\mathbf{k} = \left(k_{a_1}, k_{b_1}, \dots, k_{a_{g_{\text{M}}}}, k_{b_{g_{\text{M}}}} \right), \quad (1567)$$

$$H_{U(1)}(\mathbf{k}; \theta, K, \omega) |u_n^{U(1)}(\mathbf{k})\rangle = E_n^{U(1)}(\mathbf{k}) |u_n^{U(1)}(\mathbf{k})\rangle. \quad (1577)$$

with

For each Abelian momentum coordinate k_μ ,

$$\chi_{\mathbf{k}}(a_j) = e^{ik_{a_j}}, \quad \chi_{\mathbf{k}}(b_j) = e^{ik_{b_j}}. \quad (1568)$$

$$\partial_{k_\mu} H_{U(1)} = i \sum_{\delta} n_\mu(\delta) h(\delta) e^{i\mathbf{k}^\top n(\delta)}. \quad (1578)$$

Every coordinate is 2π -periodic:

$$k_\mu \sim k_\mu + 2\pi. \quad (1569)$$

The second derivative is

The complete Abelian generalized Brillouin space is therefore

$$\partial_{k_\mu} \partial_{k_\nu} H_{U(1)} = - \sum_{\delta} n_\mu(\delta) n_\nu(\delta) h(\delta) e^{i\mathbf{k}^\top n(\delta)}. \quad (1579)$$

$$\mathcal{B}_1 = \text{Hom}(\Gamma_{\text{M}}, U(1)) \simeq \mathbb{T}^{2g_{\text{M}}}. \quad (1570)$$

For a nondegenerate Abelian band,

Its real dimension is

$$\partial_{k_\mu} E_n^{U(1)} = \langle u_n^{U(1)} | \partial_{k_\mu} H_{U(1)} | u_n^{U(1)} \rangle. \quad (1580)$$

$$D_{\text{Ab}} = 2g_{\text{M}}. \quad (1571)$$

The exact Hessian is

Let

$$n : \Gamma_{\text{M}} \longrightarrow \mathbb{Z}^{2g_{\text{M}}} \quad (1572)$$

be the Abelianization map. Then

$$\chi_{\mathbf{k}}(\delta) = \exp[i\mathbf{k}^\top n(\delta)]. \quad (1573)$$

For the matrix-valued microscopic hopping kernel $h(\delta; \theta, K, \omega)$, the Abelian generalized Bloch Hamiltonian is

Let G_{Ab} be the restriction of the canonical Hodge metric to the Abelian character torus. The complete Abelian generalized velocity is

$$H_{U(1)}(\mathbf{k}; \theta, K, \omega) = \sum_{\delta \in \Gamma_{\text{M}}} h(\delta; \theta, K, \omega) e^{i\mathbf{k}^\top n(\delta)}. \quad (1574)$$

$$v_n^{U(1)}(\mathbf{k}) = \frac{1}{\hbar} \left\| dE_n^{U(1)}(\mathbf{k}) \right\|_{G_{\text{Ab}}^{-1}}. \quad (1582)$$

For a tracked cluster C ,

$$v_{\max,C}^{U(1)} = \sup_{\mathbf{k} \in \mathcal{B}_1} v_C^{U(1)}(\mathbf{k}). \quad (1583)$$

Let $\gamma_{\text{plot}} \subset \mathcal{B}_1$ be any one-dimensional visualization path. Since $\gamma_{\text{plot}} \subseteq \mathcal{B}_1$, the path bandwidth satisfies

$$W_C^{\text{path}} \leq W_C^{U(1)}. \quad (1584)$$

Likewise,

$$v_{\max,C}^{\text{path}} \leq v_{\max,C}^{U(1)}. \quad (1585)$$

Because a global gap is minimized over a larger domain,

$$\Delta_C^{U(1)} \leq \Delta_C^{\text{path}}. \quad (1586)$$

A one-dimensional path therefore systematically favors an apparent flat-band claim:

$$\mathcal{M}_C^{\text{path}} \text{ can exceed } \mathcal{M}_C^{U(1)}. \quad (1587)$$

We use path spectra only for visualization. Every Abelian flatness observable is evaluated over the complete $2g_M$ -dimensional torus. At a specified anchor point $(\mathbf{k}_0, \theta_0, K_0, \omega_0)$, we choose an isolated target cluster with Riesz projector

$$P_C^{U(1)}(\mathbf{k}_0) = \frac{1}{2\pi i} \oint_{\mathcal{C}} [z - H_{U(1)}(\mathbf{k}_0)]^{-1} dz. \quad (1588)$$

The cluster is continued throughout the torus and twist interval using projector overlap and contour-gap control. The global Abelian edges are

$$E_{C,-}^{U(1)} = \inf_{\mathbf{k} \in \mathcal{B}_1} \lambda_{\min} [P_C^{U(1)} H_{U(1)} P_C^{U(1)}], \quad (1589)$$

and

$$E_{C,+}^{U(1)} = \sup_{\mathbf{k} \in \mathcal{B}_1} \lambda_{\max} [P_C^{U(1)} H_{U(1)} P_C^{U(1)}]. \quad (1590)$$

The Abelian bandwidth is

$$W_C^{U(1)} = E_{C,+}^{U(1)} - E_{C,-}^{U(1)}. \quad (1591)$$

Let the lower and upper complementary Abelian families be fixed by the anchor ordering. Their global edges define

$$\Delta_{C,-}^{U(1)} = E_{C,-}^{U(1)} - E_{\text{below},+}^{U(1)}, \quad (1592)$$

and

$$\Delta_{C,+}^{U(1)} = E_{\text{above},-}^{U(1)} - E_{C,+}^{U(1)}. \quad (1593)$$

The signed Abelian isolation gap is

$$\Delta_C^{U(1)} = \min \left\{ \Delta_{C,-}^{U(1)}, \Delta_{C,+}^{U(1)} \right\}. \quad (1594)$$

The normalized Abelian cluster DOS is

$$\varrho_C^{U(1)}(E) = \frac{1}{r_C} \sum_{a \in C} \int_{\mathcal{B}_1} d\mu_H(\mathbf{k}) \delta \left[E - E_a^{U(1)}(\mathbf{k}) \right], \quad (1595)$$

where

$$d\mu_H(\mathbf{k}) = \frac{d^{2g_M} k}{(2\pi)^{2g_M}} \quad (1596)$$

is normalized Haar measure. The Abelian physical certificate is

$$\mathfrak{C}_C^{U(1)} = \left(W_C^{U(1)}, \Delta_C^{U(1)}, v_{\max,C}^{U(1)}, \Lambda_{\max,C}^{U(1)}, \varrho_{C,\eta}^{U(1),\text{coh,max}}, C_C^{U(1),\text{coh}}, \mathcal{C}_{\perp,C}^{U(1)}, \chi_{\perp,C}^{U(1)} \right). \quad (1597)$$

Let $\mathcal{P}_j \subset \mathcal{B}_1$ be a compact torus cell with center $\mathbf{k}_j$ and Hodge covering radius h_j . Define

$$L_{H,j} = \sup_{\mathbf{k} \in \mathcal{P}_j} \|DH_{U(1)}(\mathbf{k})\|_{G_{\text{Ab}}^{-1}}. \quad (1598)$$

For any $\mathbf{k} \in \mathcal{P}_j$,

$$\|H_{U(1)}(\mathbf{k}) - H_{U(1)}(\mathbf{k}_j)\| \leq L_{H,j} h_j. \quad (1599)$$

Weyl's inequality then gives

$$\left| E_n^{U(1)}(\mathbf{k}) - E_n^{U(1)}(\mathbf{k}_j) \right| \leq L_{H,j} h_j. \quad (1600)$$

For each cell, we obtain rigorous upper and lower edge bounds:

$$E_{C,+}^{U(1)} \leq \max_j \left[e_{C,+}^{U(1)}(\mathbf{k}_j) + L_{H,j} h_j \right], \quad (1601)$$

and

$$E_{C,-}^{U(1)} \geq \min_j \left[e_{C,-}^{U(1)}(\mathbf{k}_j) - L_{H,j} h_j \right]. \quad (1602)$$

Cells are recursively refined when their edge bounds overlap the current global extremal interval or when their projector-continuation bound exceeds the specified tolerance. The full-torus optimization terminates only when

$$\begin{aligned} W_C^{U(1),U} - W_C^{U(1),L} &\leq \varepsilon_W^{U(1)}, \\ \Delta_C^{U(1),U} - \Delta_C^{U(1),L} &\leq \varepsilon_{\Delta}^{U(1)}, \end{aligned} \quad (1603)$$

and the corresponding velocity and Hessian brackets satisfy their specified tolerances. The one-dimensional irreps of a finite quotient form a finite subset

$$\mathcal{X}_N^{U(1)} = \widehat{\mathcal{Q}}_N^{(1)} \subset \mathcal{B}_1. \quad (1604)$$

Define its Hodge fill distance by

$$h_{\text{Ab},N} = \sup_{\chi \in \mathcal{B}_1} \inf_{\psi \in \mathcal{X}_N^{U(1)}} d_G(\chi, \psi). \quad (1605)$$

A finite-quotient character calculation approximates the complete Abelian torus only if $h_{\text{Ab},N} \rightarrow 0$ and the sampled directions become asymptotically isotropic. Without this condition, agreement among the finite one-dimensional irreps is only a finite-quotient statement. If the target band is L_E -Lipschitz,

$$\left| W_{C,N}^{U(1)} - W_C^{U(1)} \right| \leq 2L_E h_{\text{Ab},N} + \varepsilon_{\text{spec},N}. \quad (1606)$$

For every admissible Abelian cluster, we compute the minimal and strong physical scores

$$\mathcal{M}_C^{U(1),(1)}(\theta; K, \omega), \quad \mathcal{M}_C^{U(1),(2)}(\theta; K, \omega). \quad (1607)$$

The complete Abelian landscape is

$$\mathcal{M}^{U(1)}(\theta; K, \omega) = \max_{C \in \mathfrak{C}_{\text{adm}}^{U(1)}} \mathcal{M}_C^{U(1)}(\theta; K, \omega). \quad (1608)$$

The Abelian physical flat-band angle set is

$$\Theta_{\text{flat}}^{U(1)}(K, \omega) = \arg \max_{\theta \in \Theta_{\text{tw}}} \mathcal{M}^{U(1)}(\theta; K, \omega). \quad (1609)$$

For a unique established optimum,

$$\Theta_{\text{flat}}^{U(1)} = \left\{ \theta_{\text{flat}}^{U(1)} \right\}. \quad (1610)$$

We record

$$\mathfrak{B}_{\text{flat}}^{U(1)} = \left(\theta_{\text{flat}}^{U(1)}, \mathcal{M}_*^{U(1)}, W_*^{U(1)}, \Delta_*^{U(1)}, v_*^{U(1)}, \Lambda_*^{U(1)}, \varrho_*^{U(1)}, C_*^{U(1)}, \delta \theta_*^{U(1)} \right). \quad (1611)$$

B. Full non-Abelian spectrum

On the finite quotient Q_N , the full Hamiltonian acts on

$$\mathcal{H}_N = \ell^2(Q_N) \otimes \mathbb{C}^{\nu_M}. \quad (1612)$$

The regular representation decomposes according to

$$\mathbb{C}[Q_N] \simeq \bigoplus_{\rho \in \widehat{Q}_N} V_\rho \otimes V_\rho^*. \quad (1613)$$

The exact finite Hamiltonian is therefore

$$H_N \simeq \bigoplus_{\rho \in \widehat{Q}_N} [I_{d_\rho} \otimes H_N[\rho]]. \quad (1614)$$

Each reduced block is

$$H_N[\rho] = \sum_{\delta \in Q_N} h_N(\delta) \otimes \rho(\delta), \quad (1615)$$

with dimension

$$D_\rho = \nu_M d_\rho. \quad (1616)$$

Every block eigenvalue occurs in the regular spectrum with multiplicity d_ρ . The degree-square identity

$$\sum_{\rho \in \widehat{Q}_N} d_\rho^2 = |Q_N| \quad (1617)$$

implies

$$\sum_{\rho} d_\rho D_\rho = \nu_M |Q_N| = \dim \mathcal{H}_N. \quad (1618)$$

Let

$$P_\rho(z) = \det [z I_{D_\rho} - H_N[\rho]] \quad (1619)$$

be the reduced characteristic polynomial. The full regular characteristic polynomial factorizes exactly:

$$P_N^{\text{reg}}(z) = \prod_{\rho \in \widehat{Q}_N} P_\rho(z)^{d_\rho}. \quad (1620)$$

Its Abelian factor is

$$P_N^{U(1)}(z) = \prod_{\chi \in \widehat{Q}_N^{(1)}} P_\chi(z), \quad (1621)$$

and its genuinely non-Abelian factor is

$$P_N^{\text{NA}}(z) = \prod_{\rho \in \widehat{Q}_N^{(>1)}} P_\rho(z)^{d_\rho}. \quad (1622)$$

Thus

$$P_N^{\text{reg}} = P_N^{U(1)} P_N^{\text{NA}}. \quad (1623)$$

The eigenvalues of each block can be reconstructed from exact power sums without generating every explicit matrix representation. Define

$$p_m^{(\rho)} = \text{Tr } H_N[\rho]^m, \quad m = 1, \dots, D_\rho. \quad (1624)$$

Using the group-Fourier Hamiltonian,

$$p_m^{(\rho)} = \sum_{\delta_1, \dots, \delta_m \in Q_N} \text{Tr}_{\text{orb}} [h(\delta_1) \cdots h(\delta_m)] \times \chi_\rho(\delta_1 \cdots \delta_m), \quad (1625)$$

where

$$\chi_\rho(q) = \text{Tr } \rho(q). \quad (1626)$$

Let $e_k^{(\rho)}$ denote the elementary symmetric polynomials of the D_ρ block eigenvalues, with $e_0^{(\rho)} = 1$. Newton's identities give

$$k e_k^{(\rho)} = \sum_{j=1}^k (-1)^{j-1} e_{k-j}^{(\rho)} p_j^{(\rho)}, \quad 1 \leq k \leq D_\rho. \quad (1627)$$

The exact characteristic polynomial is

$$P_\rho(z) = \sum_{k=0}^{D_\rho} (-1)^k e_k^{(\rho)} z^{D_\rho-k}. \quad (1628)$$

This route is sufficient for eigenvalue-based quantities: W , Δ , ϱ , spectral moments, but it does not determine eigenvectors. Projector coherence, layer hybridization, Hodge derivatives, and local response functions require either explicit matrices $\rho(\delta)$ or the corresponding projectors constructed directly in the full regular representation. We identify the target family before comparing sectors. At a specified reference point, usually $\omega = 0$ or a weak-coupling point with an unambiguous mono-layer ancestry, define a target projector in every retained irrep:

$$P_{C,\rho}^{(0)} = \frac{1}{2\pi i} \oint_{\mathcal{C}_\rho^{(0)}} [z - H[\rho]^{(0)}]^{-1} dz. \quad (1629)$$

We continue $P_{C,\rho}(\theta, K, \omega)$ using fixed-rank Riesz continuation. The specified branch signature is

$$\mathfrak{s}_{C,\rho} = (\rho, d_\rho, \lambda_{\text{sym}}, r_{C,\rho}, \mathfrak{a}_{\text{orb}}, \mathfrak{a}_{\text{layer}}), \quad (1630)$$

where $\mathfrak{a}_{\text{orb}}$ and $\mathfrak{a}_{\text{layer}}$ encode the orbital and layer ancestry. A representation-completed target family is

$$\mathfrak{C}_N^{\text{full}} = \{C_\rho : \rho \in \mathcal{R}_{C,N}\}, \quad (1631)$$

where the support $\mathcal{R}_{C,N} \subseteq \widehat{Q}_N$ is specified from the anchor construction rather than selected after the spectrum is inspected. The number of physical regular-representation states in the family is

$$N_{C,N} = \sum_{\rho \in \mathcal{R}_{C,N}} d_\rho r_{C,\rho}. \quad (1632)$$

Its fraction of the finite Hilbert space is

$$f_{C,N} = \frac{N_{C,N}}{\nu_M |Q_N|}. \quad (1633)$$

The representation-completeness fraction relative to the declared support is

$$C_{\text{rep},C,N} = \frac{\sum_{\rho \in \mathcal{R}_{C,N}^{\text{computed}}} d_\rho r_{C,\rho}}{\sum_{\rho \in \mathcal{R}_{C,N}^{\text{declared}}} d_\rho r_{C,\rho}}. \quad (1634)$$

A full claim requires $C_{\text{rep},C,N} = 1$. The full lower and upper target edges are

$$E_{C,-}^{\text{full}} = \min_{\rho \in \mathcal{R}_{C,N}} \min_{a \in C_\rho} E_{a,\rho}, \quad (1635)$$

and

$$E_{C,+}^{\text{full}} = \max_{\rho \in \mathcal{R}_{C,N}} \max_{a \in C_\rho} E_{a,\rho}. \quad (1636)$$

The full target bandwidth is

$$W_C^{\text{full}} = E_{C,+}^{\text{full}} - E_{C,-}^{\text{full}}. \quad (1637)$$

This width includes both dispersion within each representation component and offsets among inequivalent representation sectors. Define the sector widths

$$W_{C,\rho} = E_{C,+}^{(\rho)} - E_{C,-}^{(\rho)}. \quad (1638)$$

The intersector offset contribution is

$$W_C^{\text{inter}} = W_C^{\text{full}} - \max_{\rho \in \mathcal{R}_{C,N}} W_{C,\rho}. \quad (1639)$$

It satisfies $W_C^{\text{inter}} \geq 0$. A family can be individually narrow in every irrep while remaining broad after the sector centers are combined. The anchor ordering partitions the complementary families into $\mathfrak{C}_{\text{below}}^{\text{full}}$ and $\mathfrak{C}_{\text{above}}^{\text{full}}$. Define

$$E_{\text{below},+}^{\text{full}} = \max_{\rho, a \in \mathfrak{C}_{\text{below}}^{\text{full}}} E_{a,\rho}, \quad (1640)$$

and

$$E_{\text{above},-}^{\text{full}} = \min_{\rho, a \in \mathfrak{C}_{\text{above}}^{\text{full}}} E_{a,\rho}. \quad (1641)$$

The full lower and upper gaps are

$$\Delta_{C,-}^{\text{full}} = E_{C,-}^{\text{full}} - E_{\text{below},+}^{\text{full}}, \quad (1642)$$

and

$$\Delta_{C,+}^{\text{full}} = E_{\text{above},-}^{\text{full}} - E_{C,+}^{\text{full}}. \quad (1643)$$

The full signed isolation gap is

$$\Delta_C^{\text{full}} = \min \{ \Delta_{C,-}^{\text{full}}, \Delta_{C,+}^{\text{full}} \}. \quad (1644)$$

The family is globally isolated from all Abelian and non-Abelian states only when $\Delta_C^{\text{full}} > 0$. For an individual sector candidate C_ρ , we distinguish the sector-conditioned gap $\Delta_{C,\rho}^{\text{sec}}$, computed only inside $H[\rho]$, from the global gap

$$\Delta_{C,\rho}^{\text{glob}} = \text{gap}[C_\rho, \text{Spec } H_N \setminus C_\rho]. \quad (1645)$$

A non-Abelian sector can contain an internally isolated flat band while states from another sector overlap the same energy interval. We call it a globally isolated non-Abelian flat band only when $\Delta_{C,\rho}^{\text{glob}} > 0$. The complete target DOS, with exact regular multiplicity, is

$$\varrho_{C,N}^{\text{full}}(E) = \frac{1}{N_{C,N}} \sum_{\rho \in \mathcal{R}_{C,N}} d_\rho \sum_{a \in C_\rho} \delta(E - E_{a,\rho}). \quad (1646)$$

It satisfies

$$\int_{\mathbb{R}} \varrho_{C,N}^{\text{full}}(E) dE = 1. \quad (1647)$$

The normalized NABS-only DOS is

$$\varrho_N^{\text{NA}}(E) = \frac{\sum_{\rho \in \tilde{Q}_N^{(>1)}} d_\rho \sum_{a=1}^{D_\rho} \delta(E - E_{a,\rho})}{\sum_{\rho \in \tilde{Q}_N^{(>1)}} d_\rho D_\rho}. \quad (1648)$$

The full DOS decomposes as

$$\varrho_N^{\text{full}} = C_{\text{Ab},N} \varrho_N^{U(1)} + C_{\text{NA},N} \varrho_N^{\text{NA}}. \quad (1649)$$

This equation uses the complete block DOS normalized inside each family. For each smooth representation component $\mathcal{D}_\rho \subset \mathcal{B}_{d_\rho}$, define

$$v_{\text{max},C,\rho} = \sup_{\lambda \in \mathcal{D}_\rho} \frac{1}{\hbar} \|dE_{C,\rho}(\lambda)\|_{G_\rho}^{-1}. \quad (1650)$$

Its first-order dispersion energy is

$$V_{C,\rho} = \hbar D_{G,\rho} v_{\text{max},C,\rho}. \quad (1651)$$

The full first-order scale is

$$V_C^{\text{full}} = \max_{\rho \in \mathcal{R}_{C,N}} V_{C,\rho}. \quad (1652)$$

At a stationary point in each continuous component, define

$$\Lambda_{\text{max},C,\rho} = \max_{a,j} |\Lambda_{a,j}^{(\rho)}|. \quad (1653)$$

The full second-order dispersion energy is

$$V_C^{(2),\text{full}} = \max_{\rho \in \mathcal{R}_{C,N}} \frac{1}{2} D_{G,\rho}^2 \Lambda_{\text{max},C,\rho}. \quad (1654)$$

A finite quotient supplies a discrete set of sectors rather than a continuous representation manifold. For locally neighboring sectors ρ and σ , we use the finite-difference diagnostic

$$\Omega_{C,N}^{(\rho\sigma)} = \frac{|E_{C,\rho} - E_{C,\sigma}|}{\hbar d_G(\rho, \sigma)}. \quad (1655)$$

We promote it to a continuous generalized-velocity claim only if the sector fill distance tends to zero and the directional sampling becomes asymptotically complete. Otherwise, it remains a finite-quotient dispersion diagnostic. For a representation-completed cluster, define

$$\begin{aligned} \Phi_{v,C}^{\text{full}} &= \frac{\Delta_C^{\text{full}}}{\Delta_C^{\text{full}} + V_C^{\text{full}}}, \\ \Phi_{W,C}^{\text{full}} &= \frac{\Delta_C^{\text{full}}}{\Delta_C^{\text{full}} + W_C^{\text{full}}}, \end{aligned} \quad (1656)$$

and, when the complete Hessian is available,

$$\Phi_{2,C}^{\text{full}} = \frac{\Delta_C^{\text{full}}}{\Delta_C^{\text{full}} + V_C^{(2),\text{full}}}. \quad (1657)$$

Let

$$X_C^{\text{full}} = E_{\text{kin}} \varrho_{C,\eta}^{\text{full,coh,max}}. \quad (1658)$$

The full minimal score is

$$\mathcal{M}_{C,N}^{\text{full,(1)}} = G_{\Delta,C}^{\text{full}} \left[\Phi_{W,C}^{\text{full}} \Phi_{v,C}^{\text{full}} \times \frac{X_C^{\text{full}}}{1 + X_C^{\text{full}}} C_{C,N}^{\text{full,coh}} \right]^{1/4}.$$

The full strong score is

$$\mathcal{M}_{C,N}^{\text{full,(2)}} = G_{\Delta,C}^{\text{full}} \left[\Phi_{W,C}^{\text{full}} \Phi_{v,C}^{\text{full}} \Phi_{2,C}^{\text{full}} \times \frac{X_C^{\text{full}}}{1 + X_C^{\text{full}}} C_{C,N}^{\text{full,coh}} \right]^{1/5}.$$

The complete finite-quotient landscape is

$$\mathcal{M}_N^{\text{full}}(\theta; K, \omega) = \max_{C \in \mathfrak{C}_{\text{adm},N}^{\text{full}}} \mathcal{M}_{C,N}^{\text{full}}(\theta; K, \omega). \quad (1659)$$

Its physical flat-band angle set is

$$\Theta_{\text{flat},N}^{\text{full}}(K, \omega) = \arg \max_{\theta} \mathcal{M}_N^{\text{full}}(\theta; K, \omega). \quad (1660)$$

A thermodynamic full angle is defined only when

$$\Theta_{\text{flat},N}^{\text{full}} \longrightarrow \Theta_{\text{flat}}^{\text{full}} \quad (1661)$$

in Hausdorff distance along independent admissible cover towers. For a unique limiting optimizer, $\Theta_{\text{flat}}^{\text{full}} = \{\theta_{\text{flat}}^{\text{full}}\}$. To detect a flat manifold that is absent from all one-dimensional sectors, define

$$\mathcal{M}_N^{\text{NA}}(\theta; K, \omega) = \max_{\substack{\rho \in \tilde{Q}_N^{(>1)} \\ C_\rho \in \mathfrak{C}_{\text{adm},\rho}^{\text{full}}}} \mathcal{M}_{C,\rho}^{\text{glob}}(\theta; K, \omega), \quad (1662)$$

where the score uses the global gap $\Delta_{C,\rho}^{\text{glob}}$, not merely the sector-conditioned gap. The NABS-only flat-angle set is

$$\Theta_{\text{flat},N}^{\text{NA}} = \arg \max_{\theta} \mathcal{M}_N^{\text{NA}}(\theta). \quad (1663)$$

A finite-sector result is promoted to a non-Abelian bulk claim only after its representation signature, global gap, score, and sector weight converge on inequivalent quotient towers.

C. Core comparison

We now compare

$$\Theta_{\text{flat}}^{U(1)} \stackrel{?}{=} \Theta_{\text{flat}}^{\text{full}}. \quad (1664)$$

The comparison is performed at the level of optimizer sets, observable vectors, target ancestry, and complete uncertainty budgets. A valid comparison requires

$$\begin{aligned} h_{U(1)}(\delta) &= h_{\text{full}}(\delta), \quad L_{U(1)} = L_{\text{full}}, \quad \eta_{U(1)} = \eta_{\text{full}}, \\ E_{\text{kin}}^{U(1)} &= E_{\text{kin}}^{\text{full}}, \quad \mathbf{a}_C^{U(1)} = \mathbf{a}_C^{\text{full}}, \end{aligned} \quad (1665)$$

the same cover, twist grid, and solver tolerances are used.

Here $\mathbf{a}_C$ denotes the specified orbital and layer ancestry of the target family. We distinguish two comparisons:

$$\text{sector completion : } C^{U(1)} \longrightarrow C^{\text{full}}, \quad (1666)$$

which follows the same target ancestry into higher-dimensional sectors, and

$$\text{global optimization : } \max_{C^{U(1)}} \mathcal{M}_C^{U(1)} \quad \text{versus} \quad \max_{C^{\text{full}}} \mathcal{M}_C^{\text{full}}. \quad (1667)$$

The first determines what NABS do to one Abelian candidate. The second determines which physical family is globally optimal after all sectors are included. Define the Hausdorff angle discrepancy

$$D_\theta^{\text{NA}} = d_H \left(\Theta_{\text{flat}}^{U(1)}, \Theta_{\text{flat}}^{\text{full}} \right). \quad (1668)$$

For unique optima,

$$D_\theta^{\text{NA}} = \left| \theta_{\text{flat}}^{\text{full}} - \theta_{\text{flat}}^{U(1)} \right|. \quad (1669)$$

Let the established optimizer sets be $\hat{\Theta}_{\text{flat}}^{U(1)}$ and $\hat{\Theta}_{\text{flat}}^{\text{full}}$. A nonzero displacement is established when

$$\text{dist} \left(\hat{\Theta}_{\text{flat}}^{U(1)}, \hat{\Theta}_{\text{flat}}^{\text{full}} \right) > 0. \quad (1670)$$

For unique numerical estimates, a sufficient criterion is

$$\left| \tilde{\theta}_{\text{flat}}^{\text{full}} - \tilde{\theta}_{\text{flat}}^{U(1)} \right| > \delta\theta_{\text{full}} + \delta\theta_{U(1)} + \delta\theta_{\text{cal}}. \quad (1671)$$

Define the dimensionless angular resolvability index

$$\mathcal{R}_\theta^{\text{NA}} = \frac{\left| \tilde{\theta}_{\text{flat}}^{\text{full}} - \tilde{\theta}_{\text{flat}}^{U(1)} \right|}{\delta\theta_{\text{full}} + \delta\theta_{U(1)} + \delta\theta_{\text{cal}}}. \quad (1672)$$

A value $\mathcal{R}_\theta^{\text{NA}} > 1$ certifies a resolvable NABS-induced shift. Consider one target family for which the Abelian states form a subset of the full representation-completed family: $C^{U(1)} \subseteq C^{\text{full}}$. The full set of complementary states also contains the Abelian complement: $\mathfrak{C}_{\text{comp}}^{U(1)} \subseteq \mathfrak{C}_{\text{comp}}^{\text{full}}$. It follows directly that

$$W_C^{\text{full}} \geq W_C^{U(1)}. \quad (1673)$$

The global isolation gap obeys

$$\Delta_C^{\text{full}} \leq \Delta_C^{U(1)}. \quad (1674)$$

If the Abelian domain is isometrically embedded in the full declared representation domain, then

$$V_C^{\text{full}} \geq V_C^{U(1)}, \quad V_C^{(2),\text{full}} \geq V_C^{(2),U(1)}. \quad (1675)$$

Define

$$\Phi_W^X = \frac{\Delta^X}{\Delta^X + W^X}, \quad X \in \{U(1), \text{full}\}. \quad (1676)$$

Equations (1673) and (1674) imply

$$\Phi_W^{\text{full}} \leq \Phi_W^{U(1)}. \quad (1677)$$

Similarly,

$$\Phi_v^{\text{full}} \leq \Phi_v^{U(1)}, \quad \Phi_2^{\text{full}} \leq \Phi_2^{U(1)}. \quad (1678)$$

Thus the Abelian calculation is necessarily optimistic with respect to the bandwidth, global gap, maximum first-order dispersion, and maximum second-order dispersion of the same representation-completed family. The full coherent DOS and coherence factors are not monotone because additional sectors can redistribute spectral weight and wave-function support. Define the target broadening generated by NABS:

$$B_C^{\text{NA}} = W_C^{\text{full}} - W_C^{U(1)} \geq 0. \quad (1679)$$

Define the gap erosion:

$$G_C^{\text{NA}} = \Delta_C^{U(1)} - \Delta_C^{\text{full}} \geq 0. \quad (1680)$$

Define the first- and second-order dispersion penalties:

$$D_{C,1}^{\text{NA}} = V_C^{\text{full}} - V_C^{U(1)} \geq 0, \quad (1681)$$

and

$$D_{C,2}^{\text{NA}} = V_C^{(2),\text{full}} - V_C^{(2),U(1)} \geq 0. \quad (1682)$$

The logarithmic kinematic completion penalty for the minimal functional is

$$\Pi_{\text{kin},C}^{\text{NA},(1)} = \ln \left(\frac{\Phi_{W,C}^{U(1)} \Phi_{v,C}^{U(1)}}{\Phi_{W,C}^{\text{full}} \Phi_{v,C}^{\text{full}}} \right) \geq 0. \quad (1683)$$

For the strong functional,

$$\Pi_{\text{kin},C}^{\text{NA},(2)} = \ln \left(\frac{\Phi_{W,C}^{U(1)} \Phi_{v,C}^{U(1)} \Phi_{2,C}^{U(1)}}{\Phi_{W,C}^{\text{full}} \Phi_{v,C}^{\text{full}} \Phi_{2,C}^{\text{full}}} \right) \geq 0. \quad (1684)$$

Let

$$\Phi_\ell^X = \frac{X_\ell^X}{1 + X_\ell^X}, \quad X_\ell^X = E_{\text{kin}} \varrho_\eta^{X,\text{coh,max}}. \quad (1685)$$

For the minimal score,

$$\ln \frac{\mathcal{M}_C^{\text{full},(1)}}{\mathcal{M}_C^{U(1),(1)}} = \frac{1}{4} \left[-\Pi_{\text{kin},C}^{\text{NA},(1)} + \ln \frac{\Phi_{\ell,C}^{\text{full}}}{\Phi_{\ell,C}^{U(1)}} + \ln \frac{C_C^{\text{full,coh}}}{C_C^{U(1),\text{coh}}} \right].$$

For the strong score,

$$\ln \frac{\mathcal{M}_C^{\text{full},(2)}}{\mathcal{M}_C^{U(1),(2)}} = \frac{1}{5} \left[-\Pi_{\text{kin},C}^{\text{NA},(2)} + \ln \frac{\Phi_{\ell,C}^{\text{full}}}{\Phi_{\ell,C}^{U(1)}} + \ln \frac{C_C^{\text{full,coh}}}{C_C^{U(1),\text{coh}}} \right].$$

These identities separate the unavoidable bandwidth-gap-dispersion penalty from the possible DOS and coherence response. Let

$$\theta_A = \theta_{\text{flat}}^{U(1)}, \quad \theta_F = \theta_{\text{flat}}^{\text{full}}. \quad (1686)$$

Define

$$L_A(\theta) = \ln \mathcal{M}^{U(1)}(\theta), \quad L_F(\theta) = \ln \mathcal{M}^{\text{full}}(\theta). \quad (1687)$$

The optimized score difference has the exact decomposition

$$\begin{aligned} L_F(\theta_F) - L_A(\theta_A) &= \underbrace{L_F(\theta_F) - L_F(\theta_A)}_{\mathcal{G}_{\text{reopt}}^{\text{NA}} \geq 0} \\ &+ \underbrace{L_F(\theta_A) - L_A(\theta_A)}_{\mathcal{R}_{\text{comp}}^{\text{NA}}}. \end{aligned} \quad (1688)$$

The first term is the gain obtained by reoptimizing the twist after NABS are included. The second term is the direct completion response at the Abelian optimum. A negative $\mathcal{R}_{\text{comp}}^{\text{NA}} < 0$ means that the Abelian optimum is degraded by NABS. A sufficiently large reoptimization gain can nevertheless produce a viable full optimum at a different twist. Suppose one full branch is a smooth completion of one nondegenerate Abelian branch. Write

$$L_F(\theta) = L_A(\theta) + \delta L_{\text{NA}}(\theta). \quad (1689)$$

At the Abelian optimum,

$$L'_A(\theta_A) = 0, \quad L''_A(\theta_A) < 0. \quad (1690)$$

Let

$$\theta_F = \theta_A + \delta\theta_{\text{NA}}. \quad (1691)$$

Expansion of

$$L'_F(\theta_F) = 0 \quad (1692)$$

gives

$$0 = L''_A(\theta_A)\delta\theta_{\text{NA}} + \delta L'_{\text{NA}}(\theta_A) + O(\delta\theta_{\text{NA}}^2, \delta\theta_{\text{NA}}\|\delta L_{\text{NA}}\|_{C^2}). \quad (1693)$$

Therefore

$$\delta\theta_{\text{NA}} = -\frac{\delta L'_{\text{NA}}(\theta_A)}{L''_A(\theta_A)} + O(\|\delta L_{\text{NA}}\|_{C^2}^2). \quad (1694)$$

The NABS-induced angle shift is controlled by the angular derivative of the completion response, not merely by its value. For the minimal physical functional,

$$\begin{aligned} \partial_\theta L &= \frac{1}{4} \left[\frac{W\partial_\theta \Delta - \Delta\partial_\theta W}{\Delta(\Delta + W)} + \frac{V\partial_\theta \Delta - \Delta\partial_\theta V}{\Delta(\Delta + V)} \right. \\ &\quad \left. + \frac{\partial_\theta X}{X(1 + X)} + \partial_\theta \ln C \right], \end{aligned} \quad (1695)$$

where

$$V = \hbar D_G v_{\text{max}}, \quad X = E_{\text{kin}} \varrho_\eta^{\text{coh, max}}. \quad (1696)$$

Subtraction of the Abelian and full versions of Eq. (1695) identifies whether the angle shift is dominated by target

broadening, gap erosion, residual NABS velocity, DOS redistribution, or coherence reconstruction. Define the one-sided full-spectrum completion distance

$$\delta_{\text{spec}, N}^{\text{NA}} = \sup_{E \in S_N^{\text{full}}} \text{dist} \left(E, S_N^{U(1)} \right). \quad (1697)$$

A positive value proves that some physical finite-cover eigenvalues are absent from the Abelian theory. For the target energy interval

$$I_C^{U(1)} = \left[E_{C,-}^{U(1)}, E_{C,+}^{U(1)} \right], \quad (1698)$$

define the NABS target intrusion

$$\delta_{C,\text{in}}^{\text{NA}} = \sup_{E \in S_C^{\text{NA}}} \text{dist} \left(E, I_C^{U(1)} \right). \quad (1699)$$

This quantity measures how far the non-Abelian completion extends beyond the Abelian target interval. Let $\mathcal{G}_C^{U(1)}$ be the open Abelian isolation window between the target and its nearest complementary family. The NABS gap-filling weight is

$$\mathcal{F}_{\text{gap}, C}^{\text{NA}} = \int_{\mathcal{G}_C^{U(1)}} \varrho_{N,\eta}^{\text{NA}}(E) dE. \quad (1700)$$

After subtracting the calculated Lorentzian leakage from peaks outside the gap, a nonzero value proves that NABS fill an Abelian spectral gap.

D. Why the magic mechanism is physically non-Abelian

The adjective non-Abelian refers to the translation holonomy carried by the generalized Bloch state. It does not refer to non-Abelian particle statistics, and it must not be conflated with a non-Abelian Berry connection of an internally degenerate energy band. Let two group words satisfy

$$w_1 w_2^{-1} \in [\Gamma_M, \Gamma_M]. \quad (1701)$$

They have the same Abelianization:

$$n(w_1) = n(w_2). \quad (1702)$$

Every one-dimensional character therefore assigns the same phase:

$$\chi_{\mathbf{k}}(w_1) = \chi_{\mathbf{k}}(w_2) \quad \text{for every } \mathbf{k}. \quad (1703)$$

A higher-dimensional representation can instead satisfy

$$\text{Tr } \rho(w_1) \neq \text{Tr } \rho(w_2). \quad (1704)$$

For the generalized Hamiltonian,

$$H[\rho] = \sum_{\delta} h(\delta) \otimes \rho(\delta), \quad (1705)$$

the m -th spectral moment is

$$\begin{aligned} \text{Tr } H[\rho]^m &= \sum_{\delta_1, \dots, \delta_m} \text{Tr}_{\text{orb}} [h(\delta_1) \cdots h(\delta_m)] \\ &\times \text{Tr } \rho(\delta_1 \cdots \delta_m). \end{aligned} \quad (1706)$$

Each closed hopping process is weighted by the Wilson character of its translation word. Abelian theory groups together all paths with the same Abelianization. NABS retain the ordering of noncommuting translations. Define the normalized moment mismatch

$$\mathcal{W}_m^{\text{NA}}(\rho) = \inf_{\chi \in \mathcal{B}_1} \frac{|D_\rho^{-1} \text{Tr } H[\rho]^m - D_\chi^{-1} \text{Tr } H[\chi]^m|}{E_{\text{kin}}^m}. \quad (1707)$$

If

$$\mathcal{W}_m^{\text{NA}}(\rho) > \varepsilon_m^{\text{mom}}, \quad (1708)$$

then no Abelian sector reproduces the normalized m -th moment of that NABS. Since identical finite spectra have identical moments of every order, one resolved moment mismatch is sufficient to certify a spectral difference. Consider two hopping paths p_1 and p_2 connecting the same initial and final orbitals. Let their group words be w_1 and w_2 , and let their scalar orbital amplitudes be t_1 and t_2 . Define the relative word $c = w_1^{-1}w_2$. Assume $c \in [\Gamma_M, \Gamma_M]$. The effective path amplitude in representation ρ is

$$\begin{aligned} A_\rho &= t_1 \rho(w_1) + t_2 \rho(w_2) \\ &= \rho(w_1) [t_1 I_{d_\rho} + t_2 \rho(c)]. \end{aligned} \quad (1709)$$

Because $\rho(w_1)$ is unitary,

$$\begin{aligned} \|A_\rho\|_{\text{F}}^2 &= d_\rho (|t_1|^2 + |t_2|^2) \\ &+ 2 \text{Re} [t_1^* t_2 \text{Tr } \rho(c)]. \end{aligned} \quad (1710)$$

Define the active commutator defect

$$\eta_c(\rho) = 1 - \frac{1}{d_\rho} \text{Re} \text{Tr } \rho(c) = \frac{\|I_{d_\rho} - \rho(c)\|_{\text{F}}^2}{2d_\rho}. \quad (1711)$$

It satisfies $0 \leq \eta_c(\rho) \leq 2$. For every Abelian character, $\eta_c(\chi) = 0$. For real positive t_1 and t_2 ,

$$\|A_\rho\|_{\text{F}}^2 = d_\rho (t_1 + t_2)^2 - 2d_\rho t_1 t_2 \eta_c(\rho). \quad (1712)$$

Non-Abelian commutator holonomy reduces the coherent hopping power relative to the Abelian prediction. Let the eigenvalues of $\rho(c)$ be $e^{i\phi_1}, \dots, e^{i\phi_{d_\rho}}$. Since $t_1 I + t_2 \rho(c)$ is a normal matrix, its singular values are

$$s_j = |t_1 + t_2 e^{i\phi_j}|, \quad j = 1, \dots, d_\rho. \quad (1713)$$

For equal path amplitudes, $t_1 = t_2 = t_p$, we obtain

$$s_j = 2|t_p| \left| \cos \left(\frac{\phi_j}{2} \right) \right|. \quad (1714)$$

Every Abelian character has $\phi = 0$, so the Abelian amplitude is $2|t_p|$. A non-Abelian eigenphase near π instead produces destructive interference and a strongly suppressed effective hopping channel:

$$\phi_j \simeq \pi \implies s_j \simeq 0. \quad (1715)$$

This is a direct microscopic route to a NABS-only narrow band. Conversely, let $t_1 = -t_2 = t_p$. Abelian theory predicts exact cancellation: $A_\chi = 0$. The NABS power is

$$\|A_\rho\|_{\text{F}}^2 = 2d_\rho |t_p|^2 \eta_c(\rho). \quad (1716)$$

Any active nontrivial commutator holonomy lifts the Abelian cancellation. This supplies a microscopic mechanism by which NABS broaden an Abelian narrow band or close an Abelian gap. Let the target sector be P_ρ and the active remote sector be $Q_\rho = I - P_\rho$. The exact Feshbach self-energy is

$$\Sigma_\rho(E, \lambda) = P_\rho H Q_\rho [E - Q_\rho H Q_\rho]^{-1} Q_\rho H P_\rho. \quad (1717)$$

The representation matrices enter both the coupling vertices and the remote spectrum. Therefore

$$\Sigma_\rho \neq \Sigma_\chi \quad (1718)$$

even when the radial hopping magnitudes are identical. The energy differential has the exact decomposition

$$dE_\rho = Z_\rho (dE_\rho^{\text{bare}} + dE_\rho^\Sigma). \quad (1719)$$

A NABS-specific stationary condition is

$$dE_\rho^\Sigma = -dE_\rho^{\text{bare}} \quad (1720)$$

as a covector over every physical tangent direction. The corresponding Abelian condition need not hold because scalar characters and matrix-valued holonomies generate different path interference and remote self-energies. Let

$$A_j = \rho(a_j), \quad B_j = \rho(b_j). \quad (1721)$$

Define the normalized generator noncommutativity

$$\mathcal{N}_\rho = \max_{i,j} \frac{\|A_i B_j - B_j A_i\|_{\text{F}}}{2\sqrt{d_\rho}}. \quad (1722)$$

It satisfies $0 \leq \mathcal{N}_\rho \leq 1$. Every Abelian sector has $\mathcal{N}_\chi = 0$. Let

$$\mathcal{C}_{\text{act}} \subset [\Gamma_M, \Gamma_M] \quad (1723)$$

be the set of commutator words that occur in hopping paths with amplitudes above the specified threshold. Define

$$\eta_{\text{act}}(\rho) = \max_{c \in \mathcal{C}_{\text{act}}} \eta_c(\rho). \quad (1724)$$

A representation may be mathematically non-Abelian but spectrally insensitive to the specific hopping paths of the device. We therefore require

$$\eta_{\text{act}}(\rho) > \varepsilon_{\text{hol}} \quad (1725)$$

rather than relying on $d_\rho > 1$ alone. Let

$$\mathcal{W}_{\text{comm}} \subset \Gamma_M \quad (1726)$$

be the hopping words responsible for the resolved moment or interference witness. Define the commutator-path ablation

$$H^{(-\text{comm})}[\rho] = \sum_{\delta \notin \mathcal{W}_{\text{comm}}} h(\delta) \otimes \rho(\delta). \quad (1727)$$

The non-Abelian causal score loss is

$$\Delta \mathcal{M}_{\text{comm}} = \mathcal{M}_C[\rho] - \mathcal{M}_C^{(-\text{comm})}[\rho]. \quad (1728)$$

We also construct a fixed-hybridization-norm control:

$$\hat{T}_\rho = \frac{T_{\text{ref}}}{\|T_\rho\|_F} T_\rho, \quad (1729)$$

so that

$$\|\hat{T}_\rho\|_F = T_{\text{ref}} \quad (1730)$$

in every sector. A flatness difference that survives this control is caused by the matrix organization and holonomy of the coupling rather than its total power. We classify a flat manifold as intrinsically translation-non-Abelian only when

$$d_\rho > 1, \quad \mathcal{N}_\rho > \varepsilon_N, \quad \eta_{\text{act}}(\rho) > \varepsilon_{\text{hol}}, \\ \max_{m \leq m_{\text{max}}} \mathcal{W}_m^{\text{NA}}(\rho) > \varepsilon_{\text{mom}}, \quad \Delta_{C,\rho}^{\text{glob},L} > 0, \quad \mathcal{M}_{C,\rho}^{\text{glob},L} > \mathcal{M}_{\text{acc}}, \quad \text{Therefore}$$

the effect survives fixed-norm, shell, and cover controls, no Abelian sector supplies an equivalent admissible branch. (1731)

The complete non-Abelian magic certificate is

$$\mathfrak{C}_C^{\text{NA}} = \left(\rho, d_\rho, C_{\text{NA},N}, \mathcal{N}_\rho, \eta_{\text{act}}(\rho), \{\mathcal{W}_m^{\text{NA}}\}, W_{C,\rho}, \right. \\ \Delta_{C,\rho}^{\text{glob}}, v_{\text{max},C,\rho}, \Lambda_{\text{max},C,\rho}, \varrho_{C,\rho}^{\text{coh,max}}, \mathcal{M}_{C,\rho}, \\ \left. \Delta \mathcal{M}_{\text{comm}}, C_{\text{rep},C,N}, \varepsilon_{\text{tot}} \right). \quad (1732)$$

E. Experimentally detectable differences

We now convert the Abelian–non-Abelian comparison into a phase-sensitive circuit-QED spectroscopy protocol. The finite superconducting resonator network realizes a normal quotient Q_N , so its representation sectors can be selected by spatially patterned microwave drives. We work in frequency units and define the effective damped resolvent

$$G^R(\Omega) = \left[\Omega I - H_N + \frac{i}{2} \Gamma \right]^{-1}, \quad (1733)$$

where $\Gamma \succeq 0$ contains internal and port-induced damping. For input and output ports coupled to normalized

device vectors $|f_{\text{in}}\rangle$ and $|f_{\text{out}}\rangle$, the resonant transmission contribution is

$$S_{\text{out,in}}(\Omega) = S_{\text{bg}} - i\sqrt{\kappa_{\text{out}}\kappa_{\text{in}}} \langle f_{\text{out}} | G^R(\Omega) | f_{\text{in}} \rangle. \quad (1734)$$

The drive-resolved spectral function is

$$A_f(\Omega) = -\frac{1}{\pi} \text{Im} \langle f | G^R(\Omega) | f \rangle. \quad (1735)$$

The complete broadened DOS is

$$\varrho_\Gamma^{\text{full}}(\Omega) = -\frac{1}{\pi D_N} \text{Im Tr } G^R(\Omega). \quad (1736)$$

Choose the finite-cover basis

$$|q, s\rangle, \quad q \in Q_N, \quad s = 1, \dots, \nu_M, \quad (1737)$$

where s labels the orbitals in one common cell. For an irrep ρ of dimension d_ρ , define the phase- and amplitude-coded drive

$$|F_{\rho;\alpha\beta,s}\rangle = \sqrt{\frac{d_\rho}{|Q_N|}} \sum_{q \in Q_N} \rho(q)_{\alpha\beta}^* |q, s\rangle, \quad (1738)$$

where $\alpha, \beta = 1, \dots, d_\rho$. Schur orthogonality gives

$$\sum_{q \in Q_N} \rho(q)_{\alpha\beta}^* \sigma(q)_{\gamma\delta} = \frac{|Q_N|}{d_\rho} \delta_{\rho\sigma} \delta_{\alpha\gamma} \delta_{\beta\delta}.$$

$$\langle F_{\rho;\alpha\beta,s} | F_{\sigma;\gamma\delta,s'} \rangle = \delta_{\rho\sigma} \delta_{\alpha\gamma} \delta_{\beta\delta} \delta_{ss'}. \quad (1739)$$

For $d_\rho = 1$, the drive reduces to the familiar Abelian phase pattern

$$|F_{\chi,s}\rangle = \frac{1}{\sqrt{|Q_N|}} \sum_{q \in Q_N} \chi(q)^* |q, s\rangle. \quad (1740)$$

For $d_\rho > 1$, the d_ρ^2 mutually orthogonal patterns span the complete ρ -isotypic translation sector for the chosen orbital. The normalized sector-resolved spectral function is

$$A_\rho(\Omega) = -\frac{1}{\pi \nu_M d_\rho^2} \sum_{s=1}^{\nu_M} \sum_{\alpha,\beta=1}^{d_\rho} \text{Im} \langle F_{\rho;\alpha\beta,s} | G^R(\Omega) | F_{\rho;\alpha\beta,s} \rangle. \quad (1741)$$

It is normalized:

$$\int_{\mathbb{R}} A_\rho(\Omega) d\Omega = 1. \quad (1742)$$

The full DOS is recovered from

$$\varrho_\Gamma^{\text{full}}(\Omega) = \sum_{\rho \in \hat{Q}_N} \frac{d_\rho^2}{|Q_N|} A_\rho(\Omega). \quad (1743)$$

For an imperfect experimental drive $|f\rangle$, the sector-selection fidelity is

$$\mathcal{F}_\rho^{\text{drive}} = \frac{\langle f|P_\rho|f\rangle}{\langle f|f\rangle}. \quad (1744)$$

A sector-resolved claim requires

$$\mathcal{F}_\rho^{\text{drive}} \geq 1 - \varepsilon_{\text{drive}}. \quad (1745)$$

Let the experimentally reconstructed complex mode amplitudes be $\psi_s(q)$. Define their normalized group Fourier transform

$$\tilde{\psi}_s(\rho) = \sqrt{\frac{d_\rho}{|Q_N|}} \sum_{q \in Q_N} \psi_s(q) \rho(q)^\dagger. \quad (1746)$$

Finite-group Plancherel theory gives

$$\sum_{q,s} |\psi_s(q)|^2 = \sum_{\rho,s} \|\tilde{\psi}_s(\rho)\|_F^2. \quad (1747)$$

The measured representation weight is

$$P_\rho^{\text{mode}} = \frac{\sum_s \|\tilde{\psi}_s(\rho)\|_F^2}{\sum_{\sigma,s} \|\tilde{\psi}_s(\sigma)\|_F^2}. \quad (1748)$$

A NABS-only mode requires

$$\sum_{\rho: d_\rho > 1} P_\rho^{\text{mode}} \geq 1 - \varepsilon_{\text{rep tom}}. \quad (1749)$$

From Abelian phase-coded drives, we reconstruct $\widetilde{\mathcal{M}}^{U(1)}(\theta)$ and obtain $\tilde{\theta}_{\text{flat}}^{U(1)}$. From the complete irreducible drive set and the exact $d_\rho^2/|Q_N|$ weights, we reconstruct $\widetilde{\mathcal{M}}^{\text{full}}(\theta)$ and obtain $\tilde{\theta}_{\text{flat}}^{\text{full}}$. Let $\delta\Omega_{\text{fit}}$ be the calibrated resonance-frequency uncertainty, and let κ_C and κ_{comp} be the fitted linewidths of the target and nearest complementary modes. A conservative experimentally resolved gap is

$$\Delta_{\text{exp}}^L = \tilde{\Delta} - \frac{\kappa_C^U + \kappa_{\text{comp}}^U}{2} - 2\delta\Omega_{\text{fit}}. \quad (1750)$$

Global isolation is spectroscopically resolved only when $\Delta_{\text{exp}}^L > 0$. The conservative measured bandwidth is

$$W_{\text{exp}}^U = \widetilde{W} + 2\delta\Omega_{\text{fit}}. \quad (1751)$$

The observed angle difference is resolved when

$$\left| \tilde{\theta}_{\text{flat}}^{\text{full}} - \tilde{\theta}_{\text{flat}}^{U(1)} \right| > \delta\theta_{\text{full}} + \delta\theta_{U(1)} + \delta\theta_{\text{cal}}. \quad (1752)$$

For Outcome I, all phase-coded Abelian and NABS measurements reproduce the same target optimum within uncertainty:

$$\tilde{\theta}_{\text{flat}}^{U(1)} \simeq \tilde{\theta}_{\text{flat}}^{\text{full}}, \quad (1753)$$

and no additional NABS resonance enters the target or gap windows. For Outcome II, the same tracked target ancestry remains visible, but the full weighted spectrum optimizes at a resolvably shifted angle:

$$\left| \tilde{\theta}_{\text{flat}}^{\text{full}} - \tilde{\theta}_{\text{flat}}^{U(1)} \right| > \delta\theta_{\text{tot}}. \quad (1754)$$

For Outcome III, an Abelian drive displays a narrow isolated resonance family, while one or more NABS drives reveal either additional target dispersion or spectral weight inside the Abelian gap. Define the leakage-subtracted experimental gap filling

$$\mathcal{F}_{\text{gap}}^{\text{exp,NA}} = \int_{\mathcal{G}_C^{U(1)}} \varrho_{\text{exp}}^{\text{NA}}(\Omega) d\Omega - \mathcal{F}_{\text{leak}}^U. \quad (1755)$$

A positive lower confidence bound certifies NABS gap filling. For Outcome IV, a narrow globally isolated resonance family appears under a $d_\rho > 1$ drive but is absent under every Abelian drive. Let $\mathcal{I}_*$ be the target frequency window and define

$$I_{\text{NA}} = \int_{\mathcal{I}_*} A_{\rho_*}(\Omega) d\Omega, \quad (1756)$$

and

$$I_{\text{Ab}}^{\text{max}} = \sup_{\chi \in \widehat{Q}_N^{(1)}} \int_{\mathcal{I}_*} A_\chi(\Omega) d\Omega. \quad (1757)$$

The NABS spectral contrast is

$$\mathcal{C}_{\text{spec}}^{\text{NA}} = \frac{I_{\text{NA}} - I_{\text{Ab}}^{\text{max}}}{I_{\text{NA}} + I_{\text{Ab}}^{\text{max}}}. \quad (1758)$$

A non-Abelian-only resonance has $\mathcal{C}_{\text{spec}}^{\text{NA}} \rightarrow 1$ after drive leakage and measurement noise are removed.

For the two-path process of Eq. (1709), the experimentally measured integrated response power in sector ρ is proportional to

$$\mathcal{I}_\rho \propto \|A_\rho\|_F^2. \quad (1759)$$

The Abelian reference is

$$\mathcal{I}_{U(1)} \propto d_\rho |t_1 + t_2|^2, \quad (1760)$$

where the trivial multiplicity d_ρ is inserted only to match the normalization. For real positive path amplitudes,

$$\mathcal{I}_{U(1)} - \mathcal{I}_\rho \propto 2d_\rho t_1 t_2 \eta_c(\rho). \quad (1761)$$

Thus a phase-sensitive transmission experiment directly measures the active commutator defect. Define the normalized commutator contrast

$$\mathcal{C}_c^{\text{exp}} = \frac{\mathcal{I}_{U(1)} - \mathcal{I}_\rho}{2d_\rho |t_1 t_2|}. \quad (1762)$$

In the ideal two-path limit,

$$\mathcal{C}_c^{\text{exp}} = \eta_c(\rho). \quad (1763)$$

TABLE V. Circuit-QED signatures of the four Abelian–non-Abelian outcomes. The full measurement uses all irreps with exact regular-representation weights.

Outcome	Sector-resolved spectroscopy	Full physical measurement
I	Abelian and NABS target extrema coincide	Same flat angle and gap within uncertainty
II	NABS extrema occur at shifted twist	Full score peaks at $\theta_{\text{flat}}^{\text{full}} \neq \theta_{\text{flat}}^{U(1)}$
III	NABS show extra dispersion or gap resonances	Abelian narrow band broadens or loses global isolation
IV	A $d_\rho > 1$ drive excites an isolated narrow family	No Abelian drive contains an admissible counterpart

Agreement between the measured contrast and the independently calculated Wilson character closes the causal link between noncommuting translation holonomy and the reconstructed band spectrum. Prepare a normalized irrep-selective wave packet $|\psi_\rho(0)\rangle$ whose spectral support lies entirely inside a target interval of width $W_{C,\rho}$. The survival amplitude is

$$\mathcal{A}_\rho(t) = \langle \psi_\rho(0) | e^{-iHt/\hbar} | \psi_\rho(0) \rangle. \quad (1764)$$

After removing the midpoint energy E_0 , every phase lies in

$$\left[-\frac{W_{C,\rho}t}{2\hbar}, \frac{W_{C,\rho}t}{2\hbar} \right]. \quad (1765)$$

For

$$0 \leq \frac{W_{C,\rho}t}{2\hbar} \leq \frac{\pi}{2}, \quad (1766)$$

the real part of every phase factor is at least $\cos(W_{C,\rho}t/2\hbar)$. Therefore

$$|\mathcal{A}_\rho(t)|^2 \geq \cos^2 \left(\frac{W_{C,\rho}t}{2\hbar} \right). \quad (1767)$$

A NABS-only flat band produces a long-lived survival plateau under the matching non-Abelian drive, whereas Abelian phase patterns dephase on a shorter timescale. The corresponding dephasing scale is

$$\tau_{\text{disp},\rho} \sim \frac{\hbar}{W_{C,\rho}}. \quad (1768)$$

Let transmon qubits at resonator sites i and j couple with strengths g_i and g_j . In the dispersive regime, the photon-mediated exchange interaction at a common qubit frequency ω_q is

$$J_{ij}(\omega_q) = g_i g_j \text{Re } G_{ij}^R(\omega_q). \quad (1769)$$

The corresponding collective dissipative coupling is

$$\Gamma_{ij}^{\text{med}}(\omega_q) = -2g_i g_j \text{Im } G_{ij}^R(\omega_q). \quad (1770)$$

The resolvent inherits the exact representation decomposition:

$$G_N^R \simeq \bigoplus_{\rho \in \widehat{Q}_N} [I_{d_\rho} \otimes G_\rho^R], \quad (1771)$$

where

$$G_\rho^R(\omega_q) = \left[\omega_q I - H[\rho] + \frac{i}{2} \Gamma_\rho \right]^{-1}. \quad (1772)$$

A NABS-induced change of the flat-band position, bandwidth, or gap therefore reconstructs the spatial interaction matrix J_{ij} . Define the Abelian-truncated interaction matrix by retaining only one-dimensional sectors:

$$J^{U(1)} = \mathcal{J} \left[\bigoplus_{\chi \in \widehat{Q}_N^{(1)}} G_\chi^R \right], \quad (1773)$$

and let J^{full} use the complete resolvent. The normalized non-Abelian interaction discrepancy is

$$D_J^{\text{NA}} = \frac{\|J^{\text{full}} - J^{U(1)}\|_F}{\max\{\|J^{\text{full}}\|_F, \varepsilon_J\}}. \quad (1774)$$

A large D_J^{NA} provides an experimentally accessible many-body consequence of the non-Abelian single-particle spectrum. The dispersive approximation requires

$$\text{dist}[\omega_q, \text{Spec } H] \gg \max_i \{g_i, \kappa_i\}. \quad (1775)$$

We do not interpret a resolvent enhancement at an unresolved resonant pole as a controlled effective qubit interaction. For every measured candidate, we record

$$\mathfrak{C}_{\text{exp}}^{\text{NA}} = \left(\tilde{\theta}_{\text{flat}}^{U(1)}, \tilde{\theta}_{\text{flat}}^{\text{full}}, \mathcal{R}_\theta^{\text{NA}}, W_{\text{exp}}^U, \Delta_{\text{exp}}^L, \mathcal{F}_{\text{gap}}^{\text{exp,NA}}, \mathcal{C}_{\text{spec}}^{\text{NA}}, \right. \\ \left. \mathcal{F}_\rho^{\text{drive}}, P_\rho^{\text{mode}}, \mathcal{C}_c^{\text{exp}}, \tau_{\text{disp},\rho}, D_J^{\text{NA}}, C_{\text{rep},N}, \varepsilon_{\text{cal}}, \varepsilon_{\text{fit}} \right).$$

We call the resulting regime physically non-Abelian only when the flat-band conclusion depends on active non-commuting translation holonomies, survives the complete physical and numerical verification protocol, and produces a representation-selective response that cannot be reproduced by any one-dimensional character of the same microscopic translation group.

VIII. HODGE CANCELLATION AND GLOBAL FLATNESS

We now determine whether the Hodge-space cancellation inherited from the mathematical construction provides the microscopic mechanism of the physical flat-band optimum identified from the complete spectrum.

The first condition removes the complete first-order energy differential at one representation-space point. The second removes the complete quadratic response at that point. Neither condition alone controls the energy range over a finite generalized Brillouin domain. We therefore use the Hodge calculation as a mechanism predictor and not as the definition of a magic angle. The Hodge candidate angle is denoted by θ_H , whereas the physical optimum obtained from the complete non-Abelian spectrum and the operational flatness functional is denoted by $\theta_{\text{flat}}^{\text{full}}$. The central mechanism test is

$$\theta_H \stackrel{?}{\simeq} \theta_{\text{flat}}^{\text{full}}. \quad (1776)$$

A. Hodge/Hessian candidate point

Let

$$[\rho_*(\theta)] \in \mathcal{B}_d \quad (1777)$$

be the representation-space point at which the candidate band or spectral cluster is tested. Its physical tangent space is

$$V_\theta = H^1(\Gamma_M; \text{Ad } \rho_*(\theta)). \quad (1778)$$

The canonical intensive Hodge metric is

$$G_\theta : V_\theta \times V_\theta \longrightarrow \mathbb{R}. \quad (1779)$$

Comparing tensors at different twists requires a fixed transported space. We choose a reference angle θ_0 and a declared transport

$$\mathcal{T}_\theta : V_{\theta_0} \longrightarrow V_\theta. \quad (1780)$$

All tensors are pulled back to $V_0 := V_{\theta_0}$. We define

$$\overline{G}_\theta = \mathcal{T}_\theta^* G_\theta, \quad \overline{e}_{C,\theta} = \mathcal{T}_\theta^* dE_{C,\theta}, \quad (1781)$$

and

$$\overline{\mathcal{K}}_{C,\theta} = \mathcal{T}_\theta^* (\nabla^2 E_{C,\theta}). \quad (1782)$$

The transport, marking, metric normalization, representation rank, and target ancestry are held fixed throughout the comparison. The metric-normalized Hessian endomorphism is

$$\widehat{\mathcal{K}}_C(\theta) = \overline{G}_\theta^{-1/2} \overline{\mathcal{K}}_{C,\theta} \overline{G}_\theta^{-1/2}. \quad (1783)$$

It is self-adjoint on the Euclideanized tangent space. Let

$$\lambda_1(\theta) \leq \lambda_2(\theta) \leq \dots \leq \lambda_m(\theta) \quad (1784)$$

be its ordered principal values, where

$$m = \dim_{\mathbb{R}} V_0. \quad (1785)$$

We define the absolute principal response by

$$\Lambda_H(\theta) = \max_{1 \leq j \leq m} |\lambda_j(\theta)| = \left\| \widehat{\mathcal{K}}_C(\theta) \right\|_{\text{op}}. \quad (1786)$$

The normalized scalar response is

$$\overline{\lambda}_H(\theta) = \frac{1}{m} \text{Tr } \widehat{\mathcal{K}}_C(\theta). \quad (1787)$$

The traceless anisotropic response is

$$\widehat{\mathcal{A}}_C(\theta) = \widehat{\mathcal{K}}_C(\theta) - \overline{\lambda}_H(\theta) I_m. \quad (1788)$$

We use the two anisotropy norms

$$\mathcal{A}_H^{\text{op}}(\theta) = \left\| \widehat{\mathcal{A}}_C(\theta) \right\|_{\text{op}}, \quad (1789)$$

and

$$\mathcal{A}_H^{\text{rms}}(\theta) = \frac{1}{\sqrt{m}} \left\| \widehat{\mathcal{A}}_C(\theta) \right\|_{\text{F}}. \quad (1790)$$

The exact full Hodge-cancellation set is

$$\Theta_H = \left\{ \theta : \overline{e}_{C,\theta} = 0, \Lambda_H(\theta) = 0, \Delta_C(\theta) > 0 \right\}.$$

The condition $\Lambda_H = 0$ is equivalent to

$$\lambda_j = 0 \quad \text{for every } j. \quad (1791)$$

A trace root, $\overline{\lambda}_H = 0$, is weaker. It becomes equivalent to complete cancellation only after an independent symmetry or commutant certificate forces $\widehat{\mathcal{A}}_C = 0$. We now specialize to the natural centered surface-group bilayer. Let $V = H^1(\Gamma; \mathbb{R})$ carry the canonical Hodge metric G . Let $n : \Gamma \longrightarrow V^*$ be the Abelianization covector. For a marked first shell $s_1^{\pm 1}, \dots, s_g^{\pm 1}$, define the first-shell kinetic tensor

$$C_S = \sum_{a=1}^m dk_a \otimes dk_a. \quad (1792)$$

Here dk_a are the marked Hodge covectors associated with the oriented primitive hopping directions. Let

$$q_\gamma(\theta, K) = q_{\gamma^{-1}}(\theta, K) > 0 \quad (1793)$$

be the full-distance interlayer hopping coefficients after the common amplitude w has been factored out. Define the complete interlayer second-moment tensor

$$B_\infty(\theta, K) = \sum_{\gamma \in \Gamma} q_\gamma(\theta, K) n(\gamma) \otimes n(\gamma). \quad (1794)$$

The sum is taken in the operator norm induced by G . The derivative-weighted hopping-tail condition guarantees its convergence. For the parity-reduced layer-even character branch,

$$E_+(k; \theta, K, w) = \varepsilon(k; K) + w \tau_\infty(k; \theta, K), \quad (1795)$$

the complete Hodge Hessian at the symmetric point $k = 0$ is

$$\mathcal{H}(\theta, K, w) = 2tC_S(K) - wB_\infty(\theta, K). \quad (1796)$$

For arbitrary tangent vectors $X, Y \in V$,

$$\begin{aligned} \mathcal{K}(X, Y) = & 2t \sum_{a=1}^m dk_a(X) dk_a(Y) \\ & - w \sum_{\gamma \in \Gamma} q_\gamma n(\gamma)(X) n(\gamma)(Y). \end{aligned} \quad (1797)$$

At fixed (θ, K) , the coupling derivatives are exact:

$$\partial_w \mathcal{K} = -B_\infty, \quad \partial_w^2 \mathcal{K} = 0. \quad (1798)$$

The first shell contributes $2q_1 C_S$ to B_∞ , while every later shell contributes a positive-semidefinite form. Therefore

$$B_\infty \succeq 2q_1 C_S \succ 0. \quad (1799)$$

The normalized microscopic tensor is

$$\widehat{\mathcal{K}} = G^{-1/2} \mathcal{K} G^{-1/2}. \quad (1800)$$

Its principal eigenvalues are independent of the tangent-coordinate basis. Let $\mathcal{P}$ be the finite group of centered bilayer isometries used in the regular-octagon construction. Its tangent action is denoted by

$$U_p : V \longrightarrow V, \quad p \in \mathcal{P}. \quad (1801)$$

We impose the microscopic Hodge-isotropy conditions

$$U_p^* G = G, \quad U_p^* C_S = C_S, \quad U_p^* B_\infty = B_\infty, \quad (1802)$$

and

$$\{A = A^{*G} : AU_p = U_p A \text{ for every } p \in \mathcal{P}\} = \mathbb{R}I. \quad (1803)$$

The final relation states that the G -self-adjoint commutant of the actual microscopic point-group representation is scalar. Under these conditions, there exist positive scalars

$$c_S > 0, \quad d_\infty(\theta, K) > 0, \quad b_\infty(\theta, K) > 0 \quad (1804)$$

such that

$$C_S = c_S G, \quad B_\infty = d_\infty G, \quad (1805)$$

and

$$B_\infty = b_\infty C_S, \quad b_\infty = \frac{d_\infty}{c_S}. \quad (1806)$$

Substitution into Eq. (1796) gives

$$\mathcal{K} = c_S [2t - w b_\infty(\theta, K)] G. \quad (1807)$$

Hence

$$\widehat{\mathcal{K}} = \lambda_H(\theta, K, w) I_m, \quad (1808)$$

where

$$\lambda_H(\theta, K, w) = c_S [2t - w b_\infty(\theta, K)]. \quad (1809)$$

All principal Hessian eigenvalues are exactly locked:

$$\lambda_1 = \cdots = \lambda_m = \lambda_H. \quad (1810)$$

At fixed (θ, K) , the unique positive coupling root is

$$w_H(\theta, K) = \frac{2t}{b_\infty(\theta, K)}. \quad (1811)$$

At fixed $w = \omega t$, the Hodge candidate angle satisfies

$$b_\infty(\theta_H, K) = \frac{2}{\omega}. \quad (1812)$$

Equivalently,

$$\theta_H : \quad \lambda_{\max} [\widehat{\mathcal{K}}(\theta_H)] = 0 \quad (1813)$$

within the exact eigenvalue-locking class. Equation (1813) must not be used outside that class. Without eigenvalue locking, $\lambda_{\max} = 0$ can describe a nonzero negative-semidefinite tensor. The general condition remains $\Lambda_H = 0$. If

$$\partial_\theta b_\infty(\theta_H, K) \neq 0, \quad (1814)$$

the angle root is simple. Its principal-value slope is

$$s_H^{(\theta)} = |\partial_\theta \lambda_H(\theta_H)| = w c_S |\partial_\theta b_\infty(\theta_H)| > 0. \quad (1815)$$

The curvature and coupling responses are

$$\left. \frac{\partial \theta_H}{\partial K} = - \frac{\partial_K b_\infty}{\partial_\theta b_\infty} \right|_{\theta=\theta_H}, \quad (1816)$$

and

$$\left. \frac{\partial \theta_H}{\partial \omega} = - \frac{2}{\omega^2 \partial_\theta b_\infty} \right|_{\theta=\theta_H}. \quad (1817)$$

We now remove the scalar-commutant assumption. Define

$$A = G^{-1/2} C_S G^{-1/2}, \quad (1818)$$

and

$$D(\theta) = G^{-1/2} B_\infty(\theta) G^{-1/2}. \quad (1819)$$

The normalized Hessian is

$$\widehat{\mathcal{K}}(\theta) = 2tA - wD(\theta). \quad (1820)$$

At fixed w , a trace-root angle satisfies

$$2t \operatorname{Tr} A = w \operatorname{Tr} D(\theta_{\text{tr}}). \quad (1821)$$

At that angle,

$$\widehat{\mathcal{K}}(\theta_{\text{tr}}) = 2t \left[A - \frac{\operatorname{Tr} A}{\operatorname{Tr} D(\theta_{\text{tr}})} D(\theta_{\text{tr}}) \right]. \quad (1822)$$

Define the exact anisotropic obstruction

$$a_\infty(\theta_{\text{tr}}) = \left\| A - \frac{\text{Tr } A}{\text{Tr } D(\theta_{\text{tr}})} D(\theta_{\text{tr}}) \right\|_{\text{op}}. \quad (1823)$$

Then

$$\left\| \widehat{\mathcal{K}}(\theta_{\text{tr}}) \right\|_{\text{op}} = 2ta_\infty(\theta_{\text{tr}}). \quad (1824)$$

Full cancellation occurs at the trace root if and only if $a_\infty = 0$. This is equivalent to $B_\infty = b_\infty C_S$. When $a_\infty > 0$, the residual tensor is nonzero and traceless. A nonzero traceless self-adjoint operator cannot be sign definite. Therefore there exist tangent directions X_+ and X_- such that

$$\mathcal{K}(X_+, X_+) > 0, \quad (1825)$$

and

$$\mathcal{K}(X_-, X_-) < 0. \quad (1826)$$

The trace root is then a multidirectional saddle rather than a quadratically flat point. Let the ideal symmetry-locked tensor be

$$\widehat{\mathcal{K}}_0(\theta) = \lambda_0(\theta) I_m, \quad (1827)$$

with $\lambda_0(\theta_H) = 0$. Let the implemented or full-shell tensor be $\widetilde{\mathcal{K}}(\theta)$. On an interval

$$I_H = [\theta_H - h_\theta, \theta_H + h_\theta], \quad (1828)$$

assume

$$\sup_{\theta \in I_H} \left\| \widetilde{\mathcal{K}}(\theta) - \widehat{\mathcal{K}}_0(\theta) \right\|_{\text{op}} \leq \eta_2, \quad (1829)$$

and

$$\sup_{\theta \in I_H} \left\| \partial_\theta \widetilde{\mathcal{K}}(\theta) - \lambda'_0(\theta) I_m \right\|_{\text{op}} \leq \eta_{2,1}. \quad (1830)$$

Let

$$s_H = \inf_{\theta \in I_H} |\lambda'_0(\theta)| \quad (1831)$$

and assume $\eta_{2,1} < s_H$. Weyl stability gives, for every ordered principal eigenvalue,

$$\left| \widetilde{\lambda}_j(\theta) - \lambda_0(\theta) \right| \leq \eta_2. \quad (1832)$$

Consequently,

$$\max_{j,\ell} \left| \widetilde{\lambda}_j(\theta) - \widetilde{\lambda}_\ell(\theta) \right| \leq 2\eta_2. \quad (1833)$$

The normalized trace

$$\widetilde{\lambda}_H(\theta) = \frac{1}{m} \text{Tr } \widetilde{\mathcal{K}}(\theta) \quad (1834)$$

is strictly monotone on I_H . If

$$h_\theta > \frac{\eta_2}{s_H - \eta_{2,1}}, \quad (1835)$$

it has a unique zero $\widetilde{\theta}_{\text{tr}} \in I_H$ satisfying

$$\left| \widetilde{\theta}_{\text{tr}} - \theta_H \right| \leq \frac{\eta_2}{s_H - \eta_{2,1}}. \quad (1836)$$

At the perturbed trace root,

$$\left| \widetilde{\lambda}_j(\widetilde{\theta}_{\text{tr}}) \right| \leq 2\eta_2 \quad (1837)$$

for every j . If the perturbation preserves the scalar-commutant certificate, all principal eigenvalues remain exactly locked and $\widetilde{\theta}_{\text{tr}}$ is an exact full Hodge root. If the certificate is broken, the same angle gives only an approximate multidirectional cancellation with residual bounded by $2\eta_2$. For a calculated tensor $\widetilde{\mathcal{K}}$ with uncertainty $\varepsilon_{\mathcal{K}}$, define

$$\Lambda_H^U(\theta) = \left\| \widetilde{\mathcal{K}}(\theta) \right\|_{\text{op}} + \varepsilon_{\mathcal{K}}(\theta). \quad (1838)$$

Let

$$r_H^{(1)}(\theta) = \frac{\hbar D_{GV_C}(\rho_*; \theta)}{\Delta_C^L(\theta)}, \quad (1839)$$

and

$$r_H^{(2)}(\theta) = \frac{D_G^2 \Lambda_H^U(\theta)}{2\Delta_C^L(\theta)}. \quad (1840)$$

The established Hodge-candidate set is

$$\widehat{\Theta}_H = \left\{ \theta : \Delta_C^L(\theta) > 0, r_H^{(1),U}(\theta) \leq \varepsilon_H^{(1)}, r_H^{(2)}(\theta) \leq \varepsilon_H^{(2)} \right\}.$$

For a unique interval component, we define

$$\widetilde{\theta}_H = \arg \min_{\theta} \Lambda_H^U(\theta) \quad (1841)$$

and

$$\delta\theta_H = \frac{1}{2} \text{diam } \widehat{\Theta}_H. \quad (1842)$$

B. Comparison with the physical flat-band optimum

We compare θ_H with $\theta_{\text{flat}}^{\text{full}}$ only after fixing the same:

microscopic Hamiltonian, target projector ancestry, representation-completed spectral family, finite-cover tower and hopping shell, Hodge metric and tangent transport, spectral broadening and numerical tolerances.	(1843)
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A Hodge root computed in one character sector is not directly compared with a full-spectrum physical optimum unless the corresponding target family has been completed across every declared representation sector. Likewise, a Hessian root of one isolated eigenvalue cannot be compared with a physical optimum defined for a different multiplet. Let $\widehat{\Theta}_{\text{flat}}^{\text{full}}$ be the established full-spectrum optimizer set defined from the upper and lower physical-score envelopes. The two mechanisms are compatible when

$$\widehat{\Theta}_{\text{H}} \cap \widehat{\Theta}_{\text{flat}}^{\text{full}} \neq \emptyset. \quad (1844)$$

For unique estimates, define

$$\delta\theta_{\text{HF}} = \widetilde{\theta}_{\text{flat}}^{\text{full}} - \widetilde{\theta}_{\text{H}}. \quad (1845)$$

The combined uncertainty is

$$\delta\theta_{\text{tot}} = \delta\theta_{\text{H}} + \delta\theta_{\text{flat}} + \delta\theta_{\text{transport}} + \delta\theta_{\text{cal}}. \quad (1846)$$

The mismatch resolvability index is

$$\mathcal{R}_{\text{HF}} = \frac{|\delta\theta_{\text{HF}}|}{\delta\theta_{\text{tot}}}. \quad (1847)$$

We regard the two angles as experimentally and numerically compatible when $\mathcal{R}_{\text{HF}} \leq 1$, and as resolvably distinct when $\mathcal{R}_{\text{HF}} > 1$. The full physical score at the Hodge angle is

$$\mathcal{M}_{\text{H} \rightarrow \text{F}} = \mathcal{M}^{\text{full}}(\widetilde{\theta}_{\text{H}}). \quad (1848)$$

Let

$$\mathcal{M}_*^{\text{full}} = \mathcal{M}^{\text{full}}(\widetilde{\theta}_{\text{flat}}^{\text{full}}). \quad (1849)$$

We define the mechanism-efficiency ratio

$$\mathcal{E}_{\text{H}} = \frac{\mathcal{M}_{\text{H} \rightarrow \text{F}}}{\mathcal{M}_*^{\text{full}}}, \quad 0 \leq \mathcal{E}_{\text{H}} \leq 1. \quad (1850)$$

A value near unity shows that the Hodge predictor identifies a parameter point whose complete physical quality is nearly optimal even when the two fitted angles are not exactly equal. We also record the physical residual vector at the Hodge point:

$$\mathbf{r}_{\text{H} \rightarrow \text{F}} = \left(\frac{W_C(\theta_{\text{H}})}{\Delta_C(\theta_{\text{H}})}, \frac{\hbar D_G v_{\text{max}, C}(\theta_{\text{H}})}{\Delta_C(\theta_{\text{H}})}, \frac{D_G^2 \Lambda_{\text{max}, C}(\theta_{\text{H}})}{2\Delta_C(\theta_{\text{H}})}, \frac{1}{E_{\text{kin}} \varrho_{C, \eta}^{\text{coh}, \text{max}}(\theta_{\text{H}})}, 1 - C_C^{\text{coh}}(\theta_{\text{H}}) \right). \quad (1851)$$

A local Hodge root explains a physical magic state only when this complete vector is small and the positive-gap, representation, cover, and shell conditions remain open. Conversely, define

$$\Lambda_{\text{H} \leftarrow \text{F}} = \left\| \widehat{\mathcal{K}}_C(\widetilde{\theta}_{\text{flat}}^{\text{full}}) \right\|_{\text{op}}. \quad (1852)$$

The normalized residual is

$$\mathcal{R}_{\mathcal{K}}^{\text{flat}} = \frac{D_G^2 \Lambda_{\text{H} \leftarrow \text{F}}}{2\Delta_C(\widetilde{\theta}_{\text{flat}}^{\text{full}})}. \quad (1853)$$

A small value shows that the actual physical optimum is also quadratically soft in the complete Hodge tangent space. The two cross-evaluations, $\mathcal{E}_{\text{H}}$ and $\mathcal{R}_{\mathcal{K}}^{\text{flat}}$, are stronger than comparing the angle values alone. Let

$$L_{\text{phys}}(\theta) = \ln \mathcal{M}^{\text{full}, (1)}(\theta) \quad (1854)$$

be the differentiable minimal physical-score branch. Suppose $\theta_{\text{flat}}^{\text{full}}$ is an interior strict maximum:

$$L'_{\text{phys}}(\theta_{\text{flat}}^{\text{full}}) = 0, \quad (1855)$$

and

$$-L''_{\text{phys}}(\theta) \geq \mu_{\text{flat}} > 0 \quad (1856)$$

between θ_{H} and $\theta_{\text{flat}}^{\text{full}}$. The mean-value theorem gives

$$\theta_{\text{flat}}^{\text{full}} - \theta_{\text{H}} = -\frac{L'_{\text{phys}}(\theta_{\text{H}})}{L''_{\text{phys}}(\xi)} \quad (1857)$$

for some intermediate ξ . Therefore

$$|\theta_{\text{flat}}^{\text{full}} - \theta_{\text{H}}| \leq \frac{|L'_{\text{phys}}(\theta_{\text{H}})|}{\mu_{\text{flat}}}. \quad (1858)$$

Define

$$V_C = \hbar D_G v_{\text{max}, C}, \quad X_C = E_{\text{kin}} \varrho_{C, \eta}^{\text{coh}, \text{max}}, \quad C_C = C_C^{\text{coh}}. \quad (1859)$$

The logarithmic derivative is

$$L'_{\text{phys}} = \frac{1}{4} \left[\frac{W_C \Delta'_C - \Delta_C W'_C}{\Delta_C(\Delta_C + W_C)} + \frac{V_C \Delta'_C - \Delta_C V'_C}{\Delta_C(\Delta_C + V_C)} + \frac{X'_C}{X_C(1 + X_C)} + \frac{C'_C}{C_C} \right]. \quad (1860)$$

We define the four contributions

$$\begin{aligned} F_W &= \frac{1}{4} \frac{W_C \Delta'_C - \Delta_C W'_C}{\Delta_C(\Delta_C + W_C)}, \\ F_v &= \frac{1}{4} \frac{V_C \Delta'_C - \Delta_C V'_C}{\Delta_C(\Delta_C + V_C)}, \\ F_{\varrho} &= \frac{1}{4} \frac{X'_C}{X_C(1 + X_C)}, \quad F_{\text{coh}} = \frac{1}{4} \frac{C'_C}{C_C}. \end{aligned} \quad (1861)$$

Thus

$$L'_{\text{phys}} = F_W + F_v + F_{\varrho} + F_{\text{coh}}. \quad (1862)$$

At the Hodge angle, a nonzero value of

$$F_W + F_v + F_\varrho + F_{\text{coh}} \quad (1863)$$

quantifies the global physical force shifting the operational optimum away from the local quadratic root. The leading predictor is

$$\delta\theta_{\text{HF}}^{(1)} = - \frac{F_W + F_v + F_\varrho + F_{\text{coh}}}{L''_{\text{phys}}} \Big|_{\theta=\theta_{\text{H}}} . \quad (1864)$$

Agreement between $\delta\theta_{\text{HF}}^{(1)}$ and the directly measured $\delta\theta_{\text{HF}}$ verifies that the discrepancy is explained by the resolved observable responses rather than by branch misidentification. We classify the Hodge mechanism as quantitatively confirmed only when

$$\begin{aligned} \widehat{\Theta}_{\text{H}} \cap \widehat{\Theta}_{\text{flat}}^{\text{full}} &\neq \emptyset, \quad \mathcal{E}_{\text{H}} \geq 1 - \varepsilon_{\text{eff}}, \quad \mathcal{R}_{\mathcal{K}}^{\text{flat}} \leq \varepsilon_{\mathcal{K}}^{\text{acc}}, \\ \Delta_C^L > 0, \quad C_{\text{rep},C} &= 1 \quad \text{at every finite quotient,} \\ \text{the angle and tensor converge across independent towers,} \\ \text{the result survives the full hopping-shell limit.} \end{aligned} \quad (1865)$$

This criterion makes Hodge cancellation an independently verified explanation of the physical optimum rather than a circular definition of that optimum.

C. Local quadratic flattening

Let ρ_* be the Hodge candidate representation. We use geodesic normal coordinates $x \in T_{\rho_*}\mathcal{D}$ with $r = \|x\|_G$. The representation-space point is

$$\rho(x) = \exp_{\rho_*}(x). \quad (1866)$$

For one simple isolated branch, the covariant expansion is

$$\begin{aligned} E[\rho(x)] &= E_* + dE_*[x] + \frac{1}{2}\mathcal{K}_*[x, x] \\ &+ \frac{1}{3!}\mathcal{T}_*^{(3)}[x, x, x] \\ &+ \frac{1}{4!}\mathcal{T}_*^{(4)}[x, x, x, x] + R_5(x), \end{aligned} \quad (1867)$$

where

$$\mathcal{T}_*^{(n)} = \nabla^n E|_{\rho_*} \quad (1868)$$

is the n -th symmetric covariant derivative tensor. If

$$\sup_{\|y\|_G \leq r} \|\nabla^5 E[\rho(y)]\|_{\text{op}} \leq M_5, \quad (1869)$$

then

$$|R_5(x)| \leq \frac{M_5}{5!} r^5. \quad (1870)$$

At a fully stationary Hodge point, $dE_* = 0$. At a full quadratic cancellation point, $\mathcal{K}_* = 0$. The leading local dispersion is then

$$E[\rho(x)] - E_* = \frac{1}{3!}\mathcal{T}_*^{(3)}[x^3] + \frac{1}{4!}\mathcal{T}_*^{(4)}[x^4] + O(r^5). \quad (1871)$$

Hessian cancellation alone does not guarantee quartic-leading dispersion. A cubic term remains unless an independent symmetry removes it. Suppose a centered microscopic symmetry ι acts as tangent inversion: $U_\iota x = -x$ and preserves the target energy:

$$E[\rho(x)] = E[\rho(-x)]. \quad (1872)$$

Every odd covariant derivative then vanishes:

$$\mathcal{T}_*^{(2j+1)} = 0 \quad \text{for every } j \geq 0. \quad (1873)$$

In particular,

$$dE_* = 0, \quad \mathcal{T}_*^{(3)} = 0. \quad (1874)$$

If the Hessian also vanishes, the local normal form becomes

$$E[\rho(x)] = E_* + \frac{1}{4!}\mathcal{Q}_*[x, x, x, x] + R_6(x), \quad (1875)$$

where

$$\mathcal{Q}_* = \nabla^4 E|_{\rho_*}. \quad (1876)$$

If

$$\sup_{\|y\|_G \leq r} \|\nabla^6 E[\rho(y)]\|_{\text{op}} \leq M_6, \quad (1877)$$

then

$$|R_6(x)| \leq \frac{M_6}{6!} r^6. \quad (1878)$$

Define the quartic directional response

$$q_*(u) = \mathcal{Q}_*[u, u, u, u], \quad \|u\|_G = 1. \quad (1879)$$

Its extremal values are

$$q_*^+ = \sup_{\|u\|_G=1} q_*(u), \quad (1880)$$

and

$$q_*^- = \inf_{\|u\|_G=1} q_*(u). \quad (1881)$$

The quartic operator norm is

$$Q_* = \sup_{\|u\|_G=1} |q_*(u)| = \max\{|q_*^+|, |q_*^-|\}. \quad (1882)$$

The quartic anisotropy spread is

$$\mathcal{A}_4 = q_*^+ - q_*^-. \quad (1883)$$

On a Hodge ball $B_G(\rho_*, r_0)$ we have

$$|E[\rho(x)] - E_*| \leq \frac{Q_*}{24} r_0^4 + \frac{M_6}{720} r_0^6. \quad (1884)$$

The local energy diameter therefore satisfies

$$W_{\text{loc}}(r_0) \leq \frac{Q_*}{12} r_0^4 + \frac{M_6}{360} r_0^6. \quad (1885)$$

If the quartic tensor is positive definite, $q_*^- > 0$, the Hodge point remains a strict local minimum. If $q_*^+ > 0 > q_*^-$, the quadratically flat point is a quartic saddle. A vanishing Hessian therefore does not determine the local extremum type. For a two-dimensional Euclidean tangent space with C_4 symmetry, the quartic normal form is

$$E(\mathbf{k}) = E_* + B_1 (k_x^4 + k_y^4) + B_2 k_x^2 k_y^2 + O(k^6). \quad (1886)$$

For

$$\mathbf{k} = r \begin{pmatrix} \cos \phi \\ \sin \phi \end{pmatrix}, \quad (1887)$$

we obtain

$$E - E_* = r^4 \left[B_1 + \frac{B_2 - 2B_1}{4} \sin^2(2\phi) \right] + O(r^6). \quad (1888)$$

The axial quartic coefficient is $q_{\text{ax}} = B_1$, while the diagonal coefficient is

$$q_{\text{diag}} = \frac{2B_1 + B_2}{4}. \quad (1889)$$

Local positive definiteness requires

$$B_1 > 0, \quad 2B_1 + B_2 > 0. \quad (1890)$$

The quartic anisotropy is controlled by $B_2 - 2B_1$. Let the local tangent-space dimension be m , and use Hodge-normal coordinates with locally constant volume density $\mathcal{J}_*$. For an isotropic quadratic minimum

$$E - E_* = \frac{\lambda}{2} r^2, \quad \lambda > 0, \quad (1891)$$

the local DOS is

$$\varrho_2^{\text{loc}}(\varepsilon) = \mathcal{J}_* \frac{S_{m-1} 2^{m/2-1}}{(2\pi)^m \lambda^{m/2}} \varepsilon^{m/2-1}, \quad (1892)$$

where

$$\varepsilon = E - E_* > 0, \quad (1893)$$

and

$$S_{m-1} = \frac{2\pi^{m/2}}{\Gamma(m/2)} \quad (1894)$$

is the area of the unit $(m-1)$ -sphere. For an isotropic quartic minimum

$$E - E_* = \frac{q}{24} r^4, \quad q > 0, \quad (1895)$$

the corresponding DOS is

$$\varrho_4^{\text{loc}}(\varepsilon) = \mathcal{J}_* \frac{6S_{m-1} 24^{m/4-1}}{(2\pi)^m q^{m/4}} \varepsilon^{m/4-1}. \quad (1896)$$

The exponent changes from $\frac{m}{2} - 1$ to $\frac{m}{4} - 1$. For $m = 2$, the quartic DOS diverges as

$$\varrho_4^{\text{loc}} \propto \varepsilon^{-1/2}. \quad (1897)$$

For the four-dimensional Abelian Hodge space of a genus-two quotient, $m = 4$, the isotropic quartic DOS approaches a nonzero constant: $\varrho_4^{\text{loc}} \propto q^{-1}$. These are local Hodge-volume laws. They are promoted to a complete physical DOS statement only after the representation measure and global spectral family have been specified. We now include remote spectral modes without assuming that their contribution is negligible. Let P be the specified active target projector and $Q = I - P$. On one fixed transported fiber, write

$$H = \begin{pmatrix} H_P & V \\ V^\dagger & H_Q \end{pmatrix}. \quad (1898)$$

For

$$E \notin \text{Spec } H_Q, \quad (1899)$$

define

$$R(E) = (E - H_Q)^{-1}, \quad (1900)$$

and

$$\Sigma(E) = VR(E)V^\dagger. \quad (1901)$$

For physical tangent directions A and B , set

$$V_A = D_A V, \quad V_{AB} = D_A D_B V, \quad (1902)$$

and

$$\begin{aligned} L_A &= D_A H_Q - (D_A E) I, \\ L_{AB} &= D_A D_B H_Q - (D_A D_B E) I. \end{aligned} \quad (1903)$$

Differentiation of

$$R(E - H_Q) = I \quad (1904)$$

gives

$$D_A R = R L_A R. \quad (1905)$$

The complete first derivative is

$$D_A \Sigma = V_A R V^\dagger + V R V_A^\dagger + V R L_A R V^\dagger. \quad (1906)$$

The complete second derivative contains eleven terms:

$$\begin{aligned}
D_A D_B \Sigma &= V_{AB} R V^\dagger + V R V_{AB}^\dagger + V_A R V_B^\dagger + V_B R V_A^\dagger \\
&+ V_A R L_B R V^\dagger + V R L_B R V_A^\dagger \\
&+ V_B R L_A R V^\dagger + V R L_A R V_B^\dagger \\
&+ V R L_B R L_A R V^\dagger + V R L_{AB} R V^\dagger \\
&+ V R L_A R L_B R V^\dagger.
\end{aligned} \tag{1907}$$

No profile derivative or complementary-band derivative may be omitted. Assume $\|R\| \leq \delta^{-1}$,

$$\|V\| \leq v_0, \quad \|DV\| \leq v_1, \quad \|D^2V\| \leq v_2, \tag{1908}$$

and

$$\|L_A\| \leq \ell_1, \quad \|L_{AB}\| \leq \ell_2. \tag{1909}$$

The complete second-response bound is

$$\begin{aligned}
\|D^2\Sigma\| &\leq \frac{2v_0v_2 + 2v_1^2}{\delta} + \frac{4v_0v_1\ell_1 + v_0^2\ell_2}{\delta^2} \\
&+ \frac{2v_0^2\ell_1^2}{\delta^3}.
\end{aligned} \tag{1910}$$

This estimate is valid only for the uniformly detuned complement. Every state for which

$$\text{dist}(E, \text{Spec } H_Q) \tag{1911}$$

is comparable to the active coupling must be moved from Q into the active block P before the Hessian is evaluated. A resonant state cannot be treated as a perturbative denominator. The effective target problem is

$$[H_P + \Sigma(E)] |\psi_P\rangle = E |\psi_P\rangle. \tag{1912}$$

Its energy differential is

$$dE = Z_{\text{eff}} \langle \psi_P | dH_P + (d\Sigma)_E | \psi_P \rangle, \tag{1913}$$

where

$$Z_{\text{eff}} = [\langle \psi_P | I - \partial_E \Sigma | \psi_P \rangle]^{-1}. \tag{1914}$$

Since

$$\partial_E \Sigma = -V R^2 V^\dagger \preceq 0, \tag{1915}$$

we have $0 < Z_{\text{eff}} \leq 1$. Define the numerator covector

$$\mathcal{N} = \langle \psi_P | dH_P + (d\Sigma)_E | \psi_P \rangle. \tag{1916}$$

At a stationary point, $\mathcal{N} = 0$. Differentiation gives

$$\nabla(dE) = dZ_{\text{eff}} \otimes \mathcal{N} + Z_{\text{eff}} \nabla \mathcal{N}. \tag{1917}$$

Therefore, at stationarity,

$$\nabla^2 E = Z_{\text{eff}} \nabla \mathcal{N}. \tag{1918}$$

The positive residue rescales the Hessian but does not create a zero. Full local cancellation requires the complete direct and self-energy Hessians to cancel:

$$\mathcal{K}_P + \mathcal{K}_\Sigma = 0. \tag{1919}$$

For the exactly layer-even target in the parity-reduced natural model, layer exchange makes the two parity sectors reducing: $P_+ H P_- = 0$. The tangent derivatives also preserve parity:

$$D^j (P_+ H P_-) = 0, \quad j = 1, 2. \tag{1920}$$

Consequently,

$$\Sigma = D\Sigma = D^2\Sigma = 0 \tag{1921}$$

for this declared reduction. The tensor Eq. (1796) is then the complete Hessian of that parity branch rather than a truncated self-energy approximation.

D. Global bandwidth suppression

Let $\mathcal{D}_C$ be the complete connected representation-space domain of the tracked cluster. Let $\rho_* \in \mathcal{D}_C$ be the Hodge candidate point, and define its covering radius

$$R_* = \sup_{\rho \in \mathcal{D}_C} d_G(\rho, \rho_*). \tag{1922}$$

The domain diameter satisfies $R_* \leq D_G \leq 2R_*$. The global bandwidth is

$$W_C = \sup_{\rho \in \mathcal{D}_C} E_C(\rho) - \inf_{\rho \in \mathcal{D}_C} E_C(\rho). \tag{1923}$$

Knowing

$$dE_C(\rho_*) = 0 \tag{1924}$$

and

$$\nabla^2 E_C(\rho_*) = 0 \tag{1925}$$

does not control the derivatives away from ρ_* . Define

$$\Omega_{\text{max},C} = \frac{1}{\hbar} \sup_{\rho \in \mathcal{D}_C} \|dE_C(\rho)\|_{G^{-1}}. \tag{1926}$$

For any two points connected by a minimizing geodesic,

$$\begin{aligned}
|E_C(\rho_2) - E_C(\rho_1)| &\leq \int_{\rho_1}^{\rho_2} \|dE_C\|_{G^{-1}} ds \\
&\leq \hbar \Omega_{\text{max},C} d_G(\rho_1, \rho_2).
\end{aligned} \tag{1927}$$

Taking the supremum gives

$$W_C \leq \hbar D_G \Omega_{\text{max},C}. \tag{1928}$$

Complete generalized-velocity suppression is therefore a rigorous sufficient condition for global energetic flattening. Define the saturation ratio

$$\mathcal{J}_C^{(1)} = \frac{W_C}{\hbar D_G \Omega_{\text{max},C}}, \quad 0 \leq \mathcal{J}_C^{(1)} \leq 1. \tag{1929}$$

A small value means that the maximum derivative occurs only in a limited part of the domain and does not determine the complete energy range. The RMS generalized dispersion is

$$\Omega_{\text{rms},C} = \left[\frac{1}{\text{Vol}_G(\mathcal{D}_C)} \int_{\mathcal{D}_C} \Omega_C^2(\rho) d\mu_G(\rho) \right]^{1/2}. \quad (1930)$$

The uniformity ratio is

$$\mathcal{U}_{\text{disp},C} = \frac{\Omega_{\text{rms},C}}{\Omega_{\text{max},C}}, \quad 0 \leq \mathcal{U}_{\text{disp},C} \leq 1. \quad (1931)$$

A very small value identifies isolated residual steep sectors hidden by an otherwise weakly dispersive band. Assume

$$dE_C(\rho_*) = 0 \quad (1932)$$

and define

$$\Lambda_2 = \sup_{\rho \in \mathcal{D}_C} \left\| G_\rho^{-1/2} \nabla^2 E_C(\rho) G_\rho^{-1/2} \right\|_{\text{op}}. \quad (1933)$$

Along a unit-speed minimizing geodesic $\gamma : [0, L] \rightarrow \mathcal{D}_C$ from ρ_* , we have

$$\left| \frac{d^2}{ds^2} E_C[\gamma(s)] \right| \leq \Lambda_2. \quad (1934)$$

Since the first derivative vanishes at $s = 0$,

$$|E_C[\gamma(L)] - E_C(\rho_*)| \leq \frac{1}{2} \Lambda_2 L^2. \quad (1935)$$

Thus every energy lies in an interval of radius $\Lambda_2 R_*^2/2$, and

$$W_C \leq \Lambda_2 R_*^2. \quad (1936)$$

Define the second-order saturation ratio

$$\mathcal{J}_C^{(2)} = \frac{W_C}{\Lambda_2 R_*^2}, \quad 0 \leq \mathcal{J}_C^{(2)} \leq 1. \quad (1937)$$

Assume in addition

$$\nabla^2 E_C(\rho_*) = 0 \quad (1938)$$

and define

$$\Lambda_3 = \sup_{\rho \in \mathcal{D}_C} \left\| \nabla^3 E_C(\rho) \right\|_{\text{op}}. \quad (1939)$$

Along the same geodesic,

$$\left| \frac{d^2}{ds^2} E_C[\gamma(s)] \right| \leq \Lambda_3 s. \quad (1940)$$

Integrating twice gives

$$|E_C[\gamma(L)] - E_C(\rho_*)| \leq \frac{\Lambda_3}{6} L^3. \quad (1941)$$

Therefore

$$W_C \leq \frac{\Lambda_3}{3} R_*^3. \quad (1942)$$

If tangent inversion removes every odd derivative and

$$\Lambda_4 = \sup_{\rho \in \mathcal{D}_C} \left\| \nabla^4 E_C(\rho) \right\|_{\text{op}}, \quad (1943)$$

then

$$|E_C[\gamma(L)] - E_C(\rho_*)| \leq \frac{\Lambda_4}{24} L^4. \quad (1944)$$

Hence

$$W_C \leq \frac{\Lambda_4}{12} R_*^4. \quad (1945)$$

This is a genuine local-to-global quartic bound, but it requires a uniform fourth-derivative estimate throughout the complete domain. A quartic tensor calculated only at ρ_* does not establish Eq. (1945). We define

$$\mathcal{R}_{\text{glob}}^{(1)} = \frac{W_C}{\hbar D_G \Omega_{\text{max},C}}, \quad \mathcal{R}_{\text{glob}}^{(2)} = \frac{W_C}{\Lambda_2 R_*^2}, \quad (1946)$$

and, when the quartic hypotheses hold,

$$\mathcal{R}_{\text{glob}}^{(4)} = \frac{12W_C}{\Lambda_4 R_*^4}. \quad (1947)$$

Each ratio belongs to $[0, 1]$ when the corresponding rigorous bound has been established. A Hodge mechanism is globally closed only when the relevant derivative scale itself becomes small: $\Omega_{\text{max},C} \rightarrow 0$, or $\Lambda_2 \rightarrow 0$, or, in the quartic class,

$$\Lambda_4 R_*^4 \ll \Delta_C. \quad (1948)$$

A small saturation ratio alone does not prove flatness. Let the completed target family contain sectors $\rho \in \mathcal{R}_C$. For each sector, define its energy center and half-width:

$$c_\rho = \frac{E_\rho^+ + E_\rho^-}{2}, \quad (1949)$$

and

$$r_\rho = \frac{E_\rho^+ - E_\rho^-}{2}. \quad (1950)$$

Define

$$S_C^{\text{ctr}} = \max_\rho c_\rho - \min_\rho c_\rho, \quad (1951)$$

and

$$W_C^{\text{sec,max}} = 2 \max_\rho r_\rho = \max_\rho W_{C,\rho}. \quad (1952)$$

The full target width is exactly

$$W_C^{\text{full}} = \max_\rho (c_\rho + r_\rho) - \min_\rho (c_\rho - r_\rho). \quad (1953)$$

It obeys

$$\max \{S_C^{\text{ctr}}, W_C^{\text{sec}, \max}\} \leq W_C^{\text{full}} \leq S_C^{\text{ctr}} + W_C^{\text{sec}, \max}. \quad (1954)$$

Thus cancellation of the Hessian in every individual sector is not sufficient if the sector centers remain separated. A full globally flat representation-completed family requires

$$W_{C,\rho} \longrightarrow 0 \quad \text{uniformly in } \rho, \quad S_C^{\text{ctr}} \longrightarrow 0. \quad (1955)$$

The natural microscopic construction also supplies a global operator-norm route that does not rely solely on local Taylor expansion. Let H_* be an ideal reference operator whose target spectrum is a point island $I_* = \{E_*\}$ separated from the complementary spectrum by $g_* > 0$. Let the complete physical kernel at the Hodge root satisfy

$$\|H_\infty(\theta_H) - H_*\| \leq \varepsilon_H. \quad (1956)$$

Weyl stability gives

$$W_{\infty,C} \leq 2\varepsilon_H, \quad (1957)$$

and

$$\Delta_{\infty,C} \geq g_* - 2\varepsilon_H. \quad (1958)$$

If $\varepsilon_H < \frac{g_*}{6}$, then

$$\Delta_{\infty,C} > \frac{2g_*}{3}. \quad (1959)$$

Let

$$r_\Gamma = \frac{g_*}{3}, \quad \delta_\Gamma = r_\Gamma - \varepsilon_H > 0. \quad (1960)$$

For an additional common-space perturbation

$$\tilde{H} = H_\infty + V, \quad e = \|V\| < \delta_\Gamma, \quad (1961)$$

the Riesz projectors satisfy

$$\|\tilde{P}_C - P_C\| \leq \frac{r_\Gamma e}{\delta_\Gamma(\delta_\Gamma - e)}. \quad (1962)$$

The target rank and layer coherence persist whenever the right-hand side is smaller than one. If the total perturbation relative to the ideal root, including the angle displacement, is

$$\eta_0 = \|\tilde{H}(\tilde{\theta}_H) - H_*\|, \quad (1963)$$

then $\tilde{W}_C \leq 2\eta_0$, and

$$\tilde{\Delta}_C \geq g_* - 2\eta_0. \quad (1964)$$

This route supplies a genuine global bandwidth and gap certificate. The Hodge tensor identifies the local kinetic cancellation, while the operator-island estimate controls

the complete spectrum. Let the finite-cover target spectrum have two-sided Hausdorff error

$$\eta_N = d_H(S_{C,N}, S_{C,\infty}). \quad (1965)$$

Then

$$|W_{C,N} - W_{C,\infty}| \leq 2\eta_N, \quad (1966)$$

and

$$|\Delta_{C,N} - \Delta_{C,\infty}| \leq 2\eta_N. \quad (1967)$$

Hessian convergence requires an independent C^2 operator error $\eta_N^{(2)} \longrightarrow 0$. We require $\theta_{H,N} \longrightarrow \theta_{H,\infty}$,

$$\theta_{\text{flat},N}^{\text{full}} \longrightarrow \theta_{\text{flat},\infty}^{\text{full}}, \quad \delta\theta_{\text{HF},N} \longrightarrow \delta\theta_{\text{HF},\infty} \quad (1968)$$

on at least two inequivalent admissible cover towers.

E. Mechanism decomposition when the two angles differ

Suppose $\mathcal{R}_{\text{HF}} > 1$. We do not interpret this inequality as a failure of the microscopic Hamiltonian or of the Hodge construction. We resolve the discrepancy into five quantitatively distinct mechanisms: quartic dispersion + remote modes + multisector resonances + anisotropy + avoided crossings. At a established full Hodge root,

$$dE_* = 0, \quad \mathcal{K}_* = 0. \quad (1969)$$

If tangent inversion holds, the leading local term is quartic:

$$E - E_* = \frac{1}{24} \mathcal{Q}_*[x^4] + O(\|x\|_G^6). \quad (1970)$$

The dimensionless quartic residual is

$$r_{4,C} = \frac{Q_* R_*^4}{12\Delta_C}. \quad (1971)$$

A value $r_{4,C} \not\ll 1$ shows that complete quadratic cancellation is insufficient on the finite physical domain. Let

$$Q_M = \frac{2\pi}{\xi_M} \quad (1972)$$

be the independently determined inverse registry scale. If the quadratic continuum contribution vanishes and the quartic term is the first nonzero lattice correction, the physical bandwidth must satisfy the conditional scaling law

$$W_C^* = c_{4,C} t (aQ_M)^4 + o[t(aQ_M)^4]. \quad (1973)$$

Equivalently,

$$\frac{W_C^* \xi_M^4}{ta^4} \longrightarrow 16\pi^4 c_{4,C}. \quad (1974)$$

Failure of this collapse rules out quartic dominance in the declared scaling regime. If tangent inversion is absent, the cubic residual

$$r_{3,C} = \frac{T_{3,*} R_*^3}{3\Delta_C}, \quad (1975)$$

where

$$T_{3,*} = \sup_{\|u\|_G=1} \left| \mathcal{T}_*^{(3)}[u, u, u] \right|, \quad (1976)$$

must be tested first. We attribute the angle mismatch to higher-order dispersion only when $\Lambda_H(\theta_H)$ is uncertainty compatible with zero; $r_{3,C}$ or $r_{4,C}$ is non-negligible; the measured residual bandwidth follows the predicted power law; remote, multisector, anisotropic, and avoided-crossing corrections are smaller. The quartic interpretation is not applied to a model whose exact quadratic coefficient has no root. A nonzero quartic tensor coexisting with a strictly positive quadratic coefficient remains subleading. Let P_{act} contain the target and every mode resonant with the declared energy window. Let

$$Q_{\text{tail}} = I - P_{\text{act}} \quad (1977)$$

contain only uniformly detuned modes. Define the minimum tail detuning

$$\delta_{\text{tail}} = \inf \text{dist} [E_C, \text{Spec } H_{Q_{\text{tail}}}] . \quad (1978)$$

The second-order tail estimate is

$$\begin{aligned} \eta_{\text{tail}}^{(2)} &= \frac{2v_0 v_2 + 2v_1^2}{\delta_{\text{tail}}} + \frac{4v_0 v_1 \ell_1 + v_0^2 \ell_2}{\delta_{\text{tail}}^2} \\ &+ \frac{2v_0^2 \ell_1^2}{\delta_{\text{tail}}^3}. \end{aligned} \quad (1979)$$

The normalized remote-mode residual is

$$r_{\text{tail}} = \frac{D_G^2 \eta_{\text{tail}}^{(2)}}{2\Delta_C}. \quad (1980)$$

Let the active-shell Hodge root have slope s_H . If the tail tensor and its angle derivative satisfy

$$\|\mathcal{K}_{\text{tail}}\| \leq \eta_2, \quad (1981)$$

and

$$\|\partial_{\theta} \mathcal{K}_{\text{tail}}\| \leq \eta_{2,1} < s_H, \quad (1982)$$

then the tail-induced root displacement satisfies

$$|\delta\theta_{\text{tail}}| \leq \frac{\eta_2}{s_H - \eta_{2,1}}. \quad (1983)$$

We calculate a nested active-space sequence $P_{\text{act}}^{(1)} \subset P_{\text{act}}^{(2)} \subset \dots$ and the corresponding roots $\theta_H^{(1)}, \theta_H^{(2)}, \dots$. Remote-mode attribution requires $\theta_H^{(j)} \rightarrow \theta_H^{(\infty)}$ and agreement between the measured displacement and Eq. (1983). If $\delta_{\text{tail}} \lesssim \|V\|$, the

detuned-tail expansion is invalid. The corresponding states are reclassified as active resonant modes and are treated by exact block diagonalization. Let each representation sector possess a candidate Hodge root $\theta_{H,\rho}$. Define the root spread

$$S_{\theta}^{\text{rep}} = \max_{\rho \in \mathcal{R}_C} \theta_{H,\rho} - \min_{\rho \in \mathcal{R}_C} \theta_{H,\rho}. \quad (1984)$$

The regular-weighted mean root is

$$\bar{\theta}_H^{\text{rep}} = \frac{\sum_{\rho \in \mathcal{R}_C} d_{\rho} r_{C,\rho} \theta_{H,\rho}}{\sum_{\rho \in \mathcal{R}_C} d_{\rho} r_{C,\rho}}, \quad (1985)$$

and the weighted RMS spread is

$$\sigma_{\theta,H}^{\text{rep}} = \left[\frac{\sum_{\rho} d_{\rho} r_{C,\rho} \left(\theta_{H,\rho} - \bar{\theta}_H^{\text{rep}} \right)^2}{\sum_{\rho} d_{\rho} r_{C,\rho}} \right]^{1/2}. \quad (1986)$$

Suppose the scalar Hessian in sector ρ has an ideal simple root $\theta_{H,\rho}$, slope margin $s_{\rho} > 0$, and bounded nonlinear remainder η_{ρ} . Its possible exact root lies inside

$$I_{\rho} = \left[\theta_{H,\rho} - \frac{\eta_{\rho}}{s_{\rho}}, \theta_{H,\rho} + \frac{\eta_{\rho}}{s_{\rho}} \right]. \quad (1987)$$

A common simultaneous root requires

$$\bigcap_{\rho \in \mathcal{R}_C} I_{\rho} \neq \emptyset. \quad (1988)$$

If

$$\bigcap_{\rho} I_{\rho} = \emptyset, \quad (1989)$$

no scalar twist can cancel every declared representation sector within the established remainder bounds. Even when all sector widths are individually small, the full bandwidth contains the center spread S_C^{ctr} . We define the multisector residual

$$r_{\text{multi}} = \frac{S_C^{\text{ctr}} + W_C^{\text{sec,max}}}{\Delta_C^{\text{full}}}. \quad (1990)$$

Inequivalent irreducible sectors cannot hybridize under exact translation symmetry. Their energy or score branches can therefore cross exactly. Let $\mathcal{M}_{\rho}^*(\theta)$ be the optimized sector score. A hierarchy switch occurs at

$$\mathcal{M}_{\rho}^* = \mathcal{M}_{\sigma}^*, \quad \rho \neq \sigma. \quad (1991)$$

The global physical angle can jump from $\theta_{H,\rho}$ to $\theta_{H,\sigma}$ without any avoided crossing of the underlying symmetry-resolved eigenstates. We attribute the Hodge-physical mismatch to multisector reconstruction when

S_θ^{rep} or S_C^{ctr} is resolved; the full optimum belongs to a sector different from the seed Hodge sector; the sector crossing obeys the translation selection rule; the effect survives complete finite-quotient weighting. At a trace-root angle, define

$$\widehat{\mathcal{A}}_{\text{tr}} = \widehat{\mathcal{K}}(\theta_{\text{tr}}) - \frac{1}{m} \text{Tr} \widehat{\mathcal{K}}(\theta_{\text{tr}}) I_m. \quad (1992)$$

Since the trace is zero,

$$\widehat{\mathcal{A}}_{\text{tr}} = \widehat{\mathcal{K}}(\theta_{\text{tr}}). \quad (1993)$$

The normalized anisotropy residual is

$$r_{\text{aniso}} = \frac{D_G^2 \|\widehat{\mathcal{A}}_{\text{tr}}\|_{\text{op}}}{2\Delta_C}. \quad (1994)$$

Let

$$\theta_j : \quad \lambda_j(\theta_j) = 0 \quad (1995)$$

be the individual principal-root angles. Their spread is

$$S_\theta^{\text{dir}} = \max_j \theta_j - \min_j \theta_j. \quad (1996)$$

Under the perturbative bounds of Eqs. (1829) and (1830),

$$S_\theta^{\text{dir}} \leq \frac{2\eta_2}{s_H - \eta_{2,1}}. \quad (1997)$$

A measured spread exceeding this bound disproves the declared approximately isotropic carrier. We attribute the mismatch to anisotropy when $\bar{\lambda}_H$ crosses zero; $\mathcal{A}_H^{\text{op}}$ remains nonzero beyond uncertainty; S_θ^{dir} is resolved; no common principal-root interval exists. In that case, a scalar Hodge-trace cancellation is not the mechanism of a multidirectionally flat band. Consider two same-sector diabatic branches with effective Hamiltonian

$$H_2(x) = \bar{\varepsilon}(x) I_2 + \begin{pmatrix} \delta(x)/2 & V(x) \\ V^*(x) & -\delta(x)/2 \end{pmatrix}. \quad (1998)$$

The adiabatic energies are

$$E_\pm = \bar{\varepsilon} \pm \Omega, \quad \Omega = \sqrt{\frac{\delta^2}{4} + |V|^2}. \quad (1999)$$

At an exact resonance point

$$\delta = 0, \quad |V| > 0, \quad (2000)$$

the covariant Hessian of Ω is

$$\nabla^2 \Omega = \nabla^2 |V| + \frac{d\delta \otimes d\delta}{4|V|}. \quad (2001)$$

Therefore

$$\nabla^2 E_\pm = \frac{\nabla^2 \varepsilon_a + \nabla^2 \varepsilon_b}{2} \pm \left[\nabla^2 |V| + \frac{d\delta \otimes d\delta}{4|V|} \right]. \quad (2002)$$

The rank-one term

$$\frac{d\delta \otimes d\delta}{4|V|} \quad (2003)$$

is enhanced when the avoided-crossing gap

$$g_{\text{AC}} = 2|V| \quad (2004)$$

is small. Define the avoided-crossing Hessian scale

$$\Lambda_{\text{AC}} = \frac{\|d\delta\|_{G^{-1}}^2}{4|V|} = \frac{\|d\delta\|_{G^{-1}}^2}{2g_{\text{AC}}}. \quad (2005)$$

The normalized residual is

$$r_{\text{AC}} = \frac{D_G^2 \Lambda_{\text{AC}}}{2\Delta_C}. \quad (2006)$$

For a twist-controlled resonance, the two-level quantum metric is

$$g_{\theta\theta} = \left(\frac{d\vartheta}{d\theta} \right)^2 + \frac{1}{4} \sin^2(2\vartheta) \left(\frac{d\varphi}{d\theta} \right)^2. \quad (2007)$$

At resonance with locally constant real V ,

$$g_{\theta\theta}^{\text{res}} = \frac{(\partial_\theta \delta)^2}{16|V|^2}. \quad (2008)$$

A small gap can therefore produce an abrupt projector rotation and a large local Hessian response even though the energy branches remain continuous. Let θ_e^{AC} be the minimum-gap angle of avoided crossing e . Define

$$d_H^{\text{AC}} = \min_e |\theta_H - \theta_e^{\text{AC}}|, \quad (2009)$$

and

$$d_F^{\text{AC}} = \min_e |\theta_{\text{flat}}^{\text{full}} - \theta_e^{\text{AC}}|. \quad (2010)$$

We attribute the mismatch to an avoided-crossing mechanism when

d_H^{AC} is small,

r_{AC} is non-negligible,

$g_{\theta\theta}^{\text{res}}$ shows rapid projector rotation,

the Hodge root follows the minimum-gap trajectory,

the global physical score optimizes at a different angle. (2011)

An avoided crossing can generate local mass cancellation while failing to suppress the complete representation-space bandwidth. We collect the five normalized mismatch diagnostics into

$$\mathbf{r}_{\text{mis}} = (r_{\text{higher}}, r_{\text{tail}}, r_{\text{multi}}, r_{\text{aniso}}, r_{\text{AC}}), \quad (2012)$$

where

$$r_{\text{higher}} = \max \{r_{3,C}, r_{4,C}\}. \quad (2013)$$

Define

$$R_{\text{sum}} = r_{\text{higher}} + r_{\text{tail}} + r_{\text{multi}} + r_{\text{aniso}} + r_{\text{AC}}. \quad (2014)$$

When $R_{\text{sum}} > 0$, the normalized attribution weights are

$$\eta_j^{\text{mis}} = \frac{r_j}{R_{\text{sum}}}, \quad \sum_j \eta_j^{\text{mis}} = 1. \quad (2015)$$

These weights are diagnostic rather than probabilistic. They quantify the relative sizes of independently computed obstruction scales. The unresolved score-gradient residual is

$$F_{\text{unexpl}} = L'_{\text{phys}}(\theta_H) - \sum_j F_j^{\text{model}}, \quad (2016)$$

where F_j^{model} is the directly calculated score-gradient contribution of the five mechanisms. A complete mechanism attribution requires

$$|F_{\text{unexpl}}| \leq \varepsilon_F. \quad (2017)$$

For every candidate branch, we record

$$\begin{aligned} \mathcal{C}_{H \rightarrow \text{flat}} = & \left(\theta_H, \delta\theta_H, \theta_{\text{flat}}^{\text{full}}, \delta\theta_{\text{flat}}, \mathcal{R}_{\text{HF}}, \right. \\ & \Lambda_H(\theta_H), \mathcal{A}_H^{\text{op}}(\theta_H), s_H, \\ & W_C(\theta_H), \Delta_C(\theta_H), v_{\text{max}, C}(\theta_H), \\ & Q_*, T_{3,*}, \eta_{\text{tail}}^{(2)}, S_{\theta}^{\text{rep}}, S_C^{\text{ctr}}, \\ & S_{\theta}^{\text{dir}}, \Lambda_{\text{AC}}, \mathcal{E}_H, \mathcal{R}_{\mathcal{K}}^{\text{flat}}, \\ & \left. C_{\text{rep}, C}, \eta_N, \eta_N^{(2)}, \varepsilon_{\text{shell}} \right). \end{aligned} \quad (2018)$$

The Hodge mechanism applies when the cancellation angle agrees with the independently determined physical optimum and the local-to-global estimates hold. Local Hodge cancellation fixes the quadratic response at the selected point; the global bandwidth and isolation estimates determine whether this local mechanism accounts for the complete flat-band optimum. A resolved mismatch instead identifies a different spectral mechanism.

The logical content of the Hodge analysis is correspondingly hierarchical. The condition $\mathcal{K}(\theta_H) = 0$ establishes local quadratic cancellation of the kinetic response. If the relevant odd-symmetry constraint is also present, the cancellation of the quadratic term forces the leading local dispersion to begin at quartic order. This is still only a local statement. Global spectral flatness requires an independent control of the dispersion over the full declared domain: if $\sup_{\mathcal{D}} \|dE\| \rightarrow 0$, then the bandwidth satisfies $W \rightarrow 0$. If, in addition, $\liminf \Delta > 0$, the flat sector remains spectrally isolated and one obtains isolated energetic flatness rather than a merely collapsed bandwidth accompanied by gap closure.

The Hodge mechanism can therefore be identified with the physical magic-angle mechanism only after the local and global criteria are combined. In particular, the agreement $\theta_H \simeq \theta_{\text{flat}}^{\text{full}}$ must hold within the complete uncertainty and finite-cover error budget, and the required

local-to-global verification must be satisfied. Only under these conditions can the physical magic angle be attributed to a complete cancellation of the local kinetic response.

We therefore retain Hodge cancellation as a microscopic explanatory mechanism rather than as the definition of physical magic. Physical magic remains defined by the independently established global spectral condition. The role of the Hodge tensor is instead diagnostic: it determines whether that globally established flat sector is generated by a complete local cancellation of the kinetic response or whether a more global mechanism is required.

IX. CURVATURE-CONTROLLED FLAT-BAND LANDSCAPES AND CONDITIONAL BRANCH BIRTH

We now determine whether negative curvature merely displaces and renormalizes an existing flat-band optimum or changes the topology of the flatness landscape by creating a genuinely new stable branch. We use $\kappa = -K \geq 0$ as the nonnegative curvature magnitude. The Euclidean limit is $\kappa = 0$, whereas finite negative curvature corresponds to $\kappa > 0$. The interlayer coupling is $\omega = \frac{w}{t}$. For every continuously tracked spectral family C_ν , we distinguish three outcomes:

$$\begin{aligned} \text{regular curvature shift : } & \theta_\nu^*(0) \longrightarrow \theta_\nu^*(\kappa), \\ \text{curvature enhancement : } & \mathcal{M}_\nu^*(\kappa) > \mathcal{M}_\nu^*(0), \\ \text{curvature-induced branch birth : } & N_{\text{mag}}(\kappa) > N_{\text{mag}}(0) \quad \text{through a} \end{aligned} \quad (2019)$$

The first two statements do not imply the third. A branch can move and improve continuously without any bifurcation. We retain the phrase curvature-born flat-band branch only after verifying all of the following:

$$\begin{aligned} & \text{no matching stable Euclidean antecedent,} \\ & \text{a finite critical curvature } \kappa_c > 0, \\ & F_\nu = F_{\nu, \theta} = 0, \\ & F_{\nu, \kappa} \neq 0, \quad F_{\nu, \theta\theta} \neq 0, \\ & \Delta_\nu^L > 0, \quad C_\nu^{\text{coh}, L} > 0, \\ & |\theta_\nu^* - \theta_c| \propto |\kappa - \kappa_c|^{1/2}, \\ & \text{finite-cover, shell, and cross-tower convergence.} \end{aligned} \quad (2020)$$

A branch is called curvature-born only when every condition in Eq. (2020) holds. Other patterns describe continuous curvature control, branch enhancement, or branch restructuring.

A. (K, θ) flatness landscape

At fixed ω , we calculate the physical flatness landscape on

$$\mathcal{P}_{\kappa\theta} = [0, \kappa_{\text{max}}] \times [\theta_{\text{min}}, \theta_{\text{max}}]. \quad (2021)$$

TABLE VI. Quantitative interpretation of a discrepancy between the Hodge candidate angle and the full physical flat-band angle.

Mechanism	Primary diagnostic	Required interpretation
Quartic or higher-order dispersion	$r_{3,C}, r_{4,C}$, length-bandwidth scaling	The quadratic tensor vanishes but higher derivatives retain dispersion
Remote modes	$\eta_{\text{tail}}^{(2)}$, nested active-space root drift	Detuned complementary states renormalize the local root
Multisector resonances	$S_{\theta}^{\text{rep}}, S_C^{\text{ctr}}$	Different representation sectors optimize at different angles
Anisotropy	$a_{\infty}, S_{\theta}^{\text{dir}}$	A trace root fails to cancel every Hodge direction
Avoided crossings	$\Lambda_{AC}, g_{\theta\theta}^{\text{res}}$	Resonant level repulsion creates a local but nonglobal cancellation

The corresponding (K, θ) domain is obtained from $K = -\kappa$. For each finite cover N and tracked target family C_{ν} , define the admissible set

$$\mathcal{A}_{\nu,N} = \left\{ (\kappa, \theta) \in \mathcal{P}_{\kappa\theta} : \Delta_{\nu,N}^L > 0, C_{\nu,N}^{\text{coh},L} > 0, \right. \\ \left. \text{rank } P_{\nu,N} = r_{\nu}, C_{\text{rep},\nu,N} = 1, \right. \\ \left. \varepsilon_{\text{track},\nu,N} < \varepsilon_{\text{track}}^{\text{max}}, \varepsilon_{\text{tot},\nu,N} < \varepsilon_{\text{tot}}^{\text{max}} \right\}.$$

The hard requirements defining $\mathcal{A}_{\nu,N}$ separate a smooth stationary-point bifurcation from:

$\Delta_{\nu} = 0$	gap-mediated admissibility boundary,
$C_{\nu}^{\text{coh}} = 0$	coherence boundary,
$\text{rank } P_{\nu}$ changes	spectral-family termination,
$C_{\text{rep},\nu} < 1$	representation-incomplete calculation,
ε_{tot} large	unresolved numerical region.

(2022)

The operational quantities contain maxima and minima over generalized momentum, energy, and representation space. They are smooth only while the corresponding active extremizers remain isolated and continuously tracked. We therefore decompose $\mathcal{A}_{\nu,N}$ into smooth sheets

$$\mathcal{A}_{\nu,N} = \bigcup_{\alpha} \mathcal{S}_{\nu,N}^{(\alpha)}, \quad (2023)$$

such that on every sheet: the same upper and lower band-edge extremizers are active; the same complementary states determine the gap; the same generalized-momentum point determines v_{max} , the same broadened-DOS peak is followed; the representation and point-group signatures are fixed. For a fold calculation we require

$$L_{\nu} \in C^3 \left(\mathcal{S}_{\nu,N}^{(\alpha)} \right), \quad (2024)$$

and for a cusp calculation we require

$$L_{\nu} \in C^4 \left(\mathcal{S}_{\nu,N}^{(\alpha)} \right). \quad (2025)$$

A switch of the active extremizer produces a piecewise-smooth envelope event. We do not classify such a switch as a fold unless a separate smooth branch decomposition

resolves an actual stationary-point collision. Define

$$V_{1,\nu} = \hbar D_{G,\nu} v_{\text{max},\nu}, \\ V_{2,\nu} = \frac{1}{2} D_{G,\nu}^2 \Lambda_{\text{max},\nu}, \quad (2026) \\ X_{\nu} = E_{\text{kin}} \varrho_{\nu,\eta}^{\text{coh,max}},$$

and $C_{\nu} = C_{\nu}^{\text{coh}}$. Inside an admissible sheet, the minimal operational score is

$$\mathcal{M}_{\nu}^{(1)} = \left[\frac{\Delta_{\nu}}{\Delta_{\nu} + W_{\nu}} \frac{\Delta_{\nu}}{\Delta_{\nu} + V_{1,\nu}} \times \frac{X_{\nu}}{1 + X_{\nu}} C_{\nu} \right]^{1/4}.$$

The strong score is

$$\mathcal{M}_{\nu}^{(2)} = \left[\frac{\Delta_{\nu}}{\Delta_{\nu} + W_{\nu}} \frac{\Delta_{\nu}}{\Delta_{\nu} + V_{1,\nu}} \right. \\ \left. \times \frac{\Delta_{\nu}}{\Delta_{\nu} + V_{2,\nu}} \frac{X_{\nu}}{1 + X_{\nu}} C_{\nu} \right]^{1/5}.$$

The score used for branch topology is fixed before the bifurcation scan. We denote the selected score by $\mathcal{M}_{\nu}$. Changing from $\mathcal{M}_{\nu}^{(1)}$ to $\mathcal{M}_{\nu}^{(2)}$ after observing a branch is treated as a new analysis rather than as a continuation of the same specified landscape. Define

$$L_{\nu} = \ln \mathcal{M}_{\nu}. \quad (2027)$$

For the minimal score,

$$4L_{\nu}^{(1)} = 2 \ln \Delta_{\nu} - \ln(\Delta_{\nu} + W_{\nu}) \\ - \ln(\Delta_{\nu} + V_{1,\nu}) + \ln X_{\nu} \\ - \ln(1 + X_{\nu}) + \ln C_{\nu}. \quad (2028)$$

For the strong score,

$$5L_{\nu}^{(2)} = 3 \ln \Delta_{\nu} - \ln(\Delta_{\nu} + W_{\nu}) \\ - \ln(\Delta_{\nu} + V_{1,\nu}) - \ln(\Delta_{\nu} + V_{2,\nu}) \\ + \ln X_{\nu} - \ln(1 + X_{\nu}) + \ln C_{\nu}. \quad (2029)$$

For a positive smooth scalar function $Y(\theta)$, write

$$Y_j = \partial_{\theta}^j Y. \quad (2030)$$

The first four logarithmic derivatives are

$$\begin{aligned}\mathfrak{D}_1[Y] &= \frac{Y_1}{Y}, \\ \mathfrak{D}_2[Y] &= \frac{Y_2}{Y} - \frac{Y_1^2}{Y^2}, \\ \mathfrak{D}_3[Y] &= \frac{Y_3}{Y} - \frac{3Y_1Y_2}{Y^2} + \frac{2Y_1^3}{Y^3},\end{aligned}\quad (2031)$$

and

$$\begin{aligned}\mathfrak{D}_4[Y] &= \frac{Y_4}{Y} - \frac{4Y_1Y_3 + 3Y_2^2}{Y^2} \\ &\quad + \frac{12Y_1^2Y_2}{Y^3} - \frac{6Y_1^4}{Y^4}.\end{aligned}\quad (2032)$$

The stationary equation is

$$F_\nu = \partial_\theta L_\nu = 0. \quad (2033)$$

For the minimal score,

$$\begin{aligned}4F_\nu &= 2\mathfrak{D}_1[\Delta_\nu] - \mathfrak{D}_1[\Delta_\nu + W_\nu] \\ &\quad - \mathfrak{D}_1[\Delta_\nu + V_{1,\nu}] + \mathfrak{D}_1[X_\nu] \\ &\quad - \mathfrak{D}_1[1 + X_\nu] + \mathfrak{D}_1[C_\nu].\end{aligned}\quad (2034)$$

Equivalently,

$$\begin{aligned}4F_\nu &= \frac{2\Delta_{\nu,\theta}}{\Delta_\nu} - \frac{\Delta_{\nu,\theta} + W_{\nu,\theta}}{\Delta_\nu + W_\nu} \\ &\quad - \frac{\Delta_{\nu,\theta} + V_{1,\nu,\theta}}{\Delta_\nu + V_{1,\nu}} + \frac{X_{\nu,\theta}}{X_\nu(1 + X_\nu)} + \frac{C_{\nu,\theta}}{C_\nu}.\end{aligned}\quad (2035)$$

The transverse stability derivative is

$$\begin{aligned}4F_{\nu,\theta} &= 2\mathfrak{D}_2[\Delta_\nu] - \mathfrak{D}_2[\Delta_\nu + W_\nu] \\ &\quad - \mathfrak{D}_2[\Delta_\nu + V_{1,\nu}] + \mathfrak{D}_2[X_\nu] \\ &\quad - \mathfrak{D}_2[1 + X_\nu] + \mathfrak{D}_2[C_\nu].\end{aligned}\quad (2036)$$

A stable interior branch satisfies

$$F_\nu = 0, \quad F_{\nu,\theta} < 0. \quad (2037)$$

The positive transverse flatness curvature is

$$\kappa_{\perp,\nu} = -F_{\nu,\theta} > 0. \quad (2038)$$

The next two derivatives required for fold and cusp verification are

$$\begin{aligned}4F_{\nu,\theta\theta} &= 2\mathfrak{D}_3[\Delta_\nu] - \mathfrak{D}_3[\Delta_\nu + W_\nu] \\ &\quad - \mathfrak{D}_3[\Delta_\nu + V_{1,\nu}] + \mathfrak{D}_3[X_\nu] \\ &\quad - \mathfrak{D}_3[1 + X_\nu] + \mathfrak{D}_3[C_\nu],\end{aligned}\quad (2039)$$

and

$$\begin{aligned}4F_{\nu,\theta\theta\theta} &= 2\mathfrak{D}_4[\Delta_\nu] - \mathfrak{D}_4[\Delta_\nu + W_\nu] \\ &\quad - \mathfrak{D}_4[\Delta_\nu + V_{1,\nu}] + \mathfrak{D}_4[X_\nu] \\ &\quad - \mathfrak{D}_4[1 + X_\nu] + \mathfrak{D}_4[C_\nu].\end{aligned}\quad (2040)$$

For two variables θ and κ , define

$$\mathfrak{D}_{\theta\kappa}[Y] = \frac{Y_{\theta\kappa}}{Y} - \frac{Y_\theta Y_\kappa}{Y^2}. \quad (2041)$$

The mixed third derivative required at a cusp is

$$\begin{aligned}\mathfrak{D}_{\theta\theta\kappa}[Y] &= \frac{Y_{\theta\theta\kappa}}{Y} - \frac{Y_{\theta\theta}Y_\kappa + 2Y_\theta Y_{\theta\kappa}}{Y^2} \\ &\quad + \frac{2Y_\theta^2 Y_\kappa}{Y^3}.\end{aligned}\quad (2042)$$

The corresponding derivative with respect to ω is obtained by the exact replacement $\kappa \rightarrow \omega$. For the minimal landscape,

$$\begin{aligned}4F_{\nu,\kappa} &= 2\mathfrak{D}_{\theta\kappa}[\Delta_\nu] - \mathfrak{D}_{\theta\kappa}[\Delta_\nu + W_\nu] \\ &\quad - \mathfrak{D}_{\theta\kappa}[\Delta_\nu + V_{1,\nu}] + \mathfrak{D}_{\theta\kappa}[X_\nu] \\ &\quad - \mathfrak{D}_{\theta\kappa}[1 + X_\nu] + \mathfrak{D}_{\theta\kappa}[C_\nu],\end{aligned}\quad (2043)$$

and

$$\begin{aligned}4F_{\nu,\theta\kappa} &= 2\mathfrak{D}_{\theta\theta\kappa}[\Delta_\nu] - \mathfrak{D}_{\theta\theta\kappa}[\Delta_\nu + W_\nu] \\ &\quad - \mathfrak{D}_{\theta\theta\kappa}[\Delta_\nu + V_{1,\nu}] + \mathfrak{D}_{\theta\theta\kappa}[X_\nu] \\ &\quad - \mathfrak{D}_{\theta\theta\kappa}[1 + X_\nu] + \mathfrak{D}_{\theta\theta\kappa}[C_\nu].\end{aligned}\quad (2044)$$

The formulas for $F_{\nu,\omega}$ and $F_{\nu,\theta\omega}$ follow by replacing every κ -derivative by the corresponding ω -derivative. For the strong score, Eqs. (2034)–(2044) are replaced by

$$\begin{aligned}5\partial_\theta^n L_\nu^{(2)} &= 3\mathfrak{D}_n[\Delta_\nu] - \mathfrak{D}_n[\Delta_\nu + W_\nu] \\ &\quad - \mathfrak{D}_n[\Delta_\nu + V_{1,\nu}] - \mathfrak{D}_n[\Delta_\nu + V_{2,\nu}] \\ &\quad + \mathfrak{D}_n[X_\nu] - \mathfrak{D}_n[1 + X_\nu] + \mathfrak{D}_n[C_\nu],\end{aligned}\quad (2045)$$

for $n = 1, 2, 3, 4$, with the same mixed-derivative replacement. The two-dimensional local Hessian of the logarithmic score is

$$\mathbb{H}_\nu^L = \begin{pmatrix} L_{\nu,\theta\theta} & L_{\nu,\theta\kappa} \\ L_{\nu,\theta\kappa} & L_{\nu,\kappa\kappa} \end{pmatrix} = \begin{pmatrix} F_{\nu,\theta} & F_{\nu,\kappa} \\ F_{\nu,\kappa} & L_{\nu,\kappa\kappa} \end{pmatrix}. \quad (2046)$$

Curvature is an external control coordinate, so a stable magic branch requires only $F_{\nu,\theta} < 0$. A simultaneous local maximum in both θ and κ would additionally require $L_{\nu,\kappa\kappa} < 0$ and

$$\det \mathbb{H}_\nu^L > 0. \quad (2047)$$

We do not impose these stronger conditions on a control-parameter branch. At fixed (κ, ω) , define the number of admissible stable interior branches by

$$\begin{aligned}N_{\text{mag},N}(\kappa, \omega) &= \sum_\nu \# \left\{ \theta \in (\theta_{\min}, \theta_{\max}) : F_\nu = 0, \right. \\ &\quad \left. F_{\nu,\theta} < 0, \Delta_{\nu,N}^L > 0, C_{\nu,N}^{\text{coh},L} > 0 \right\}.\end{aligned}$$

This integer is locally constant away from bifurcation, admissibility, and physical-domain boundaries. We distinguish the following loci:

$$\begin{aligned}\mathcal{L}_{\text{fold}} &= \{F_\nu = 0, \quad F_{\nu,\theta} = 0\}, \\ \mathcal{L}_{\text{cusp}} &= \{F_\nu = F_{\nu,\theta} = F_{\nu,\theta\theta} = 0\}, \\ \mathcal{L}_{\text{gap}} &= \{\Delta_\nu = 0\}, \quad \mathcal{L}_{\text{coh}} = \{C_\nu^{\text{coh}} = 0\}, \\ \mathcal{L}_{\text{hier},\mu\nu} &= \{L_\mu^* = L_\nu^*\},\end{aligned}\quad (2048)$$

and

$$\mathcal{L}_{\text{cryst}} = \{g\theta \in \text{Comm}_G(\Gamma)\}. \quad (2049)$$

These sets are plotted and established separately. Let the conservative observable bounds be

$$\begin{aligned} W_\nu^L &\leq W_\nu \leq W_\nu^U, & \Delta_\nu^L &\leq \Delta_\nu \leq \Delta_\nu^U, \\ V_{1,\nu}^L &\leq V_{1,\nu} \leq V_{1,\nu}^U, \\ X_\nu^L &\leq X_\nu \leq X_\nu^U, & C_\nu^L &\leq C_\nu \leq C_\nu^U. \end{aligned} \quad (2050)$$

The established lower score is

$$\mathcal{M}_\nu^L = \left[\frac{\Delta_\nu^L}{\Delta_\nu^L + W_\nu^U} \frac{\Delta_\nu^L}{\Delta_\nu^L + V_{1,\nu}^U} \times \frac{X_\nu^L}{1 + X_\nu^L} C_\nu^L \right]^{1/4},$$

where it is set to zero if either $\Delta_\nu^L \leq 0$ or $C_\nu^L \leq 0$. The corresponding upper score is

$$\mathcal{M}_\nu^U = \left[\frac{\Delta_\nu^U}{\Delta_\nu^U + W_\nu^L} \frac{\Delta_\nu^U}{\Delta_\nu^U + V_{1,\nu}^L} \times \frac{X_\nu^U}{1 + X_\nu^U} C_\nu^U \right]^{1/4}.$$

Monotonicity gives

$$\mathcal{M}_\nu^L \leq \mathcal{M}_\nu \leq \mathcal{M}_\nu^U. \quad (2051)$$

At fixed curvature, every possible global optimizer lies in

$$\hat{\Theta}_{\text{opt}}(\kappa) = \left\{ \theta : \mathcal{M}^U(\kappa, \theta) \geq \sup_{\vartheta} \mathcal{M}^L(\kappa, \vartheta) \right\}. \quad (2052)$$

The envelope identifies the admissible angular uncertainty of the physical optimum without relying on one fitted maximum.

B. Branch tracking

At fixed ω , a regular stationary branch satisfies

$$F_\nu[\theta_\nu^*(\kappa), \kappa] = 0. \quad (2053)$$

When $F_{\nu,\theta} \neq 0$, the implicit-function theorem gives a locally unique smooth solution. Differentiation gives the exact branch velocity

$$\frac{d\theta_\nu^*}{d\kappa} = -\frac{F_{\nu,\kappa}}{F_{\nu,\theta}}. \quad (2054)$$

In the original curvature coordinate $K = -\kappa$,

$$\frac{d\theta_\nu^*}{dK} = \frac{F_{\nu,\kappa}}{F_{\nu,\theta}}. \quad (2055)$$

A second differentiation gives

$$\begin{aligned} 0 &= F_{\nu,\theta} \frac{d^2\theta_\nu^*}{d\kappa^2} + F_{\nu,\theta\theta} \left(\frac{d\theta_\nu^*}{d\kappa} \right)^2 \\ &\quad + 2F_{\nu,\theta\kappa} \frac{d\theta_\nu^*}{d\kappa} + F_{\nu,\kappa\kappa}. \end{aligned} \quad (2056)$$

Therefore

$$\frac{d^2\theta_\nu^*}{d\kappa^2} = -\frac{F_{\nu,\kappa\kappa} + 2F_{\nu,\theta\kappa}\theta_\nu^{*'} + F_{\nu,\theta\theta}(\theta_\nu^{*'})^2}{F_{\nu,\theta}}. \quad (2057)$$

The optimized branch score is

$$L_\nu^*(\kappa) = L_\nu[\theta_\nu^*(\kappa), \kappa]. \quad (2058)$$

Since $L_{\nu,\theta} = 0$ along the branch, the envelope theorem gives

$$\frac{dL_\nu^*}{d\kappa} = L_{\nu,\kappa} \Big|_{\theta=\theta_\nu^*}. \quad (2059)$$

The second derivative is

$$\frac{d^2L_\nu^*}{d\kappa^2} = L_{\nu,\kappa\kappa} - \frac{L_{\nu,\theta\kappa}^2}{L_{\nu,\theta\theta}} \quad (2060)$$

on every regular branch. At a regular point (κ_j, θ_j^*) , the second-order predictor is

$$\theta_{j+1}^{\text{pred}} = \theta_j^* + \theta_\nu^{*'}(\kappa_j)\delta\kappa + \frac{1}{2}\theta_\nu^{*''}(\kappa_j)\delta\kappa^2.$$

The corrector applies Newton iteration:

$$\theta^{(r+1)} = \theta^{(r)} - \frac{F_\nu(\theta^{(r)}, \kappa_{j+1})}{F_{\nu,\theta}(\theta^{(r)}, \kappa_{j+1})}. \quad (2061)$$

The correction terminates only when

$$|F_\nu| \leq \varepsilon_F, \quad |\theta^{(r+1)} - \theta^{(r)}| \leq \varepsilon_\theta, \quad (2062)$$

and the spectral projector satisfies the identity tests below. A step is rejected and subdivided when

$$|F_{\nu,\theta}| \leq m_\theta^{\text{reg}}, \quad (2063)$$

because ordinary parameter continuation becomes ill conditioned near a fold. Let

$$z = \begin{pmatrix} \theta \\ \kappa \end{pmatrix}. \quad (2064)$$

At a point satisfying $F_\nu(z) = 0$, a tangent vector $t = (t_\theta, t_\kappa)^\top$ obeys

$$F_{\nu,\theta}t_\theta + F_{\nu,\kappa}t_\kappa = 0. \quad (2065)$$

We choose

$$t = \frac{1}{\sqrt{F_{\nu,\theta}^2 + F_{\nu,\kappa}^2}} \begin{pmatrix} F_{\nu,\kappa} \\ -F_{\nu,\theta} \end{pmatrix}. \quad (2066)$$

The arclength predictor is

$$z_{\text{pred}} = z_j + \delta s t_j. \quad (2067)$$

The corrector solves

$$\begin{cases} F_\nu(z) = 0, \\ t_j^\top(z - z_{\text{pred}}) = 0. \end{cases} \quad (2068)$$

The Newton matrix is

$$J_{\text{arc}} = \begin{pmatrix} F_{\nu,\theta} & F_{\nu,\kappa} \\ t_{j,\theta} & t_{j,\kappa} \end{pmatrix}. \quad (2069)$$

At a generic fold,

$$F_{\nu,\theta} = 0, \quad F_{\nu,\kappa} \neq 0, \quad (2070)$$

so the arclength Jacobian remains invertible. Pseudo-arclength continuation therefore follows both the stable maximum and the unstable minimum through their collision. Let $P_{\nu,j}$ and $P_{\mu,j+1}$ be equal-rank projectors at consecutive control points after transport to one Hilbert space. Define

$$\mathcal{O}_{\nu\mu}^P = \frac{1}{r_\nu} \text{Tr}(P_{\nu,j} P_{\mu,j+1}). \quad (2071)$$

The largest principal-angle distance is

$$d_{\nu\mu}^P = \|P_{\nu,j} - P_{\mu,j+1}\|. \quad (2072)$$

We require

$$\mathcal{O}_{\nu\mu}^P \geq \mathcal{O}_{\min}, \quad d_{\nu\mu}^P \leq d_P^{\max}. \quad (2073)$$

Among compatible candidates, the branch assignment maximizes $\mathcal{O}_{\nu\mu}^P$. Each branch carries the signature

$$\mathfrak{s}_\nu = \left(r_\nu, d_{\rho,\nu}, \lambda_{\text{sym},\nu}, q_{M,\nu}, g_{M,\nu}, \mathfrak{a}_{\text{orb},\nu}, \mathfrak{a}_{\text{layer},\nu} \right). \quad (2074)$$

A regular continuation requires this signature to remain compatible. A discontinuous change of $d_{\rho,\nu}$, r_ν , $\lambda_{\text{sym},\nu}$, $q_{M,\nu}$ is recorded as representation or crystallographic reconstruction, not as ordinary motion of one fixed branch. Let $P_1^{\text{ref}}, \dots, P_{n_{\text{ref}}}^{\text{ref}}$ be specified reference projectors. The ancestry vector is

$$\mathbf{a}_\nu = \left(\frac{\text{Tr } P_\nu P_1^{\text{ref}}}{r_\nu}, \dots, \frac{\text{Tr } P_\nu P_{n_{\text{ref}}}^{\text{ref}}}{r_\nu} \right). \quad (2075)$$

A rapid exchange of the components of $\mathbf{a}_\nu$ while P_ν remains continuous describes branch transmutation through an avoided crossing. It is not branch creation. We represent the complete stationary structure by a branch graph

$$\mathfrak{G}_{\text{br}} = (\mathfrak{V}_{\text{br}}, \mathfrak{E}_{\text{br}}). \quad (2076)$$

Each vertex stores

$$\mathbf{v}_{\nu,j} = \left(\kappa_j, \theta_{\nu,j}^*, L_{\nu,j}^*, F_{\nu,\theta}, P_{\nu,j}, \mathfrak{s}_{\nu,j}, \mathbf{a}_{\nu,j}, \Delta_{\nu,j}, C_{\nu,j}^{\text{coh}} \right). \quad (2077)$$

An edge joins two vertices only when projector, signature, and continuation residuals satisfy their tolerances. At a generic fold, two edges terminate at one critical vertex: stable maximum $\longleftrightarrow$ fold $\longleftrightarrow$ unstable minimum. A new isolated numerical maximum with no connected unstable partner is not accepted as evidence of a saddle-node birth.

C. Euclidean control

At $\kappa = 0$, we evaluate the same Hamiltonian, target prescription, representation completion, broadening, hopping shell, and physical score used at finite curvature. The Euclidean stable-branch set is

$$\mathcal{B}_E = \left\{ (\theta_{\mu,E}^*, P_{\mu,E}, \mathfrak{s}_{\mu,E}) : F_\mu(\theta, 0, \omega) = 0, \right. \\ \left. F_{\mu,\theta}(\theta, 0, \omega) < 0, \Delta_\mu^L > 0, \quad C_\mu^L > 0 \right\}. \quad (2078)$$

Its cardinality is

$$N_{\text{mag}}^E = \#\mathcal{B}_E. \quad (2079)$$

The complete Euclidean control must cover the full physical twist interval. Absence of a branch on a sparse angular grid does not certify

$$\theta_{\mu,E}^* \notin [\theta_{\min}, \theta_{\max}]. \quad (2080)$$

We therefore require either interval exclusion of $F_\mu = 0$ or complete continuation of every Euclidean stationary point. For an interval $I \subset \Theta_{\text{tw}}$, a sufficient no-stationary-root condition is

$$\inf_{\theta \in I} |F_\mu(\theta, 0, \omega)| > \varepsilon_F^U. \quad (2081)$$

If roots exist but all are unstable, we require $F_{\mu,\theta}^L > 0$ at every established root in I . Transport a finite-curvature branch projector to the Euclidean reference fiber and denote it by $\bar{P}_\nu(\kappa)$. For a Euclidean branch μ , define

$$D_{\nu\mu}^E(\kappa)^2 = \frac{[\theta_\nu^*(\kappa) - \theta_{\mu,E}^*]^2}{\theta_{\text{sc}}^2} \\ + \frac{\|\bar{P}_\nu(\kappa) - P_{\mu,E}\|_F^2}{2r_\nu} \\ + \frac{[E_\nu^*(\kappa) - E_{\mu,E}^*]^2}{E_{\text{kin}}^2} + \mathfrak{d}_{\text{sig}}^2. \quad (2082)$$

Here $\mathfrak{d}_{\text{sig}} = 0$ for compatible signatures and is assigned a positive exclusion penalty otherwise. A branch is Euclidean continuable if there exists $\mu \in \mathcal{B}_E$ such that

$$\lim_{\kappa \rightarrow 0^+} D_{\nu\mu}^E(\kappa) = 0. \quad (2083)$$

It is not curvature born even if its angle or score changes rapidly at finite curvature. A necessary condition for curvature-born classification is

$$\inf_{\mu \in \mathcal{B}_E} \liminf_{\kappa \rightarrow 0^+} D_{\nu\mu}^E(\kappa) > 0, \quad (2084)$$

together with a finite-curvature fold. The distance condition alone is insufficient because a branch can enter through a gap boundary, coherence boundary, or physical angular endpoint. Let

$$\zeta = \frac{a\sqrt{\kappa}}{2}, \quad (2085)$$

and let $\Lambda_\nu > 1$ be the fixed dimensionless root parameter of the geometry-only model. The predicted half-angle is

$$s_\nu^{\text{geo}}(\zeta) = \sin \left[\frac{\theta_\nu^{\text{geo}}(\zeta)}{2} \right] = \frac{\sinh \zeta}{\sinh(\Lambda_\nu \zeta)}. \quad (2086)$$

Differentiation gives

$$\frac{d \ln s_\nu^{\text{geo}}}{d\zeta} = \coth \zeta - \Lambda_\nu \coth(\Lambda_\nu \zeta). \quad (2087)$$

Define

$$f(x) = x \coth x. \quad (2088)$$

Its derivative is

$$\begin{aligned} f'(x) &= \coth x - x \operatorname{csch}^2 x \\ &= \frac{\sinh x \cosh x - x}{\sinh^2 x} > 0 \end{aligned} \quad (2089)$$

for every $x > 0$, because

$$\sinh x \cosh x = \frac{1}{2} \sinh(2x) > x. \quad (2090)$$

Since $\Lambda_\nu \zeta > \zeta$,

$$\Lambda_\nu \coth(\Lambda_\nu \zeta) > \coth \zeta. \quad (2091)$$

Therefore

$$\frac{ds_\nu^{\text{geo}}}{d\zeta} < 0. \quad (2092)$$

Every Euclidean geometry-only seed consequently has one unique, single-valued, monotone continuation. Pure replacement of the Euclidean registry scale by the hyperbolic registry scale cannot create an interior fold. The weak-curvature expansion is

$$s_\nu^{\text{geo}} = \frac{1}{\Lambda_\nu} \left[1 - \frac{\Lambda_\nu^2 - 1}{6} \zeta^2 + O(\zeta^4) \right]. \quad (2093)$$

Let

$$\theta_{\nu,E}^* = 2 \arcsin(\Lambda_\nu^{-1}). \quad (2094)$$

Using $\zeta^2 = a^2 \kappa / 4$, we obtain

$$\theta_\nu^{\text{geo}}(\kappa) = \theta_{\nu,E}^* - \frac{a^2 \kappa}{12} \cot \left(\frac{\theta_{\nu,E}^*}{2} \right) + O(a^4 \kappa^2). \quad (2095)$$

In terms of $K = -\kappa$,

$$\theta_\nu^{\text{geo}}(K) = \theta_{\nu,E}^* + \frac{a^2 K}{12} \cot \left(\frac{\theta_{\nu,E}^*}{2} \right) + O(a^4 K^2). \quad (2096)$$

The nongeometric angular residual is

$$\delta \theta_\nu^{\text{ng}}(\kappa) = \theta_\nu^*(\kappa) - \theta_\nu^{\text{geo}}(\kappa). \quad (2097)$$

A systematic nonzero value diagnoses microscopic tunnelling, representation, or spectral-root reconstruction beyond local registry remapping. Let $\gamma_1(\kappa, \theta)$ be the active normalized tunnelling harmonic and let $\alpha_\nu^*(\kappa, \theta)$ be the dimensionless root extracted from the complete microscopic problem. The pure-geometry ablation freezes

$$\gamma_1(\kappa, \theta) \longrightarrow \gamma_1(0, \theta_{\nu,E}^*), \quad (2098)$$

and

$$\alpha_\nu^*(\kappa, \theta) \longrightarrow \alpha_\nu^*(0, \theta_{\nu,E}^*), \quad (2099)$$

while retaining the exact hyperbolic registry map. The ablated branch number must satisfy

$$N_{\text{mag}}^{\text{geo}}(\kappa) = N_{\text{mag}}^{\text{E}} \quad (2100)$$

throughout every region where the original Euclidean branches remain inside the physical twist interval. Any additional branch in the full model must disappear under this ablation if it is generated by curvature-dependent microscopic reconstruction.

D. Curvature-induced branch birth

Fix ω . A candidate finite-curvature fold occurs at

$$(\theta_c, \kappa_c), \quad \kappa_c > 0, \quad \theta_c \in (\theta_{\min}, \theta_{\max}), \quad (2101)$$

and satisfies

$$F_c = 0, \quad (F_\theta)_c = 0. \quad (2102)$$

The generic nondegeneracy conditions are

$$(F_\kappa)_c \neq 0, \quad (F_{\theta\theta})_c \neq 0. \quad (2103)$$

The physical conditions must remain open:

$$\Delta_c^L > 0, \quad C_c^L > 0, \quad C_{\text{rep},c} = 1. \quad (2104)$$

Introduce

$$x = \theta - \theta_c, \quad u = \kappa - \kappa_c. \quad (2105)$$

Define the local coefficients

$$\alpha_f = (F_\kappa)_c, \quad \beta_f = (F_{\theta\theta})_c, \quad \gamma_f = (F_{\theta\kappa})_c, \quad (2106)$$

and

$$\delta_f = (F_{\theta\theta\theta})_c. \quad (2107)$$

The local expansion is

$$\begin{aligned} F(x, u) &= \alpha_f u + \frac{\beta_f}{2} x^2 + \gamma_f x u \\ &\quad + \frac{\delta_f}{6} x^3 + \frac{1}{2} (F_{\kappa\kappa})_c u^2 \\ &\quad + O(|x|^4 + |x|^2 |u| + |x| |u|^2 + |u|^3). \end{aligned} \quad (2108)$$

At leading order,

$$x^2 = -\frac{2\alpha_f}{\beta_f}u. \quad (2109)$$

The two stationary branches exist on the side satisfying

$$-\frac{\alpha_f u}{\beta_f} > 0. \quad (2110)$$

Their next-order expansions are

$$x_{\pm}(u) = \pm \sqrt{-\frac{2\alpha_f}{\beta_f}u} + \left[-\frac{\gamma_f}{\beta_f} + \frac{\alpha_f \delta_f}{3\beta_f^2} \right] u + O(|u|^{3/2}). \quad (2111)$$

The $O(u)$ term shifts the midpoint of the pair but does not modify the leading branch separation. The stationary-branch separation is

$$\Delta\theta_{\text{fold}} = |\theta_+ - \theta_-| = 2\sqrt{-\frac{2\alpha_f}{\beta_f}u} + O(|u|^{3/2}). \quad (2112)$$

Therefore

$$(\Delta\theta_{\text{fold}})^2 = A_{\text{fold}}|u| + o(|u|), \quad (2113)$$

where

$$A_{\text{fold}} = 8 \left| \frac{\alpha_f}{\beta_f} \right|. \quad (2114)$$

The midpoint drift is

$$\frac{\theta_+ + \theta_-}{2} - \theta_c = \left[-\frac{\gamma_f}{\beta_f} + \frac{\alpha_f \delta_f}{3\beta_f^2} \right] u + O(|u|^{3/2}). \quad (2115)$$

This term distinguishes asymmetric microscopic unfolding from the symmetric leading normal form. Near the fold,

$$F_{\theta} = \beta_f x + \gamma_f u + \frac{\delta_f}{2} x^2 + O(|u|^{3/2}). \quad (2116)$$

The branch satisfying $\beta_f x < 0$ is a local maximum, whereas the opposite branch is a local minimum. On the stable branch,

$$\kappa_{\perp} = -F_{\theta} = \sqrt{2|\alpha_f \beta_f u|} + O(|u|). \quad (2117)$$

Thus

$$\kappa_{\perp} \propto |u|^{1/2}. \quad (2118)$$

The branch susceptibility diverges:

$$\frac{d\theta_{\pm}}{d\kappa} = \pm \frac{1}{2} \sqrt{-\frac{2\alpha_f}{\beta_f}} |u|^{-1/2} + O(1). \quad (2119)$$

The divergence does not indicate a singular microscopic Hamiltonian. It is the geometric response of the stationary-point locus. For a fixed allowed logarithmic

score reduction $\delta L_{\text{acc}} > 0$, the local angular half-width of the stable maximum is

$$\delta\theta_{\text{magic}} \simeq \sqrt{\frac{2\delta L_{\text{acc}}}{\kappa_{\perp}}}. \quad (2120)$$

Consequently,

$$\delta\theta_{\text{magic}} = \sqrt{2\delta L_{\text{acc}}} (2|\alpha_f \beta_f|)^{-1/4} |u|^{-1/4} + o(|u|^{-1/4}). \quad (2121)$$

The newly created maximum becomes broad and shallow as the fold is approached. Since $F = \partial_{\theta} L$, integration of the leading fold normal form gives

$$L = L_c + L_{\kappa,c} u + \alpha_f u x + \frac{\beta_f}{6} x^3 + \dots \quad (2122)$$

At a stationary point,

$$\alpha_f u = -\frac{\beta_f}{2} x^2 + O(|u|^{3/2}), \quad (2123)$$

so the branch-dependent part is

$$L_{\text{branch}} = -\frac{\beta_f}{3} x^3 + O(u^2). \quad (2124)$$

The score difference between the stationary maximum and minimum is

$$|\Delta L_{\text{fold}}| = \frac{2^{5/2}}{3} \frac{|\alpha_f|^{3/2}}{|\beta_f|^{1/2}} |u|^{3/2} + o(|u|^{3/2}). \quad (2125)$$

Equivalently,

$$|\Delta \ln \mathcal{M}| \propto |\kappa - \kappa_c|^{3/2}. \quad (2126)$$

The generic fold therefore has the universal exponent set

$$\begin{aligned} |\theta^* - \theta_c| &\propto |u|^{1/2}, & \Delta\theta_{\text{fold}} &\propto |u|^{1/2}, \\ \kappa_{\perp} &\propto |u|^{1/2}, & \delta\theta_{\text{magic}} &\propto |u|^{-1/4}, \\ |\Delta \ln \mathcal{M}| &\propto |u|^{3/2}. \end{aligned} \quad (2127)$$

We test the normal form without freely fitting critical exponents. Define

$$Q_{\theta}(u) = \frac{(\Delta\theta_{\text{fold}})^2}{|u|}. \quad (2128)$$

The fold prediction is $Q_{\theta}(u) \rightarrow A_{\text{fold}}$. Define

$$Q_{\perp}(u) = \frac{\kappa_{\perp}^2}{|u|}. \quad (2129)$$

Then

$$Q_{\perp}(u) \rightarrow 2|\alpha_f \beta_f|. \quad (2130)$$

Define

$$Q_L(u) = \frac{|\Delta L_{\text{fold}}|}{|u|^{3/2}}. \quad (2131)$$

The predicted limit is

$$Q_L(u) \longrightarrow \frac{2^{5/2} |\alpha_f|^{3/2}}{3 |\beta_f|^{1/2}}. \quad (2132)$$

Finally,

$$Q_{\text{width}}(u) = \delta\theta_{\text{magic}} |u|^{1/4} \quad (2133)$$

must converge to

$$Q_{\text{width}} \longrightarrow \sqrt{2\delta L_{\text{acc}}} (2|\alpha_f \beta_f|)^{-1/4}. \quad (2134)$$

Agreement of all four compensated observables is substantially stronger than a log-log fit of one branch displacement. We now identify the microscopic structure required to violate the geometry-only no-bifurcation result. Define

$$\zeta = \frac{a\sqrt{\kappa}}{2}, \quad b_\zeta = \sinh \zeta, \quad (2135)$$

and

$$s_\theta = \sin\left(\frac{\theta}{2}\right), \quad c_\theta = \cos\left(\frac{\theta}{2}\right). \quad (2136)$$

The reduced branch equation is

$$G_\nu(\theta, \kappa, \omega) = \text{arsinh}\left(\frac{b_\zeta}{s_\theta}\right) - \frac{2\pi\zeta}{n_c} \sqrt{\frac{d\alpha_\nu^*(\kappa, \theta)}{\gamma_1(\kappa, \theta)\omega}} = 0.$$

On the branch, define

$$L_M = \text{arsinh}\left(\frac{b_\zeta}{s_\theta}\right). \quad (2137)$$

The geometric derivative is

$$\frac{\partial}{\partial \theta} \text{arsinh}\left(\frac{b_\zeta}{s_\theta}\right) = -\frac{b_\zeta c_\theta}{2s_\theta \sqrt{s_\theta^2 + b_\zeta^2}}. \quad (2138)$$

On a solution of $G_\nu = 0$, differentiation gives

$$G_{\nu,\theta} = -\frac{b_\zeta c_\theta}{2s_\theta \sqrt{s_\theta^2 + b_\zeta^2}} - \frac{L_M}{2} [\partial_\theta \ln \alpha_\nu^* - \partial_\theta \ln \gamma_1].$$

A reduced-model fold requires $G_{\nu,\theta} = 0$. Therefore

$$\partial_\theta \ln\left(\frac{\gamma_1}{\alpha_\nu^*}\right) = \frac{b_\zeta c_\theta}{s_\theta \sqrt{s_\theta^2 + b_\zeta^2} L_M}. \quad (2139)$$

The right-hand side is strictly positive for $0 < \theta < \pi$. Define the fold-balance residual

$$\mathcal{R}_{\text{fold}}^{\text{micro}} = \partial_\theta \ln\left(\frac{\gamma_1}{\alpha_\nu^*}\right) - \frac{b_\zeta c_\theta}{s_\theta \sqrt{s_\theta^2 + b_\zeta^2} L_M}. \quad (2140)$$

A necessary reduced-model fold condition is $\mathcal{R}_{\text{fold}}^{\text{micro}} = 0$ If

$$\inf_{(\kappa, \theta) \in \mathcal{R}} |\mathcal{R}_{\text{fold}}^{\text{micro}}| > \varepsilon_{\text{micro}} \quad (2141)$$

on a candidate region $\mathcal{R}$, the reduced microscopic mechanism cannot produce a fold there. Because pure geometry keeps both γ_1 and α_ν^* fixed,

$$\partial_\theta \ln\left(\frac{\gamma_1}{\alpha_\nu^*}\right) = 0, \quad (2142)$$

and can never satisfy Eq. (2139). A verified fold therefore requires angular reconstruction of the active tunnelling harmonic, the dimensionless spectral root, or both. A branch can first become visible at finite curvature without a stationary-point bifurcation. A gap-mediated onset satisfies $\Delta_\nu \rightarrow 0^+$, while $F_{\nu,\theta} \neq 0$. It is an admissibility transition rather than a fold. A coherence-mediated onset satisfies $C_\nu^{\text{coh}} \rightarrow 0^+$ with a regular stationary equation. A physical-boundary onset satisfies

$$\theta_\nu^* = \theta_{\min} \quad \text{or} \quad \theta_\nu^* = \theta_{\max}. \quad (2143)$$

The optimum can enter the interior continuously from a constrained boundary maximum. A crystallographic onset occurs when

$$g_\theta \in \text{Comm}_G(\Gamma) \quad (2144)$$

first becomes available along an exact sequence. It changes the exact finite-cell realization but does not by itself create a local spectral fold. We classify a branch as curvature born only when

$$\Delta_c^L > 0, \quad C_c^L > 0, \quad \theta_c \in (\theta_{\min}, \theta_{\max}), \quad (2145)$$

$$F_c = F_{\theta,c} = 0, \quad F_{\kappa,c} F_{\theta\theta,c} \neq 0.$$

Assume that the stable branch exists for $\kappa > \kappa_c$ after orienting the local curvature coordinate. Define the branch order parameter

$$B_\nu(\kappa) = |\theta_\nu^*(\kappa) - \theta_c|. \quad (2146)$$

The fold law gives

$$B_\nu(\kappa) = B_{\nu,0}(\kappa - \kappa_c)^{1/2} + o\left[(\kappa - \kappa_c)^{1/2}\right], \quad (2147)$$

with

$$B_{\nu,0} = \sqrt{2 \left| \frac{F_{\nu,\kappa}}{F_{\nu,\theta\theta}} \right|_c}. \quad (2148)$$

The branch is curvature born when

$$N_{\text{mag}}(0, \omega) < N_{\text{mag}}(\kappa, \omega) \quad (2149)$$

and the increase is generated by the finite-curvature fold

$$F = F_\theta = 0, \quad F_\kappa \neq 0, \quad F_{\theta\theta} \neq 0. \quad (2150)$$

For every candidate branch we record

$$\mathfrak{B}_\nu = \left(\theta_\nu^*, \mathcal{M}_\nu^*, \kappa_{\perp,\nu}, W_\nu, \Delta_\nu, \right. \\ \left. v_{\text{max},\nu}, \Lambda_{\text{max},\nu}, \varrho_\nu^{\text{coh,max}}, C_\nu^{\text{coh}}, \right. \\ \left. q_{\text{M},\nu}, d_{\rho,\nu}, r_\nu, \mathfrak{s}_\nu \right). \quad (2151)$$

E. Fold and cusp verification

Whenever analytic or automatic differentiation of the complete observable chain is available, we use it as the primary derivative method. We independently verify it with symmetric finite differences. For a step $h_\theta > 0$,

$$\widehat{F}_{h_\theta} = \frac{L(\theta + h_\theta) - L(\theta - h_\theta)}{2h_\theta} \quad (2152)$$

has an $O(h_\theta^2)$ truncation error. The transverse derivative is

$$\widehat{F}_{\theta, h_\theta} = \frac{L(\theta + h_\theta) - 2L(\theta) + L(\theta - h_\theta)}{h_\theta^2}. \quad (2153)$$

The fold nondegeneracy derivative is

$$\widehat{F}_{\theta\theta, h_\theta} = \frac{L(\theta + 2h_\theta) - 2L(\theta + h_\theta) + 2L(\theta - h_\theta) - L(\theta - 2h_\theta)}{2h_\theta^3} \quad (2154)$$

The cusp nondegeneracy derivative is

$$\widehat{F}_{\theta\theta\theta, h_\theta} = \frac{L(\theta - 2h_\theta) - 4L(\theta - h_\theta) + 6L(\theta) - 4L(\theta + h_\theta) + L(\theta + 2h_\theta)}{h_\theta^4} \quad (2155)$$

For a curvature step h_κ ,

$$\widehat{F}_\kappa = \frac{1}{4h_\theta h_\kappa} \left[L(\theta + h_\theta, \kappa + h_\kappa) - L(\theta + h_\theta, \kappa - h_\kappa) - L(\theta - h_\theta, \kappa + h_\kappa) + L(\theta - h_\theta, \kappa - h_\kappa) \right].$$

The derivative $F_{\theta\kappa}$ is obtained by applying the centered κ -difference to $\widehat{F}_{\theta, h_\theta}$. For every second-order accurate stencil D_h , define the Richardson value

$$D_R = \frac{4D_{h/2} - D_h}{3}. \quad (2156)$$

The step-refinement discrepancy

$$\varepsilon_D^{\text{step}} = \frac{|D_{h/2} - D_h|}{3} \quad (2157)$$

is added to the solver, observable, projector, and finite-cover uncertainties. Let the estimated critical point be $(\widehat{\theta}_c, \widehat{\kappa}_c)$. It satisfies the zero conditions when $|\widehat{F}_c| \leq \varepsilon_F$, and

$$|\widehat{F}_{\theta, c}| \leq \varepsilon_{F_\theta}. \quad (2158)$$

The nonzero derivative margins are

$$|\widehat{F}_{\kappa, c}| - \varepsilon_{F_\kappa} > 0, \quad (2159)$$

and

$$|\widehat{F}_{\theta\theta, c}| - \varepsilon_{F_{\theta\theta}} > 0. \quad (2160)$$

The physical margins are

$$\widetilde{\Delta}_c - \varepsilon_{\Delta, c} > 0, \quad (2161)$$

and

$$\widetilde{C}_c^{\text{coh}} - \varepsilon_{C, c} > 0. \quad (2162)$$

A conservative interval for the predicted fold coefficient is

$$A_{\text{fold}}^L = 8 \frac{|\widetilde{F}_{\kappa, c}| - \varepsilon_{F_\kappa}}{|\widetilde{F}_{\theta\theta, c}| + \varepsilon_{F_{\theta\theta}}}, \quad (2163)$$

and

$$A_{\text{fold}}^U = 8 \frac{|\widetilde{F}_{\kappa, c}| + \varepsilon_{F_\kappa}}{|\widetilde{F}_{\theta\theta, c}| - \varepsilon_{F_{\theta\theta}}}. \quad (2164)$$

For sampled two-branch points κ_j , define

$$A_j^{\text{meas}} = \frac{[\theta_{+, j} - \theta_{-, j}]^2}{|\kappa_j - \widetilde{\kappa}_c|}. \quad (2165)$$

The fit-free fold residual is

$$\mathcal{R}_{\text{fold}}^{\text{sep}} = \max_j \frac{\text{dist}[A_j^{\text{meas}}, [A_{\text{fold}}^L, A_{\text{fold}}^U]]}{A_{\text{fold}}^U}. \quad (2166)$$

A established fold requires $\mathcal{R}_{\text{fold}}^{\text{sep}} \rightarrow 0$ as the fitting window is contracted while remaining above the numerical resolution scale. The branch separation, score splitting, and transverse curvature must all dominate their uncertainties:

$$\Delta\theta_{\text{fold}} > 2\varepsilon_\theta, \quad |\Delta L_{\text{fold}}| > 2\varepsilon_L, \quad (2167)$$

and $\kappa_\perp > \varepsilon_{\kappa_\perp}$. The complete fold certificate is

$$\mathfrak{C}_{\text{fold}} = \left(\kappa_c, \theta_c, F_c, F_{\theta, c}, F_{\kappa, c}, F_{\theta\theta, c}, \Delta_c, C_c^{\text{coh}}, A_{\text{fold}}, B_{\nu, 0}, Q_\theta, Q_\perp, Q_L, Q_{\text{width}}, P_c, \mathfrak{s}_c, C_{\text{rep}, c}, \varepsilon_{\text{tot}, c} \right). \quad (2168)$$

A cusp requires two independent controls. We use

$$p_1 = \kappa, \quad p_2 = \omega. \quad (2169)$$

A candidate cusp occurs at $(\theta_c, \kappa_c, \omega_c)$ and satisfies

$$F_c = F_{\theta, c} = F_{\theta\theta, c} = 0. \quad (2170)$$

The quartic nondegeneracy condition is $F_{\theta\theta\theta, c} \neq 0$. The two controls must independently unfold the constant and linear terms. Define

$$D_{\text{unf}} = \det \begin{pmatrix} F_\kappa & F_\omega \\ F_{\theta\kappa} & F_{\theta\omega} \end{pmatrix}_c. \quad (2171)$$

A generic cusp requires $D_{\text{unf}} \neq 0$. Let

$$x = \theta - \theta_c, \quad \delta\kappa = \kappa - \kappa_c, \quad \delta\omega = \omega - \omega_c. \quad (2172)$$

Define

$$\begin{aligned} a_\kappa &= F_{\kappa,c}, & a_\omega &= F_{\omega,c}, \\ b_\kappa &= F_{\theta\kappa,c}, & b_\omega &= F_{\theta\omega,c}, \end{aligned} \quad (2173)$$

and $c_3 = F_{\theta\theta,c}$. The leading expansion is

$$\begin{aligned} F &= a_\kappa \delta\kappa + a_\omega \delta\omega \\ &+ (b_\kappa \delta\kappa + b_\omega \delta\omega) x + \frac{c_3}{6} x^3 + O(4). \end{aligned} \quad (2174)$$

Define the canonical control coordinates

$$U = -\frac{6}{c_3} (b_\kappa \delta\kappa + b_\omega \delta\omega), \quad (2175)$$

and

$$V = -\frac{6}{c_3} (a_\kappa \delta\kappa + a_\omega \delta\omega). \quad (2176)$$

The nonzero determinant D_{unf} makes the transformation from $(\delta\kappa, \delta\omega)$ to (U, V) locally invertible. After multiplication by $-6/c_3$, the stationary equation takes the canonical form

$$-x^3 + Ux + V + O(4) = 0. \quad (2177)$$

To leading order,

$$x^3 - Ux - V = 0. \quad (2178)$$

Its discriminant is

$$\mathcal{D}_{\text{cusp}} = 4U^3 - 27V^2. \quad (2179)$$

The local root structure is

$$\begin{aligned} \mathcal{D}_{\text{cusp}} < 0 : & \quad \text{one stationary point,} \\ \mathcal{D}_{\text{cusp}} = 0 : & \quad \text{one double and one simple stationary point,} \\ \mathcal{D}_{\text{cusp}} > 0 : & \quad \text{three distinct stationary points.} \end{aligned} \quad (2180)$$

The fold boundaries are

$$V_\pm = \pm \frac{2}{3\sqrt{3}} U^{3/2}, \quad U \geq 0. \quad (2181)$$

The canonical stability derivative is

$$F_\theta \propto -3x^2 + U. \quad (2182)$$

Inside the three-root region, the two outer roots are stable maxima and the central root is an unstable minimum. The canonical logarithmic-score potential is

$$\tilde{L} = -\frac{x^4}{4} + \frac{Ux^2}{2} + Vx + \dots \quad (2183)$$

At

$$V = 0, \quad U > 0, \quad (2184)$$

the stationary points are

$$x_0 = 0, \quad x_\pm = \pm\sqrt{U}. \quad (2185)$$

The two stable maxima have equal score and angle separation

$$\Delta\theta_{\text{hier}} = 2\sqrt{U}. \quad (2186)$$

For small V ,

$$\tilde{L}_+^* - \tilde{L}_-^* = 2\sqrt{U}V + O(V^2). \quad (2187)$$

Thus the sign of V selects the globally preferred branch. A candidate cusp must satisfy the compensated fold-line relation

$$Q_{\text{cusp}} = \frac{|V|}{U^{3/2}} \longrightarrow \frac{2}{3\sqrt{3}} \quad (2188)$$

on both fold boundaries. Along the hierarchy-switching section $V = 0$,

$$Q_{\text{jump}} = \frac{\Delta\theta_{\text{hier}}}{\sqrt{U}} \longrightarrow 2. \quad (2189)$$

The root count measured from the actual landscape must agree with $\text{sgn } \mathcal{D}_{\text{cusp}}$. The same tracked projector families must form the two stable outer branches. Three unrelated numerical maxima do not constitute a cusp. The complete cusp certificate is

$$\begin{aligned} \mathfrak{C}_{\text{cusp}} = & \left(\kappa_c, \omega_c, \theta_c, F_c, F_{\theta,c}, F_{\theta\theta,c}, \right. \\ & F_{\theta\theta\theta,c}, D_{\text{unf}}, U, V, \mathcal{D}_{\text{cusp}}, \\ & Q_{\text{cusp}}, Q_{\text{jump}}, \Delta_c, C_c^{\text{coh}}, \\ & \left. P_{+,c}, P_{-,c}, \mathfrak{s}_{+,c}, \mathfrak{s}_{-,c}, \varepsilon_{\text{tot},c} \right). \end{aligned} \quad (2190)$$

Let $(\kappa_{c,N}, \theta_{c,N})$ be the fold point on cover N . A bulk fold requires

$$\begin{aligned} \kappa_{c,N} &\longrightarrow \kappa_{c,\infty}, & \theta_{c,N} &\longrightarrow \theta_{c,\infty}, \\ F_{\kappa,c}^{(N)} &\longrightarrow F_{\kappa,c}^{(\infty)}, & F_{\theta\theta,c}^{(N)} &\longrightarrow F_{\theta\theta,c}^{(\infty)}. \end{aligned} \quad (2191)$$

Consequently,

$$A_{\text{fold}}^{(N)} = 8 \left| \frac{F_{\kappa,c}^{(N)}}{F_{\theta\theta,c}^{(N)}} \right| \longrightarrow A_{\text{fold}}^{(\infty)}. \quad (2192)$$

The same convergence is required on two inequivalent finite-cover towers A and B :

$$\begin{aligned} |\kappa_{c,N}^{(A)} - \kappa_{c,M}^{(B)}| &\longrightarrow 0, \\ |\theta_{c,N}^{(A)} - \theta_{c,M}^{(B)}| &\longrightarrow 0, \end{aligned} \quad (2193)$$

and

$$|A_{\text{fold},N}^{(A)} - A_{\text{fold},M}^{(B)}| \longrightarrow 0. \quad (2194)$$

The branch observables must converge simultaneously:

$$\begin{aligned} \liminf_{N \rightarrow \infty} \Delta_{\nu, N} &> 0, \\ \liminf_{N \rightarrow \infty} C_{\nu, N}^{\text{coh}} &> 0, \end{aligned} \quad (2195)$$

and

$$C_{\text{rep}, \nu, N} = 1 \quad (2196)$$

at every finite level entering the full-spectrum claim. The hopping shell must satisfy

$$L_N \rightarrow \infty, \quad \frac{L_N}{r_{\text{inj}, N}} \rightarrow 0, \quad (2197)$$

with

$$\varepsilon_N^{(0)} \rightarrow 0, \quad \varepsilon_N^{(1)} \rightarrow 0, \quad \varepsilon_N^{(2)} \rightarrow 0. \quad (2198)$$

Fold derivatives require at least the declared smoothness tier of the complete score. Spectral C^0 convergence alone cannot certify F_κ , $F_{\theta\theta}$, or $F_{\theta\theta\theta}$. Let the stable optimized branch scores be ordered as

$$L_{\pi_1}^* \geq L_{\pi_2}^* \geq \dots \geq L_{\pi_{N_{\text{mag}}}}^*. \quad (2199)$$

The hierarchy permutation is

$$\Pi_{\text{mag}} = (\pi_1, \dots, \pi_{N_{\text{mag}}}). \quad (2200)$$

The rank of branch ν is

$$r_\nu^{\text{mag}} = 1 + \sum_{\mu \neq \nu} \mathbf{1} [L_\mu^* > L_\nu^*]. \quad (2201)$$

For two control points p_1 and p_2 , define

$$R_{\text{hier}}(p_1, p_2) = \frac{1}{2} \sum_\nu |r_\nu^{\text{mag}}(p_2) - r_\nu^{\text{mag}}(p_1)|. \quad (2202)$$

A value $R_{\text{hier}} = 0$ means the ordering is unchanged. A positive value measures score reordering even when $\Delta N_{\text{mag}} = 0$. For two stable branches μ and ν , define

$$\Phi_{\mu\nu} = L_\mu^* - L_\nu^*. \quad (2203)$$

A hierarchy wall satisfies $\Phi_{\mu\nu} = 0$. If

$$\partial_\omega \Phi_{\mu\nu} \neq 0, \quad (2204)$$

its slope is

$$\frac{d\omega_{\mu\nu}^{\text{wall}}}{d\kappa} = -\frac{\partial_\kappa \Phi_{\mu\nu}}{\partial_\omega \Phi_{\mu\nu}}. \quad (2205)$$

Because each score is already optimized over θ ,

$$\partial_\kappa L_\nu^* = L_{\nu, \kappa} \Big|_{\theta=\theta_\nu^*}, \quad (2206)$$

and

$$\partial_\omega L_\nu^* = L_{\nu, \omega} \Big|_{\theta=\theta_\nu^*}. \quad (2207)$$

A hierarchy switch can change the globally selected angle discontinuously while every individual branch remains smooth. We do not classify such a switch as branch birth unless N_{mag} also changes through a fold. Define the Euclidean-exclusion condition

$$G_E = \begin{cases} 1, & \text{no matching stable Euclidean antecedent exists,} \\ 0, & \text{otherwise,} \end{cases} \quad (2208)$$

the fold condition

$$G_F = \begin{cases} 1, & F = F_\theta = 0, \quad F_\kappa F_{\theta\theta} \neq 0, \\ 0, & \text{otherwise,} \end{cases} \quad (2209)$$

the physical condition

$$G_P = \begin{cases} 1, & \Delta^L > 0, \quad C^L > 0, \quad C_{\text{rep}} = 1, \\ 0, & \text{otherwise,} \end{cases} \quad (2210)$$

the scaling condition

$$G_S = \begin{cases} 1, & Q_\theta, Q_\perp, Q_L, Q_{\text{width}} \text{ converge to their predicted values,} \\ 0, & \text{otherwise,} \end{cases} \quad (2211)$$

and the bulk condition

$$G_B = \begin{cases} 1, & \text{critical data converge across covers and towers,} \\ 0, & \text{otherwise.} \end{cases} \quad (2212)$$

The curvature-born statement condition is

$$G_{\text{birth}} = G_E G_F G_P G_S G_B. \quad (2213)$$

We use the phrase curvature-born flat-band branch if and only if $G_{\text{birth}} = 1$. If $G_{\text{birth}} = 0$ but a regular stable branch is continuously tracked, the strongest supported result is

$$\frac{d\theta_\nu^*}{d\kappa} = -\frac{F_{\nu, \kappa}}{F_{\nu, \theta}}, \quad F_{\nu, \theta} < 0, \quad (2214)$$

together with the measured curvature response of $\mathcal{M}_\nu^*$, W_ν , Δ_ν , $v_{\text{max}, \nu}$, $\Lambda_{\text{max}, \nu}$, $\varrho_\nu^{\text{coh}, \text{max}}$. We then describe the result as continuous curvature control of the flat-band optimum.

X. FINITE-COVER BULK LIMIT

Every numerical calculation replaces the infinite hyperbolic bilayer by a finite-dimensional operator. A large finite matrix is not, by itself, a controlled approximation to a bulk Hamiltonian. We therefore separate six logically independent properties:

finite-device exactness :	H_N is the exact Hamiltonian of one finite circuit,
local geometric exactness :	finite-radius neighborhoods lift isometrically,
weak trace convergence :	$\nu_N \Rightarrow \nu_{\text{bulk}}$,
no spectral loss :	$\Sigma_{\text{bulk}} \subset (S_N)_{\ell_N}$,
no spectral pollution :	$S_N \subset (\Sigma_{\text{bulk}})_{p_N}$,
representation completeness :	all declared finite-quotient sectors are included.

Here

$$\Sigma_{\text{bulk}} = \text{Spec } H_{\text{bulk}} \quad (2216)$$

is the spectrum of the declared infinite-volume bulk operator, whereas

$$S_N = \text{Spec } H_N^{\text{ret}} \quad (2217)$$

is the spectrum of the complete finite circuit or of a specified invariant retained sector. The two one-sided spectral errors are

$$\ell_N = \sup_{\lambda \in \Sigma_{\text{bulk}}} \text{dist}(\lambda, S_N), \quad (2218)$$

and

$$p_N = \sup_{s \in S_N} \text{dist}(s, \Sigma_{\text{bulk}}). \quad (2219)$$

Their maximum is the Hausdorff error:

$$\eta_N = d_H(S_N, \Sigma_{\text{bulk}}) = \max\{\ell_N, p_N\}. \quad (2220)$$

The fundamental verification hierarchy is

$$\begin{aligned} \Gamma_M &\longrightarrow \Gamma_N \triangleleft \Gamma_M \longrightarrow Q_N = \Gamma_M / \Gamma_N \longrightarrow \mathcal{G}_N \\ &\longrightarrow H_N \longrightarrow (r_{\text{inj}, N}, L_N, \varepsilon_{\text{tail}, N}) \longrightarrow (\ell_N, p_N, C_{\text{rep}, N}) \longrightarrow \text{bulk bandwidth, gap, DOS, velocity, and Hessian.} \\ &\quad (2221) \end{aligned}$$

We do not infer any arrow in Eq. (2221) from matrix size alone.

A. Finite circuit graphs

Let Γ_M be the exact hyperbolic moiré translation group of a commensurate bilayer. One exact moiré cell contains

$$\nu_M = 2n_b q_M \quad (2222)$$

physical resonator or orbital degrees of freedom. The infinite one-particle Hilbert space is

$$\mathcal{H}_\infty = \mathbb{C}^{\nu_M} \otimes \ell^2(\Gamma_M). \quad (2223)$$

We use the basis

$$|\alpha, \gamma\rangle, \quad \alpha = 1, \dots, \nu_M, \quad \gamma \in \Gamma_M. \quad (2224)$$

Let

$$h_{\mathbf{p}}(\delta) \in M_{\nu_M}(\mathbb{C}), \quad \delta \in \Gamma_M, \quad (2225)$$

be the matrix-valued hopping kernel at physical control point $\mathbf{p} = (\theta, K, \omega)$. Hermiticity requires

$$h_{\mathbf{p}}(\delta^{-1}) = h_{\mathbf{p}}(\delta)^\dagger. \quad (2226)$$

Let $\lambda_{\Gamma_M}(\delta)$ denote the left regular representation:

$$\lambda_{\Gamma_M}(\delta)|\gamma\rangle = |\delta\gamma\rangle. \quad (2227)$$

The reduced regular bulk Hamiltonian is

$$H_{\text{bulk}}(\mathbf{p}) = \sum_{\delta \in \Gamma_M} h_{\mathbf{p}}(\delta) \otimes \lambda_{\Gamma_M}(\delta). \quad (2228)$$

For a finite-range kernel, define

$$\mathcal{S}_H = \{\delta \in \Gamma_M : h_{\mathbf{p}}(\delta) \neq 0\}, \quad (2229)$$

with $|\mathcal{S}_H| < \infty$. For an exponentially decaying kernel, we assume

$$\|h_{\mathbf{p}}(\delta)\| \leq A_0 \exp[-\mu d_{\mathbb{H}}(o, \delta o)], \quad (2230)$$

uniformly on the declared control domain. If the orbit-counting exponent is h_Γ , boundedness requires the positive summability margin

$$\gamma = \mu - h_\Gamma > 0. \quad (2231)$$

For the hyperbolic-plane geometric estimate,

$$h_\Gamma = \frac{1}{R} \quad (2232)$$

up to constants fixed by the chosen cocompact lattice. The propagation-truncated operator is

$$H_{\text{bulk}}^{(L)} = \sum_{d_{\mathbb{H}}(o, \delta o) \leq L} h_{\mathbf{p}}(\delta) \otimes \lambda_{\Gamma_M}(\delta). \quad (2233)$$

Its tail obeys

$$\varepsilon_{\text{tail}, \infty}^{(0)}(L) = \|H_{\text{bulk}} - H_{\text{bulk}}^{(L)}\| \leq C_0 e^{-\gamma L}. \quad (2234)$$

Derivative observables require the separately weighted bounds

$$\varepsilon_{\text{tail}, \infty}^{(j)}(L) \leq C_j (1 + L)^{s_j} e^{-\gamma_j L}, \quad j = 1, 2. \quad (2235)$$

Choose a finite-index normal subgroup $\Gamma_N \triangleleft \Gamma_M$. Define the quotient map

$$q_N : \Gamma_M \longrightarrow Q_N = \Gamma_M / \Gamma_N, \quad (2236)$$

and the cover degree

$$d_N = |Q_N| = [\Gamma_M : \Gamma_N]. \quad (2237)$$

The compact periodic surface is

$$\Sigma_N = \Gamma_N \backslash \mathbb{H}_R^2. \quad (2238)$$

Its area is

$$A_N = d_N A_M^{\text{sc}} = d_N q_M A_0. \quad (2239)$$

For a torsion-free closed quotient,

$$g_N = 1 + d_N (g_M - 1) = 1 + d_N q_M (g_0 - 1). \quad (2240)$$

The finite Hilbert space is

$$\mathcal{H}_N = \mathbb{C}^{\nu_M} \otimes \ell^2(Q_N), \quad (2241)$$

with dimension

$$D_N = \nu_M d_N = 2n_b q_M d_N. \quad (2242)$$

Let $R_N(q)$ be the right regular representation:

$$R_N(q)|x\rangle = |xq^{-1}\rangle. \quad (2243)$$

It satisfies

$$R_N(q_1)R_N(q_2) = R_N(q_1q_2), \quad (2244)$$

and

$$R_N(q)^\dagger = R_N(q^{-1}). \quad (2245)$$

The exact finite-cover Hamiltonian is

$$H_N(\mathbf{p}) = \sum_{\delta \in \Gamma_M} h_{\mathbf{p}}(\delta) \otimes R_N[q_N(\delta)]. \quad (2246)$$

For finite-range hopping,

$$H_N = \sum_{\delta \in \mathcal{S}_H} h_{\mathbf{p}}(\delta) \otimes R_N[q_N(\delta)]. \quad (2247)$$

Using Eqs. (2226) and (2245),

$$\begin{aligned} H_N^\dagger &= \sum_{\delta} h_{\mathbf{p}}(\delta)^\dagger \otimes R_N[q_N(\delta)]^\dagger \\ &= \sum_{\delta} h_{\mathbf{p}}(\delta^{-1}) \otimes R_N[q_N(\delta^{-1})] \\ &= H_N. \end{aligned} \quad (2248)$$

Thus $H_N^\dagger = H_N$. We define the weighted finite circuit graph

$$\mathcal{G}_N = (\mathcal{V}_N, \mathcal{E}_N, \mathbf{W}_N). \quad (2249)$$

Its vertices are

$$\mathcal{V}_N = \{(x, \alpha) : x \in Q_N, \alpha = 1, \dots, \nu_M\}. \quad (2250)$$

Therefore $|\mathcal{V}_N| = D_N$. For every nonzero matrix element

$$[h_{\mathbf{p}}(\delta)]_{\alpha\beta} \neq 0, \quad (2251)$$

and every $x \in Q_N$, we add the directed weighted edge

$$(x, \beta) \longrightarrow (xq_N(\delta)^{-1}, \alpha) \quad (2252)$$

with weight

$$\mathbf{W}_N[(x, \beta), (xq_N(\delta)^{-1}, \alpha)] = [h_{\mathbf{p}}(\delta)]_{\alpha\beta}. \quad (2253)$$

Hermiticity pairs this edge with its reverse. Onsite frequencies correspond to

$$\delta = e, \quad \alpha = \beta. \quad (2254)$$

Internal-cell couplings correspond to

$$\delta = e, \quad \alpha \neq \beta. \quad (2255)$$

Intercell couplings have $\delta \neq e$. The matrix of the weighted adjacency-plus-onsite operator of $\mathcal{G}_N$ is exactly H_N . The finite circuit graph is therefore not a visualization extracted after diagonalization; it is the primary finite Hamiltonian object. Suppose

$$\Gamma_M = \langle s_1, \dots, s_r \mid \mathcal{R}_1, \dots, \mathcal{R}_p \rangle. \quad (2256)$$

The quotient is specified by

$$q_\mu = q_N(s_\mu) \in Q_N. \quad (2257)$$

Before constructing H_N , we verify

$$\mathcal{R}_a(q_1, \dots, q_r) = e_{Q_N}, \quad a = 1, \dots, p. \quad (2258)$$

For a genus- g_M surface group, this includes

$$\prod_{j=1}^{g_M} [q_{a_j}, q_{b_j}] = e_{Q_N}. \quad (2259)$$

We also verify

$$\begin{aligned} R_N(q_1)R_N(q_2) &= R_N(q_1q_2), \\ R_N(q)^\dagger R_N(q) &= I_{d_N}, \end{aligned} \quad (2260)$$

and

$$\text{Tr } R_N(q) = \begin{cases} d_N, & q = e, \\ 0, & q \neq e. \end{cases} \quad (2261)$$

Define the directed local channel count

$$z_{\text{eff}} = \max_{\beta} \sum_{\delta \in \mathcal{S}_H} \# \{ \alpha : [h(\delta)]_{\alpha\beta} \neq 0 \}. \quad (2262)$$

Then every matrix column has at most z_{eff} off-diagonal channels, apart from onsite terms. Consequently,

$$\text{nnz}(H_N) \leq D_N (z_{\text{eff}} + 1). \quad (2263)$$

For fixed finite-range hopping,

$$\text{nnz}(H_N) = O(D_N). \quad (2264)$$

If distinct hopping words remain distinct in the quotient over the declared support, the sharper identity is

$$\text{nnz}(H_N) = d_N \sum_{\delta \in \mathcal{S}_H} \text{nnz} h(\delta). \quad (2265)$$

If two distinct words have the same quotient image, their matrix contributions add and Eq. (2263) remains the safe statement. For long-range hopping truncated at distance L , hyperbolic growth gives

$$z_{\text{eff}}(L) \leq C_z e^{h_{\Gamma} L}. \quad (2266)$$

Hence

$$\text{nnz} \left(H_N^{(L)} \right) = O \left[D_N e^{h_{\Gamma} L} \right]. \quad (2267)$$

The cutoff controls both the physical approximation error and the computational cost. Let $N_{\text{coup},N}^{\text{dir}}$ be the number of directed off-diagonal nonzero matrix entries. The number of reciprocal physical coupler pairs satisfies

$$N_{\text{coup},N} \leq \frac{1}{2} D_N z_{\text{eff}}. \quad (2268)$$

Equality holds when every physical coupling is represented by exactly one Hermitian pair and no parallel channels are combined. Before extracting spectral data, we evaluate the normalized Hermiticity defect

$$\varepsilon_{\text{Herm},N} = \frac{\|H_N - H_N^\dagger\|_{\text{F}}}{\max\{1, \|H_N\|_{\text{F}}\}}. \quad (2269)$$

For every tested symmetry $U_{g,N}$, define

$$\varepsilon_{\text{sym},N}(g) = \frac{\|H_N U_{g,N} - U_{g,N} H_N\|_{\text{F}}}{\max\{1, \|H_N\|_{\text{F}}\}}. \quad (2270)$$

The maximal symmetry defect is

$$\varepsilon_{\text{sym},N}^{\max} = \max_{g \in \mathcal{G}_{\text{test}}} \varepsilon_{\text{sym},N}(g). \quad (2271)$$

For a normalized approximate eigenpair $(\tilde{E}_j, |\tilde{\psi}_j\rangle)$, define

$$r_j = H_N |\tilde{\psi}_j\rangle - \tilde{E}_j |\tilde{\psi}_j\rangle. \quad (2272)$$

Hermiticity gives

$$\text{dist} \left(\tilde{E}_j, \text{Spec } H_N \right) \leq \|r_j\|. \quad (2273)$$

For complete diagonalization, we verify

$$\sum_{j=1}^{D_N} E_j = \text{Tr } H_N, \quad (2274)$$

and

$$\sum_{j=1}^{D_N} E_j^2 = \text{Tr } H_N^2. \quad (2275)$$

Higher moments may be checked through

$$\varepsilon_{m,N}^{\text{mom}} = \frac{\left| \sum_j E_j^m - \text{Tr } H_N^m \right|}{\max\{1, |\text{Tr } H_N^m|\}}. \quad (2276)$$

For a generic incommensurate twist,

$$[\Gamma : \Gamma \cap g_\theta \Gamma g_\theta^{-1}] = \infty. \quad (2277)$$

No finite common periodic bilayer quotient exists. We then choose a finite cover of the first layer, $\Gamma_N^{(1)} \triangleleft \Gamma$, and its conjugate cover

$$\Gamma_N^{(2)} = g_\theta \Gamma_N^{(1)} g_\theta^{-1}. \quad (2278)$$

The two finite layers are lifted to the common universal cover when interlayer distances are evaluated. For an observable supported inside radius L_{obs} and a finite hopping range L_H , exact local representation requires

$$r_{\text{inj},N}^{(1)} > L_{\text{obs}} + L_H, \quad r_{\text{inj},N}^{(2)} > L_{\text{obs}} + L_H. \quad (2279)$$

This construction conditions local Hamiltonian and local spectral observables. It does not create a nonexistent common translation group or a complete global periodic spectrum.

B. Injectivity radius

The systole of the compact cover is

$$\ell_{\text{sys},N} = \inf_{\gamma \in \Gamma_N \setminus \{e\}} \ell(\gamma), \quad (2280)$$

where

$$\ell(\gamma) = \inf_{x \in \mathbb{H}_R^2} d_{\mathbb{H}}(x, \gamma x) \quad (2281)$$

is the translation length. The global geometric injectivity radius is

$$r_{\text{inj},N}^{\text{geo}} = \frac{1}{2} \ell_{\text{sys},N} \quad (2282)$$

For a chosen base point o , the based injectivity radius is

$$r_{\text{inj},N}(o) = \frac{1}{2} \min_{\gamma \in \Gamma_N \setminus \{e\}} d_{\mathbb{H}}(o, \gamma o). \quad (2283)$$

Let $|\gamma|_{\mathcal{S}}$ be the word length for a fixed finite symmetric generating set $\mathcal{S}$. The word injectivity radius is

$$r_{\text{inj},N}^{\text{word}} = \frac{1}{2} \min_{\gamma \in \Gamma_N \setminus \{e\}} |\gamma|_{\mathcal{S}}. \quad (2284)$$

Cocompactness implies a quasi-isometry between word and physical distance. Thus there exist constants

$$A_q \geq 1, \quad B_q \geq 0 \quad (2285)$$

such that

$$A_q^{-1}|\gamma|_S - B_q \leq d_{\mathbb{H}}(o, \gamma o) \leq A_q|\gamma|_S + B_q. \quad (2286)$$

Consequently,

$$r_{\text{inj},N}^{\text{word}} \longrightarrow \infty \iff r_{\text{inj},N}(o) \longrightarrow \infty. \quad (2287)$$

A normal residual tower is

$$\Gamma_M = \Gamma_0 \triangleright \Gamma_1 \triangleright \Gamma_2 \triangleright \cdots, \quad \bigcap_{N \geq 1} \Gamma_N = \{e\}. \quad (2288)$$

For every fixed finite word ball

$$B_S(e, L) = \{\gamma : |\gamma|_S \leq L\}, \quad (2289)$$

residuality gives

$$\Gamma_N \cap [B_S(e, L) \setminus \{e\}] = \emptyset \quad (2290)$$

for all sufficiently large N . Therefore

$$r_{\text{inj},N}^{\text{word}} \longrightarrow \infty. \quad (2291)$$

Normality makes every rooted ball of radius smaller than $r_{\text{inj},N}^{\text{word}}$ exactly identical to the corresponding Cayley-graph ball. The statement is uniform over all vertices of the quotient. For a general finite-index subgroup, let

$$X_N = \Gamma_N \backslash \Gamma_M \quad (2292)$$

carry the right group action. A sequence is Farber when, for every $\gamma \neq e$,

$$\frac{\#\{x \in X_N : x\gamma = x\}}{|X_N|} \longrightarrow 0. \quad (2293)$$

Let $b_N(L)$ be the fraction of roots whose labelled L -ball does not agree with the universal cover. Farber convergence gives

$$\boxed{b_N(L) \longrightarrow 0} \quad (2294)$$

for every fixed L . A Farber sequence therefore gives typical-root convergence. It need not give an injective ball at every root. A normal residual tower gives the stronger global statement. Let

$$L_H = \sup \{d_{\mathbb{H}}(x_\alpha, \delta x_\beta) : h_{\alpha\beta}(\delta) \neq 0\} \quad (2295)$$

be the physical propagation radius of a finite-range Hamiltonian. For an observable supported inside a ball of radius L_{obs} , all matrix elements entering that observable lie within radius $L_{\text{obs}} + L_H$. Therefore

$$r_{\text{inj},N}^{\text{geo}} > L_{\text{obs}} + L_H \quad (2296)$$

implies exact equality between the finite-cover and infinite-system local observable. We define the established causal radius

$$L_{\text{cert},N} = r_{\text{inj},N}^{\text{geo}} - L_H. \quad (2297)$$

Every strictly local observable supported inside $L_{\text{obs}} < L_{\text{cert},N}$ is represented exactly. A product of m hopping operators has propagation radius at most mL_H . A quotient-closed path can differ from a contractible element of Γ_N . Therefore aliasing is absent when

$$2r_{\text{inj},N}^{\text{geo}} > mL_H. \quad (2298)$$

Define the moment aliasing margin

$$\mathfrak{a}_N^{(m)} = 2r_{\text{inj},N}^{\text{geo}} - mL_H. \quad (2299)$$

Exact stabilization of the m -th moment requires $\mathfrak{a}_N^{(m)} > 0$. Let

$$\tau_\infty(A) = \frac{1}{\nu_M} \sum_{\alpha=1}^{\nu_M} \langle \alpha, e | A | \alpha, e \rangle \quad (2300)$$

be the canonical trace per exact moiré cell, and let

$$\tau_N(A) = \frac{1}{D_N} \text{Tr } A \quad (2301)$$

be the finite-cover normalized trace. Under Eq. (2298),

$$\tau_N(H_N^m) = \tau_\infty(H_{\text{bulk}}^m). \quad (2302)$$

For each fixed m , the equality becomes exact at finite cover depth. It is not merely an asymptotic fit. Define the finite empirical spectral measure

$$\nu_N = \frac{1}{D_N} \sum_{j=1}^{D_N} \delta_{E_j^{(N)}}. \quad (2303)$$

The bulk regular-trace measure is defined by

$$\int_{\mathbb{R}} f(E) d\nu_{\text{bulk}}(E) = \tau_\infty[f(H_{\text{bulk}})]. \quad (2304)$$

Uniform boundedness places all measures inside a common compact interval. Exact stabilization of every fixed polynomial moment therefore gives

$$\nu_N \Rightarrow \nu_{\text{bulk}}. \quad (2305)$$

Equivalently,

$$\lim_{N \rightarrow \infty} \frac{1}{D_N} \text{Tr } f(H_N) = \tau_\infty[f(H_{\text{bulk}})] \quad (2306)$$

for every continuous f on the common spectral interval. Weak convergence controls normalized trace observables. It does not control isolated outliers whose multiplicity is

$o(D_N)$. A geodesic ball of radius $r_{\text{inj},N}^{\text{geo}}$ embeds isometrically into Σ_N . Its area is

$$A_{\mathbb{H}}(r) = 2\pi R^2 \left[\cosh\left(\frac{r}{R}\right) - 1 \right]. \quad (2307)$$

Therefore

$$A_N \geq A_{\mathbb{H}}\left(r_{\text{inj},N}^{\text{geo}}\right). \quad (2308)$$

Using $A_N = d_N A_M^{\text{sc}}$, we obtain

$$d_N \geq \frac{2\pi R^2}{A_M^{\text{sc}}} \left[\cosh\left(\frac{r_{\text{inj},N}^{\text{geo}}}{R}\right) - 1 \right]. \quad (2309)$$

The Hilbert-space dimension must therefore satisfy

$$D_N \geq \nu_M \frac{2\pi R^2}{A_M^{\text{sc}}} \left[\cosh\left(\frac{r_{\text{inj},N}^{\text{geo}}}{R}\right) - 1 \right]. \quad (2310)$$

For $r_{\text{inj},N} \gg R$,

$$D_N \gtrsim \frac{\pi \nu_M R^2}{A_M^{\text{sc}}} \exp\left(\frac{r_{\text{inj},N}}{R}\right). \quad (2311)$$

Increasing the established physical radius therefore requires an exponentially growing number of resonators in negative curvature. For an exponential hopping kernel, choose a cover-dependent cutoff $L_N \rightarrow \infty$ satisfying

$$\frac{L_N}{r_{\text{inj},N}} \rightarrow 0. \quad (2312)$$

This simultaneously removes the physical hopping tail and leaves an asymptotically large injectivity buffer. An explicit admissible schedule is

$$L_N = \frac{1}{\gamma} \ln(1 + \gamma r_{\text{inj},N}). \quad (2313)$$

It obeys

$$L_N \rightarrow \infty, \quad \frac{L_N}{r_{\text{inj},N}} \rightarrow 0, \quad (2314)$$

and

$$e^{-\gamma L_N} = \frac{1}{1 + \gamma r_{\text{inj},N}}. \quad (2315)$$

For an angular approximant $\theta_N \rightarrow \theta$, hyperbolic displacement amplification additionally requires

$$|\theta_N - \theta| \exp\left(\frac{L_N}{R}\right) \rightarrow 0. \quad (2316)$$

Ordinary angular convergence without Eq. (2316) is insufficient on a growing hyperbolic domain.

C. No-loss

Let

$$\Sigma = \text{Spec } H_{\text{bulk}}, \quad S_N = \text{Spec } H_N^{\text{ret}}. \quad (2317)$$

The no-loss error is

$$\ell_N = \sup_{\lambda \in \Sigma} \text{dist}(\lambda, S_N). \quad (2318)$$

The no-loss condition is $\ell_N \rightarrow 0$. No loss means that every infinite-system spectral point is approximated by finite-cover spectrum. It does not exclude additional finite-cover spectral points. For a bounded self-adjoint operator,

$$\Sigma = \sigma_{\text{ap}}(H_{\text{bulk}}), \quad (2319)$$

where σ_{ap} denotes the approximate point spectrum. Thus for every $\lambda \in \Sigma$ and every $\varepsilon > 0$, there exists a normalized vector ψ satisfying

$$\|(H_{\text{bulk}} - \lambda)\psi\| < \varepsilon. \quad (2320)$$

Finitely supported vectors are dense in $\mathcal{H}_{\infty}$. Since $H_{\text{bulk}} - \lambda$ is bounded, we may choose ψ with finite support. Suppose the support of ψ and its L_H -neighborhood lift isometrically to the cover. Let ψ_N be the normalized lift. Then

$$\|(H_N - \lambda)\psi_N\| = \|(H_{\text{bulk}} - \lambda)\psi\|. \quad (2321)$$

For any self-adjoint operator A and unit vector u ,

$$\text{dist}(\lambda, \text{Spec } A) \leq \|(A - \lambda)u\|. \quad (2322)$$

Therefore

$$\text{dist}(\lambda, S_N) < \varepsilon. \quad (2323)$$

Compactness of Σ converts pointwise approximation into uniform no loss. Choose an $\varepsilon/3$ -net

$$\{\lambda_1, \dots, \lambda_J\} \subset \Sigma. \quad (2324)$$

Lift one finite Weyl vector for each net point. At a sufficiently deep cover all of them lift simultaneously. Since $\text{dist}(\lambda, S_N)$ is one-Lipschitz in λ , $\ell_N < \varepsilon$ for all sufficiently large N . We introduce a quantitative no-loss modulus. For $R > 0$, let $\mathcal{V}_R$ be the set of normalized vectors whose support has physical diameter at most $2R$:

$$\mathcal{V}_R = \{\psi : \|\psi\| = 1, \text{diam supp } \psi \leq 2R\}. \quad (2325)$$

Define

$$\mathbf{w}_H(R) = \sup_{\lambda \in \Sigma} \inf_{\psi \in \mathcal{V}_R} \|(H_{\text{bulk}} - \lambda)\psi\|. \quad (2326)$$

The function is nonincreasing:

$$R_2 \geq R_1 \implies \mathbf{w}_H(R_2) \leq \mathbf{w}_H(R_1). \quad (2327)$$

The finite-Weyl-vector argument and compactness of Σ give

$$\mathfrak{w}_H(R) \longrightarrow 0 \quad (R \rightarrow \infty). \quad (2328)$$

If

$$R + L_H < r_{\text{inj},N}, \quad (2329)$$

every vector in the minimizing finite family used to approximate $\mathfrak{w}_H(R)$ lifts without aliasing. Consequently,

$$\ell_N \leq \mathfrak{w}_H(r_{\text{inj},N} - L_H). \quad (2330)$$

This formula converts injectivity radius directly into a one-sided spectral-approximation bound whenever the Weyl modulus is estimated. Suppose a set

$$\Lambda_\delta = \{\lambda_1, \dots, \lambda_J\} \subset \Sigma \quad (2331)$$

is a established δ_λ -net of Σ :

$$\sup_{\lambda \in \Sigma} \min_j |\lambda - \lambda_j| \leq \delta_\lambda. \quad (2332)$$

Let $\psi_{j,N}$ be normalized finite-cover vectors with residuals

$$r_{j,N} = \|(H_N - \lambda_j)\psi_{j,N}\|. \quad (2333)$$

Then

$$\ell_N \leq \delta_\lambda + \max_{1 \leq j \leq J} r_{j,N}. \quad (2334)$$

The net must cover the independently defined infinite spectral set. A dense sampling of S_N alone does not prove this hypothesis. Let $H_{\text{bulk}}^{(L)}$ and $H_N^{(L)}$ be matched finite-range truncations. Define

$$\varepsilon_\infty^{\text{tr}}(L) = \|H_{\text{bulk}} - H_{\text{bulk}}^{(L)}\|, \quad (2335)$$

and

$$\varepsilon_N^{\text{tr}}(L) = \|H_N - H_N^{(L)}\|. \quad (2336)$$

Let

$$\ell_N^{(L)} = \sup_{\lambda \in \text{Spec } H_{\text{bulk}}^{(L)}} \text{dist}[\lambda, \text{Spec } H_N^{(L)}]. \quad (2337)$$

The triangle inequality for one-sided spectral distance gives

$$\ell_N \leq \varepsilon_\infty^{\text{tr}}(L) + \ell_N^{(L)} + \varepsilon_N^{\text{tr}}(L). \quad (2338)$$

The middle term is controlled by lifted Weyl vectors once the truncated propagation radius is smaller than the injectivity buffer. If the bulk trace is faithful on the C^* -algebra generated by H_{bulk} , then

$$\text{supp } \nu_{\text{bulk}} = \Sigma. \quad (2339)$$

Together with weak convergence $\nu_N \Rightarrow \nu_{\text{bulk}}$, every open interval meeting Σ eventually contains finite-cover spectrum. Compactness again upgrades this to no loss. Faithfulness and weak convergence still do not exclude finite-cover outliers of vanishing normalized weight. Let $I_C \subset \Sigma$ be an intrinsic spectral island. Define

$$\ell_{C,N} = \sup_{\lambda \in I_C} \text{dist}(\lambda, S_N), \quad (2340)$$

and

$$\ell_{C,N}^{\text{near}} = \sup_{\lambda \in (\Sigma \setminus I_C) \cap \mathcal{N}_{\text{gap}}} \text{dist}(\lambda, S_N), \quad (2341)$$

where $\mathcal{N}_{\text{gap}}$ is a fixed neighborhood containing the two nearest complementary spectral edges. No loss for the target alone does not certify its isolation gap. The nearest complementary spectrum must also be transported.

D. No-pollution

The pollution error is

$$p_N = \sup_{s \in S_N} \text{dist}(s, \Sigma). \quad (2342)$$

No pollution means $p_N \longrightarrow 0$. The implications

$$\begin{aligned} \nu_N \Rightarrow \nu_{\text{bulk}} &\not\Rightarrow p_N \rightarrow 0, \\ \ell_N \rightarrow 0 &\not\Rightarrow p_N \rightarrow 0, \end{aligned} \quad (2343)$$

and

$$r_{\text{inj},N} \rightarrow \infty \not\Rightarrow p_N \rightarrow 0 \quad (2344)$$

are all false in general. Let $A_{\mathbb{T}_d}$ be adjacency on the infinite d -regular tree, $d \geq 3$. Its spectrum is

$$\text{Spec } A_{\mathbb{T}_d} = [-2\sqrt{d-1}, 2\sqrt{d-1}]. \quad (2345)$$

Every finite d -regular quotient graph has the constant eigenvector with eigenvalue d . Therefore

$$p_N \geq d - 2\sqrt{d-1} = (\sqrt{d-1} - 1)^2 > 0. \quad (2346)$$

This remains true even when the quotient girth and injectivity radius diverge. The normalized multiplicity of the outlier is

$$\frac{1}{|V_N|} \longrightarrow 0. \quad (2347)$$

Weak DOS convergence therefore remains valid while global edge convergence fails. For a nonamenable group with symmetric generating set $\mathcal{S}$, the finite quotient adjacency has trivial-sector eigenvalue $|\mathcal{S}|$, whereas

$$\left\| \sum_{s \in \mathcal{S}} \lambda_\Gamma(s) \right\| < |\mathcal{S}|. \quad (2348)$$

Removing the trivial sector eliminates this particular eigenvalue but does not prove that every remaining quotient sector is tempered. The strongest direct certificate is

$$S_N \subset \Sigma_{\varepsilon_N} = \{x : \text{dist}(x, \Sigma) \leq \varepsilon_N\}, \quad (2349)$$

with $\varepsilon_N \rightarrow 0$. It immediately gives $p_N \leq \varepsilon_N$. For a finite-dimensional Hermitian matrix,

$$\inf_{\|v\|=1} \|(H_N - \lambda)v\| = \text{dist}(\lambda, S_N). \quad (2350)$$

A sufficient no-pollution certificate is

$$\inf_{\|v\|=1} \|(H_N - \lambda)v\| \geq \text{dist}(\lambda, \Sigma) - \varepsilon_N \quad (2351)$$

for every real λ in a common spectral interval. Setting $\lambda = s \in S_N$ gives

$$0 \geq \text{dist}(s, \Sigma) - \varepsilon_N, \quad (2352)$$

and therefore $p_N \leq \varepsilon_N$. Outside the ε_N -neighborhood of Σ , the equivalent resolvent bound is

$$\|(H_N - \lambda)^{-1}\| \leq \frac{1}{\text{dist}(\lambda, \Sigma) - \varepsilon_N}. \quad (2353)$$

Let

$$\mathcal{F}_\varepsilon = [-M, M] \setminus \Sigma_\varepsilon \quad (2354)$$

be the forbidden compact energy set. Choose a grid

$$\Lambda_h = \{\lambda_1, \dots, \lambda_J\} \subset \mathcal{F}_\varepsilon \quad (2355)$$

with covering radius

$$h_\lambda = \sup_{\lambda \in \mathcal{F}_\varepsilon} \min_j |\lambda - \lambda_j|. \quad (2356)$$

The smallest singular value is one-Lipschitz:

$$|\sigma_{\min}(H_N - \lambda I) - \sigma_{\min}(H_N - \lambda' I)| \leq |\lambda - \lambda'|. \quad (2357)$$

Suppose a established lower bound

$$\underline{\sigma}_j \leq \sigma_{\min}(H_N - \lambda_j I) \quad (2358)$$

is available at every grid point. Then, for every $\lambda \in \mathcal{F}_\varepsilon$,

$$\sigma_{\min}(H_N - \lambda I) \geq \min_j \underline{\sigma}_j - h_\lambda. \quad (2359)$$

Therefore

$$\min_j \underline{\sigma}_j > h_\lambda \implies S_N \cap \mathcal{F}_\varepsilon = \emptyset. \quad (2360)$$

Consequently, $p_N \leq \varepsilon$. This converts a continuous forbidden-energy verification into finitely many established singular-value calculations. Let

$$\mathcal{F} \subset [-M, M] \setminus \Sigma \quad (2361)$$

be a compact forbidden set. Choose a real polynomial p satisfying

$$\sup_{\lambda \in \Sigma} |p(\lambda)| \leq a, \quad (2362)$$

and

$$\inf_{\lambda \in \mathcal{F}} |p(\lambda)| \geq b, \quad b > a. \quad (2363)$$

If

$$\|p(H_N^{\text{ret}})\| < b, \quad (2364)$$

then

$$S_N \cap \mathcal{F} = \emptyset. \quad (2365)$$

Indeed, an eigenvalue $s \in \mathcal{F}$ would imply

$$\|p(H_N^{\text{ret}})\| \geq |p(s)| \geq b. \quad (2366)$$

An asymptotic sufficient condition is

$$\limsup_{N \rightarrow \infty} \|p(H_N^{\text{ret}})\| \leq \|p(H_{\text{bulk}})\| \quad (2367)$$

for every real polynomial p . This is imposed on the entire declared retained sector, not merely after removal of one known outlier. Suppose the finite and infinite operators act on different Hilbert spaces. Let

$$J_N : \mathcal{H}_\infty \longrightarrow \mathcal{H}_N^{\text{ret}}, \quad K_N : \mathcal{H}_N^{\text{ret}} \longrightarrow \mathcal{H}_\infty \quad (2368)$$

be contractions satisfying

$$\begin{aligned} \|J_N u\| &\geq (1 - a_N) \|u\|, \\ \|K_N v\| &\geq (1 - a_N) \|v\|, \end{aligned} \quad (2369)$$

and

$$\begin{aligned} \|H_N^{\text{ret}} J_N - J_N H_{\text{bulk}}\| &\leq b_N, \\ \|H_{\text{bulk}} K_N - K_N H_N^{\text{ret}}\| &\leq b_N. \end{aligned} \quad (2370)$$

Then

$$d_H(S_N, \Sigma) \leq \frac{b_N}{1 - a_N}. \quad (2371)$$

The forward map supplies no loss. The reverse map is the explicit no-pollution half. Let

$$\mathcal{K}_N \subset \mathcal{H}_N \quad (2372)$$

be a specified invariant subspace:

$$H_N \mathcal{K}_N \subseteq \mathcal{K}_N. \quad (2373)$$

Define

$$H_N^{\text{ret}} = H_N|_{\mathcal{K}_N}. \quad (2374)$$

For a representation-sector decomposition,

$$\mathcal{K}_N = \bigoplus_{\rho \in \mathcal{R}_N} \text{Ran } P_{\rho,N}. \quad (2375)$$

The sectorwise pollution errors are

$$p_{N,\rho} = \sup_{s \in \text{Spec } H_N[\rho]} \text{dist}(s, \Sigma). \quad (2376)$$

The retained error is

$$p_N^{\text{ret}} = \max_{\rho \in \mathcal{R}_N} p_{N,\rho}. \quad (2377)$$

Deleting the trivial representation establishes only that one block has been removed. It does not bound $p_{N,\rho}$ for the remaining blocks. Let $I_C \subset \Sigma$ be a proper compact spectral island with

$$\Delta_C = \text{dist}(I_C, \Sigma \setminus I_C) > 0. \quad (2378)$$

Its bandwidth is

$$W_C = \text{diam } I_C. \quad (2379)$$

Assume

$$\eta_N < \frac{\Delta_C}{3}. \quad (2380)$$

Use the fixed intrinsic neighborhood

$$I_{C,N} = \left\{ s \in S_N : \text{dist}(s, I_C) \leq \frac{\Delta_C}{3} \right\}. \quad (2381)$$

Let

$$B_{C,N} = S_N \setminus I_{C,N}. \quad (2382)$$

Then

$$d_H(I_{C,N}, I_C) \leq \eta_N, \quad (2383)$$

and

$$d_H(B_{C,N}, \Sigma \setminus I_C) \leq \eta_N. \quad (2384)$$

Consequently,

$$|W_{C,N} - W_C| \leq 2\eta_N, \quad (2385)$$

and

$$|\Delta_{C,N} - \Delta_C| \leq 2\eta_N. \quad (2386)$$

In particular,

$$\Delta_{C,N} \geq \Delta_C - 2\eta_N > 0. \quad (2387)$$

The fixed neighborhood is determined by the intrinsic island. It prevents post-selection of whichever finite-cover

eigenvalue cluster appears narrowest. Let the finite-range truncated pair satisfy

$$\eta_N^{\text{fr}}(L) = d_H \left[\text{Spec } H_N^{(L),\text{ret}}, \text{Spec } H_{\text{bulk}}^{(L)} \right]. \quad (2388)$$

Then

$$\begin{aligned} \eta_N^{\text{full}}(L) &\leq \varepsilon_{\text{tail},N}^{(0)}(L) + \eta_N^{\text{fr}}(L) \\ &\quad + \varepsilon_{\text{tail},\infty}^{(0)}(L) + \varepsilon_{\text{param},N}^{(0)}(L). \end{aligned} \quad (2389)$$

The parameter term includes angular approximant, geometry transport, and other common-space matching errors. If the physical claim is normalized by a scale $E_{\text{ref},N} > 0$, the required condition is

$$\frac{\eta_N^{\text{full}}(L_N)}{E_{\text{ref},N}} \longrightarrow 0. \quad (2390)$$

An absolute error tending to zero can remain physically large if $E_{\text{ref},N}$ vanishes more rapidly.

E. Representation completeness

We distinguish four representation-theoretic objects. First, the finite quotient dual is

$$\widehat{Q}_N = \{\text{irreducible unitary representations of } Q_N\} / \simeq. \quad (2391)$$

Second, the tower samples the finite-image representations

$$\widehat{\Gamma}_{\text{fin}}^{\mathcal{T}} = \bigcup_N q_N^* \left(\widehat{Q}_N \right), \quad (2392)$$

where

$$q_N^* \rho = \rho \circ q_N. \quad (2393)$$

Third, the finite-dimensional automorphic hierarchy is

$$\mathcal{B}_{\text{fd}} = \bigsqcup_{d \geq 1} \text{Hom}_{\text{irr}}(\Gamma_M, U(d)) / U(d). \quad (2394)$$

Fourth, the reduced regular bulk is defined by

$$C_r^*(\Gamma_M), \quad L(\Gamma_M), \quad \tau, \quad (2395)$$

and by the regular Hamiltonian

$$H_{\text{bulk}} \in M_{\nu_M} [C_r^*(\Gamma_M)]. \quad (2396)$$

The implications

$$\widehat{Q}_N \text{ complete} \not\Rightarrow \mathcal{B}_{\text{fd}} \text{ complete}, \quad (2397)$$

and

$$\bigcup_N q_N^* \widehat{Q}_N \not\Rightarrow \text{a Plancherel decomposition of } H_{\text{bulk}} \quad (2398)$$

are essential scope restrictions. For every finite quotient,

$$\ell^2(Q_N) \simeq \bigoplus_{\rho \in \hat{Q}_N} V_\rho \otimes V_\rho^*. \quad (2399)$$

If

$$d_\rho = \dim V_\rho, \quad (2400)$$

then

$$|Q_N| = \sum_{\rho \in \hat{Q}_N} d_\rho^2. \quad (2401)$$

Push the group kernel to Q_N and define

$$H_N[\rho] = \sum_{\delta \in \Gamma_M} h(\delta) \otimes \rho[q_N(\delta)]. \quad (2402)$$

Its dimension is

$$D_{\rho,N} = \nu_M d_\rho. \quad (2403)$$

The complete finite-cover Hamiltonian decomposes as

$$H_N \simeq \bigoplus_{\rho \in \hat{Q}_N} [H_N[\rho] \otimes I_{d_\rho}]. \quad (2404)$$

Each block eigenvalue occurs with multiplicity d_ρ . The total number of physical states carried by the ρ -isotypic component is

$$N_{\rho,N} = \nu_M d_\rho^2. \quad (2405)$$

Summation gives

$$\sum_{\rho} N_{\rho,N} = \nu_M |Q_N| = D_N. \quad (2406)$$

Let

$$\chi_\rho(q) = \text{Tr } \rho(q) \quad (2407)$$

be the character. The exact isotypic projector is

$$P_{\rho,N} = \frac{d_\rho}{|Q_N|} \sum_{q \in Q_N} \chi_\rho(q^{-1}) (I_{\nu_M} \otimes R_N(q)). \quad (2408)$$

Character orthogonality gives $P_{\rho,N}^2 = P_{\rho,N}$,

$$P_{\rho,N}^\dagger = P_{\rho,N}, \quad P_{\rho,N} P_{\sigma,N} = 0 \quad (\rho \neq \sigma), \quad (2409)$$

and

$$\sum_{\rho \in \hat{Q}_N} P_{\rho,N} = I_{D_N}. \quad (2410)$$

Using Eq. (2261),

$$\text{Tr } P_{\rho,N} = \frac{d_\rho}{|Q_N|} \chi_\rho(e) \nu_M |Q_N|$$

$$= \nu_M d_\rho^2. \quad (2411)$$

Thus

$$\text{rank } P_{\rho,N} = \nu_M d_\rho^2. \quad (2412)$$

Define the reduced characteristic polynomial

$$P_{\rho,N}(z) = \det [z I_{\nu_M d_\rho} - H_N[\rho]]. \quad (2413)$$

The full finite-cover characteristic polynomial factorizes exactly:

$$\det [z I_{D_N} - H_N] = \prod_{\rho \in \hat{Q}_N} P_{\rho,N}(z)^{d_\rho}. \quad (2414)$$

For every continuous function f ,

$$\frac{1}{D_N} \text{Tr } f(H_N) = \sum_{\rho \in \hat{Q}_N} \frac{d_\rho^2}{|Q_N|} \frac{1}{\nu_M d_\rho} \text{Tr } f(H_N[\rho]). \quad (2415)$$

The physical regular-representation weight is

$$w_{\rho,N} = \frac{d_\rho^2}{|Q_N|}, \quad \sum_{\rho} w_{\rho,N} = 1. \quad (2416)$$

The complete finite-cover DOS is

$$\nu_N^{\text{full}} = \sum_{\rho \in \hat{Q}_N} \frac{d_\rho^2}{|Q_N|} \nu_{\rho,N}, \quad (2417)$$

where

$$\nu_{\rho,N} = \frac{1}{\nu_M d_\rho} \sum_{a=1}^{\nu_M d_\rho} \delta_{E_{a,\rho}^{(N)}}. \quad (2418)$$

Equal weighting of inequivalent irreps is generally incorrect. For a computed subset

$$\mathcal{R}_N \subseteq \hat{Q}_N, \quad (2419)$$

define

$$C_{\text{rep},N} = \frac{1}{|Q_N|} \sum_{\rho \in \mathcal{R}_N} d_\rho^2. \quad (2420)$$

Exact finite-quotient completeness is $C_{\text{rep},N} = 1$. The omitted state fraction is $1 - C_{\text{rep},N}$. Let μ_N^{ret} be the retained DOS normalized to unit mass, and let μ_N^{miss} be the normalized omitted DOS. Then

$$\mu_N^{\text{full}} = C_{\text{rep},N} \mu_N^{\text{ret}} + (1 - C_{\text{rep},N}) \mu_N^{\text{miss}}. \quad (2421)$$

For every bounded f ,

$$\left| \int f d\mu_N^{\text{full}} - \int f d\mu_N^{\text{ret}} \right| \leq 2(1 - C_{\text{rep},N}) \|f\|_\infty. \quad (2422)$$

If the retained contribution is left unnormalized, the error satisfies

$$\left| \int f d\mu_N^{\text{full}} - C_{\text{rep},N} \int f d\mu_N^{\text{ret}} \right| \leq (1 - C_{\text{rep},N}) \|f\|_{\infty}. \quad (2423)$$

These bounds control bounded trace observables. They do not control spectral extrema. A single omitted sector of arbitrarily small weight can contain the global minimum, global maximum, or a gap-filling eigenvalue. Let $C_{\rho,N}$ contain $r_{C,\rho,N}$ reduced block eigenvalues in sector ρ . The number of physical regular-representation states is

$$N_{C,N}^{\text{full}} = \sum_{\rho \in \mathcal{R}_{C,N}^{\text{decl}}} d_{\rho} r_{C,\rho,N}. \quad (2424)$$

If only $\mathcal{R}_{C,N}^{\text{comp}}$ is computed, define

$$C_{\text{rep},C,N} = \frac{\sum_{\rho \in \mathcal{R}_{C,N}^{\text{comp}}} d_{\rho} r_{C,\rho,N}}{\sum_{\rho \in \mathcal{R}_{C,N}^{\text{decl}}} d_{\rho} r_{C,\rho,N}}. \quad (2425)$$

A full target-band conclusion requires $C_{\text{rep},C,N} = 1$. For a normalized eigenvector $|\psi_n\rangle$, define

$$w_{n,\rho} = \langle \psi_n | P_{\rho,N} | \psi_n \rangle. \quad (2426)$$

The exact sum rule is

$$\sum_{\rho} w_{n,\rho} = 1. \quad (2427)$$

An ideally pure state has $w_{n,\rho_*} = 1$ for one ρ_* . The representation-purity defect is

$$\varepsilon_{\text{rep},n} = 1 - \max_{\rho} w_{n,\rho}. \quad (2428)$$

A symmetry-resolved result requires $\varepsilon_{\text{rep},n} \leq \varepsilon_{\text{rep}}^{\text{tol}}$. For a degenerate cluster projector P_C , define

$$w_{C,\rho} = \frac{\text{Tr}(P_C P_{\rho,N})}{\text{Tr } P_C}. \quad (2429)$$

This remains invariant under basis rotations inside the cluster. For any finite-dimensional unitary representation

$$\rho : \Gamma_M \longrightarrow U(d_{\rho}), \quad (2430)$$

the automorphic block

$$H[\rho] = \sum_{\gamma} h(\gamma) \otimes \rho(\gamma) \quad (2431)$$

is an exact finite-dimensional flat-bundle Hamiltonian. It is a reduced-bulk block only when $\rho \prec \lambda_{\Gamma_M}$, equivalently,

$$\|\rho(a)\| \leq \|\lambda_{\Gamma_M}(a)\| \quad (2432)$$

for every $a \in \mathbb{C}\Gamma_M$. Under this condition,

$$\text{Spec } H[\rho] \subseteq \text{Spec } H_{\text{bulk}}. \quad (2433)$$

Without it, $H[\rho]$ remains an exact automorphic probe but is not automatically part of the reduced regular bulk spectrum. We therefore use the following terminology:

- complete finite-cover spectrum : $\text{Spec } H_N$ with every $\rho \in$
- complete finite-quotient coverage : $C_{\text{rep},N} = 1$,
- declared automorphic-sector spectrum : $\text{Spec } H[\rho]$ for specified ρ
- established retained-sector approximation : $\text{Spec } H_N^{\text{ret}}$ with no loss a
- reduced regular bulk spectrum : $\text{Spec } H_{\text{bulk}}$.

$$(2434)$$

Complete finite-level representation coverage and reduced-bulk convergence are independent conditions.

F. Cross-cover convergence

Let $\mathcal{P}$ be a compact physical parameter domain. Consider the normalized finite-propagation family

$$\hat{H}(\mathbf{p}) = \sum_{\gamma \in \mathcal{F}} A_{\gamma}(\mathbf{p}) \otimes \lambda_{\Gamma_M}(\gamma), \quad \mathbf{p} \in \mathcal{P}, \quad (2435)$$

where $\mathcal{F} = \mathcal{F}^{-1}$ is finite and

$$A_{\gamma^{-1}}(\mathbf{p}) = A_{\gamma}(\mathbf{p})^{\dagger}. \quad (2436)$$

Define the coefficient modulus

$$\omega_H(\delta) = \sup_{d_{\mathcal{P}}(\mathbf{p}, \mathbf{p}') \leq \delta} \sum_{\gamma \in \mathcal{F}} \|A_{\gamma}(\mathbf{p}) - A_{\gamma}(\mathbf{p}')\|. \quad (2437)$$

We assume

$$\omega_H(\delta) \longrightarrow 0 \quad (\delta \rightarrow 0). \quad (2438)$$

Then

$$\|\hat{H}(\mathbf{p}) - \hat{H}(\mathbf{p}')\| \leq \omega_H[d_{\mathcal{P}}(\mathbf{p}, \mathbf{p}')]. \quad (2439)$$

The finite-cover families inherit the same modulus uniformly in N . Let

$$\mathcal{K}_N \subseteq \mathcal{H}_N \quad (2440)$$

be a declared invariant retained sector, with projector P_N . Define

$$\hat{H}_N^{\text{ret}}(\mathbf{p}) = \hat{H}_N(\mathbf{p})|_{\mathcal{K}_N}. \quad (2441)$$

We call a tower spectrally admissible on $\mathcal{P}$ when the following three conditions hold. First, the subgroup sequence is Farber:

$$\frac{\#\{x \in X_N : x\gamma = x\}}{|X_N|} \longrightarrow 0 \quad (2442)$$

for every $\gamma \neq e$. Second, the retained sector is locally exhaustive. For every finite set $E \subset \Gamma_M$, let

$$E^+ = E \cup E\mathcal{F}. \quad (2443)$$

There exist rooted local isometries

$$J_{N,E^+} : \ell^2(E^+) \otimes \mathbb{C}^{\nu_M} \longrightarrow \mathcal{H}_N \quad (2444)$$

such that

$$\|(I - P_N)J_{N,E^+}\| \longrightarrow 0. \quad (2445)$$

Third, the retained sector obeys uniform polynomial norm domination:

$$\limsup_{N \rightarrow \infty} \sup_{\mathbf{p} \in \mathcal{P}} \left[\|p(\hat{H}_N^{\text{ret}}(\mathbf{p}))\| - \|p(\hat{H}(\mathbf{p}))\| \right] \leq 0 \quad (2446)$$

for every real polynomial p . An independently verified uniform residual lower bound or direct upper spectral inclusion on the same retained sector may replace the third condition. Local exhaustivity does not replace no pollution. Define

$$\Sigma(\mathbf{p}) = \text{Spec } \hat{H}(\mathbf{p}), \quad (2447)$$

and

$$S_N^{\text{ret}}(\mathbf{p}) = \text{Spec } \hat{H}_N^{\text{ret}}(\mathbf{p}). \quad (2448)$$

For every spectrally admissible tower,

$$\eta_N^{\mathcal{T}} = \sup_{\mathbf{p} \in \mathcal{P}} d_H[S_N^{\text{ret}}(\mathbf{p}), \Sigma(\mathbf{p})] \longrightarrow 0. \quad (2449)$$

Local exhaustivity and lifted Weyl vectors provide the no-loss half. Polynomial norm domination excludes a uniformly separated polluted subsequence. Uniform coefficient continuity converts finite parameter nets into convergence over the complete compact domain. Suppose

$$I_C(\mathbf{p}) \subset \Sigma(\mathbf{p}) \quad (2450)$$

is a continuously tracked proper spectral island with uniform gap

$$\Delta_0 = \inf_{\mathbf{p} \in \mathcal{P}} \text{dist}[I_C(\mathbf{p}), \Sigma(\mathbf{p}) \setminus I_C(\mathbf{p})] > 0. \quad (2451)$$

Define the intrinsic functions

$$\Phi_W(\mathbf{p}) = \text{diam } I_C(\mathbf{p}), \quad (2452)$$

and

$$\Phi_{\Delta}(\mathbf{p}) = \text{dist}[I_C(\mathbf{p}), \Sigma(\mathbf{p}) \setminus I_C(\mathbf{p})]. \quad (2453)$$

For tower $\mathcal{T}$, once

$$\eta_N^{\mathcal{T}} < \frac{\Delta_0}{3}, \quad (2454)$$

define

$$I_{C,N}^{\mathcal{T}}(\mathbf{p}) = \left\{ s \in S_N^{\mathcal{T}}(\mathbf{p}) : \text{dist}[s, I_C(\mathbf{p})] \leq \frac{\Delta_0}{3} \right\}, \quad (2455)$$

and its complement

$$B_{C,N}^{\mathcal{T}}(\mathbf{p}) = S_N^{\mathcal{T}}(\mathbf{p}) \setminus I_{C,N}^{\mathcal{T}}(\mathbf{p}). \quad (2456)$$

The finite-cover functions are

$$\Phi_{W,N}^{\mathcal{T}}(\mathbf{p}) = \text{diam } I_{C,N}^{\mathcal{T}}(\mathbf{p}), \quad (2457)$$

and

$$\Phi_{\Delta,N}^{\mathcal{T}}(\mathbf{p}) = \text{dist}[I_{C,N}^{\mathcal{T}}(\mathbf{p}), B_{C,N}^{\mathcal{T}}(\mathbf{p})]. \quad (2458)$$

They satisfy

$$\sup_{\mathbf{p} \in \mathcal{P}} |\Phi_{W,N}^{\mathcal{T}}(\mathbf{p}) - \Phi_W(\mathbf{p})| \leq 2\eta_N^{\mathcal{T}}, \quad (2459)$$

and

$$\sup_{\mathbf{p} \in \mathcal{P}} |\Phi_{\Delta,N}^{\mathcal{T}}(\mathbf{p}) - \Phi_{\Delta}(\mathbf{p})| \leq 2\eta_N^{\mathcal{T}}. \quad (2460)$$

Let

$$\mathcal{T}_A = \{\Gamma_N^A\}, \quad \mathcal{T}_B = \{\Gamma_M^B\} \quad (2461)$$

be inequivalent spectrally admissible towers. Define the cross-cover bandwidth residual

$$R_{W,N,M}^{A,B} = \sup_{\mathbf{p} \in \mathcal{P}} |\Phi_{W,N}^A(\mathbf{p}) - \Phi_{W,M}^B(\mathbf{p})|. \quad (2462)$$

Define the cross-cover gap residual

$$R_{\Delta,N,M}^{A,B} = \sup_{\mathbf{p} \in \mathcal{P}} |\Phi_{\Delta,N}^A(\mathbf{p}) - \Phi_{\Delta,M}^B(\mathbf{p})|. \quad (2463)$$

The triangle inequality gives

$$R_{W,N,M}^{A,B} \leq 2\eta_N^A + 2\eta_M^B, \quad (2464)$$

and

$$R_{\Delta,N,M}^{A,B} \leq 2\eta_N^A + 2\eta_M^B. \quad (2465)$$

Both residuals therefore vanish. More generally, let $\mathfrak{F}$ be a spectral-set functional satisfying

$$|\mathfrak{F}(A) - \mathfrak{F}(B)| \leq L_{\mathfrak{F}} d_H(A, B). \quad (2466)$$

Then

$$|\mathfrak{F}(S_N^A) - \mathfrak{F}(S_M^B)| \leq L_{\mathfrak{F}} (\eta_N^A + \eta_M^B). \quad (2467)$$

Homogeneity in energy does not by itself imply Hausdorff continuity. For tower $T \in \{A, B\}$, define

$$\begin{aligned} \eta_N^{T,\text{exp}} &\leq \varepsilon_{\text{tail},N}^{T,(0)}(L_N^T) + \eta_N^{T,\text{fr}}(L_N^T) \\ &\quad + \varepsilon_{\text{tail},\infty}^{(0)}(L_N^T) + \varepsilon_{\text{param},N}^{T,(0)}. \end{aligned} \quad (2468)$$

The cross-cover bandwidth error becomes

$$R_{W,N,M}^{A,B} \leq 2\eta_N^{A,\text{exp}} + 2\eta_M^{B,\text{exp}}, \quad (2469)$$

and the same bound holds for the gap. Each tower must satisfy its own

$$L_N^T \rightarrow \infty, \quad \frac{L_N^T}{r_{\text{inj},N}^T} \rightarrow 0, \quad (2470)$$

and its own no-pollution certificate at the growing cutoff. Let $\nu_{N,\mathbf{p}}^T$ be the full normalized finite-cover measure, and let $\nu_{\mathbf{p}}$ be the regular-trace bulk measure. For every continuous f on a common compact spectral interval, Farber convergence gives

$$\sup_{\mathbf{p} \in \mathcal{P}} \left| \int f d\nu_{N,\mathbf{p}}^T - \int f d\nu_{\mathbf{p}} \right| \rightarrow 0. \quad (2471)$$

For a bounded jointly continuous broadening kernel

$$K_\eta(E - s), \quad \eta > 0, \quad (2472)$$

define

$$\varrho_\eta^{T,N}(\mathbf{p}, E) = \int K_\eta(E - s) d\nu_{N,\mathbf{p}}^T(s). \quad (2473)$$

Then

$$\sup_{(\mathbf{p}, E) \in \mathcal{P} \times \mathcal{E}} |\varrho_\eta^{T,N}(\mathbf{p}, E) - \varrho_\eta(\mathbf{p}, E)| \rightarrow 0 \quad (2474)$$

for every compact energy interval $\mathcal{E}$. The safe order of limits is

$$N \rightarrow \infty \quad \text{at fixed } \eta > 0, \quad \text{then } \eta \rightarrow 0^+. \quad (2475)$$

Weak convergence does not justify the unsmoothed DOS, its pointwise maximum, or an interchange of these limits without a uniform regularity estimate. Spectral Hausdorff convergence is a C^0 statement. A generalized velocity requires convergence of the differentiated Hamiltonian on identified fibers. For first derivatives, we require

$$\begin{aligned} & \sup_{\mathbf{p} \in \mathcal{P}} \left[\|\hat{H}_N^T(\mathbf{p}) - \hat{H}(\mathbf{p})\| \right. \\ & \left. + \|D\hat{H}_N^T(\mathbf{p}) - D\hat{H}(\mathbf{p})\|_{\text{op,dual}} \right] \rightarrow 0. \end{aligned} \quad (2476)$$

Together with a uniform isolation gap and uniform convergence of the canonical metric, this transfers

$$dE_{C,N}^T \rightarrow dE_C, \quad (2477)$$

and

$$v_{\text{max},C,N}^T \rightarrow v_{\text{max},C}. \quad (2478)$$

For Hessians and effective masses, we additionally require

$$\sup_{\mathbf{p} \in \mathcal{P}} \|D^2\hat{H}_N^T(\mathbf{p}) - D^2\hat{H}(\mathbf{p})\| \rightarrow 0. \quad (2479)$$

Uniform C^0 convergence alone cannot control either derivative tier. Let $P_{C,N}^T(\mathbf{p})$ and $P_C(\mathbf{p})$ be defined by one fixed contour family with uniform distance $\delta_{\mathcal{C}} > 0$ from the complementary spectrum. If

$$\varepsilon_{H,N}^T = \sup_{\mathbf{p}} \|H_N^T(\mathbf{p}) - H(\mathbf{p})\| < \delta_{\mathcal{C}}, \quad (2480)$$

then

$$\sup_{\mathbf{p}} \|P_{C,N}^T(\mathbf{p}) - P_C(\mathbf{p})\| \leq \frac{|\mathcal{C}|}{2\pi} \frac{\varepsilon_{H,N}^T}{\delta_{\mathcal{C}} (\delta_{\mathcal{C}} - \varepsilon_{H,N}^T)}. \quad (2481)$$

Coherence observables additionally require convergence of the real-space cell partition, symmetry covariance, and representation coverage. They are not functions of the spectral set alone. Let $\mathcal{M}_N^T(\theta)$ be a finite-cover physical flatness landscape, and let $\mathcal{M}_\infty(\theta)$ be its bulk limit. Suppose

$$\|\mathcal{M}_N^T - \mathcal{M}_\infty\|_\infty \rightarrow 0. \quad (2482)$$

If the bulk maximizer is unique,

$$\theta_* = \arg \max_{\theta} \mathcal{M}_\infty(\theta), \quad (2483)$$

then every exact finite-cover maximizer converges: $\theta_{*,N}^T \rightarrow \theta_*$. For any open neighborhood $U \ni \theta_*$, define the optimizer margin

$$m_U = \mathcal{M}_\infty(\theta_*) - \sup_{\theta \notin U} \mathcal{M}_\infty(\theta) > 0. \quad (2484)$$

If

$$\|\mathcal{M}_N^T - \mathcal{M}_\infty\|_\infty < \frac{m_U}{3}, \quad (2485)$$

every finite-cover maximizer lies inside U . If the logarithmic bulk score

$$L_\infty = \ln \mathcal{M}_\infty \quad (2486)$$

satisfies the strong-concavity bound

$$L_\infty(\theta_*) - L_\infty(\theta) \geq \frac{\mu_\theta}{2} |\theta - \theta_*|^2, \quad (2487)$$

and

$$\|L_N^T - L_\infty\|_\infty \leq \varepsilon_{L,N}^T, \quad (2488)$$

then

$$|\theta_{*,N}^T - \theta_*| \leq 2\sqrt{\frac{\varepsilon_{L,N}^T}{\mu_\theta}}. \quad (2489)$$

Without uniqueness, uniform convergence controls the maximal score value and the cluster points of optimizer sets, but it does not select one unique angle. For an observable O_N , define the normalized successive-cover difference

$$\Delta_{O,N} = \frac{|O_{N+1} - O_N|}{\max\{O_{\text{ref}}, |O_{N+1}|, |O_N|\}}. \quad (2490)$$

The nonzero reference scale prevents an artificial divergence when the observable approaches zero. For two inequivalent covers A and B , define

$$\Delta_{O,N,M}^{A,B} = \frac{|O_N^A - O_M^B|}{\max\{O_{\text{ref}}, |O_N^A|, |O_M^B|\}}. \quad (2491)$$

These quantities are convergence diagnostics. They become bulk certificates only when accompanied by the no-loss and no-pollution bounds that identify both sequences with the same intrinsic object. Cover dependence is plotted against both D_N and $r_{\text{inj},N}$. Two covers of similar dimension can have different systoles and different finite-size errors. Correlation with injectivity radius is therefore more physically informative than correlation with site count alone. For tower T at level N , we record

$$\begin{aligned} \mathfrak{C}_{\text{bulk},N}^T = & \left(Q_N^T, d_N^T, D_N^T, g_N^T, r_{\text{inj},N}^T, \right. \\ & L_N^T, L_N^T/r_{\text{inj},N}^T, \varepsilon_{\text{Herm},N}^T, \varepsilon_{\text{sym},N}^T, \\ & \varepsilon_{\text{tail},N}^{T,(0)}, \varepsilon_{\text{tail},N}^{T,(1)}, \varepsilon_{\text{tail},N}^{T,(2)}, \\ & \varepsilon_{\text{param},N}^T, \varepsilon_{\text{solver},N}^T, \varepsilon_{\text{fp},N}^T, \\ & \ell_N^T, p_N^T, \eta_N^T, C_{\text{rep},N}^T, C_{\text{rep},C,N}^T, \\ & W_{C,N}^T, \Delta_{C,N}^T, v_{\text{max},C,N}^T, \Lambda_{\text{max},C,N}^T, \\ & \left. \varrho_{C,N,\eta}^T, C_{C,N}^{T,\text{coh}}, \theta_{\text{flat},N}^T \right). \quad (2492) \end{aligned}$$

The finite graph defines the exact finite-device problem. Local bulk agreement holds when

$$r_{\text{inj},N} > L_{\text{obs}} + L_H.$$

Fixed-moment or Farber convergence gives the weak density-of-states limit, and complete finite-quotient representation coverage is expressed by $C_{\text{rep},N} = 1$. A retained sector converges spectrally when its loss and pollution indicators both tend to zero; the corresponding full-bulk statement requires these limits on the complete finite Hilbert space. Finite-level representation completeness and reduced-bulk convergence are therefore separate properties.

The final verification logic is deliberately hierarchical. The condition $r_{\text{inj},N} \rightarrow \infty$ first guarantees exact agreement with the infinite system on every fixed-radius neighborhood. More specifically, whenever $2r_{\text{inj},N} > mL_H$, all closed paths relevant to the m -th spectral moment remain inside an injective neighborhood, and the corresponding fixed moment is therefore reproduced exactly by the finite cover. Stabilization of every fixed moment then yields weak convergence of the normalized spectral measure, $\nu_N \Rightarrow \nu_{\text{bulk}}$, under the stated moment or Farber hypotheses. This weak convergence controls the density of states; individual edges, gaps, and isolated components require the two-sided spectral estimates below. Spectral inclusion requires two further and logically independent directions. Lifted Weyl vectors provide the lower-inclusion estimate and imply $\ell_N \rightarrow 0$, thereby excluding asymptotic loss of the declared bulk spectrum. An

independent upper-inclusion argument, such as a direct operator estimate or a suitable polynomial-domination bound, is required to establish $p_N \rightarrow 0$ and hence exclude spectral pollution. Once both quantities vanish, the combined spectral error η_N also satisfies $\eta_N \rightarrow 0$. If, for a declared isolated spectral component C , this error eventually obeys $\eta_N < \Delta_C/3$, then its finite-cover bandwidth and gap are stable in the quantitative sense

$$|W_{C,N} - W_C| \leq 2\eta_N, \quad |\Delta_{C,N} - \Delta_C| \leq 2\eta_N.$$

Thus the spectral component itself, rather than merely the normalized density of states, is then under established control. Representation completeness forms a separate condition in this logic. The condition $C_{\text{rep},N} = 1$ guarantees complete coverage of the finite-quotient state space, but it does not replace the lower- and upper-spectral-inclusion estimates required for bulk convergence. Likewise, derivative observables require stronger convergence than spectral values alone. C^1 convergence is required to promote the finite-cover band structure to a convergent generalized velocity, whereas C^2 convergence is required for Hessians and effective masses. Finally, agreement along a single admissible sequence of covers establishes convergence only for that tower. Cover independence requires the same limiting quantities to be obtained on at least two inequivalent admissible towers. Several implications are therefore explicitly prohibited. Growth of the finite dimension D_N does not imply growth of the injectivity radius $r_{\text{inj},N}$. Likewise, $r_{\text{inj},N} \rightarrow \infty$ does not imply $p_N \rightarrow 0$, because local exactness does not by itself exclude global spectral pollution. Weak convergence $\nu_N \Rightarrow \nu_{\text{bulk}}$ does not imply convergence of spectral edges or gaps, and the no-loss condition $\ell_N \rightarrow 0$ does not imply the no-pollution condition $p_N \rightarrow 0$. Complete finite-quotient representation coverage, $C_{\text{rep},N} = 1$, likewise does not imply convergence to the reduced or tempered bulk sector. Convergence along a single cover tower does not establish tower independence. Finally, C^0 spectral convergence controls only spectral values and cannot be promoted, without additional regularity estimates, to convergence of generalized velocities, Hessians, effective masses, or coherence quantities. A finite-cover flat-band candidate is promoted to a genuine bulk result only when all required conditions are satisfied for the same specified target family. The mesoscopic scale L_N must diverge while remaining asymptotically smaller than the injectivity radius,

$$L_N \rightarrow \infty, \quad \frac{L_N}{r_{\text{inj},N}} \rightarrow 0.$$

On the same declared sector, both spectral loss and spectral pollution must vanish,

$$\ell_N \rightarrow 0, \quad p_N \rightarrow 0,$$

and the representation-completeness condition $C_{\text{rep},C,N} = 1$ must hold. If first- or second-derivative

observables are invoked, the corresponding truncation or tail errors must also vanish to the required order,

$$\varepsilon_{\text{tail},N}^{(0,1,2)} \rightarrow 0,$$

with only the derivative orders actually used being required. In addition, the finite-cover bandwidth and isolation gap must converge,

$$W_{C,N} \rightarrow W_C, \quad \Delta_{C,N} \rightarrow \Delta_C > 0,$$

and the same limiting quantities must be reproduced on inequivalent admissible cover towers. Only after all of these conditions are satisfied do the finite-cover bandwidth, spectral gap, density of states, generalized velocity, effective mass, Hodge-cancellation diagnostic, and physical flat-band angle acquire a cover-independent bulk interpretation. Before that point, each quantity must retain the scope of the strongest verification level that has actually been established.

XI. ROBUSTNESS TO CIRCUIT IMPERFECTIONS

We now determine whether the physical flat-band regime survives the imperfections of a finite superconducting-resonator implementation. We distinguish seven perturbation channels:

$$\begin{aligned} \text{resonator-frequency disorder : } & \delta\Omega_i, \\ \text{coupling disorder : } & \delta J_{ij}, \\ \text{layer detuning : } & \delta L, \\ \text{photon loss : } & \mathbf{\Gamma}, \\ \text{hopping cutoff : } & L_C, \\ \text{multimode corrections : } & \Sigma_{\text{mm}}(z), \\ \text{magic-angle displacement : } & \delta\theta_{\text{flat}}. \end{aligned} \quad (2493)$$

All single-particle matrices in this section are written in angular-frequency units. The corresponding physical energy operator is obtained by multiplying by $\hbar$. In the co-rotating layer frame, the nominal bilayer matrix is

$$\mathbf{H}_0(\theta, K) = \begin{pmatrix} \mathbf{h}_0(K) & \mathbf{T}(\theta, K) \\ \mathbf{T}^\dagger(\theta, K) & \mathbf{h}_0(K) \end{pmatrix}. \quad (2494)$$

The ideal layer blocks are identical, and every rigid-twist derivative is carried by the interlayer block:

$$\partial_\theta \mathbf{H}_0 = \begin{pmatrix} 0 & \partial_\theta \mathbf{T} \\ \partial_\theta \mathbf{T}^\dagger & 0 \end{pmatrix}. \quad (2495)$$

The complete effective single-excitation operator is written as

$$\begin{aligned} \mathbf{H}_{\text{eff}}(z) = & \mathbf{H}_0 + \mathbf{V}_\Omega + \mathbf{V}_J + \mathbf{V}_L + \mathbf{V}_{\text{cut}} \\ & + \Sigma_{\text{mm}}(z) - \frac{i}{2} \mathbf{\Gamma}. \end{aligned} \quad (2496)$$

The first five corrections are Hermitian. The multi-mode self-energy is Hermitian for real z lying outside the eliminated lossless-mode spectrum, but becomes complex when lossy parasitic modes are integrated out. The final term contains irreversible photon loss. For a static Hermitian imperfection class X , define the uniform C^j operator budgets

$$\varepsilon_X^{(j)} = \sup_{\theta \in \Theta_{\text{scan}}} \left\| \partial_\theta^j \mathbf{V}_X(\theta) \right\|, \quad j = 0, 1, 2. \quad (2497)$$

The complete Hermitian budgets are

$$\varepsilon_{\text{H}}^{(j)} = \sum_X \varepsilon_X^{(j)}. \quad (2498)$$

The C^0 budget controls energies, spectral edges, bandwidths, gaps, and Riesz projectors. The C^1 budget is required for generalized velocities and stationary-point displacement. The C^2 budget is required for Hessians, effective masses, and transverse magic-ridge curvature. Let I_C be the nominal target spectral island and let

$$\Delta_C^{(0)} = \text{dist}[I_C, \text{Spec } \mathbf{H}_0 \setminus I_C] > 0. \quad (2499)$$

For a Hermitian perturbation $\mathbf{V}$ with

$$\|\mathbf{V}\| \leq \varepsilon_0, \quad (2500)$$

self-adjoint spectral stability gives

$$\delta_{\text{H}}[\text{Spec}(\mathbf{H}_0 + \mathbf{V}), \text{Spec } \mathbf{H}_0] \leq \varepsilon_0. \quad (2501)$$

For the continuously tracked target family,

$$\left| \widetilde{W}_C - W_C^{(0)} \right| \leq 2\varepsilon_0, \quad (2502)$$

and

$$\widetilde{\Delta}_C \geq \Delta_C^{(0)} - 2\varepsilon_0. \quad (2503)$$

Let $\mathcal{C}$ be a nominal Riesz contour of length $\ell_{\mathcal{C}}$, with

$$\delta_{\mathcal{C}} = \text{dist}[\mathcal{C}, \text{Spec } \mathbf{H}_0] > 0. \quad (2504)$$

If $\varepsilon_0 < \delta_{\mathcal{C}}$, the perturbed projector satisfies

$$\left\| \widetilde{P}_C - P_C \right\| \leq \frac{\ell_{\mathcal{C}}}{2\pi} \frac{\varepsilon_0}{\delta_{\mathcal{C}}(\delta_{\mathcal{C}} - \varepsilon_0)}. \quad (2505)$$

If the right-hand side is smaller than one, the target rank is preserved. These deterministic bounds are worst-case statements. Typical random imperfections can be substantially smaller because an extended eigenmode averages over many fabrication errors. We calculate both the worst-case operator radius and the statistical mode-dependent response.

A. Resonator-frequency disorder

Let the nominal angular frequency of resonator i be $\Omega_i^{(0)}$. Fabrication disorder gives

$$\Omega_i = \Omega_i^{(0)} + \delta\Omega_i. \quad (2506)$$

The single-particle perturbation is

$$\mathbf{V}_\Omega = \sum_{i=1}^D \delta\Omega_i |i\rangle\langle i|. \quad (2507)$$

Its exact operator norm is

$$\|\mathbf{V}_\Omega\| = \max_{1 \leq i \leq D} |\delta\Omega_i|. \quad (2508)$$

Separate the common-mode frequency shift from the spatially varying component:

$$\overline{\delta\Omega} = \frac{1}{D} \sum_{i=1}^D \delta\Omega_i, \quad (2509)$$

and

$$\delta\Omega_i^c = \delta\Omega_i - \overline{\delta\Omega}. \quad (2510)$$

Then

$$\mathbf{V}_\Omega = \overline{\delta\Omega} \mathbf{I} + \mathbf{D}_\Omega, \quad \mathbf{D}_\Omega = \sum_i \delta\Omega_i^c |i\rangle\langle i|. \quad (2511)$$

The scalar term translates the complete spectrum:

$$E_n \mapsto E_n + \overline{\delta\Omega}. \quad (2512)$$

It leaves invariant W_C , Δ_C , P_C , $v_{\max,C}$, $\Lambda_{\max,C}$, $\mathcal{M}_C$. Only the centered disorder $\mathbf{D}_\Omega$ degrades flatness and isolation. Define the disorder oscillation

$$\text{osc}(\delta\Omega) = \max_i \delta\Omega_i - \min_i \delta\Omega_i. \quad (2513)$$

Since the spectrum of a compression lies inside the convex hull of the full operator spectrum, the first-order splitting of any exactly degenerate cluster satisfies

$$\text{diam Spec}(P_C \mathbf{V}_\Omega P_C) \leq \text{osc}(\delta\Omega). \quad (2514)$$

Let

$$\mathbf{H}_0 |u_n\rangle = E_n^{(0)} |u_n\rangle \quad (2515)$$

be a simple normalized mode. The first-order frequency shift is

$$\delta E_n^{(1)} = \langle u_n | \mathbf{V}_\Omega | u_n \rangle = \sum_{i=1}^D p_{n,i} \delta\Omega_i, \quad (2516)$$

where

$$p_{n,i} = |\langle i | u_n \rangle|^2, \quad \sum_i p_{n,i} = 1. \quad (2517)$$

Thus the frequency shift is a spatially weighted average of the fabrication errors. The second-order correction is

$$\delta E_n^{(2)} = \sum_{m \neq n} \frac{\left| \sum_i u_{m,i}^* \delta\Omega_i u_{n,i} \right|^2}{E_n^{(0)} - E_m^{(0)}}. \quad (2518)$$

If

$$g_n = \min_{m \neq n} |E_n^{(0)} - E_m^{(0)}|, \quad (2519)$$

then

$$|\delta E_n^{(2)}| \leq \frac{\|\mathbf{D}_\Omega\|^2}{g_n}. \quad (2520)$$

This perturbative expression is used only when

$$\|\mathbf{D}_\Omega\| \ll g_n. \quad (2521)$$

Near a degeneracy, we diagonalize the complete compressed disorder matrix $P_C \mathbf{D}_\Omega P_C$ instead. Let $\mathbb{E}[\delta\Omega_i] = 0$, and define the covariance matrix

$$\mathbf{C}_{\Omega,ij} = \mathbb{E}[\delta\Omega_i \delta\Omega_j]. \quad (2522)$$

Equation (2516) gives

$$\text{Var}(\delta E_n^{(1)}) = \mathbf{p}_n^\top \mathbf{C}_\Omega \mathbf{p}_n. \quad (2523)$$

For two modes,

$$\text{Var}(\delta E_n^{(1)} - \delta E_m^{(1)}) = (\mathbf{p}_n - \mathbf{p}_m)^\top \mathbf{C}_\Omega (\mathbf{p}_n - \mathbf{p}_m). \quad (2524)$$

For independent identically distributed onsite disorder,

$$\mathbf{C}_\Omega = \sigma_\Omega^2 \mathbf{I}, \quad (2525)$$

and hence

$$\text{Var}(\delta E_n^{(1)}) = \sigma_\Omega^2 \text{IPR}_n, \quad (2526)$$

where

$$\text{IPR}_n = \sum_{i=1}^D p_{n,i}^2. \quad (2527)$$

A mode uniformly extended over N_{eff} resonators has

$$\text{IPR}_n \simeq \frac{1}{N_{\text{eff}}}, \quad (2528)$$

and therefore

$$\sqrt{\text{Var}(\delta E_n^{(1)})} \simeq \frac{\sigma_\Omega}{\sqrt{N_{\text{eff}}}}. \quad (2529)$$

This statistical self-averaging does not replace the worst-case operator bound. Rare resonators with large errors

can still close a global gap. If each $\delta\Omega_i$ is sub-Gaussian with scale σ_Ω , then

$$\mathbb{P} [|\delta\Omega_i| > x] \leq 2 \exp \left[-\frac{x^2}{2\sigma_\Omega^2} \right]. \quad (2530)$$

The union bound gives, with probability at least $1 - \alpha$,

$$\|\mathbf{V}_\Omega\| \leq \sigma_\Omega \sqrt{2 \ln \left(\frac{2D}{\alpha} \right)}. \quad (2531)$$

Consequently, a sufficient high-probability gap condition is

$$2\sigma_\Omega \sqrt{2 \ln \left(\frac{2D}{\alpha} \right)} < \Delta_C^{(0)}. \quad (2532)$$

For corresponding resonators i in the two layers, define

$$\delta\Omega_i^+ = \frac{\delta\Omega_{1i} + \delta\Omega_{2i}}{2}, \quad \delta\Omega_i^- = \frac{\delta\Omega_{1i} - \delta\Omega_{2i}}{2}. \quad (2533)$$

The disorder operator becomes

$$\begin{aligned} \mathbf{V}_\Omega &= I_L \otimes \sum_i \delta\Omega_i^+ |i\rangle\langle i| \\ &\quad + Z_L \otimes \sum_i \delta\Omega_i^- |i\rangle\langle i|. \end{aligned} \quad (2534)$$

The layer-symmetric part commutes with layer exchange. The layer-antisymmetric part anticommutes with it and acts as a spatially inhomogeneous layer detuning. We therefore report two independent disorder scales:

$$\sigma_{\Omega,+}^2 = \frac{1}{N} \sum_i (\delta\Omega_i^+)^2, \quad \sigma_{\Omega,-}^2 = \frac{1}{N} \sum_i (\delta\Omega_i^-)^2. \quad (2535)$$

The first primarily broadens the target manifold. The second additionally breaks layer-exchange symmetries and reduces layer coherence. Define

$$\varepsilon_\Omega = \|\mathbf{D}_\Omega\|. \quad (2536)$$

The target island remains spectrally isolated when

$$\varepsilon_\Omega < \frac{\Delta_C^{(0)}}{2}. \quad (2537)$$

The normalized gap and bandwidth sensitivities are

$$r_{\Omega,\Delta} = \frac{2\varepsilon_\Omega}{\Delta_C^{(0)}}, \quad r_{\Omega,W} = \frac{2\varepsilon_\Omega}{\max \{W_C^{(0)}, W_{\text{ref}}\}}. \quad (2538)$$

The first ratio determines whether isolation can be established. The second determines whether a finite nonzero bandwidth remains experimentally distinguishable from disorder broadening.

B. Coupling disorder

Let $J_{ij}^{(0)}$ be a nominal reciprocal coupling. Fabrication imperfections give

$$J_{ij} = J_{ij}^{(0)} + \delta J_{ij}, \quad \delta J_{ji} = \delta J_{ij}^*. \quad (2539)$$

The corresponding perturbation is

$$\mathbf{V}_J = \sum_{i < j} [\delta J_{ij} |i\rangle\langle j| + \delta J_{ij}^* |j\rangle\langle i|]. \quad (2540)$$

The Hermitian Schur bound gives

$$\|\mathbf{V}_J\| \leq \max_i \sum_{j \neq i} |\delta J_{ij}|. \quad (2541)$$

In layer-block form,

$$\mathbf{V}_J = \begin{pmatrix} \delta\mathbf{h}_1 & \delta\mathbf{T} \\ \delta\mathbf{T}^\dagger & \delta\mathbf{h}_2 \end{pmatrix}. \quad (2542)$$

Separating diagonal and off-diagonal layer blocks gives

$$\|\mathbf{V}_J\| \leq \max \{ \|\delta\mathbf{h}_1\|, \|\delta\mathbf{h}_2\| \} + \|\delta\mathbf{T}\|. \quad (2543)$$

Parameterize a programmed complex coupling by

$$\tilde{J}_{ij} = J_{ij}^{(0)} (1 + \eta_{ij}) e^{i\phi_{ij}}. \quad (2544)$$

The exact deviation is

$$\delta J_{ij} = J_{ij}^{(0)} [(1 + \eta_{ij}) e^{i\phi_{ij}} - 1]. \quad (2545)$$

Using

$$(1 + \eta) e^{i\phi} - 1 = \eta e^{i\phi} + (e^{i\phi} - 1), \quad (2546)$$

we obtain

$$|\delta J_{ij}| \leq |J_{ij}^{(0)}| \left[|\eta_{ij}| + 2 \left| \sin \left(\frac{\phi_{ij}}{2} \right) \right| \right]. \quad (2547)$$

If

$$|\eta_{ij}| \leq \eta_{\max}, \quad |\phi_{ij}| \leq \phi_{\max}, \quad (2548)$$

then

$$\|\mathbf{V}_J\| \leq \max_i \sum_{j \neq i} |J_{ij}^{(0)}| \left[\eta_{\max} + 2 \sin \left(\frac{\phi_{\max}}{2} \right) \right]. \quad (2549)$$

For small errors,

$$\delta J_{ij} = J_{ij}^{(0)} (\eta_{ij} + i\phi_{ij}) + O(\eta_{ij}^2, \eta_{ij}\phi_{ij}, \phi_{ij}^2). \quad (2550)$$

Amplitude errors perturb the magnitude of a hopping path. Phase errors perturb interference among paths and can therefore be more damaging near a cancellation-induced flat band. For a simple nominal mode,

$$\delta E_{n,J}^{(1)} = \langle u_n | \mathbf{V}_J | u_n \rangle = 2 \operatorname{Re} \sum_{i < j} \delta J_{ij} u_{n,i}^* u_{n,j}.$$

For independent real bond errors with

$$\mathbb{E}[\delta J_{ij}] = 0, \quad \text{Var}(\delta J_{ij}) = \sigma_{ij}^2, \quad (2551)$$

the variance is

$$\text{Var}(\delta E_{n,J}^{(1)}) = 4 \sum_{i < j} \sigma_{ij}^2 [\text{Re}(u_{n,i}^* u_{n,j})]^2. \quad (2552)$$

This expression distinguishes modes that occupy the same sites but carry different relative phases on the bonds. A common rescaling of all couplings has a different physical meaning from a distortion of the coupling texture. Let the nominal coupling matrix be $\mathbf{J}_0$ and its perturbation be $\delta \mathbf{J}$. Define the best-fit scalar component

$$a_J = \frac{\text{Re Tr}(\mathbf{J}_0^\dagger \delta \mathbf{J})}{\|\mathbf{J}_0\|_F^2}. \quad (2553)$$

The perturbation decomposes as

$$\delta \mathbf{J} = a_J \mathbf{J}_0 + \delta \mathbf{J}_{\text{tex}}, \quad (2554)$$

with

$$\text{Re Tr}(\mathbf{J}_0^\dagger \delta \mathbf{J}_{\text{tex}}) = 0. \quad (2555)$$

The scalar part moves the system along an already existing coupling coordinate. The texture part changes the relative strengths and phases of different hopping paths. For uniform intralayer and interlayer scale errors,

$$t \mapsto t(1 + \eta_t), \quad w \mapsto w(1 + \eta_w), \quad (2556)$$

the effective ratio changes according to

$$\delta \ln \omega = \delta \ln \left(\frac{w}{t} \right) = \eta_w - \eta_t + O(\eta^2). \quad (2557)$$

If $\eta_w = \eta_t$, the complete Hamiltonian is rescaled to first order. Dimensionless bandwidth-to-gap ratios and the normalized magic score are unchanged, although all absolute frequencies and linewidth ratios to a fixed external loss rate change. The irreducible texture-disorder ratio is

$$r_{\text{tex}} = \frac{\|\delta \mathbf{J}_{\text{tex}}\|_F}{\|\mathbf{J}_0\|_F}. \quad (2558)$$

Let $\mathcal{G}$ be an exact finite symmetry group represented by U_g . The symmetry-preserving projection of a perturbation is

$$\Pi_{\mathcal{G}}(\mathbf{V}_J) = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} U_g \mathbf{V}_J U_g^\dagger. \quad (2559)$$

Define

$$\mathbf{V}_J^\parallel = \Pi_{\mathcal{G}}(\mathbf{V}_J), \quad (2560)$$

and

$$\mathbf{V}_J^\perp = \mathbf{V}_J - \mathbf{V}_J^\parallel. \quad (2561)$$

Then

$$[\mathbf{V}_J^\parallel, U_g] = 0 \quad (2562)$$

for every g , while $\mathbf{V}_J^\perp$ carries all symmetry-breaking matrix elements. Inside a multidimensional irreducible symmetry factor, Schur's lemma gives

$$\mathbf{V}_J^\parallel = I_{d_\rho} \otimes \delta \mathbf{v}_\rho. \quad (2563)$$

It cannot split the symmetry-enforced d_ρ -fold multiplicity. The symmetry-breaking part can split it. For a degenerate doublet projector P_D , write

$$P_D \mathbf{V}_J^\perp P_D = v_0 I + v_x \sigma_x + v_y \sigma_y + v_z \sigma_z. \quad (2564)$$

The first-order splitting is

$$\Delta E_D = 2 \sqrt{v_x^2 + v_y^2 + v_z^2}. \quad (2565)$$

For the centered microscopic class, the nominal Hodge Hessian is

$$\mathcal{K} = 2t C_S - w B_\infty. \quad (2566)$$

Coupling disorder changes both scalar amplitudes and tensor textures:

$$\begin{aligned} \delta \mathcal{K} &= 2\delta t C_S + 2t \delta C_S \\ &\quad - \delta w B_\infty - w \delta B_\infty + O(\delta^2). \end{aligned} \quad (2567)$$

The trace part moves the scalar cancellation point. The traceless part produces residual anisotropy:

$$\delta \mathcal{K}_{\text{aniso}} = \delta \mathcal{K} - \frac{\text{Tr}(G^{-1} \delta \mathcal{K})}{m} G. \quad (2568)$$

Let the nominal normalized principal response be

$$\lambda_H(\theta_H) = 0, \quad (2569)$$

with slope

$$s_H = |\partial_\theta \lambda_H(\theta_H)| > 0. \quad (2570)$$

The leading scalar-root displacement is

$$\delta \theta_H = -\frac{\delta \bar{\lambda}_H(\theta_H)}{\partial_\theta \lambda_H(\theta_H)} + O(\|\delta \mathcal{K}\|^2), \quad (2571)$$

where

$$\delta \bar{\lambda}_H = \frac{1}{m} \text{Tr}(G^{-1} \delta \mathcal{K}). \quad (2572)$$

If

$$\sup_\theta \|\delta \widehat{\mathcal{K}}\| \leq \eta_2, \quad (2573)$$

and

$$\sup_\theta \|\partial_\theta \delta \widehat{\mathcal{K}}\| \leq \eta_{2,1} < s_H, \quad (2574)$$

then the continued trace root obeys

$$|\tilde{\theta}_H - \theta_H| \leq \frac{\eta_2}{s_H - \eta_{2,1}}. \quad (2575)$$

At that point, every principal Hessian value is bounded by $|\tilde{\lambda}_j| \leq 2\eta_2$. Exact simultaneous cancellation survives only when the disorder preserves the scalar-commutant structure. Otherwise, the best possible statement is approximate multidirectional flattening.

C. Layer detuning

Let the mean frequencies of the two layers differ by δ_L . After removing their common average, the perturbation is

$$\mathbf{V}_L = \frac{\delta_L}{2} Z_L, \quad Z_L = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}. \quad (2576)$$

Its exact norm is

$$\|\mathbf{V}_L\| = \frac{|\delta_L|}{2}. \quad (2577)$$

The generic gap bound becomes

$$\tilde{\Delta}_C \geq \Delta_C^{(0)} - |\delta_L|. \quad (2578)$$

The layer-exchange operator is

$$P_L = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}. \quad (2579)$$

It anticommutes with the polarization:

$$\{P_L, Z_L\} = 0. \quad (2580)$$

Therefore

$$P_L \mathbf{V}_L P_L^{-1} = -\mathbf{V}_L. \quad (2581)$$

A nonzero layer detuning breaks every exact symmetry whose spatial action must be completed by layer exchange. For a centered dihedral bilayer, the symmetry reduction is $D_m^{\text{BL}} \rightarrow C_m$. At a parameter point where exact layer exchange reduces the nominal Hamiltonian, define

$$U_P = \frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ I & -I \end{pmatrix}. \quad (2582)$$

Then

$$U_P^\dagger P_L U_P = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad (2583)$$

whereas

$$U_P^\dagger Z_L U_P = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}. \quad (2584)$$

If the interlayer matrix is Hermitian at that point, the complete Hamiltonian becomes

$$U_P^\dagger (\mathbf{H}_0 + \mathbf{V}_L) U_P = \begin{pmatrix} \mathbf{h}_0 + \mathbf{T} & \delta_L I/2 \\ \delta_L I/2 & \mathbf{h}_0 - \mathbf{T} \end{pmatrix}. \quad (2585)$$

Layer detuning therefore hybridizes the nominal even and odd sectors. Suppose a normalized orbital channel satisfies

$$\mathbf{h}_0 |u_n\rangle = \varepsilon_n |u_n\rangle, \quad (2586)$$

and

$$\mathbf{T} |u_n\rangle = \tau_n |u_n\rangle. \quad (2587)$$

The layer-space Hamiltonian is

$$\mathbf{H}_n = \varepsilon_n I_2 + \begin{pmatrix} \delta_L/2 & \tau_n \\ \tau_n^* & -\delta_L/2 \end{pmatrix}. \quad (2588)$$

Its exact eigenfrequencies are

$$E_{n,\pm} = \varepsilon_n \pm \sqrt{|\tau_n|^2 + \frac{\delta_L^2}{4}}. \quad (2589)$$

The layer polarization is

$$\langle Z_L \rangle_{n,\pm} = \pm \frac{\delta_L}{\sqrt{\delta_L^2 + 4|\tau_n|^2}}. \quad (2590)$$

The squared transverse layer coherence is

$$\mathcal{C}_{L,n} = \frac{4|\tau_n|^2}{\delta_L^2 + 4|\tau_n|^2}. \quad (2591)$$

It obeys

$$\mathcal{C}_{L,n} + \langle Z_L \rangle_{n,\pm}^2 = 1. \quad (2592)$$

For weak detuning,

$$E_{n,\pm} = \varepsilon_n \pm |\tau_n| \pm \frac{\delta_L^2}{8|\tau_n|} + O\left(\frac{\delta_L^4}{|\tau_n|^3}\right).$$

There is no linear energy shift at the exactly layer-symmetric point. The leading energy correction is quadratic, while the layer polarization is linear:

$$\langle Z_L \rangle_{n,\pm} = \pm \frac{\delta_L}{2|\tau_n|} + O\left(\frac{\delta_L^3}{|\tau_n|^3}\right). \quad (2593)$$

In the large-detuning regime, $|\delta_L| \gg \|\mathbf{T}\|$, the layers become perturbatively decoupled. Eliminating the lower-frequency layer gives

$$\mathbf{H}_{\text{eff},1} = \mathbf{h}_0 + \frac{\delta_L}{2} I + \frac{\mathbf{T}\mathbf{T}^\dagger}{\delta_L} + O\left(\frac{\|\mathbf{T}\|^3}{\delta_L^2}\right). \quad (2594)$$

The interlayer reconstruction is then reduced from a first-order resonant hybridization to a second-order self-energy. For a required minimum layer-coherence fraction

$0 < C_{\min} < 1$, Eq. (2591) gives the exact admissible detuning interval

$$|\delta_L| \leq 2|\tau_n| \sqrt{\frac{1 - C_{\min}}{C_{\min}}}. \quad (2595)$$

This is a channel-resolved physical threshold rather than a purely operator-norm condition. Layer exchange maps

$$\mathbf{H}(\theta, \delta_L) \mapsto \mathbf{H}(\theta, -\delta_L) \quad (2596)$$

whenever all other layer parameters are exchanged simultaneously. Every layer-exchange-invariant scalar observable is therefore even in δ_L :

$$\mathcal{O}(\theta, \delta_L) = \mathcal{O}(\theta, -\delta_L). \quad (2597)$$

Let

$$L(\theta, \delta_L) = \ln \mathcal{M}(\theta, \delta_L), \quad (2598)$$

and

$$F(\theta, \delta_L) = \partial_\theta L. \quad (2599)$$

The magic angle satisfies

$$F[\theta_*(\delta_L), \delta_L] = 0. \quad (2600)$$

At the symmetric point,

$$\left. \frac{d\theta_*}{d\delta_L} \right|_{\delta_L=0} = 0. \quad (2601)$$

Define the nominal transverse score curvature

$$\kappa_\perp = -F_{\theta}(\theta_*, 0) > 0. \quad (2602)$$

Differentiating the root equation twice gives

$$\left. \frac{d^2\theta_*}{d\delta_L^2} \right|_0 = \left. \frac{F_{\delta_L\delta_L}}{\kappa_\perp} \right|_{(\theta_*, 0)}. \quad (2603)$$

Consequently,

$$\theta_*(\delta_L) = \theta_*(0) + \frac{F_{\delta_L\delta_L}}{2\kappa_\perp} \delta_L^2 + O(\delta_L^4). \quad (2604)$$

A measured linear displacement under a nominally symmetric detuning sweep signals an additional layer-asymmetric imperfection or a branch-tracking error.

D. Photon loss

Let $\mathbf{a} = (a_1, \dots, a_D)^\top$ collect the resonator annihilation operators. The coherent Hamiltonian is

$$\hat{H} = \hbar \mathbf{a}^\dagger \mathbf{H}_H \mathbf{a}, \quad (2605)$$

where $\mathbf{H}_H = \mathbf{H}_H^\dagger$. For linear loss channels

$$L_\mu = \ell_\mu^\dagger \mathbf{a}, \quad (2606)$$

the master equation is

$$\dot{\rho} = -\frac{i}{\hbar} [\hat{H}, \rho] + \sum_\mu \left[L_\mu \rho L_\mu^\dagger - \frac{1}{2} \{ L_\mu^\dagger L_\mu, \rho \} \right]. \quad (2607)$$

Define the positive-semidefinite damping matrix

$$\mathbf{\Gamma} = \sum_\mu \ell_\mu \ell_\mu^\dagger \succeq 0. \quad (2608)$$

The first moments obey

$$\frac{d}{dt} \langle \mathbf{a} \rangle = \left(-i\mathbf{H}_H - \frac{1}{2}\mathbf{\Gamma} \right) \langle \mathbf{a} \rangle + \mathbf{f}_{\text{in}}(t). \quad (2609)$$

The effective non-Hermitian matrix is therefore

$$\mathbf{H}_{\text{nh}} = \mathbf{H}_H - \frac{i}{2}\mathbf{\Gamma}. \quad (2610)$$

The retarded Green operator is

$$\mathbf{G}^R(\Omega) = \left[\Omega I - \mathbf{H}_H + \frac{i}{2}\mathbf{\Gamma} \right]^{-1}. \quad (2611)$$

For drive and readout vectors $|f_{\text{in}}\rangle$ and $|f_{\text{out}}\rangle$, the resonant transmission amplitude is proportional to

$$\mathcal{S}_{\text{out,in}}(\Omega) \propto \langle f_{\text{out}} | \mathbf{G}^R(\Omega) | f_{\text{in}} \rangle. \quad (2612)$$

The drive-resolved spectral function is

$$A_f(\Omega) = -\frac{1}{\pi} \text{Im} \langle f | \mathbf{G}^R(\Omega) | f \rangle. \quad (2613)$$

For spatially uniform loss, $\mathbf{\Gamma} = \gamma_0 I$, we have

$$\mathbf{H}_{\text{nh}} = \mathbf{H}_H - \frac{i\gamma_0}{2} I. \quad (2614)$$

Every complex eigenfrequency is

$$z_n = E_n - \frac{i\gamma_0}{2}. \quad (2615)$$

Uniform loss leaves the real spectrum, eigenvectors, bandwidth, isolation gap, Hodge Hessian, and intrinsic physical magic angle unchanged. It only gives every mode the same linewidth. For an additional numerical Lorentzian broadening η ,

$$\eta_{\text{eff}} = \eta + \frac{\gamma_0}{2}. \quad (2616)$$

The measured DOS is therefore a uniformly broadened version of the Hermitian DOS. If $[\mathbf{H}_H, \mathbf{\Gamma}] = 0$, the two matrices possess a common orthonormal eigenbasis:

$$\mathbf{H}_H |u_n\rangle = E_n |u_n\rangle, \quad \mathbf{\Gamma} |u_n\rangle = \gamma_n |u_n\rangle. \quad (2617)$$

Then

$$z_n = E_n - \frac{i\gamma_n}{2}. \quad (2618)$$

The real band remains unchanged, but different modes acquire different linewidths. When $[\mathbf{H}_H, \mathbf{\Gamma}] \neq 0$, loss also reconstructs the real spectrum at second order. For a simple Hermitian mode, define

$$\Gamma_{mn} = \langle u_m | \mathbf{\Gamma} | u_n \rangle. \quad (2619)$$

Non-Hermitian perturbation theory gives

$$z_n = E_n - \frac{i}{2}\Gamma_{nn} - \frac{1}{4} \sum_{m \neq n} \frac{|\Gamma_{mn}|^2}{E_n - E_m} + O\left(\frac{\|\mathbf{\Gamma}\|^3}{g_n^2}\right).$$

The first-order term is purely dissipative. The leading real-frequency shift is quadratic:

$$|\delta E_{n,\Gamma}^{(2)}| \leq \frac{\|\mathbf{\Gamma}\|^2}{4g_n}. \quad (2620)$$

Let

$$\gamma_{\min} = \lambda_{\min}(\mathbf{\Gamma}), \quad \gamma_{\max} = \lambda_{\max}(\mathbf{\Gamma}). \quad (2621)$$

For every right eigenvector of $\mathbf{H}_{nh}$,

$$-\frac{\gamma_{\max}}{2} \leq \text{Im } z_n \leq -\frac{\gamma_{\min}}{2}. \quad (2622)$$

Moreover,

$$\text{dist}[z_n, \text{Spec } \mathbf{H}_H] \leq \frac{1}{2} \|\mathbf{\Gamma}\|. \quad (2623)$$

Let

$$\mathbf{H}_{nh} |u_n^R\rangle = z_n |u_n^R\rangle, \quad (2624)$$

and

$$\langle u_n^L | \mathbf{H}_{nh} = z_n \langle u_n^L|. \quad (2625)$$

Choose $\langle u_n^L | u_n^R \rangle = 1$. For any control parameter λ ,

$$\frac{dz_n}{d\lambda} = \left\langle u_n^L \left| \frac{d\mathbf{H}_{nh}}{d\lambda} \right| u_n^R \right\rangle. \quad (2626)$$

Define the eigenvalue condition factor

$$\chi_n = \|u_n^L\| \|u_n^R\|. \quad (2627)$$

Then

$$\left| \frac{dz_n}{d\lambda} \right| \leq \chi_n \left\| \frac{d\mathbf{H}_{nh}}{d\lambda} \right\|. \quad (2628)$$

The Petermann factor is $K_n^P = \chi_n^2$. A large value signals nonnormal amplification and proximity to an exceptional-point regime. We do not use ordinary Hermitian eigenvector perturbation bounds when χ_n is unresolved or diverging. Consider

$$\mathbf{H}_2^{nh} = \begin{pmatrix} \varepsilon_1 - i\gamma_1/2 & V \\ V^* & \varepsilon_2 - i\gamma_2/2 \end{pmatrix}. \quad (2629)$$

Define

$$\bar{\varepsilon} = \frac{\varepsilon_1 + \varepsilon_2}{2}, \quad \Delta\varepsilon = \varepsilon_1 - \varepsilon_2, \quad (2630)$$

and

$$\bar{\gamma} = \frac{\gamma_1 + \gamma_2}{2}, \quad \Delta\gamma = \gamma_1 - \gamma_2. \quad (2631)$$

The exact complex eigenfrequencies are

$$z_{\pm} = \bar{\varepsilon} - \frac{i\bar{\gamma}}{2} \pm \sqrt{|V|^2 + \frac{1}{4} \left(\Delta\varepsilon - \frac{i\Delta\gamma}{2} \right)^2}. \quad (2632)$$

At real-frequency resonance, $\Delta\varepsilon = 0$, an exceptional point occurs when

$$|V| = \frac{|\Delta\gamma|}{4}. \quad (2633)$$

The coherent avoided crossing is spectrally resolved only when

$$|V| > \frac{|\Delta\gamma|}{4} \quad (2634)$$

with an additional experimental margin set by the mean linewidth. For every complex mode, define its full linewidth

$$\gamma_n^{\text{mode}} = -2 \text{Im } z_n. \quad (2635)$$

For a target mode n and complementary mode m , define the linewidth-corrected separation

$$\Delta_{nm}^{\text{res}} = |\text{Re } z_n - \text{Re } z_m| - \frac{\gamma_n^{\text{mode}} + \gamma_m^{\text{mode}}}{2}. \quad (2636)$$

The target-family resolved gap is

$$\Delta_C^{\text{res}} = \min_{n \in C, m \notin C} \Delta_{nm}^{\text{res}}. \quad (2637)$$

A directly resolvable isolated band requires $\Delta_C^{\text{res}} > 0$. Define the spectral-resolution ratio

$$\mathcal{Q}_{\Delta,C} = \min_{n \in C, m \notin C} \frac{2|\text{Re } z_n - \text{Re } z_m|}{\gamma_n^{\text{mode}} + \gamma_m^{\text{mode}}}. \quad (2638)$$

The condition $\mathcal{Q}_{\Delta,C} > 1$ is equivalent to nonoverlap of the nominal half-maximum linewidth intervals. Uniform photon loss can leave the intrinsic Hermitian magic angle exact while making it experimentally unresolved. We therefore report both $\theta_{\text{flat}}^{\text{Herm}}$ and $\theta_{\text{flat}}^{\text{res}}$, where the latter includes the resolved-gap and linewidth conditions.

E. Hopping cutoff

Let the number of group elements in the n -th physical shell satisfy

$$\#\mathcal{S}_n \leq C\Gamma e^{h_{\Gamma} n \rho}. \quad (2639)$$

Suppose the j -th parameter derivative of the hopping kernel obeys

$$\left\| \partial_\theta^j h_{xy} \right\| \leq A_j (1 + d_{xy})^{s_j} e^{-\mu_j d_{xy}}, \quad j = 0, 1, 2. \quad (2640)$$

Define the positive margins

$$\gamma_j = \mu_j - h_\Gamma > 0. \quad (2641)$$

For a cutoff L , let

$$N_L = \left\lfloor \frac{L}{\rho} \right\rfloor. \quad (2642)$$

The omitted C^j operator tail satisfies

$$\varepsilon_{\text{tail}}^{(j)}(L) \leq A_j C_\Gamma \sum_{n=N_L}^{\infty} [1 + (n+1)\rho]^{s_j} \times e^{-\gamma_j n \rho}.$$

It also admits a closed bound

$$\varepsilon_{\text{tail}}^{(j)}(L) \leq C_j e^{-\gamma_j L/2}. \quad (2643)$$

For separate intralayer and interlayer tails, $\varepsilon_{\parallel,1}^{(j)}$, $\varepsilon_{\parallel,2}^{(j)}$, $\varepsilon_{\perp}^{(j)}$, the bilayer block estimate is

$$\varepsilon_{\text{cut}}^{(j)} \leq \max \left\{ \varepsilon_{\parallel,1}^{(j)}, \varepsilon_{\parallel,2}^{(j)} \right\} + \varepsilon_{\perp}^{(j)}. \quad (2644)$$

At the C^0 level,

$$|W_C^{(L)} - W_C^{(\infty)}| \leq 2\varepsilon_{\text{cut}}^{(0)}(L), \quad (2645)$$

and

$$|\Delta_C^{(L)} - \Delta_C^{(\infty)}| \leq 2\varepsilon_{\text{cut}}^{(0)}(L). \quad (2646)$$

For a fixed contour,

$$\left\| P_C^{(L)} - P_C^{(\infty)} \right\| \leq \frac{\ell_{\mathcal{C}}}{2\pi} \frac{\varepsilon_{\text{cut}}^{(0)}(L)}{\delta_{\mathcal{C}} - \varepsilon_{\text{cut}}^{(0)}(L)}. \quad (2647)$$

Velocity convergence additionally requires $\varepsilon_{\text{cut}}^{(1)}(L)$ and Hessian or effective-mass convergence requires $\varepsilon_{\text{cut}}^{(2)}(L) \rightarrow 0$. A hard cutoff has the form

$$h_{ij}^{(L)}(\theta) = h_{ij}(\theta) \Theta[L - d_{ij}(\theta)]. \quad (2648)$$

If a bond crosses the cutoff surface, $d_{ij}(\theta_0) = L$, the retained support changes discontinuously. The derivative contains a distributional term:

$$\begin{aligned} \partial_\theta h_{ij}^{(L)} &= (\partial_\theta h_{ij}) \Theta(L - d_{ij}) \\ &\quad - h_{ij} \delta(L - d_{ij}) \partial_\theta d_{ij}. \end{aligned} \quad (2649)$$

A hard-cutoff scan can therefore produce artificial jumps in velocity, Hessian, or the stationary equation even when the physical infinite-range Hamiltonian is smooth. We

use one of two admissible procedures. First, we may freeze a common bond superset

$$\mathcal{S}_{\text{sup}} = \bigcup_{\theta \in \Theta_{\text{scan}}} \{(i, j) : d_{ij}(\theta) \leq L + s\} \quad (2650)$$

before the scan. Second, we may replace the hard cutoff by a smooth taper. Let

$$\chi(x) \in C^2(\mathbb{R}) \quad (2651)$$

satisfy

$$\chi(x) = 1 \quad (x \leq 0), \quad \chi(x) = 0 \quad (x \geq 1). \quad (2652)$$

Define

$$\chi_{L,s}(d) = \chi\left(\frac{d-L}{s}\right), \quad s > 0. \quad (2653)$$

The smoothly truncated hopping is

$$h_{ij}^{(L,s)}(\theta) = h_{ij}(\theta) \chi_{L,s}[d_{ij}(\theta)]. \quad (2654)$$

Define

$$\chi'_d = \frac{1}{s} \chi' \left(\frac{d-L}{s} \right), \quad \chi''_d = \frac{1}{s^2} \chi'' \left(\frac{d-L}{s} \right). \quad (2655)$$

The first twist derivative is

$$\partial_\theta h_{ij}^{(L,s)} = [h'_{ij}(d) \chi_{L,s}(d) + h_{ij}(d) \chi'_d] d_{ij,\theta}. \quad (2656)$$

The second derivative is

$$\begin{aligned} \partial_\theta^2 h_{ij}^{(L,s)} &= [h''_{ij} \chi_{L,s} + 2h'_{ij} \chi'_d + h_{ij} \chi''_d] d_{ij,\theta}^2 \\ &\quad + [h'_{ij} \chi_{L,s} + h_{ij} \chi'_d] d_{ij,\theta\theta}. \end{aligned} \quad (2657)$$

All taper-induced derivative terms are included explicitly in the C^1 and C^2 error budgets. Let $r_{\text{inj},N}$ be the finite-cover injectivity radius. A controlled sequence satisfies

$$L_N \rightarrow \infty, \quad \frac{L_N}{r_{\text{inj},N}} \rightarrow 0. \quad (2658)$$

For an incommensurate angular approximant θ_N , we additionally require

$$|\theta_N - \theta| \exp\left(\frac{L_N}{R}\right) \rightarrow 0. \quad (2659)$$

A fixed physical cutoff does not define the infinite-kernel magic angle. The normalized cutoff error must vanish relative to every shrinking spectral scale used in the claim:

$$\frac{\varepsilon_{\text{cut}}^{(0)}(L_N)}{E_{\text{ref},N}} \rightarrow 0. \quad (2660)$$

F. Multimode effects

A fabricated resonator possesses a sequence of electromagnetic modes. Let $a_{i\mu}$ annihilate mode μ of resonator i . In the rotating-wave single-particle approximation,

$$\hat{H}_{\text{multi}} = \hbar \sum_{i,\mu} \Omega_{i\mu} a_{i\mu}^\dagger a_{i\mu} + \hbar \sum_{i\mu,j\nu} J_{i\mu,j\nu} a_{i\mu}^\dagger a_{j\nu}.$$

The target single-mode model retains one designated mode per physical resonator. Let P project onto this fundamental-mode space and let $Q = I - P$ project onto all higher and parasitic modes. The full one-particle matrix is

$$\mathbf{H}_{\text{multi}} = \begin{pmatrix} \mathbf{H}_P & \mathbf{V} \\ \mathbf{V}^\dagger & \mathbf{H}_Q \end{pmatrix}. \quad (2661)$$

Let $z \notin \text{Spec } \mathbf{H}_Q$. Eliminating the Q -component gives the exact nonlinear eigenvalue problem

$$[\mathbf{H}_P + \boldsymbol{\Sigma}_{\text{mm}}(z)] |\psi_P\rangle = z |\psi_P\rangle, \quad (2662)$$

where

$$\boldsymbol{\Sigma}_{\text{mm}}(z) = \mathbf{V} (z - \mathbf{H}_Q)^{-1} \mathbf{V}^\dagger. \quad (2663)$$

Let I_C be the target energy interval and define the parasitic-mode detuning

$$d_{\text{mm}} = \text{dist}[I_C, \text{Spec } \mathbf{H}_Q]. \quad (2664)$$

If $d_{\text{mm}} > 0$, then

$$\|(z - \mathbf{H}_Q)^{-1}\| \leq \frac{1}{d_{\text{mm}}} \quad (2665)$$

throughout the target interval. Consequently,

$$\|\boldsymbol{\Sigma}_{\text{mm}}(z)\| \leq \frac{\|\mathbf{V}\|^2}{d_{\text{mm}}}. \quad (2666)$$

Define

$$\mathbf{R} = (z - \mathbf{H}_Q)^{-1}. \quad (2667)$$

Along a self-consistent eigenvalue branch, set

$$\mathbf{L}_\theta = \partial_\theta \mathbf{H}_Q - (\partial_\theta z) I, \quad (2668)$$

and

$$\mathbf{L}_{\theta\theta} = \partial_\theta^2 \mathbf{H}_Q - (\partial_\theta^2 z) I. \quad (2669)$$

Differentiation gives

$$\partial_\theta \mathbf{R} = \mathbf{R} \mathbf{L}_\theta \mathbf{R}. \quad (2670)$$

Write

$$\mathbf{V}_\theta = \partial_\theta \mathbf{V}, \quad \mathbf{V}_{\theta\theta} = \partial_\theta^2 \mathbf{V}. \quad (2671)$$

The complete first derivative is

$$\partial_\theta \boldsymbol{\Sigma}_{\text{mm}} = \mathbf{V}_\theta \mathbf{R} \mathbf{V}^\dagger + \mathbf{V} \mathbf{R} \mathbf{V}_\theta^\dagger + \mathbf{V} \mathbf{R} \mathbf{L}_\theta \mathbf{V}^\dagger.$$

The complete second derivative contains eleven terms:

$$\begin{aligned} \partial_\theta^2 \boldsymbol{\Sigma}_{\text{mm}} = & \mathbf{V}_{\theta\theta} \mathbf{R} \mathbf{V}^\dagger + \mathbf{V} \mathbf{R} \mathbf{V}_{\theta\theta}^\dagger + \mathbf{V}_\theta \mathbf{R} \mathbf{V}_\theta^\dagger + \mathbf{V}_\theta \mathbf{R} \mathbf{V}_\theta^\dagger \\ & + \mathbf{V}_\theta \mathbf{R} \mathbf{L}_\theta \mathbf{R} \mathbf{V}^\dagger + \mathbf{V} \mathbf{R} \mathbf{L}_\theta \mathbf{R} \mathbf{V}_\theta^\dagger + \mathbf{V}_\theta \mathbf{R} \mathbf{L}_\theta \mathbf{R} \mathbf{V}^\dagger \\ & + \mathbf{V} \mathbf{R} \mathbf{L}_\theta \mathbf{R} \mathbf{V}_\theta^\dagger + \mathbf{V} \mathbf{R} \mathbf{L}_\theta \mathbf{R} \mathbf{V}_\theta^\dagger + \mathbf{V} \mathbf{R} \mathbf{L}_{\theta\theta} \mathbf{R} \mathbf{V}^\dagger \\ & + \mathbf{V} \mathbf{R} \mathbf{L}_{\theta\theta} \mathbf{R} \mathbf{V}^\dagger. \end{aligned}$$

The repeated mixed terms in Eq. (2672) represent the two distinct product-rule placements and are retained before combining equal expressions. Equivalently, collecting identical terms gives

$$\begin{aligned} \partial_\theta^2 \boldsymbol{\Sigma}_{\text{mm}} = & \mathbf{V}_{\theta\theta} \mathbf{R} \mathbf{V}^\dagger + \mathbf{V} \mathbf{R} \mathbf{V}_{\theta\theta}^\dagger + 2 \mathbf{V}_\theta \mathbf{R} \mathbf{V}_\theta^\dagger \\ & + 2 \mathbf{V}_\theta \mathbf{R} \mathbf{L}_\theta \mathbf{R} \mathbf{V}^\dagger + 2 \mathbf{V} \mathbf{R} \mathbf{L}_\theta \mathbf{R} \mathbf{V}_\theta^\dagger \\ & + 2 \mathbf{V} \mathbf{R} \mathbf{L}_{\theta\theta} \mathbf{R} \mathbf{V}^\dagger + \mathbf{V} \mathbf{R} \mathbf{L}_{\theta\theta} \mathbf{R} \mathbf{V}^\dagger, \end{aligned}$$

when the same tangent direction is used twice. Assume

$$\|\mathbf{V}\| \leq v_0, \quad \|\mathbf{V}_\theta\| \leq v_1, \quad \|\mathbf{V}_{\theta\theta}\| \leq v_2, \quad (2672)$$

and

$$\|\mathbf{L}_\theta\| \leq \ell_1, \quad \|\mathbf{L}_{\theta\theta}\| \leq \ell_2. \quad (2673)$$

Then

$$\begin{aligned} \|\partial_\theta^2 \boldsymbol{\Sigma}_{\text{mm}}\| \leq & \frac{2v_0v_2 + 2v_1^2}{d_{\text{mm}}} + \frac{4v_0v_1\ell_1 + v_0^2\ell_2}{d_{\text{mm}}^2} \\ & + \frac{2v_0^2\ell_1^2}{d_{\text{mm}}^3}. \end{aligned} \quad (2674)$$

This is the multimode contribution to the effective-mass and Hodge-Hessian error budget. The energy derivative of the self-energy is

$$\partial_z \boldsymbol{\Sigma}_{\text{mm}} = -\mathbf{V} (z - \mathbf{H}_Q)^{-2} \mathbf{V}^\dagger. \quad (2675)$$

For a normalized effective-space eigenvector, define

$$\mathbf{Z}_{\text{mm}} = [1 - \langle \psi_P | \partial_z \boldsymbol{\Sigma}_{\text{mm}} | \psi_P \rangle]^{-1}. \quad (2676)$$

For a lossless detuned complement, $0 < \mathbf{Z}_{\text{mm}} \leq 1$. The physical parameter derivative is

$$\frac{dz}{d\lambda} = \mathbf{Z}_{\text{mm}} \langle \psi_P | \partial_\lambda [\mathbf{H}_P + \boldsymbol{\Sigma}_{\text{mm}}]_z | \psi_P \rangle. \quad (2677)$$

The residue renormalizes the effective velocity and curvature even when the static self-energy is small. The eliminated component is

$$|\psi_Q\rangle = (z - \mathbf{H}_Q)^{-1} \mathbf{V}^\dagger |\psi_P\rangle. \quad (2678)$$

Therefore

$$\|\psi_Q\| \leq \frac{v_0}{d_{\text{mm}}} \|\psi_P\|. \quad (2679)$$

For a normalized full mode, the retained fundamental-mode weight satisfies

$$\langle P \rangle_\psi \geq \frac{1}{1 + (v_0/d_{\text{mm}})^2}. \quad (2680)$$

A required minimum weight P_{min} therefore imposes

$$\frac{v_0}{d_{\text{mm}}} \leq \sqrt{\frac{1 - P_{\text{min}}}{P_{\text{min}}}}. \quad (2681)$$

Choose a reference frequency z_0 inside the target interval. The resolvent identity gives

$$\begin{aligned} \Sigma_{\text{mm}}(z) - \Sigma_{\text{mm}}(z_0) &= (z_0 - z)\mathbf{V}(z - \mathbf{H}_Q)^{-1} \\ &\quad \times (z_0 - \mathbf{H}_Q)^{-1}\mathbf{V}^\dagger. \end{aligned} \quad (2682)$$

If both resolvents have distance at least d_{mm} , then

$$\|\Sigma_{\text{mm}}(z) - \Sigma_{\text{mm}}(z_0)\| \leq \frac{v_0^2}{d_{\text{mm}}^2} |z - z_0|. \quad (2683)$$

A frequency-independent renormalized Hamiltonian is valid only when the right-hand side is small relative to the target bandwidth, isolation gap, and Hessian scale. If a parasitic mode satisfies

$$\text{dist}[I_C, \text{Spec } \mathbf{H}_Q] \lesssim \|\mathbf{V}\|, \quad (2684)$$

the perturbative elimination is invalid. The resonant mode must be moved from Q into the active projector:

$$P \mapsto P + P_{\text{res}}. \quad (2685)$$

The enlarged block is diagonalized nonperturbatively. A fixed-rank single-mode model is not continued across a parasitic-mode crossing. A physical capacitive or inductive coupler can also generate counter-rotating terms,

$$\hat{V}_{\text{cr}} = \hbar \sum_{\alpha\beta} g_{\alpha\beta}^{\text{cr}} (a_\alpha a_\beta + a_\alpha^\dagger a_\beta^\dagger). \quad (2686)$$

Let

$$\Delta_{\text{cr}} = \min_{\alpha\beta} (\Omega_\alpha + \Omega_\beta) \quad (2687)$$

over the active virtual processes, and let

$$v_{\text{cr}} = \|\hat{V}_{\text{cr}}/\hbar\|. \quad (2688)$$

For $v_{\text{cr}} \ll \Delta_{\text{cr}}$, the Bloch-Siegert-type correction obeys the second-order scale

$$\|\delta \mathbf{H}_{\text{cr}}\| = O\left(\frac{v_{\text{cr}}^2}{\Delta_{\text{cr}}}\right). \quad (2689)$$

This term is included in the multimode C^0 , C^1 , and C^2 budgets whenever the required derivatives are used.

G. Magic-angle stability

Let $\mathcal{M}(\theta, \boldsymbol{\xi})$ be the physical flatness score, where

$$\boldsymbol{\xi} = (\delta \boldsymbol{\Omega}, \delta \mathbf{J}, \delta L, \boldsymbol{\Gamma}, L_c^{-1}, \boldsymbol{\xi}_{\text{mm}}) \quad (2690)$$

collects all imperfection coordinates. Define

$$L(\theta, \boldsymbol{\xi}) = \ln \mathcal{M}(\theta, \boldsymbol{\xi}), \quad (2691)$$

and

$$F(\theta, \boldsymbol{\xi}) = \partial_\theta L(\theta, \boldsymbol{\xi}). \quad (2692)$$

The retuned magic angle satisfies

$$F[\theta_*(\boldsymbol{\xi}), \boldsymbol{\xi}] = 0. \quad (2693)$$

At the nominal point, $F(\theta_0, 0) = 0$, and the transverse stability is

$$\kappa_{\perp,0} = -\partial_\theta F(\theta_0, 0) = -\partial_\theta^2 L(\theta_0, 0) > 0. \quad (2694)$$

Differentiating Eq. (2693) with respect to an imperfection coordinate ξ_a gives

$$F_\theta \frac{\partial \theta_*}{\partial \xi_a} + F_{\xi_a} = 0. \quad (2695)$$

Therefore

$$\frac{\partial \theta_*}{\partial \xi_a} = -\frac{F_{\xi_a}}{F_\theta} = \frac{F_{\xi_a}}{\kappa_\perp}. \quad (2696)$$

The leading total displacement is

$$\delta \theta_*^{(1)} = \frac{1}{\kappa_{\perp,0}} \sum_a F_{\xi_a} \xi_a. \quad (2697)$$

For a Banach-space imperfection s ,

$$D_s \theta_* = -\frac{D_s F}{F_\theta}. \quad (2698)$$

If

$$\|D_s F\| \leq L_s, \quad (2699)$$

then

$$\|D_s \theta_*\| \leq \frac{L_s}{\kappa_\perp}. \quad (2700)$$

Differentiate the stationary equation twice. The exact mixed second derivative is

$$\frac{\partial^2 \theta_*}{\partial \xi_a \partial \xi_b} = -\frac{1}{F_\theta} \left[F_{\xi_a \xi_b} + F_{\theta \xi_a} \frac{\partial \theta_*}{\partial \xi_b} + F_{\theta \xi_b} \frac{\partial \theta_*}{\partial \xi_a} + F_{\theta \theta} \frac{\partial \theta_*}{\partial \xi_a} \frac{\partial \theta_*}{\partial \xi_b} \right].$$

At the nominal maximum,

$$F_\theta = -\kappa_{\perp,0}, \quad (2701)$$

so every second-order contribution is divided by the same transverse stability scale. Let the small imperfection vector have mean $\mathbb{E}[\boldsymbol{\xi}] = 0$ and covariance

$$\boldsymbol{\Sigma}_\xi = \mathbb{E}[\boldsymbol{\xi}\boldsymbol{\xi}^\top]. \quad (2702)$$

Define the linear sensitivity vector

$$g_a = \frac{F_{\xi_a}}{\kappa_{\perp,0}}. \quad (2703)$$

The leading magic-angle variance is

$$\sigma_\theta^2 = \mathbf{g}^\top \boldsymbol{\Sigma}_\xi \mathbf{g}. \quad (2704)$$

The second-order mean bias is

$$b_\theta = \mathbb{E}[\theta_*(\boldsymbol{\xi}) - \theta_0] = \frac{1}{2} \sum_{a,b} \theta_{*,ab} (\boldsymbol{\Sigma}_\xi)_{ab} + O(\mathbb{E}\|\boldsymbol{\xi}\|^3). \quad (2705)$$

Let

$$I_\theta = [\theta_0 - h_\theta, \theta_0 + h_\theta] \quad (2706)$$

lie inside the physical twist interval. Suppose

$$-F_{0,\theta}(\theta) \geq \kappa_0 > 0 \quad (\theta \in I_\theta). \quad (2707)$$

Assume the perturbed stationary equation obeys

$$\|F(\cdot, \boldsymbol{\xi}) - F_0\|_{\infty, I_\theta} \leq \varepsilon_F < \kappa_0 h_\theta, \quad (2708)$$

and

$$\|F_\theta(\cdot, \boldsymbol{\xi}) - F_{0,\theta}\|_{\infty, I_\theta} \leq \varepsilon_{F_\theta} < \kappa_0. \quad (2709)$$

Then there is exactly one stationary point $\tilde{\theta}_*$ in I_θ , with

$$|\tilde{\theta}_* - \theta_0| \leq \frac{\varepsilon_F}{\kappa_0}. \quad (2710)$$

Its transverse curvature satisfies

$$\tilde{\kappa}_\perp \geq \kappa_0 - \varepsilon_{F_\theta} > 0. \quad (2711)$$

Thus the continued point remains a strict local maximum. Let s be a normed imperfection coordinate with nominal value $s = 0$. Assume

$$\begin{aligned} \|F(\cdot, s) - F_0\|_\infty &\leq L_F \|s\|, \\ \|F_\theta(\cdot, s) - F_{0,\theta}\|_\infty &\leq L_{F_\theta} \|s\|, \\ \sup_\theta \|H(\theta, s) - H_0(\theta)\| &\leq L_H \|s\|, \end{aligned} \quad (2712)$$

and

$$\|C^{\text{coh}}(\cdot, s) - C_0^{\text{coh}}\|_\infty \leq L_C \|s\|. \quad (2713)$$

Let the nominal coherence margin be

$$C_0^{\text{coh}} \geq C_{\min} > 0. \quad (2714)$$

Let the nominal multimode detuning and resolved open-system gap be

$$d_{\text{mm},0} > 0, \quad \Delta_{\text{res},0} > 0, \quad (2715)$$

and suppose their changes are bounded by

$$|d_{\text{mm}}(s) - d_{\text{mm},0}| \leq L_{\text{mm}} \|s\|, \quad (2716)$$

and

$$|\Delta_{\text{res}}(s) - \Delta_{\text{res},0}| \leq L_{\text{res}} \|s\|. \quad (2717)$$

Define the joint admissibility radius

$$r_{\text{adm}} = \min \left\{ r_0, \frac{\kappa_0 h_\theta}{L_F}, \frac{\kappa_0}{2L_{F_\theta}}, \frac{\Delta_C^{(0)}}{2L_H}, \frac{C_{\min}}{L_C}, \frac{d_{\text{mm},0}}{2L_{\text{mm}}}, \frac{\Delta_{\text{res},0}}{2L_{\text{res}}} \right\}.$$

For $\|s\| < r_{\text{adm}}$, the following hold simultaneously:

$$\begin{aligned} |\theta_*(s) - \theta_0| &\leq \frac{L_F}{\kappa_0} \|s\|, \quad \kappa_\perp(s) \geq \frac{\kappa_0}{2}, \\ \Delta_C(s) &\geq \Delta_C^{(0)} - 2L_H \|s\| > 0, \\ C^{\text{coh}}(s) &\geq C_{\min} - L_C \|s\| > 0, \\ d_{\text{mm}}(s) &> \frac{d_{\text{mm},0}}{2}, \quad \Delta_{\text{res}}(s) > \frac{\Delta_{\text{res},0}}{2}. \end{aligned} \quad (2718)$$

For the minimal physical score, define

$$V_C = \hbar D_G v_{\text{max},C}, \quad X_C = E_{\text{kin}} \varrho_{C,\eta}^{\text{coh,max}}, \quad C_C = C_C^{\text{coh}}. \quad (2719)$$

Its logarithmic differential under any imperfection is

$$\begin{aligned} dL = \frac{1}{4} \left[\frac{W_C d\Delta_C - \Delta_C dW_C}{\Delta_C(\Delta_C + W_C)} + \frac{V_C d\Delta_C - \Delta_C dV_C}{\Delta_C(\Delta_C + V_C)} \right. \\ \left. + \frac{dX_C}{X_C(1 + X_C)} + \frac{dC_C}{C_C} \right]. \end{aligned} \quad (2720)$$

The first term measures bandwidth and gap reconstruction. The second measures generalized-velocity and gap reconstruction. The third measures coherent-DOS reconstruction. The fourth measures loss of bulk coherence. For imperfection coordinate ξ_a , the force moving the magic angle is

$$F_{\xi_a} = \partial_\theta [\partial_{\xi_a} L]. \quad (2721)$$

Substitution of Eq. (2720) gives an explicit decomposition

$$F_{\xi_a} = F_{\xi_a}^{W\Delta} + F_{\xi_a}^{v\Delta} + F_{\xi_a}^{\varrho} + F_{\xi_a}^{\text{coh}}. \quad (2722)$$

The corresponding angle contribution is

$$\delta\theta_{*,a} = \frac{F_{\xi_a}}{\kappa_{\perp,0}} \xi_a. \quad (2723)$$

A fabricated device can be operated either at the nominal design angle θ_0 or at the retuned optimum $\theta_*(\boldsymbol{\xi})$. The untuned score is

$$L_{\text{fix}}(\boldsymbol{\xi}) = L(\theta_0, \boldsymbol{\xi}), \quad (2724)$$

whereas the retuned score is

$$L_{\text{ret}}(\boldsymbol{\xi}) = L[\theta_*(\boldsymbol{\xi}), \boldsymbol{\xi}]. \quad (2725)$$

Expanding around the nominal maximum gives

$$L_{\text{ret}} - L_{\text{fix}} = \frac{1}{2\kappa_{\perp,0}} \left(\sum_a F_{\xi_a} \xi_a \right)^2 + O(\|\boldsymbol{\xi}\|^3). \quad (2726)$$

The leading retuning gain is nonnegative. It compensates the component of the imperfection that moves the optimum, but it cannot restore a gap, coherence, or spectral island that has already been destroyed. The optimized-score Hessian with respect to imperfections is

$$\frac{\partial^2 L_{\text{ret}}}{\partial \xi_a \partial \xi_b} = L_{\xi_a \xi_b} - \frac{L_{\theta \xi_a} L_{\theta \xi_b}}{L_{\theta \theta}} \quad (2727)$$

evaluated at the stationary point. Since $L_{\theta \theta} < 0$, the second term is positive semidefinite and quantifies the improvement available through angle retuning. For an allowed logarithmic score reduction $\delta L_{\text{acc}} > 0$, the nominal magic half-width is

$$\delta \theta_{\text{magic}} = \sqrt{\frac{2\delta L_{\text{acc}}}{\kappa_{\perp,0}}}. \quad (2728)$$

A sharp optimum has large $\kappa_{\perp,0}$ and small tolerance to angle errors. A broad plateau has small $\kappa_{\perp,0}$, but its optimum location is correspondingly harder to estimate precisely. Assume the fabricated magic-angle displacement is approximately Gaussian:

$$\delta \theta_* \sim \mathcal{N}(b_\theta, \sigma_\theta^2). \quad (2729)$$

The probability that the unretuned device remains inside a tolerance θ_{tol} is

$$Y_{\text{fab}} = \Phi\left(\frac{\theta_{\text{tol}} - b_\theta}{\sigma_\theta}\right) - \Phi\left(\frac{-\theta_{\text{tol}} - b_\theta}{\sigma_\theta}\right), \quad (2730)$$

where Φ is the standard normal cumulative distribution. For zero bias,

$$Y_{\text{fab}} = \text{erf}\left(\frac{\theta_{\text{tol}}}{\sqrt{2}\sigma_\theta}\right). \quad (2731)$$

For confidence level $1 - \alpha$, define the fabrication robustness index

$$\mathcal{R}_{\text{fab}} = \frac{|b_\theta| + z_{1-\alpha/2}\sigma_\theta + \delta\theta_{\text{sys}}}{\delta\theta_{\text{magic}}}. \quad (2732)$$

Here $\delta\theta_{\text{sys}}$ contains deterministic calibration and numerical errors. The sufficient yield condition is $\mathcal{R}_{\text{fab}} < 1$. Define the total C^0 Hermitian error

$$\begin{aligned} \varepsilon_{\text{stat}}^{(0)} &= \varepsilon_{\Omega}^{(0)} + \varepsilon_J^{(0)} + \frac{|\delta L|}{2} \\ &+ \varepsilon_{\text{cut}}^{(0)} + \varepsilon_{\text{mm}}^{(0)} + \varepsilon_{\text{cr}}^{(0)}. \end{aligned} \quad (2733)$$

The complete intrinsic-gap lower bound is

$$\Delta_C^L = \Delta_C^{(0)} - 2\varepsilon_{\text{stat}}^{(0)}. \quad (2734)$$

The bandwidth upper bound is

$$W_C^U = W_C^{(0)} + 2\varepsilon_{\text{stat}}^{(0)}. \quad (2735)$$

The projector error is

$$\varepsilon_P = \frac{\ell_{\mathcal{C}}}{2\pi} \frac{\varepsilon_{\text{stat}}^{(0)}}{\delta_{\mathcal{C}} [\delta_{\mathcal{C}} - \varepsilon_{\text{stat}}^{(0)}]}. \quad (2736)$$

The complete magic-angle displacement bound is

$$\delta \theta_*^U = \frac{\varepsilon_F}{\kappa_0}, \quad (2737)$$

and the transverse-stability lower bound is

$$\kappa_{\perp}^L = \kappa_0 - \varepsilon_{F_\theta}. \quad (2738)$$

For every nominal magic candidate, we record

$$\begin{aligned} \mathfrak{C}_{\text{rob}} = & \left(\theta_0, \tilde{\theta}_*, \delta \theta_*^U, \kappa_{\perp}^L, \delta \theta_{\text{magic}}, \right. \\ & \varepsilon_{\Omega}^{(0)}, \varepsilon_J^{(0)}, |\delta L|/2, \|\Gamma\|/2, \\ & \varepsilon_{\text{cut}}^{(0,1,2)}, \varepsilon_{\text{mm}}^{(0,1,2)}, \varepsilon_{\text{cr}}^{(0,1,2)}, \\ & W_C^U, \Delta_C^L, \Delta_C^{\text{res}}, \varepsilon_P, C_C^{\text{coh},L}, \\ & d_{\text{mm}}^L, P_{\text{fund}}^L, \mathcal{Q}_{\Delta,C}, \mathcal{R}_{\text{fab}}, Y_{\text{fab}}, \\ & \left. r_{\text{inj}}, L_c, C_{\text{rep},C}, \varepsilon_{\text{solver}} \right). \end{aligned} \quad (2739)$$

A fabricated device realizes robust magic when the spectral and resolved gaps remain positive, the transverse curvature is positive, the projector error is below one, the coherence and multimode margins remain positive, and $\mathcal{R}_{\text{fab}} < 1$. A shifted optimum can be recovered by retuning when the intrinsic island survives. If loss or broadening obscures an otherwise intact island, the limitation is spectroscopic resolution. A device outside the established perturbation radius lies outside the range of the ideal-device approximation.

The complete robustness analysis proceeds through a fixed causal chain from fabrication uncertainties to the final device classification. The fabrication and open-system parameters first determine the effective uncertainty objects

$$\mathbf{V}_{\Omega}, \mathbf{V}_J, \mathbf{V}_L, \mathbf{\Gamma}, \mathbf{V}_{\text{cut}}, \mathbf{\Sigma}_{\text{mm}}.$$

These quantities determine the established perturbation budgets $\varepsilon^{(0)}$, $\varepsilon^{(1)}$, $\varepsilon^{(2)}$, which respectively control spectral values, first derivatives, and second derivatives to the order required by the observable under consideration. The resulting perturbation bounds are then propagated to the physical spectral and coherence observables

$$W_C, \Delta_C, P_C, v_{\text{max},C}, \Lambda_{\text{max},C}, C_C^{\text{coh}}.$$

TABLE VII. Quantitative effect of the seven circuit-imperfection channels on the physical flat-band certificate. A check mark denotes a quantity that can change at leading order; a parenthesized check mark denotes a change that begins only at second order in a symmetric reference device.

Imperfection	Real spectrum	Gap	Projector	Symmetry	Linewidth	Magic angle
Common frequency shift	shift only	no	no	no	no	no
Centered frequency disorder	✓	✓	✓	generic	no	✓
Coupling-amplitude disorder	✓	✓	✓	generic	no	✓
Coupling-phase disorder	✓	✓	✓	generic	no	✓
Uniform layer detuning	(✓)	✓	✓	$D_m^{\text{BL}} \rightarrow C_m$	no	(✓)
Uniform photon loss	no	unresolved experimentally	no	no	✓	intrinsic no
Nonuniform photon loss	(✓)	✓	biorthogonal	generic	✓	✓
Finite hopping cutoff	✓	✓	✓	support dependent	no	✓
Off-resonant higher modes	(✓)	✓	✓	device dependent	optional	✓

These established observables enter the robustness functional $F(\theta, \xi)$, from which we determine the displacement of the physical optimum, $\delta\theta_*$, the transverse stability scale $\kappa_\perp$, and the resulting magic-angle uncertainty $\delta\theta_{\text{magic}}$. These quantities in turn determine the experimentally resolved gap Δ_C^{res} , the fabrication-risk measure $\mathcal{R}_{\text{fab}}$, and the fabrication yield Y_{fab} . These quantities determine the experimentally accessible operating region. A device is robust when all spectral, coherence, and fabrication margins remain positive; a displaced optimum can be recovered by retuning; and an intrinsic island obscured by loss remains a spectroscopic-resolution problem.

XII. MICROWAVE SPECTROSCOPY PROTOCOL

We now convert the finite-cover Hamiltonian, the representation-resolved band structure, and the operational flat-band certificate into a complete microwave-spectroscopy protocol. The finite circuit contains

$$D = \nu_M |Q_N| \quad (2740)$$

internal single-photon modes and M calibrated microwave ports. We work in angular-frequency units. Multiplication by $\hbar$ restores physical energies. The protocol is constructed to determine the following quantities from measured scattering data: complete resonance positions and linewidths; global and local spectral densities; layer polarization and interlayer coherence; real-space mode and projector profiles; Abelian and non-Abelian representation weights; bandwidth, isolation gap, and flat-band angle; the experimental conditions that falsify each claim. We remain in the linear-response regime. If the resonators possess weak self-Kerr nonlinearities K_i^{Kerr} , the applied coherent power is restricted by

$$\epsilon_{\text{nl}} = \max_i \frac{|K_i^{\text{Kerr}}| \bar{n}_i}{\min\{\gamma_i, \Delta_C, g_{\text{mm}}\}} \ll 1, \quad (2741)$$

where $\bar{n}_i$ is the mean photon number, γ_i is the loaded linewidth, Δ_C is the target isolation gap, and g_{mm} is the nearest multimode detuning. Measurements that violate

Eq. (2741) are not compared with the one-particle theory. The complete experimental logic is

$$\begin{aligned} \text{calibrated ports} &\longrightarrow \mathbf{S}(\Omega) \longrightarrow \mathbf{G}^R(\Omega) \\ &\longrightarrow \begin{cases} \text{global transmission,} \\ \text{local spectral matrix,} \\ \text{layer-resolved response,} \\ \text{pole residues} \end{cases} \\ &\longrightarrow \begin{cases} E_n, \gamma_n, \\ P_C, \\ Q_x(P_C), \\ w_{C,\rho} \end{cases} \\ &\longrightarrow (W_C, \Delta_C, v_{\text{max},C}, \varrho_C^{\text{coh}}, C_C^{\text{coh}}) \\ &\longrightarrow \theta_{\text{flat}}^{\text{exp}} \longrightarrow \text{claim-specific falsification tests.} \end{aligned} \quad (2742)$$

A. Global transmission

Let $\mathbf{a} = (a_1 \cdots a_D)^\top$ collect the internal annihilation operators. The Hermitian circuit Hamiltonian is

$$\hat{H} = \hbar \mathbf{a}^\dagger \mathbf{H} \mathbf{a}, \quad \mathbf{H} = \mathbf{H}^\dagger. \quad (2743)$$

The calibrated port-coupling matrix is

$$\mathbf{K} = (|k_1\rangle \cdots |k_M\rangle) \in \mathbb{C}^{D \times M}. \quad (2744)$$

The a -th column has the microscopic form

$$|k_a\rangle = \sum_{i=1}^D \sqrt{\kappa_{ia}} e^{i\phi_{ia}} |i\rangle, \quad (2745)$$

where κ_{ia} is the calibrated external coupling rate and ϕ_{ia} is the programmed or measured cable phase. The external damping matrix is

$$\mathbf{\Gamma}_{\text{ext}} = \mathbf{K} \mathbf{K}^\dagger. \quad (2746)$$

Internal material, dielectric, radiative, and unmonitored losses are collected in $\mathbf{\Gamma}_{\text{int}}$. The total damping is

$$\mathbf{\Gamma} = \mathbf{\Gamma}_{\text{ext}} + \mathbf{\Gamma}_{\text{int}}. \quad (2747)$$

After removal of calibrated cable delays and the direct background path, the Heisenberg–Langevin equation is

$$\frac{d\mathbf{a}}{dt} = -i\mathbf{H}\mathbf{a} - \frac{1}{2}\mathbf{\Gamma}\mathbf{a} + \mathbf{K}\mathbf{s}_{\text{in}}(t). \quad (2748)$$

The input–output relation is

$$\mathbf{s}_{\text{out}} = \mathbf{s}_{\text{in}} - \mathbf{K}^\dagger \mathbf{a}. \quad (2749)$$

For a harmonic convention

$$\mathbf{s}_{\text{in}}(t) = \mathbf{s}_{\text{in}}(\Omega)e^{-i\Omega t}, \quad (2750)$$

define the retarded Green operator

$$\mathbf{G}^R(\Omega) = \left[\Omega I - \mathbf{H} + \frac{i}{2}\mathbf{\Gamma} \right]^{-1}. \quad (2751)$$

Equation (2748) gives

$$\mathbf{a}(\Omega) = i\mathbf{G}^R(\Omega)\mathbf{K}\mathbf{s}_{\text{in}}(\Omega). \quad (2752)$$

Substitution into Eq. (2749) gives the exact multiport scattering matrix

$$\mathbf{S}(\Omega) = I_M - i\mathbf{K}^\dagger \mathbf{G}^R(\Omega)\mathbf{K}. \quad (2753)$$

If a nontrivial calibrated direct scattering matrix $\mathbf{S}_{\text{dir}}$ remains, we first transform the reference planes so that $\mathbf{S}_{\text{dir}} \rightarrow I_M$. A frequency-dependent unremoved background is included as a nuisance model and is never interpreted as a circuit pole. Define

$$\mathbf{G}^A(\Omega) = \mathbf{G}^R(\Omega)^\dagger. \quad (2754)$$

The matrix spectral density is

$$\mathbf{A}(\Omega) = -\frac{1}{\pi} \text{Im } \mathbf{G}^R(\Omega) = \frac{i}{2\pi} [\mathbf{G}^R(\Omega) - \mathbf{G}^A(\Omega)]. \quad (2755)$$

The resolvent identity gives

$$\begin{aligned} \mathbf{G}^R - \mathbf{G}^A &= \mathbf{G}^R [(\mathbf{G}^A)^{-1} - (\mathbf{G}^R)^{-1}] \mathbf{G}^A \\ &= -i\mathbf{G}^R \mathbf{\Gamma} \mathbf{G}^A. \end{aligned} \quad (2756)$$

Hence

$$\mathbf{A}(\Omega) = \frac{1}{2\pi} \mathbf{G}^R \mathbf{\Gamma} \mathbf{G}^A \succeq 0. \quad (2757)$$

The same matrix can be written as

$$\mathbf{A} = \frac{1}{2\pi} \mathbf{G}^A \mathbf{\Gamma} \mathbf{G}^R. \quad (2758)$$

For a stable finite-dimensional passive network,

$$\int_{-\infty}^{\infty} \mathbf{A}(\Omega) d\Omega = I_D. \quad (2759)$$

The normalized global density of states is

$$\varrho_{\text{glob}}(\Omega) = \frac{1}{D} \text{Tr } \mathbf{A}(\Omega). \quad (2760)$$

It satisfies

$$\int_{-\infty}^{\infty} \varrho_{\text{glob}}(\Omega) d\Omega = 1. \quad (2761)$$

Using Eq. (2753), direct multiplication gives the passivity identity

$$I_M - \mathbf{S}^\dagger \mathbf{S} = \mathbf{K}^\dagger \mathbf{G}^A \mathbf{\Gamma}_{\text{int}} \mathbf{G}^R \mathbf{K} \succeq 0. \quad (2762)$$

A calibrated passive device must therefore obey $\mathbf{S}^\dagger \mathbf{S} \preceq I_M$. A statistically significant violation indicates a calibration error, gain, nonlinearity, or an incomplete port model. Let $|e_a\rangle$ be a unit input in port a , and let

$$|k_a\rangle = \mathbf{K}|e_a\rangle. \quad (2763)$$

The resonantly scattered power relative to the calibrated direct path is

$$P_{\text{sc},a}(\Omega) = \|\mathbf{S}(\Omega) - I\| |e_a\rangle\|^2. \quad (2764)$$

Equation (2753) gives

$$P_{\text{sc},a} = \langle k_a | \mathbf{G}^A \mathbf{\Gamma}_{\text{ext}} \mathbf{G}^R | k_a \rangle. \quad (2765)$$

The power absorbed into unmonitored internal channels is

$$P_{\text{abs},a} = \langle e_a | (I - \mathbf{S}^\dagger \mathbf{S}) | e_a \rangle, \quad (2766)$$

and Eq. (2762) gives

$$P_{\text{abs},a} = \langle k_a | \mathbf{G}^A \mathbf{\Gamma}_{\text{int}} \mathbf{G}^R | k_a \rangle. \quad (2767)$$

The total resonant response is therefore

$$\begin{aligned} P_{\text{res},a} &= P_{\text{sc},a} + P_{\text{abs},a} \\ &= \langle k_a | \mathbf{G}^A \mathbf{\Gamma} \mathbf{G}^R | k_a \rangle. \end{aligned} \quad (2768)$$

Using Eq. (2757),

$$P_{\text{res},a}(\Omega) = 2\pi \langle k_a | \mathbf{A}(\Omega) | k_a \rangle. \quad (2769)$$

Thus the sum of monitored resonant scattering and calibrated absorption measures a positive spectral quadratic form exactly. A single global transmission trace need not observe every circuit mode. Let

$$|f_a\rangle = \frac{|k_a\rangle}{\|k_a\|} \quad (2770)$$

be a normalized internal drive pattern. Suppose the measured patterns and positive weights w_a form the frame operator

$$\mathbf{F}_{\text{port}} = \sum_{a=1}^M w_a |f_a\rangle \langle f_a|. \quad (2771)$$

An exact tight frame satisfies $\mathbf{F}_{\text{port}} = I_D$. An approximate frame satisfies

$$\|\mathbf{F}_{\text{port}} - I_D\| \leq \varepsilon_{\text{frame}}. \quad (2772)$$

Define the reconstructed global DOS

$$\hat{\varrho}_{\text{glob}}(\Omega) = \frac{1}{D} \sum_{a=1}^M w_a \langle f_a | \mathbf{A}(\Omega) | f_a \rangle. \quad (2773)$$

Since $\mathbf{A} \succeq 0$,

$$\begin{aligned} |\hat{\varrho}_{\text{glob}} - \varrho_{\text{glob}}| &= \frac{1}{D} |\text{Tr}[(\mathbf{F}_{\text{port}} - I)\mathbf{A}]| \\ &\leq \varepsilon_{\text{frame}} \varrho_{\text{glob}}. \end{aligned} \quad (2774)$$

Therefore

$$\frac{|\hat{\varrho}_{\text{glob}} - \varrho_{\text{glob}}|}{\varrho_{\text{glob}}} \leq \varepsilon_{\text{frame}} \quad (2775)$$

whenever the denominator is nonzero. If $\mathbf{F}_{\text{port}}$ is rank deficient, the measurement returns the projected spectral density

$$\varrho_{\text{port}} = \frac{1}{D} \text{Tr}(\mathbf{F}_{\text{port}} \mathbf{A}), \quad (2776)$$

not the complete DOS. Let P_C be the target spectral projector. The port observability operator on the target is

$$\mathbf{O}_C = P_C \mathbf{F}_{\text{port}} P_C. \quad (2777)$$

Define its minimum target eigenvalue

$$o_C^{\min} = \lambda_{\min}[\mathbf{O}_C|_{\text{Ran } P_C}]. \quad (2778)$$

If $o_C^{\min} > 0$, every normalized target state has nonzero total coupling to the measured drive frame:

$$\sum_a w_a |\langle f_a | u \rangle|^2 \geq o_C^{\min}, \quad u \in \text{Ran } P_C. \quad (2779)$$

If $o_C^{\min} = 0$, the target contains a dark direction. Absence of a transmission peak cannot then be interpreted as absence of a mode. Define the loaded non-Hermitian operator

$$\mathbf{H}_{\text{nh}} = \mathbf{H} - \frac{i}{2} \mathbf{\Gamma}. \quad (2780)$$

Suppose it is diagonalizable:

$$\begin{aligned} \mathbf{H}_{\text{nh}} |u_n^R\rangle &= z_n |u_n^R\rangle, \\ \langle u_n^L | \mathbf{H}_{\text{nh}} &= z_n \langle u_n^L|, \end{aligned} \quad (2781)$$

with

$$\langle u_m^L | u_n^R \rangle = \delta_{mn}. \quad (2782)$$

The poles are

$$z_n = E_n - \frac{i}{2} \gamma_n. \quad (2783)$$

The Green operator has the expansion

$$\mathbf{G}^R(\Omega) = \sum_n \frac{|u_n^R\rangle \langle u_n^L|}{\Omega - z_n}. \quad (2784)$$

Therefore

$$\mathbf{S}(\Omega) = I - i \sum_n \frac{\mathbf{K}^\dagger |u_n^R\rangle \langle u_n^L| \mathbf{K}}{\Omega - z_n}. \quad (2785)$$

The scattering residue is

$$\mathbf{R}_n^S = \text{Res}_{\Omega=z_n} \mathbf{S}(\Omega) = -i \mathbf{K}^\dagger |u_n^R\rangle \langle u_n^L| \mathbf{K}. \quad (2786)$$

For uniform or sufficiently weak normal loss,

$$|u_n^R\rangle = |u_n\rangle, \quad \langle u_n^L| = \langle u_n|, \quad (2787)$$

and

$$\mathbf{S}_{ba}(\Omega) = \delta_{ba} - i \sum_n \frac{g_{bn} g_{an}^*}{\Omega - E_n + i\gamma_n/2}, \quad (2788)$$

where $g_{an} = \langle k_a | u_n \rangle$. For a tracked target pole set C , define the measured real-frequency bandwidth

$$W_C^{\text{pole}} = \max_{n \in C} \text{Re } z_n - \min_{n \in C} \text{Re } z_n. \quad (2789)$$

The intrinsic center-to-center gap is

$$\Delta_C^{\text{pole}} = \min_{n \in C, m \notin C} |\text{Re } z_n - \text{Re } z_m|. \quad (2790)$$

The experimentally resolvable gap subtracts the half-width intervals:

$$\Delta_C^{\text{res}} = \min_{n \in C, m \notin C} \left[|\text{Re } z_n - \text{Re } z_m| - \frac{\gamma_n + \gamma_m}{2} \right].$$

Direct spectral isolation requires $\Delta_C^{\text{res}} > 0$. Let $\varepsilon_{E,n}$ and $\varepsilon_{\gamma,n}$ be established fit uncertainties. Conservative bounds are

$$W_C^U = W_C^{\text{pole}} + 2 \max_{n \in C} \varepsilon_{E,n}, \quad (2791)$$

and

$$\begin{aligned} \Delta_C^{\text{res},L} = \min_{n \in C, m \notin C} & \left[|\tilde{E}_n - \tilde{E}_m| - \varepsilon_{E,n} - \varepsilon_{E,m} \right. \\ & \left. - \frac{\tilde{\gamma}_n + \tilde{\gamma}_m + \varepsilon_{\gamma,n} + \varepsilon_{\gamma,m}}{2} \right]. \end{aligned} \quad (2792)$$

For each control point (θ, K, ω) , we perform the following operations with all thresholds fixed before the flat-band scan:

The global microwave-spectroscopy protocol is performed in a fixed sequence. We first calibrate the cable delays, direct transmission paths, channel gains, and

relative port phases so that the subsequent multiport response is placed on a common complex reference. We then measure the complete complex $M \times M$ scattering matrix over the frequency window of interest and verify that the measured response is consistent with passivity and with the required spectral sum rules. After these calibration and consistency checks, the resonant structure is extracted by jointly fitting the poles across all measured ports. From the joint fit, we determine the corresponding pole-residue matrices and use them to quantify the observability of each mode in the accessible measurement channels. The target spectral component is then identified by projector continuation rather than by frequency ordering alone, which allows the same component to be tracked through parameter variation and possible avoided crossings. Once the target projector has been identified, the experimentally established upper bandwidth W_C^U , lower spectral gap Δ_C^L , and lower resolved gap $\Delta_C^{\text{res},L}$ are calculated from the extracted pole data and their associated uncertainty budgets. Finally, the complete measurement and extraction procedure is repeated under local, layer-selective, and representation-selective drives in order to test the robustness of the target assignment and to resolve how different sectors contribute to the observed spectral response.

B. Local density of states

For internal basis states $|i\rangle$ and $|j\rangle$, define

$$\mathcal{A}_{ij}(\Omega) = \langle i | \mathbf{A}(\Omega) | j \rangle. \quad (2793)$$

The local density of states at site i is

$$\varrho_i(\Omega) = \mathcal{A}_{ii}(\Omega) = -\frac{1}{\pi} \text{Im } G_{ii}^R(\Omega). \quad (2794)$$

It obeys

$$\int_{-\infty}^{\infty} \varrho_i(\Omega) d\Omega = 1. \quad (2795)$$

Summing over all sites recovers the global DOS:

$$\frac{1}{D} \sum_{i=1}^D \varrho_i(\Omega) = \varrho_{\text{glob}}(\Omega). \quad (2796)$$

Let a calibrated probe port couple locally to site i :

$$|k_p\rangle = \sqrt{\kappa_p} |i\rangle. \quad (2797)$$

The loaded reflection coefficient is

$$S_{pp}(\Omega) = 1 - i\kappa_p G_{ii}^R(\Omega). \quad (2798)$$

Therefore the loaded LDOS is obtained directly from the calibrated in-phase reflection quadrature:

$$\varrho_i^{\text{load}}(\Omega) = \frac{1 - \text{Re } S_{pp}(\Omega)}{\pi \kappa_p}. \quad (2799)$$

The dispersive quadrature gives

$$\text{Re } G_{ii}^R(\Omega) = -\frac{\text{Im } S_{pp}(\Omega)}{\kappa_p}. \quad (2800)$$

The probe adds the rank-one damping

$$\delta \mathbf{\Gamma}_p = \kappa_p |i\rangle \langle i|. \quad (2801)$$

Let $\mathbf{G}_0^R$ denote the Green operator without the probe and $\mathbf{G}_p^R$ the measured Green operator. Their inverses obey

$$(\mathbf{G}_p^R)^{-1} = (\mathbf{G}_0^R)^{-1} + \frac{i\kappa_p}{2} |i\rangle \langle i|. \quad (2802)$$

The resolvent identity gives

$$\mathbf{G}_p^R - \mathbf{G}_0^R = -\frac{i\kappa_p}{2} \mathbf{G}_0^R |i\rangle \langle i| \mathbf{G}_p^R. \quad (2803)$$

On a contour whose distance from the unprobed spectrum is $\delta_p > \kappa_p/2$,

$$\|\mathbf{G}_p^R - \mathbf{G}_0^R\| \leq \frac{\kappa_p/2}{\delta_p(\delta_p - \kappa_p/2)}. \quad (2804)$$

We either de-embed this known probe loading or include it explicitly in the fitted Hamiltonian. The complete local spectral matrix contains phase information absent from the diagonal LDOS. Let

$$|f_\phi^{(ij)}\rangle = \frac{1}{\sqrt{2}} (|i\rangle + e^{i\phi} |j\rangle). \quad (2805)$$

Define

$$A_\phi^{(ij)} = \langle f_\phi^{(ij)} | \mathbf{A} | f_\phi^{(ij)} \rangle. \quad (2806)$$

Direct expansion gives

$$A_\phi^{(ij)} = \frac{\mathcal{A}_{ii} + \mathcal{A}_{jj}}{2} + \text{Re} [e^{i\phi} \mathcal{A}_{ij}]. \quad (2807)$$

Consequently,

$$\text{Re } \mathcal{A}_{ij} = \frac{A_0^{(ij)} - A_{\pi}^{(ij)}}{2}, \quad (2808)$$

and

$$\text{Im } \mathcal{A}_{ij} = -\frac{A_{\pi/2}^{(ij)} - A_{3\pi/2}^{(ij)}}{2}. \quad (2809)$$

Four phase-coded measurements therefore reconstruct one complex off-diagonal matrix element. If local ports p and q couple to sites i and j , the off-diagonal Green function can also be obtained directly from transmission:

$$G_{ij}^R = \frac{ie^{-i(\phi_{jq} - \phi_{ip})}}{\sqrt{\kappa_{ip}\kappa_{jq}}} S_{pq}, \quad p \neq q, \quad (2810)$$

after subtraction of the calibrated direct crosstalk. Let Π_x project onto all internal orbitals in finite-cover cell x . The cell-resolved DOS is

$$\varrho_x(\Omega) = \frac{1}{\nu_M} \text{Tr} [\Pi_x \mathbf{A}(\Omega)]. \quad (2811)$$

Let P_1 and P_2 be the two layer projectors. The layer-resolved DOS is

$$\varrho_\ell(\Omega) = \frac{1}{D_\ell} \text{Tr} [P_\ell \mathbf{A}(\Omega)], \quad \ell = 1, 2. \quad (2812)$$

For equal layers,

$$D_1 = D_2 = \frac{D}{2}. \quad (2813)$$

For a set of sites $\mathcal{R}$, define

$$P_{\mathcal{R}} = \sum_{i \in \mathcal{R}} |i\rangle\langle i|. \quad (2814)$$

The regional DOS is

$$\varrho_{\mathcal{R}}(\Omega) = \frac{1}{\text{rank } P_{\mathcal{R}}} \text{Tr} [P_{\mathcal{R}} \mathbf{A}(\Omega)]. \quad (2815)$$

For an open-cluster control, we use a central region $\mathcal{R}_{\text{core}}$ and compare it with an outer shell. For a periodic finite cover, $\partial\Sigma_N = \emptyset$, so no boundary-state subtraction is required. Let

$$I_C = [\Omega_C^-, \Omega_C^+] \quad (2816)$$

contain the complete target island and no complementary pole in the zero-linewidth limit. The ideal Hermitian projector is

$$P_C = \lim_{\eta \rightarrow 0^+} \int_{I_C} \mathbf{A}_\eta(\Omega) d\Omega. \quad (2817)$$

For one Lorentzian mode of linewidth γ_n , centered at E_n , the fraction contained in a symmetric window $[E_n - m_n, E_n + m_n]$ is

$$f_n^{\text{win}} = \frac{2}{\pi} \arctan \left(\frac{2m_n}{\gamma_n} \right). \quad (2818)$$

The missing fraction is

$$\epsilon_n^{\text{miss}} = 1 - f_n^{\text{win}}. \quad (2819)$$

For a complementary pole $m \notin C$, define its leakage fraction into the target window by

$$f_{m \rightarrow C}^{\text{leak}} = \frac{1}{\pi} \left[\arctan \left(\frac{2(\Omega_C^+ - E_m)}{\gamma_m} \right) - \arctan \left(\frac{2(\Omega_C^- - E_m)}{\gamma_m} \right) \right]. \quad (2820)$$

For mutually orthogonal normal modes, the window-projector error obeys

$$\|\hat{P}_C^{\text{win}} - P_C\| \leq \max \left\{ \max_{n \in C} \epsilon_n^{\text{miss}}, \max_{m \notin C} f_{m \rightarrow C}^{\text{leak}} \right\}. \quad (2821)$$

For overlapping or nonnormal modes, we use the fitted pole-residue projector or a complex contour rather than the raw real-frequency window. For a rank- r_C target projector, define the normalized cell weight

$$Q_x(P_C) = \frac{\text{Tr}(\Pi_x P_C)}{r_C}. \quad (2822)$$

It satisfies

$$Q_x(P_C) \geq 0, \quad \sum_x Q_x(P_C) = 1. \quad (2823)$$

The projector cell-participation factor is

$$p_{\text{cell}}(P_C) = \frac{1}{N_c \sum_x Q_x(P_C)^2}. \quad (2824)$$

It obeys

$$\frac{1}{N_c} \leq p_{\text{cell}}(P_C) \leq 1. \quad (2825)$$

A translation-covariant target projector has

$$Q_x(P_C) = \frac{1}{N_c}, \quad (2826)$$

and hence $p_{\text{cell}}(P_C) = 1$. The measured coherent DOS is constructed from the same projector-based coherence rule used in the physical magic certificate:

$$\varrho_{C,\eta}^{\text{coh}}(\Omega) = C_C^{\text{coh}} \varrho_{C,\eta}(\Omega). \quad (2827)$$

The coherence factor is calculated from the reconstructed projector, rather than from an arbitrarily selected eigenvector inside a degenerate multiplet. The local measurements must satisfy

$$\begin{aligned} \sum_i \varrho_i(\Omega) &= D \varrho_{\text{glob}}(\Omega), \\ \sum_x \nu_M \varrho_x(\Omega) &= D \varrho_{\text{glob}}(\Omega), \end{aligned} \quad (2828)$$

and

$$D_1 \varrho_1(\Omega) + D_2 \varrho_2(\Omega) = D \varrho_{\text{glob}}(\Omega). \quad (2829)$$

Violations beyond the propagated calibration uncertainty indicate an incomplete port frame, frequency-window truncation, gain mismatch, or an incorrect internal-mode assignment.

C. Layer-selective excitation

Let

$$P_1 + P_2 = I, \quad P_1 P_2 = 0 \quad (2830)$$

be the layer decomposition. For each spatial drive pattern α , define matched layer drives

$$|f_{1,\alpha}\rangle \in \text{Ran } P_1, \quad |f_{2,\alpha}\rangle \in \text{Ran } P_2. \quad (2831)$$

The layer frames satisfy

$$\sum_{\alpha} w_{\alpha} |f_{\ell,\alpha}\rangle \langle f_{\ell,\alpha}| = P_{\ell} \quad (2832)$$

in the exactly complete case. The layer-selective spectral response is

$$A_{\ell,\alpha}(\Omega) = \langle f_{\ell,\alpha} | \mathbf{A}(\Omega) | f_{\ell,\alpha} \rangle. \quad (2833)$$

Summing over a complete layer frame gives

$$\sum_{\alpha} w_{\alpha} A_{\ell,\alpha}(\Omega) = \text{Tr} [P_{\ell} \mathbf{A}(\Omega)]. \quad (2834)$$

For a simple normalized Hermitian mode $|u_n\rangle$, define

$$w_{\ell,n} = \langle u_n | P_{\ell} | u_n \rangle. \quad (2835)$$

Using a complete layer drive frame, the integrated pole area satisfies

$$I_{\ell,n} = \sum_{\alpha} w_{\alpha} |\langle f_{\ell,\alpha} | u_n \rangle|^2 = w_{\ell,n}. \quad (2836)$$

Since

$$w_{1,n} + w_{2,n} = 1, \quad (2837)$$

the measured layer polarization is

$$P_{L,n} = \frac{I_{1,n} - I_{2,n}}{I_{1,n} + I_{2,n}} = w_{1,n} - w_{2,n}. \quad (2838)$$

The effective number of occupied layers is

$$\mathcal{P}_{L,n} = \frac{2}{1 + P_{L,n}^2}. \quad (2839)$$

A balanced state has

$$P_{L,n} = 0, \quad \mathcal{P}_{L,n} = 2. \quad (2840)$$

A fully layer-polarized state has

$$|P_{L,n}| = 1, \quad \mathcal{P}_{L,n} = 1. \quad (2841)$$

For one matched drive pair, define the modal amplitudes

$$a_{n,\alpha} = \langle f_{1,\alpha} | u_n \rangle, \quad b_{n,\alpha} = \langle f_{2,\alpha} | u_n \rangle. \quad (2842)$$

Use the coherent superposition

$$|f_{\alpha,\phi}\rangle = \frac{1}{\sqrt{2}} (|f_{1,\alpha}\rangle + e^{i\phi} |f_{2,\alpha}\rangle). \quad (2843)$$

The integrated pole intensity is

$$I_{n,\alpha}(\phi) = |\langle f_{\alpha,\phi} | u_n \rangle|^2 = \frac{1}{2} [|a_{n,\alpha}|^2 + |b_{n,\alpha}|^2] + \text{Re} [e^{-i\phi} a_{n,\alpha}^* b_{n,\alpha}]. \quad (2844)$$

The real interlayer coherence is

$$2 \text{Re} (a_{n,\alpha}^* b_{n,\alpha}) = I_{n,\alpha}(0) - I_{n,\alpha}(\pi). \quad (2845)$$

The imaginary interlayer coherence is

$$2 \text{Im} (a_{n,\alpha}^* b_{n,\alpha}) = I_{n,\alpha}(\pi/2) - I_{n,\alpha}(3\pi/2). \quad (2846)$$

Define the normalized complex layer coherence in this channel by

$$\mathcal{C}_{12,n,\alpha} = \frac{2a_{n,\alpha}^* b_{n,\alpha}}{|a_{n,\alpha}|^2 + |b_{n,\alpha}|^2}. \quad (2847)$$

It obeys $|\mathcal{C}_{12,n,\alpha}| \leq 1$. For a degenerate target projector, define the cluster layer polarization

$$P_{L,C} = \frac{1}{r_C} \text{Tr} [(P_1 - P_2) P_C]. \quad (2848)$$

A basis-independent interlayer coherence is

$$\mathcal{C}_{L,C}^{\text{tr}} = \frac{2}{r_C} \|P_1 P_C P_2\|_1, \quad (2849)$$

where $\|\cdot\|_1$ is the trace norm. The projector block geometry gives $0 \leq \mathcal{C}_{L,C}^{\text{tr}} \leq 1$. This quantity is invariant under every unitary rotation inside $\text{Ran } P_C$. Suppose an exact unitary layer symmetry

$$\mathcal{P}_L^2 = I, \quad [\mathbf{H}, \mathcal{P}_L] = 0, \quad [\mathbf{\Gamma}, \mathcal{P}_L] = 0 \quad (2850)$$

is present. Define

$$P_{\pm} = \frac{1}{2} (I \pm \mathcal{P}_L). \quad (2851)$$

For one matched pair related by the symmetry,

$$|f_{\pm,\alpha}\rangle = \frac{1}{\sqrt{2}} (|f_{1,\alpha}\rangle \pm |f_{2,\alpha}\rangle). \quad (2852)$$

Then

$$P_{\mp} |f_{\pm,\alpha}\rangle = 0. \quad (2853)$$

The exact selection rule is

$$\langle f_{-,\beta} | \mathbf{G}^R(\Omega) | f_{+,\alpha} \rangle = 0. \quad (2854)$$

At a general centered twist, the relevant symmetry may be a combined layer-exchanging spatial operation

$$\tilde{\sigma} = \mathcal{P}_L U_{\sigma}, \quad (2855)$$

rather than bare layer exchange. The corresponding drives are eigenvectors of $\tilde{\sigma}$:

$$|f_{\pm}\rangle = \frac{1}{\sqrt{2}}(|f\rangle \pm \tilde{\sigma}|f\rangle). \quad (2856)$$

A uniform layer detuning is

$$V_L = \frac{\delta_L}{2} Z_L. \quad (2857)$$

For one nominal even-odd pair with splitting $2|\tau_n|$, the opposite parity admixture is

$$w_{\text{forbidden},n} = \frac{1}{2} \left[1 - \frac{2|\tau_n|}{\sqrt{4|\tau_n|^2 + \delta_L^2}} \right]. \quad (2858)$$

For weak detuning,

$$w_{\text{forbidden},n} = \frac{\delta_L^2}{16|\tau_n|^2} + O\left(\frac{\delta_L^4}{|\tau_n|^4}\right). \quad (2859)$$

The measured leakage under nominally forbidden drives therefore provides an independent calibration of residual layer asymmetry.

D. Mode-profile reconstruction

Introduce

$$\mathbf{B} = \mathbf{K}^\dagger : \mathbb{C}^D \longrightarrow \mathbb{C}^M. \quad (2860)$$

For a simple normal mode, define the port-space mode vector

$$|c_n\rangle = \mathbf{B}|u_n\rangle. \quad (2861)$$

Equation (2786) becomes

$$i\mathbf{R}_n^S = |c_n\rangle\langle c_n|. \quad (2862)$$

The matrix on the right is positive semidefinite and rank one. Let $\sigma_{1,n} \geq \sigma_{2,n} \geq \dots$ be the singular values of $i\mathbf{R}_n^S$. The isolated-mode rank diagnostic is

$$r_{\text{rank},n} = \frac{\sigma_{2,n}}{\sigma_{1,n}}. \quad (2863)$$

A value significantly above the propagated fit uncertainty indicates overlapping poles, nonnormality, residual background, or an incorrectly chosen frequency window. The dominant eigenvector of $i\mathbf{R}_n^S$ reconstructs $|c_n\rangle$ up to one global phase. That phase has no physical effect on a single isolated mode. For a diagonalizable nonnormal loaded operator,

$$i\mathbf{R}_n^S = |c_n^R\rangle\langle c_n^L|, \quad (2864)$$

where

$$|c_n^R\rangle = \mathbf{B}|u_n^R\rangle, \quad \langle c_n^L| = \langle u_n^L|\mathbf{K}. \quad (2865)$$

A singular-value decomposition determines the two port vectors up to the biorthogonal gauge

$$|c_n^R\rangle \mapsto \alpha|c_n^R\rangle, \quad \langle c_n^L| \mapsto \alpha^{-1}\langle c_n^L|. \quad (2866)$$

We report left and right profiles separately whenever non-normality is resolved. If $\mathbf{B}$ has full column rank,

$$\text{rank } \mathbf{B} = D, \quad (2867)$$

the internal mode is reconstructed by

$$|u_n\rangle = \mathbf{B}^+|c_n\rangle, \quad (2868)$$

where $\mathbf{B}^+$ is the Moore–Penrose pseudoinverse. More generally, suppose only the target subspace is observable. Define

$$\mathbf{B}_C = \mathbf{B}P_C : \text{Ran } P_C \longrightarrow \mathbb{C}^M. \quad (2869)$$

If

$$\sigma_{\text{obs},C} = \sigma_{\min}(\mathbf{B}_C) > 0, \quad (2870)$$

then

$$|u_n\rangle = \mathbf{B}_C^+|c_n\rangle, \quad n \in C. \quad (2871)$$

For reconstruction errors δc_n and calibration error $\delta \mathbf{B}_C$, the leading bound is

$$\|\delta u_n\| \leq \frac{\|\delta c_n\|}{\sigma_{\text{obs},C}} + \frac{\|\delta \mathbf{B}_C\| \|c_n\|}{\sigma_{\text{obs},C}^2} + O(\delta^2).$$

Poor port observability therefore amplifies both measurement noise and port-calibration errors. When a movable or multiplexed local probe is available, one fixed reference port can reconstruct all relative mode amplitudes. Let port r couple to a reference site with $u_{n,r} \neq 0$. Near an isolated weak-loss pole,

$$G_{ir}^R(\Omega) \simeq \frac{u_{n,i}u_{n,r}^*}{\Omega - E_n + i\gamma_n/2}. \quad (2872)$$

Thus

$$\frac{u_{n,i}}{u_{n,j}} = \frac{\text{Res}_{E_n} G_{ir}^R}{\text{Res}_{E_n} G_{jr}^R}. \quad (2873)$$

The result reconstructs both relative amplitude and phase. A different reference is used if the selected mode is dark at site r . Individual eigenvectors inside an exactly or nearly degenerate multiplet are gauge dependent. The physical object is the cluster projector

$$P_C = \sum_{n \in C} |u_n\rangle\langle u_n|. \quad (2874)$$

The sum of scattering residues gives the port-projected projector:

$$\mathbf{M}_C^{\text{port}} = i \sum_{n \in C} \mathbf{R}_n^S = \mathbf{B}P_C\mathbf{B}^\dagger. \quad (2875)$$

Equivalently,

$$\mathbf{M}_C^{\text{port}} = \frac{1}{2\pi i} \oint_{\mathcal{C}_C} \mathbf{B} \mathbf{G}^R(z) \mathbf{B}^\dagger dz. \quad (2876)$$

If $\mathbf{B}$ has full column rank,

$$P_C = \mathbf{B}^+ \mathbf{M}_C^{\text{port}} (\mathbf{B}^+)^{\dagger}. \quad (2877)$$

The reconstruction error obeys

$$\begin{aligned} \|\delta P_C\| &\lesssim \|\mathbf{B}^+\|^2 \|\delta \mathbf{M}_C^{\text{port}}\| \\ &+ 2\|\mathbf{B}^+\|^3 \|\mathbf{M}_C^{\text{port}}\| \|\delta \mathbf{B}\|. \end{aligned} \quad (2878)$$

If only a projected subspace is observable, we report $\mathbf{B} P_C \mathbf{B}^\dagger$ and do not claim a complete real-space projector. Suppose

$$\|\widehat{\mathbf{G}}^R(z) - \mathbf{G}^R(z)\| \leq \varepsilon_G \quad (2879)$$

uniformly on a contour $\mathcal{C}_C$ of length ℓ_C . The reconstructed spectral projector satisfies

$$\|\widehat{P}_C - P_C\| \leq \frac{\ell_C}{2\pi} \varepsilon_G. \quad (2880)$$

If the right-hand side is smaller than one, the reconstructed and exact projectors have the same rank. From a reconstructed simple mode, we calculate

$$\begin{aligned} p_{n,i} &= |u_{n,i}|^2, \quad \text{IPR}_n = \sum_i p_{n,i}^2, \\ P_{L,n} &= \langle u_n | (P_1 - P_2) | u_n \rangle, \end{aligned} \quad (2881)$$

and

$$W_{\perp,n} = \langle u_n | \mathbf{H}_{\perp} | u_n \rangle. \quad (2882)$$

From a cluster projector, we calculate

$$\begin{aligned} Q_x(P_C) &= \frac{\text{Tr}(\Pi_x P_C)}{r_C}, \\ p_{\text{cell}}(P_C) &= \left[N_c \sum_x Q_x(P_C)^2 \right]^{-1}, \end{aligned} \quad (2883)$$

and

$$P_{L,C} = \frac{1}{r_C} \text{Tr}[(P_1 - P_2) P_C]. \quad (2884)$$

Let U_g be a known microscopic symmetry permutation or unitary. For a simple mode, define

$$\chi_n(g) = \langle u_n | U_g | u_n \rangle. \quad (2885)$$

For a degenerate cluster, the gauge-invariant normalized character is

$$\chi_C(g) = \frac{1}{r_C} \text{Tr}(P_C U_g). \quad (2886)$$

Let

$$\Psi_C = (|u_1\rangle \cdots |u_{r_C}\rangle) \quad (2887)$$

be any orthonormal basis of $\text{Ran } P_C$. The reconstructed symmetry matrices are

$$\rho_C(g) = \Psi_C^\dagger U_g \Psi_C. \quad (2888)$$

Under a basis change $\Psi_C \mapsto \Psi_C V$,

$$\rho_C(g) \mapsto V^\dagger \rho_C(g) V. \quad (2889)$$

Characters, eigenvalues, and commutator norms are therefore basis-independent. A necessary invariance test is

$$\varepsilon_{C,\text{inv}}(g) = \|(I - P_C) U_g P_C\|. \quad (2890)$$

A representation label is assigned only when

$$\max_g \varepsilon_{C,\text{inv}}(g) \leq \varepsilon_{\text{inv}}^{\text{tol}}. \quad (2891)$$

The steady-state field at one real frequency is

$$|\psi_{\text{drv}}(\Omega)\rangle = i \mathbf{G}^R(\Omega) |f\rangle. \quad (2892)$$

Near several overlapping resonances,

$$|\psi_{\text{drv}}\rangle = i \sum_n \frac{|u_n^R\rangle \langle u_n^L| f\rangle}{\Omega - z_n}. \quad (2893)$$

A spatial field image taken at one frequency is therefore generally a coherent superposition of modes. We identify an eigenmode profile from pole residues or a cluster projector, not directly from a single steady-state image.

E. Abelian and non-Abelian experimental signatures

Let the finite normal cover have translation quotient

$$Q_N = \Gamma_M / \Gamma_N. \quad (2894)$$

Use the basis

$$|q, s\rangle, \quad q \in Q_N, \quad s = 1, \dots, \nu_M. \quad (2895)$$

For an irreducible unitary representation

$$\rho : Q_N \longrightarrow U(d_\rho), \quad (2896)$$

define the phase- and amplitude-coded drive

$$|F_{\rho;\alpha\beta,s}\rangle = \sqrt{\frac{d_\rho}{|Q_N|}} \sum_{q \in Q_N} \rho(q)_{\alpha\beta}^* |q, s\rangle, \quad (2897)$$

where $1 \leq \alpha, \beta \leq d_\rho$. Schur orthogonality gives

$$\sum_{q \in Q_N} \rho(q)_{\alpha\beta}^* \sigma(q)_{\gamma\delta} = \frac{|Q_N|}{d_\rho} \delta_{\rho\sigma} \delta_{\alpha\gamma} \delta_{\beta\delta}. \quad (2898)$$

Consequently,

$$\langle F_{\rho;\alpha\beta,s} | F_{\sigma;\gamma\delta,s'} \rangle = \delta_{\rho\sigma} \delta_{\alpha\gamma} \delta_{\beta\delta} \delta_{ss'}. \quad (2899)$$

The complete drive set resolves the full finite-cover Hilbert space:

$$\sum_{\rho \in \widehat{Q}_N} \sum_{\alpha,\beta=1}^{d_\rho} \sum_{s=1}^{\nu_M} |F_{\rho;\alpha\beta,s}\rangle \langle F_{\rho;\alpha\beta,s}| = I_D. \quad (2900)$$

The state-count identity is

$$\sum_{\rho \in \widehat{Q}_N} d_\rho^2 = |Q_N|. \quad (2901)$$

For $d_\rho = 1$, write the representation as a character

$$\chi : Q_N \longrightarrow U(1). \quad (2902)$$

Equation (2897) becomes

$$|F_{\chi,s}\rangle = \frac{1}{\sqrt{|Q_N|}} \sum_{q \in Q_N} \chi(q)^* |q,s\rangle. \quad (2903)$$

This is the finite-cover analogue of a phase-coded Bloch drive. The spatially uniform drive is

$$|F_{\text{triv},s}\rangle = \frac{1}{\sqrt{|Q_N|}} \sum_{q \in Q_N} |q,s\rangle. \quad (2904)$$

It transforms in the trivial representation and is exactly orthogonal to every nontrivial irrep drive. Therefore absence from uniform transmission $\not\Rightarrow$ absence from the physical spectrum. A uniform global drive can be completely blind to non-Abelian sectors. For one drive vector, define

$$A_{\rho;\alpha\beta,s}(\Omega) = \langle F_{\rho;\alpha\beta,s} | A(\Omega) | F_{\rho;\alpha\beta,s} \rangle. \quad (2905)$$

The normalized sector spectral function is

$$A_\rho(\Omega) = \frac{1}{\nu_M d_\rho^2} \sum_{s=1}^{\nu_M} \sum_{\alpha,\beta=1}^{d_\rho} A_{\rho;\alpha\beta,s}(\Omega). \quad (2906)$$

It satisfies

$$\int_{-\infty}^{\infty} A_\rho(\Omega) d\Omega = 1. \quad (2907)$$

The complete normalized finite-cover DOS is reconstructed exactly as

$$\varrho_N^{\text{full}}(\Omega) = \sum_{\rho \in \widehat{Q}_N} \frac{d_\rho^2}{|Q_N|} A_\rho(\Omega). \quad (2908)$$

The weights are the physical regular-representation state fractions. Equal weighting of inequivalent irreps is incorrect. Define

$$\widehat{Q}_N^{(1)} = \{\rho : d_\rho = 1\}, \quad (2909)$$

and

$$\widehat{Q}_N^{(>1)} = \{\rho : d_\rho > 1\}. \quad (2910)$$

Their state fractions are

$$C_{\text{Ab},N} = \sum_{\rho \in \widehat{Q}_N^{(1)}} \frac{d_\rho^2}{|Q_N|}, \quad (2911)$$

and

$$C_{\text{NA},N} = 1 - C_{\text{Ab},N}. \quad (2912)$$

The normalized Abelian aggregate is

$$A_{\text{Ab}}(\Omega) = \frac{1}{C_{\text{Ab},N}} \sum_{\rho \in \widehat{Q}_N^{(1)}} \frac{d_\rho^2}{|Q_N|} A_\rho(\Omega). \quad (2913)$$

The normalized non-Abelian aggregate is

$$A_{\text{NA}}(\Omega) = \frac{1}{C_{\text{NA},N}} \sum_{\rho \in \widehat{Q}_N^{(>1)}} \frac{d_\rho^2}{|Q_N|} A_\rho(\Omega). \quad (2914)$$

The full DOS decomposes as

$$\varrho_N^{\text{full}} = C_{\text{Ab},N} A_{\text{Ab}} + C_{\text{NA},N} A_{\text{NA}}. \quad (2915)$$

If the Hamiltonian and damping preserve the quotient translation group, their resolvents commute with every quotient translation. Hence inequivalent isotypic sectors do not mix:

$$P_\rho G^R P_\sigma = 0, \quad \rho \neq \sigma. \quad (2916)$$

Therefore

$$\langle F_{\sigma;\gamma\delta,s'} | G^R | F_{\rho;\alpha\beta,s} \rangle = 0, \quad \rho \neq \sigma. \quad (2917)$$

Define the integrated cross-sector leakage

$$\mathcal{L}_{\rho \rightarrow \sigma} = \frac{\int_{\mathcal{I}} \sum_{\alpha\beta\gamma\delta ss'} |\langle F_{\sigma;\gamma\delta,s'} | G^R | F_{\rho;\alpha\beta,s} \rangle|^2 d\Omega}{\int_{\mathcal{I}} \sum_{\alpha\beta ss'} |\langle F_{\rho;\alpha\beta,s} | G^R | F_{\rho;\alpha\beta,s} \rangle|^2 d\Omega}. \quad (2918)$$

For exact symmetry and exact drive synthesis,

$$\mathcal{L}_{\rho \rightarrow \sigma} = 0. \quad (2919)$$

A nonzero measured value quantifies translation-symmetry breaking, drive leakage, or port-calibration error. For an implemented normalized drive $|f\rangle$, define its irrep fidelity

$$\mathcal{F}_\rho^{\text{drv}} = \langle f | P_\rho | f \rangle. \quad (2920)$$

Since the resolvent is block diagonal,

$$\langle f | G^R | f \rangle = \sum_{\sigma} \langle P_\sigma f | G^R | P_\sigma f \rangle. \quad (2921)$$

The measured response is therefore a weighted sector mixture whenever $\mathcal{F}_\rho^{\text{drv}} < 1$. A sector-specific conclusion requires

$$\mathcal{F}_\rho^{\text{drv},L} \geq 1 - \varepsilon_{\text{drv}}. \quad (2922)$$

For a reconstructed simple mode,

$$w_{n,\rho} = \langle u_n | P_\rho | u_n \rangle. \quad (2923)$$

For a target projector,

$$w_{C,\rho} = \frac{1}{r_C} \text{Tr}(P_C P_\rho). \quad (2924)$$

The completeness sum is

$$\sum_{\rho \in \hat{Q}_N} w_{C,\rho} = 1. \quad (2925)$$

The total non-Abelian weight is

$$w_C^{\text{NA}} = \sum_{\rho: d_\rho > 1} w_{C,\rho}. \quad (2926)$$

Let the reconstructed mode amplitudes be

$$\psi_s(q) = \langle q, s | \psi \rangle. \quad (2927)$$

The normalized finite-group Fourier transform is

$$\tilde{\psi}_s(\rho) = \sqrt{\frac{d_\rho}{|Q_N|}} \sum_{q \in Q_N} \psi_s(q) \rho(q)^\dagger. \quad (2928)$$

Plancherel theory gives

$$\sum_{q,s} |\psi_s(q)|^2 = \sum_{\rho,s} \left\| \tilde{\psi}_s(\rho) \right\|_F^2. \quad (2929)$$

The representation weight is

$$w_\rho^{\text{FT}} = \frac{\sum_s \left\| \tilde{\psi}_s(\rho) \right\|_F^2}{\sum_{\sigma,s} \left\| \tilde{\psi}_s(\sigma) \right\|_F^2}. \quad (2930)$$

This provides an independent reconstruction of Eq. (2923). Let

$$\rho_C(g) = \Psi_C^\dagger U_g \Psi_C \quad (2931)$$

be the measured translation matrices on an invariant cluster. Define

$$\mathcal{N}_C = \max_{g,h \in \mathcal{G}_{\text{gen}}} \frac{\left\| [\rho_C(g), \rho_C(h)] \right\|_F}{2\sqrt{r_C}}. \quad (2932)$$

It obeys $0 \leq \mathcal{N}_C \leq 1$. Every one-dimensional Abelian sector has $\mathcal{N}_C = 0$. A resolved value $\mathcal{N}_C^L > 0$ is a direct mode-tomography witness of a noncommuting translation image. Choose two physical hopping paths connecting the same initial and final channels. Let their group

words be w_1 and w_2 , and define the relative commutator word

$$c = w_1^{-1} w_2 \in [Q_N, Q_N]. \quad (2933)$$

Let their amplitudes be t_1 and t_2 , with a controllable relative phase ϕ . In irrep ρ , the path operator is

$$A_\rho(\phi) = t_1 I_{d_\rho} + e^{i\phi} t_2 \rho(c). \quad (2934)$$

The representation-averaged response power is

$$\begin{aligned} \mathcal{I}_\rho(\phi) &= \|A_\rho(\phi)\|_F^2 = d_\rho (|t_1|^2 + |t_2|^2) \\ &\quad + 2 \text{Re} [e^{i\phi} t_1^* t_2 \chi_\rho(c)]. \end{aligned} \quad (2935)$$

Here

$$\chi_\rho(c) = \text{Tr} \rho(c). \quad (2936)$$

For every Abelian character, $\chi(c) = 1$. Define the active commutator defect

$$\eta_c(\rho) = 1 - \frac{1}{d_\rho} \text{Re} \chi_\rho(c). \quad (2937)$$

For equal real path amplitudes $t_1 = t_2 = t_p$,

$$\mathcal{I}_{\text{Ab}}(0) - \mathcal{I}_\rho(0) = 2d_\rho |t_p|^2 \eta_c(\rho). \quad (2938)$$

The fringe visibility is

$$\mathcal{V}_\rho = \frac{\mathcal{I}_\rho^{\text{max}} - \mathcal{I}_\rho^{\text{min}}}{\mathcal{I}_\rho^{\text{max}} + \mathcal{I}_\rho^{\text{min}}} = \frac{2|t_1 t_2| |\chi_\rho(c)|}{d_\rho (|t_1|^2 + |t_2|^2)}. \quad (2939)$$

The fringe phase gives

$$\arg [t_1^* t_2 \chi_\rho(c)]. \quad (2940)$$

A matrix-resolved drive set can additionally reconstruct the eigenphases of $\rho(c)$, rather than only its trace. For the same target ancestry, define the measured Abelian bandwidth

$$W_C^{\text{Ab}} = \max_{\chi \in \hat{Q}_N^{(1)}, n \in C_\chi} E_{n,\chi} - \min_{\chi \in \hat{Q}_N^{(1)}, n \in C_\chi} E_{n,\chi}. \quad (2941)$$

The complete representation-resolved bandwidth is

$$W_C^{\text{full}} = \max_{\rho \in \hat{Q}_N, n \in C_\rho} E_{n,\rho} - \min_{\rho \in \hat{Q}_N, n \in C_\rho} E_{n,\rho}. \quad (2942)$$

The corresponding global gaps are denoted by Δ_C^{Ab} , Δ_C^{full} . The experimentally reconstructed flat-angle sets are

$$\hat{\Theta}_{\text{flat}}^{\text{Ab}} = \arg \max_{\theta} \widehat{\mathcal{M}}^{\text{Ab}}(\theta), \quad (2943)$$

and

$$\hat{\Theta}_{\text{flat}}^{\text{full}} = \arg \max_{\theta} \widehat{\mathcal{M}}^{\text{full}}(\theta). \quad (2944)$$

For unique optima, the measured non-Abelian shift is

$$\delta\theta_{\text{NA}}^{\text{exp}} = \hat{\theta}_{\text{flat}}^{\text{full}} - \hat{\theta}_{\text{flat}}^{\text{Ab}}. \quad (2945)$$

It is resolved when

$$|\delta\theta_{\text{NA}}^{\text{exp}}| > \delta\theta_{\text{full}} + \delta\theta_{\text{Ab}} + \delta\theta_{\text{cal}}. \quad (2946)$$

Let $\mathcal{G}_C^{\text{Ab}}$ be the frequency interval identified as an Abelian isolation gap. The non-Abelian gap-filling weight is

$$\mathcal{F}_{\text{gap},C}^{\text{NA}} = C_{\text{NA},N} \int_{\mathcal{G}_C^{\text{Ab}}} A_{\text{NA}}(\Omega) d\Omega. \quad (2947)$$

Let $\mathcal{F}_{\text{tail}}^U$ be the maximum calculated Lorentzian leakage from poles lying outside the gap. Genuine gap filling is established when

$$\mathcal{F}_{\text{gap},C}^{\text{NA},L} > \mathcal{F}_{\text{tail}}^U. \quad (2948)$$

Let $\mathcal{I}_C$ be a candidate non-Abelian target interval. Define

$$I_{\text{NA}} = \int_{\mathcal{I}_C} A_{\text{NA}}(\Omega) d\Omega, \quad (2949)$$

and

$$I_{\text{Ab}}^{\text{max}} = \max_{\chi \in \hat{\mathcal{Q}}_N^{(1)}} \int_{\mathcal{I}_C} A_{\chi}(\Omega) d\Omega. \quad (2950)$$

The spectral contrast is

$$\mathcal{C}_{\text{spec}}^{\text{NA}} = \frac{I_{\text{NA}} - I_{\text{Ab}}^{\text{max}}}{I_{\text{NA}} + I_{\text{Ab}}^{\text{max}}}. \quad (2951)$$

It satisfies $-1 \leq \mathcal{C}_{\text{spec}}^{\text{NA}} \leq 1$. A non-Abelian-only resonance requires a positive lower confidence bound near unity, together with $w_C^{\text{NA},L} > 0$ and $\Delta_C^{\text{res},L} > 0$.

For every candidate, we record

$$\begin{aligned} \mathfrak{C}_{\text{NA}}^{\text{exp}} = & \left(\rho, d_{\rho}, \mathcal{F}_{\rho}^{\text{drv}}, w_C^{\text{NA}}, \mathcal{N}_C, \right. \\ & \eta_c(\rho), \mathcal{V}_{\rho}, \mathcal{L}_{\rho \rightarrow \sigma}, \\ & W_C^{\text{Ab}}, W_C^{\text{full}}, \Delta_C^{\text{Ab}}, \Delta_C^{\text{full}}, \\ & \theta_{\text{flat}}^{\text{Ab}}, \theta_{\text{flat}}^{\text{full}}, \delta\theta_{\text{NA}}^{\text{exp}}, \\ & \mathcal{F}_{\text{gap},C}^{\text{NA}}, \mathcal{C}_{\text{spec}}^{\text{NA}}, \\ & \left. \varepsilon_{\text{frame}}, \varepsilon_P, \varepsilon_{\text{cal}}, C_{\text{rep},N} \right). \end{aligned} \quad (2952)$$

XIII. PHOTON-MEDIATED QUBIT INTERACTIONS

We now couple transmon qubits to the twisted hyperbolic resonator bilayer and determine the coherent and dissipative interactions generated by virtual-photon exchange. The resulting qubit–qubit interaction structure

is analysed independently of the photonic flat-band classification, because spectral flatness alone does not determine the spatial range, focusing properties, frustration pattern, or representation content of the mediated interaction. Photonic flatness refers specifically to the existence of a spectrally isolated target band C with $W_C \ll \Delta_C$. Target-band mediation is a stronger and separate statement: it requires the qubit self-energy to be dominated by the projector onto that band, so that

$$\Sigma_q \simeq \frac{\mathbf{V}_q P_C \mathbf{V}_q^\dagger}{\Delta_{qC}}.$$

Only after this projector-dominance condition is established can the effective qubit interaction be attributed primarily to virtual photons within the declared photonic sector. The spatial extent of the induced interaction is diagnosed independently through the interaction range R_J . A large R_J constitutes evidence for long-range mediation, whereas a flat photonic band by itself does not imply such a range enhancement. Likewise, interaction focusing is identified only when the pair-averaged interaction profile develops a resolved interior spatial maximum. This criterion distinguishes genuine focusing from either uniform coupling or a monotonic spatial decay and therefore cannot be inferred from photonic flatness alone. Interaction frustration is characterised separately through gauge-invariant cycle phases of the effective coupling matrix. The interaction is classified as frustrated only when these cycle phases are nontrivial; neither a narrow photonic bandwidth nor a long interaction range is sufficient to establish frustration. Finally, non-Abelian mediation requires a representation-sensitive effect: higher-dimensional translation sectors must modify the effective interaction matrix $\mathbf{J}$ by an amount that exceeds the complete theoretical and experimental uncertainty budget. In the absence of such a resolved change, the mediated interaction cannot be attributed specifically to non-Abelian photonic sectors. Accordingly, photonic flatness, target-band mediation, long-range interaction, interaction focusing, interaction frustration, and non-Abelian mediation are treated as logically distinct physical claims. None of the latter properties is inferred solely from the existence of a photonic flat band; each requires its own independent diagnostic and verification.

The implication

$$W_C \rightarrow 0 \implies \Sigma_q \propto \mathbf{V}_q P_C \mathbf{V}_q^\dagger \quad (2953)$$

holds only under a controlled target-band detuning and a suppressed complementary-band remainder. It fixes the energy denominator but does not determine whether the sampled projector is local, extended, shell-enhanced, anisotropic, or frustrated. All Hamiltonians below are written in angular-frequency units. Physical energies are obtained by multiplication by $\hbar$.

TABLE VIII. Experimentally distinguishable Abelian and non-Abelian spectral outcomes. All rows require calibrated drive fidelity, projector continuation, and complete regular-representation weighting.

Outcome	Microwave signature	Supported conclusion
I	Abelian and complete weighted spectra peak at the same twist	Abelian sectors are sufficient for the declared t
II	The complete weighted optimum is displaced beyond angular uncertainty	Non-Abelian sectors shift the physical flat-ban
III	Non-Abelian drives reveal added width or spectral weight in an Abelian gap	The Abelian candidate is not a globally isolate
IV	A $d_\rho > 1$ drive excites an isolated narrow target absent from all character drives	The measured target is intrinsically translation

A. Transmon placement

Let the finite resonator circuit contain D_γ single-photon modes. The photonic Hamiltonian is

$$\hat{H}_\gamma = \hbar \mathbf{a}^\dagger \mathbf{H}_\gamma \mathbf{a}, \quad \mathbf{H}_\gamma = \mathbf{H}_\gamma^\dagger, \quad (2954)$$

where

$$\mathbf{a} = (a_1 \ \cdots \ a_{D_\gamma})^\top. \quad (2955)$$

We introduce N_q transmons. The weakly anharmonic transmon Hamiltonian is

$$\frac{\hat{H}_q}{\hbar} = \sum_{a=1}^{N_q} \left[\omega_a b_a^\dagger b_a - \frac{\alpha_a}{2} b_a^{\dagger 2} b_a^2 \right], \quad (2956)$$

where $\alpha_a > 0$ is the magnitude of the negative transmon anharmonicity. The bare qubit-frequency matrix is

$$\boldsymbol{\Omega}_q = \text{diag}(\omega_1, \dots, \omega_{N_q}). \quad (2957)$$

The rotating-wave light-matter coupling is

$$\frac{\hat{H}_{q\gamma}}{\hbar} = \sum_{a=1}^{N_q} [b_a^\dagger \langle v_a | \mathbf{a} + \mathbf{a}^\dagger | v_a \rangle b_a], \quad (2958)$$

where

$$\langle v_a | = \sum_{i=1}^{D_\gamma} g_{ai} \langle i | \quad (2959)$$

contains the calibrated transmon-resonator coupling amplitudes. The complete placement matrix is

$$\mathbf{V}_q = \sum_{a=1}^{N_q} |a\rangle \langle v_a| \in \mathbb{C}^{N_q \times D_\gamma}. \quad (2960)$$

Its entries are $[\mathbf{V}_q]_{ai} = g_{ai}$. For a transmon coupled dominantly to one resonator mode

$$i_a = (q_a, \ell_a, s_a), \quad (2961)$$

where q_a is a finite-cover cell, $\ell_a \in \{1, 2\}$ is the layer, and s_a is the intracell orbital, we write

$$\langle v_a | = g_a \langle i_a |. \quad (2962)$$

The local coupling strength is

$$g_a = |g_a| e^{i\varphi_a}. \quad (2963)$$

The phase φ_a can be absorbed into the transmon operator when the transmon participates in only one coupling channel. Relative phases remain physical when one transmon couples to several resonators. For a distributed capacitive coupler,

$$\langle v_a | = \sum_{i \in \mathcal{N}_a} g_{ai} \langle i |, \quad (2964)$$

where $\mathcal{N}_a$ is the explicitly specified coupling neighborhood. No pointlike approximation is used when the measured values $\{g_{ai}\}$ are available. For corresponding resonator modes i in the two layers, define $|i, 1\rangle$, $|i, 2\rangle$. The layer-even and layer-odd coupling vectors are

$$|v_{a,+}\rangle = \frac{g_a}{\sqrt{2}} (|i, 1\rangle + |i, 2\rangle), \quad (2965)$$

and

$$|v_{a,-}\rangle = \frac{g_a}{\sqrt{2}} (|i, 1\rangle - |i, 2\rangle). \quad (2966)$$

At an exact layer-parity point, these two transmons couple to disjoint photonic parity sectors. At a generic centered twist, the exact symmetry can be a combined layer-exchanging spatial operation

$$\tilde{\sigma} = P_L U_\sigma. \quad (2967)$$

A symmetry-adapted coupling vector then satisfies

$$\tilde{\sigma} |v_{a,\lambda}\rangle = \lambda |v_{a,\lambda}\rangle, \quad |\lambda| = 1. \quad (2968)$$

We distinguish two placement classes. A sparse probe configuration contains a fixed number of transmons: $N_q = O(1)$ as the finite-cover size grows. It measures local matrix elements of the photonic resolvent. A translation-complete configuration places one identical transmon on every site in a declared quotient orbit:

$$\mathcal{Q}_{\text{orb}} = \{(q, s_q) : q \in \mathcal{Q}_N\}. \quad (2969)$$

For identical local coupling g , the placement intertwines the finite translation representations:

$$\mathbf{V}_q U_\gamma^{(\gamma)} = U_\gamma^{(q)} \mathbf{V}_q. \quad (2970)$$

The resulting qubit interaction matrix is exactly translation covariant. A sparse configuration need not preserve translation symmetry. It can still sample non-Abelian photonic sectors because a local site state is a superposition of all finite-quotient irreducible representations. Let P_C be a rank- r_C target photonic projector. For a nonzero coupling vector, define

$$|\widehat{v}_a\rangle = \frac{|v_a\rangle}{\|v_a\|}. \quad (2971)$$

The target-band placement weight is

$$p_{a,C} = \langle \widehat{v}_a | P_C | \widehat{v}_a \rangle. \quad (2972)$$

It obeys $0 \leq p_{a,C} \leq 1$. A placement satisfying $p_{a,C} \simeq 0$ is dark to the target band even when the global photonic DOS is large. For two transmons, define the target projector bridge

$$B_{ab,C} = \langle v_a | P_C | v_b \rangle. \quad (2973)$$

The Cauchy–Schwarz inequality gives

$$|B_{ab,C}|^2 \leq \langle v_a | P_C | v_a \rangle \langle v_b | P_C | v_b \rangle. \quad (2974)$$

A nonzero local DOS at both transmon sites is therefore necessary but not sufficient for a large off-diagonal interaction. The relative phase and projector coherence between the two placements remain essential. Define the target placement matrix

$$\mathbf{V}_{q,C} = \mathbf{V}_q P_C. \quad (2975)$$

Its positive-semidefinite Gram matrix is

$$\mathbf{K}_C = \mathbf{V}_q P_C \mathbf{V}_q^\dagger = \mathbf{V}_{q,C} \mathbf{V}_{q,C}^\dagger. \quad (2976)$$

The diagonal entries are

$$[\mathbf{K}_C]_{aa} = \langle v_a | P_C | v_a \rangle, \quad (2977)$$

and the off-diagonal entries are

$$[\mathbf{K}_C]_{ab} = B_{ab,C}. \quad (2978)$$

Let $s_{1,C} \geq s_{2,C} \geq \dots$ be the singular values of $\mathbf{V}_{q,C}$. The nonzero eigenvalues of $\mathbf{K}_C$ are $s_{\mu,C}^2$. The number of independently addressable target channels is

$$r_{\text{addr},C} = \text{rank } \mathbf{V}_{q,C} \leq \min \{N_q, r_C\}. \quad (2979)$$

To address every target direction, a necessary condition is $N_q \geq r_C$. A sufficient condition is $s_{r_C,C} > 0$. The target placement condition number is

$$\kappa_{\text{pl},C} = \frac{s_{1,C}}{s_{r_{\text{addr},C},C}}. \quad (2980)$$

A large value indicates that some target combinations are nearly dark and that the mediated-interaction eigenvalues will be strongly nonuniform even for an exactly flat photonic band. Let

$$\Omega_{\min} = \min \{\omega_a, E_n\} \quad (2981)$$

over the retained transmon and photon modes. The rotating-wave approximation requires

$$\frac{\|\mathbf{V}_q\|}{\Omega_{\min}} \ll 1. \quad (2982)$$

Counter-rotating corrections are included as an additional Bloch–Siegert self-energy if this ratio is not negligible. Projection of each transmon onto its two lowest levels requires

$$\max \{ \|\mathbf{J}\|, \|\mathbf{\Gamma}^{\text{med}}\|, \Omega_{\text{drive}} \} \ll \min_a \alpha_a, \quad (2983)$$

unless the full Duffing Hilbert space is retained explicitly.

B. Effective interaction

In the one-excitation sector, use the ordered basis

$$(|1_q\rangle, \dots, |N_{q,q}\rangle; |1_\gamma\rangle, \dots, |D_{\gamma,\gamma}\rangle). \quad (2984)$$

The exact block Hamiltonian is

$$\mathbb{H}_1 = \begin{pmatrix} \Omega_q & \mathbf{V}_q \\ \mathbf{V}_q^\dagger & \mathbf{H}_\gamma \end{pmatrix}. \quad (2985)$$

Let an eigenvector be

$$|\Psi\rangle = \begin{pmatrix} |c_q\rangle \\ |\psi_\gamma\rangle \end{pmatrix}. \quad (2986)$$

The equation

$$\mathbb{H}_1 |\Psi\rangle = z |\Psi\rangle \quad (2987)$$

gives

$$(z - \Omega_q) |c_q\rangle = \mathbf{V}_q |\psi_\gamma\rangle, \quad (2988)$$

and

$$(z - \mathbf{H}_\gamma) |\psi_\gamma\rangle = \mathbf{V}_q^\dagger |c_q\rangle. \quad (2989)$$

For

$$z \notin \text{Spec } \mathbf{H}_\gamma, \quad (2990)$$

define

$$\mathbf{G}_\gamma(z) = (z - \mathbf{H}_\gamma)^{-1}. \quad (2991)$$

Then

$$|\psi_\gamma\rangle = \mathbf{G}_\gamma(z) \mathbf{V}_q^\dagger |c_q\rangle. \quad (2992)$$

Substitution gives the exact nonlinear qubit-space problem

$$[z - \Omega_q - \Sigma_q(z)] |c_q\rangle = 0, \quad (2993)$$

where

$$\Sigma_q(z) = \mathbf{V}_q \mathbf{G}_\gamma(z) \mathbf{V}_q^\dagger. \quad (2994)$$

Equation (2994) is the exact single-excitation photon-mediated interaction before any static, dispersive, flat-band, or continuum approximation. For pointlike transmon couplings,

$$\langle v_a | = g_a \langle i_a |, \quad (2995)$$

the self-energy matrix element is

$$[\Sigma_q(z)]_{ab} = g_a g_b^* G_{i_a i_b}(z). \quad (2996)$$

The diagonal element $[\Sigma_q]_{aa}$ is the photonic Lamb shift. The off-diagonal element

$$[\Sigma_q]_{ab}, \quad a \neq b, \quad (2997)$$

is the photon-mediated exchange amplitude. Thus the frequently used formula

$$J_{ab}(\omega_q) = g_a g_b^* G_{i_a i_b}(\omega_q) \quad (2998)$$

is valid as a coherent Hermitian interaction only when ω_q lies in the resolvent set of the lossless photonic Hamiltonian. In a lossy or resonant system, the same Green function generates both coherent exchange and collective decay. Let

$$\mathbf{H}_\gamma |u_n\rangle = E_n |u_n\rangle, \quad \langle u_n | u_m \rangle = \delta_{nm}. \quad (2999)$$

The Green operator is

$$\mathbf{G}_\gamma(z) = \sum_n \frac{|u_n\rangle \langle u_n|}{z - E_n}. \quad (3000)$$

Define the qubit-mode coupling

$$g_{an} = \langle v_a | u_n \rangle. \quad (3001)$$

The exact self-energy becomes

$$[\Sigma_q(z)]_{ab} = \sum_n \frac{g_{an} g_{bn}^*}{z - E_n}. \quad (3002)$$

For local couplings,

$$g_{an} = g_a u_n(i_a). \quad (3003)$$

Therefore

$$[\Sigma_q(z)]_{ab} = g_a g_b^* \sum_n \frac{u_n(i_a) u_n^*(i_b)}{z - E_n}. \quad (3004)$$

A large global DOS does not guarantee a large J_{ab} , because the modal amplitudes and their relative phases at the two transmon positions can cancel. For unequal qubit frequencies, define

$$\Delta_{an} = \omega_a - E_n. \quad (3005)$$

Assume

$$\epsilon_{\text{disp}} = \max_{a,n} \frac{|g_{an}|}{|\Delta_{an}|} \ll 1 \quad (3006)$$

for every retained virtual mode. To second order, the Lamb shift is

$$\delta\omega_a = \sum_n \frac{|g_{an}|^2}{\omega_a - E_n}. \quad (3007)$$

The Hermitian exchange between two nonidentical transmons is

$$J_{ab}^{\text{SW}} = \frac{1}{2} \sum_n g_{an} g_{bn}^* \left[\frac{1}{\omega_a - E_n} + \frac{1}{\omega_b - E_n} \right], \quad a \neq b. \quad (3008)$$

For identical qubit frequencies

$$\omega_a = \omega_b = \omega_q, \quad (3009)$$

Eq. (3008) reduces to

$$J_{ab}^{\text{SW}} = [\Sigma_q(\omega_q)]_{ab} = g_a g_b^* G_{i_a i_b}(\omega_q). \quad (3010)$$

The exact self-energy is energy dependent. For two frequencies z and z_0 in the photonic resolvent set, the resolvent identity gives

$$\mathbf{G}_\gamma(z) - \mathbf{G}_\gamma(z_0) = (z_0 - z) \mathbf{G}_\gamma(z) \mathbf{G}_\gamma(z_0). \quad (3011)$$

Therefore

$$\|\Sigma_q(z) - \Sigma_q(z_0)\| \leq \frac{\|\mathbf{V}_q\|^2 |z - z_0|}{\delta_\gamma(z) \delta_\gamma(z_0)}, \quad (3012)$$

where

$$\delta_\gamma(z) = \text{dist}[z, \text{Spec } \mathbf{H}_\gamma]. \quad (3013)$$

A frequency-independent effective interaction is controlled only when the right-hand side is small relative to the interaction, qubit detuning, and target spectral gaps. Equation (2992) gives

$$\|\psi_\gamma\| \leq \frac{\|\mathbf{V}_q\|}{\delta_\gamma(z)} \|c_q\|. \quad (3014)$$

Define

$$\epsilon_{q\gamma} = \frac{\|\mathbf{V}_q\|}{\delta_\gamma(z)}. \quad (3015)$$

For a normalized full eigenstate, the qubit-sector weight satisfies

$$\|c_q\|^2 \geq \frac{1}{1 + \epsilon_{q\gamma}^2}. \quad (3016)$$

The perturbative spin description requires $\epsilon_{q\gamma} \ll 1$. Let the photonic retarded Green operator be

$$\mathbf{G}_\gamma^R(\omega) = \left[\omega I - \mathbf{H}_\gamma + \frac{i}{2} \mathbf{\Gamma}_\gamma \right]^{-1}, \quad (3017)$$

where $\mathbf{\Gamma}_\gamma \succeq 0$. The retarded qubit self-energy is

$$\Sigma_q^R(\omega) = \mathbf{V}_q \mathbf{G}_\gamma^R(\omega) \mathbf{V}_q^\dagger. \quad (3018)$$

We decompose it as

$$\mathbf{\Sigma}_q^R = \mathbf{J} - \frac{i}{2} \mathbf{\Gamma}^{\text{med}}, \quad (3019)$$

where

$$\mathbf{J} = \frac{1}{2} [\mathbf{\Sigma}_q^R + \mathbf{\Sigma}_q^{R\dagger}], \quad (3020)$$

and

$$\mathbf{\Gamma}^{\text{med}} = i [\mathbf{\Sigma}_q^R - \mathbf{\Sigma}_q^{R\dagger}]. \quad (3021)$$

The Green-function identity

$$\mathbf{G}_\gamma^R - \mathbf{G}_\gamma^A = -i \mathbf{G}_\gamma^R \mathbf{\Gamma}_\gamma \mathbf{G}_\gamma^A \quad (3022)$$

gives

$$\mathbf{\Gamma}^{\text{med}} = \mathbf{V}_q \mathbf{G}_\gamma^R \mathbf{\Gamma}_\gamma \mathbf{G}_\gamma^A \mathbf{V}_q^\dagger \succeq 0. \quad (3023)$$

The collective decay matrix is therefore positive semidefinite. After projecting each transmon to its two lowest levels, the coherent effective Hamiltonian is

$$\frac{\hat{H}_q^{\text{eff}}}{\hbar} = \sum_a \tilde{\omega}_a \sigma_a^+ \sigma_a^- + \sum_{a \neq b} J_{ab} \sigma_a^+ \sigma_b^-,$$

where the diagonal self-energy has been absorbed into

$$\tilde{\omega}_a = \omega_a + J_{aa}. \quad (3024)$$

The reduced qubit master equation is

$$\begin{aligned} \dot{\rho}_q = & -i \left[\hat{H}_q^{\text{eff}} / \hbar, \rho_q \right] \\ & + \sum_{a,b} \Gamma_{ab}^{\text{med}} \left[\sigma_b^- \rho_q \sigma_a^+ - \frac{1}{2} \{ \sigma_a^+ \sigma_b^-, \rho_q \} \right] \\ & + \mathcal{L}_{\text{intr}}[\rho_q], \end{aligned} \quad (3025)$$

where $\mathcal{L}_{\text{intr}}$ contains the independently measured intrinsic transmon relaxation and dephasing. The off-diagonal coherent interaction satisfies $J_{ab} = J_{ba}^*$, whereas the collective decay matrix satisfies $\Gamma_{ab}^{\text{med}} = \Gamma_{ba}^{\text{med}*}$. Define the off-diagonal interaction matrix

$$\mathbf{J}_{\text{off}} = \mathbf{J} - \text{diag}(\mathbf{J}). \quad (3026)$$

The RMS coherent exchange is

$$J_{\text{rms}} = \left[\frac{1}{N_q(N_q - 1)} \sum_{a \neq b} |J_{ab}|^2 \right]^{1/2}. \quad (3027)$$

The RMS collective decay is

$$\Gamma_{\text{rms}}^{\text{med}} = \left[\frac{1}{N_q(N_q - 1)} \sum_{a \neq b} |\Gamma_{ab}^{\text{med}}|^2 \right]^{1/2}. \quad (3028)$$

The coherent-to-dissipative ratio is

$$\mathcal{Q}_J = \frac{\|\mathbf{J}_{\text{off}}\|_{\text{F}}}{\max \{ \|\mathbf{\Gamma}_{\text{off}}^{\text{med}}\|_{\text{F}}, \Gamma_{\text{ref}} \}}. \quad (3029)$$

A large interaction amplitude does not define a useful coherent interaction if $\mathcal{Q}_J$ remains small. Let P_C project onto the photonic target cluster and define

$$Q_C = I - P_C. \quad (3030)$$

Because P_C is a spectral projector,

$$[P_C, \mathbf{H}_\gamma] = 0. \quad (3031)$$

The Green operator decomposes exactly:

$$\mathbf{G}_\gamma(z) = \mathbf{G}_C(z) + \mathbf{G}_{\bar{C}}(z), \quad (3032)$$

where

$$\mathbf{G}_C(z) = P_C (z - \mathbf{H}_\gamma)^{-1} P_C, \quad (3033)$$

and

$$\mathbf{G}_{\bar{C}}(z) = Q_C (z - \mathbf{H}_\gamma)^{-1} Q_C. \quad (3034)$$

The qubit self-energy decomposes as

$$\mathbf{\Sigma}_q = \mathbf{\Sigma}_C + \mathbf{\Sigma}_{\bar{C}}, \quad (3035)$$

with

$$\mathbf{\Sigma}_C = \mathbf{V}_q \mathbf{G}_C \mathbf{V}_q^\dagger, \quad (3036)$$

and

$$\mathbf{\Sigma}_{\bar{C}} = \mathbf{V}_q \mathbf{G}_{\bar{C}} \mathbf{V}_q^\dagger. \quad (3037)$$

Define the target spectral edges E_C^- , E_C^+ , the bandwidth

$$W_C = E_C^+ - E_C^-, \quad (3038)$$

the midpoint

$$\bar{E}_C = \frac{E_C^+ + E_C^-}{2}, \quad (3039)$$

and the half-width

$$r_C = \frac{W_C}{2}. \quad (3040)$$

Inside the target subspace, write

$$P_C \mathbf{H}_\gamma P_C = \bar{E}_C P_C + \mathbf{K}_C^{(\gamma)}, \quad (3041)$$

where

$$\|\mathbf{K}_C^{(\gamma)}\| \leq r_C. \quad (3042)$$

Define the qubit detuning from the target midpoint:

$$\Delta_{qC} = \omega_q - \bar{E}_C. \quad (3043)$$

If $|\Delta_{qC}| > r_C$, the target Green operator has the convergent expansion

$$\begin{aligned} \mathbf{G}_C(\omega_q) &= [\Delta_{qC} P_C - \mathbf{K}_C^{(\gamma)}]^{-1} \\ &= \frac{1}{\Delta_{qC}} \sum_{m=0}^{\infty} \left[\frac{\mathbf{K}_C^{(\gamma)}}{\Delta_{qC}} \right]^m P_C. \end{aligned} \quad (3044)$$

Therefore

$$\mathbf{G}_C(\omega_q) = \frac{P_C}{\Delta_{qC}} + \mathbf{R}_C^{(\text{disp})}, \quad (3045)$$

with

$$\|\mathbf{R}_C^{(\text{disp})}\| \leq \frac{r_C}{|\Delta_{qC}| (|\Delta_{qC}| - r_C)}. \quad (3046)$$

The target-mediated interaction is consequently

$$\Sigma_C(\omega_q) = \frac{\mathbf{V}_q P_C \mathbf{V}_q^\dagger}{\Delta_{qC}} + \mathbf{R}_{q,C}^{(\text{disp})}, \quad (3047)$$

where

$$\|\mathbf{R}_{q,C}^{(\text{disp})}\| \leq \|\mathbf{V}_q\|^2 \frac{r_C}{|\Delta_{qC}| (|\Delta_{qC}| - r_C)}. \quad (3048)$$

Let

$$\delta_{q\bar{C}} = \text{dist}[\omega_q, \text{Spec}(Q_C \mathbf{H}_\gamma Q_C)]. \quad (3049)$$

The complementary-band contribution obeys

$$\|\Sigma_{\bar{C}}\| \leq \frac{\|\mathbf{V}_q\|^2}{\delta_{q\bar{C}}}. \quad (3050)$$

Combining the two estimates gives

$$\left\| \Sigma_q - \frac{\mathbf{V}_q P_C \mathbf{V}_q^\dagger}{\Delta_{qC}} \right\| \leq \|\mathbf{V}_q\|^2 \left[\frac{r_C}{|\Delta_{qC}| (|\Delta_{qC}| - r_C)} + \frac{1}{\delta_{q\bar{C}}} \right].$$

If the target band is exactly flat,

$$P_C \mathbf{H}_\gamma P_C = E_C P_C, \quad (3051)$$

then

$$\mathbf{G}_C(\omega_q) = \frac{P_C}{\omega_q - E_C}. \quad (3052)$$

The exact target-mediated qubit interaction is

$$\Sigma_C(\omega_q) = \frac{\mathbf{V}_q P_C \mathbf{V}_q^\dagger}{\omega_q - E_C}. \quad (3053)$$

Its rank satisfies

$$\text{rank } \Sigma_C \leq \min\{N_q, r_C\}. \quad (3054)$$

Its nonzero eigenvalues are

$$\lambda_\mu(\Sigma_C) = \frac{s_{\mu,C}^2}{\omega_q - E_C}. \quad (3055)$$

Thus exact photonic flatness removes the variation of the energy denominator. The complete interaction pattern is the sampled target projector. A local projector produces local interactions; an extended projector can produce extended interactions; a phase-structured projector can produce frustration. None of these spatial outcomes follows from $W_C = 0$ alone. Suppose the isolated target has a uniform modal linewidth:

$$P_C \mathbf{\Gamma}_\gamma P_C = \gamma_C P_C. \quad (3056)$$

For an exactly flat target,

$$\mathbf{G}_C^R(\omega_q) = \frac{P_C}{\Delta_{qC} + i\gamma_C/2}. \quad (3057)$$

The target Gram matrix

$$\mathbf{K}_C = \mathbf{V}_q P_C \mathbf{V}_q^\dagger \quad (3058)$$

generates both coherent exchange and collective decay:

$$\mathbf{J}_C = \frac{\Delta_{qC}}{\Delta_{qC}^2 + (\gamma_C/2)^2} \mathbf{K}_C, \quad (3059)$$

and

$$\mathbf{\Gamma}_C^{\text{med}} = \frac{\gamma_C}{\Delta_{qC}^2 + (\gamma_C/2)^2} \mathbf{K}_C. \quad (3060)$$

Whenever $\mathbf{K}_C \neq 0$,

$$\frac{\|\mathbf{J}_C\|}{\|\mathbf{\Gamma}_C^{\text{med}}\|} = \frac{|\Delta_{qC}|}{\gamma_C}. \quad (3061)$$

Moving the qubits closer to the photonic flat band increases the total response but simultaneously reduces its coherent-to-dissipative ratio. Define the off-diagonal target and complementary interactions: $\mathbf{J}_C^{\text{off}}$, $\mathbf{J}_{\bar{C}}^{\text{off}}$. The target-band selectivity is

$$\mathcal{S}_C^J = \frac{\|\mathbf{J}_C^{\text{off}}\|_F}{\|\mathbf{J}_C^{\text{off}}\|_F + \|\mathbf{J}_{\bar{C}}^{\text{off}}\|_F}. \quad (3062)$$

It obeys $0 \leq \mathcal{S}_C^J \leq 1$. A flat target band does not dominate the interaction unless $\mathcal{S}_C^J \simeq 1$. After photons are eliminated, two Duffing transmons have the effective exchange

$$J_{ab} b_a^\dagger b_b + J_{ab}^* b_b^\dagger b_a. \quad (3063)$$

Define their renormalized detuning

$$\Delta_{ab}^q = \tilde{\omega}_a - \tilde{\omega}_b. \quad (3064)$$

If

$$|J_{ab}| \ll \min\{|\Delta_{ab}^q|, |\alpha_a - \Delta_{ab}^q|, |\alpha_b + \Delta_{ab}^q|\}, \quad (3065)$$

the leading computational-subspace cross-Kerr shift is

$$\zeta_{ab} = 2|J_{ab}|^2 \left[\frac{1}{\alpha_a - \Delta_{ab}^q} + \frac{1}{\alpha_b + \Delta_{ab}^q} \right] + O(J_{ab}^4). \quad (3066)$$

This expression is not applied at a resonant qubit pair. A resonant pair is diagonalized in the full retained transmon manifold.

C. Generic twist

For an exact commensurate twist, let $\Gamma_M(\theta)$ be the common translation group and

$$Q_N(\theta) = \Gamma_M(\theta)/\Gamma_N \quad (3067)$$

a normal finite quotient. The finite-cover photonic Hamiltonian is $H_{\gamma,N}(\theta)$, and the exact finite-device interaction is

$$\Sigma_{q,N}(z; \theta) = V_{q,N} [z - H_{\gamma,N}(\theta)]^{-1} V_{q,N}^\dagger. \quad (3068)$$

This quantity is exact for the finite circuit. Its promotion to a bulk interaction requires the finite-cover verification appropriate to the declared qubit placement and frequency. For a generic twist,

$$[\Gamma : \Gamma \cap g_\theta \Gamma g_\theta^{-1}] = \infty. \quad (3069)$$

No finite exact common translation quotient exists. We therefore define the interaction locally. Choose a fixed finite transmon set

$$\mathcal{Q}_0 = \{a = 1, \dots, N_q\} \quad (3070)$$

whose placements lie inside a ball of radius $L_q < \infty$. The two layer covers are lifted to the common universal hyperbolic plane, where the interlayer hopping matrix is evaluated. For a strictly finite-range photonic Hamiltonian of range L_H , the microscopic Hamiltonian patch is exact whenever

$$r_{\text{inj},N} > L_q + L_H. \quad (3071)$$

A resolvent is not strictly finite range. Local interaction convergence therefore also requires an off-spectrum decay estimate or an independent strong-resolvent theorem. Suppose a transported finite Hamiltonian satisfies

$$\|H_{\gamma,N} - H_{\gamma,\infty}\| \leq \varepsilon_{H,N} \quad (3072)$$

on one identified Hilbert space. Let

$$\delta_q = \text{dist}[\omega_q, \text{Spec } H_{\gamma,\infty}] > 0. \quad (3073)$$

If $\varepsilon_{H,N} < \delta_q$, the resolvent identity gives

$$\|G_{\gamma,N}(\omega_q) - G_{\gamma,\infty}(\omega_q)\| \leq \frac{\varepsilon_{H,N}}{\delta_q(\delta_q - \varepsilon_{H,N})}. \quad (3074)$$

Consequently,

$$\|\Sigma_{q,N} - \Sigma_{q,\infty}\| \leq \|V_q\|^2 \frac{\varepsilon_{H,N}}{\delta_q(\delta_q - \varepsilon_{H,N})} + \varepsilon_{V,N}, \quad (3075)$$

where $\varepsilon_{V,N}$ contains the separately measured placement and coupling-vector transport errors. For two local transmons,

$$\left| J_{ab}^{(N)} - J_{ab}^{(\infty)} \right| \leq \|v_a\| \|v_b\| \frac{\varepsilon_{H,N}}{\delta_q(\delta_q - \varepsilon_{H,N})} + \varepsilon_{V,ab,N}. \quad (3076)$$

Let $d(i, j)$ be the physical or graph metric on the photonic lattice. For $\mu > 0$, define the exponentially weighted hopping variation

$$\mathcal{B}_\mu = \sup_i \sum_j |[H_\gamma]_{ij}| \left[e^{\mu d(i,j)} - 1 \right]. \quad (3077)$$

Let

$$\delta_z = \text{dist}[z, \text{Spec } H_\gamma]. \quad (3078)$$

If $\mathcal{B}_\mu < \delta_z$, the weighted resolvent estimate gives

$$|G_{ij}(z)| \leq \frac{e^{-\mu d(i,j)}}{\delta_z - \mathcal{B}_\mu}. \quad (3079)$$

For local transmons,

$$|J_{ab}(z)| \leq \frac{|g_a g_b| e^{-\mu d(i_a, i_b)}}{\delta_z - \mathcal{B}_\mu}. \quad (3080)$$

The bound holds at frequencies lying in the photonic resolvent set. It does not impose exponential decay when the qubits are tuned inside a continuum of propagating photonic states. Let the number of qubit sites in a shell of radius r satisfy

$$N_{\text{sh}}(r) \leq C_{\text{sh}} e^{h_\Gamma r}. \quad (3081)$$

If the pair interaction obeys

$$|J_{ab}| \leq J_0 e^{-\mu r_{ab}}, \quad (3082)$$

then the shell-summed absolute interaction satisfies

$$\sum_{b: r_{ab} \simeq r} |J_{ab}| \lesssim C_{\text{sh}} J_0 e^{(h_\Gamma - \mu)r}. \quad (3083)$$

Absolute summability requires $\mu > h_\Gamma$. The shell-summed interaction power satisfies

$$\sum_{b: r_{ab} \simeq r} |J_{ab}|^2 \lesssim C_{\text{sh}} J_0^2 e^{(h_\Gamma - 2\mu)r}. \quad (3084)$$

Square summability requires $2\mu > h_\Gamma$. A pointwise decaying interaction can therefore carry increasing shell-integrated weight because the number of sites grows exponentially. Let H and $H^{(L)}$ agree on a radius- L neighborhood of the transmon placements, and define

$$V_L = H - H^{(L)}. \quad (3085)$$

Suppose V_L is supported outside a buffer of width B from both transmon sites. Define the shell tail

$$C_{\mu,h}(B) = C_{\text{sh}} \sum_{n=0}^{\infty} e^{h_\Gamma(B+n\rho)} e^{-2\mu(B+n\rho)}. \quad (3086)$$

When $2\mu > h_\Gamma$, the sum is

$$C_{\mu,h}(B) = \frac{C_{\text{sh}} e^{-(2\mu - h_\Gamma)B}}{1 - e^{-(2\mu - h_\Gamma)\rho}}. \quad (3087)$$

Using

$$\mathbf{G} - \mathbf{G}^{(L)} = \mathbf{G} \mathbf{V}_L \mathbf{G}^{(L)}, \quad (3088)$$

the local matrix-element error obeys

$$\left| G_{i_a i_b} - G_{i_a i_b}^{(L)} \right| \leq \frac{\|\mathbf{V}_L\| C_{\mu,h}(B)}{(\delta_z - \mathcal{B}_\mu)^2}. \quad (3089)$$

Therefore

$$\left| J_{ab} - J_{ab}^{(L)} \right| \leq |g_a g_b| \frac{\|\mathbf{V}_L\| C_{\mu,h}(B)}{(\delta_z - \mathcal{B}_\mu)^2}. \quad (3090)$$

The paired-cover interaction converges exponentially with the available buffer whenever the off-spectrum decay exponent dominates half the hyperbolic growth exponent. Let a generic target twist θ be approximated by exact commensurate angles $\theta_N \rightarrow \theta$. On a physical radius L_N , hyperbolic displacement amplification requires

$$|\theta_N - \theta| e^{L_N/R} \rightarrow 0. \quad (3091)$$

The interaction limit additionally requires

$$r_{\text{inj},N} - L_N - L_H \rightarrow \infty, \quad (3092)$$

and a uniform qubit-photon resolvent distance

$$\inf_N \delta_{q,N} > 0. \quad (3093)$$

The safe limit for a fixed transmon pair is

$$L \text{ fixed} \rightarrow N \rightarrow \infty \rightarrow \theta_N \rightarrow \theta \rightarrow L \rightarrow \infty. \quad (3094)$$

For a fixed finite qubit set, $N_q = O(1)$, convergence of all matrix elements $J_{ab}^{(N)} \rightarrow J_{ab}^{(\infty)}$ defines a local generic-twist interaction. For a qubit array whose size grows with the finite cover,

$$N_q = N_q(N) \rightarrow \infty, \quad (3095)$$

pairwise local convergence is insufficient. An extensive interaction claim requires an operator topology such as

$$\|\mathbf{J}_N - \mathbf{J}_\infty\| \rightarrow 0, \quad (3096)$$

together with the no-loss, no-pollution, representation, and cutoff conditions required by the photonic mediator.

D. Physical flat-band angle

Let $\Theta_{\text{flat}}^{\text{full}}(K, \omega)$ be the independently established physical photonic flat-angle set. We do not define an interaction angle without specifying the interaction observable. For any declared scalar interaction diagnostic $\mathcal{O}_J(\theta)$, define

$$\Theta_{\mathcal{O}_J} = \arg \text{ext}_\theta \mathcal{O}_J(\theta), \quad (3097)$$

where $\arg \text{ext}$ denotes either a maximum or a minimum fixed before the scan. Examples include $\Theta_{J_{\text{rms}}}$, $\Theta_{\mathcal{Q}_J}$, $\Theta_{S_C^J}$, Θ_{R_J} , $\Theta_{\mathcal{F}_J}$. No single one of these sets is designated as the physical interaction optimum without an application-specific objective. A twist scan can be performed under two physically different qubit frequency protocols. The fixed-laboratory-frequency protocol uses

$$\omega_q(\theta) = \omega_q^{\text{lab}}. \quad (3098)$$

In this protocol, interaction changes include both photonic wave-function reconstruction and motion of the target-band center relative to the qubits. The co-moving-detuning protocol uses

$$\omega_q(\theta) = \overline{E}_C(\theta) + \Delta_{qC}^{(0)}. \quad (3099)$$

It keeps the target midpoint detuning fixed:

$$\Delta_{qC}(\theta) = \Delta_{qC}^{(0)}. \quad (3100)$$

This protocol isolates changes in the target projector, bandwidth, complementary spectrum, and photon loss from the trivial effect of a moving band center. Both protocols are calculated. A claimed flat-band-mediated reconstruction must survive the co-moving-detuning control. Let

$$\mathbf{G}^R(\theta) = \left[\omega_q(\theta) I - \mathbf{H}_\gamma(\theta) + \frac{i}{2} \mathbf{\Gamma}_\gamma(\theta) \right]^{-1}. \quad (3101)$$

Define

$$\mathbf{B}_\theta = \partial_\theta \mathbf{H}_\gamma - \omega'_q(\theta) I - \frac{i}{2} \partial_\theta \mathbf{\Gamma}_\gamma. \quad (3102)$$

Differentiation gives

$$\partial_\theta \mathbf{G}^R = \mathbf{G}^R \mathbf{B}_\theta \mathbf{G}^R. \quad (3103)$$

Define

$$\mathbf{B}_{\theta\theta} = \partial_\theta^2 \mathbf{H}_\gamma - \omega''_q(\theta) I - \frac{i}{2} \partial_\theta^2 \mathbf{\Gamma}_\gamma. \quad (3104)$$

The second derivative is

$$\partial_\theta^2 \mathbf{G}^R = \mathbf{G}^R \mathbf{B}_{\theta\theta} \mathbf{G}^R + 2 \mathbf{G}^R \mathbf{B}_\theta \mathbf{G}^R \mathbf{B}_\theta \mathbf{G}^R. \quad (3105)$$

The self-energy is

$$\Sigma_q^R = \mathbf{V}_q \mathbf{G}^R \mathbf{V}_q^\dagger. \quad (3106)$$

Its first derivative is

$$\begin{aligned} \partial_\theta \Sigma_q^R &= \mathbf{V}_{q,\theta} \mathbf{G}^R \mathbf{V}_q^\dagger + \mathbf{V}_q \mathbf{G}^R \mathbf{V}_{q,\theta}^\dagger \\ &\quad + \mathbf{V}_q \mathbf{G}^R \mathbf{B}_\theta \mathbf{G}^R \mathbf{V}_q^\dagger. \end{aligned} \quad (3107)$$

The complete second derivative is

$$\partial_\theta^2 \Sigma_q^R = \mathbf{V}_{q,\theta\theta} \mathbf{G}^R \mathbf{V}_q^\dagger + \mathbf{V}_q \mathbf{G}^R \mathbf{V}_{q,\theta\theta}^\dagger + 2 \mathbf{V}_{q,\theta} \mathbf{G}^R \mathbf{V}_{q,\theta}^\dagger$$

$$\begin{aligned}
& + 2V_{q,\theta} G^R B_\theta G^R V_q^\dagger + 2V_q G^R B_\theta G^R V_{q,\theta}^\dagger \\
& + V_q G^R B_{\theta\theta} G^R V_q^\dagger + 2V_q G^R B_\theta G^R B_\theta G^R V_q^\dagger.
\end{aligned}$$

For fixed transmon couplings and fixed laboratory frequency,

$$V_{q,\theta} = 0, \quad \omega'_q = 0, \quad \Gamma_{\gamma,\theta} = 0, \quad (3108)$$

so

$$\partial_\theta \Sigma_q = V_q G_\gamma H_{\gamma,\theta} G_\gamma V_q^\dagger. \quad (3109)$$

The response depends on two Green propagators and can be strongly enhanced near a photonic resonance. At $\theta = \theta_{\text{flat}}^{\text{full}}$, we record the complete interaction vector

$$\begin{aligned}
\mathbf{I}_{\text{flat}} = & \left(J_{\text{rms}}, \Gamma_{\text{rms}}^{\text{med}}, Q_J, \mathcal{S}_C^J, \right. \\
& R_J, P_J, \mathcal{F}_J, \mathcal{F}_{\text{cyc}}, \\
& \lambda_{\min}(J_{\text{off}}), \lambda_{\max}(J_{\text{off}}), \\
& \left. D_J^{\text{NA}}, \varepsilon_{J,\text{proj}}, \varepsilon_{J,\text{cover}} \right). \quad (3110)
\end{aligned}$$

No entry is inferred from the photonic bandwidth alone. Define the leading target-projector prediction

$$J_{\text{proj}} = \frac{V_q P_C V_q^\dagger}{\Delta_{qC}} \quad (3111)$$

for a lossless dispersive calculation. The normalized projector-control residual is

$$\varepsilon_{J,\text{proj}} = \frac{\|J - J_{\text{proj}}\|_F}{\max\{\|J\|_F, J_{\text{ref}}\}}. \quad (3112)$$

A small value verifies that the flat-band projector, rather than remote bands or accidental detuning, controls the interaction matrix. For any declared interaction observable, define

$$D_\theta^{(\mathcal{O}_J)} = d_H(\Theta_{\text{flat}}^{\text{full}}, \Theta_{\mathcal{O}_J}). \quad (3113)$$

For unique extrema,

$$D_\theta^{(\mathcal{O}_J)} = |\theta_{\mathcal{O}_J} - \theta_{\text{flat}}^{\text{full}}|. \quad (3114)$$

Let the established angle uncertainties be $\delta\theta_{\text{flat}}$, $\delta\theta_{\mathcal{O}_J}$, $\delta\theta_{\text{cal}}$. The angular mismatch is resolved when

$$D_\theta^{(\mathcal{O}_J)} > \delta\theta_{\text{flat}} + \delta\theta_{\mathcal{O}_J} + \delta\theta_{\text{cal}}. \quad (3115)$$

We classify the relation between the photonic flat-band optimum and the interaction optimum according to the degree of angular agreement resolved by the specified diagnostics. We use the term *flat-band interaction locking* when the angular discrepancy $D_\theta^{(\mathcal{O}_J)}$ remains unresolved, within the complete uncertainty budget, for every specified interaction diagnostic. This corresponds to the case in which the interaction extrema are statistically

compatible with the independently determined photonic flat-band optimum. We assign the weaker classification *partial locking* when only a subset of the specified diagnostics aligns with the photonic optimum, while one or more remaining observables do not exhibit the same angular coincidence. In this case, the data support only a diagnostic-dependent correspondence rather than a complete locking of the interaction response to the flat-band angle. The classification *interaction displacement* is used when the relevant extrema are individually identifiable and their angular separation is larger than the combined uncertainty, so that the interaction optimum and the photonic flat-band optimum are resolvably distinct. Finally, we use *interaction indeterminacy* when one or more of the extrema cannot be localized with sufficient precision to decide whether locking or displacement occurs. In this regime, no mechanistic conclusion concerning the angular relation is promoted beyond the available resolution. Define an interaction-focusing diagnostic only after specifying a source, a physical distance, and the normalization against hyperbolic shell growth. We do not include a focusing factor in the definition of the photonic flat-band angle. The statement $\theta_{\text{flat}}^{\text{full}} = \theta_{\text{focus}}$ is treated as a testable numerical outcome, not as a hypothesis of the model.

A valid positive conclusion requires: an interior maximum of the pair-averaged interaction profile; a lower confidence bound exceeding the local baseline; persistence under fixed target detuning; persistence under cover and cutoff enlargement; a projector-level mechanism or a resolved alternative mechanism.

E. Interaction range and frustration

Let

$$r_{ab} = d_{\mathbb{H}}(x_a, x_b) \quad (3116)$$

be the physical geodesic distance between transmons. Choose fixed radial bins

$$\mathcal{B}_m = [r_m, r_m + \Delta r). \quad (3117)$$

For source transmon a , define the shell population

$$N_a(r_m) = \#\{b \neq a : r_{ab} \in \mathcal{B}_m\}. \quad (3118)$$

The pair-averaged absolute interaction is

$$j_{1,a}(r_m) = \frac{1}{N_a(r_m)} \sum_{b:r_{ab} \in \mathcal{B}_m} |J_{ab}|. \quad (3119)$$

The pair-averaged RMS interaction is

$$j_{2,a}(r_m) = \left[\frac{1}{N_a(r_m)} \sum_{b:r_{ab} \in \mathcal{B}_m} |J_{ab}|^2 \right]^{1/2}. \quad (3120)$$

The shell-integrated interaction power is

$$\mathcal{P}_{2,a}(r_m) = \sum_{b:r_{ab} \in \mathcal{B}_m} |J_{ab}|^2 = N_a(r_m) j_{2,a}^2(r_m). \quad (3121)$$

The distinction between $j_{2,a}$ and $\mathcal{P}_{2,a}$ is essential. A peak in $\mathcal{P}_{2,a}$ can be generated solely by exponential shell multiplicity even when the interaction per pair decreases monotonically. Define the source-resolved quadratic range

$$R_{J,a}^2 = \frac{\sum_{b \neq a} r_{ab}^2 |J_{ab}|^2}{\sum_{b \neq a} |J_{ab}|^2}. \quad (3122)$$

The global RMS range is

$$R_J^2 = \frac{\sum_{a \neq b} r_{ab}^2 |J_{ab}|^2}{\sum_{a \neq b} |J_{ab}|^2}. \quad (3123)$$

The cumulative tail fraction beyond radius R_0 is

$$T_J(R_0) = \frac{\sum_{a \neq b: r_{ab} > R_0} |J_{ab}|^2}{\sum_{a \neq b} |J_{ab}|^2}. \quad (3124)$$

monotone local :	$j_2(r)$ decreases and $T_J(R_0)$ is small,	
extended monotone :	$j_2(r)$ decreases slowly and R_J is large,	
geometric shell accumulation :	$\mathcal{F}_{\text{shell}} > 1$ but $\mathcal{F}_{\text{pair}} \leq 1$,	(3132)
pairwise interaction focusing :	$\mathcal{F}_{\text{pair}} > 1$ beyond uncertainty,	
multifocal :	$j_2(r)$ has several resolved interior maxima.	

Only the fourth class is called interaction focusing. Under a local qubit phase transformation

$$\sigma_a^+ \mapsto e^{i\chi_a} \sigma_a^+, \quad (3133)$$

It satisfies $0 \leq T_J(R_0) \leq 1$. For source a , define the interaction participation number

$$P_{J,a} = \frac{\left(\sum_{b \neq a} |J_{ab}|^2 \right)^2}{\sum_{b \neq a} |J_{ab}|^4}. \quad (3125)$$

It obeys $1 \leq P_{J,a} \leq N_q - 1$. The normalized participation is

$$p_{J,a} = \frac{P_{J,a}}{N_q - 1}. \quad (3126)$$

A large $R_{J,a}$ with small $P_{J,a}$ indicates coupling to a small number of distant targets. A large $R_{J,a}$ with large $P_{J,a}$ indicates broadly distributed interaction weight. In the exact target-flat-band limit,

$$J_{ab}^{(C)} = \frac{B_{ab,C}}{\Delta_{qC}}. \quad (3127)$$

The detuning is a common scalar. Consequently,

$$R_J[J_C] = R_J[K_C], \quad (3128)$$

and

$$P_J[J_C] = P_J[K_C]. \quad (3129)$$

The interaction range and participation are independent of the magnitude of Δ_{qC} in this ideal limit. Changing the detuning rescales the interaction but does not change its spatial profile. Let r_{nn} denote the first nonempty distance shell. Define the interior pairwise-focusing ratio

$$\mathcal{F}_{\text{pair}} = \frac{\max_{r_m > r_{\text{nn}}} j_2(r_m)}{\max \{j_2(r_{\text{nn}}), J_{\text{floor}}\}}. \quad (3130)$$

Define the shell-accumulation ratio

$$\mathcal{F}_{\text{shell}} = \frac{\max_{r_m > r_{\text{nn}}} \mathcal{P}_2(r_m)}{\max \{\mathcal{P}_2(r_{\text{nn}}), P_{\text{floor}}\}}. \quad (3131)$$

We use the following outcome classes:

the exchange amplitudes transform as

$$J_{ab} \mapsto e^{i(\chi_a - \chi_b)} J_{ab}. \quad (3134)$$

The magnitudes $|J_{ab}|$ are gauge invariant, but an individual edge phase is not. Let

$$\mathcal{C} = (a_1, a_2, \dots, a_m, a_1) \quad (3135)$$

be an oriented cycle in the qubit interaction graph. The interaction Wilson phase is

$$e^{i\Phi_{\mathcal{C}}^J} = \prod_{r=1}^m \frac{J_{a_r a_{r+1}}}{|J_{a_r a_{r+1}}|}. \quad (3136)$$

The cycle flux is

$$\Phi_{\mathcal{C}}^J = \arg \prod_{r=1}^m J_{a_r a_{r+1}} \mod 2\pi. \quad (3137)$$

All local qubit phases cancel around the cycle, so $\Phi_{\mathcal{C}}^J$ is gauge invariant. For real reciprocal interactions, $J_{ab} \in \mathbb{R}$, the cycle phase is either 0 or π . The signed interaction graph is balanced if and only if

$$\prod_{(a,b) \in \mathcal{C}} \text{sgn } J_{ab} = +1 \quad (3138)$$

for every cycle. This is equivalent to the existence of local signs $\eta_a = \pm 1$ such that

$$\eta_a J_{ab} \eta_b \geq 0 \quad (3139)$$

on every occupied edge. Define

$$J_1 = \sum_{a < b} |J_{ab}|. \quad (3140)$$

For $J_1 > 0$, define

$$\mathcal{F}_J = 1 - \frac{1}{J_1} \max_{\{\chi_a\}} \sum_{a < b} \text{Re} \left[J_{ab} e^{i(\chi_b - \chi_a)} \right]. \quad (3141)$$

Since the maximum is no larger than J_1 and is nonnegative, $0 \leq \mathcal{F}_J \leq 1$. We have $\mathcal{F}_J = 0$ if and only if all occupied interaction phases can be removed by local qubit rephasing. Choose a fixed fundamental cycle basis $\mathcal{C}_{\text{fund}}$. With positive weights $w_{\mathcal{C}}$, define

$$\mathcal{F}_{\text{cyc}} = \frac{\sum_{\mathcal{C} \in \mathcal{C}_{\text{fund}}} w_{\mathcal{C}} [1 - \cos \Phi_{\mathcal{C}}^J]}{2 \sum_{\mathcal{C} \in \mathcal{C}_{\text{fund}}} w_{\mathcal{C}}}. \quad (3142)$$

It obeys $0 \leq \mathcal{F}_{\text{cyc}} \leq 1$. The cycle basis and its weights are fixed before the twist scan. Diagonalize the coherent interaction matrix:

$$\mathbf{J} = \sum_{\mu=1}^{N_q} \lambda_{\mu}^J |\phi_{\mu}^J\rangle \langle \phi_{\mu}^J|. \quad (3143)$$

The collective qubit excitation

$$\mathbf{S}_{\mu}^+ = \sum_a \phi_{\mu,a}^J \sigma_a^+ \quad (3144)$$

has the interaction shift λ_{μ}^J . A spatially extended interaction matrix can nevertheless have only a small number of nonzero collective channels when it is mediated by a low-rank flat-band projector.

F. Abelian and non-Abelian interaction predictions

For a normal finite quotient Q_N , the photonic Hilbert space decomposes as

$$\ell^2(Q_N) \simeq \bigoplus_{\rho \in \hat{Q}_N} V_{\rho} \otimes V_{\rho}^*. \quad (3145)$$

The photonic Hamiltonian satisfies

$$\mathbf{H}_{\gamma,N} \simeq \bigoplus_{\rho \in \hat{Q}_N} [I_{d_{\rho}} \otimes \mathbf{H}_{\gamma,N}[\rho]]. \quad (3146)$$

The representation block is

$$\mathbf{H}_{\gamma,N}[\rho] = \sum_{\delta \in Q_N} h_N(\delta) \otimes \rho(\delta). \quad (3147)$$

The Green operator decomposes exactly:

$$\mathbf{G}_{\gamma,N}(z) \simeq \bigoplus_{\rho \in \hat{Q}_N} [I_{d_{\rho}} \otimes \mathbf{G}_{\gamma,N}[\rho](z)], \quad (3148)$$

where

$$\mathbf{G}_{\gamma,N}[\rho](z) = [zI - \mathbf{H}_{\gamma,N}[\rho]]^{-1}. \quad (3149)$$

Define the Abelian isotypic projector

$$P_{\text{Ab},N} = \sum_{\rho \in \hat{Q}_N: d_{\rho}=1} P_{\rho,N}, \quad (3150)$$

and the non-Abelian projector

$$P_{\text{NA},N} = \sum_{\rho \in \hat{Q}_N: d_{\rho}>1} P_{\rho,N}. \quad (3151)$$

They satisfy

$$P_{\text{Ab},N} + P_{\text{NA},N} = I, \quad P_{\text{Ab},N} P_{\text{NA},N} = 0. \quad (3152)$$

Because they commute with the photonic Hamiltonian,

$$\mathbf{G}_{\gamma,N} = \mathbf{G}_{\text{Ab},N} + \mathbf{G}_{\text{NA},N}, \quad (3153)$$

where

$$\mathbf{G}_{\text{Ab},N} = P_{\text{Ab},N} \mathbf{G}_{\gamma,N} P_{\text{Ab},N}, \quad (3154)$$

and

$$\mathbf{G}_{\text{NA},N} = P_{\text{NA},N} \mathbf{G}_{\gamma,N} P_{\text{NA},N}. \quad (3155)$$

The full qubit self-energy splits exactly:

$$\Sigma_q^{\text{full}} = \Sigma_q^{\text{Ab}} + \Sigma_q^{\text{NA}}, \quad (3156)$$

with

$$\Sigma_q^{\text{Ab}} = \mathbf{V}_q \mathbf{G}_{\text{Ab}} \mathbf{V}_q^{\dagger}, \quad (3157)$$

and

$$\Sigma_q^{\text{NA}} = \mathbf{V}_q \mathbf{G}_{\text{NA}} \mathbf{V}_q^\dagger. \quad (3158)$$

The coherent interactions obey

$$\mathbf{J}^{\text{full}} = \mathbf{J}^{\text{Ab}} + \mathbf{J}^{\text{NA}}, \quad (3159)$$

and the mediated decay matrices obey

$$\mathbf{\Gamma}^{\text{med,full}} = \mathbf{\Gamma}^{\text{med,Ab}} + \mathbf{\Gamma}^{\text{med,NA}}. \quad (3160)$$

Define the normalized coherent-interaction discrepancy

$$D_J^{\text{NA}} = \frac{\|\mathbf{J}^{\text{full}} - \mathbf{J}^{\text{Ab}}\|_{\text{F}}}{\max\{\|\mathbf{J}^{\text{full}}\|_{\text{F}}, J_{\text{ref}}\}}. \quad (3161)$$

The dissipative discrepancy is

$$D_{\Gamma}^{\text{NA}} = \frac{\|\mathbf{\Gamma}^{\text{med,full}} - \mathbf{\Gamma}^{\text{med,Ab}}\|_{\text{F}}}{\max\{\|\mathbf{\Gamma}^{\text{med,full}}\|_{\text{F}}, \Gamma_{\text{ref}}\}}. \quad (3162)$$

The range discrepancy is

$$D_R^{\text{NA}} = \frac{|R_J^{\text{full}} - R_J^{\text{Ab}}|}{\max\{R_J^{\text{full}}, R_{\text{ref}}\}}. \quad (3163)$$

The frustration discrepancy is

$$D_{\mathcal{F}}^{\text{NA}} = |\mathcal{F}_J^{\text{full}} - \mathcal{F}_J^{\text{Ab}}|. \quad (3164)$$

Suppose one identical transmon is placed on each quotient translate of a fixed intracell orbital. The effective interaction matrix commutes with the finite translation group and has its own exact Wedderburn decomposition:

$$\mathbf{J}_N \simeq \bigoplus_{\rho \in \hat{Q}_N} [I_{d_\rho} \otimes \mathbf{J}_N[\rho]]. \quad (3165)$$

The sector interaction is

$$\mathbf{J}_N[\rho] = \text{Herm}\{\mathbf{V}_q[\rho] \mathbf{G}_{\gamma,N}[\rho] \mathbf{V}_q[\rho]^\dagger\}. \quad (3166)$$

Define the collective qubit operator

$$S_{\rho;\alpha\beta,s}^+ = \sqrt{\frac{d_\rho}{|Q_N|}} \sum_{q \in Q_N} \rho(q)_{\alpha\beta}^* \sigma_{q,s}^+. \quad (3167)$$

Different irreducible representations are orthogonal in the single-excitation qubit Hilbert space. A translation-invariant effective interaction cannot couple collective qubit excitations in inequivalent irreducible sectors. For a local qubit at (q, s) , $|q, s\rangle$ has nonzero projections onto many irreducible sectors. Its mediated interaction is the coherent sum

$$J_{ab}^{\text{full}} = \sum_{\rho \in \hat{Q}_N} J_{ab}^{(\rho)}. \quad (3168)$$

A local qubit experiment can therefore detect non-Abelian mediator sectors without preparing a collective non-Abelian qubit state.

Consider two photonic propagation paths from transmon b to transmon a , carrying group words w_1 and w_2 . Suppose

$$c = w_1^{-1} w_2 \in [\Gamma_M, \Gamma_M]. \quad (3169)$$

Let their frequency-dependent scalar path amplitudes be $A_1(z)$, $A_2(z)$. In representation ρ , their contribution is

$$\begin{aligned} J_{ab}^{(12)}[\rho] &= g_a g_b^* [A_1 \rho(w_1) + A_2 \rho(w_2)] \\ &= g_a g_b^* \rho(w_1) [A_1 I + A_2 \rho(c)]. \end{aligned} \quad (3170)$$

Every Abelian character satisfies $\chi(c) = 1$, and therefore predicts

$$J_{ab}^{(12)}[\chi] = g_a g_b^* \chi(w_1) (A_1 + A_2). \quad (3171)$$

Define

$$\eta_c(\rho) = 1 - \frac{1}{d_\rho} \text{Re Tr } \rho(c) = \frac{\|I - \rho(c)\|_{\text{F}}^2}{2d_\rho}. \quad (3172)$$

The deviation of the second path from the Abelianized prediction is

$$\Delta J_{ab}^{(c)} = g_a g_b^* A_2 \rho(w_1) [\rho(c) - I]. \quad (3173)$$

Its Frobenius norm is

$$\|\Delta J_{ab}^{(c)}\|_{\text{F}} = |g_a g_b A_2| \sqrt{2d_\rho \eta_c(\rho)}. \quad (3174)$$

A null-homology commutator can therefore modify the mediated interaction even though every Abelian momentum assigns it the trivial phase.

G. Hodge and homology interpretation

For $|z| > \|\mathbf{H}_\gamma\|$, the resolvent has the norm-convergent Neumann expansion

$$\mathbf{G}_\gamma(z) = \frac{1}{z} \sum_{n=0}^{\infty} \left(\frac{\mathbf{H}_\gamma}{z} \right)^n. \quad (3175)$$

For two photonic sites i and j ,

$$G_{ij}(z) = \sum_{n=0}^{\infty} \frac{1}{z^{n+1}} \sum_{p: j \rightarrow i, |p|=n} A(p), \quad (3176)$$

where

$$A(p) = \prod_{r=0}^{n-1} H_{v_{r+1} v_r} \quad (3177)$$

is the ordered hopping amplitude along

$$p = (v_0 = j, v_1, \dots, v_n = i). \quad (3178)$$

TABLE IX. Operational comparison between Abelian and complete photon-mediated qubit interactions. Every comparison uses identical transmon placements, qubit frequencies, photonic losses, and numerical resolutions.

Outcome	Required observation	Physical interpretation
I	Full and Abelian interaction matrices agree within uncertainty	Abelian sectors are sufficient for the declared qubit configuration
II	$D_J^{\text{NA}} > 0$ or $D_R^{\text{NA}} > 0$	Non-Abelian sectors renormalize interaction strength or spatial
III	$D_F^{\text{NA}} > 0$ or new cycle fluxes appear	Non-Abelian propagation reconstructs interaction frustration
IV	A nonzero interaction eigenchannel belongs only to $d_\rho > 1$	The collective qubit coupling is intrinsically translation non-Abelian

The local qubit interaction is

$$J_{ab}(z) = g_a g_b^* \sum_{n=0}^{\infty} \frac{1}{z^{n+1}} \sum_{p: i_b \rightarrow i_a, |p|=n} A(p). \quad (3179)$$

Choose reference lifts of the two transmon sites to the universal cover. Every propagation path determines a relative group word $\gamma \in \Gamma_M$. Let $\mathcal{P}_{ab}(\gamma)$ be the set of paths from b to a in the class γ . Define the class-resolved propagator coefficient

$$\mathcal{G}_{ab}(\gamma; z) = \sum_{p \in \mathcal{P}_{ab}(\gamma)} \frac{A(p)}{z^{|p|+1}}. \quad (3180)$$

In representation ρ , the propagator is

$$\mathbf{G}_{ab}[\rho](z) = \sum_{\gamma \in \Gamma_M} \mathcal{G}_{ab}(\gamma; z) \rho(\gamma). \quad (3181)$$

The interaction is

$$\mathbf{J}_{ab}[\rho](z) = g_a g_b^* \sum_{\gamma} \mathcal{G}_{ab}(\gamma; z) \rho(\gamma). \quad (3182)$$

The Abelianization map is

$$n : \Gamma_M \longrightarrow H_1(\Sigma_M; \mathbb{Z}) \simeq \mathbb{Z}^{2g_M}. \quad (3183)$$

For an Abelian character,

$$\chi_{\mathbf{k}}(\gamma) = e^{i\mathbf{k}^T n(\gamma)}. \quad (3184)$$

Define the homology-resolved propagator

$$\mathcal{G}_{ab}(\mathbf{h}; z) = \sum_{\gamma: n(\gamma)=\mathbf{h}} \mathcal{G}_{ab}(\gamma; z). \quad (3185)$$

Then

$$G_{ab}^{\text{Ab}}(\mathbf{k}; z) = \sum_{\mathbf{h} \in \mathbb{Z}^{2g_M}} \mathcal{G}_{ab}(\mathbf{h}; z) e^{i\mathbf{k}^T \mathbf{h}}. \quad (3186)$$

The mediated interaction is

$$J_{ab}^{\text{Ab}}(\mathbf{k}; z) = g_a g_b^* \sum_{\mathbf{h}} \mathcal{G}_{ab}(\mathbf{h}; z) e^{i\mathbf{k}^T \mathbf{h}}. \quad (3187)$$

The first homology derivative is

$$\partial_{k_\mu} J_{ab}^{\text{Ab}} = i g_a g_b^* \sum_{\mathbf{h}} h_\mu \mathcal{G}_{ab}(\mathbf{h}; z) e^{i\mathbf{k}^T \mathbf{h}}. \quad (3188)$$

The second derivative is

$$\partial_{k_\mu} \partial_{k_\nu} J_{ab}^{\text{Ab}} = -g_a g_b^* \sum_{\mathbf{h}} h_\mu h_\nu \mathcal{G}_{ab}(\mathbf{h}; z) e^{i\mathbf{k}^T \mathbf{h}}. \quad (3189)$$

The interaction Hessian is therefore a second homology moment of the photon propagation amplitudes. Two paths with

$$n(w_1) = n(w_2) \quad (3190)$$

are indistinguishable in every Abelian character. Their relative word lies in the commutator subgroup:

$$w_1^{-1} w_2 \in [\Gamma_M, \Gamma_M]. \quad (3191)$$

A higher-dimensional representation can satisfy

$$\rho(w_1) \neq \rho(w_2). \quad (3192)$$

The hierarchy is therefore

$$\begin{aligned} \text{Euclidean momentum : } & \mathbb{Z}^2 \text{ path displacement,} \\ \text{hyperbolic Abelian sector : } & H_1(\Sigma_M; \mathbb{Z}) \text{ path homology,} \\ \text{hyperbolic non-Abelian sector : } & \pi_1(\Sigma_M) \text{ ordered path holonomy.} \end{aligned} \quad (3193)$$

At a smooth irreducible representation

$$\rho : \Gamma_M \longrightarrow U(d_\rho), \quad (3194)$$

the physical tangent space is

$$T_{[\rho]} \mathcal{B}_{d_\rho} \simeq H^1(\Gamma_M; \text{Ad } \rho). \quad (3195)$$

Let

$$X \in H^1(\Gamma_M; \text{Ad } \rho) \quad (3196)$$

be represented by the infinitesimal holonomy

$$\dot{\rho}_X(\gamma) = X(\gamma) \rho(\gamma). \quad (3197)$$

The photonic Hamiltonian derivative is

$$D_X \mathbf{H}_\gamma[\rho] = \sum_{\gamma} h(\gamma) \otimes X(\gamma) \rho(\gamma). \quad (3198)$$

At fixed qubit frequency,

$$D_X \mathbf{G}_\gamma[\rho] = \mathbf{G}_\gamma[\rho] (D_X \mathbf{H}_\gamma[\rho]) \mathbf{G}_\gamma[\rho]. \quad (3199)$$

For fixed covariantly transported transmon placements,

$$D_X \mathbf{V}_q = 0, \quad (3200)$$

and therefore

$$D_X \Sigma_q[\rho] = \mathbf{V}_q \mathbf{G}_\gamma (D_X \mathbf{H}_\gamma) \mathbf{G}_\gamma \mathbf{V}_q^\dagger. \quad (3201)$$

For two tangent directions X and Y ,

$$D_X D_Y \Sigma_q = \mathbf{V}_q \left[\mathbf{G} \mathbf{H}_{XY} \mathbf{G} + \mathbf{G} \mathbf{H}_X \mathbf{G} \mathbf{H}_Y \mathbf{G} + \mathbf{G} \mathbf{H}_Y \mathbf{G} \mathbf{H}_X \mathbf{G} \right] \mathbf{V}_q^\dagger. \quad (3202)$$

Here

$$\mathbf{H}_X = D_X \mathbf{H}, \quad \mathbf{H}_{XY} = D_X D_Y \mathbf{H}. \quad (3203)$$

Let G_ρ^{can} be the canonical intensive Hodge metric. For one pair interaction, define the covector

$$dJ_{ab} \in T_{[\rho]}^* \mathcal{B}_{d_\rho}. \quad (3204)$$

Its Hodge-normalized magnitude is

$$\Omega_{J,ab}^H = \|dJ_{ab}\|_{(G_\rho^{\text{can}})^{-1}}. \quad (3205)$$

Let $D_{G,\rho}$ be the declared Hodge diameter of the tested representation domain. The dimensionless interaction-variation ratio is

$$\chi_{J,ab}^H = \frac{D_{G,\rho} \Omega_{J,ab}^H}{\max\{|J_{ab}|, J_{\text{ref}}\}}. \quad (3206)$$

For a simple spectral decomposition,

$$\mathbf{G}(z) = \sum_n \frac{P_n}{z - E_n}. \quad (3207)$$

Differentiation gives

$$D_X \mathbf{G} = \sum_n \left[\frac{D_X P_n}{z - E_n} + \frac{P_n D_X E_n}{(z - E_n)^2} \right]. \quad (3208)$$

At a Hodge energy-cancellation point,

$$D_X E_n = 0 \quad (3209)$$

for the target energy branch, but the projector response $D_X P_n$ can remain nonzero. Therefore

$$D_X E_C = 0 \not\Rightarrow D_X J_{ab} = 0. \quad (3210)$$

A Hodge-stationary interaction additionally requires

$$\mathbf{V}_q (D_X P_C) \mathbf{V}_q^\dagger = 0 \quad (3211)$$

or a cancellation against the complementary-band response. Let the qubit interaction network be equipped with a fixed cellular embedding. Choose one orientation

for every occupied interaction edge. Let N_V , N_E , N_F be the numbers of vertices, oriented edges, and faces. Let

$$\mathbf{B}_1 \in \mathbb{R}^{N_V \times N_E} \quad (3212)$$

be the vertex-edge incidence matrix, and let

$$\mathbf{B}_2 \in \mathbb{R}^{N_E \times N_F} \quad (3213)$$

be the edge-face incidence matrix. They satisfy $\mathbf{B}_1 \mathbf{B}_2 = 0$. Choose an unwrapped edge-phase vector

$$\phi \in \mathbb{R}^{N_E}, \quad \phi_{ab} = \arg J_{ab}, \quad (3214)$$

on a declared continuous phase branch. The edge Hodge Laplacian is

$$\Delta_1 = \mathbf{B}_1^\top \mathbf{B}_1 + \mathbf{B}_2 \mathbf{B}_2^\top. \quad (3215)$$

The orthogonal Hodge decomposition is

$$\phi = \mathbf{B}_1^\top \chi + \mathbf{B}_2 \eta + \mathbf{h}, \quad (3216)$$

where

$$\mathbf{h} \in \ker \Delta_1. \quad (3217)$$

The exact component $\mathbf{B}_1^\top \chi$ is removable by local qubit rephasing. The coexact component $\mathbf{B}_2 \eta$ contains contractible plaquette flux. The harmonic component $\mathbf{h}$ contains flux through noncontractible homology cycles. For a closed genus- g cellular surface,

$$\dim \ker \Delta_1 = 2g. \quad (3218)$$

Remove the exact gauge component:

$$\phi_\perp = \mathbf{B}_2 \eta + \mathbf{h}. \quad (3219)$$

For $\|\phi_\perp\| > 0$, define the local frustration fraction

$$f_{\text{loc}}^J = \frac{\|\mathbf{B}_2 \eta\|^2}{\|\phi_\perp\|^2}, \quad (3220)$$

and the homological frustration fraction

$$f_{\text{hom}}^J = \frac{\|\mathbf{h}\|^2}{\|\phi_\perp\|^2}. \quad (3221)$$

Orthogonality gives

$$f_{\text{loc}}^J + f_{\text{hom}}^J = 1. \quad (3222)$$

The decomposition is applied only after phase unwrapping is established. The exact modulo- 2π observables remain the cycle Wilson phases. Choose the canonical homology basis $a_1, b_1, \dots, a_g, b_g$. Define

$$\Phi_H^J = (\Phi_{a_1}^J \quad \Phi_{b_1}^J \quad \dots \quad \Phi_{a_g}^J \quad \Phi_{b_g}^J)^\top. \quad (3223)$$

With the canonical Hodge metric $\mathbf{G}_H$, define the coordinate-invariant homological flux norm

$$\mathcal{F}_H^J = [(\Phi_H^J)^\top \mathbf{G}_H^{-1} \Phi_H^J]^{1/2}. \quad (3224)$$

This quantity measures interaction frustration along the noncontractible cycles of the compact hyperbolic quotient. For a representation-resolved collective qubit channel, the interaction between cells can be matrix valued. Let the polar decomposition of a nonsingular interaction block be

$$J_{ab}^{(\rho)} = P_{ab}^{(\rho)} U_{ab}^{(\rho)}, \quad (3225)$$

where

$$P_{ab}^{(\rho)} \succeq 0, \quad U_{ab}^{(\rho)} \in U(d_\rho). \quad (3226)$$

For an oriented qubit cycle $\mathcal{C}$, define the matrix Wilson loop

$$W_{\mathcal{C}}^{J,\rho} = \prod_{(a,b) \in \mathcal{C}} U_{ab}^{(\rho)}. \quad (3227)$$

The normalized Wilson character is

$$w_{\mathcal{C}}^{J,\rho} = \frac{1}{d_\rho} \text{Tr } W_{\mathcal{C}}^{J,\rho}. \quad (3228)$$

It is invariant under local representation-basis conjugations. For $d_\rho = 1$, Eq. (3228) reduces to the scalar cycle phase. For $d_\rho > 1$, the eigenphases of $W_{\mathcal{C}}^{J,\rho}$ contain information that cannot be compressed into one Abelian flux. For every qubit configuration and control point, we record

$$\begin{aligned} \mathfrak{C}_{q-q} = & \left(\{x_a, \ell_a, s_a, g_{ai}, \omega_a, \alpha_a\}_{a=1}^{N_q}, \right. \\ & \delta_{qC}, \delta_{q\bar{C}}, \epsilon_{q\gamma}, r_{\text{addr}, C}, \kappa_{\text{pl}, C}, \\ & J, \Gamma^{\text{med}}, J_{\text{rms}}, \Gamma_{\text{rms}}^{\text{med}}, \mathcal{Q}_J, \\ & \mathcal{S}_C^J, \varepsilon_{J, \text{proj}}, R_J, T_J(R_0), P_J, \\ & \mathcal{F}_{\text{pair}}, \mathcal{F}_{\text{shell}}, \mathcal{F}_J, \mathcal{F}_{\text{cyc}}, \\ & D_J^{\text{NA}}, D_\Gamma^{\text{NA}}, D_R^{\text{NA}}, D_{\mathcal{F}}^{\text{NA}}, \\ & \mathcal{N}_C, \eta_c, \mathcal{F}_H^J, f_{\text{loc}}^J, f_{\text{hom}}^J, \\ & \theta_{\text{flat}}^{\text{full}}, \{\Theta_{\mathcal{O}_J}\}, \varepsilon_{J, \text{cover}}, \varepsilon_{J, \text{cut}}, \\ & \left. r_{\text{inj}}, C_{\text{rep}, N}, \text{classification} \right). \quad (3229) \end{aligned}$$

Coherent mediation requires $\epsilon_{q\gamma} \ll 1$, $\mathcal{Q}_J$ above its stated threshold, and $\Delta_{qC} > 0$. Projector control additionally bounds $\varepsilon_{J, \text{proj}}$ and satisfies $\mathcal{S}_C^J$. A resolved value of D_J^{NA} , or an independent cycle witness, identifies a non-Abelian contribution. Pairwise focusing corresponds to $\mathcal{F}_{\text{pair}} > 1$; shell accumulation has $\mathcal{F}_{\text{shell}} > 1$ with $\mathcal{F}_{\text{pair}} \leq 1$; and nonzero $\mathcal{F}_J$ or $\mathcal{F}_{\text{cyc}}$ identifies frustration. The effective interaction description applies inside the dispersive and finite-cover regime used to derive it.

The allowed positive implications are deliberately restricted to those supported by the corresponding spectral and resolvent controls. If the target photonic bandwidth satisfies $W_C \rightarrow 0$ while the qubit-band detuning remains outside the band according to

$$|\Delta_{qC}| > \frac{W_C}{2},$$

then the target-band Green operator approaches the projector-dominated form

$$G_C \rightarrow \frac{P_C}{\Delta_{qC}}.$$

Under this condition, the corresponding coherent qubit interaction inherits the same projector structure,

$$J_C \rightarrow \frac{V_q P_C V_q^\dagger}{\Delta_{qC}}.$$

This establishes projector-controlled target-band mediation, but does not by itself determine the spatial range, focusing profile, or frustration structure of the resulting interaction. Spatial decay requires an additional resolvent-locality condition. If the relevant detuning margin satisfies $\delta_q > 0$ and the off-diagonal propagation bound obeys $\mathcal{B}_\mu < \delta_q$, then the induced pair interaction is exponentially bounded as

$$|J_{ab}| \leq C e^{-\mu r_{ab}}.$$

Thus, the spatial range of the effective interaction is controlled by the corresponding Green-function locality rather than by flatness alone. Likewise, a non-Abelian reconstruction of the interaction is supported only when $D_J^{\text{NA}} > 0$ is resolved together with complete representation-sector control. Interaction frustration has a separate gauge-invariant criterion: a nontrivial cycle phase,

$$\Phi_C^J \not\equiv 0 \pmod{2\pi},$$

is sufficient to establish a frustrated interaction structure on that cycle. Several stronger implications are explicitly prohibited. The flat-band limit $W_C \rightarrow 0$ does not imply that the interaction range R_J diverges, nor does it imply pairwise focusing, $\mathcal{F}_{\text{pair}} > 1$. Similarly, a divergent or strongly enhanced photonic density of states, $\varrho_C^{\text{max}} \rightarrow \infty$, does not by itself imply $|J_{ab}| \rightarrow \infty$, because the mediated interaction also depends on detuning, projector overlaps, coupling geometry, and dissipative corrections. A large matrix element $|J_{ab}|$ does not imply a large coherent-interaction quality factor $\mathcal{Q}_J$, and a long interaction range R_J does not imply the absence of frustration. Representation-theoretic statements require the same caution. The existence of a higher-dimensional irreducible representation $d_\rho > 1$ does not imply a nonzero interaction diagnostic D_J^{NA} , because the corresponding sector may have no resolved effect on the qubit-mediated coupling. Likewise, $D_X E_C = 0$ for a photonic spectral quantity does not imply $D_X J_{ab} = 0$, since the effective qubit interaction depends on both the photonic spectrum and the coupling projectors. Finally, an interaction matrix extracted from a single finite cover cannot by itself be promoted to either a generic-twist interaction or a cover-independent bulk interaction. The complete causal construction begins with the photonic control parameters (θ, K, ω) , which determine the photonic Hamiltonian H_γ .

The latter fixes the target projector P_C , bandwidth W_C , isolation gap Δ_C , and photonic Green operator $\mathbf{G}_\gamma$. Independently, the placement and local coupling geometry of the transmon qubits determine the qubit-photon coupling matrix $\mathbf{V}_q$. The combined data $(\mathbf{H}_\gamma, \mathbf{V}_q, \omega_q)$ then determine the retarded qubit self-energy Σ_q^R , from which both the coherent interaction matrix $\mathbf{J}$ and the mediated dissipative matrix $\mathbf{\Gamma}^{\text{med}}$ are extracted. These effective interaction matrices determine the spatial and structural diagnostics

$$R_J, \quad P_J, \quad \mathcal{F}_{\text{pair}}, \quad \mathcal{F}_J,$$

which respectively characterize the interaction range, spatial distribution, pairwise focusing, and frustration structure. The same interaction data are then analysed through the representation-sensitive quantities

$$D_J^{\text{NA}}, \quad \mathcal{F}_H^J, \quad \eta_c,$$

which test whether higher-dimensional translation sectors make a resolved contribution to the effective qubit dynamics. The final interaction classification is therefore obtained only after propagating the complete sequence

$$(\theta, K, \omega) \longrightarrow \mathbf{H}_\gamma \longrightarrow (P_C, W_C, \Delta_C, \mathbf{G}_\gamma),$$

together with the independent transmon-placement map $\mathbf{V}_q$, through the retarded self-energy Σ_q^R , the coherent and dissipative effective interactions $(\mathbf{J}, \mathbf{\Gamma}^{\text{med}})$, and finally the spatial, focusing, frustration, and representation-sensitive diagnostics. Depending on which of these established signatures is resolved, the induced qubit interaction is classified as local, extended, shell-accumulated, focused, frustrated, or as exhibiting a resolved non-Abelian contribution. We therefore treat the photonic flat-band angle as an independently determined operating point at which the induced qubit interaction is subsequently tested. The photonic flat band is not defined through the qubit interaction, and interaction focusing, long-range mediation, frustration, or non-Abelian reconstruction is never inferred solely from photonic flatness. Each of these properties remains an independent physical claim requiring its own diagnostic and verification.

XIV. DISCUSSION

We now assemble the geometric, crystallographic, representation-theoretic, spectral, finite-cover, spectroscopic, and interaction results into one physical hierarchy. Our central conclusion is not that negative curvature automatically creates a magic angle. Negative curvature changes the geometric and algebraic architecture in which a magic-band mechanism may occur. The existence, position, stability, and physical meaning of a flat-band optimum remain properties of a declared microscopic Hamiltonian and of a declared spectral family.

The complete hierarchy contains five distinct layers:

$$\begin{aligned} \mathfrak{H}_{\text{phys}} = & \left(\xi_M, A_M^{\text{eff}}, N_M^{\text{geo}}; q_M, N_M^{\text{exact}}, D_N; \right. \\ & \Gamma_M, \{\rho\}, \mathcal{B}_d; W_C, \Delta_C, \Omega_C^{\text{max}}, \Lambda_C^{\text{max}}, \varrho_C, C_C^{\text{coh}}; \\ & \left. \mathbf{J}, \mathbf{\Gamma}^{\text{med}}, R_J, \mathcal{F}_J \right). \end{aligned} \quad (3230)$$

The first line contains local moiré geometry. The second contains exact crystallographic and finite-device complexity. The third contains the generalized translation representation structure. The fourth contains the physical single-particle spectral certificate. The final line contains photon-mediated interaction observables. These layers are connected by the directed chain

$$\begin{aligned} (K, \theta, a) & \longrightarrow (\xi_M, A_M^{\text{eff}}), \\ (\Gamma, g_\theta) & \longrightarrow (\Gamma_M, q_M), \\ (\Gamma_M, \rho, h) & \longrightarrow H[\rho], \\ H[\rho] & \longrightarrow (W_C, \Delta_C, \Omega_C^{\text{max}}, \Lambda_C^{\text{max}}), \\ (H_\gamma, \mathbf{V}_q, \omega_q) & \longrightarrow (\mathbf{J}, \mathbf{\Gamma}^{\text{med}}). \end{aligned} \quad (3231)$$

No reverse implication in Eq. (3231) is automatic. In particular,

$$\begin{aligned} A_M^{\text{eff}} & \not\Rightarrow q_M, \\ q_M & \not\Rightarrow \text{spectral flatness}, \\ d_\rho > 1 & \not\Rightarrow \text{a flat band}, \\ \nabla^2 E = 0 & \not\Rightarrow W_C = 0, \\ W_C \ll \Delta_C & \not\Rightarrow \text{interaction focusing}. \end{aligned} \quad (3232)$$

A. What negative curvature actually changes

We parameterize constant negative Gaussian curvature by

$$K = -\frac{1}{R^2} < 0, \quad (3233)$$

where R is the curvature radius. Let

$$s_\theta = \left| \sin \left(\frac{\theta}{2} \right) \right|. \quad (3234)$$

The exact displacement between a point at geodesic radius r and its copy rotated through the relative angle θ is

$$d_\theta^{(\text{H})}(r) = 2R \operatorname{arsinh} \left[s_\theta \sinh \left(\frac{r}{R} \right) \right]. \quad (3235)$$

We define the local moiré length by the fixed microscopic registry condition

$$d_\theta^{(\text{H})}(\xi_M) = a. \quad (3236)$$

Solving exactly gives

$$\xi_M^{(\text{H})} = R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{s_\theta} \right]. \quad (3237)$$

Introduce the exact crossover variable

$$\chi = \frac{\sinh(a/2R)}{s_\theta}. \quad (3238)$$

Then

$$\xi_M^{(\mathbb{H})} = R \operatorname{arsinh} \chi. \quad (3239)$$

The associated geodesic-disk area is

$$A_M^{\text{eff}, \mathbb{H}} = 2\pi R^2 \left(\sqrt{1 + \chi^2} - 1 \right). \quad (3240)$$

For a primitive single-layer quotient area $A_0(K)$ and n_b orbitals per layer, the area-normalized effective microscopic count is

$$N_M^{\text{geo}, \mathbb{H}} = 2n_b \frac{A_M^{\text{eff}, \mathbb{H}}}{A_0(K)}. \quad (3241)$$

The Euclidean limit is obtained at fixed a and fixed nonzero θ :

$$\lim_{R \rightarrow \infty} \sinh \left(\frac{a}{2R} \right) = \frac{a}{2R}, \quad (3242)$$

and therefore

$$\lim_{R \rightarrow \infty} \xi_M^{(\mathbb{H})} = \frac{a}{2s_\theta} = \xi_M^{(\mathbb{E})}. \quad (3243)$$

The corresponding Euclidean effective area is

$$A_M^{\text{eff}, \mathbb{E}} = \pi \left(\xi_M^{(\mathbb{E})} \right)^2 = \frac{\pi a^2}{4s_\theta^2}. \quad (3244)$$

For small twist at fixed negative curvature, $\chi \rightarrow \infty$, and

$$\operatorname{arsinh} \chi = \ln(2\chi) + O(\chi^{-2}). \quad (3245)$$

Consequently,

$$\xi_M^{(\mathbb{H})} = R \ln \left[\frac{2 \sinh(a/2R)}{s_\theta} \right] + O(Rs_\theta^2). \quad (3246)$$

Since

$$s_\theta = \frac{|\theta|}{2} + O(|\theta|^3), \quad (3247)$$

we obtain

$$\xi_M^{(\mathbb{H})} = R \ln |\theta|^{-1} + O(R). \quad (3248)$$

For the area,

$$\sqrt{1 + \chi^2} - 1 = \chi + O(1), \quad (3249)$$

so

$$A_M^{\text{eff}, \mathbb{H}} \sim \frac{4\pi R^2 \sinh(a/2R)}{|\theta|}. \quad (3250)$$

In contrast,

$$A_M^{\text{eff}, \mathbb{E}} \sim \frac{\pi a^2}{|\theta|^2}. \quad (3251)$$

Define the running geometric exponent

$$\beta_{\text{geo}} = -\frac{d \ln N_M^{\text{geo}}}{d \ln |\theta|}. \quad (3252)$$

Then

$$\beta_{\text{geo}}^{(\mathbb{E})} \longrightarrow 2, \quad \beta_{\text{geo}}^{(\mathbb{H})} \longrightarrow 1. \quad (3253)$$

This exponent change is the most robust geometric consequence of negative curvature. It follows from exponential geodesic separation and exponential area growth. It does not require an exact supercell, a Hamiltonian, or a spectral root. Negative curvature does not imply that the effective area is monotone in $|K|$ at every fixed twist. Define

$$\alpha = \frac{a}{2R}. \quad (3254)$$

The exact logarithmic curvature response can be written as

$$\frac{d \ln A_M^{\text{eff}}}{d \ln |K|} = \frac{1}{2} [\alpha \coth \alpha \beta_{\text{geo}} - 2]. \quad (3255)$$

Its sign is not fixed universally. Increasing negative curvature contracts the registry radius but also increases the area carried by a fixed geodesic radius. The two effects compete. The exact crossover is controlled by $\chi \simeq 1$. For $a/R \ll 1$, this gives the approximate angular scale

$$|\theta_\times| \sim \frac{a}{R}. \quad (3256)$$

The regime $|\theta| \gg |\theta_\times|$ is locally Euclidean, whereas $|\theta| \ll |\theta_\times|$ enters the curvature-dominated logarithmic regime. Local registry does not determine exact commensurability. For a single-layer Fuchsian translation group Γ , the exact common translation group is

$$\Gamma_M(\theta) = \Gamma \cap g_\theta \Gamma g_\theta^{-1}. \quad (3257)$$

An exact finite moiré quotient exists only when

$$q_M = [\Gamma : \Gamma_M] < \infty. \quad (3258)$$

The exact number of physical orbitals in one bilayer moiré quotient is

$$N_M^{\text{exact}} = 2n_b q_M. \quad (3259)$$

Define the geometric–arithmetic locking factor

$$L_M = \frac{N_M^{\text{exact}}}{N_M^{\text{geo}}} = \frac{q_M A_0(K)}{A_M^{\text{eff}}}. \quad (3260)$$

The exact count factorizes as

$$N_M^{\text{exact}} = L_M N_M^{\text{geo}}. \quad (3261)$$

All deviation of exact device complexity from local registry geometry is contained in L_M . Along an exact small-angle sequence, define

$$\beta_{\text{arith}} = \lim \frac{\ln L_M}{\ln |\theta|^{-1}} \quad (3262)$$

when the limit exists. The exact crystallographic exponent is then

$$\beta_{\text{exact}} = \beta_{\text{geo}} + \beta_{\text{arith}}. \quad (3263)$$

In the curvature-dominated hyperbolic regime, $\beta_{\text{geo}} = 1$. The explicit centered arithmetic sequence used in the established $\beta_{\text{exact}} = 4$, calculation has and hence $\beta_{\text{arith}} = 3$. This result is a property of the declared arithmetic sequence. It is not a universal exponent for every hyperbolic commensurate family. If the common quotient has genus g_M , a torsion-free exact cover satisfies

$$g_M = 1 + q_M(g_0 - 1). \quad (3264)$$

Thus exact arithmetic complexity also changes the topology of the compact quotient on which generalized Bloch sectors are defined. A numerical or physical finite cover introduces a second complexity factor. Let

$$Q_N = \Gamma_M / \Gamma_N, \quad d_N = |Q_N|. \quad (3265)$$

The finite one-particle Hilbert-space dimension is

$$D_N = 2n_b q_M d_N. \quad (3266)$$

The exact moiré index q_M specifies the number of orbitals in one common cell. The cover degree d_N specifies how many common cells are required to reach a chosen injectivity radius. A geodesic ball of radius $r_{\text{inj},N}$ embeds into a compact cover only if the cover area is at least

$$2\pi R^2 \left[\cosh \left(\frac{r_{\text{inj},N}}{R} \right) - 1 \right]. \quad (3267)$$

Consequently,

$$D_N \geq 2n_b q_M \frac{2\pi R^2}{A_M^{\text{sc}}} \left[\cosh \left(\frac{r_{\text{inj},N}}{R} \right) - 1 \right]. \quad (3268)$$

For $r_{\text{inj},N} \gg R$, this becomes

$$D_N \gtrsim C_{\text{dev}} \exp \left(\frac{r_{\text{inj},N}}{R} \right). \quad (3269)$$

Negative curvature therefore makes increasing local bulk fidelity exponentially expensive in physical device size. This resource law is independent of whether the target spectrum is flat. An exact compact hyperbolic quotient has a non-Abelian surface-group translation structure:

$$\Gamma_M = \left\langle a_1, b_1, \dots, a_{g_M}, b_{g_M} \left| \prod_{j=1}^{g_M} [a_j, b_j] = e \right. \right\rangle. \quad (3270)$$

Its Abelianization is

$$\Gamma_M^{\text{ab}} \simeq \mathbb{Z}^{2g_M}. \quad (3271)$$

The one-dimensional sectors therefore form the Abelian character torus

$$\mathcal{B}_1 \simeq \mathbb{T}^{2g_M}. \quad (3272)$$

The full group also admits higher-dimensional irreducible unitary representations

$$\rho : \Gamma_M \longrightarrow U(d_\rho), \quad d_\rho > 1. \quad (3273)$$

At a smooth irreducible point, the local representation-space dimension is

$$\dim_{\mathbb{R}} \mathcal{B}_{d_\rho} = (2g_M - 2)d_\rho^2 + 2. \quad (3274)$$

Negative curvature therefore changes the reciprocal-space problem from one ordinary two-dimensional momentum torus to a hierarchy:

$$\mathbb{T}^2 \longrightarrow \mathbb{T}^{2g_M} \longrightarrow \mathcal{B}_{d_\rho > 1}. \quad (3275)$$

The generalized Bloch Hamiltonian is

$$H[\rho] = \sum_{\delta \in \Gamma_M} h(\delta; K, \theta, \omega) \otimes \rho(\delta). \quad (3276)$$

Curvature can therefore enter the spectral problem through three separate channels: the microscopic kernel $h(\delta; K, \theta, \omega)$, the exact common group $\Gamma_M(K, \theta)$, the available representation sectors $[\rho] \in \mathcal{B}_d$. A generic incommensurate twist has

$$[\Gamma : \Gamma_M] = \infty. \quad (3277)$$

There is then no finite compact common quotient and no global finite representation-space Bloch decomposition of the commensurate type. The local Hamiltonian remains well defined, but its correct infinite-volume interpretation is local or approximant based. For a tracked target cluster C , we use the physical spectral vector

$$\mathfrak{S}_C = \left(W_C, \Delta_C, \hbar D_G \Omega_C^{\text{max}}, \frac{1}{2} D_G^2 \Lambda_C^{\text{max}}, \varrho_{C,\eta}^{\text{coh,max}}, C_C^{\text{coh}} \right). \quad (3278)$$

Negative curvature can modify every component of $\mathfrak{S}_C$, but no component has a universal monotonic sign as a function of K . For a simple branch,

$$\partial_K E_n = \langle u_n | \partial_K H | u_n \rangle. \quad (3279)$$

If $H_\omega = tV_\perp$, the mixed curvature-hybridization response is

$$\begin{aligned} \partial_K \partial_\omega E_n &= \langle n | H_{K\omega} | n \rangle \\ &+ 2 \operatorname{Re} \sum_{m \neq n} \frac{\langle n | H_K | m \rangle \langle m | H_\omega | n \rangle}{E_n - E_m}. \end{aligned} \quad (3280)$$

The first term measures direct curvature dependence of the tunnelling matrix. The second measures curvature-induced reconstruction of the eigenvectors and energy denominators. Their signs are not fixed. The exact square five-state first-star model and the natural centered surface-group model exhibit a genuinely model-dependent dichotomy. In the square five-state first-star construction, the quadratic magic condition admits no root. By contrast, the natural centered surface-group model possesses a positive simple root, while the regular-octagon subclass realizes the still stronger condition of complete Hodge–Hessian cancellation. These outcomes must be interpreted within their respective Hamiltonians. In particular, the positive root found in the centered surface-group model does not repair, modify, or retrospectively reinterpret the absence of a quadratic magic root in the square five-state model; the two statements concern different Hamiltonians and therefore constitute independent spectral results. The resulting physical conclusion is correspondingly more restricted than a universal curvature-induced magic condition. Negative curvature changes the space of available spectral mechanisms and can permit cancellation structures that are absent in the corresponding Euclidean or first-star construction, but curvature alone does not force such a cancellation to occur. In particular, the existence of negative curvature cannot by itself be promoted to the existence of a spectral root. The presence, simplicity, and possible enhancement of such a root remain properties of the specific Hamiltonian and its representation structure. For transmon placement matrix $\mathbf{V}_q$, qubit frequency ω_q , and photonic loss matrix $\mathbf{\Gamma}_\gamma$, the retarded photon-mediated self-energy is

$$\Sigma_q^R = \mathbf{V}_q \left[\omega_q I - H_\gamma + \frac{i}{2} \mathbf{\Gamma}_\gamma \right]^{-1} \mathbf{V}_q^\dagger. \quad (3281)$$

Its Hermitian and anti-Hermitian parts define

$$\Sigma_q^R = \mathbf{J} - \frac{i}{2} \mathbf{\Gamma}^{\text{med}}. \quad (3282)$$

For fixed placement, fixed qubit frequency, and curvature-independent loss, the exact curvature derivative is

$$\partial_K \Sigma_q^R = \mathbf{V}_q \mathbf{G}_\gamma^R (\partial_K H_\gamma) \mathbf{G}_\gamma^R \mathbf{V}_q^\dagger. \quad (3283)$$

This response depends on two propagators and can therefore be enhanced near a photonic resonance. The same enhancement also increases photon admixture and dissipative sensitivity. In the isolated exactly flat target-band limit,

$$P_C H_\gamma P_C = E_C P_C, \quad (3284)$$

the target contribution is

$$\mathbf{J}_C = \frac{\mathbf{V}_q P_C \mathbf{V}_q^\dagger}{\omega_q - E_C}. \quad (3285)$$

Curvature can change P_C , the detuning, the complementary-band self-energy, and the spatial metric. It does not determine the interaction pattern from the bandwidth alone. The physical distinction is

flat denominator :	$E_n \simeq E_C,$	
interaction profile :	$\mathbf{V}_q P_C \mathbf{V}_q^\dagger,$	
interaction range :	$R_J,$	
interaction focusing :	an interior maximum of pair-averaged $ J_{ab} ,$	
shell accumulation :	growth caused by hyperbolic shell multiplicity	(3286)

For pairwise decay

$$|J_{ab}| \lesssim J_0 e^{-\mu r_{ab}}, \quad (3287)$$

and hyperbolic shell growth

$$N_{\text{sh}}(r) \sim e^{h_{\text{r}} r}, \quad (3288)$$

the shell-summed interaction scales as

$$\sum_{b: r_{ab} \simeq r} |J_{ab}| \sim e^{(h_{\text{r}} - \mu)r}. \quad (3289)$$

A growing shell sum can therefore coexist with monotonically decreasing pairwise interactions. We call the former shell accumulation and reserve interaction focusing for a genuine nonmonotonic pair-averaged profile.

B. What is genuinely non-Abelian

The surface group is non-Abelian because there exist $\gamma_1, \gamma_2 \in \Gamma_{\text{M}}$ satisfying

$$[\gamma_1, \gamma_2] = \gamma_1 \gamma_2 \gamma_1^{-1} \gamma_2^{-1} \neq e. \quad (3290)$$

This is a property of the translation group. It does not imply that a particular state probes noncommuting holonomies. Every one-dimensional character satisfies

$$\chi([\gamma_1, \gamma_2]) = 1. \quad (3291)$$

Therefore one-dimensional generalized Bloch sectors read only the Abelianized path data. Let

$$\rho : \Gamma_{\text{M}} \longrightarrow U(d_\rho) \quad (3292)$$

be irreducible. We use the following exact criterion:

$$d_\rho > 1 \iff \rho(\Gamma_{\text{M}}) \text{ is noncommutative.} \quad (3293)$$

The reverse direction is immediate because a one-dimensional image is commutative. For the forward direction, suppose every matrix in $\rho(\Gamma_{\text{M}})$ commutes. The matrices are unitary and therefore normal. A commuting family of normal matrices is simultaneously diagonalizable. Every common one-dimensional eigenspace is invariant, contradicting irreducibility when $d_\rho > 1$. Therefore an irreducible higher-dimensional sector necessarily

TABLE X. Five physically distinct effects of negative curvature. The final column records an implication that is not supported without an additional bridge theorem or numerical certificate.

Layer	Primary object	What negative curvature changes	What does not follow automatically
Local geometry	ξ_M, A_M^{eff}	Geodesic separation and effective-count exponent	Exact commensurate cell size
Exact complexity	q_M, D_N, r_{inj}	Arithmetic possibilities and finite-cover resources	A flat spectral branch
Representation structure	$\Gamma_M, \mathcal{B}_d$	Non-Abelian translation sectors and Hodge tangent space	Spectral relevance of every sector
Single-particle spectrum	$W_C, \Delta_C, \Omega_C^{\text{max}}$	Hybridization matrix and self-energy reconstruction	A universal magic angle
Interaction response	J, Γ^{med}	Photon propagator, projector, range, and cycle phases	Long-range or focused interactions

has a noncommuting image. Equivalently, for every irreducible $d_\rho > 1$ sector there exists at least one commutator word $c \in [\Gamma_M, \Gamma_M]$ such that $\rho(c) \neq I_{d_\rho}$. Define the gauge-invariant commutator defect

$$\eta_c(\rho) = 1 - \frac{1}{d_\rho} \text{Re Tr } \rho(c). \quad (3294)$$

For a unitary matrix,

$$\begin{aligned} \|I - \rho(c)\|_F^2 &= \text{Tr} [2I - \rho(c) - \rho(c)^\dagger] \\ &= 2d_\rho - 2 \text{Re Tr } \rho(c), \end{aligned} \quad (3295)$$

so

$$\eta_c(\rho) = \frac{\|I - \rho(c)\|_F^2}{2d_\rho}. \quad (3296)$$

Thus $0 \leq \eta_c(\rho) \leq 2$, and

$$\eta_c(\rho) = 0 \iff \rho(c) = I. \quad (3297)$$

Representation-level noncommutativity becomes physically relevant only when it changes a spectral invariant. The m -th spectral moment of Eq. (3276) is

$$\begin{aligned} \text{Tr } H[\rho]^m &= \sum_{\delta_1, \dots, \delta_m} \text{Tr}_{\text{orb}} [h(\delta_1) \cdots h(\delta_m)] \\ &\times \chi_\rho(\delta_1 \cdots \delta_m), \end{aligned} \quad (3298)$$

where

$$\chi_\rho(w) = \text{Tr } \rho(w) \quad (3299)$$

is the Wilson character of the propagation word. Define the normalized moment-separation witness

$$\mathcal{W}_m^{\text{NA}}(\rho) = \inf_{\chi} \frac{|D_\rho^{-1} \text{Tr } H[\rho]^m - D_\chi^{-1} \text{Tr } H[\chi]^m|}{E_{\text{kin}}^m}, \quad (3300)$$

where the infimum is over all one-dimensional characters in the declared Abelian comparison set. If $\mathcal{W}_m^{\text{NA}}(\rho) > 0$ beyond uncertainty, no Abelian sector has the same normalized m -th moment. Since identical finite spectra have identical moments, one resolved moment difference proves spectral inequivalence. For two paths with words w_1 and w_2 satisfying

$$w_1^{-1} w_2 = c \in [\Gamma_M, \Gamma_M], \quad (3301)$$

the representation-space amplitude is

$$A_\rho = \rho(w_1) [t_1 I + t_2 \rho(c)]. \quad (3302)$$

The corresponding Frobenius power is

$$\|A_\rho\|_F^2 = d_\rho (|t_1|^2 + |t_2|^2) + 2 \text{Re} [t_1^* t_2 \text{Tr } \rho(c)]. \quad (3303)$$

Every Abelian character replaces $\rho(c)$ by one. A measured difference is therefore a direct commutator-sensitive spectral effect. For a target projector P_C , define the total higher-rank weight

$$w_C^{\text{NA}} = \frac{1}{r_C} \sum_{\rho: d_\rho > 1} \text{Tr}(P_C P_\rho). \quad (3304)$$

Define the reconstructed translation noncommutativity

$$\mathcal{N}_C = \max_{g, h} \frac{\|[\rho_C(g), \rho_C(h)]\|_F}{2\sqrt{r_C}}. \quad (3305)$$

Define the full-versus-Abelian spectral discrepancy

$$D_{\text{spec}, C}^{\text{NA}} = \frac{d_H(S_C^{\text{full}}, S_C^{\text{Ab}})}{E_{\text{kin}}}. \quad (3306)$$

For photon-mediated interactions, define

$$D_J^{\text{NA}} = \frac{\|J^{\text{full}} - J^{\text{Ab}}\|_F}{\max\{\|J^{\text{full}}\|_F, J_{\text{ref}}\}}. \quad (3307)$$

We call a target physically translation non-Abelian only when

$$\begin{aligned} C_{\text{rep}, C} &= 1, \quad w_C^{\text{NA}, L} > 0, \quad \mathcal{N}_C^L > 0, \quad \eta_{\text{act}}^L > 0, \\ &\text{a spectral or interaction discrepancy is resolved,} \\ &\text{no Abelian sector supplies an equivalent admissible target.} \end{aligned} \quad (3308)$$

Here η_{act} is the largest commutator defect among propagation words that carry non-negligible microscopic hopping amplitude. The first two lines establish representation support. The third and fourth establish active noncommutativity. The fifth establishes a physical consequence. The final line excludes a relabeling of an Abelian target. We do not use the term translation non-Abelian for any of the following properties by itself:

$$K < 0, \quad g_M > 1, \quad \mathcal{B}_1 \simeq \mathbb{T}^{2g_M}, \quad (3309)$$

a multidimensional point-group irrep; a layer pseudospin doublet; a degenerate spectral multiplet; a matrix-valued Berry connection; a nonnormal lossy Hamiltonian; a matrix-valued tight-binding Hamiltonian. A higher-genus Abelian character remains Abelian because every group element is represented by a scalar phase. A point-group multiplet can be non-Abelian under a finite spatial symmetry group while remaining Abelian under translation. A non-Abelian Berry connection describes parallel transport inside a degenerate band bundle; it is conceptually different from a higher-dimensional irreducible representation of the translation group. Nonnormality from photon loss is a property of the dynamical generator and does not establish noncommuting translation holonomy.

C. Difference from ordinary Euclidean magic-angle physics

Euclidean and hyperbolic moiré systems share one essential physical principle: band folding is not magic-band flattening. Folding changes the representation of the spectrum in a reduced reciprocal domain. Genuine flattening requires dynamical reconstruction by interlayer hybridization or another microscopic coupling. For an exactly commensurate Euclidean bilayer with coincidence index Σ ,

$$A_M^{\text{sc}} = \Sigma a^2, \quad (3310)$$

and the moiré Brillouin-zone area is

$$A_{\text{mBZ}} = \frac{A_{\text{BZ}}}{\Sigma}. \quad (3311)$$

Each monolayer band folds into Σ branches. Without interlayer coupling, the folded branches are

$$E_\alpha^{(0)}(\mathbf{k}) = \varepsilon_0(\mathbf{k} + \mathbf{Q}_\alpha), \quad (3312)$$

and

$$\nabla_{\mathbf{k}} E_\alpha^{(0)} = \nabla_{\mathbf{q}} \varepsilon_0. \quad (3313)$$

The intrinsic velocity is unchanged. After interlayer tunnelling is introduced,

$$H_{\text{fold}}(\mathbf{k}) = \begin{pmatrix} E_1^{\text{fold}}(\mathbf{k}) & T(\mathbf{k}) \\ T^\dagger(\mathbf{k}) & E_2^{\text{fold}}(\mathbf{k}) \end{pmatrix}. \quad (3314)$$

At an isolated two-state resonance,

$$\Delta_{\text{hyb}} = 2|V|. \quad (3315)$$

Away from resonance,

$$\delta E = \frac{|V|^2}{\Delta} + O(|V|^4). \quad (3316)$$

The Euclidean hybridization-causality test requires the same tracked family to satisfy

$$W_C(\omega) < W_C(0), \quad v_{\text{max},C}(\omega) < v_{\text{max},C}(0), \quad (3317)$$

and $N_{\text{part},k} > 1$. The final condition proves that the eigenstate contains a nontrivial superposition of folded momenta. In a Dirac-type Euclidean model, suppression of a Dirac velocity can be an important local signature. In the square-lattice benchmark, no Dirac cone is assumed. We therefore retain the more general operational conditions

$$W_C \ll \Delta_C, \quad \hbar D_G \Omega_C^{\text{max}} \ll \Delta_C, \quad (3318)$$

supplemented by coherent participation and hybridization causality. The Euclidean local scale is algebraic:

$$\xi_M^{(\mathbb{E})} \sim a|\theta|^{-1}, \quad N_M^{\text{geo},\mathbb{E}} \sim |\theta|^{-2}. \quad (3319)$$

The curvature-dominated hyperbolic scale is logarithmic in length and linear in effective area:

$$\xi_M^{(\mathbb{H})} \sim R \ln |\theta|^{-1}, \quad N_M^{\text{geo},\mathbb{H}} \sim |\theta|^{-1}. \quad (3320)$$

The two problems therefore belong to different geometric asymptotic classes even though the hyperbolic construction has a continuous Euclidean limit at fixed nonzero twist. Suppose the relevant effective coupling has the generic form

$$\alpha_{\text{eff}} \sim \omega \left(\frac{\xi_M}{a} \right)^p, \quad p > 0, \quad (3321)$$

and suppose a microscopic model has an independently established root $\alpha_{\text{eff}} = \alpha_*$. In the Euclidean class,

$$\alpha_{\text{eff}}^{(\mathbb{E})} \sim \omega |\theta|^{-p}, \quad (3322)$$

so

$$|\theta_*^{(\mathbb{E})}| \sim C_{\mathbb{E}} \omega^{1/p}. \quad (3323)$$

In the curvature-dominated hyperbolic class,

$$\alpha_{\text{eff}}^{(\mathbb{H})} \sim \omega [\ln |\theta|^{-1}]^p, \quad (3324)$$

and therefore

$$|\theta_*^{(\mathbb{H})}| \sim \exp \left[-C_{\mathbb{H}}(K) \omega^{-1/p} \right]. \quad (3325)$$

For the square-type kinetic scaling $p = 2$,

$$|\theta_*^{(\mathbb{E})}| \sim \sqrt{\omega}, \quad |\theta_*^{(\mathbb{H})}| \sim \exp \left[-C_{\mathbb{H}}(K) \omega^{-1/2} \right]. \quad (3326)$$

These relations are conditional images of an independently established microscopic root. Geometry determines the map from α_* to θ_* ; geometry does not prove that α_* exists. The Euclidean translation group is $\Gamma_{\mathbb{E}} = \mathbb{Z}^2$. Every finite-dimensional irreducible unitary representation is a one-dimensional character

$$\rho_{\mathbf{k}}(\mathbf{R}) = e^{i\mathbf{k} \cdot \mathbf{R}}. \quad (3327)$$

The band problem is globally parameterized by $\mathbf{k} \in \mathbb{T}^2$. The hyperbolic surface group instead supplies

$$\mathcal{B}_1 \simeq \mathbb{T}^{2g_M} \quad (3328)$$

TABLE XI. Hierarchy of non-Abelian statements. Each row requires the preceding mathematical object but does not automatically imply the next physical row.

Level	Certificate	Physical meaning
Group	$[\gamma_1, \gamma_2] \neq e$	The translation group is noncommutative
Representation	$d_\rho > 1, \rho(c) \neq I$	A state carries noncommuting translation holonomy
Spectral	$\mathcal{W}_m^{\text{NA}} > 0$ or $D_{\text{spec}}^{\text{NA}} > 0$	Noncommutativity changes the one-particle spectrum
Experimental	$w_C^{\text{NA}} > 0, \mathcal{N}_C > 0$, active commutator response	The measured target probes a non-Abelian sector
Interaction	$D_J^{\text{NA}} > 0$ or a matrix Wilson-loop difference	Non-Abelian propagation changes the qubit interaction

and higher-rank spaces $\mathcal{B}_{d_\rho > 1}$. The ordinary Euclidean derivative $\nabla_{\mathbf{k}} E$ is therefore replaced by the covector

$$dE \in T_{[\rho]}^* \mathcal{B}_{d_\rho}. \quad (3329)$$

Its norm requires the canonical Hodge metric. A Euclidean periodic supercell has a boundary-to-volume ratio that vanishes in the conventional thermodynamic limit. An open hyperbolic disk retains an extensive boundary. We therefore use boundaryless finite covers and injectivity radius rather than open-cluster site count as the primary approximation variable. The implication

$$D_N \rightarrow \infty \implies r_{\text{inj}, N} \rightarrow \infty \quad (3330)$$

is false without a controlled cover sequence. Even when $r_{\text{inj}, N} \rightarrow \infty$, weak local or DOS convergence does not exclude global spectral pollution. The hyperbolic calculation therefore introduces verification layers that are normally less prominent in an ordinary Euclidean Bloch calculation: local lifting + no loss + no pollution + representation completeness + cross-tower convergence. The exact square five-state first-star Hamiltonian is root-free. The absence of the root is not repaired by: changing the normalization; embedding the model in a larger finite cover; relabeling the sector by representation theory; sampling an arithmetic sequence; invoking the geometric exponent crossover. A root in the richer centered surface-group Hamiltonian demonstrates that a physically enlarged hopping and hybridization structure can produce a cancellation absent from the five-state truncation. It does not turn the truncated square model into a magic model.

D. Hodge cancellation and global flattening

At a smooth representation point, let

$$V_\rho = T_{[\rho]} \mathcal{B}_{d_\rho} \quad (3331)$$

be the physical tangent space, equipped with canonical metric G_ρ . For one simple energy branch, the first differential is $dE \in V_\rho^*$, and the covariant Hessian is

$$\mathcal{K} = \nabla^2 E \in \text{Sym}^2 V_\rho^*. \quad (3332)$$

The global bandwidth is

$$W_C = \sup_{\mathcal{D}_C} E_C - \inf_{\mathcal{D}_C} E_C. \quad (3333)$$

We maintain the strict distinction

$$\begin{aligned} dE(\rho_*) &= 0 : && \text{local stationarity,} \\ \mathcal{K}(\rho_*) &= 0 : && \text{local quadratic cancellation,} \\ W_C &\ll \Delta_C : && \text{global energetic flatness and isolation.} \end{aligned} \quad (3334)$$

The metric-normalized Hessian is

$$\widehat{\mathcal{K}} = G_\rho^{-1/2} \mathcal{K} G_\rho^{-1/2}. \quad (3335)$$

Let its principal values be $\lambda_1, \dots, \lambda_m$. Complete Hodge cancellation means

$$\lambda_j = 0 \quad \text{for every } j. \quad (3336)$$

This condition is independent of tangent-coordinate choices. A vanishing trace, $\text{Tr } \widehat{\mathcal{K}} = 0$ is insufficient. It can describe a saddle with positive and negative principal values. For the natural centered surface-group bilayer, the Hodge Hessian has the explicit form

$$\mathcal{K}(w) = 2tC_S - wB_\infty. \quad (3337)$$

Here C_S is the intralayer kinetic tensor and B_∞ is the full-distance interlayer second-moment tensor. In the established regular-octagon symmetry class, the self-adjoint commutant is scalar, forcing

$$B_\infty = b_\infty C_S. \quad (3338)$$

The common root is

$$w_H = \frac{2t}{b_\infty}. \quad (3339)$$

At this coupling,

$$\mathcal{K}(w_H) = 0. \quad (3340)$$

All principal values vanish simultaneously and cross with one common nonzero slope. Outside the proportionality class, define

$$A = G^{-1/2} C_S G^{-1/2}, \quad D = G^{-1/2} B_\infty G^{-1/2}. \quad (3341)$$

The trace-root coupling is

$$w_{\text{tr}} = 2t \frac{\text{Tr } A}{\text{Tr } D}. \quad (3342)$$

TABLE XII. Comparison of ordinary Euclidean and hyperbolic magic-angle frameworks. The final row records the physical mechanism common to both settings.

Property	Euclidean bilayer	Hyperbolic bilayer
Local moiré length	$\xi_M \sim \theta ^{-1}$	$\xi_M \sim R \ln \theta ^{-1}$
Effective count	$N_M^{\text{geo}} \sim \theta ^{-2}$	$N_M^{\text{geo}} \sim \theta ^{-1}$
Translation group	$\mathbb{Z}^2$	Surface or Fuchsian group
Reciprocal domain	$\mathbb{T}^2$	$\mathbb{T}^{2g_M}$ plus higher-rank moduli
Irreducible sectors	All one dimensional	One- and higher-dimensional sectors
Generic incommensurate twist	No exact finite supercell	No exact finite common quotient
Finite approximation	Periodic supercell or open sample	Boundaryless finite covers and paired local covers
Differential dispersion	$\nabla_k E$	dE with canonical Hodge norm
Magic mechanism	Hybridization-induced dispersion reconstruction	Hybridization- and holonomy-dependent dispersion reconstruction

At this point,

$$\widehat{\mathcal{K}}(w_{\text{tr}}) = 2t \left[A - \frac{\text{Tr } A}{\text{Tr } D} D \right]. \quad (3343)$$

Define the anisotropic obstruction

$$a_\infty = \left\| A - \frac{\text{Tr } A}{\text{Tr } D} D \right\|_{\text{op}}. \quad (3344)$$

Then

$$\|\widehat{\mathcal{K}}(w_{\text{tr}})\|_{\text{op}} = 2t a_\infty. \quad (3345)$$

Complete cancellation occurs if and only if $a_\infty = 0$. In geodesic normal coordinates $x \in V_{\rho_*}$, the energy has the covariant expansion

$$\begin{aligned} E(x) &= E_* + dE_*[x] + \frac{1}{2} \mathcal{K}_*[x, x] \\ &+ \frac{1}{3!} \mathcal{T}_*^{(3)}[x, x, x] + \frac{1}{4!} \mathcal{T}_*^{(4)}[x, x, x, x] + \dots \end{aligned} \quad (3346)$$

At a complete stationary quadratic root,

$$dE_* = 0, \quad \mathcal{K}_* = 0, \quad (3347)$$

so

$$E(x) - E_* = \frac{1}{3!} \mathcal{T}_*^{(3)}[x^3] + \frac{1}{4!} \mathcal{T}_*^{(4)}[x^4] + \dots \quad (3348)$$

A tangent-inversion symmetry can force $\mathcal{T}_*^{(3)} = 0$, leaving quartic-leading local dispersion. Hodge cancellation therefore removes the quadratic term. It does not remove higher-order dispersion. The complete generalized velocity satisfies

$$\Omega_C(\rho) = \frac{1}{\hbar} \|dE_C(\rho)\|_{G^{-1}}. \quad (3349)$$

For a connected representation domain of Hodge diameter D_G ,

$$W_C \leq \hbar D_G \Omega_C^{\text{max}}. \quad (3350)$$

This is a global bound because the maximum is taken over the complete domain. If a stationary point ρ_* has covering radius

$$R_* = \sup_{\rho \in \mathcal{D}_C} d_G(\rho, \rho_*), \quad (3351)$$

and

$$\Lambda_2 = \sup_{\rho \in \mathcal{D}_C} \|G^{-1/2} \nabla^2 E_C(\rho) G^{-1/2}\|_{\text{op}}, \quad (3352)$$

then

$$W_C \leq \Lambda_2 R_*^2. \quad (3353)$$

If the central Hessian vanishes, tangent inversion removes odd orders, and a uniform fourth-derivative bound Λ_4 holds, then

$$W_C \leq \frac{\Lambda_4}{12} R_*^4. \quad (3354)$$

These bounds identify the extra hypotheses needed to turn local cancellation into global suppression. Let θ_H denote the angle or coupling inferred from complete Hodge cancellation, and let $\theta_{\text{flat}}^{\text{full}}$ denote the optimum of the full operational spectral certificate. Define

$$D_{\text{HF}} = d_H(\Theta_H, \Theta_{\text{flat}}^{\text{full}}). \quad (3355)$$

Hodge cancellation is identified as the microscopic explanation of the physical flat-band optimum only when the Hodge angle and the independently determined full flat-band optimum are statistically compatible. In particular, the discrepancy D_{HF} must remain unresolved within the combined theoretical and numerical uncertainty budget. At the same angle θ_H , the target band must also satisfy a strong flatness condition, with W_C/Δ_C small, while the geometric-dispersion scale must remain perturbative in the sense that

$$\frac{\hbar D_G \Omega_C^{\text{max}}}{\Delta_C}$$

is likewise small. These local and spectral conditions are not sufficient unless the target sector remains physically well defined. The protecting gap must stay open,

$\Delta_C > 0$, and the corresponding coherence measure must remain strictly positive, $C_C^{\text{coh}} > 0$. Finally, the same conclusion must survive completion of the representation content, enlargement of the relevant shell structure, and the finite-cover convergence analysis. Only after all of these conditions are satisfied do we interpret Hodge cancellation as the mechanism responsible for the established physical flat-band optimum. If $\mathcal{K}(\theta_H) = 0$ but the global spectral conditions fail, we obtain a local Hodge cancellation without a physical global flat band. If a global flat band exists but Θ_H and $\Theta_{\text{flat}}^{\text{full}}$ are separated, Hodge cancellation is not its dominant mechanism. Let

$$G(z) = \sum_n \frac{P_n}{z - E_n} \quad (3356)$$

be the photonic Green operator. Along a Hodge tangent X ,

$$D_X G = \sum_n \left[\frac{D_X P_n}{z - E_n} + \frac{P_n D_X E_n}{(z - E_n)^2} \right]. \quad (3357)$$

At an energy-stationary point, $D_X E_C = 0$, but $D_X P_C$ can remain nonzero. Therefore

$$D_X E_C = 0 \not\Rightarrow D_X J = 0. \quad (3358)$$

The interaction can remain highly sensitive to holonomy through projector rotation even when the target energy is stationary.

E. Experimental realizability

The most direct implementation is a synthetic circuit-QED or superconducting-resonator network. The resonators need not be physically embedded on a negatively curved substrate. The hyperbolic geometry is encoded in: the graph connectivity; the finite quotient relations; the calibrated hopping amplitudes; the programmed hopping phases; the layer-dependent interconnection pattern. The device Hamiltonian is

$$\frac{\hat{H}_{\text{dev}}}{\hbar} = \sum_i \Omega_i a_i^\dagger a_i + \sum_{i \neq j} J_{ij} a_i^\dagger a_j, \quad (3359)$$

with $J_{ji} = J_{ij}^*$. A distance-dependent microscopic design uses

$$J_{ij} = J_0 \exp\left(-\frac{d_{ij}}{\lambda}\right) e^{i\phi_{ij}}, \quad (3360)$$

supplemented by separately programmed intralayer and interlayer channels. The relevant experimental statement is therefore not that a physical material surface has constant Gaussian curvature. The statement is that a finite network realizes the same single-particle operator and symmetry relations as the declared hyperbolic lattice model. At an exact commensurate twist, the natural

finite device is a normal finite quotient of the common translation group:

$$Q_N = \Gamma_M / \Gamma_N. \quad (3361)$$

Its vertex set is

$$\mathcal{V}_N = \{(q, \alpha) : q \in Q_N, \quad \alpha = 1, \dots, \nu_M\}. \quad (3362)$$

The exact finite Hamiltonian is

$$H_N = \sum_\delta h(\delta) \otimes R_N[q_N(\delta)]. \quad (3363)$$

Each nonzero microscopic matrix element specifies one physical coupling between resonators in quotient-related cells. Before spectroscopy, the device is checked for group relations, Hermiticity, state counting, point-group covariance, and port-coupling calibration. At a generic incommensurate twist, no exact finite common quotient exists. We therefore implement paired finite covers of the two individual layers and evaluate interlayer couplings from their common lifts to the universal cover. For an observable of support radius L_{obs} and hopping range L_H , exact local representation requires

$$r_{\text{inj},N} > L_{\text{obs}} + L_H. \quad (3364)$$

For exponentially decaying hopping, L_H is replaced by a growing buffer whose omitted tail is explicitly bounded. If a commensurate angle θ_N approximates a target angle θ on a growing radius L_N , the angular error must satisfy

$$|\theta_N - \theta| \exp\left(\frac{L_N}{R}\right) \rightarrow 0. \quad (3365)$$

A large approximant with slowly decreasing angular error is not a controlled generic-twist realization. For port-coupling matrix K , the measured scattering matrix is

$$S(\Omega) = I - iK^\dagger \left[\Omega I - H_N + \frac{i}{2} \Gamma \right]^{-1} K. \quad (3366)$$

The local spectral density is

$$\varrho_i(\Omega) = -\frac{1}{\pi} \text{Im } G_{ii}^R(\Omega). \quad (3367)$$

The complete target projector can be reconstructed from pole residues or from a resolved spectral contour. Representation-selective drives have the form

$$|F_{\rho;\alpha\beta,s}\rangle = \sqrt{\frac{d_\rho}{|Q_N|}} \sum_{q \in Q_N} \rho(q)_{\alpha\beta}^* |q, s\rangle. \quad (3368)$$

For $d_\rho = 1$, these reduce to phase-coded Abelian drives. For $d_\rho > 1$, they address non-Abelian translation sectors. Layer-even and layer-odd drives are

$$|f_\pm\rangle = \frac{1}{\sqrt{2}} (|f_1\rangle \pm |f_2\rangle). \quad (3369)$$

TABLE XIII. Physical interpretation of Hodge-related outcomes.

Observed result	Supported interpretation	Excluded interpretation
$dE = 0$ along one direction	Directional stationarity	Complete generalized flattening
$\text{Tr } \widehat{\mathcal{K}} = 0$	Scalar trace cancellation	Full Hessian cancellation without anisotropy control
$\widehat{\mathcal{K}} = 0$	Complete local quadratic cancellation	Global flatness without higher-order control
$\widehat{\mathcal{K}} = 0$ and small global derivative bounds	Hodge mechanism for global suppression	Automatic positive isolation gap
$\theta_H \simeq \theta_{\text{flat}}^{\text{full}}$ with all conditions	Hodge explanation of the physical optimum	A universal mechanism for every model
Global flatness with $D_{\text{HF}} > 0$	A non-Hodge global mechanism	Failure of the flat-band observation

Their relative spectral weights determine layer polarization and interlayer coherence. Transmons coupled through placement matrix $\mathbf{V}_q$ provide a second physical probe. In the dispersive regime, the coherent interaction matrix is

$$\mathbf{J} = \text{Herm} \{ \mathbf{V}_q \mathbf{G}_\gamma^R(\omega_q) \mathbf{V}_q^\dagger \}. \quad (3370)$$

This measurement tests the photonic Green operator rather than only its spectrum. It is sensitive to target-projector geometry, propagation phases, complementary bands, and non-Abelian holonomy. Let the total static Hermitian implementation error be $\varepsilon_{\text{stat}}^{(0)}$. The conservative spectral margins are

$$W_C^U = W_C^{(0)} + 2\varepsilon_{\text{stat}}^{(0)}, \quad (3371)$$

and

$$\Delta_C^L = \Delta_C^{(0)} - 2\varepsilon_{\text{stat}}^{(0)}. \quad (3372)$$

The intrinsic island survives only when $\Delta_C^L > 0$. Let

$$\kappa_\perp = -\partial_\theta^2 \ln \mathcal{M}|_{\theta_*} > 0 \quad (3373)$$

be the transverse score curvature. If the stationary equation changes by at most ε_F , the angle displacement obeys

$$|\delta\theta_*| \leq \frac{\varepsilon_F}{\kappa_\perp}. \quad (3374)$$

The intrinsic angular half-width for an allowed logarithmic score loss δL_{acc} is

$$\delta\theta_{\text{magic}} = \sqrt{\frac{2\delta L_{\text{acc}}}{\kappa_\perp}}. \quad (3375)$$

Photon loss introduces a distinct resolution condition. For target pole n and complementary pole m ,

$$\Delta_{nm}^{\text{res}} = |\text{Re } z_n - \text{Re } z_m| - \frac{\gamma_n + \gamma_m}{2}. \quad (3376)$$

A Hermitian flat band can survive while becoming spectroscopically unresolved. The experimentally accessible claims form a hierarchy. At the first stage, a small finite circuit tests exact quotient relations, symmetry multiplicities, mode counting, and the finite-device spectrum. This stage does not require a thermodynamic interpretation. At the second stage, several increasing covers

test local convergence as a function of injectivity radius. Local Green functions and local DOS are the primary observables. At the third stage, complete multiport and representation-selective spectroscopy tests Abelian versus non-Abelian spectral content. At the fourth stage, two or more inequivalent cover towers test whether the target bandwidth, gap, and projector are cover independent. At the fifth stage, transmon probes test whether the reconstructed photonic target produces the predicted coherent and dissipative interaction matrices. The minimum logical requirements are summarized in Table XIV.

The most immediate physical target is a finite synthetic lattice whose Hamiltonian, symmetry, and spectroscopy agree with the declared model. The strongest and most expensive target is a cover-independent full-spectrum bulk statement. The exponential resource law in Eq. (3269) implies that only modest injectivity radii may be reachable in an initial experiment. Such a device can still test: the finite-group spectral decomposition; the positive-root mechanism; Hodge-component locking; commutator-sensitive spectroscopy; projector-controlled qubit mediation. It cannot by size alone establish full thermodynamic spectral convergence. We do not identify a specific condensed-matter material whose microscopic parameters realize the complete model. The current experimental interpretation is a programmable synthetic-platform proposal.

1. Physical conclusion

Negative curvature places the bilayer in a distinct local moiré scaling class and permits non-Abelian representation sectors absent from the Euclidean Bloch description. In the circuit-QED realization these sectors enter the measurable resonance spectrum, local density of states, layer coherence, mode profiles, and photon-mediated qubit interactions.

The microscopic comparison separates geometry from dynamics. The square five-state model is root-free, whereas the centered hyperbolic surface-group Hamiltonian has a positive simple root and a stable isolated spectral target. Regular-octagon symmetry promotes the scalar cancellation to the complete Hodge Hessian. The finite-cover analysis then controls the connection between these local mechanisms and the bulk spectral observables.

Together, the positive hyperbolic construction and the

TABLE XIV. Staged route from a finite synthetic circuit to a bulk and interaction-level claim.

Stage	Primary hardware	Primary measurement	Supported conclusion
Finite device	One exact quotient circuit	Complete pole spectrum and symmetry analysis	Exact finite-network re
Local geometry	Increasing injectivity radii	Local Green function and LDOS	Cover-independent loca
Representation spectroscopy	Phase- and amplitude-coded drives	Sector-resolved scattering	Abelian or non-Abelian
Bulk verification	Two admissible cover towers	Edges, gaps, projectors, and no-pollution tests	Retained-sector bulk co
Qubit mediation	Transmon probes or arrays	Exchange and collective decay	Photon-mediated intera

Euclidean obstruction show that curvature enlarges the available interference and representation structure, while the microscopic hopping law selects the realized magic state. This combination provides a concrete route from curved-space twistronics to spectroscopy and interaction measurements in superconducting resonator arrays.

XV. APPENDIX

A. Construction of square and hyperbolic lattices

We define the primitive square lattice by

$$\Lambda = a\mathbb{Z}^2 = \left\{ a \begin{pmatrix} j \\ k \end{pmatrix} : j, k \in \mathbb{Z} \right\}, \quad (3377)$$

with primitive vectors

$$\mathbf{a}_1 = a \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{a}_2 = a \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (3378)$$

The two layers are

$$\Lambda_1 = \Lambda, \quad \Lambda_2 = R_\theta \Lambda + \mathbf{d}, \quad (3379)$$

where

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad (3380)$$

and $\mathbf{d}$ is a fixed in-plane translation. For a three-dimensional bilayer representation, the physical coordinates are

$$\mathbf{X}_{1,\mathbf{n}} = \begin{pmatrix} a\mathbf{n} \\ -h/2 \end{pmatrix}, \quad \mathbf{X}_{2,\mathbf{n}} = \begin{pmatrix} R_\theta a\mathbf{n} + \mathbf{d} \\ +h/2 \end{pmatrix}. \quad (3381)$$

The layer label remains an independent microscopic degree of freedom even when the two in-plane coordinates coincide. We parameterize every exact square-lattice commensurate twist by coprime integers m, n :

$$\tan \left(\frac{\theta}{2} \right) = \frac{n}{m}, \quad \gcd(m, n) = 1. \quad (3382)$$

The corresponding trigonometric values are evaluated exactly as

$$\cos \theta = \frac{m^2 - n^2}{m^2 + n^2}, \quad \sin \theta = \frac{2mn}{m^2 + n^2}. \quad (3383)$$

Define the parity divisor

$$d_{mn} = \begin{cases} 1, & m, n \text{ have opposite parity,} \\ 2, & m, n \text{ are both odd.} \end{cases} \quad (3384)$$

The exact coincidence index is

$$\Sigma(m, n) = \frac{m^2 + n^2}{d_{mn}}. \quad (3385)$$

The exact bilayer supercell therefore contains

$$N_{\text{sc}} = 2\Sigma \quad (3386)$$

microscopic layer-labelled sites. For opposite parity, we use the coincidence basis

$$\mathbf{T}_1 = a \begin{pmatrix} m \\ n \end{pmatrix}, \quad \mathbf{T}_2 = a \begin{pmatrix} -n \\ m \end{pmatrix}. \quad (3387)$$

Its area is

$$|\det(\mathbf{T}_1 \ \mathbf{T}_2)| = a^2(m^2 + n^2) = \Sigma a^2. \quad (3388)$$

When m, n are both odd, we use the primitive reduced basis

$$\mathbf{T}_1 = \frac{a}{2} \begin{pmatrix} m+n \\ n-m \end{pmatrix}, \quad \mathbf{T}_2 = \frac{a}{2} \begin{pmatrix} m-n \\ m+n \end{pmatrix}. \quad (3389)$$

Its determinant is

$$|\det(\mathbf{T}_1 \ \mathbf{T}_2)| = \frac{a^2}{2}(m^2 + n^2) = \Sigma a^2. \quad (3390)$$

Let

$$\mathbf{T}_M = (\mathbf{T}_1 \ \mathbf{T}_2). \quad (3391)$$

We represent one supercell by the half-open parallelogram

$$\mathcal{P}_M = \{\mathbf{T}_M \mathbf{u} : 0 \leq u_1 < 1, \quad 0 \leq u_2 < 1\}. \quad (3392)$$

The half-open convention prevents double counting of sites on opposite cell boundaries. For layer ℓ , a candidate coordinate $\mathbf{r}_\ell$ is retained when

$$\mathbf{u}_\ell = \mathbf{T}_M^{-1} \mathbf{r}_\ell \quad (3393)$$

satisfies

$$-\varepsilon_{\text{cell}} \leq u_{\ell,a} < 1 - \varepsilon_{\text{cell}}, \quad a = 1, 2. \quad (3394)$$

The tolerance is used only to resolve floating-point boundary representations. The integer lattice coordinates and layer indices are stored exactly and determine the unique site identity. The retained site counts are checked through

$$N_1 = N_2 = \Sigma, \quad N_1 + N_2 = 2\Sigma. \quad (3395)$$

Failure of this equality stops Hamiltonian construction. The reciprocal coincidence basis is

$$\mathbf{B}_M = 2\pi (\mathbf{T}_M^{-1})^\top. \quad (3396)$$

The moiré Brillouin zone is the corresponding reciprocal parallelogram, and its area is

$$A_{\text{mBZ}} = \frac{1}{\Sigma} \left(\frac{2\pi}{a} \right)^2. \quad (3397)$$

We use the hyperbolic plane of curvature

$$K = -\frac{1}{R^2} < 0. \quad (3398)$$

Geodesic polar coordinates centered at the twist point o carry the metric

$$ds^2 = dr^2 + R^2 \sinh^2 \left(\frac{r}{R} \right) d\phi^2. \quad (3399)$$

We implement the same geometry independently in the hyperboloid and Poincaré-disk models. In the hyperboloid model, a point is represented by

$$X(r, \phi) = R \begin{pmatrix} \cosh(r/R) \\ \sinh(r/R) \cos \phi \\ \sinh(r/R) \sin \phi \end{pmatrix}. \quad (3400)$$

The Lorentzian inner product is

$$\langle X, Y \rangle_L = -X_0 Y_0 + X_1 Y_1 + X_2 Y_2. \quad (3401)$$

The geodesic distance is evaluated as

$$d_{\mathbb{H}}(X, Y) = R \operatorname{arcosh} \left[-\frac{\langle X, Y \rangle_L}{R^2} \right]. \quad (3402)$$

In the Poincaré disk, the same point is represented by

$$z(r, \phi) = \tanh \left(\frac{r}{2R} \right) e^{i\phi}. \quad (3403)$$

For $z, w \in \mathbb{D}$, the distance is

$$d_{\mathbb{H}}(z, w) = 2R \operatorname{artanh} \left| \frac{z - w}{1 - \bar{z}w} \right|. \quad (3404)$$

Every generated distance is calculated in both coordinate models. We define the coordinate-consistency residual

$$\varepsilon_{\text{coord}} = \max_{i,j} \frac{|d_{\mathbb{H}}^{\text{hyp}}(i, j) - d_{\mathbb{H}}^{\text{disk}}(i, j)|}{\max \left\{ R, d_{\mathbb{H}}^{\text{hyp}}(i, j) \right\}}. \quad (3405)$$

Production geometry is accepted only when this residual is below its specified floating-point or high-precision threshold. The centered relative twist is

$$g_\theta : (r, \phi) \mapsto (r, \phi + \theta). \quad (3406)$$

In the hyperboloid model,

$$g_\theta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}, \quad (3407)$$

whereas in the disk model $z \mapsto e^{i\theta} z$. For a point at radius r , the exact twist-induced displacement is

$$d_\theta(r) = 2R \operatorname{arsinh} \left[\sinh \left(\frac{r}{R} \right) \left| \sin \left(\frac{\theta}{2} \right) \right| \right]. \quad (3408)$$

We calculate $d_\theta(r)$ in three independent ways:

1. by direct evaluation of Eq. (3408);
2. by rotating the hyperboloid coordinate and applying Eq. (3402);
3. by rotating the disk coordinate and applying Eq. (3404).

The local moiré radius is defined by the fixed registry threshold

$$d_\theta(\xi_M) = a. \quad (3409)$$

Its analytical value is

$$\xi_M = R \operatorname{arsinh} \left[\frac{\sinh(a/2R)}{|\sin(\theta/2)|} \right]. \quad (3410)$$

We also solve Eq. (3409) numerically with a bracketed monotone root solver. The analytical and numerical results are compared before any spectral calculation is performed. For the validation model, we use the genus-two surface group

$$\Gamma = \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = e \rangle. \quad (3411)$$

The fundamental polygon is a regular hyperbolic octagon with interior angle $\pi/4$. For a regular $\{p, q\}$ polygon centered at o , the circumradius is set by

$$\cosh \left(\frac{r_v}{R} \right) = \cot \left(\frac{\pi}{p} \right) \cot \left(\frac{\pi}{q} \right). \quad (3412)$$

The regular-octagon realization uses $p = q = 8$. The vertices are

$$z_j = \tanh \left(\frac{r_v}{2R} \right) \exp \left[i \left(\phi_0 + \frac{2\pi j}{8} \right) \right], \quad j = 0, \dots, 7. \quad (3413)$$

Opposite sides are paired by orientation-preserving hyperbolic isometries. Numerically, each side-pairing generator is represented by an $SU(1, 1)$ matrix

$$M_\gamma = \begin{pmatrix} \alpha_\gamma & \beta_\gamma \\ \beta_\gamma & \bar{\alpha}_\gamma \end{pmatrix}, \quad |\alpha_\gamma|^2 - |\beta_\gamma|^2 = 1, \quad (3414)$$

acting through

$$z \mapsto \frac{\alpha_\gamma z + \beta_\gamma}{\beta_\gamma z + \bar{\alpha}_\gamma}. \quad (3415)$$

The matrix is obtained from the unique isometry mapping the ordered endpoints and midpoint of one side to the oppositely oriented endpoints and midpoint of its paired side. The exact arithmetic generator data, rather than rounded disk coordinates, define the group element. Before orbit generation, we verify

$$M_{a_1} M_{b_1} M_{a_1}^{-1} M_{b_1}^{-1} M_{a_2} M_{b_2} M_{a_2}^{-1} M_{b_2}^{-1} = I \quad (3416)$$

within exact arithmetic or the declared high-precision residual. The single-layer lattice is the center orbit $\Lambda = \Gamma o$. Each orbit point is indexed primarily by a reduced group word, not by a rounded coordinate. Coordinates are secondary geometric data. We generate a finite visualization patch by breadth-first traversal of the Cayley graph:

$$\mathcal{W}_L = \{\gamma \in \Gamma : |\gamma|_S \leq L\}, \quad (3417)$$

where

$$\mathcal{S} = \{a_1^{\pm 1}, b_1^{\pm 1}, a_2^{\pm 1}, b_2^{\pm 1}\}. \quad (3418)$$

Inverse-letter cancellation is performed before a word is queued. Exact group or quotient identities are used whenever available. Coordinate proximity alone is never used to identify two group elements in production calculations. The two layers are

$$\Lambda_1 = \Gamma o, \quad \Lambda_2 = g_\theta \Gamma o. \quad (3419)$$

They are embedded in $\mathbb{H}_R^2 \times \mathbb{R}$ at normal coordinates $-h/2$ and $+h/2$. For sites $x_i \in \Lambda_1$ and $y_j \in \Lambda_2$, the in-plane mismatch is

$$d_{ij}^\parallel = d_{\mathbb{H}}(x_i, y_j). \quad (3420)$$

The product-space distance used in the scalar tunnelling kernel is

$$D_{ij} = \sqrt{h^2 + (d_{ij}^\parallel)^2}. \quad (3421)$$

We also define the excess separation

$$r_{ij} = D_{ij} - h. \quad (3422)$$

Every distance table is checked for

$$D_{ij} \geq h, \quad D_{ij} = D_{ji}, \quad (3423)$$

and invariance under every exact symmetry included in the model. Exact hyperbolic commensurability is defined by

$$\Gamma_M(\theta) = \Gamma \cap g_\theta \Gamma g_\theta^{-1}, \quad (3424)$$

with finite coincidence index

$$q_M(\theta) = [\Gamma : \Gamma_M(\theta)]. \quad (3425)$$

For the centered arithmetic validation sequence, we use

$$\theta_j = 2 \arctan \left(\frac{\sqrt{3}}{j} \right), \quad (3426)$$

restricted to the specified admissible values of j . The common subgroup and its index are computed independently of the local registry-length calculation. For a generic incommensurate twist, no finite common quotient is introduced artificially. We use a finite cover $\Gamma_N^{(1)}$ of the first layer and its conjugate

$$\Gamma_N^{(2)} = g_\theta \Gamma_N^{(1)} g_\theta^{-1} \quad (3427)$$

for the second layer. Interlayer distances are evaluated after both covers are lifted to the common universal cover. For an observation radius L_{obs} and propagation radius L_H , a local paired-cover calculation is accepted only when

$$r_{\text{inj},N}^{(1)} > L_{\text{obs}} + L_H, \quad r_{\text{inj},N}^{(2)} > L_{\text{obs}} + L_H. \quad (3428)$$

If exact angles θ_N are used to approximate an incommensurate target θ on a growing radius L_N , we enforce

$$|\theta_N - \theta| \exp \left(\frac{L_N}{R} \right) \rightarrow 0. \quad (3429)$$

B. Hamiltonian assembly

In the co-rotating layer frame, the one-particle Hamiltonian is

$$H_{\text{BL}}(\theta, K) = \begin{pmatrix} H_1(K) & T_\theta(K) \\ T_\theta^\dagger(K) & H_2(K) \end{pmatrix}. \quad (3430)$$

For identical monolayers,

$$H_1 = H_2 = H_0. \quad (3431)$$

All rigid-twist dependence is placed in the interlayer block:

$$\partial_\theta H_0 = 0, \quad \partial_\theta H_{\text{BL}} = \begin{pmatrix} 0 & \partial_\theta T_\theta \\ \partial_\theta T_\theta^\dagger & 0 \end{pmatrix}. \quad (3432)$$

For every directed hopping channel, $t_{\alpha\beta}(\delta)$ connects orbital β in the reference cell to orbital α in the translated cell δ . Hermiticity is enforced at the kernel level:

$$t_{\beta\alpha}(\delta^{-1}) = t_{\alpha\beta}(\delta)^*. \quad (3433)$$

We insert each Hermitian pair once and construct the reverse entry by conjugation. We do not independently evaluate both directions and then average them. For one

square-lattice orbital per microscopic site, the monolayer nearest-neighbour Hamiltonian is

$$\hat{H}_0 = -t \sum_{\langle i,j \rangle} (c_i^\dagger c_j + c_j^\dagger c_i). \quad (3434)$$

The primitive dispersion is

$$\varepsilon_0(\mathbf{q}) = -2t [\cos(q_x a) + \cos(q_y a)]. \quad (3435)$$

Let $\alpha = (\ell, s)$ denote a layer and a site inside the exact moiré cell. In the periodic Bloch gauge, the supercell matrix is

$$[H(\mathbf{k})]_{\alpha\beta} = \sum_{\mathbf{R}_M} t_{\alpha\beta}(\mathbf{R}_M) e^{i\mathbf{k} \cdot \mathbf{R}_M}. \quad (3436)$$

Here $t_{\alpha\beta}(\mathbf{R}_M)$ connects orbital β in the reference cell to orbital α in cell $\mathbf{R}_M$. Nearest-neighbour intralayer bonds are generated in exact integer coordinates. For each basis site, we add the four primitive displacements $\pm a\mathbf{e}_x$, $\pm a\mathbf{e}_y$, reduce the endpoint modulo the coincidence cell, and record the corresponding moiré translation $\mathbf{R}_M$. The interlayer matrix element is evaluated from the physical three-dimensional distance

$$D_{ij} = \sqrt{h^2 + |\mathbf{r}_{1i} - \mathbf{r}_{2j}|^2}. \quad (3437)$$

The scalar exponential profile is

$$T_{ij} = w \exp \left[-\frac{D_{ij} - h}{\lambda_\perp} \right], \quad (3438)$$

whereas a Gaussian control uses

$$T_{ij}^G = w \exp \left[-\frac{D_{ij}^2 - h^2}{2\lambda_\perp^2} \right]. \quad (3439)$$

The kernel type is fixed in the analysis plan record. For the analytical square first-star benchmark, define

$$\mathbf{p} = \frac{\mathbf{k}}{G_M}, \quad \alpha = \frac{w_1}{AG_M^2}, \quad (3440)$$

where $A > 0$ is the quadratic kinetic coefficient and w_1 is the first-star coupling. In the ordered basis

$$(|c\rangle, |x+\rangle, |x-\rangle, |y+\rangle, |y-\rangle), \quad (3441)$$

define

$$\begin{aligned} d_0 &= p_x^2 + p_y^2, \\ d_{x\pm} &= p_x^2 + p_y^2 + 1 \pm 2p_x, \end{aligned} \quad (3442)$$

and

$$d_{y\pm} = p_x^2 + p_y^2 + 1 \pm 2p_y. \quad (3443)$$

The exact normalized Hamiltonian is

$$\hat{H}_\square(\mathbf{p}, \alpha) = \begin{pmatrix} d_0 & \alpha & \alpha & \alpha & \alpha \\ \alpha & d_{x+} & 0 & 0 & 0 \\ \alpha & 0 & d_{x-} & 0 & 0 \\ \alpha & 0 & 0 & d_{y+} & 0 \\ \alpha & 0 & 0 & 0 & d_{y-} \end{pmatrix}. \quad (3444)$$

No satellite-satellite matrix element is inserted. The no-root test is performed on this exact matrix, not on a truncated perturbative expansion of its eigenvalues. For the centered genus-two model, define the symmetric first shell

$$\mathcal{S} = \{s_a, s_a^{-1}\}_{a=1}^4. \quad (3445)$$

Let λ_Γ denote the left regular representation. The first-shell adjacency is

$$A_S = \sum_{a=1}^4 [\lambda_\Gamma(s_a) + \lambda_\Gamma(s_a)^\dagger]. \quad (3446)$$

The intralayer Hamiltonian is

$$h_0 = -tA_S. \quad (3447)$$

For $\gamma \in \Gamma$, define

$$D_\gamma = \sqrt{h^2 + d_{\mathbb{H}}(x_1, \gamma x_2)^2}, \quad r_\gamma = D_\gamma - h. \quad (3448)$$

The full-distance tunnelling coefficient is

$$q_\gamma = e^{-\mu r_\gamma}, \quad \mu = \lambda_\perp^{-1}. \quad (3449)$$

We explicitly enforce $q_{\gamma^{-1}} = q_\gamma$. The first-shell and full interlayer operators are

$$Q_1 = I + q_1 A_S, \quad (3450)$$

and

$$Q_\infty = \sum_{\gamma \in \Gamma} q_\gamma \lambda_\Gamma(\gamma). \quad (3451)$$

The scalar bilayer Hamiltonian is

$$H_Q(w) = \begin{pmatrix} h_0 & wQ \\ wQ & h_0 \end{pmatrix}, \quad Q \in \{Q_1, Q_\infty\}. \quad (3452)$$

Layer exchange is represented by

$$P_L = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}. \quad (3453)$$

Since $[H_Q, P_L] = 0$, the even and odd blocks are

$$H_\pm = h_0 \pm wQ. \quad (3454)$$

The positive-root calculation uses the specified even sector. The sector is not changed after the response crosses zero. At each hyperbolic site i , we use the local orbital basis

$$\{|s_i\rangle, |p_{i1}\rangle, |p_{i2}\rangle\}, \quad (3455)$$

where $\{\mathbf{e}_{i1}, \mathbf{e}_{i2}\}$ is an orthonormal tangent frame. Let

$$\mathcal{P}_{j \rightarrow i} : T_{x_j} \mathbb{H}_R^2 \longrightarrow T_{x_i} \mathbb{H}_R^2 \quad (3456)$$

be Levi–Civita parallel transport along the unique shortest geodesic. Let

$$\hat{\mathbf{d}}_{ij} \in T_{x_i} \mathbb{H}_R^2 \quad (3457)$$

be the unit tangent at i pointing toward j . The covariant intralayer Slater–Koster entries are

$$\begin{aligned} H_{ss,ij} &= V_{ss\sigma}(d_{ij}), \\ H_{sp_b,ij} &= V_{sp\sigma}(d_{ij}) \left\langle \hat{\mathbf{d}}_{ij}, \mathcal{P}_{j \rightarrow i} \mathbf{e}_{jb} \right\rangle, \end{aligned} \quad (3458)$$

and

$$\begin{aligned} H_{p_a p_b, ij} &= V_{pp\pi}(d_{ij}) \left\langle \mathbf{e}_{ia}, \mathcal{P}_{j \rightarrow i} \mathbf{e}_{jb} \right\rangle \\ &\quad + [V_{pp\sigma}(d_{ij}) - V_{pp\pi}(d_{ij})] \left\langle \mathbf{e}_{ia}, \hat{\mathbf{d}}_{ij} \right\rangle \\ &\quad \times \left\langle \hat{\mathbf{d}}_{ij}, \mathcal{P}_{j \rightarrow i} \mathbf{e}_{jb} \right\rangle. \end{aligned} \quad (3459)$$

The reverse directed block is set by

$$H_{ji} = H_{ij}^\dagger. \quad (3460)$$

The interlayer orbital block is

$$T_{\perp, ij}^{\alpha\beta} = W_{\alpha\beta} \left(D_{ij}, \hat{\mathbf{D}}_{ij} \right) \exp \left[-\frac{D_{ij} - h}{\lambda_{\perp}} \right]. \quad (3461)$$

The angular factor $W_{\alpha\beta}$ is evaluated in transported local frames. We use the scalar model as the symmetry-preserving reference subspace. Bond anisotropy, orbital splitting, s - p mixing, and unequal $pp\sigma$ and $pp\pi$ channels are introduced one at a time and in specified combinations. Each perturbation amplitude is stored explicitly; no unregistered orbital parameter is absorbed into a single fitted effective coupling. For a finite-range Hamiltonian, the support is

$$\mathcal{S}_H = \{\delta : h(\delta) \neq 0\}. \quad (3462)$$

For a long-range kernel, we define

$$H^{(L)} = \sum_{d(\mathbf{e}, \delta) \leq L} h(\delta) \otimes U(\delta). \quad (3463)$$

If shell multiplicity obeys

$$\#\mathcal{S}_n \leq C_{\Gamma} e^{h_{\Gamma} n \rho} \quad (3464)$$

and the kernel satisfies

$$\|h(\delta)\| \leq A_0 e^{-\mu d(\mathbf{e}, \delta)}, \quad (3465)$$

we require

$$\gamma_0 = \mu - h_{\Gamma} > 0. \quad (3466)$$

The operator tail is bounded by

$$\varepsilon_{\text{tail}}^{(0)}(L) = \|H - H^{(L)}\| \leq A_0 C_{\Gamma} \sum_{n \geq \lceil L/\rho \rceil} e^{-\gamma_0 n \rho}.$$

Thus

$$\varepsilon_{\text{tail}}^{(0)}(L) \leq \frac{A_0 C_{\Gamma} e^{\gamma_0 \rho}}{1 - e^{-\gamma_0 \rho}} e^{-\gamma_0 L}. \quad (3467)$$

For derivative order $r = 1, 2$, we separately evaluate

$$\varepsilon_{\text{tail}}^{(r)}(L) = \sup_{\mathbf{p}} \left\| \partial_{\mathbf{p}}^r H - \partial_{\mathbf{p}}^r H^{(L)} \right\|. \quad (3468)$$

A C^0 tail is never reused as a C^1 or C^2 bound. Before diagonalization, every finite matrix must satisfy the normalized Hermiticity test

$$\varepsilon_{\text{Herm}} = \frac{\|H - H^\dagger\|_{\text{F}}}{\max\{1, \|H\|_{\text{F}}\}}. \quad (3469)$$

For each exact symmetry U_g , we compute

$$\varepsilon_{\text{sym}}(g) = \frac{\|H U_g - U_g H\|_{\text{F}}}{\max\{1, \|H\|_{\text{F}}\}}. \quad (3470)$$

The tested set contains generators of the translation group, residual point group, layer exchange, and every combined layer–spatial symmetry used later. For a completely diagonalized matrix,

$$\sum_{j=1}^D E_j = \text{Tr } H, \quad (3471)$$

and

$$\sum_{j=1}^D E_j^2 = \text{Tr } H^2. \quad (3472)$$

The second identity is used to detect missing, duplicated, or incorrectly weighted hopping entries.

C. Finite-quotient and representation decomposition

Let $\Gamma_N \triangleleft \Gamma_M$ be a finite-index normal subgroup. We define

$$Q_N = \Gamma_M / \Gamma_N, \quad d_N = |Q_N|. \quad (3473)$$

The quotient is supplied by a surjective homomorphism

$$\varphi_N : \Gamma_M \longrightarrow Q_N, \quad \ker \varphi_N = \Gamma_N. \quad (3474)$$

If

$$\Gamma_M = \langle s_1, \dots, s_r \mid \mathcal{R}_1, \dots, \mathcal{R}_p \rangle, \quad (3475)$$

the quotient input consists of

$$q_{\mu} = \varphi_N(s_{\mu}). \quad (3476)$$

Before any spectral calculation, we verify

$$\mathcal{R}_a(q_1, \dots, q_r) = e_{Q_N}, \quad a = 1, \dots, p. \quad (3477)$$

For the genus- g surface relation, this becomes

$$\prod_{j=1}^g [\varphi_N(a_j), \varphi_N(b_j)] = e_{Q_N}. \quad (3478)$$

The quotient-cell basis is

$$\{|x\rangle : x \in Q_N\}. \quad (3479)$$

The right-regular representation is

$$R_N(q)|x\rangle = |xq^{-1}\rangle. \quad (3480)$$

We verify

$$\begin{aligned} R_N(q_1)R_N(q_2) &= R_N(q_1q_2), \\ R_N(q)^\dagger &= R_N(q^{-1}), \end{aligned} \quad (3481)$$

and

$$\text{Tr } R_N(q) = \begin{cases} d_N, & q = e, \\ 0, & q \neq e. \end{cases} \quad (3482)$$

For ν internal orbitals per quotient cell, the full finite Hamiltonian is

$$H_N^{\text{reg}} = \sum_{\delta \in \Gamma_M} h(\delta) \otimes R_N[\varphi_N(\delta)]. \quad (3483)$$

Its dimension is

$$D_N = \nu d_N. \quad (3484)$$

The full regular matrix is constructed before any representation block is selected. Let $\hat{Q}_N$ denote the set of inequivalent irreducible unitary representations of Q_N . For $\rho \in \hat{Q}_N$, let

$$d_\rho = \dim V_\rho. \quad (3485)$$

Finite Peter–Weyl theory gives

$$\ell^2(Q_N) \simeq \bigoplus_{\rho \in \hat{Q}_N} V_\rho \otimes V_\rho^*, \quad (3486)$$

with

$$\sum_{\rho \in \hat{Q}_N} d_\rho^2 = |Q_N|. \quad (3487)$$

The reduced representation block is

$$H_N[\rho] = \sum_{\delta} h(\delta) \otimes \rho[\varphi_N(\delta)]. \quad (3488)$$

Its dimension is

$$D_\rho = \nu d_\rho. \quad (3489)$$

The full Hamiltonian decomposes exactly as

$$H_N^{\text{reg}} \simeq \bigoplus_{\rho \in \hat{Q}_N} [H_N[\rho] \otimes I_{d_\rho}]. \quad (3490)$$

Every block eigenvalue therefore occurs with multiplicity d_ρ in the full regular spectrum. The exact physical state weight of an irrep is

$$w_{\rho,N} = \frac{d_\rho^2}{|Q_N|}. \quad (3491)$$

We never average inequivalent irreps with equal weights unless their dimensions and multiplicities make those weights equal. Let

$$\chi_\rho(q) = \text{Tr } \rho(q) \quad (3492)$$

be the character. The exact isotypic projector on the full regular space is

$$P_{\rho,N} = \frac{d_\rho}{|Q_N|} \sum_{q \in Q_N} \chi_\rho(q^{-1}) [I_\nu \otimes R_N(q)]. \quad (3493)$$

We verify

$$\begin{aligned} P_{\rho,N}^2 &= P_{\rho,N}, & P_{\rho,N}^\dagger &= P_{\rho,N}, \\ P_{\rho,N} P_{\sigma,N} &= 0 & (\rho \neq \sigma), \end{aligned} \quad (3494)$$

and

$$\sum_{\rho \in \hat{Q}_N} P_{\rho,N} = I_{D_N}. \quad (3495)$$

The expected rank is

$$\text{rank } P_{\rho,N} = \nu d_\rho^2. \quad (3496)$$

A mismatch between the numerical rank and Eq. (3496) is treated as an implementation failure. When explicit block matrices are expensive but exact character data are available, we reconstruct their characteristic polynomials from traces. For a block of dimension D_ρ , define

$$p_k^{(\rho)} = \text{Tr } [H_N[\rho]^k], \quad k = 1, \dots, D_\rho. \quad (3497)$$

Set $e_0^{(\rho)} = 1$. Newton identities give

$$k e_k^{(\rho)} = \sum_{j=1}^k (-1)^{j-1} e_{k-j}^{(\rho)} p_j^{(\rho)}. \quad (3498)$$

The characteristic polynomial is

$$\begin{aligned} \mathcal{P}_\rho(z) &= z^{D_\rho} - e_1^{(\rho)} z^{D_\rho-1} + e_2^{(\rho)} z^{D_\rho-2} \\ &\quad + \dots + (-1)^{D_\rho} e_{D_\rho}^{(\rho)}. \end{aligned} \quad (3499)$$

Its roots are compared with direct block diagonalization on all tractable levels. For a computed subset $\mathcal{R}_N \subseteq \hat{Q}_N$, the finite-level representation coverage is

$$C_{\text{rep},N} = \sum_{\rho \in \mathcal{R}_N} \frac{d_\rho^2}{|Q_N|}. \quad (3500)$$

A complete finite-quotient calculation requires $C_{\text{rep},N} = 1$. For a target expected to contain r_* reduced branches in every sector, let

$$r_C(\rho) = \frac{1}{d_\rho} \text{rank}(P_C P_{\rho,N}). \quad (3501)$$

The target coverage is

$$C_{\text{rep},C,N} = \sum_{\rho \in \tilde{Q}_N} \frac{d_\rho^2}{|\tilde{Q}_N|} \mathbf{1}[r_C(\rho) \geq r_*]. \quad (3502)$$

A character-sector result, a complete finite-quotient result, a retained projective-sector result, and a reduced-regular bulk result are stored under different scope labels. Finite Peter–Weyl completeness is not used as an ordinary irreducible Bloch decomposition of the infinite non-type-I surface group.

D. Band and projector tracking

Before the interacting scan, we choose an anchor

$$\mathbf{p}_0 = (K_0, \theta_0, \omega = 0, \dots) \quad (3503)$$

and define

$$H_{\text{ref}} = H(\mathbf{p}_0). \quad (3504)$$

A closed contour $\mathcal{C}_\nu$ encloses the declared reference cluster and no other reference eigenvalue. The specified projector is

$$P_\nu^0 = \frac{1}{2\pi i} \oint_{\mathcal{C}_\nu} (z - H_{\text{ref}})^{-1} dz. \quad (3505)$$

Its rank and center are

$$r_\nu = \text{Tr } P_\nu^0, \quad E_\nu^0 = \frac{\text{Tr}(P_\nu^0 H_{\text{ref}})}{r_\nu}. \quad (3506)$$

We freeze the following target signature:

$$\mathfrak{s}_\nu^0 = \left(r_\nu, E_\nu^0, \text{layer parity, point-group irrep, translation-sector content, energy window, orbital ancestry} \right). \quad (3507)$$

No component of this signature is redefined after a narrow band is observed. For small matrices, we compute the full spectrum by dense Hermitian diagonalization:

$$HV = VD, \quad V^\dagger V = I. \quad (3508)$$

For large sparse matrices, extremal eigenvalues are obtained by Lanczos or block Lanczos. The Krylov space is

$$\mathcal{K}_m(H, V_1) = \text{span} \{V_1, HV_1, \dots, H^{m-1}V_1\}. \quad (3509)$$

Whenever a representation sector is declared, starting vectors are projected:

$$V_{1,\rho} = P_{\rho,N} V_1. \quad (3510)$$

Since $[H, P_{\rho,N}] = 0$, the complete Krylov space remains inside the same isotypic sector. Interior eigenvalues near shift σ are calculated with

$$A_\sigma = (H - \sigma I)^{-1}. \quad (3511)$$

Every recovered eigenpair is recertified with the original Hamiltonian:

$$\mathbf{r}_n = H|\tilde{\psi}_n\rangle - \tilde{E}_n|\tilde{\psi}_n\rangle. \quad (3512)$$

For a normalized Ritz vector,

$$\text{dist}(\tilde{E}_n, \text{Spec } H) \leq \|\mathbf{r}_n\|. \quad (3513)$$

The Krylov orthogonality defect is

$$\varepsilon_{\text{orth}} = \|Q_m^\dagger Q_m - I\|. \quad (3514)$$

Full or selective reorthogonalization is enabled whenever closely spaced or symmetry-degenerate eigenvalues are resolved. Let $|\psi_n(\lambda_j)\rangle$ and $|\psi_m(\lambda_{j+1})\rangle$ be normalized states at neighboring parameter values after transport to one common fiber. The overlap matrix is

$$M_{nm}^{(j)} = |\langle \psi_n(\lambda_j) | \mathcal{T}_{j+1 \rightarrow j} \psi_m(\lambda_{j+1}) \rangle|^2. \quad (3515)$$

For a set of simple branches, the assignment permutation π_j maximizes

$$\pi_j = \arg \max_{\pi} \sum_n M_{n,\pi(n)}^{(j)}. \quad (3516)$$

We solve this linear-sum assignment problem exactly. Energy ordering is used only as a secondary tie breaker after overlap, symmetry, and target ancestry have been considered. Let

$$U_j = (|u_{1,j}\rangle \cdots |u_{r,j}\rangle) \quad (3517)$$

be an orthonormal basis of a rank- r cluster. Define $P_j = U_j U_j^\dagger$. The transported overlap matrix is

$$\mathbf{O}_j = U_j^\dagger \mathcal{T}_{j+1 \rightarrow j} U_{j+1}. \quad (3518)$$

$$\mathbf{O}_j = \mathbf{U}_j \mathbf{\Sigma}_j \mathbf{V}_j^\dagger \quad (3519)$$

be its singular-value decomposition. The singular values are

$$\sigma_a = \cos \vartheta_a, \quad (3520)$$

where ϑ_a are the principal angles. The largest subspace-angle distance is

$$d_{\text{sub},j} = \sin \left(\max_a \vartheta_a \right) = \left\| P_j - \mathcal{T}_{j+1 \rightarrow j} P_{j+1} \mathcal{T}_{j+1 \rightarrow j}^\dagger \right\|. \quad (3521)$$

The mean projector overlap is

$$\mathcal{O}_{P,j} = \frac{1}{r} \text{Tr} \left[P_j \mathcal{T}_{j+1 \rightarrow j} P_{j+1} \mathcal{T}_{j+1 \rightarrow j}^\dagger \right] = \frac{1}{r} \sum_{a=1}^r \sigma_a^2. \quad (3522)$$

For visualization or differentiated calculations, we fix the within-cluster gauge by the Procrustes rotation

$$U_{j+1}^{\text{aligned}} = U_{j+1} \mathbf{V}_j \mathbf{U}_j^\dagger. \quad (3523)$$

All physical cluster observables are nevertheless calculated from P_j , not from the selected basis. For a circular contour

$$z(\phi) = E_c + r_c e^{i\phi}, \quad (3524)$$

the projector is

$$P_C = \frac{1}{2\pi} \int_0^{2\pi} r_c e^{i\phi} [z(\phi) - H]^{-1} d\phi. \quad (3525)$$

With $N_{\mathcal{C}}$ equispaced nodes

$$\phi_q = \frac{2\pi(q + 1/2)}{N_{\mathcal{C}}}, \quad (3526)$$

we use

$$\tilde{P}_C = \frac{1}{N_{\mathcal{C}}} \sum_{q=0}^{N_{\mathcal{C}}-1} r_c e^{i\phi_q} [z(\phi_q) - H]^{-1}. \quad (3527)$$

The projector quadrature is refined until the following quantities stabilize:

$$\varepsilon_{\text{id}} = \|\tilde{P}_C^2 - \tilde{P}_C\|, \quad \varepsilon_{\text{P,Herm}} = \|\tilde{P}_C - \tilde{P}_C^\dagger\|, \quad (3528)$$

and

$$\varepsilon_{\text{comm}} = \|H\tilde{P}_C - \tilde{P}_C H\|. \quad (3529)$$

The numerical rank is obtained by thresholding the spectrum of the symmetrized projector

$$\frac{1}{2} (\tilde{P}_C + \tilde{P}_C^\dagger) \quad (3530)$$

around $1/2$. The rank must equal the specified rank. Let $P_1^{\text{ref}}, \dots, P_{n_{\text{ref}}}^{\text{ref}}$ be specified reference subspaces. The ancestry vector of a tracked cluster is

$$\mathbf{a}_C = \left(\frac{\text{Tr } P_C P_1^{\text{ref}}}{r_C}, \dots, \frac{\text{Tr } P_C P_{n_{\text{ref}}}^{\text{ref}}}{r_C} \right). \quad (3531)$$

At an avoided crossing, we distinguish:

1. the adiabatic eigenvalue branches, ordered by the exact eigenvalues of H ;
2. the diabatic ancestry branches, defined by maximal overlap with the fixed reference projectors;

3. the complete resonant cluster projector, which remains smooth when individual one-dimensional projectors exchange ancestry.

A target is not continued through a point where its isolation gap closes unless the target rank is enlarged before the crossing. If rank enlargement was not specified, the branch is terminated and the subsequent point is classified separately.

For a normalized state,

$$w_{n,\rho} = \langle \psi_n | P_{\rho,N} | \psi_n \rangle. \quad (3532)$$

The purity defect is

$$\varepsilon_{\text{rep},n} = 1 - \max_{\rho} w_{n,\rho}. \quad (3533)$$

For a cluster,

$$w_{C,\rho} = \frac{\text{Tr}(P_C P_{\rho,N})}{r_C}. \quad (3534)$$

The translation-covariance defect is

$$\varepsilon_{\text{cov}}[P_C] = \max_{g \in \mathcal{G}_{\text{test}}} \frac{\|[P_C, U_g]\|_{\text{F}}}{2\|P_C\|_{\text{F}}}. \quad (3535)$$

These quantities are evaluated at every point used in a representation-resolved or coherence-weighted claim.

E. Bandwidth, velocity and Hessian evaluation

For a continuously tracked target family C over a declared parameter domain $\mathcal{D}_C$, define

$$E_C^- = \inf_{\mathbf{p} \in \mathcal{D}_C, n \in C} E_n(\mathbf{p}), \quad (3536)$$

and

$$E_C^+ = \sup_{\mathbf{p} \in \mathcal{D}_C, n \in C} E_n(\mathbf{p}). \quad (3537)$$

The bandwidth is

$$W_C = E_C^+ - E_C^-. \quad (3538)$$

The lower and upper complementary edges are

$$E_{\text{below}} = \sup \{ E_m < E_C^- : m \notin C \}, \quad (3539)$$

and

$$E_{\text{above}} = \inf \{ E_m > E_C^+ : m \notin C \}. \quad (3540)$$

The isolation gap is

$$\Delta_C = \min \{ E_C^- - E_{\text{below}}, E_{\text{above}} - E_C^+ \}. \quad (3541)$$

The target is spectrally admissible only when $\Delta_C > 0$. For a quasi-uniform sampling mesh $\mathcal{D}_C^{(h)}$, define the covering radius

$$\delta_h = \sup_{\mathbf{p} \in \mathcal{D}_C} \min_{\mathbf{p}_j \in \mathcal{D}_C^{(h)}} d_G(\mathbf{p}, \mathbf{p}_j). \quad (3542)$$

If

$$\|dE_n\|_{G^{-1}} \leq L_{E,n} \quad (3543)$$

on the domain, then

$$|E_{n,\max} - E_{n,\max}^{(h)}| \leq L_{E,n} \delta_h, \quad (3544)$$

and similarly for the minimum. Therefore

$$|W_C - W_C^{(h)}| \leq 2L_{E,C} \delta_h. \quad (3545)$$

All reported continuous-domain edges are refined locally and carry this mesh remainder. In the periodic Bloch gauge of Eq. (3436),

$$\partial_{k_a} H(\mathbf{k}) = \sum_{\mathbf{R}_M} iR_{M,a} t(\mathbf{R}_M) e^{i\mathbf{k} \cdot \mathbf{R}_M}. \quad (3546)$$

The second derivative is

$$\partial_{k_a} \partial_{k_b} H(\mathbf{k}) = - \sum_{\mathbf{R}_M} R_{M,a} R_{M,b} t(\mathbf{R}_M) e^{i\mathbf{k} \cdot \mathbf{R}_M}. \quad (3547)$$

For a simple normalized band,

$$H(\mathbf{k})|u_n\rangle = E_n|u_n\rangle, \quad (3548)$$

the velocity is

$$v_{n,a} = \frac{1}{\hbar} \langle u_n | \partial_{k_a} H | u_n \rangle. \quad (3549)$$

The maximum target speed is

$$v_{\max,C} = \sup_{\mathbf{k}, n \in C} \left[\sum_a |v_{n,a}|^2 \right]^{1/2}. \quad (3550)$$

Let $\rho(\mathbf{u})$ be a local chart on a representation space with coordinates u^a . The Hamiltonian derivatives are

$$H_a = \partial_{u^a} H[\rho] = \sum_{\delta} h(\delta) \otimes \partial_{u^a} \rho(\delta). \quad (3551)$$

For a tangent vector

$$X = X^a \partial_{u^a}, \quad (3552)$$

we write

$$H_X = X^a H_a. \quad (3553)$$

The energy differential is

$$dE_n[X] = \langle u_n | H_X | u_n \rangle. \quad (3554)$$

Let G_{ab} be the fixed canonical Hodge metric. The generalized speed is

$$\Omega_n = \frac{1}{\hbar} [(\partial_a E_n) G^{ab} (\partial_b E_n)]^{1/2}. \quad (3555)$$

The complete target maximum is

$$\Omega_{\max,C} = \sup_{[\rho] \in \mathcal{D}_C, n \in C} \Omega_n[\rho]. \quad (3556)$$

The metric and its normalization are kept fixed throughout every comparison. Let

$$Q_n = I - |u_n\rangle\langle u_n| \quad (3557)$$

and define the reduced resolvent

$$R_n = Q_n (E_n - H)^{-1} Q_n. \quad (3558)$$

For a simple isolated eigenvalue, the exact mixed Hessian is

$$\begin{aligned} \mathcal{H}_{ab}^{(n)} &= \langle u_n | H_{ab} | u_n \rangle \\ &+ \langle u_n | H_a R_n H_b + H_b R_n H_a | u_n \rangle. \end{aligned} \quad (3559)$$

Equivalently,

$$\begin{aligned} \mathcal{H}_{ab}^{(n)} &= \langle u_n | H_{ab} | u_n \rangle \\ &+ 2 \operatorname{Re} \sum_{m \neq n} \frac{\langle u_n | H_a | u_m \rangle \langle u_m | H_b | u_n \rangle}{E_n - E_m}. \end{aligned} \quad (3560)$$

Every state whose denominator becomes comparable to its coupling is moved into the active cluster before Eq. (3560) is used. For an isolated rank- r_C cluster, define its mean energy

$$\bar{E}_C = \frac{1}{r_C} \operatorname{Tr}(P_C H). \quad (3561)$$

The first derivative is

$$\partial_a \bar{E}_C = \frac{1}{r_C} \operatorname{Tr}(P_C H_a). \quad (3562)$$

The projector derivative is

$$\partial_a P_C = \frac{1}{2\pi i} \oint_{\mathcal{C}_C} R(z) H_a R(z) dz, \quad (3563)$$

where

$$R(z) = (z - H)^{-1}. \quad (3564)$$

The cluster-mean Hessian is

$$\partial_a \partial_b \bar{E}_C = \frac{1}{r_C} \operatorname{Tr} [P_C H_{ab} + (\partial_a P_C) H_b]. \quad (3565)$$

The symmetrized numerical value is

$$\mathcal{H}_{ab}^C \leftarrow \frac{1}{2} (\mathcal{H}_{ab}^C + \mathcal{H}_{ba}^C). \quad (3566)$$

The antisymmetric remainder is stored as a differentiation error diagnostic rather than discarded silently. The coordinate-invariant normalized Hessian is

$$\widehat{\mathcal{H}} = G^{-1/2} \mathcal{H} G^{-1/2}. \quad (3567)$$

Its ordered eigenvalues $\lambda_1 \leq \dots \leq \lambda_m$ are the principal kinetic responses. We record

$$\begin{aligned}\Lambda_{\max} &= \max_j |\lambda_j|, \\ \bar{\lambda} &= \frac{1}{m} \sum_{j=1}^m \lambda_j,\end{aligned}\quad (3568)$$

and

$$\mathcal{A}_{\mathcal{K}} = \left\| \widehat{\mathcal{K}} - \bar{\lambda} I \right\|_{\text{op}}. \quad (3569)$$

A trace zero is not classified as complete cancellation unless $\Lambda_{\max}$ is compatible with zero. In an ordinary Euclidean momentum coordinate, a nonzero principal curvature defines

$$m_j^* = \frac{\hbar^2}{\lambda_j}. \quad (3570)$$

In a general Hodge coordinate, we report λ_j directly and use the term effective mass only after a physical momentum-length normalization has been declared. Analytic or automatic derivatives of the assembled Hamiltonian are used whenever available. Finite differences provide an independent analysis. For a scalar or matrix-valued function $F(x)$, the fourth-order centered first derivative is

$$D_h^{(1)} F = \frac{-F(x+2h) + 8F(x+h) - 8F(x-h) + F(x-2h)}{12h}. \quad (3571)$$

The fourth-order centered second derivative is

$$D_h^{(2)} F = \frac{-F(x+2h) + 16F(x+h) - 30F(x) + 16F(x-h) - F(x-2h)}{12h^2}. \quad (3572)$$

For two coordinates,

$$D_{h_a, h_b}^{(ab)} F = \frac{F_{++} - F_{+-} - F_{-+} + F_{--}}{4h_a h_b}, \quad (3573)$$

where

$$F_{\pm\pm} = F(x_a \pm h_a, x_b \pm h_b). \quad (3574)$$

For a fourth-order stencil, the Richardson-extrapolated value is

$$D_R = \frac{16D_{h/2} - D_h}{15}. \quad (3575)$$

The step-refinement estimate is

$$\varepsilon_{\text{diff}} = \frac{\|D_{h/2} - D_h\|}{15}. \quad (3576)$$

We calculate a geometric sequence of steps and retain only a stability plateau on which: the derivative changes by less than its tolerance; the target projector rank is constant; the isolation contour remains open; round-off growth has not begun; No derivative is evaluated across a

target switch, gap closure, fold, cusp, or change of active extremizer. Define

$$\begin{aligned}V_{1,C} &= \hbar D_G \Omega_{\max,C}, \\ V_{2,C} &= \frac{1}{2} D_G^2 \Lambda_{\max,C},\end{aligned}\quad (3577)$$

and

$$X_C = E_{\text{kin}} \varrho_{C,\eta}^{\text{coh,max}}. \quad (3578)$$

The minimal operational score is

$$\mathcal{M}_C^{(1)} = \left[\frac{\Delta_C}{\Delta_C + W_C} \frac{\Delta_C}{\Delta_C + V_{1,C}} \times \frac{X_C}{1 + X_C} C_C^{\text{coh}} \right]^{1/4}.$$

The strong score is

$$\begin{aligned}\mathcal{M}_C^{(2)} &= \left[\frac{\Delta_C}{\Delta_C + W_C} \frac{\Delta_C}{\Delta_C + V_{1,C}} \frac{\Delta_C}{\Delta_C + V_{2,C}} \right. \\ &\quad \left. \times \frac{X_C}{1 + X_C} C_C^{\text{coh}} \right]^{1/5}.\end{aligned}$$

The selected score and every threshold are fixed before the $(K, \theta, w/t)$ landscape is evaluated.

F. DOS calculations

For a completely diagonalized D -dimensional Hermitian matrix, the normalized empirical spectral measure is

$$d\nu_D(E) = \frac{1}{D} \sum_{j=1}^D \delta(E - E_j) dE. \quad (3579)$$

A broadened density is

$$\varrho_\eta(E) = \frac{1}{D} \sum_{j=1}^D K_\eta(E - E_j), \quad (3580)$$

with either the Lorentzian kernel

$$K_\eta^L(x) = \frac{1}{\pi} \frac{\eta}{x^2 + \eta^2}, \quad (3581)$$

or the Gaussian kernel

$$K_\eta^G(x) = \frac{1}{\sqrt{2\pi}\eta} e^{-x^2/2\eta^2}. \quad (3582)$$

The kernel type is kept fixed across compared covers and methods. Let established spectral bounds satisfy

$$E_{\min}^L \leq \text{Spec } H \leq E_{\max}^U. \quad (3583)$$

We define

$$c_E = \frac{E_{\max}^U + E_{\min}^L}{2}, \quad (3584)$$

and choose a safety factor $0 < \delta_K < 1$:

$$a_E = \frac{E_{\max}^U - E_{\min}^L}{2(1 - \delta_K)}. \quad (3585)$$

The rescaled Hamiltonian is

$$\tilde{H} = \frac{H - c_E I}{a_E}, \quad (3586)$$

so that

$$\text{Spec } \tilde{H} \subset [-1 + \delta_K, 1 - \delta_K]. \quad (3587)$$

The normalized Chebyshev moments are

$$\mu_m = \frac{1}{D} \text{Tr } T_m(\tilde{H}), \quad m = 0, \dots, M. \quad (3588)$$

For a normalized random vector $|r_s\rangle$ satisfying

$$\mathbb{E}[|r_s\rangle\langle r_s|] = \frac{1}{D} I, \quad (3589)$$

we estimate

$$\hat{\mu}_m = \frac{1}{N_r} \sum_{s=1}^{N_r} \langle r_s | T_m(\tilde{H}) | r_s \rangle. \quad (3590)$$

For each vector, the Chebyshev recursion is

$$|t_0^{(s)}\rangle = |r_s\rangle, \quad |t_1^{(s)}\rangle = \tilde{H}|r_s\rangle, \quad (3591)$$

and

$$|t_{m+1}^{(s)}\rangle = 2\tilde{H}|t_m^{(s)}\rangle - |t_{m-1}^{(s)}\rangle. \quad (3592)$$

For highest polynomial order M , define

$$\vartheta_M = \frac{\pi}{M+2}. \quad (3593)$$

We use the Jackson coefficients

$$g_m^J = \frac{(M - m + 2) \cos(m\vartheta_M) + \sin(m\vartheta_M) \cot \vartheta_M}{M + 2}, \quad (3594)$$

for $m = 0, \dots, M$. The rescaled DOS is

$$\tilde{\varrho}_M(x) = \frac{1}{\pi\sqrt{1-x^2}} \left[g_0^J \hat{\mu}_0 + 2 \sum_{m=1}^M g_m^J \hat{\mu}_m T_m(x) \right]. \quad (3595)$$

The physical density is

$$\varrho_M(E) = \frac{1}{a_E} \tilde{\varrho}_M\left(\frac{E - c_E}{a_E}\right). \quad (3596)$$

For each moment, define the sample value

$$x_{s,m} = \langle r_s | T_m(\tilde{H}) | r_s \rangle. \quad (3597)$$

The empirical standard error is

$$\text{SE}(\hat{\mu}_m) = \left[\frac{1}{N_r(N_r - 1)} \sum_{s=1}^{N_r} |x_{s,m} - \hat{\mu}_m|^2 \right]^{1/2}. \quad (3598)$$

We repeat the calculation under $M \mapsto 2M$ and $N_r \mapsto 2N_r$. Polynomial-order and stochastic convergence are accepted separately. For each normalized random vector $|r_s\rangle$, we run an m_L -step fully reorthogonalized Lanczos process:

$$H Q_s = Q_s T_s + \beta_{m_L+1} |q_{m_L+1}\rangle \mathbf{e}_{m_L}^\top. \quad (3599)$$

Diagonalize

$$T_s = Y_s \text{diag}(\vartheta_{s1}, \dots, \vartheta_{sm_L}) Y_s^\dagger. \quad (3600)$$

The quadrature weights are

$$w_{sj} = |(Y_s)_{1j}|^2, \quad \sum_j w_{sj} = 1. \quad (3601)$$

The SLQ spectral measure is

$$d\hat{\nu}_{\text{SLQ}}(E) = \frac{1}{N_r} \sum_{s=1}^{N_r} \sum_{j=1}^{m_L} w_{sj} \delta(E - \vartheta_{sj}) dE. \quad (3602)$$

The corresponding cumulative distribution is

$$F_{\text{SLQ}}(E) = \frac{1}{N_r} \sum_{s,j} w_{sj} \mathbf{1}[\vartheta_{sj} \leq E]. \quad (3603)$$

The tridiagonal depth and number of probe vectors are doubled independently. The KPM cumulative distribution is obtained by numerically integrating the normalized KPM density:

$$F_{\text{KPM}}(E) = \int_{-\infty}^E \varrho_{\text{KPM}}(E') dE'. \quad (3604)$$

The independent method discrepancy is

$$\kappa_{\text{method}} = \sup_E |F_{\text{KPM}}(E) - F_{\text{SLQ}}(E)|. \quad (3605)$$

For cover N , let F_{ref} be the specified deeper cover, cross-tower, or independently established reference CDF. We define

$$\kappa_N = \sup_E |F_N(E) - F_{\text{ref}}(E)|. \quad (3606)$$

The fixed finite-broadening schedule is

$$\eta_N = \sqrt{\kappa_N} \quad (3607)$$

in the dimensionless spectral convention used by the validation run. We use this schedule only to test finite-broadening consistency. Convergence of ϱ_{η_N} is not promoted to pointwise unsmoothed DOS convergence without a separate tower-uniform regularity theorem. For site i , the local density of states is

$$\varrho_i(E) = \langle i | \delta(E - H) | i \rangle. \quad (3608)$$

For cell projector Π_x , the normalized cell density is

$$\varrho_x(E) = \frac{1}{\nu} \text{Tr} [\Pi_x \delta(E - H)]. \quad (3609)$$

For a target projector P_C ,

$$\varrho_C(E) = \frac{1}{r_C} \text{Tr} [P_C \delta(E - H)]. \quad (3610)$$

Let $c_n \in [0, 1]$ be a specified coherence weight obtained from the projector, layer, or cell-coherence definition. The coherence-weighted density is

$$\varrho_C^{\text{coh}}(E) = \frac{\sum_{n \in C} c_n \delta(E - E_n)}{\sum_{n \in C} c_n}. \quad (3611)$$

The same weights are used at every cover and every parameter point. They are not refitted to sharpen a peak.

G. Green-function and qubit-interaction calculations

For a lossy photonic Hamiltonian, define

$$A(\Omega) = \Omega I - H_\gamma + \frac{i}{2} \Gamma_\gamma. \quad (3612)$$

The retarded Green operator is

$$G_\gamma^R(\Omega) = A(\Omega)^{-1}. \quad (3613)$$

We do not form the dense inverse unless the matrix is small. For a matrix of required right-hand sides

$$B = (\mathbf{b}_1 \ \cdots \ \mathbf{b}_s), \quad (3614)$$

we solve

$$A(\Omega)X(\Omega) = B. \quad (3615)$$

Sparse direct factorization is used when fill-in remains tractable. Otherwise, a preconditioned complex Krylov solver is used. The linear-system residual is

$$R_{\text{lin}} = B - A\tilde{X}. \quad (3616)$$

If a established lower singular-value bound

$$\sigma_{\min} \leq \sigma_{\min}[A(\Omega)] \quad (3617)$$

is available, then

$$\|\tilde{X} - A^{-1}B\| \leq \frac{\|R_{\text{lin}}\|}{\sigma_{\min}}. \quad (3618)$$

For nonnormal matrices, a distance to the eigenvalue set is not used as a substitute for $\sigma_{\min}$. Let

$$K \in \mathbb{C}^{D_\gamma \times M} \quad (3619)$$

contain the calibrated port-coupling vectors. The external damping is $\Gamma_{\text{ext}} = KK^\dagger$. After direct-path de-embedding, the scattering matrix is

$$S(\Omega) = I_M - iK^\dagger G_\gamma^R(\Omega)K. \quad (3620)$$

It is evaluated by solving

$$A(\Omega)X_K = K \quad (3621)$$

and setting

$$S = I_M - iK^\dagger X_K. \quad (3622)$$

Let the transmon placement matrix be

$$V_q \in \mathbb{C}^{N_q \times D_\gamma}. \quad (3623)$$

At qubit frequency ω_q , solve

$$A(\omega_q)X_q = V_q^\dagger. \quad (3624)$$

The exact retarded qubit self-energy is

$$\Sigma_q^R(\omega_q) = V_q X_q = V_q G_\gamma^R(\omega_q) V_q^\dagger. \quad (3625)$$

We decompose it as

$$\Sigma_q^R = J - \frac{i}{2} \Gamma^{\text{med}}, \quad (3626)$$

with

$$J = \frac{1}{2} (\Sigma_q^R + \Sigma_q^{R\dagger}), \quad (3627)$$

and

$$\Gamma^{\text{med}} = i (\Sigma_q^R - \Sigma_q^{R\dagger}). \quad (3628)$$

We verify

$$\Gamma^{\text{med}} \succeq -\varepsilon_\Gamma I. \quad (3629)$$

For local placements

$$\langle v_a | = g_a \langle i_a |, \quad (3630)$$

the matrix element is

$$[\Sigma_q^R]_{ab} = g_a g_b^* G_{i_a i_b}^R(\omega_q). \quad (3631)$$

On small systems, we diagonalize

$$H_\gamma |u_n\rangle = E_n |u_n\rangle \quad (3632)$$

and calculate

$$[\Sigma_q(z)]_{ab} = \sum_n \frac{g_{an} g_{bn}^*}{z - E_n}, \quad (3633)$$

where

$$g_{an} = \langle v_a | u_n \rangle. \quad (3634)$$

The direct mode sum and sparse linear solve must agree within the combined eigenpair and linear-solver uncertainty. Let P_C be an isolated photonic target with midpoint $\bar{E}_C$ and half-width

$$r_C = \frac{W_C}{2}. \quad (3635)$$

Define

$$\Delta_{qC} = \omega_q - \bar{E}_C. \quad (3636)$$

When $|\Delta_{qC}| > r_C$, we compare the exact target contribution with

$$\Sigma_{q,C}^{\text{proj}} = \frac{\mathbf{V}_q P_C \mathbf{V}_q^\dagger}{\Delta_{qC}}. \quad (3637)$$

The finite-bandwidth remainder satisfies

$$\left\| \Sigma_{q,C} - \Sigma_{q,C}^{\text{proj}} \right\| \leq \|\mathbf{V}_q\|^2 \frac{r_C}{|\Delta_{qC}| (|\Delta_{qC}| - r_C)}. \quad (3638)$$

Let

$$\delta_{q\bar{C}} = \text{dist}[\omega_q, \text{Spec}(Q_C H_\gamma Q_C)]. \quad (3639)$$

The complementary contribution obeys

$$\|\Sigma_{q,\bar{C}}\| \leq \frac{\|\mathbf{V}_q\|^2}{\delta_{q\bar{C}}}. \quad (3640)$$

For transmon positions x_a , define

$$r_{ab} = d_{\mathbb{H}}(x_a, x_b). \quad (3641)$$

The quadratic interaction range is

$$R_J^2 = \frac{\sum_{a \neq b} r_{ab}^2 |J_{ab}|^2}{\sum_{a \neq b} |J_{ab}|^2}. \quad (3642)$$

The participation number is

$$P_J = \frac{\left(\sum_{a \neq b} |J_{ab}|^2 \right)^2}{\sum_{a \neq b} |J_{ab}|^4}. \quad (3643)$$

For an oriented cycle $\mathcal{C} = (a_1, \dots, a_m, a_1)$, the gauge-invariant interaction flux is

$$\Phi_{\mathcal{C}}^J = \arg \prod_{j=1}^m J_{a_j a_{j+1}} \mod 2\pi. \quad (3644)$$

Let

$$\mathbf{A}(\lambda) \mathbf{X}(\lambda) = \mathbf{B}(\lambda). \quad (3645)$$

Differentiation gives

$$\mathbf{A} \mathbf{X}_\lambda = \mathbf{B}_\lambda - \mathbf{A}_\lambda \mathbf{X}. \quad (3646)$$

The second derivative satisfies

$$\mathbf{A} \mathbf{X}_{\lambda\lambda} = \mathbf{B}_{\lambda\lambda} - \mathbf{A}_{\lambda\lambda} \mathbf{X} - 2\mathbf{A}_\lambda \mathbf{X}_\lambda. \quad (3647)$$

We reuse the same factorization or preconditioner at each frequency. This avoids finite-differencing a densely formed inverse.

H. Convergence and uncertainty analysis

For derivative order $r = 0, 1, 2$, we write the Hamiltonian-level budget as

$$\begin{aligned} \varepsilon_H^{(r)} &= \varepsilon_{\text{model}}^{(r)} + \varepsilon_{\text{cut}}^{(r)} + \varepsilon_{\text{angle}}^{(r)} \\ &\quad + \varepsilon_{\text{cover}}^{(r)} + \varepsilon_{\text{assembly}}^{(r)} + \varepsilon_{\text{fp}}^{(r)}. \end{aligned} \quad (3648)$$

Solver error, stochastic-trace error, differentiation error, and statistical fitting error are stored separately:

$$\varepsilon_{\text{num}} = \varepsilon_{\text{solver}} + \varepsilon_{\text{stoch}} + \varepsilon_{\text{diff}} + \varepsilon_{\text{fit}}. \quad (3649)$$

No cancellation between independently bounded error channels is assumed. Let L_H be the physical propagation radius. An observable supported inside radius L_{obs} is exactly represented on a normal finite cover whenever

$$r_{\text{inj},N} > L_{\text{obs}} + L_H. \quad (3650)$$

A product of m hopping operators propagates at most mL_H . Therefore the m -th normalized spectral moment is exact when

$$2r_{\text{inj},N} > mL_H. \quad (3651)$$

We record the aliasing margin

$$\mathbf{a}_N^{(m)} = 2r_{\text{inj},N} - mL_H. \quad (3652)$$

Let

$$\Sigma_\infty = \text{Spec } H_\infty \quad (3653)$$

and

$$S_N = \text{Spec } H_N^{\text{ret}}. \quad (3654)$$

The no-loss error is

$$u_N = \sup_{\lambda \in \Sigma_\infty} \text{dist}(\lambda, S_N). \quad (3655)$$

The no-pollution error is

$$b_N = \sup_{s \in S_N} \text{dist}(s, \Sigma_\infty). \quad (3656)$$

The two-sided spectral error is

$$\eta_N^{\text{spec}} = \max\{u_N, b_N\}. \quad (3657)$$

For the long-range calculation, the supported budget is

$$\eta_N^{\text{supported}} = u_N^{(L_N)} + b_N^{(L_N)} + \tau_\infty(L_N) + \tau_N(L_N) + \varepsilon_{\text{solver},N}. \quad (3658)$$

If b_N is unresolved, the result is not reported as strong full-spectrum convergence even when weak spectral measures agree. For each cover, let $r_{\text{inj},N}^L$ be the established injectivity-radius lower bound. The specified shell law is

$$L_N = \left\lfloor \sqrt{r_{\text{inj},N}^L} \right\rfloor. \quad (3659)$$

It is designed to satisfy $L_N \rightarrow \infty$, and

$$\frac{L_N}{r_{\text{inj},N}^L} \rightarrow 0. \quad (3660)$$

We evaluate

$$\varepsilon_N^{(0)}(L_N), \quad \varepsilon_N^{(1)}(L_N), \quad \varepsilon_N^{(2)}(L_N) \quad (3661)$$

independently. A decreasing but nonzero extrapolated floor is retained explicitly in the error budget. For a Hermitian perturbation

$$\tilde{H} = H + \Delta H, \quad \|\Delta H\| \leq \varepsilon_H^{(0)}, \quad (3662)$$

the spectral Hausdorff error satisfies

$$d_H(\text{Spec } \tilde{H}, \text{Spec } H) \leq \varepsilon_H^{(0)}. \quad (3663)$$

For a matched target island,

$$|\tilde{W}_C - W_C| \leq 2\varepsilon_H^{(0)}. \quad (3664)$$

The gap obeys

$$\tilde{\Delta}_C \geq \Delta_C - 2\varepsilon_H^{(0)}. \quad (3665)$$

For a contour of length $\ell_{\mathcal{C}}$ at distance $\delta_{\mathcal{C}}$ from the unperturbed spectrum, if

$$\varepsilon_H^{(0)} < \delta_{\mathcal{C}}, \quad (3666)$$

then

$$\|\tilde{P}_C - P_C\| \leq \frac{\ell_{\mathcal{C}}}{2\pi} \frac{\varepsilon_H^{(0)}}{\delta_{\mathcal{C}}(\delta_{\mathcal{C}} - \varepsilon_H^{(0)})}. \quad (3667)$$

Suppose the implemented matrix is

$$\tilde{H} = H + \Delta H \quad (3668)$$

and a normalized numerical state satisfies

$$\tilde{H}|\tilde{\psi}\rangle = \tilde{E}|\tilde{\psi}\rangle + |\mathbf{r}\rangle. \quad (3669)$$

Then

$$\text{dist}(\tilde{E}, \text{Spec } H) \leq \|\Delta H\| + \|\mathbf{r}\|. \quad (3670)$$

We define

$$\varepsilon_E = \varepsilon_H^{(0)} + \varepsilon_{\text{solver}}. \quad (3671)$$

A finite nonzero bandwidth claim requires $W_C \gg 2\varepsilon_E$. An exact-flatness claim instead requires an analytical identity or symmetry mechanism forcing $W_C = 0$. For an angular error

$$\delta\theta_N = \theta_N - \theta, \quad (3672)$$

the exact geometric displacement at radius L is

$$\varepsilon_{\text{geo}}(L, \delta\theta_N) = 2R \text{arsinh}[\sinh(L/R) |\sin(\delta\theta_N/2)|]. \quad (3673)$$

If the hopping kernel is Lipschitz with constant Λ_t and the local channel count is bounded by z_{eff} , we use

$$\varepsilon_{\text{angle}}^{(0)} \leq z_{\text{eff}} \Lambda_t \varepsilon_{\text{geo}}. \quad (3674)$$

For $L \gg R$ and small $\delta\theta_N$,

$$\varepsilon_{\text{geo}} \simeq \frac{R}{2} e^{L/R} |\delta\theta_N|. \quad (3675)$$

We use the implication hierarchy

C^0 operator control $\rightarrow$ energies, spectral sets, bandwidths, gaps,

C^1 operator control $\rightarrow$ velocities and stationary equations,

C^2 operator control $\rightarrow$ Hessians, effective masses, Hodge tensors.

(3676)
No C^1 or C^2 conclusion is inferred from a decreasing C^0 error. Let a root be defined by $F(\theta_*) = 0$ with

$$|F'(\theta)| \geq s_F > 0 \quad (3677)$$

on a specified interval. If

$$\|\tilde{F} - F\|_{\infty} \leq \varepsilon_F \quad (3678)$$

and

$$\|\tilde{F}' - F'\|_{\infty} \leq \varepsilon_{F'} < s_F, \quad (3679)$$

then

$$|\tilde{\theta}_* - \theta_*| \leq \frac{\varepsilon_F}{s_F - \varepsilon_{F'}}. \quad (3680)$$

For a unique strict maximum of

$$L(\theta) = \ln \mathcal{M}(\theta) \quad (3681)$$

satisfying

$$-L''(\theta) \geq \mu_{\theta} > 0, \quad (3682)$$

and a uniform score error

$$\|\tilde{L} - L\|_{\infty} \leq \varepsilon_L, \quad (3683)$$

the optimizer obeys

$$|\tilde{\theta}_* - \theta_*| \leq 2\sqrt{\frac{\varepsilon_L}{\mu_{\theta}}}. \quad (3684)$$

For observable O_N , define the successive-cover residual

$$\Delta_{O,N} = \frac{|O_{N+1} - O_N|}{\max\{O_{\text{ref}}, |O_{N+1}|, |O_N|\}}. \quad (3685)$$

For inequivalent towers A and B ,

$$\Delta_{O,N,M}^{A,B} = \frac{|O_N^A - O_M^B|}{\max\{O_{\text{ref}}, |O_N^A|, |O_M^B|\}}. \quad (3686)$$

Covers are matched primarily by injectivity radius, not solely by matrix dimension. A bulk result requires convergence on at least two inequivalent towers or an independently proved tower-independence theorem.

TABLE XV. Error topology required by the numerical observables. A stronger row does not follow from a weaker row without an independently converged derivative or regularity estimate.

Observable	Required control	Primary analysis
Eigenvalue and spectral edge	C^0 operator error	Original-Hamiltonian residual and Weyl bound
Bandwidth and isolation gap	Two-sided spectral and target-island control	No loss, no pollution, fixed target
Projector and coherence	Uniform contour isolation	Riesz error and rank preservation
Velocity	C^1 operator convergence	Analytic-finite-difference agreement
Hessian and effective mass	C^2 operator convergence	Principal-value and Richardson stability
Weak DOS	Moment or CDF convergence	KPM-SLQ and cross-tower CDF tests
Unsmoothed DOS	Tower-uniform local regularity	Not inferred from weak convergence
Green function	Resolvent-set or singular-value margin	Linear-system residual
Qubit interaction	Green-function and placement convergence	Direct solve versus mode-sum cross-check

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